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# Proven C Library Complete Manual (v0.11.0)

The manual of proven, a C23 systems library. Every chapter is one file under `manual/`; the Korean edition mirrors it under `manual-ko/`.

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- [Appendix A - Alias index](#)

## Copyright

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- Repository: [github.com/rubidus-api/proven\\_c\\_lib](https://github.com/rubidus-api/proven_c_lib)
- Edition: v0.11.0
- License: MIT - Copyright (c) 2026 rubidus-api. The library and this manual are under the same license; the text is in `LICENSE` at the repository root.
- Every code example in this manual is compiled, and the quoted ones are run, by the build.

## Tutorial: the library in nine short programs

**Part I - Start here.** No prerequisites beyond one introductory C book. **After lesson 6** you can read the greeting program in [Chapter 0](#) line by line, and the reference chapters stop looking like a wall of new words. **After lesson 9** you have used an arena, a growable container and a file - the first things most programs need once they can handle text.

### Why this exists

Chapter 0 shows a working program on its first page. That program is thirty lines long and every one of them is a new idea: an allocator you pass in, a result you have to check, a view that carries its length, an append that refuses rather than truncates, a destroy paired with the allocator that created it. If you have finished one C book and no more, five new ideas arriving together is four too many.

So this tutorial takes them one at a time. Nine programs, each introducing exactly one thing. The first six lead up to the Chapter 0 program; the last three go past it, into arenas, containers and files. Every one of them is a real file under `manual/examples/` that the build compiles and runs, so what you read here is what actually ran.

Build any of them the way you build any C program - the library is source you compile with your own code, so there is nothing to install:

```
cc -std=c23 -Iinclude your_program.c src/proven/*.c platform/*.c -o your_program
```

If that line is unfamiliar, read [Chapter 0 section 4](#) first and come back.

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## Lesson 1 - printing, and a build that works

**The one new thing:** `proven_println`, and the fact that its arguments are checked.

`printf("%d", 3.5)` compiles. It is wrong, and on a bad day it prints garbage or crashes, because the format string is a *string* - nothing connects it to the arguments that follow. `proven_println` uses `{}` as the placeholder and wraps each argument in `PROVEN_ARG`, which records the argument's type. The two cannot disagree, because the type travels with the value.

```
/*
 * Lesson 1 - get one line to build and run.
 *
 * Nothing here is about the library's ideas yet. The only question this program
 * answers is "does my build command work?", and it is worth answering on its
 * own, because every later lesson assumes it.
 *
 * printf(fmt, ...) is not checked: the compiler cannot tell you that "%d" was
 * handed a double. proven_println checks, because every argument is wrapped by
 * PROVEN_ARG and carries its own type.
```

```

*/

int main(void) {
    proven_println("hello from proven");

    /* {} is the placeholder. The value goes through PROVEN_ARG, which records
     * what type it is, so the format string and the argument cannot disagree. */
    proven_println("one number: {}", PROVEN_ARG(42));
    proven_println("two of them: {} and {}", PROVEN_ARG(1), PROVEN_ARG(2));

    return EXAMPLE_OK();
}

```

Running it prints three lines. If it built and ran, your include path and your source list are right, and everything after this is about the library rather than about your build.

**Common first stumble.** {} is not %s. There is no letter to get wrong, and there is nothing to remember about long versus int: PROVEN\_ARG knows.

---

## Lesson 2 - text that knows how long it is

**The one new thing:** the *view* - a pointer and a size that travel together.

In the C you know, a string is a pointer, and its length is wherever the first zero byte happens to be. Every function that touches it has to walk the bytes to find out how much there is; if the zero is missing, it walks off the end. That single design decision is behind a large share of the security advisories in the language's history.

A view makes the missing half explicit. It does not own the bytes - it borrows them, and it never frees anything.

```

/*
 * Lesson 2 - text that knows how long it is.
 *
 * In the C you already know, a string is a pointer and the length is wherever
 * the first zero byte happens to be. Every function has to walk the bytes to
 * find out how much there is, and if the zero is missing it walks off the end.
 *
 * A view is the pair that C leaves implicit: a pointer AND a size, travelling
 * together. It borrows - it does not own the bytes and never frees them.
 */

int main(void) {
    /* PROVEN_LIT makes a view from a literal. The size is computed at compile
     * time, so no scan happens here - unlike strlen. */
    proven_u8str_view_t hello = PROVEN_LIT("hello");

    EXAMPLE_REQUIRE(hello.size == 5, "the view already knows its own length");
    EXAMPLE_REQUIRE(hello.ptr != NULL, "and it points at the literal's bytes");

    /* Because the length travels with the pointer, comparing is a size check
     * plus a memcmp. No walking, and no way to run off the end. */
    EXAMPLE_REQUIRE(proven_u8str_view_eq(hello, PROVEN_LIT("hello")), "same text");
    EXAMPLE_REQUIRE(!proven_u8str_view_eq(hello, PROVEN_LIT("hell")), "shorter text differs");

    /* A view can name PART of something without copying it. Here is the middle
     * of the literal - still borrowed, still no allocation. */
    proven_u8str_view_t ell = { .ptr = hello.ptr + 1, .size = 3 };
    EXAMPLE_REQUIRE(proven_u8str_view_eq(ell, PROVEN_LIT("ell")), "a window onto the same bytes");
}

```

```

    proven_println("whole: {} (size {})", PROVEN_ARG(hello), PROVEN_ARG(hello.size));
    proven_println("part : {} (size {})", PROVEN_ARG(ell), PROVEN_ARG(ell.size));

    return EXAMPLE_OK();
}

```

**What to notice.** `PROVEN_LIT` computes the size at compile time, so it is not `strlen`. And the third view in the program names the *middle* of the literal without copying it: a view can describe part of something, which is why splitting and parsing in this library allocate nothing.

**Common first stumble.** A view is borrowed. If the bytes it points at go away - a local buffer that leaves scope, a string you destroyed - the view is dangling, exactly like any other C pointer. The library never extends a lifetime behind your back.

## Lesson 3 - a call that can fail says so

**The one new thing:** `proven_err_t`, and refusal instead of truncation.

C reports failure in three unrelated ways: a magic return value, a null pointer, or the global `errno` that the next call overwrites. All three are easy not to look at, and nothing complains when you do not. Here, a call that can fail returns an error *value* - and these functions are marked `[[nodiscard]]`, so throwing it away is a compiler warning rather than a habit.

```

/*
 * Lesson 3 - a call that can fail says so, in its return value.
 *
 * The C you know reports failure in three different ways: a magic return value
 * (-1), a null pointer, or a global called errno that the next call overwrites.
 * All three are easy to not look at, and nothing complains when you don't.
 *
 * Here a fallible call returns proven_err_t. It is an ordinary value: you can
 * store it, compare it, and pass it upward. And [[nodiscard]] on these
 * functions means ignoring it is a compiler warning, not a habit.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* Room for exactly 8 bytes. (Chapter 2 is about where that room comes
     * from; for now, notice only that we said how much.) */
    proven_result_u8str_t s = proven_u8str_create(alloc, 8);
    EXAMPLE_REQUIRE(proven_is_ok(s.err), "8 bytes should be available");

    /* This fits. */
    proven_err_t err = proven_u8str_append(&s.value, PROVEN_LIT("12345678"));
    EXAMPLE_REQUIRE(proven_is_ok(err), "eight bytes into eight bytes fits exactly");

    /* This does not - and the library REFUSES it. It does not append the part
     * that would fit, because half a word is not a shorter word, it is a
     * different one. */
    proven_err_t too_much = proven_u8str_append(&s.value, PROVEN_LIT("9"));
    EXAMPLE_REQUIRE(!proven_is_ok(too_much), "one byte more than capacity must fail");
    EXAMPLE_REQUIRE(too_much == PROVEN_ERR_OUT_OF_BOUNDS, "and it says why: out of bounds");

    /* The refusal changed nothing. The string is exactly as it was, which is
     * what "failure-atomic" means: a failed call leaves no half-done state. */
    EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&s.value), PROVEN_LIT("12345678")),
        "the refused append must not have written anything");

    proven_println("after the refusal, the string is still: {}",

```

```

        PROVEN_ARG(proven_u8str_as_view(&s.value)));

    proven_u8str_destroy(alloc, &s.value);
    return EXAMPLE_OK();
}

```

**What to notice.** The append that does not fit changes *nothing*. It does not store the part that would have fit. This is called failure atomicity, and the reason for it is that a truncated string is not a shorter message, it is a different one - a truncated path names a different file, and a truncated command is a different command.

**Common first stumble.** `proven_is_ok(err)` is the check; `err == PROVEN_OK` says the same thing. What you must not do is ignore it and read the value anyway.

---

## Lesson 4 - the value and the error arrive together

**The one new thing:** result structs - `.err` and `.value` in one return.

When a call has nothing to give back, an error code is enough. When it does have something, the library returns both in one small struct, and the rule is one sentence: **value means nothing until you have looked at err.**

```

/*
 * Lesson 4 - when the call has a value to give back, the value and the error
 * arrive together.
 *
 * proven_err_t alone is enough when there is nothing to return. When there IS
 * something, you get a small struct with two fields: `err` and `value`. The
 * rule is one sentence: `value` means nothing until you have looked at `err`.
 *
 * This is the same discipline as checking malloc for NULL, except the checking
 * place is part of the type instead of a convention you have to remember.
 */

/* A function of your own can return one too. Nothing in the library is magic. */
static proven_result_size_t safe_div(proven_size_t a, proven_size_t b) {
    proven_result_size_t res = {0};
    if (b == 0) {
        res.err = PROVEN_ERR_INVALID_ARG;    /* value stays 0 - and means nothing */
        return res;
    }
    res.err = PROVEN_OK;
    res.value = a / b;
    return res;
}

int main(void) {
    proven_result_size_t ok = safe_div(10, 2);
    EXAMPLE_REQUIRE(proven_is_ok(ok.err), "dividing by 2 is fine");
    EXAMPLE_REQUIRE(ok.value == 5, "and only now may we read the value");

    proven_result_size_t bad = safe_div(10, 0);
    EXAMPLE_REQUIRE(!proven_is_ok(bad.err), "dividing by zero must fail");
    EXAMPLE_REQUIRE(bad.err == PROVEN_ERR_INVALID_ARG, "and say which rule was broken");
    /* bad.value is 0, but that is not an answer. It is the absence of one. */

    /* The library's own calls have the same shape. proven_u8str_create hands
     * back the string it made together with the error that guards it. */
    proven_allocator_t alloc = proven_heap_allocator();
    proven_result_u8str_t s = proven_u8str_create(alloc, 16);
}

```

```

    if (!proven_is_ok(s.err)) return 1;      /* nothing was created; nothing to destroy */

    proven_println("10 / 2 = {}", PROVEN_ARG(ok.value));

    proven_u8str_destroy(alloc, &s.value);
    return EXAMPLE_OK();
}

```

**What to notice.** `safe_div` is *your* function, not the library's. The pattern is a plain struct return; there is no machinery underneath, which is why you can use it in your own code the moment you have read this page.

**Common first stumble.** On failure the `value` field usually holds a zero. That zero is not an answer - it is the absence of one. Checking `err` is what tells them apart.

---

## Lesson 5 - who gives out the memory is an argument

**The one new thing:** `proven_allocator_t`, passed in rather than assumed.

`malloc` is a decision made for you: one heap, one strategy, and nothing at the call site says so. In this library, anything that needs memory takes an allocator and uses only that one. Two things follow, and both are the point: you can always answer "who allocated this?" by reading the call, and you can hand the same code a different allocator without changing it.

```

/*
 * Lesson 5 - who gives out the memory is an argument, not a global.
 *
 * malloc is a global decision made for you: one heap, one strategy, invisible
 * at the call site. Here, anything that needs memory takes a
 * proven_allocator_t and uses only that. Two consequences follow, and both are
 * the point.
 *
 * - You can always answer "who allocated this?" by looking at the call.
 * - You can hand a different allocator to the same code without changing it.
 */

/* This function does not know or care where the memory comes from. */
static proven_result_u8str_t make_greeting(proven_allocator_t alloc,
                                           proven_u8str_view_t name) {
    proven_result_u8str_t out = proven_u8str_create(alloc, 64);
    if (!proven_is_ok(out.err)) return out;

    proven_err_t err = proven_u8str_append(&out.value, PROVEN_LIT("hello, "));
    if (proven_is_ok(err)) err = proven_u8str_append(&out.value, name);
    if (!proven_is_ok(err)) {
        proven_u8str_destroy(alloc, &out.value); /* undo what we made */
        out.err = err;
        out.value = (proven_u8str_t){0};
    }
    return out;
}

int main(void) {
    /* (a) the ordinary heap - malloc and free underneath */
    proven_allocator_t heap = proven_heap_allocator();

    proven_result_u8str_t a = make_greeting(heap, PROVEN_LIT("world"));
    EXAMPLE_REQUIRE(proven_is_ok(a.err), "the heap should be able to give 64 bytes");
    proven_println("from the heap : {}", PROVEN_ARG(proven_u8str_as_view(&a.value)));
    /* Destroyed with the SAME allocator that created it. That pairing is the

```

```

    * whole ownership rule of this library. */
    proven_u8str_destroy(heap, &a.value);

    /* (b) an arena - one block of memory handed out in order, freed all at once.
     * Note that make_greeting above did not change at all. */
    alignas(PROVEN_MAX_ALIGN) proven_byte_t backing[512];
    proven_arena_t arena = proven_arena_create((proven_mem_mut_t){
        .ptr = backing, .size = sizeof backing });
    proven_allocator_t from_arena = proven_arena_as_allocator(&arena);

    proven_result_u8str_t b = make_greeting(from_arena, PROVEN_LIT("arena"));
    EXAMPLE_REQUIRE(proven_is_ok(b.err), "the arena has room for this too");
    proven_println("from an arena : {}", PROVEN_ARG(proven_u8str_as_view(&b.value)));

    /* No destroy loop here: an arena frees everything by being reset. That is
     * lesson 7's subject; the point now is only that the caller chose. */
    proven_arena_reset(&arena);

    return EXAMPLE_OK();
}

```

**What to notice.** `make_greeting` is written once and runs against the heap and against an arena – a block of memory you own, handed out by bumping a pointer, and released all at once. The function did not change, and nothing global was configured.

**The ownership rule, in one line:** whatever created it destroys it, with the *same* allocator.

**Common first stumble.** With an arena, `destroy` reclaims nothing - an arena frees by being reset. Call it anyway: the pairing is what makes the code correct when the allocator later changes.

---

## Lesson 6 - the Chapter 0 program, read line by line

**The one new thing:** nothing. That is the exercise.

```

/*
 * Lesson 6 - the program from Chapter 0, read line by line.
 *
 * Nothing new is introduced here. This is the same greeting program the manual
 * opens with, and the point of the lesson is that you can now name every part
 * of it: the allocator you pass in (lesson 5), the result you must check
 * (lesson 4), the error a refusal returns (lesson 3), the view that carries its
 * own length (lesson 2), and the printing that checks its arguments (lesson 1).
 *
 * If that reads as ordinary now, the tutorial has done its job and the
 * reference chapters are open to you.
 */

int main(void) {
    /* lesson 5 - the caller decides where memory comes from */
    proven_allocator_t alloc = proven_heap_allocator();

    /* lesson 4 - the string and the error that guards it, together */
    proven_result_u8str_t greeting = proven_u8str_create(alloc, 64);
    if (!proven_is_ok(greeting.err)) return 1;

    /* lesson 2 - borrowed text that knows its own size */
    proven_u8str_view_t name = PROVEN_LIT("world");

    /* lesson 3 - each append either fits or refuses; none of them truncates */
    proven_err_t err = proven_u8str_append(&greeting.value, PROVEN_LIT("hello, "));
}

```

```

if (proven_is_ok(err)) err = proven_u8str_append(&greeting.value, name);
if (proven_is_ok(err)) err = proven_u8str_append(&greeting.value, PROVEN_LIT("!"));

if (!proven_is_ok(err)) {
    proven_u8str_destroy(alloc, &greeting.value);
    return 1;
}

EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&greeting.value),
    PROVEN_LIT("hello, world!")),
    "the three appends should have built the whole greeting");

/* lesson 1 - the format string and the argument cannot disagree */
proven_printf("{", PROVEN_ARG(proven_u8str_as_view(&greeting.value)));

/* lesson 5 again - destroyed with the allocator that created it */
proven_u8str_destroy(alloc, &greeting.value);
return EXAMPLE_OK();
}

```

If those comments now read as labels for things you know rather than as new information, you are ready for the reference chapters. The next three lessons take the first steps into them - arenas, containers and files - in the same shape: one new idea each.

---

## Lesson 7 - memory that is freed all at once

**The one new thing:** `proven_arena_reset` - one call that gives back everything an arena handed out.

Lesson 5 passed an arena to `make_greeting` and reset it at the end without saying what the reset did. Here is the whole idea. An arena hands out pieces of a block of memory you own by moving a pointer forward. It has no way to give back one piece - destroy on an arena reclaims nothing. It gives back *all* of them, at once, when you reset it.

That fits a shape most programs have: a loop where each round builds some temporary text, uses it, and is finished with it. Allocate freely during the round, reset at the end, and there is nothing to track and nothing to leak.

```

/*
 * Lesson 7 - memory that is freed all at once.
 *
 * Lesson 5 handed an arena to make_greeting and then reset it, without saying
 * what the reset was. This lesson is about that one call.
 *
 * An arena bumps a pointer through a block of memory you own. There is no
 * per-object free: destroy on an arena reclaims nothing. Everything comes back
 * at once, with proven_arena_reset. That sounds like a restriction, and it is
 * the reason to use one - a loop that builds scratch text for each piece of
 * work can drop all of it with one statement, however many strings it made.
 */

int main(void) {
    /* The arena does not own this array - we do. It only hands out pieces. */
    alignas(PROVEN_MAX_ALIGN) proven_byte_t backing[256];
    proven_arena_t arena = proven_arena_create((proven_mem_mut_t){
        .ptr = backing, .size = sizeof backing });
    proven_allocator_t scratch = proven_arena_as_allocator(&arena);

    static const proven_u8str_view_t names[] = {
        PROVEN_LIT("ada"), PROVEN_LIT("grace"), PROVEN_LIT("barbara"),
    };
}

```

```

for (proven_size_t i = 0; i < sizeof names / sizeof names[0]; ++i) {
    /* Two strings per round, both out of the arena. */
    proven_result_u8str_t line = proven_u8str_create(scratch, 32);
    proven_result_u8str_t note = proven_u8str_create(scratch, 32);
    EXAMPLE_REQUIRE(proven_is_ok(line.err) && proven_is_ok(note.err),
        "each round fits easily in 256 bytes");
    if (!proven_is_ok(line.err) || !proven_is_ok(note.err)) return 1;

    proven_err_t err = proven_u8str_append(&line.value, PROVEN_LIT("hello, "));
    if (proven_is_ok(err)) err = proven_u8str_append(&line.value, names[i]);
    if (proven_is_ok(err)) err = proven_u8str_append(&note.value, PROVEN_LIT("(scratch)"));
    EXAMPLE_REQUIRE(proven_is_ok(err), "the appends fit their capacity");

    /* The first allocation of every round lands at the start of `backing`:
     * the reset at the end of the previous round gave all of it back. */
    EXAMPLE_REQUIRE((const void *)proven_u8str_as_view(&line.value).ptr == (const void *)backing,
        "each round starts again at the beginning of the block");

    proven_println("round {}: {} {} ({} bytes in use)",
        PROVEN_ARG(i), PROVEN_ARG(proven_u8str_as_view(&line.value)),
        PROVEN_ARG(proven_u8str_as_view(&note.value)),
        PROVEN_ARG(arena.offset));

    /* One statement, and both strings are gone. No destroy loop, nothing to
     * forget. `line` and `note` point at reclaimed memory from here on. */
    proven_arena_reset(&arena);
    EXAMPLE_REQUIRE(arena.offset == 0, "reset reclaims everything the round allocated");
}

/* An arena cannot grow. Asking for more than the block holds is an error
 * value, the same kind lesson 3 showed - not a crash, and not a silent
 * fallback to malloc. */
proven_result_u8str_t too_big = proven_u8str_create(scratch, 1024);
EXAMPLE_REQUIRE(too_big.err == PROVEN_ERR_NOMEM, "a 256-byte arena refuses 1 KiB");

proven_arena_destroy(&arena);
return EXAMPLE_OK();
}

```

**What to notice.** Every round reports the same number of bytes in use, and the first string of every round lands at the start of `backing` - the program checks that. The reset did not free two strings; it moved one number back to zero, which costs the same whether the round made two strings or two thousand. And when the arena is too small the answer is `PROVEN_ERR_NOMEM`, an error value like lesson 3's, not a crash.

**Common first stumble.** After the reset, `line` and `note` still hold their old pointers, and those pointers now point at memory the next round will overwrite. A reset leaves every pointer the arena ever gave out dangling, all at once. Keep nothing from a round that you have not copied out of the arena first.

---

## Lesson 8 - a container remembers its allocator

**The one new thing:** a container takes the allocator once, at creation, and keeps it.

The string in lesson 3 had a fixed room, and you passed the allocator both to create and to destroy. A growable array cannot work like that: a push that outgrows its room must allocate again, at a moment the caller did not plan for. So the array keeps the allocator it was created with. You hand it over once; after that, push, get and destroy take only the array.

```

/*
 * Lesson 8 - a container remembers its allocator.
 *
 * A string is created with an allocator and destroyed with the same one; you
 * pass it both times. A growable array has to allocate again later, whenever a
 * push outgrows its storage, so it keeps the allocator it was created with.
 * That is the one new thing: you hand the allocator over once, at creation,
 * and after that push, get and destroy take nothing but the array.
 *
 * The PROVEN_ARRAY_* macros take the element type as an argument and build the
 * value as that type, so a value with no conversion to it - a struct pushed
 * into an array of int - does not compile. They cannot tell whether the type
 * you name is the one the array was created with: name the same one each time.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* 2 is a starting capacity, not a limit. */
    proven_result_array_t made = PROVEN_ARRAY_INIT(alloc, int, 2);
    EXAMPLE_REQUIRE(proven_is_ok(made.err), "creating an empty array must succeed");
    if (!proven_is_ok(made.err)) return 1;
    proven_array_t squares = made.value;

    /* Ten pushes into room for two. Each push that does not fit grows the
     * storage through the allocator the array remembered - no allocator here. */
    for (int i = 1; i <= 10; ++i) {
        proven_err_t err = PROVEN_ARRAY_PUSH(&squares, int, i * i);
        EXAMPLE_REQUIRE(proven_is_ok(err), "the heap can grow a ten-int array");
        if (!proven_is_ok(err)) {
            PROVEN_ARRAY_DESTROY(&squares);
            return 1;
        }
    }
    EXAMPLE_REQUIRE(squares.len == 10, "ten pushes, ten elements");

    /* get returns a pointer to the element, or NULL when the index is past the
     * end. An index out of range is an answer you can test, not a wild read. */
    const int *third = PROVEN_ARRAY_GET(&squares, int, 2);
    EXAMPLE_REQUIRE(third && *third == 9, "element 2 is 3 * 3");
    EXAMPLE_REQUIRE(PROVEN_ARRAY_GET(&squares, int, 10) == NULL, "index 10 is past the end");

    int sum = 0;
    for (proven_size_t i = 0; i < squares.len; ++i) {
        sum += *PROVEN_ARRAY_GET(&squares, int, i);
    }
    EXAMPLE_REQUIRE(sum == 385, "1 + 4 + 9 + ... + 100");
    proven_println("{} squares, sum {}", PROVEN_ARG(squares.len), PROVEN_ARG(sum));

    /* Destroy takes only the array: it frees through the allocator it kept. */
    PROVEN_ARRAY_DESTROY(&squares);
    return EXAMPLE_OK();
}

```

**What to notice.** The array was created with room for two and took ten pushes, and not one of them mentions an allocator. `PROVEN_ARRAY_GET` past the end is `NULL` - an answer you can check, not a read of whatever lies beyond the storage. The macros take the element type (`int`) as an argument, so a value that cannot convert to it, such as a struct, does not compile; naming the same type the array was created with is still your job.

**Common first stumble.** The pointer `PROVEN_ARRAY_GET` returns points *into* the array's storage. The next push may move that storage to a bigger block, and the old pointer then dangles. Fetch it, use it, and do not keep it across a push; [Chapter 4](#) shows the mistake in full. It is lesson 7's stumble again, for the same reason: something other than your pointer decides when that memory moves.

---

## Lesson 9 - the outside world fails too, and says so the same way

**The one new thing:** failures that are not your program's fault arrive as the same error values.

Until now every failure had a cause inside the program - not enough room, a bad argument. Files bring failures from outside: a name that does not exist, a directory you may not write to, a full disk. None of them is a bug, and all of them happen to working programs. The library reports them with the same `proven_err_t` values as before, so there is no second error system to learn.

The calls are the whole-file ones: `proven_fs_write_file_atomic` writes a file in one call, and `proven_fs_read_all_u8str` reads one back as an owned string. There is no open, no read loop, and no close to forget.

```
/*
 * Lesson 9 - the outside world fails too, and says so the same way.
 *
 * Everything so far could only fail for reasons inside the program: not enough
 * room, a bad argument. A file can be missing, unreadable, or on a full disk,
 * and none of that is a bug in your code. The library reports those failures
 * with the same error values as before, so the checking you already learned
 * is all the checking there is.
 *
 * The calls here are the whole-file ones: one call writes a file, one call
 * reads it back. No open, no read loop, no close to forget.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* A name in the current directory; the program removes it before it ends. */
    proven_u8str_view_t path = PROVEN_LIT("proven_tutorial_notes.tmp");
    proven_u8str_view_t text = PROVEN_LIT("buy milk\ncall home\n");

    /* Atomic: a reader sees the old file or the new one, never half of it.
     * The allocator is scratch space the call may need for the path. */
    proven_err_t err = proven_fs_write_file_atomic(alloc, path, proven_mem_view_from_u8(text));
    EXAMPLE_REQUIRE(proven_is_ok(err), "writing a small file in the current directory should work");
    if (!proven_is_ok(err)) return 1;

    /* Reading gives back an owned string - lesson 4's shape, lesson 5's rule:
     * it was made with `alloc`, so it is destroyed with `alloc`. */
    proven_result_u8str_t back = proven_fs_read_all_u8str(alloc, path);
    EXAMPLE_REQUIRE(proven_is_ok(back.err), "the file we just wrote can be read");
    if (proven_is_ok(back.err)) {
        EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&back.value), text),
            "the bytes come back exactly as written");
        proven_println("read {} bytes back", PROVEN_ARG(proven_u8str_as_view(&back.value).size));
        proven_u8str_destroy(alloc, &back.value);
    }

    err = proven_fs_remove(alloc, path);
    EXAMPLE_REQUIRE(proven_is_ok(err), "removing the file we created should work");

    /* Now it is gone, and reading it is a failure from outside the program.
```

```

    * It arrives as an ordinary value, and it says which failure it is. */
    proven_result_u8str_t missing = proven_fs_read_all_u8str(alloc, path);
    EXAMPLE_REQUIRE(missing.err == PROVEN_ERR_NOT_FOUND, "a removed file reads as NOT_FOUND");
    if (proven_is_ok(missing.err)) proven_u8str_destroy(alloc, &missing.value);
    proven_println("reading it again: not found, as expected");

    return EXAMPLE_OK();
}

```

**What to notice.** The string `proven_fs_read_all_u8str` returns is owned: made with `alloc`, destroyed with `alloc` - lesson 5's rule, unchanged. Reading the file after removing it gives `PROVEN_ERR_NOT_FOUND`, the same kind of value as lesson 3's refusal, and it names what went wrong. *Atomic* in the write's name means a reader sees the old file or the new one and never a mix; [Chapter 5](#) says what it does not promise, such as surviving a power cut.

**Common first stumble.** The allocator handed to the write and to `proven_fs_remove` is scratch space for handling the path, not the owner of anything: those calls give back whatever they took before they return. Only the read, which hands you a string, gives you something to destroy.

---

## Where to go next

| If you want to                            | Read                         |
|-------------------------------------------|------------------------------|
| the full version of everything above      | <a href="#">Chapter 0</a>    |
| the types, errors and contracts in detail | <a href="#">Chapter 1</a>    |
| arenas, pools and buffers properly        | <a href="#">Chapter 2</a>    |
| owned strings, splitting, encodings       | <a href="#">Chapter 3</a>    |
| arrays, maps, lists, hashing              | <a href="#">Chapter 4</a>    |
| files, streams, time, randomness          | <a href="#">Chapter 5</a>    |
| a word you do not recognise               | <a href="#">the glossary</a> |

# Chapter 0: Start Here

**Part I - Start here.** No prerequisites beyond one introductory C book. **After this chapter** you can build a program against proven, read any other chapter, and look up a term you do not recognise.

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- 

## 1. Who this manual is for

This manual assumes you have finished one introductory C book. Concretely, it assumes you are comfortable with variables and control flow, functions, arrays, struct, pointers and `*/&`, `malloc` and `free`, `printf`, `char *` strings with `strlen`, and compiling a program with `gcc main.c -o main`. If all of that is familiar, you have enough.

It does **not** assume you have met ownership as a discipline, borrowed versus owned data, arenas or pools, function-pointer tables used as an interface, C23 attributes, undefined behaviour as something a compiler actively exploits, alignment beyond "it works", atomics, or the idea that a library might *refuse* an operation instead of doing its best. Every one of those is explained where it is first used, and every one is in the glossary in section 13.

This is not a C tutorial. It will not explain what a pointer is. It will explain, at length, why this library hands you a struct containing an error instead of setting `errno`, because that is a design decision you are entitled to disagree with, and you cannot disagree with a decision nobody explained.

---

## 2. Why this library exists

C gives you almost nothing and trusts you completely. That is its great strength - there is no runtime, no hidden allocation, no cost you did not write - and it is why C is still the language of operating systems, embedded devices and everything that has to be small and predictable.

It is also why the same five bugs have been shipping for fifty years. This library is a set of answers to those five bugs. Each answer costs something, and this section says what.

### The string functions do not know how big anything is

```
char buf[64];
strcpy(buf, name);           /* wrong: how long is name? strcpy never asks */
strcat(buf, ", welcome!");   /* wrong: and how much room is left now? */
```

strcpy receives a destination pointer and a source pointer. Nothing in that signature carries the size of the destination, so nothing can check it. The function will happily write the 200th byte into a 64-byte buffer, and what it corrupts is whatever the compiler happened to put next -- often the return address. This is not a rare mistake by careless people; it is the single most exploited class of bug in the history of the language, and the API makes it the *default* behaviour.

strncpy is the traditional answer and it is a trap of its own: it does not always NUL-terminate, so the "safe" version silently produces a string that is not a string.

**What this library does instead.** A string carries its length. `proven_u8str_view_t` is a pointer *and* a size, together, always. An append checks the destination's capacity because it knows the capacity, and when the text does not fit it **returns an error and writes nothing** -- it does not truncate, because a truncated path is a wrong path and a truncated command is a different command. See [Chapter 3](#).

**What it costs.** Every string is two words instead of one, and you cannot pass a proven string straight to a `printf("%s")` without asking for a NUL-terminated form.

### Nothing makes you check for failure

```
char *p = malloc(n);
p[0] = 'x';           /* wrong: malloc returns NULL when it fails */
```

`malloc` reports failure by returning `NULL`, and C will not say a word if you never look. The same is true of `fopen`, of `realloc`, of every function that returns a sentinel. The error is *available*; noticing it is optional.

`errno` is worse, because it is a global that survives the call. You must check it at exactly the right moment, before any other library call has a chance to overwrite it, and you must remember that a successful call is allowed to set it to garbage.

**What this library does instead.** A function that can fail returns its error *as a value*, and when it also has a result, the two come back together in one struct:

```
proven_result_u8str_t s = proven_u8str_create(alloc, 64);
if (!proven_is_ok(s.err)) return 1;    /* s.value means nothing until you check */
```

Functions whose only job can fail are marked `[[nodiscard]]`, which means the compiler refuses to build code that throws the error away. You can still ignore it deliberately - `(void)` in front of the call - and having to type that is the point. See [Chapter 1](#).

**What it costs.** More `ifs`. There is no exception mechanism to jump you out of a deep failure, so error paths are visible in the shape of the code. That visibility is the feature.

### printf believes whatever you tell it

```
printf("%d\n", 3.0);           /* wrong: %d with a double. This compiles. */
printf("%s\n", 42);           /* wrong: and this one crashes */
```

The format string is checked by nothing at runtime. Modern compilers warn about literal formats, which helps exactly until the format is a variable, and then you are back to a function that reads whatever bytes the `varargs` stack happens to hold, in whatever shape the string demanded.

**What this library does instead.** `{}` is a placeholder with no type in it, and the type comes from the argument, resolved at compile time by `_Generic`:

```
proven_println("{} is {}", PROVEN_ARG(name), PROVEN_ARG(count));
```

There is no `%d`-versus-`double` mismatch available, because you never wrote the type twice. See [Chapter 3 section 3](#) for the tutorial and [Chapter 8](#) for the full grammar.

**What it costs.** `PROVEN_ARG` around each argument, and a format language that is not the one in your fingers.

## Who frees this?

```
char *s = build_message();    /* do I free this? the type does not say */
```

A `char *` returned from a function might be freshly allocated, might point into a caller's buffer, might be a string literal in read-only memory, or might be a pointer into a static buffer that the next call will overwrite. The type is identical in all four cases. The answer lives in documentation, and documentation drifts.

**What this library does instead.** Ownership is in the type name and in the signature. A `proven_u8str_t` is **owned** - you got it from a `_create`, and you must `_destroy` it with the same allocator. A `proven_u8str_view_t` is **borrowed** - it points at bytes someone else owns, you never destroy it, and it stops being valid when the owner does. Any function that might allocate takes the allocator as a parameter, so a signature without an allocator cannot allocate.

**What it costs.** Two types where C has one, and the discipline of asking "who owns this?" at every boundary - which you were paying anyway, just later and in a debugger.

## The comparison function nobody can typecheck

```
qsort(a, n, sizeof *a, cmp); /* cmp takes const void*; get it wrong and it is UB */
```

`qsort` takes a comparator through a `void *` interface, so a comparator with the wrong parameter types still compiles. The classic version of this bug is comparing the pointers instead of what they point at, and it produces a program that runs, sorts nothing correctly, and never crashes.

**What this library does instead.** The same `void *` shape - this is C, there is no other way -- but the library documents the contract precisely, gives you working comparators to copy, and `proven_array_sort` is an introsort with an  $O(n \log n)$  guarantee rather than a quicksort that degrades to  $O(n^2)$  on the input an attacker chooses. See [Chapter 4](#).

## The bytes have a type, even when you did not choose one

You think of memory as bytes. C's abstract machine does not: it treats memory as *typed*, and reading the same bytes through pointers of two different types is undefined behaviour - which the compiler is allowed to exploit, silently, only when optimisations are on. This is called **strict aliasing**, and it is the trap under every hand-written parser that reads a byte buffer through pointers of different widths:

```
void *buf = malloc(8);
uint32_t *w = buf;    /* the same memory, seen as 32-bit - no cast, no warning */
uint16_t *h = buf;    /* the same memory, seen as 16-bit */
*w = 0xAAAAAAAAu;
*h = 0x1234;          /* change the low half */
printf("%08x\n", *w); /* wrong to expect aaaa1234: at -O2 this prints aaaaaaaa */
```

Compiled at `-O0` it prints `aaaa1234`; at `-O2` it prints `aaaaaaaa`, because the compiler assumed a `uint16_t` write and a `uint32_t` read cannot touch the same memory and dropped the write. No warning is given, and the program passed every test you ran in a debug build. This is the class of bug the Linux kernel avoids by compiling with `-fno-strict-aliasing` - a whole flag, for one rule.

**What this library does instead.** Raw memory is `proven_byte_t`, an alias of `unsigned char` -- the one type the rule explicitly exempts, because the standard lets you inspect any object's bytes through it. The ordinary API never quietly reinterprets your bytes as a wider type, so the bug above cannot be written through it. (Strict aliasing has a subtler sibling, *provenance*, that the library is named after; [Chapter 6 section 3](#) and the project README cover it.)

## What this library is not

It is not a framework, and it does not want to own your `main`. It has no global state to initialise, starts no threads, registers no `atexit` handler, and allocates nothing you did not hand it an allocator

for. Every module is usable on its own. Most of it runs with no operating system at all - see [freestanding mode](#).

| The problem              | What C gives you                                                           | What proven gives you                                                                                        | Cost                              |
|--------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Buffer overruns          | strcpy, strcat - no size anywhere                                          | Views carry a length; writes refuse rather than truncate                                                     | Two words per string              |
| Unchecked failure        | NULL returns and errno                                                     | Errors returned as values, [[nodiscard]] on the ones you must not drop                                       | More ifs                          |
| Format mismatch          | printf trusts the format string                                            | { } with the type taken from the argument                                                                    | PROVEN_ARG at each call           |
| Unclear ownership        | char * means four different things                                         | Owned and borrowed are different types                                                                       | Two types instead of one          |
| Hidden allocation        | Anything may call malloc                                                   | Only functions taking an allocator can allocate (one bounded exception: proven_println on an over-long line) | The parameter is on the signature |
| Bytes with a hidden type | Reinterpreting memory through a wider pointer is UB the optimiser exploits | Raw bytes go through proven_byte_t, the type the rule exempts                                                | -                                 |

### 3. Your first program

This is the whole of it. Every line is one of the contracts in section 5, and the build compiles and runs this exact file, so it cannot quietly stop being true.

```
/*
 * The first program. It is deliberately small, and every line of it is one of
 * the five contracts you will meet on every page of this manual.
 *
 * Compare it with the C you already know:
 *
 *   char buf[64];
 *   strcpy(buf, name);           <- how big is name? strcpy does not ask.
 *   strcat(buf, ", welcome!");  <- and now? strcat does not ask either.
 *   printf("%s\n", buf);
 *
 * That program is correct until the day `name` is longer than you assumed, and
 * then it is a security advisory. The version below cannot do that: every write
 * knows the size of its destination, and every operation that could fail hands
 * you back an error you are not allowed to ignore silently.
 */

int main(void) {
    /* (1) You pass the allocator in. The library never reaches for a global
     *     malloc behind your back, so you always know who allocated what. */
    proven_allocator_t alloc = proven_heap_allocator();
```

```

/* (2) Anything that can fail returns its error WITH its value. There is no
 *     errno to remember to check, and `greeting.value` means nothing until
 *     you have looked at `greeting.err`. */
proven_result_u8str_t greeting = proven_u8str_create(alloc, 64);
if (!proven_is_ok(greeting.err)) return 1;

/* (3) A view is borrowed text that knows its own length. PROVEN_LIT builds
 *     one from a literal at compile time - no strlen scan happens here. */
proven_u8str_view_t name = PROVEN_LIT("world");

/* (4) The append refuses rather than truncates. If "hello, " and the name
 *     did not fit in the 64 bytes asked for above, this returns
 *     PROVEN_ERR_OUT_OF_BOUNDS and writes nothing - it never quietly stores
 *     half a word and lets you carry on. */
proven_err_t err = proven_u8str_append(&greeting.value, PROVEN_LIT("hello, "));
if (proven_is_ok(err)) err = proven_u8str_append(&greeting.value, name);
if (proven_is_ok(err)) err = proven_u8str_append(&greeting.value, PROVEN_LIT("!"));

if (!proven_is_ok(err)) {
    proven_u8str_destroy(alloc, &greeting.value);
    return 1;
}

EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&greeting.value),
                                     PROVEN_LIT("hello, world!")),
               "the three appends should have built the whole greeting");

proven_println("{} ", PROVEN_ARG(proven_u8str_as_view(&greeting.value)));

/* (5) You created it with `alloc`, so you destroy it with the SAME `alloc`.
 *     Owning things are destroyed exactly once; borrowed things - like
 *     `name` above - are never destroyed at all. */
proven_u8str_destroy(alloc, &greeting.value);

return EXAMPLE_OK();
}

```

EXAMPLE\_REQUIRE and EXAMPLE\_OK are not part of the library - they come from `manual/examples/example.h` and exist so the build can check that every example in this manual still does what the text says it does. Your own program would use neither.

Three details worth pausing on, because they recur everywhere:

- `proven_u8str_create(alloc, 64)` **asks for 64 bytes of capacity**, and that capacity includes the NUL terminator. It does not grow on its own. `proven_u8str_append` is the fixed-capacity form: it fails when the text does not fit. When you want growth, you call `proven_u8str_append_grow` and pass the allocator, and the signature tells you at a glance which one you are using.
- `PROVEN_LIT("hello, ")` **costs nothing at runtime**. It is a compile-time `sizeof` on a string literal. `proven_u8str_view_from_cstr(s)` exists for the case where the text comes from elsewhere, and that one really does scan for the NUL - the two are spelled differently because one is free and the other is  $O(n)$ .
- **The destroy takes the allocator again**. The string does not remember which allocator made it; you must pass the same one. That keeps the string small and makes the dependency visible, and it is a real hazard - see section 5.

---

## 4. Building and including

There is one header for everything:

```
#include "proven.h"
```

It pulls in the whole public API. If you prefer to include only what you use, every module has its own header under `include/proven/` - `#include "proven/u8str.h"`, `#include "proven/fs.h"` and so on - and section 14 below maps every header to the chapter that documents it.

Compiling by hand, which is all this library needs:

```
gcc -std=c2x -Iinclude -Iplatform your_program.c src/proven/*.c platform/*.c -o your_program
```

`src/proven/` is the portable library. `platform/` is the small layer that talks to the operating system - the **PAL**, or platform abstraction layer. Everything that makes a syscall lives there and nowhere else, which is why the rest of the library can be compiled for a target with no operating system at all.

The repository builds itself with a C program rather than a build system:

```
cc -std=c2x -o nob nob.c      # build the build driver, once
./nob build                  # compile everything and run the whole test suite
./nob release                # the same, optimised (NDEBUG: misuse checks compiled out)
./nob hardened               # optimised, misuse checks kept in
```

`./nob` with no arguments lists the other modes - sanitizers, freestanding, cross-compilation and the benchmark. There is no `make`, no `CMake`, and nothing to install.

**A C23 compiler is required.** The library uses C23 features deliberately, `[[nodiscard]]` most visibly. GCC 13+ and Clang 16+ are verified, and on Windows mingw-w64 GCC (x86-64 and i686, the full suite run natively). MSVC and clang-cl are not supported: their C23 support is incomplete, and support may come later. The build driver probes for `-std=c23` and falls back to `-std=c2x` for slightly older compilers.

---

## 5. The five contracts you will meet on every page

These five rules explain most of the library's shape. Each is stated once here and assumed everywhere else. section 8 below gives the formal versions; these are the plain-language ones.

| # | Contract                                     | In one line                                                                               | Chapter |
|---|----------------------------------------------|-------------------------------------------------------------------------------------------|---------|
| 1 | <b>Errors are values</b>                     | Fallible calls return <code>proven_err_t</code> , or a <code>{ err, value }</code> struct | 1       |
| 2 | <b>Views are borrowed</b>                    | A view is a pointer + a length into memory someone else owns                              | 3       |
| 3 | <b>Allocation is a parameter</b>             | If a function can allocate, it takes an allocator; if it cannot, it does not              | 2       |
| 4 | <b>Caller-owned state must not be copied</b> | Some structs point into themselves; copying them creates dangling pointers                | 5       |
| 5 | <b>Refuse, never truncate</b>                | An operation that does not fit fails and writes nothing                                   | 3       |

### 1. Errors are values

Every fallible function hands the error back. When there is nothing else to return it is a bare `proven_err_t`; when there is a value, the value and the error travel together and the value is meaningless until you have checked the error. `PROVEN_OK` is zero, and `proven_is_ok(err)` reads better than `err == 0`.

Wrong - reading the value before the error:

```
proven_result_u8str_t s = proven_u8str_create(alloc, 64);
proven_u8str_append(&s.value, text); /* wrong: s.value is garbage if create failed */
```

The whole point of pairing them is that the value is only valid on the success path. Check first, every time.

## 2. Views are borrowed

`proven_u8str_view_t` is a pointer and a size. It owns nothing, allocates nothing, and is free to copy - but it stops being valid the moment the thing it points into is destroyed or moved.

Wrong - a view that outlives what it points at:

```
proven_u8str_view_t name;
{
    proven_result_u8str_t owned = proven_u8str_create(alloc, 16);
    (void)proven_u8str_append(&owned.value, PROVEN_LIT("temp"));
    name = proven_u8str_as_view(&owned.value);
    proven_u8str_destroy(alloc, &owned.value); /* the bytes are gone */
}
use(name); /* wrong: dangling view */
```

This is the same lifetime bug as a dangling pointer, and it is worth stating separately because a view *looks* like a value. It is not; it is a pointer wearing a struct.

## 3. Allocation is a parameter

Read the signature. `proven_u8str_append(str, data)` cannot allocate, so it fails when the text does not fit. `proven_u8str_append_grow(alloc, str, data)` takes an allocator, so it can grow. Nothing in this library calls `malloc` behind your back, which is what makes it usable in an arena, in a pool, or on a device with no heap.

The corollary is the hazard: **you must destroy with the allocator you created with.** The object does not remember.

Wrong - mismatched allocators:

```
proven_result_u8str_t s = proven_u8str_create(arena_alloc, 64);
proven_u8str_destroy(heap_alloc, &s.value); /* wrong: heap free on arena memory */
```

Nothing checks this today. It is heap corruption that surfaces later, somewhere else.

## 4. Caller-owned state must not be copied

Some objects are structs you own and pass by pointer - buffered writers, line readers, directory iterators. Several of them contain a pointer to one of their own fields, so copying the struct copies a pointer that still points at the *original*.

Wrong - copying a state struct:

```
proven_writer_buf_t a = ...;
proven_writer_buf_t b = a; /* wrong: b's internals still point into a */
```

section 9.2 below lists all twenty-one of these types. The rule is simple: create it where it lives, pass `&it`, and do not assign it.

## 5. Refuse, never truncate

When a result does not fit, this library fails the operation and leaves the destination unchanged. It does not write "as much as fits". A truncated path opens the wrong file, a truncated command runs the wrong command, and a truncated number is a different number - and every one of those is worse than an error you can see.

Where truncation genuinely is what you want, there is a separate, differently named function that tells you how much it wrote:

```
proven_result_size_t r = proven_u8str_append_partial(&s, huge);
/* r.value is how many bytes were actually appended. Reading it is the point. */
```

Wrong - assuming the two behave alike:

```
(void)proven_u8str_append_partial(&s, huge); /* wrong: the count WAS the result */
```

---

Everything up to here is the plain-language introduction. The rest of this chapter is the reference the other chapters lean on: the library's intent and build model stated formally, the global contracts, the ownership matrix, the behaviour classes, the reading order, platform support, and the appendices. It used to be a document of its own, the manual's front page; it lives here now so that the front page is nothing but the table of contents.

## 6. Intent and design philosophy

proven is a compact C23 systems foundation library. It is intended for C programs that want practical infrastructure without hiding memory ownership, error control flow, or platform access behind global state.

It is not a libc replacement. It provides a focused set of allocator-driven memory tools, byte views, containers, strings, formatting, scanning, hashing (FNV, SipHash, CRC-32, SHA-256), hex/Base64 encoding, OS-strength randomness, filesystem helpers, buffered streams, time helpers, memory mapping, stackless coroutine macros, and a bounded job system.

Core design principles:

- C23 first: the build driver uses `-std=c23`.
- Explicit errors: fallible functions return `proven_err_t` or `proven_result_*_t`.
- Explicit ownership: owned objects have clear destroy functions and allocator rules.
- Failure atomicity: grow/realloc-style APIs preserve the old object on allocation failure unless documented otherwise.
- Pointer provenance discipline: raw object access uses `proven_byte_t` and bounded views.
- PAL isolation: hosted OS services live under `platform/` and are called through public wrappers.
- Core containers do not add hidden locks. Shared mutation requires caller synchronization.
- The build system is a single checked-in `nob.c`; tests are plain C executables.

## 7. Build and include model

### There is nothing to install

This library has no `configure`, no `CMake`, no package to fetch, and no shared object to place somewhere. It is C source: you compile it into your program alongside your own files.

That is a deliberate choice and it costs something. You do not get a system package, and updating means pulling new source rather than bumping a version constraint. What you get is that the library cannot be a different version than the one you are looking at, cannot pick up a build flag you did not choose, and cannot fail to link because a distribution built it with different options. For a library that is meant to run on hosted systems *and* on bare metal, "compile it with your program" is the only model that works in both places.

Two directories matter: `src/proven/` is the portable library - no OS calls anywhere. `platform/` is the small layer that makes syscalls, and it is the only part a new target has to replace. A freestanding build simply leaves the hosted files out; see [the freestanding guide](#).

The build driver, `nob.c`, is a C program rather than a build system, and you compile it the same way you compile everything else here. It is checked into the repository, so there is no bootstrap step and no version of it to install either.

**A C23 compiler is required** - GCC 13+ or Clang 16+ (mingw-w64 GCC on Windows; MSVC is not supported). The driver probes `-std=c23` and falls back to `-std=c2x` for compilers that still use the transitional spelling.

Build and run the hosted test suite:

```
cc nob.c -o nob
./nob build
```

Common validation commands:

```
./nob release
./nob strict
./nob strict-error
./nob asan
./nob ubsan
./nob tsan
./nob regression
./nob regression-asan
./nob regression-ubsan
./nob freestanding
./nob cross -build-root build-out/proven_c_lib
```

Use the umbrella header when you want the full hosted API:

```
#include "proven.h"
```

Use smaller includes when you want a narrower translation unit:

```
#include "proven/heap.h"
#include "proven/u8str.h"
#include "proven/fmt.h"
```

A direct hosted application build can follow the same source layout as nob.c:

```
cc -std=c23 -D_DEFAULT_SOURCE -D_POSIX_C_SOURCE=200809L \
-Iinclude -Iplatform \
app.c src/proven/*.c platform/proven_sys/*.c \
-pthread -o app
```

## 8. Global contracts

### 8.1 Result contract

A result value is usable only when `err == PROVEN_OK`.

Correct:

```
proven_result_mem_mut_t r = alloc.alloc_fn(alloc.ctx, 128, PROVEN_DEFAULT_ALIGNMENT);
if (proven_is_ok(r.err)) {
    /* only now does r.value mean anything */
    proven_mem_mut_t mem = r.value;
    (void)proven_mem_copy(mem.ptr, mem.size, proven_mem_view_from_u8(PROVEN_LIT("hi")));
    alloc.free_fn(alloc.ctx, mem.ptr);
}
```

Wrong (does not compile as shown - the value is read before the error is checked):

```
proven_result_mem_mut_t r = alloc.alloc_fn(alloc.ctx, 128, PROVEN_DEFAULT_ALIGNMENT);
use_bytes(r.value.ptr, r.value.size); /* wrong: r.err was not checked */
```

### 8.2 Public struct contract

Many public structs expose layout for C usability. This does not mean arbitrary field mutation is supported. Treat direct mutation of internal fields as caller misuse unless the header explicitly allows it.

Wrong:

```
arr.len = 999; /* wrong: breaks array invariants */
str.internal.cap=0; /* wrong: breaks string invariants */
```

Use the functions and macros that maintain invariants.

### 8.3 Borrowed view contract

Views do not own memory. A `proven_u8str_view_t`, `proven_u16str_view_t`, `proven_mem_view_t`, or `proven_mem_mut_t` is valid only while the referenced storage remains alive and unmoved.

Wrong:

```
proven_u8str_view_t v = proven_u8str_as_view(&s);
proven_u8str_append_grow(&alloc, &s, PROVEN_LIT("more"));
use_view(v); /* wrong: growth may reallocate s */
```

### 8.4 Allocator contract

A `proven_allocator_t` is valid only when all three function pointers are present. Reallocation must be failure-atomic: if it fails, the old allocation remains valid.

```
proven_allocator_t heap = proven_heap_allocator();
if (!proven_alloc_is_valid(heap)) {
    /* no usable allocator here: e.g. the freestanding heap stub */
    proven_panic("no heap allocator on this target");
}
```

### 8.5 PAL boundary contract

Code under `src/proven/` should not depend on OS headers directly. Hosted services should route through `platform/proven_sys_*.ch` and public wrappers such as `proven_fs_*`, `proven_sysio_*`, `proven_time_*`, and `proven_mmap_*`.

Application code should prefer public APIs. Direct PAL calls are for porting and platform integration.

## 9. Ownership and destruction matrix

There are **two kinds of object** in this library, and the difference decides what you must do with them.

**Owning objects** hold storage they allocated, and you must destroy them. They are the table immediately below.

**Caller-owned state objects** allocate nothing. They are scratch structs you declare - usually on the stack - and hand to a constructor, which returns a small by-value handle that points *into* them. They have **no destroy function**, because there is nothing to free. What they have instead is a rule that owning objects do not have, and it is the one that bites:

**A caller-owned state object must not be copied or moved once a handle has been made from it.** The handle holds a pointer into the struct. Copy the struct and the handle still addresses the original; let the original go out of scope and the handle points at dead memory.

They are listed in [section 4.2](#).

### 9.1 Owning objects - you must destroy these

| Object                      | Owns storage                       | Stores allocator | Destroy function                                      | Notes                                                                                                                                 |
|-----------------------------|------------------------------------|------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_arena_t</code> | no - caller owns the backing slice | no               | <code>proven_arena_destroy(&amp;arena)</code> (no-op) | Return pointer over caller memory: <code>alloc</code> advances an offset, <code>free</code> is a no-op, <code>reset</code> rewinds to |

|                                      |                           |                            |                                                                |                                                                                                                                                                                                                                                                               |
|--------------------------------------|---------------------------|----------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      |                           |                            |                                                                | empty. The caller owns/frees the backing block.                                                                                                                                                                                                                               |
| proven_pool_t                        | yes - items + recycle bin | yes (base_alloc)           | proven_pool_destroy(&pool)                                     | Fixed item size. You use it <b>through the allocator trait</b> (proven_pool_as_allocator): the trait's free_fn returns a slot to the recycle bin for reuse rather than freeing it. There is no proven_pool_free - freeing goes through the trait, like every other allocator. |
| proven_buf_t                         | yes                       | no                         | proven_buf_destroy(&buf)                                       | Caller must pass the matching allocator.                                                                                                                                                                                                                                      |
| proven_u8str_t                       | yes                       | no                         | proven_u8str_destroy(&str)                                     | Always NUL-terminated when valid.                                                                                                                                                                                                                                             |
| proven_u16str_t                      | yes                       | no                         | proven_u16str_destroy(&str)                                    | Tracks byte length internally; API length is in proven_u16 units.                                                                                                                                                                                                             |
| proven_array_t                       | yes                       | yes                        | proven_array_destroy(&arr)<br>OR<br>PROVEN_ARRAY_DESTROY(&arr) | Prints into elements may be invalidated by growth.                                                                                                                                                                                                                            |
| proven_ring_t                        | yes                       | yes                        | proven_ring_destroy(&ring)<br>OR<br>PROVEN_RING_DESTROY(&ring) | Fixed capacity; no growth.                                                                                                                                                                                                                                                    |
| proven_map_t                         | yes                       | yes                        | proven_map_destroy(&map)<br>OR<br>PROVEN_MAP_DESTROY(&map)     | Borrowed U8 keys are not copied.                                                                                                                                                                                                                                              |
| proven_array_t from proven_fs_list() | yes                       | yes plus owned entry names | proven_fs_list_destroy(&list)                                  | Do not use plain array destroy; entry names need cleanup.                                                                                                                                                                                                                     |
| proven_mmap_t                        | OS mapping                | OS handle state            | proven_mmap_destroy(&map)                                      | Maps into the mapping die with the mapping.                                                                                                                                                                                                                                   |

|                    |     |          |                                                            |                                       |
|--------------------|-----|----------|------------------------------------------------------------|---------------------------------------|
| proven_job_sys_t * | yes | internal | proven_job_system_close then proven_job_system_destroy(5s) | Destroy must not race with producers. |
|--------------------|-----|----------|------------------------------------------------------------|---------------------------------------|

## 9.2 Caller-owned state - no destroy, do not copy

These allocate nothing and free nothing. You declare one, pass its address to a constructor, and use the small handle you get back. The struct must **outlive the handle**, and must **not be copied or moved** while the handle is alive.

| State object                | Constructed by                                              | The handle it backs               | Notes                                                                                                                       |
|-----------------------------|-------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| proven_sha256_t             | proven_sha256_init                                          | (used directly)                   | A hashing context. Safe to copy <i>before</i> you start, meaningless to copy mid-stream unless you intend to fork the hash. |
| proven_sha1_t, proven_md5_t | proven_sha1_init, proven_md5_init                           | (used directly)                   | The legacy digest contexts; the same rules as proven_sha256_t.                                                              |
| proven_xoshiro256ss_t       | proven_xoshiro256ss_seed                                    | proven_rng_t                      | Copying it <b>clones the sequence</b> - deliberate and useful for a replay, a bug anywhere else.                            |
| proven_chacha_rng_t         | proven_chacha_rng_seed / _seed_from_entropy                 | proven_rng_t                      | Copying it clones the keystream: two "independent" tokens become the same token.                                            |
| proven_writer_buf_t         | proven_writer_from_buffer                                   | proven_writer_t                   | <b>Do not copy.</b>                                                                                                         |
| proven_writer_u8str_t       | proven_writer_from_u8str                                    | proven_writer_t                   | <b>Do not copy.</b> The string and allocator must also outlive it.                                                          |
| proven_writer_buffered_t    | proven_writer_buffered                                      | proven_writer_t                   | <b>Do not copy.</b> You must proven_writer_flush before it or its buffer dies.                                              |
| proven_reader_view_t        | proven_reader_from_view                                     | proven_reader_t                   | <b>Do not copy.</b>                                                                                                         |
| proven_reader_buffered_t    | proven_reader_buffered                                      | proven_reader_t                   | <b>Do not copy.</b> Views returned by proven_reader_read_line point <i>into</i> its buffer.                                 |
| proven_sysio_std_t          | proven_sysio_stdout_writer / _stderr_writer / _stdin_reader | proven_writer_t / proven_reader_t | <b>Do not copy.</b> Holds the standard handle the writer points at.                                                         |
| proven_sysio_out_t          | proven_sysio_stdout_buffered / _file_buffered               | proven_writer_t                   | <b>Do not copy.</b> Must be flushed.                                                                                        |

|                           |                                                      |                                      |                                                                                                                                        |
|---------------------------|------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| proven_sysio_lines_t      | proven_sysio_lines_open /<br>_stdin_lines            | (used via<br>proven_sysio_read_line) | The one exception:<br>proven_sysio_read_line<br>re-binds it on every<br>call, so this one <b>may</b><br>be moved.                      |
| proven_sysio_scanner_t    | proven_sysio_scanner_init                            | (used directly)                      | The exception in the<br>other direction: this<br>one <b>does</b> own a<br>buffer, and you must<br>call<br>proven_sysio_scanner_deinit. |
| proven_transport_stream_t | proven_transport_reader /<br>proven_transport_writer | proven_reader_t /<br>proven_writer_t | <b>Do not copy.</b> The<br>transport it wraps<br>must outlive it too.                                                                  |
| proven_http_body_t        | proven_http_body_init                                | (used directly)                      | A body being decoded.<br>Copying it mid-body<br>forks the decoder; do<br>not.                                                          |
| proven_url_query_iter_t   | proven_url_query_iter                                | (used directly)                      | A position in a query<br>string; a copy is an<br>independent position.                                                                 |

Wrong - the copy looks harmless and is a use-after-free:

```
proven_sysio_out_t out;
proven_writer_t w = proven_sysio_stdout_buffered(&out, buf);

proven_sysio_out_t saved = out; /* wrong: `w` still points into `out`, not `saved` */
use_elsewhere(&saved);          /* and if `out` goes out of scope, `w` is dangling */
```

Wrong - returning the state by value from a factory function does the same thing:

```
proven_sysio_out_t make_logger(void) {
    proven_sysio_out_t out;
    proven_writer_t w = proven_sysio_stdout_buffered(&out, buf);
    (void)w;
    return out; /* wrong: any writer made from `out` addresses this dead frame */
}
```

Correct - the state stays put, and the handle travels:

```
proven_byte_t buf[512];
proven_sysio_out_t out; /* lives as long as `w` */
proven_writer_t w = proven_sysio_stdout_buffered(&out,
    (proven_mem_mut_t){ .ptr = buf, .size = sizeof buf });

(void)proven_fprintln(w, "one syscall, not {}", PROVEN_ARG(100));
(void)proven_writer_flush(w); /* or the bytes never happened */
```

## 10. Operation behavior classes

### One operation, three honest answers

"Append this text to that string" has three defensible behaviours when the text does not fit, and most libraries pick one and hide the choice. This library exposes all three, gives them different names, and puts the difference in the signature - because which one you want depends on what the text is, and only the caller knows that.

Consider appending to a filesystem path:

- If the path does not fit, **truncating is catastrophic**. `documents/report.pdf` becomes `documents/rep`, which is a different, possibly existing, file. The operation must fail and change nothing.
- Now consider appending to a log line. If it does not fit, **truncating is fine** - you would rather have most of the message than none of it - as long as you are told how much was written.
- And appending to a buffer you are building up, where you would simply like it to **grow**.

The three classes below are those three answers. The one you get is decided by the function name and the presence of an allocator parameter, never by a flag or a global:

- `proven_u8str_append(str, data)` - no allocator, so it cannot grow: **refuses**.
- `proven_u8str_append_partial(str, data)` - returns a count: **truncates and tells you**.
- `proven_u8str_append_grow(alloc, str, data)` - takes an allocator: **grows**.

The default across the library is the first one, and [Chapter 0 section 5](#) explains why: a truncated path opens the wrong file, a truncated command runs the wrong command, and a truncated number is a different number.

Wrong - ignoring the count from the truncating form:

```
(void)proven_u8str_append_partial(&s, huge); /* wrong: the count WAS the answer */
```

Several APIs intentionally expose three behavior classes:

| Class                         | Example                                                                                                                                                | Behavior on insufficient capacity                                                                                                                                                                                                                                                                               |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Atomic fixed-capacity         | <code>proven_u8str_append</code> ,<br><code>proven_u16str_append</code> ,<br><code>proven_u8str_append_fmt</code>                                      | Return an error and leave the old object unchanged.                                                                                                                                                                                                                                                             |
| Best-effort truncating        | <code>proven_u8str_append_partial</code> ,<br><code>proven_u16str_append_partial</code> ,<br><code>proven_u8str_append_fmt_trunc</code>                | Write as much as fits, preserve a valid object, report how much was written.                                                                                                                                                                                                                                    |
| Atomic growable               | <code>proven_u8str_append_grow</code> ,<br><code>proven_u16str_append_grow</code> ,<br><code>proven_u8str_append_fmt_grow</code>                       | Grow with an allocator; on allocation failure, leave the old object unchanged.                                                                                                                                                                                                                                  |
| <b>Refuse, never truncate</b> | <code>proven_hex_encode</code> ,<br><code>proven_base64_encode</code> ,<br><code>proven_base64_decode</code> ,<br><code>proven_reader_read_line</code> | Write <b>nothing</b> and return <code>PROVEN_ERR_OUT_OF_BOUNDS</code> . A half-encoded string or a shortened line is a wrong answer that looks like a right one, so these do not produce one. Size the buffer with the module's own size function ( <code>proven_base64_encoded_size, ...</code> ), not by eye. |

Choose the class deliberately. Do not treat a truncating function as an all-or-nothing function.

Wrong - assuming the truncating and the atomic form behave alike:

```
/* `_partial` wrote what fit and told you so; the error you did not read is the  
   difference between "all of it" and "some of it". */  
(void)proven_u8str_append_partial(&s, huge); /* wrong: the result was the point */
```

## 11. Manual chapters

**Just finished an introductory C book? Start with the [tutorial](#).** It is nine short programs, each introducing exactly one idea: the sixth is the greeting program from this chapter read line by line,

and the last three go on to arenas, containers and files. This chapter shows that program on its first page and it carries five new ideas at once; the tutorial hands them over one at a time.

**Otherwise start with this chapter.** It is the only chapter that assumes nothing: why the library exists, argued from the C bugs it is answering; a hello-world program; how to build; the five contracts the rest of the manual takes for granted; and a glossary plus a libc-to-proven table. Every other chapter assumes it.

### The reading order

The chapters are grouped into parts, and the parts are ordered so that each one only needs the ones before it. That order is not the same as the header dependency graph, and it is not arbitrary: strings need allocators, containers need allocators, and hosted services need strings, so the sequence below is what the material itself requires.

The chapter *numbers* are stable identifiers, not a reading sequence. Two of them are out of order on purpose: the alias index is an appendix you look things up in, and Chapter 8 is a reference you read after Chapter 3 has introduced the subject.

| Part                                          | Read                                                                          | Prerequisites                 | You can then                                                                                                         |
|-----------------------------------------------|-------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>I - Start here</b>                         | <a href="#">tutorial</a> -> 0                                                 | One introductory C book       | Build against the library and read anything below                                                                    |
| <b>II - The vocabulary every program uses</b> | <a href="#">1</a> -> <a href="#">2</a> -> <a href="#">3</a>                   | Chapter 0                     | Handle errors as values, own memory deliberately, hold text safely                                                   |
| <b>III - Data structures</b>                  | <a href="#">4</a>                                                             | Part II                       | Arrays, maps, lists, rings, sorting, searching, hashing, encoding                                                    |
| <b>IV - Text in and out</b>                   | <a href="#">8</a>                                                             | Chapter 3 section 3-section 4 | Format and parse anything, and teach the formatter your own types                                                    |
| <b>V - Talking to the operating system</b>    | <a href="#">5</a> -> <a href="#">9</a> -> <a href="#">10</a>                  | Part II                       | Files, directories, streams, standard I/O, time, randomness, mapping; sockets with deadlines; URLs and HTTP messages |
| <b>VI - Going further</b>                     | <a href="#">6</a> -> <a href="#">freestanding</a>                             | Parts II-V                    | Coroutines, jobs, thread-safety, bare metal, cross builds                                                            |
| <b>Appendices</b>                             | <a href="#">A</a> , <a href="#">B</a> , <a href="#">C</a> , <a href="#">D</a> | -                             | Look things up                                                                                                       |

### The chapters

- **Tutorial:** the library in nine short programs, one new idea at a time - *Part I, optional on-ramp*
  1. **Start here:** why this exists, hello world, the five contracts, glossary, [libc map](#) - *Part I*
  2. **Foundation:** types, errors, memory views, alignment, version, [panic](#) - *Part II*
  3. **Allocation:** heap, arena, pool, byte buffers, and the allocator trait - *Part II*

4. **Strings and text:** U8, U16, and an introduction to formatting and scanning - *Part II; the tutorial half of the text material*
5. **Containers and algorithms:** array, list, ring, map, sort/search, hashing, encoding - *Part III*
6. **Hosted services:** filesystem, tree walk, streams, sysio, environment, randomness, mmap, time - *Part V*
7. **Execution and platform:** coroutines, jobs, thread-safety, aliases, PAL, cross builds - *Part VI*
8. **Appendix A - Alias index:** every `alias_xcv.h` spelling - *reference only; not reading material*
9. **Formatting and scanning:** the full `fmt.h` and `scan.h` reference - *Part IV; the reference half of the text material*
10. **Networking:** addresses, TCP, UDP, deadlines, readiness, transports - *Part V; after Chapter 5*
11. **URLs and HTTP messages:** URL parsing and safe paths, the HTTP/1.1 head parser, body framing, writers, dates - *Part V; pure text handling, freestanding-available*

**Chapters 3 and 8 both cover the formatter and the scanner, and the division is deliberate.**

Chapter 3 introduces them alongside strings, with the everyday cases and enough to be productive. Chapter 8 is the complete reference: the full format grammar, every argument constructor, the scanner's error codes and recovery rules, and how to teach the formatter a type of your own. Read Chapter 3 first; reach for Chapter 8 when you need the exact behaviour of a specifier or a failure.

## 12. Platform support and verification

Primary verified hosted target:

- Linux x86\_64 with GCC or Clang in C23 mode.

Compile-only cross coverage exists when the corresponding toolchains are installed:

- Linux AArch64.
- Linux ARM hard-float.
- Linux i686 through `i686-linux-gnu-gcc` OR `gcc -m32 multilib`.
- Windows x86\_64 and i686 through MinGW/WinAPI paths.
- ARM Cortex-M freestanding.
- RISC-V ELF freestanding.

The cross matrix checks compilation, public header visibility, and target ABI assumptions. It does not replace runtime validation on the target platform.

Freestanding mode builds a reduced subset with OS-backed services removed. See `manual-freestanding.md` and Chapter 6 for details.

## 13. Appendix B: glossary

Terms this manual uses as if they were ordinary words. They are not ordinary C words, and every one of them is load-bearing.

Two rules the chapters follow, so that nothing here has to be guessed at:

- **No private vocabulary.** Where a widely used word exists for an idea, the manual uses that word. Where the library has invented something, it is defined here.
- **Every abbreviation is written out the first time it appears in a chapter**, and every one of them is listed below as well.

The Korean edition adds one more: an English term appears with its original spelling in parentheses the first time each chapter uses it - 뷰(view) - so a reader who learned the idea in English can find it, and a reader who did not is never handed an untranslated word.

| Term | Meaning here |
|------|--------------|
|------|--------------|

|                                 |                                                                                                                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>owned</b>                    | You are responsible for destroying it. Comes from a <code>_create</code> , goes to a <code>_destroy</code> , exactly once.                                                                                                          |
| <b>borrowed</b>                 | Points at memory someone else owns. Never destroyed. Valid only while the owner is.                                                                                                                                                 |
| <b>view</b>                     | A borrowed pointer + length pair, e.g. <code>proven_u8str_view_t</code> . Copyable, non-owning, no allocation.                                                                                                                      |
| <b>allocator</b>                | A value carrying four things: a context pointer and three function pointers ( <code>alloc</code> , <code>realloc</code> , <code>free</code> ). Passed by value into anything that may allocate.                                     |
| <b>arena</b>                    | An allocator that hands out memory by bumping a pointer through one block. Individual frees do nothing; you reset or destroy the whole arena at once. Fast, and perfect for "many small things with the same lifetime".             |
| <b>pool</b>                     | An allocator for many objects of one fixed size, with a free list, so freeing really does recycle a slot.                                                                                                                           |
| <b>trait</b>                    | A struct of function pointers used as an interface - C's answer to a virtual table. <code>proven_allocator_t</code> , <code>proven_writer_t</code> and <code>proven_rng_t</code> are traits. Not a C keyword; borrowed terminology. |
| <b>PAL</b>                      | Platform abstraction layer: the code under <code>platform/</code> that makes actual syscalls. The only OS-dependent part.                                                                                                           |
| <b>freestanding</b>             | A build with no operating system and no libc - bare metal. <code>PROVEN_FREESTANDING</code> selects it.                                                                                                                             |
| <b>failure atomicity</b>        | If an operation fails, it changes nothing. A failed grow leaves your old data intact and valid.                                                                                                                                     |
| <b>provenance</b>               | Which allocation a pointer came from. C's optimizer assumes pointers from different allocations never overlap; violating that is undefined behaviour, not merely surprising. Chapter 6 covers it.                                   |
| <b>UB (undefined behaviour)</b> | Not "unpredictable output" - the standard imposes no requirement at all, and optimizers are allowed to assume it never happens. This is why UB can delete your if.                                                                  |
| <b>[[nodiscard]]</b>            | A C23 attribute. The compiler errors if you throw the return value away. Used on every function whose error you must not drop.                                                                                                      |
| <b>fixed-capacity</b>           | Will not grow. Fails when full. Takes no allocator.                                                                                                                                                                                 |
| <b>growable</b>                 | Will reallocate when full. Takes an allocator. Always spelled <code>_grow</code> in the name.                                                                                                                                       |
| <b>CSPRNG</b>                   | Cryptographically secure pseudo-random number generator: output an attacker cannot predict even after seeing earlier output.                                                                                                        |

|                                                |                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>intrusive</b>                               | The list's links live <i>inside</i> your struct rather than in separately allocated nodes. No allocation per element.                                                                                                                                                                              |
| <b>code unit</b>                               | One element of an encoding: a byte in UTF-8, a 16-bit value in UTF-16. Not a character - one character can take several.                                                                                                                                                                           |
| <b>code point</b>                              | One character's number in Unicode. A code point takes one to four bytes in UTF-8, and one or two code units in UTF-16 - which is why counting either one is not counting characters.                                                                                                               |
| <b>API (application programming interface)</b> | The set of functions and types a library exposes for other programs to call. In this library, everything declared in <code>include/proven/</code> .                                                                                                                                                |
| <b>hosted</b>                                  | A build that has an operating system and a C standard library under it - the opposite of <i>freestanding</i> .                                                                                                                                                                                     |
| <b>heap</b>                                    | The general-purpose pool of memory <code>malloc</code> hands out from. <code>proven_heap_allocator()</code> is the allocator that uses it.                                                                                                                                                         |
| <b>slice</b>                                   | A borrowed pointer + length pair you may <b>write</b> through, e.g. <code>proven_mem_mut_t</code> . A <i>view</i> is the read-only form of the same idea.                                                                                                                                          |
| <b>result</b>                                  | A small struct carrying an error code and a value together, e.g. <code>proven_result_u8str_t</code> . The value means nothing until the error beside it has been checked.                                                                                                                          |
| <b>panic</b>                                   | The deliberate stop taken when a failure has no caller to return to. <code>proven_set_panic_handler()</code> chooses what happens; the default stops the program.                                                                                                                                  |
| <b>reserve</b>                                 | Raise a container's capacity <b>now</b> , so later growth does not reallocate. Saves copying on the heap, and dead storage in an arena.                                                                                                                                                            |
| <b>dangling</b>                                | A pointer to memory that has been freed or moved. Using one is undefined behaviour; the usual cause here is holding a pointer across a call that may reallocate.                                                                                                                                   |
| <b>use-after-free</b>                          | Reading or writing through a dangling pointer. The sanitizers below detect it.                                                                                                                                                                                                                     |
| <b>sanitizer</b>                               | A compiler mode that adds run-time checks: <b>ASan</b> (AddressSanitizer) finds memory errors, <b>UBSan</b> (UndefinedBehaviorSanitizer) finds undefined behaviour, <b>TSan</b> (ThreadSanitizer) finds data races. <code>./nob asan</code> , <code>./nob ubsan</code> , <code>./nob tsan</code> . |
| <b>partial write / short read</b>              | A single write or read that moved <i>fewer</i> bytes than asked for. Normal, not an error - and treating a short read as end of input is the classic way to lose the tail of a file.                                                                                                               |

|                          |                                                                                                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EOF</b> (end of file) | There is no more input. Reported as <code>PROVEN_ERR_EOF</code> , never as a zero-byte success, so it cannot be confused with "nothing arrived yet".                                                                                                         |
| <b>flush</b>             | Push a buffered writer's accumulated bytes onward. Nothing in this library flushes on your behalf at exit.                                                                                                                                                   |
| <b>back-pressure</b>     | Slowing a producer down because the consumer cannot keep up. <code>proven_writer_write_partial()</code> is the call that lets you notice and react.                                                                                                          |
| <b>durability</b>        | The guarantee that data survives a power cut. Reaching the operating system is not enough: <code>proven_fs_sync()</code> puts a file's bytes on the storage device, and <code>proven_fs_sync_dir()</code> does the same for a rename.                        |
| <b>atomic rename</b>     | Replacing a file by renaming a finished temporary over it. A reader sees the whole old file or the whole new one, never a half-written mixture.                                                                                                              |
| <b>advisory lock</b>     | A lock that excludes only the processes that also ask for it ( <code>proven_fs_lock()</code> ). It is a convention between cooperating programs, not access control.                                                                                         |
| <b>hard link</b>         | A second <b>name</b> for the same file. There is no original; the data lives until the last name is removed. Same filesystem only.                                                                                                                           |
| <b>symbolic link</b>     | A small file that holds a path. It may cross filesystems, and it may point at nothing - following it then fails.                                                                                                                                             |
| <b>memory mapping</b>    | Making a file's contents appear at an address, so the processor reads it as memory ( <code>mmap.h</code> ). <b>SHARED</b> mappings write back to the file; <b>PRIVATE</b> ones are <i>copy-on-write</i> : the writes exist in your process and nowhere else. |
| <b>copy-on-write</b>     | Sharing memory until somebody writes, at which point that writer gets a private copy. What <code>PROVEN_MMAP_PRIVATE</code> does.                                                                                                                            |
| <b>cursor</b>            | The scanner's position in the text it is reading ( <code>proven_scan_t.cursor</code> ). Scanning advances it; <code>proven_scan_skip_*</code> moves it deliberately.                                                                                         |
| <b>locale</b>            | The system's idea of local conventions - including whether the decimal separator is <code>.</code> or <code>,</code> . This library's number parsing is <b>locale-free</b> : a comma is never a decimal point, on any machine.                               |
| <b>entropy</b>           | Genuinely unpredictable bits, from the operating system or from hardware. A generator is <i>seeded</i> from entropy; a clock or a counter is not entropy, however random it looks.                                                                           |
| <b>seed</b>              | The starting value of a generator. The same seed replays the same sequence - essential for a reproducible test, and fatal for a key.                                                                                                                         |

|                                       |                                                                                                                                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PRNG / CSPRNG</b>                  | A pseudo-random number generator computes a sequence from a seed. A <b>C*ryptographically S*ecure</b> one (CSPRNG) is additionally unpredictable to an attacker who has seen earlier output.                                     |
| <b>HashDoS</b>                        | An attack that feeds a hash table keys chosen to collide, turning $O(1)$ lookups into $O(n^2)$ .<br>proven_map_create() defends against it with a keyed hash; proven_map_create_trusted() opts out for keys you choose yourself. |
| <b>open addressing</b>                | The map's layout: entries live in one flat bucket array and a collision moves to the next slot, rather than following a chain of separately allocated nodes.                                                                     |
| <b>tombstone</b>                      | The marker left where a map entry was removed, so lookups keep probing past it. Tombstones count towards the load factor, which is why heavy remove traffic still triggers a rehash.                                             |
| <b>rehash</b>                         | Rebuilding the bucket array at a new size. It invalidates every pointer a previous get_mut returned, which is why you hold the key rather than the pointer.                                                                      |
| <b>checksum</b>                       | A short value that detects accidental corruption - proven_crc32(). It detects accidents, never tampering.                                                                                                                        |
| <b>digest / hash</b>                  | A fixed-size value computed from data. <b>SHA-256</b> is a cryptographic digest (tamper-evident); <b>FNV-1a</b> and <b>SipHash-2-4</b> are table hashes, and only SipHash is keyed.                                              |
| <b>hex / Base64 / Base64URL</b>       | Ways to write bytes as text. Hex is two characters per byte; Base64 packs three bytes into four characters with + / =; Base64URL uses - _ and no padding, so it is safe in a URL or filename.                                    |
| <b>padding</b>                        | The = characters Base64 adds so the output length is a multiple of four. Base64URL leaves them out.                                                                                                                              |
| <b>BMP (Basic Multilingual Plane)</b> | The first 65,536 Unicode code points. A character outside it - an emoji, many rarer CJK characters - needs two UTF-16 code units.                                                                                                |
| <b>NUL terminator</b>                 | The zero byte that marks the end of a C string. A <i>view</i> does not have one, which is exactly why it carries a length instead.                                                                                               |
| <b>shortest round trip</b>            | Printing the fewest digits that read back as exactly the same floating-point value.<br>PROVEN_FLOAT_FORMAT_MODE_SHORTEST asks for it; it is what a serialiser wants.                                                             |
| <b>dispatch macro</b>                 | A macro built on C11 _Generic that picks a function from an argument's type - PROVEN_ARG(x) and PROVEN_SCAN_ARG(&x). It chooses among the named constructors; it is not itself one.                                              |

|                             |                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>identity constructor</b> | <code>proven_arg_identity()</code> / <code>proven_scan_arg_identity()</code> : they take an argument that has already been built and pass it through, so macro-driven code can accept one.                                                |
| <b>scratch allocator</b>    | An allocator passed for temporary working memory only, separate from the one that owns the result - e.g. the <code>scratch</code> parameter of <code>proven_map_set_with_scratch()</code> .                                               |
| <b>stackless coroutine</b>  | A function that can suspend and resume without a stack of its own: its state lives in a struct you hold. <code>coro.h</code> implements it with a switch, so it costs no thread and no allocation.                                        |
| <b>bounded queue</b>        | A queue with a fixed capacity that refuses when full rather than growing. The job system uses one, so a producer that outruns the workers is told so instead of exhausting memory.                                                        |
| <b>monotonic clock</b>      | A clock that only moves forward, for measuring how long something took. Distinct from the wall clock, which can jump when the system time is corrected - measuring a duration with the wall clock is how a negative elapsed time happens. |

## 14. Appendix C: public header map

### How to find things

There are 35 public headers and one umbrella. `#include "proven.h"` pulls in everything and is what the examples in this manual do; including individual headers is for when you care about compile time or want the dependency to be visible in the file.

Two things this table tells you that the file names do not:

- **Which chapter documents it.** Every header has exactly one chapter that explains it, and the build enforces that every public function is named somewhere under `manual/` - so if a symbol is not in the chapter you expect, it is in the manual somewhere and this map says where.
- **Whether it survives a freestanding build.** The headers assigned to Chapter 5 are the hosted ones: they need a filesystem, standard streams, a clock, virtual memory or threads. Everything else compiles with no operating system. [The freestanding guide](#) has the authoritative per-module table.

| Header                   | Main purpose                                                                                                                              | Chapter          |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <code>proven.h</code>    | Umbrella include                                                                                                                          | This file        |
| <code>types.h</code>     | Fixed-width aliases, checked arithmetic, error enum                                                                                       | Chapter 1        |
| <code>error.h</code>     | Error predicate helpers                                                                                                                   | Chapter 1        |
| <code>memory.h</code>    | Byte views, slicing, range checks, <code>memcpy</code>                                                                                    | Chapter 1        |
| <code>align.h</code>     | Alignment constants and align-up helpers                                                                                                  | Chapter 1        |
| <code>version.h</code>   | Version macros                                                                                                                            | Chapter 1        |
| <code>panic.h</code>     | Registerable panic handler                                                                                                                | Chapter 1        |
| <code>config.h</code>    | Compile-time feature toggles ( <code>PROVEN_FREESTANDING</code> , <code>PROVEN_FMT_NO_FLOAT</code> , <code>PROVEN_NO_U16STR</code> , ...) | Chapters 1 and 6 |
| <code>allocator.h</code> | Allocator trait                                                                                                                           | Chapter 2        |

|                |                                                                                                                                                                                                   |                  |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| heap.h         | PAL-backed heap allocator                                                                                                                                                                         | Chapter 2        |
| alloc_check.h  | An allocator wrapper that knows its own blocks: catches a foreign free, a double free and a wrong realloc at the call (tests and debugging; off unless PROVEN_ALLOC_CHECK)                        | Chapter 2        |
| arena.h        | Bump allocator                                                                                                                                                                                    | Chapter 2        |
| pool.h         | Fixed-size recycler allocator                                                                                                                                                                     | Chapter 2        |
| buffer.h       | Fixed-capacity byte buffer                                                                                                                                                                        | Chapter 2        |
| u8str.h        | Owned U8 string and borrowed U8 views                                                                                                                                                             | Chapter 3        |
| u16str.h       | Owned U16 string and borrowed U16 views                                                                                                                                                           | Chapter 3        |
| utf.h          | Strict UTF-8 <-> UTF-16 transcoding: measure, convert, convert in pieces, grow                                                                                                                    | Chapter 3        |
| fmt.h          | Structural formatter and format arguments                                                                                                                                                         | Chapter 3        |
| scan.h         | Structural scanner and typed scan destinations                                                                                                                                                    | Chapter 3        |
| float_parse.h  | Locale-free decimal -> double/float parser (proven_strtod, proven_parse_double_ascii)                                                                                                             | Chapter 8        |
| float_format.h | double/float -> decimal formatter (fixed %f/%e, shortest)                                                                                                                                         | Chapter 8        |
| float_config.h | Float-engine tuning (PROVEN_FLOAT_BIGINT_LIMBS, precision caps)                                                                                                                                   | Chapters 6 and 8 |
| array.h        | Generic growable vector                                                                                                                                                                           | Chapter 4        |
| list.h         | Intrusive doubly-linked list                                                                                                                                                                      | Chapter 4        |
| ring.h         | Fixed-capacity FIFO ring                                                                                                                                                                          | Chapter 4        |
| map.h          | Open-addressing map                                                                                                                                                                               | Chapter 4        |
| algorithm.h    | Array sort and search helpers                                                                                                                                                                     | Chapter 4        |
| hash.h         | FNV-1a, SipHash-2-4, CRC-32, SHA-256, by use case                                                                                                                                                 | Chapter 4        |
| hash_legacy.h  | SHA-1 and MD5, for formats that fix them                                                                                                                                                          | Chapter 4        |
| encode.h       | Hex and Base64 (standard + URL-safe), bytes to text and back                                                                                                                                      | Chapter 4        |
| fs.h           | Files, directories, metadata, links, locks, read-all, tree walk                                                                                                                                   | Chapter 5        |
| stream.h       | Buffered writers, readers, and a line reader - and, through sysio.h, the standard streams (hosted-only)                                                                                           | Chapter 5        |
| net.h          | Addresses, TCP, UDP, Unix-domain streams, deadlines, readiness, transports (hosted-only)                                                                                                          | Chapter 9        |
| url.h          | URL parsing, percent-coding, and a request path resolved to stay inside its root                                                                                                                  | Chapter 10       |
| http.h         | HTTP/1.1 codec: head parser, body framing and decoder, writers, dates                                                                                                                             | Chapter 10       |
| sysio.h        | Standard streams as writers/readers, line input from stdin, buffered output, printing, scanning, environment access                                                                               | Chapter 5        |
| random.h       | Randomness by use case: xoshiro256** (reproducible), ChaCha20 (cryptographic), the OS CSPRNG, and unbiased range/shuffle helpers. The generators work freestanding; only the OS source is hosted. | Chapter 5        |
| mmap.h         | Memory-mapped file regions                                                                                                                                                                        | Chapter 5        |
| time.h         | Timestamp, datetime, sleep, datetime formatting                                                                                                                                                   | Chapter 5        |
| coro.h         | Stackless coroutine macros                                                                                                                                                                        | Chapter 6        |

|             |                                                       |                  |
|-------------|-------------------------------------------------------|------------------|
| job.h       | Bounded worker-thread job system                      | Chapter 6        |
| alias_xcv.h | Optional short alias layer and generated spelling map | Chapters 6 and 7 |

## 15. Appendix D: the libc map

If you already write C, this is the fastest way in. Every row links to the chapter that explains the trade.

| You would write        | Use instead                                                | Why it differs                                                                                  |
|------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| malloc / free          | proven_heap_allocator() + _create / _destroy               | The allocator is a parameter, so the same code runs on an arena or a pool. <a href="#">Ch 2</a> |
| strcpy, strcat         | proven_u8str_append, _append_grow                          | Sizes are known, so overruns are refused instead of performed. <a href="#">Ch 3</a>             |
| strlen                 | view.size                                                  | The length is already there; nothing scans. <a href="#">Ch 3</a>                                |
| strcmp for equality    | proven_u8str_view_eq                                       | Works on text with embedded NULs, and does not walk past the end. <a href="#">Ch 3</a>          |
| strstr                 | proven_u8str_view_find                                     | Returns an index or PROVEN_INDEX_NOT_FOUND; the search is not naive. <a href="#">Ch 3</a>       |
| strtok                 | <i>(no equivalent yet)</i>                                 | strtok mutates its input and cannot be nested. A view-based splitter is designed in RFC-0004.   |
| printf                 | proven_println("{ }", PROVEN_ARG(x))                       | Types come from the arguments, not from the format string. <a href="#">Ch 8</a>                 |
| sprintf                | proven_u8str_append_fmt                                    | Writes into a sized destination and refuses to overrun it. <a href="#">Ch 8</a>                 |
| sscanf                 | proven_scan_*, proven_scan_fmt                             | Reports which field failed and where the cursor stopped. <a href="#">Ch 8</a>                   |
| strtod                 | proven_parse_f64_ascii                                     | Correctly rounded, locale-independent, no errno. <a href="#">Ch 8</a>                           |
| fopen / fread / fclose | proven_fs_open, _read, _close, OR proven_fs_read_all_u8str | Explicit errors, no hidden buffering, one call for whole files. <a href="#">Ch 5</a>            |
| fgets                  | proven_sysio_read_line, proven_reader_read_line            | A line that exactly fills the buffer is returned, not lost. <a href="#">Ch 5</a>                |
| qsort                  | proven_array_sort                                          | Introsort: $O(n \log n)$ guaranteed, not quicksort's worst case. <a href="#">Ch 4</a>           |
| bsearch                | proven_array_binary_search                                 | Same shape, same comparator contract. <a href="#">Ch 4</a>                                      |

|                                       |                                                                         |                                                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| rand                                  | proven_xoshiro256ss_* OR<br>proven_random_bytes                         | Reproducible and fast, or<br>unpredictable and secure - you<br>pick, deliberately. <a href="#">Ch 5</a>             |
| time / clock                          | proven_time_now,<br>proven_time_monotonic_now,<br>proven_time_breakdown | Nanoseconds, with the wall<br>clock and the monotonic clock<br>as two separate calls. <a href="#">Ch 5</a>          |
| socket / connect / send / recv / poll | proven_net_connect, _read,<br>_write_all, proven_net_poll               | Every wait takes a deadline,<br>and refused, reset and end-of-<br>input are different answers. <a href="#">Ch 9</a> |
| assert                                | proven_panic + a panic hook                                             | Works in freestanding builds<br>and is overridable. <a href="#">Ch 1</a>                                            |

## 16. Where to go next

The chapters are ordered so that each one only needs the ones before it.

| Part              | Read                                                         | For                                                                                          |
|-------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>I</b>          | This chapter                                                 | The contracts and the vocabulary                                                             |
| <b>II</b>         | <a href="#">1</a> -> <a href="#">2</a> -> <a href="#">3</a>  | Errors, memory, and text: what every program<br>uses                                         |
| <b>III</b>        | <a href="#">4</a>                                            | Arrays, maps, lists, rings, sorting, hashing,<br>encoding                                    |
| <b>IV</b>         | <a href="#">8</a>                                            | Formatting and scanning in full, once Chapter 3<br>has introduced them                       |
| <b>V</b>          | <a href="#">5</a> -> <a href="#">9</a> -> <a href="#">10</a> | Files, streams, standard I/O, time, randomness,<br>mapping; then sockets; then URLs and HTTP |
| <b>VI</b>         | <a href="#">6</a> -> <a href="#">freestanding</a>            | Coroutines, jobs, thread-safety, bare metal, cross<br>builds                                 |
| <b>Appendices</b> | <a href="#">A: alias index</a> , B and D above               | Looking things up                                                                            |

If you are an experienced C programmer in a hurry, read section 15 above, then [Chapter 1](#), then whichever chapter covers the thing you need. If you are newer, read Parts I and II in order - they are short, and everything later assumes them.

# Chapter 1: Foundation - Errors, Types, and Memory Views

**Part II - The vocabulary every program uses. Prerequisite: [Chapter 0](#).** After this chapter you can handle every failure this library reports, describe a region of memory without losing track of its size, and do arithmetic on sizes that cannot silently wrap.

This chapter covers `types.h`, `error.h`, `memory.h`, `align.h`, `version.h`, and `panic.h` - the pieces every other chapter is built out of. Nothing here allocates, and nothing here talks to the operating system. It is the shortest chapter that is genuinely required reading.

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## 1. Errors are values

### Why this is first

Every other page of this manual assumes you know how failure is reported here, so this is the one section you cannot skip.

C has never agreed with itself about how a function should report failure. `malloc` returns `NULL`. `fopen` returns `NULL`. `read` returns `-1`. `strtol` returns `0` and *maybe* sets `errno`, but only if you cleared `errno` first, because a successful call is permitted to leave junk in it. `printf` returns a negative number that almost nobody checks. Four conventions, and the only thing they share is that **ignoring them is silent and legal**:

```
char *p = malloc(n);
p[0] = 'x';                /* wrong: malloc returns NULL on failure */

FILE *f = fopen(path, "r");
fread(buf, 1, n, f);       /* wrong: f may be NULL */

long v = strtol(s, NULL, 10); /* wrong: 0 could be the value or the failure */
```

The deeper problem is not that these are easy to get wrong. It is that **the type does not say anything is possible**. `char *` is the same type whether or not it can be `NULL`, so nothing -- not the compiler, not a reviewer skimming, not you at 2 a.m. - is prompted to check.

`errno` makes it worse by making the error *global and temporary*. It must be read at exactly the right moment; any intervening library call may overwrite it; and it is per-thread, so the fix for one bug class opened another.

### What this library does instead

The error is a return value, and it has a type of its own.

When a function's only outcome is success or failure, it returns `proven_err_t` - a plain enum where `PROVEN_OK` is `0`:

```
proven_err_t err = proven_u8str_append(&s, PROVEN_LIT("hello"));
if (!proven_is_ok(err)) { /* handle it */ }
```

When a function has something to give back, the value and the error travel together in one struct, and **the value is meaningless until the error has been checked**:

```

proven_result_u8str_t s = proven_u8str_create(alloc, 64);
if (!proven_is_ok(s.err)) return;      /* s.value is not a string; do not touch it */

```

This is the same idea as `Result` in Rust or `std::expected` in C++23, expressed in the only way C allows - a struct returned by value. There is no allocation, no indirection, and no hidden control flow. What you give up is the ability to skip error handling silently, which is the point.

Three properties are worth stating explicitly because the rest of the library relies on them:

- **PROVEN\_OK is zero**, so a result struct zero-initialised with `{0}` starts out meaning "success, empty".
- **Fallible functions are** `[[nodiscard]]` - a C23 attribute that makes the compiler reject code that throws the return value away. You can still ignore an error, but you have to write `(void)` in front of the call and thereby say that you meant it.
- **Failure is failure-atomic** unless a function documents otherwise: if an operation fails, it changed nothing. A failed append leaves your string exactly as it was.

## Reference

```

typedef enum {
    PROVEN_OK = 0,
    PROVEN_ERR_NOMEM,
    PROVEN_ERR_OUT_OF_BOUNDS,
    PROVEN_ERR_INVALID_ENCODING,
    PROVEN_ERR_INVALID_ARG,
    PROVEN_ERR_IO,
    PROVEN_ERR_NOT_FOUND,
    PROVEN_ERR_INVALID_STATE,
    PROVEN_ERR_NEED_MORE,
    PROVEN_ERR_OVERFLOW,
    PROVEN_ERR_UNSUPPORTED,
    PROVEN_ERR_AGAIN,
    PROVEN_ERR_EOF,
    PROVEN_ERR_BUSY,
    PROVEN_ERR_PERMISSION,
    PROVEN_ERR_INVALID_FORMAT,
    PROVEN_ERR_EXISTS,
    PROVEN_ERR_TIMEOUT,
    PROVEN_ERR_REFUSED,
    PROVEN_ERR_RESET,
    PROVEN_ERR_UNREACHABLE,
    PROVEN_ERR_UNTRUSTED
} proven_err_t;

```

```

#define PROVEN_ERR_LAST      PROVEN_ERR_UNTRUSTED      /* moves when a code is added */
#define PROVEN_ERR_RESERVED_END 0x1000                /* never reached, in any version */

```

| Error                       | Typical meaning                                                                | You usually respond by                                |
|-----------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------|
| PROVEN_OK                   | Success.                                                                       | Carrying on.                                          |
| PROVEN_ERR_NOMEM            | The allocator could not provide memory.                                        | Giving up on this operation; the object is unchanged. |
| PROVEN_ERR_OUT_OF_BOUNDS    | An index, size, capacity, or range is invalid - including "this does not fit". | Growing the destination, or refusing the input.       |
| PROVEN_ERR_INVALID_ENCODING | Encoded text failed validation (hex, Base64).                                  | Rejecting the input as malformed.                     |

|                           |                                                                                                                                    |                                                                                       |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| PROVEN_ERR_INVALID_ARG    | A null pointer, an invalid allocator, an impossible mode, a wrong argument count.                                                  | Fixing the call - this one usually means a bug in your code, not bad data.            |
| PROVEN_ERR_IO             | The OS or device failed the operation.                                                                                             | Retrying, or reporting to the user.                                                   |
| PROVEN_ERR_NOT_FOUND      | A file, key, substring, or resource does not exist.                                                                                | Taking the "absent" branch; often not an error at all.                                |
| PROVEN_ERR_INVALID_STATE  | The object's state does not allow this operation.                                                                                  | Fixing the sequence of calls.                                                         |
| PROVEN_ERR_NEED_MORE      | The input ends inside something more input would complete - a split UTF-8 character, a number cut by a read.                       | Keeping what you have and reading more; for a complete text it is malformed.          |
| PROVEN_ERR_OVERFLOW       | Integer conversion or size arithmetic overflowed.                                                                                  | Refusing the size; see section 4.                                                     |
| PROVEN_ERR_UNSUPPORTED    | Unavailable on this platform or build profile.                                                                                     | Taking a different route (e.g. a pipe cannot seek).                                   |
| PROVEN_ERR_AGAIN          | Not now; retry later.                                                                                                              | Retrying, usually after waiting.                                                      |
| PROVEN_ERR_EOF            | End of input.                                                                                                                      | Stopping the read loop - expected, not exceptional.                                   |
| PROVEN_ERR_BUSY           | A queue, lock, or resource is busy.                                                                                                | Backing off.                                                                          |
| PROVEN_ERR_PERMISSION     | Access denied.                                                                                                                     | Reporting; retrying will not help.                                                    |
| PROVEN_ERR_INVALID_FORMAT | A format or scan template is itself malformed.                                                                                     | Fixing the format string - a bug, not bad data.                                       |
| PROVEN_ERR_EXISTS         | A call asked to create a name found it already there: an exclusive create (PROVEN_FS_CREATE_NEW), proven_fs_mkdir, proven_fs_link. | Choosing another name, or using what is there - after checking it is what you expect. |
| PROVEN_ERR_TIMEOUT        | A deadline you gave passed before the operation could finish.                                                                      | Retrying, giving up, or closing - the object is still usable; nothing broke.          |
| PROVEN_ERR_REFUSED        | The other end answered and said no: nothing is listening at that address.                                                          | Trying another address, or reporting; an immediate retry gets the same answer.        |
| PROVEN_ERR_RESET          | The peer ended the connection abruptly, or it was aborted on the way.                                                              | Closing your end; whatever was in flight may not have arrived.                        |
| PROVEN_ERR_UNREACHABLE    | No route to that network or host.                                                                                                  | Reporting, or retrying later - the cause is outside the program.                      |
| PROVEN_ERR_UNTRUSTED      | A peer's identity could not be verified. Reserved for certificate checks; nothing in this version returns it.                      | Refusing to continue.                                                                 |

**Room for your own codes.** Every `proven_err_t` value is below `PROVEN_ERR_RESERVED_END` (0x1000), and that is a promise for every later version, not a description of this one: proven will never define a

code at or above it. `PROVEN_ERR_LAST` names the highest code today and moves when one is added - compare against it, never against whichever code you happen to know is last. A program with failures of its own can therefore carry proven's codes unchanged in a wider integer and number its own from 0x1000 up: no remapping table, and a proven code it passes on still means exactly what the table above says.

```
/* An application's own codes, above everything proven will ever use. */
enum { APP_ERR_FIRST = PROVEN_ERR_RESERVED_END, APP_ERR_BAD_GLYPH = APP_ERR_FIRST, APP_ERR_TOO_DEEP };
_Static_assert(PROVEN_ERR_LAST < APP_ERR_FIRST, "proven's codes stay below ours");
int app_err = APP_ERR_TOO_DEEP; /* a proven_err_t converts into the same int unchanged */
(void)app_err;

static inline int proven_is_ok(proven_err_t err);
#define PROVEN_IS_OK(err) proven_is_ok(err)
```

`proven_is_ok` returns non-zero when `err == PROVEN_OK`. `PROVEN_IS_OK` is the macro spelling for places where you want it to look like a macro; they do the same thing.

Correct:

```
proven_result_u8str_t s = proven_u8str_create(alloc, 32);
if (!proven_is_ok(s.err)) {
    return; /* nothing was created, so there is nothing to destroy */
}
(void)proven_u8str_append(&s.value, PROVEN_LIT("ready"));
proven_u8str_destroy(alloc, &s.value);
```

Wrong - testing the error as a bare truth value:

```
proven_result_u8str_t s = proven_u8str_create(alloc, 32);
if (s.err) {
    /* wrong: works today only because PROVEN_OK is 0. It reads as "if there is
       an error" to someone who already knows that, and as nothing at all to
       everyone else. Say what you mean: !proven_is_ok(s.err). */
    return s.err;
}
```

Wrong - the mistake this whole design exists to prevent:

```
proven_result_u8str_t s = proven_u8str_create(alloc, 32);
(void)proven_u8str_append(&s.value, text); /* wrong: s.value may be garbage */
proven_u8str_destroy(alloc, &s.value); /* wrong: destroying what was never created */
```

Reading `.value` before checking `.err` is the one error this convention cannot catch for you. The compiler will not complain - the struct exists either way - so it is worth making a habit of writing the check on the line immediately after the call.

### Worked example: errors as values

This program is compiled and run by the test suite, so it cannot fall out of date. It shows the two shapes a fallible call takes - a bare `proven_err_t` when there is nothing to hand back, and a `proven_result_*_t` when there is - and why the value inside a result means nothing until you have checked the error beside it.

```
/*
 * Errors are values in proven: a fallible call hands back either an error or a
 * result, and the compiler makes you look at it. There is nothing to unwind and
 * nothing global to consult.
 */

/* A fallible operation returns proven_err_t when it has no value to give back. */
static proven_err_t write_greeting(proven_u8str_t *out) {
    return proven_u8str_append(out, PROVEN_LIT("hello"));
}
```

```

/* When there IS a value, it comes wrapped with the error that guards it. The
 * value is only meaningful once you have checked `err`. */
static proven_result_size_t half(proven_size_t n) {
    proven_result_size_t res = {0};
    if (n % 2 != 0) {
        res.err = PROVEN_ERR_INVALID_ARG; /* leave res.value at 0: it means nothing */
        return res;
    }
    res.err = PROVEN_OK;
    res.value = n / 2;
    return res;
}

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- checking a plain proven_err_t ----- */
    proven_result_u8str_t s = proven_u8str_create(alloc, 32);
    EXAMPLE_REQUIRE(proven_is_ok(s.err), "creating a 32-byte string should succeed");

    proven_err_t err = write_greeting(&s.value);
    if (!proven_is_ok(err)) {
        /* Nothing was appended, and the string is still valid: proven's
         * grow-style operations are failure-atomic. */
        proven_u8str_destroy(alloc, &s.value);
        return 1;
    }
    EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&s.value), PROVEN_LIT("hello")),
        "the greeting should have been appended");

    /* --- checking a result struct ----- */
    proven_result_size_t ok = half(10);
    EXAMPLE_REQUIRE(proven_is_ok(ok.err), "10 is even, so halving it must succeed");
    EXAMPLE_REQUIRE(ok.value == 5, "half of 10 is 5");

    proven_result_size_t bad = half(7);
    EXAMPLE_REQUIRE(bad.err == PROVEN_ERR_INVALID_ARG, "7 is odd, so halving it must fail");
    /* bad.value is NOT to be read. It is 0 here, but that is an implementation
     * detail of this function, not a promise of the result type. */

    /* --- the error is impossible to drop by accident ----- */
    /* proven_u8str_append is [[nodiscard]], so this would be a compile error:
     *
     *     proven_u8str_append(&s.value, PROVEN_LIT("!"));
     *
     * If you really do want to ignore a failure, you have to say so: */
    (void)proven_u8str_append(&s.value, PROVEN_LIT("!"));

    printf("greeting: %s\n", proven_u8str_as_cstr(&s.value));

    proven_u8str_destroy(alloc, &s.value);
    return EXAMPLE_OK();
}

```

## 2. The types, and why they are spelled differently

### Why not just use `int` and `char`?

C's built-in types are famously vague. `int` is at least 16 bits, `long` is at least 32, and `char` may be signed or unsigned depending on the compiler and the target. Code that assumes otherwise works on your machine and breaks on someone else's, and the break is usually silent.

C99 fixed the width problem with `<stdint.h>` (`int32_t`, `uint64_t` and friends). This library uses those underneath and gives them its own names, for one reason: **a single spelling that means the same thing on every target the library supports**, including the freestanding ones where `<stdint.h>` may be the only header available.

There is a second, subtler reason, and it is the one that matters in practice: `proven_u8` and `proven_byte_t` are both 8 bits, and they mean different things.

- `proven_u8` is a **number** between 0 and 255. Add it, compare it, print it.
- `proven_byte_t` is a **byte of some object's representation**. It is an alias of `unsigned char`, which is the one type C allows you to use to inspect the raw bytes of anything without invoking undefined behaviour.

Using `proven_byte_t` for raw memory is not a style preference. C's strict-aliasing rules say that reading an object through a pointer of the wrong type is undefined, with `unsigned char` as the explicit exception. Every buffer, view, and hash input in this library is `proven_byte_t` for exactly that reason.

### Reference

| Name                                                                                                 | Meaning                           | Notes                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_i8</code> , <code>proven_i16</code> , <code>proven_i32</code> , <code>proven_i64</code> | Signed fixed-width integers       | Aliases of the standard fixed-width types.                                                                                                                           |
| <code>proven_u8</code> , <code>proven_u32</code> , <code>proven_u64</code>                           | Unsigned fixed-width integers     | <code>proven_u8</code> is <b>numeric</b> ; use <code>proven_byte_t</code> for raw object bytes.                                                                      |
| <code>proven_u16</code>                                                                              | 16-bit code unit type             | Uses <code>char16_t</code> when <code>&lt;uchar.h&gt;</code> is available, otherwise <code>uint_least16_t</code> . The U16 string APIs use it for UTF-16 code units. |
| <code>proven_byte_t</code>                                                                           | Byte-level object representation  | Alias of <code>unsigned char</code> ; the type you may legally inspect any object's bytes through.                                                                   |
| <code>proven_size_t</code>                                                                           | Size and index type               | Alias of <code>size_t</code> . Unsigned - see section 4 before you subtract two of them.                                                                             |
| <code>proven_ptrdiff_t</code>                                                                        | Pointer difference, signed offset | Alias of <code>ptrdiff_t</code> .                                                                                                                                    |
| <code>proven_intptr_t</code> , <code>proven_uintptr_t</code>                                         | Pointer-sized integers            | Only for explicit pointer-to-integer work, such as the arena's range checks.                                                                                         |

```
typedef struct {
    proven_err_t err;
    proven_size_t value;
} proven_result_size_t;
```

`proven_result_size_t` returns either a size or an error; value is valid only when `err == PROVEN_OK`. It is what you get back from partial appends, file reads and writes, and size queries - anywhere the answer is "how many", and the operation could fail.

Wrong - treating a byte as a small number's storage:

```
proven_u8 *raw = (proven_u8 *)&some_struct; /* wrong type for raw inspection */
for (size_t i = 0; i < sizeof some_struct; i++) hash ^= raw[i];
```

It happens to work on every mainstream compiler, and it is still the wrong type to have written: `proven_byte_t` is the one with the aliasing guarantee behind it, and using it documents that you are looking at representation rather than at a number.

### 3. Memory views: a pointer and a length, together

#### The problem: pointers forget

A pointer says where something starts and nothing else. Everything C has ever done about the "where does it end" half has been a convention layered on top:

- **A NUL terminator**, for strings - which costs an  $O(n)$  scan every time you want the length, cannot represent text containing a zero byte, and produces a buffer overrun the moment the terminator is missing.
- **A separate length parameter**, for everything else - which works right up until someone passes the wrong one, because nothing ties the two arguments together.

```
void process(const unsigned char *data, size_t len);
process(buf, sizeof buf); /* fine */
process(buf, sizeof(buf) - 1); /* also compiles; now the last byte is invisible */
process(other, sizeof buf); /* also compiles; now it reads past the end */
```

Every one of those is a legal call. The relationship between the pointer and the length exists only in the programmer's head.

#### What this library does instead

A **view** is a pointer and a size in one struct, passed by value. They cannot get separated, because they are one thing.

```
typedef struct { const proven_byte_t *ptr; proven_size_t size; } proven_mem_view_t; /* read-only */
typedef struct {      proven_byte_t *ptr; proven_size_t size; } proven_mem_mut_t; /* writable */
typedef struct {      proven_byte_t *ptr; proven_size_t size; } proven_mem_t;      /* owned */
```

They are two words. Copying one is free, and there is no allocation anywhere in sight. What a view does *not* do is own anything: it points at bytes that belong to someone else, and it becomes invalid the moment they do. That rule is contract 2 from [Chapter 0](#).

The three spellings differ only in intent, and the intent is the point:

- `proven_mem_view_t` - **borrowed, read-only**. The `const` is on the pointed-to bytes, so the compiler enforces it.
- `proven_mem_mut_t` - **borrowed, writable**. You may write through it; you still do not own it.
- `proven_mem_t` - **owned**. Somebody must free this with the allocator that produced it. The type does not enforce that; it announces it.

And two result wrappers, for the operations that can fail:

```
typedef struct { proven_err_t err; proven_mem_mut_t value; } proven_result_mem_mut_t;
typedef struct { proven_err_t err; proven_mem_view_t value; } proven_result_mem_view_t;
```

`proven_result_mem_mut_t` is what an allocator hands back. `proven_result_mem_view_t` is what a *checked* slice hands back.

## Slicing, checked and unchecked

Taking a sub-range of a view is the most common thing you will do with one, and it comes in two forms on purpose:

| API                                                                      | Purpose                                                                                                                | Return                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_mem_view_from_owned(mem)</code>                             | Read-only view over an owned block.                                                                                    | <code>proven_mem_view_t</code> .                                                                                                                                                                                           |
| <code>proven_mem_mut_from_owned(mem)</code>                              | Writable view over an owned block.                                                                                     | <code>proven_mem_mut_t</code> .                                                                                                                                                                                            |
| <code>proven_mem_view_slice_checked(view, offset, size)</code>           | Sub-view, validated.                                                                                                   | <code>proven_result_mem_view_t</code> .                                                                                                                                                                                    |
| <code>proven_mem_view_slice_unchecked(view, offset, size)</code>         | Sub-view, <b>not</b> validated.                                                                                        | <code>proven_mem_view_t</code> .                                                                                                                                                                                           |
| <code>proven_mem_mut_slice_checked(mut, offset, size)</code>             | Writable sub-slice, validated.                                                                                         | <code>proven_result_mem_mut_t</code> .                                                                                                                                                                                     |
| <code>proven_mem_mut_slice_unchecked(mut, offset, size)</code>           | Writable sub-slice, <b>not</b> validated.                                                                              | <code>proven_mem_mut_t</code> .                                                                                                                                                                                            |
| <code>proven_range_contains_ptr(base, cap, ptr, size, out_offset)</code> | Is this pointer range inside that allocation? Integer address comparison, no pointer arithmetic on unrelated pointers. | <code>_Bool</code> .                                                                                                                                                                                                       |
| <code>proven_memcmp(s1, s2, size)</code>                                 | Compare raw memory.                                                                                                    | Zero if equal; sign by byte order (unsigned).                                                                                                                                                                              |
| <code>proven_mem_copy(dst, dst_cap, src)</code>                          | Bounded copy of a view into <code>dst</code> .                                                                         | <code>PROVEN_OK</code> ; <code>PROVEN_ERR_OUT_OF_BOUNDS</code> if it would not fit - <b>and nothing is written</b> ; <code>PROVEN_ERR_INVALID_ARG</code> on a null pointer with a non-zero size. Regions must not overlap. |
| <code>proven_mem_move(dst, dst_cap, src)</code>                          | As <code>proven_mem_copy</code> , but the regions may overlap.                                                         | As above.                                                                                                                                                                                                                  |

The checked form's exact behaviour:

- `view.size > 0 && view.ptr == NULL -> PROVEN_ERR_INVALID_ARG` (a size with no memory behind it is a bug, not an empty view).
- The requested range falls outside the view -> `PROVEN_ERR_OUT_OF_BOUNDS`.
- `size == 0 -> an empty view (ptr == NULL, size == 0)` with `PROVEN_OK`.

**Use the checked form by default.** The unchecked form exists for the case where you have already proved the range in the lines above - inside a loop whose bounds you computed, for instance - and the second check would be pure cost. That is a real case, and it is narrower than it feels.

Wrong - unchecked slicing on numbers that came from outside:

```
proven_mem_view_t part = proven_mem_view_slice_unchecked(view, user_offset, user_size);
/* wrong: user_offset and user_size were never validated. This is the buffer
   overrun from the top of section 3, reintroduced through a safer-looking API. */
```

Wrong - rejecting the empty view:

```
if (!view.ptr) return PROVEN_ERR_INVALID_ARG; /* wrong: {NULL, 0} is legal */
```

An empty view is a normal value: a zero-length slice, a file with no content, a string that was just reset. The test for a *malformed* view is a null pointer with a non-zero size:

```
proven_mem_view_t view = {0}; /* size 0, ptr NULL: a legal empty view */
proven_err_t err = PROVEN_OK;

if (view.size > 0 && !view.ptr) {
    err = PROVEN_ERR_INVALID_ARG; /* only a null pointer with bytes behind it is a bug */
}
(void)err;
```

### Passing a view to a macro

Several entry points are function-like macros: `PROVEN_ARG`, `proven_scan_fmt`, the `print` and `append_fmt` families, and the `PROVEN_MAP_*_U8*` wrappers. The preprocessor splits macro arguments at every comma that is not inside parentheses - and the comma inside a compound literal's braces is not. Passing `(proven_u8str_view_t){ p, n }` therefore hands the macro two arguments, and the compiler's error names an internal function you never called. Pass a variable, a `PROVEN_LIT`, or wrap the compound literal in parentheses:

```
const proven_byte_t bytes[] = { '4', '2' };
proven_u8str_view_t digits = { bytes, sizeof bytes };
proven_println("{", PROVEN_ARG(digits)); /* a variable */
proven_println("{", PROVEN_ARG(((proven_u8str_view_t){ bytes, 1 }))); /* parentheses */
```

## 4. Size arithmetic that cannot wrap

### Why a multiplication needs a function call

`proven_size_t` is unsigned, and unsigned arithmetic in C does not overflow - it *wraps*, silently and legally, modulo  $2^{64}$ . That single rule is behind a large fraction of the industry's allocation bugs:

```
proven_size_t total = count * sizeof(item_t); /* wrong: wraps for large count */
void *p = malloc(total); /* succeeds, and is far too small */
items[count - 1] = x; /* writes way past the end */
```

With `count =  $2^{61}$`  and a 16-byte item, `total` is 0. `malloc(0)` succeeds. Every subsequent write is out of bounds, and nothing anywhere reported an error. The same shape appears whenever a size is computed from data that came from outside the program - a file header, a network packet, a user-supplied count.

Subtraction has the mirror problem, and it is easier to hit:

```
proven_size_t remaining = view.size - offset; /* wrong when offset > size: a huge number */
```

### What this library does instead

Three macros that do the arithmetic and tell you whether it fit.

```
#define PROVEN_CKD_ADD(res, a, b) /* true on overflow */
#define PROVEN_CKD_SUB(res, a, b)
#define PROVEN_CKD_MUL(res, a, b)
```

| Macro                                  | Computes           | Returns                                                 | Writes *res       |
|----------------------------------------|--------------------|---------------------------------------------------------|-------------------|
| <code>PROVEN_CKD_ADD(res, a, b)</code> | <code>a + b</code> | <b>true on overflow</b> , false on success              | Only when it fits |
| <code>PROVEN_CKD_SUB(res, a, b)</code> | <code>a - b</code> | <b>true on overflow</b> (including unsigned underflow)  | Only when it fits |
| <code>PROVEN_CKD_MUL(res, a, b)</code> | <code>a * b</code> | <b>true on overflow</b>                                 | Only when it fits |
| <code>PROVEN_SIZE_MAX</code>           | -                  | The largest value a <code>proven_size_t</code> can hold | -                 |

- `res` is a **pointer** to where the answer goes.

- They return **true on overflow**, false on success. That reads backwards the first time; the convention comes from C23's <stdckdint.h>, which these use when the compiler provides it, and from the GCC/Clang `__builtin*_overflow` family otherwise. Read the call as *"did this overflow?"* and it comes out right.
- `PROVEN_SIZE_MAX` is there for the cases where you want to compare against the limit directly rather than compute and check.

Correct:

```
typedef struct { int id; double weight; } item_t;

proven_size_t count = 1024;
proven_size_t total = 0;
proven_err_t err = PROVEN_OK;

if (PROVEN_CKD_MUL(&total, count, sizeof(item_t))) {
    err = PROVEN_ERR_OVERFLOW; /* refuse to allocate a size that wrapped */
} else {
    proven_result_mem_mut_t block = alloc.alloc_fn(alloc.ctx, total, alignof(item_t));
    err = block.err;
    if (proven_is_ok(err)) {
        alloc.free_fn(alloc.ctx, block.value.ptr);
    }
}
(void)err;
```

Wrong:

```
proven_size_t total = count * sizeof(item_t); /* wrong: may wrap silently */
```

You do not need these for every arithmetic expression in your program. You need them wherever a size is **computed from a value you did not choose yourself** - and that is precisely where the bugs are.

## 5. Alignment

### What alignment is, and why the library talks about it

Every type in C has an alignment: an address it must start on for the hardware to load it efficiently, or at all. A `double` typically wants an address divisible by 8. Some architectures fault on a misaligned load; x86 merely makes it slow; and either way, C says accessing an object through a misaligned pointer is undefined behaviour.

You have not had to think about this, because `malloc` returns memory suitably aligned for anything. That guarantee is exactly what disappears the moment you hand out memory yourself - and handing out memory yourself is what [Chapter 2](#)'s arenas and pools do. An arena that bumps a pointer by 13 bytes and hands you the result has just given you a misaligned `double`.

So the helpers here exist for the allocators, and for you if you write one.

### Reference

| Macro                                 | Meaning                                                                  |
|---------------------------------------|--------------------------------------------------------------------------|
| <code>PROVEN_DEFAULT_ALIGNMENT</code> | The default the library uses when you do not say otherwise. Currently 8. |
| <code>PROVEN_MAX_ALIGN</code>         | <code>alignof(max_align_t)</code> - enough for any built-in type.        |

| Function                       | Purpose                                                           | Return              |
|--------------------------------|-------------------------------------------------------------------|---------------------|
| <code>proven_is_pow2(x)</code> | Is <code>x</code> a non-zero power of two?<br>Alignments must be. | <code>bool</code> . |

|                                                   |                                                                         |                                                                                             |
|---------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <code>proven_mem_align_up(addr, align)</code>     | Round a size or address up to the next multiple of <code>align</code> . | The aligned value, or <b>0</b> if <code>align</code> is invalid or the rounding overflowed. |
| <code>proven_uintptr_align_up(addr, align)</code> | The same, for a <code>proven_uintptr_t</code> address.                  | As above.                                                                                   |

Note the error convention: these return **0 on failure**, because they have no room for an error code. Zero is never a valid aligned address, so testing for it is unambiguous.

Correct:

```
typedef struct { int id; double weight; } item_t;

proven_size_t size = 100;
proven_size_t aligned = proven_mem_align_up(size, alignof(item_t));
proven_err_t err = (aligned == 0) ? PROVEN_ERR_OVERFLOW : PROVEN_OK;
(void)err;
```

Wrong - the classic bit-twiddling version:

```
proven_size_t aligned = (size + align - 1) & ~(align - 1); /* wrong: may overflow */
```

That line is in a great deal of production code and is correct for every input except the ones near the top of the range, where `size + align - 1` wraps to a small number and the result is an address *below* where you started.

### Worked example: views, slicing, and alignment together

The three ideas in this chapter - a view that carries its own length, a slice that is checked before it is taken, and a boundary that has to be a power of two -- only show their point when a program uses them at once. This one keeps a small table of fixed-size records in a single block it owns, and then does the four things any such program has to do: hand the table to a reader that must not modify it, hand one row to code that may modify only that row, delete a row by moving the rows after it down over it, and decide whether a pointer that arrived from somewhere else even points into the table.

Two of those are the places C is most often quietly wrong. Moving overlapping bytes is not what `proven_mem_copy` promises, so it has a separate call, `proven_mem_move`. And comparing two pointers with `<` when they may belong to different objects is undefined behaviour - the compiler is entitled to assume it never happens - so "is this pointer inside my buffer?" is asked with `proven_range_contains_ptr` rather than with `>=` and `<`.

This program is compiled and run by the test suite, so it cannot fall out of date.

```
#include <string.h>

/*
 * One owned buffer, three ways of talking about it.
 *
 * The program holds a small table of fixed-size records in a single block of
 * memory it owns, and then does the four things every such program has to do:
 *
 * 1. hand the table to a reader that must not modify it (a read-only view),
 * 2. hand one row to a writer that may modify only that row (a slice),
 * 3. delete a row by moving the rows after it down over it (an overlapping
 *    copy, which plain copying is not allowed to do),
 * 4. take a pointer somebody else returned and decide whether it even points
 *    into the table before using it as a row index.
 *
 * Every step is a place where a bare `char *` loses the length and the program
```

```

* finds out later. The types here carry the length with the pointer, so the
* bounds question is answered where the mistake would be made.
*/

#define ROW_SIZE 8u
#define ROW_COUNT 4u

/* A reader gets a view: it can read every byte and write none of them. The
* const in the type is not advice - assigning through it does not compile. */
static proven_size_t count_nonzero(proven_mem_view_t table) {
    proven_size_t n = 0;
    for (proven_size_t i = 0; i < table.size; ++i) {
        if (table.ptr[i] != 0) {
            ++n;
        }
    }
    return n;
}

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* Allocate the table. The allocator returns a read-write slice, which is
    * what an owner holds: a pointer and the size that goes with it. */
    proven_result_mem_mut_t got = alloc.alloc_fn(alloc.ctx, ROW_SIZE * ROW_COUNT, alignof(proven_u32));
    EXAMPLE_REQUIRE(proven_is_ok(got.err), "allocating the record table must succeed");
    if (!proven_is_ok(got.err)) {
        return 1;
    }

    /* proven_mem_t is the "I own this" type. Keeping the owned block in one
    * variable, and handing out views and slices derived from it, is the whole
    * ownership discipline this library asks for. */
    proven_mem_t owned = { .ptr = got.value.ptr, .size = got.value.size };
    proven_mem_mut_t table = proven_mem_mut_from_owned(owned);

    /* Fill row i with the byte value i + 1, so a moved row is recognisable. */
    for (proven_u32 row = 0; row < ROW_COUNT; ++row) {
        proven_result_mem_mut_t slot = proven_mem_mut_slice_checked(table, row * ROW_SIZE, ROW_SIZE);
        EXAMPLE_REQUIRE(proven_is_ok(slot.err), "every row of the table must be in bounds");
        if (!proven_is_ok(slot.err)) {
            alloc.free_fn(alloc.ctx, owned.ptr);
            return 1;
        }
        memset(slot.value.ptr, (int)(row + 1), slot.value.size);
    }

    /* 1. Read-only use. The reader cannot modify the table and cannot run off
    * the end of it, because it was handed both facts at once. */
    proven_mem_view_t read_only = proven_mem_view_from_owned(owned);
    EXAMPLE_REQUIRE(count_nonzero(read_only) == ROW_SIZE * ROW_COUNT,
        "every byte of every row was written");

    /* 2. Asking for a row that does not exist is an error you can handle, not a
    * crash you have to debug. Row 4 of a 4-row table starts one row past
    * the end. */
    proven_result_mem_mut_t past_end = proven_mem_mut_slice_checked(table, ROW_COUNT * ROW_SIZE, ROW_SIZE);
    EXAMPLE_REQUIRE(past_end.err == PROVEN_ERR_OUT_OF_BOUNDS,
        "slicing past the end must report OUT_OF_BOUNDS, not return a bad slice");

```

```

/* 3. Delete row 1 by moving rows 2 and 3 down one row. Source and
 * destination overlap, so this is the one case plain copying is not
 * allowed to handle: proven_mem_copy documents non-overlapping regions,
 * proven_mem_move is the one that accepts them. */
proven_mem_view_t tail = {
    .ptr = table.ptr + 2 * ROW_SIZE,
    .size = 2 * ROW_SIZE
};
proven_err_t moved = proven_mem_move(table.ptr + 1 * ROW_SIZE, table.size - 1 * ROW_SIZE, tail);
EXAMPLE_REQUIRE(proven_is_ok(moved), "moving the tail of the table down must succeed");
EXAMPLE_REQUIRE(table.ptr[1 * ROW_SIZE] == 3, "row 2 moved into row 1's place");
EXAMPLE_REQUIRE(table.ptr[2 * ROW_SIZE] == 4, "row 3 moved into row 2's place");

/* The move is bounded like everything else: a destination too small to hold
 * the source is refused, and nothing is written. */
proven_err_t refused = proven_mem_move(table.ptr, ROW_SIZE, tail);
EXAMPLE_REQUIRE(refused == PROVEN_ERR_OUT_OF_BOUNDS,
    "a move that would not fit must be refused before it writes");

/* 4. A pointer from elsewhere. Comparing unrelated pointers with < is
 * undefined behaviour in C - the compiler is allowed to assume it never
 * happens - so the question "does this pointer point into my buffer?"
 * has to be asked with a function that does the comparison correctly. */
proven_size_t offset = 0;
const proven_byte_t *from_elsewhere = table.ptr + 2 * ROW_SIZE;
bool inside = proven_range_contains_ptr(table.ptr, table.size, from_elsewhere, ROW_SIZE, &offset);
EXAMPLE_REQUIRE(inside, "a pointer to row 2 is inside the table");
EXAMPLE_REQUIRE(offset / ROW_SIZE == 2, "and its offset recovers the row index");

proven_u8 unrelated[ROW_SIZE] = { 0 };
EXAMPLE_REQUIRE(!proven_range_contains_ptr(table.ptr, table.size, unrelated, ROW_SIZE, NULL),
    "a pointer into a different object is not inside the table");

/* A row that starts inside but ends outside is also outside: the check is
 * about the whole range, not just where it begins. */
EXAMPLE_REQUIRE(!proven_range_contains_ptr(table.ptr, table.size,
    table.ptr + table.size - 1, ROW_SIZE, NULL),
    "a range that starts inside but runs past the end is refused");

/* 5. Once the range check has proved the row is in bounds, the unchecked
 * slice is the right call: it does no work, and the proof is the line
 * above it rather than a comment. Never write it without that proof. */
proven_mem_mut_t row2 = proven_mem_mut_slice_unchecked(table, offset, ROW_SIZE);
EXAMPLE_REQUIRE(row2.size == ROW_SIZE && row2.ptr[0] == 4,
    "the unchecked slice names the row the checked test just proved");

/* 6. Alignment, in the one place a caller meets it: laying two differently
 * aligned things out inside one block. The address of the second one has
 * to be rounded up, and rounding up is only defined for a power-of-two
 * boundary - so the boundary is checked first. */
proven_size_t want_align = alignof(proven_u32);
EXAMPLE_REQUIRE(proven_is_pow2(want_align), "an alignment boundary must be a power of two");
EXAMPLE_REQUIRE(!proven_is_pow2(24u), "24 is not a power of two, so it is not a valid boundary");

proven_uintptr_t raw = (proven_uintptr_t)(table.ptr + 1); /* deliberately odd */
proven_uintptr_t aligned = proven_uintptr_align_up(raw, want_align);
EXAMPLE_REQUIRE(aligned >= raw, "aligning up never moves backwards");
EXAMPLE_REQUIRE(aligned % want_align == 0, "and the result is on the boundary asked for");
EXAMPLE_REQUIRE(aligned - raw < want_align, "the padding is never more than one boundary");

```

```

/* A boundary that is not a power of two returns 0 rather than a plausible
 * wrong address, so the mistake stops here instead of downstream. */
EXAMPLE_REQUIRE(proven_uintptr_align_up(raw, 24u) == 0,
                "aligning to a non-power-of-two returns 0, not a wrong address");

alloc.free_fn(alloc.ctx, owned.ptr);
return EXAMPLE_OK();
}

```

Wrong - the same deletion written with the copy that forbids overlap:

```

/* rows 2..3 moved down onto row 1: source and destination overlap */
proven_mem_copy(table.ptr + ROW_SIZE, table.size - ROW_SIZE, tail); /* wrong */

```

`proven_mem_copy` documents non-overlapping regions. It will not report this; it will simply be free to copy in whatever order is fastest, and on the day that order changes the table quietly ends up holding two copies of one row.

Wrong - the bounds question asked with plain pointer comparison:

```

if (ptr >= table.ptr && ptr + size <= table.ptr + table.size) { /* wrong */ }

```

If `ptr` came from a different allocation, this comparison is undefined behaviour; not merely a comparison that returns the wrong answer.

## 6. Panic: when there is no one left to return an error to

### Why a library that returns errors has a panic at all

Everything in this chapter has argued that failure should be a value you can inspect. A panic is the admission that this is not always possible.

Some APIs have no error channel by construction. `proven_arena_alloc_or_panic()` returns a memory block, not a result, because it exists for the setup phase of a program where an allocation failure means the program cannot run at all and there is nothing sensible to return. In a freestanding build there may be no `abort()`, no `stderr`, and nothing to unwind to.

So the library raises a panic by calling `proven_panic()`, which dispatches to a handler you can replace. It does not call `exit`, print to `stderr`, or assume an operating system.

```

typedef void (*proven_panic_handler_t)(const char *msg);

```

```

void proven_panic(const char *msg);
void proven_set_panic_handler(proven_panic_handler_t handler);

```

| API                                            | Purpose                                             | Notes                                                                                                                    |
|------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>proven_panic(msg)</code>                 | Raise a panic. Dispatches to the installed handler. | Called by the library's <code>_or_panic</code> APIs; you can call it too.                                                |
| <code>proven_set_panic_handler(handler)</code> | Install a handler.                                  | Pass <code>NULL</code> to restore the default. Not thread-safe - install it during start-up, before other threads exist. |
| <code>proven_panic_handler_t</code>            | <code>void (*)(const char *msg)</code>              | The handler must not return.                                                                                             |

The default handler traps: `__builtin_trap()` on GCC and Clang, and an infinite loop on any other compiler. The dispatch goes through a function pointer rather than a weak symbol, so it links the same way on ELF and on PE/COFF (Windows, mingw-w64) toolchains.

Installing your own (a file-scope function, so this is a listing rather than something you paste inside another function):

```
static void my_panic(const char *msg) {
    log_critical(msg);          /* your logger */
    for (;;) {
        /* reset, halt, or wait for a debugger */
    }
}
```

```
proven_set_panic_handler(my_panic); /* pass NULL to restore the default */
```

**A panic handler must not return.** If it does, the `_or_panic` family has nothing to give back and the validity of its result is not guaranteed.

Wrong - a handler that returns:

```
static void my_panic(const char *msg) {
    fprintf(stderr, "%s\n", msg); /* wrong: this returns, and then the caller
                                   proceeds with a block that was never allocated */
}
```

Wrong - reaching for `_or_panic` in ordinary code:

```
proven_mem_mut_t block = proven_arena_alloc_or_panic(&arena, n);
/* wrong for anything that handles input: an arena that is merely full has just
   killed the process. Use proven_arena_alloc and read the error. */
```

## 7. Version macros

Compile-time identification, for diagnostics and for code that must adapt to the library version.

```
#define PROVEN_VERSION_MAJOR 0
#define PROVEN_VERSION_MINOR 11
#define PROVEN_VERSION_PATCH 0
#define PROVEN_VERSION_STRING "proven_c_lib-v0.11.0"
#define PROVEN_VERSION_ENCODE(major, minor, patch) ((major) * 1000000L + (minor) * 1000L + (patch))
#define PROVEN_VERSION_NUM PROVEN_VERSION_ENCODE(PROVEN_VERSION_MAJOR, PROVEN_VERSION_MINOR,
PROVEN_VERSION_PATCH)
```

The version is semantic - MAJOR.MINOR.PATCH, from v0.0.1. A PATCH release fixes or documents and changes nothing a caller has to; a MINOR release adds without breaking an existing call; a MAJOR release may break one. While MAJOR is 0 the API may still move between minors.

PROVEN\_VERSION\_STRING is what you print in a build report or a `--version` flag. The three numbers are for code that must adapt to the library version, and PROVEN\_VERSION\_NUM folds them into one integer for `#if` - compare it with PROVEN\_VERSION\_ENCODE, never with a literal: `#if PROVEN_VERSION_NUM >= PROVEN_VERSION_ENCODE(0, 1, 0)`.

Releases before v0.0.1 were numbered by date (v26.MM.DDx), and PROVEN\_VERSION\_NUM was that date. The number restarted with the scheme, so a comparison against an old date value is wrong on every current release; rewrite it with PROVEN\_VERSION\_ENCODE.

The numbers and the string are checked against `include/proven/version.h` by the build, and the string is checked against the numbers, so the manual cannot drift from the header. (It did, once, and this sentence is why it will not again: the string was gated and the rest was not.)

## 8. Examples and misuse cases

### Safe result handling

```
proven_byte_t record[16] = {0};
proven_mem_view_t view = { .ptr = record, .size = sizeof record };

proven_result_mem_view_t part = proven_mem_view_slice_checked(view, 4, 8);
if (!proven_is_ok(part.err)) {
    return; /* the range was not inside the view */
}
```

```
}
```

```
proven_byte_t payload[8];
proven_err_t err = proven_mem_copy(payload, sizeof payload, part.value);
(void)err;
```

### Unchecked slicing only after proof

Correct when the caller already proved the range:

```
proven_byte_t record[16] = {0};
proven_mem_view_t view = { .ptr = record, .size = sizeof record };
proven_size_t offset = 4;
proven_size_t size = 8;

if (offset <= view.size && size <= view.size - offset) {
    proven_mem_view_t part = proven_mem_view_slice_unchecked(view, offset, size);
    (void)part; /* proved in range: safe to read */
}
```

Note the shape of that test. `size <= view.size - offset` is written that way, rather than the more natural `offset + size <= view.size`, precisely because of section 4: the sum can wrap and the difference cannot, given the `offset <= view.size` check that precedes it.

Wrong:

```
proven_mem_view_t part = proven_mem_view_slice_unchecked(view, user_offset, user_size);
/* wrong: user_offset and user_size were not validated */
```

### Empty views are allowed

A zero-size view may have a null pointer. Do not reject it blindly.

Wrong:

```
if (!view.ptr) return PROVEN_ERR_INVALID_ARG; /* wrong for empty views */
```

Better:

```
proven_mem_view_t view = {0}; /* size 0, ptr NULL: a legal empty view */
proven_err_t err = PROVEN_OK;

if (view.size > 0 && !view.ptr) {
    err = PROVEN_ERR_INVALID_ARG; /* only a null pointer with bytes behind it is a bug */
}
(void)err;
```

### Where to go next

[Chapter 2](#) is where these types start doing work: allocators produce `proven_mem_mut_t`, arenas use the alignment helpers from section 5, and every growable container in the library takes the allocator as a parameter for the reasons section 1 gave for taking the error as a return value - the cost should be visible in the signature.

## Chapter 2: Allocation - Heap, Arenas, Pools, and Buffers

**Part II - The vocabulary every program uses. Prerequisite: [Chapter 1](#).** After this chapter you can choose an allocation strategy deliberately instead of reaching for `malloc` by reflex, and you will know which of the three costs you are paying.

This chapter covers `heap.h`, `arena.h`, `pool.h`, `allocator.h`, and `buffer.h`. The thread-safety and pointer-provenance material that used to be section 7 here now lives in [Chapter 6](#), with the rest of the concurrency subject - it was the hardest material in the book sitting in one of the first chapters.

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### 1. Why allocation is a parameter, and the heap allocator

#### The problem with `malloc`

`malloc` is a global. Any function may call it, nothing in a signature says whether a function does, and every call goes to the same general-purpose allocator no matter what the memory is for.

That has four consequences you have probably met:

- **You cannot tell from a signature whether a function allocates.** `char *build_message(int n)` might allocate, might return a pointer into a static buffer, might return a literal. The type is identical in all three cases, so the caller cannot know whether to free, and the answer lives in a comment that may be wrong.
- **You cannot change the strategy for one part of a program.** If a parser makes ten thousand small allocations that all die at the end of the parse, `malloc` and `free` will do ten thousand general-purpose allocations and ten thousand frees. A bump allocator would do one. There is no way to say so without rewriting every call.
- **You cannot test the failure path.** Making `malloc` fail on demand means intercepting it globally - `LD_PRELOAD`, a linker trick, a `#define malloc my_malloc` that also catches the library's internal calls. Meanwhile the branch you most want to test is the one that runs when memory runs out.
- **You cannot use it at all where there is no heap.** Firmware, a kernel, a bootloader: no `malloc`, and every library that assumes one is unusable.

`malloc` is not badly designed. It is a fixed policy, hardcoded into every call site by the fact that it takes no parameter saying otherwise.

#### What this library does instead

**An allocator is a value, and it is a parameter.** A function that can allocate takes a `proven_allocator_t`; a function that cannot, does not. That is the whole idea, and every consequence follows from it:

```
proven_err_t proven_u8str_append(proven_u8str_t *str, proven_u8str_view_t data);
proven_err_t proven_u8str_append_grow(proven_allocator_t alloc, proven_u8str_t *str, proven_u8str_view_t data);
```

Read the signatures. The first cannot grow the string, so it fails when the text does not fit. The second can, and says so by taking the allocator. You never have to wonder which one you called.

The same code now works with three different strategies, chosen by the caller:

| Allocator    | Get one from                                                                                                             | Frees individually?                                               | Use it when                                                                          |
|--------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Heap</b>  | <code>proven_heap_allocator()</code>                                                                                     | Yes                                                               | The general case. Objects with unrelated lifetimes.                                  |
| <b>Arena</b> | <code>proven_arena_create(backing),</code><br>then<br><code>proven_arena_as_allocator(&amp;a)</code>                     | <b>No</b> - free is a no-op; you reset or destroy the whole thing | Many allocations that all die at the same moment: one request, one frame, one parse. |
| <b>Pool</b>  | <code>proven_pool_init(&amp;p, base, size, align, bin_cap),</code> then<br><code>proven_pool_as_allocator(&amp;p)</code> | Yes, into a free list                                             | Many objects of <b>one fixed size</b> , allocated and freed repeatedly.              |

Start with the heap. Reach for the other two when you have a reason, and the reason is usually a measurement.

### The heap allocator

```
proven_allocator_t proven_heap_allocator(void);
```

This is `malloc`, `realloc` and `free` wearing the library's interface. It is what you should use unless you have a specific reason not to, and there is nothing clever about it - which is the point. Every example in this manual that does not have a reason to do otherwise uses it:

```
proven_allocator_t heap = proven_heap_allocator();

proven_result_u8str_t s = proven_u8str_create(heap, 64);
if (!proven_is_ok(s.err)) {
    return; /* nothing was created, so there is nothing to destroy */
}
(void)proven_u8str_append(&s.value, PROVEN_LIT("ready"));
proven_u8str_destroy(heap, &s.value); /* the SAME allocator */
```

The value is four words - a context pointer and three function pointers - and it is passed by value. Copying it is free, storing it in your own struct is fine, and it does not need to be destroyed.

**In freestanding builds** `proven_heap_allocator` **does not exist**. There is no `malloc` to wrap. That is not a limitation to work around; it is the reason the whole library takes allocators as parameters, and it is why an arena over a static array makes the same code run on a microcontroller. See [freestanding mode](#).

Wrong - destroying with a different allocator than you created with:

```
proven_result_u8str_t s = proven_u8str_create(arena_alloc, 64);
proven_u8str_destroy(heap_alloc, &s.value); /* wrong: heap free on arena memory */
```

The object does not remember which allocator produced it - that is what keeps it small. Nothing checks this today, and the failure is heap corruption that surfaces somewhere else, later. Pair them by construction: keep the allocator next to the object, or pass both together.

## 2. Arena: many things, one lifetime

### The problem: many small allocations with the same death date

Consider parsing a configuration file. You allocate a string for each key, a string for each value, a node for each section - a few thousand small allocations. Then the parse finishes, you build your result, and every one of those allocations becomes garbage at the same instant.

With `malloc` you pay for that twice. Each allocation searches a free list, updates bookkeeping, and returns memory with a header attached; each `free` returns it and possibly coalesces neighbours. And you must not miss one, because a leak is one forgotten free away.

The observation an arena makes is that **all of those objects have the same lifetime**, so individual frees are pointless work. If they die together, they can be freed together.

### How an arena works

An arena is a block of memory and an offset. Allocating means rounding the offset up to the alignment you asked for, returning the address, and moving the offset along. That is the entire algorithm:

```
[####used####|                free                ]
      ^ offset
```

- **Allocation is a few instructions.** No search, no free list, no per-allocation header.
- **free does nothing at all.** It is a no-op, deliberately.
- **You reclaim by resetting or destroying the whole arena**, which sets the offset back to zero. One operation frees ten thousand objects.

The memory comes from you. `proven_arena_create` takes a `proven_mem_mut_t` - a block you got from the heap, or a static array, or a region on the stack. The arena never allocates on its own, which is what lets it work with no heap underneath.

### What you give up

An arena is not a general-purpose allocator, and the trade is real:

- **You cannot free one object.** If the lifetimes are not actually shared, an arena leaks by design - memory is only reclaimed at reset.
- **Running out is a hard limit.** The backing block is fixed. `PROVEN_ERR_NOMEM` here does not mean the machine is out of memory, it means this arena is.
- **Every pointer into it dies at reset**, all at once, with nothing to warn you. A view that outlives the reset is a dangling view; see contract 2 in [Chapter 0](#).

Use an arena when the shared lifetime is a fact about your program, not a hope.

```
proven_arena_t
typedef struct {
    proven_mem_mut_t backing;
    proven_size_t offset;
} proven_arena_t;
```

Fields:

- `backing`: mutable memory range owned by the caller.
- `offset`: next allocation position within `backing`.

### Arena functions and helpers

| API                                                                           | Intent                                                  | Parameters                                          | Return                                              |
|-------------------------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| <code>proven_arena_create(backing)</code>                                     | Initialize an arena over a backing slice.               | <code>backing</code> : caller-owned memory.         | <code>proven_arena_t</code> .                       |
| <code>proven_arena_reset(arena)</code>                                        | Discard all arena allocations.                          | <code>arena</code> .                                | <code>void</code> .                                 |
| <code>proven_arena_destroy(arena)</code>                                      | Formal cleanup.                                         | <code>arena</code> .                                | <code>void</code> ; no-op for caller-backed arenas. |
| <code>proven_arena_alloc_aligned(size, align)</code>                          | Allocate with explicit alignment.                       | <code>arena</code> , byte size, power-of-two align. | <code>proven_result_mem_mut_t</code> .              |
| <code>proven_arena_realloc_aligned(old_ptr, old_size, new_size, align)</code> | Reallocate; can extend/shrink tail allocation in place. | old allocation details and new size.                | <code>proven_result_mem_mut_t</code> .              |

|                                         |                                         |                     |                                                    |
|-----------------------------------------|-----------------------------------------|---------------------|----------------------------------------------------|
| proven_arena_alloc(arena, size)         | Allocate with PROVEN_DEFAULT_ALIGNMENT. | arena, byte size.   | proven_result_mem_mut_t.                           |
| proven_arena_alloc_aligned(size, align) | Allocate or call panic hook.            | arena, size, align. | proven_mem_mut_t.                                  |
| proven_arena_alloc_or_panic(size)       | Default-aligned panic allocation.       | arena, size.        | proven_mem_mut_t.                                  |
| proven_arena_as_allocator(arena)        | Expose arena as proven_allocator_t.     | arena.              | allocator trait, or zero allocator for null arena. |

Trait adapter helpers:

- proven\_arena\_alloc\_trait
- proven\_arena\_realloc\_trait
- proven\_arena\_free\_trait

These are exposed because the allocator trait needs function pointers. Application code normally calls proven\_arena\_as\_allocator() instead.

Example:

```
alignas(max_align_t) proven_byte_t storage[4096];
proven_arena_t arena = proven_arena_create((proven_mem_mut_t){
    .ptr = storage,
    .size = sizeof storage,
});

proven_result_mem_mut_t r = proven_arena_alloc(&arena, 64);
if (!proven_is_ok(r.err)) {
    return; /* the arena cannot grow: it reports NOMEM instead */
}

proven_arena_reset(&arena); /* reclaims r and everything else at once */
proven_arena_destroy(&arena);
```

### Arena with growable containers

Growable containers can use an arena allocator, but growth may abandon earlier blocks because arena free is a no-op. Reserve capacity up front when possible.

Correct:

```
alignas(max_align_t) proven_byte_t storage[4096];
proven_arena_t arena = proven_arena_create((proven_mem_mut_t){
    .ptr = storage,
    .size = sizeof storage,
});
proven_allocator_t a = proven_arena_as_allocator(&arena);

/* Ask for the capacity up front, so growth never has to abandon a block. */
proven_result_array_t ar = PROVEN_ARRAY_INIT(a, int, 128);
if (!proven_is_ok(ar.err)) {
    return;
}
(void)PROVEN_ARRAY_PUSH(&ar.value, int, 10);
PROVEN_ARRAY_DESTROY(&ar.value); /* correct, but arena free reclaims nothing */
proven_arena_reset(&arena); /* this is what gives the bytes back */
```

Wrong - growing into an arena from a tiny initial capacity:

```

proven_result_array_t ar = PROVEN_ARRAY_INIT(a, int, 1);
for (int i = 0; i < 10000; ++i) {
    PROVEN_ARRAY_PUSH(&ar.value, int, i); /* wrong: every regrow abandons the old block */
}

```

Each regrow asks the arena for a bigger block and then "frees" the old one - which, in an arena, does nothing. The array ends up correct and the arena ends up holding every intermediate size it ever allocated: 1, 2, 4, 8 ... 8192 elements' worth of abandoned space, on top of the one buffer you wanted. Reserve the capacity up front, as the correct version above does.

Wrong - a view that outlives the reset:

```

proven_u8str_view_t name = /* ... built in the arena ... */;
proven_arena_reset(&arena);
use(name); /* wrong: those bytes are now free space */

```

This is the arena's sharpest edge. Reset does not touch the memory it reclaims, so the bytes are usually still there and the bug usually does not show up in testing - right up until the next allocation writes over them.

### 3. Pool: many things, one size

#### The problem an arena does not solve

An arena assumes shared lifetimes. Plenty of workloads have the opposite shape: objects of one type, created and destroyed continuously, in no particular order. A linked list of events. Nodes in a tree that grows and shrinks. Connection records that come and go.

An arena cannot do this - it never reclaims a single object, so a long-running program would grow without bound. The heap can, and that is exactly what it costs: every allocation searches, every free updates bookkeeping, and the general-purpose allocator does that general-purpose work for a request it already made a thousand times.

The observation a pool makes is that **all these objects are the same size**. If they are the same size, a freed one fits a future request exactly, so it can be handed straight back out with no search at all.

#### How a pool works

A pool keeps a small stack of freed blocks - the *bin* - and does the obvious thing:

- **Allocate:** if the bin has anything in it, pop one and return it. That is a pointer read and a decrement. Only when the bin is empty does it go to the underlying allocator.
- **Free:** if the bin has room, push the block onto it for reuse. If the bin is full, hand the memory back to the underlying allocator so the pool does not park memory forever.

`bin_cap` is therefore a dial: how much memory this pool is allowed to keep in reserve.

#### What you give up

- **One pool serves exactly one size and alignment.** A request for anything else is refused with `PROVEN_ERR_INVALID_ARG` - not served from somewhere else. A stricter alignment than the pool was built with is refused too; a looser one is fine, because the block already satisfies it.
- **The pool does not track live objects.** `proven_pool_destroy` frees what is in the bin, not what you are still holding. Free everything you allocated first, or you have leaked it *and* it is now dangling.

Arena versus pool, in one line each: an arena is for *allocate many, free all at once*; a pool is for *allocate and free the same size, over and over, cheaply*.

```

proven_pool_t
typedef struct {
    proven_allocator_t base_alloc;
    proven_size_t item_size;
    proven_size_t item_align;
}

```

```

    void **bin;
    proven_size_t bin_cap;
    proven_size_t bin_len;
} proven_pool_t;

```

Fields:

- **base\_alloc**: allocator used for new blocks and the recycle bin.
- **item\_size**: exact object size accepted by the pool. Any other size is rejected with `PROVEN_ERR_INVALID_ARG`.
- **item\_align**: the alignment every block is allocated with. A request for a *stricter* alignment is rejected with `PROVEN_ERR_INVALID_ARG`; a looser one is served, since the block already meets it.
- **bin**: cached free-list array.
- **bin\_cap**: maximum cached block count.
- **bin\_len**: current cached block count.

### How recycling works

The pool does not own one big slab; it allocates each item individually from `base_alloc` and keeps a small **stack of freed blocks** (the bin, an array of up to `bin_cap` pointers):

- **Allocate**. If `bin_len > 0`, pop the top pointer (O(1), no `base_alloc` call) -- this is the whole point of the pool. Otherwise fall through to `base_alloc`.
- **Free**. If `bin_len < bin_cap`, push the pointer onto the bin for reuse; otherwise (bin full) free it straight back to `base_alloc`. So the bin caps how much memory the pool keeps parked for reuse.
- **Destroy**. `proven_pool_destroy` frees every pointer still in the bin and the bin array itself - but it does **not** track live (handed-out) items, so you must free everything you allocated before destroying the pool.

This makes the pool a churn optimizer for short-lived same-type objects (nodes, events), not a region allocator. Use an arena when you want "allocate many, free all at once"; use a pool when you want "allocate/free the same size over and over cheaply".

### Counter-examples

```

/* WRONG: the pool only handles its configured size/alignment. */
proven_allocator_t a = proven_pool_as_allocator(&pool); /* item_size == sizeof(Node) */
a.alloc_fn(a.ctx, sizeof(BigThing), alignof(BigThing)); /* not the pool's item -> rejected */

/* WRONG: destroying while items are still live leaks (and dangles) them. */
void *p = a.alloc_fn(a.ctx, sizeof(Node), alignof(Node)).value.ptr;
proven_pool_destroy(&pool); /* `p` is NOT freed and is now dangling */
/* RIGHT: free every handed-out item first, then destroy. */

```

### Pool functions

| API                                                                             | Intent                                     | Return                            |
|---------------------------------------------------------------------------------|--------------------------------------------|-----------------------------------|
| <code>proven_pool_init(pool, base_alloc, item_size, item_align, bin_cap)</code> | Initialize a fixed-size pool.              | <code>PROVEN_OK</code> or error.  |
| <code>proven_pool_as_allocator(pool)</code>                                     | Return allocator trait backed by the pool. | <code>proven_allocator_t</code> . |
| <code>proven_pool_destroy(pool)</code>                                          | Free cached blocks and bin storage.        | <code>void</code> .               |

Example:

```

typedef struct Node { int value; } Node;

proven_pool_t pool = {0};
proven_err_t e = proven_pool_init(&pool, alloc, sizeof(Node), alignof(Node), 64);
if (!proven_is_ok(e)) {

```

```

    return;
}

proven_allocator_t node_alloc = proven_pool_as_allocator(&pool);
proven_result_mem_mut_t n = node_alloc.alloc_fn(node_alloc.ctx, sizeof(Node), alignof(Node));
if (proven_is_ok(n.err)) {
    /* Hand it back before destroy: the pool does not track live items. */
    node_alloc.free_fn(node_alloc.ctx, n.value.ptr);
}

proven_pool_destroy(&pool);

```

Wrong:

```

node_alloc.alloc_fn(node_alloc.ctx, sizeof(LargerObject), alignof(LargerObject));
/* wrong: one pool is for one fixed object size and alignment */

```

## 4. The allocator trait

### Why this section is fourth and not first

You have now used three allocators without seeing the interface they share, and that was deliberate. The interface is the least interesting part of the idea and the hardest thing to read cold: three function-pointer typedefs and an alignment contract mean very little until you have seen a heap, an arena and a pool behind them.

You need this section for two things: writing an allocator of your own, and understanding the rules every allocator in this library promises to follow. If you are only ever going to *use* the three above, `proven_heap_allocator()`, `proven_arena_as_allocator()` and `proven_pool_as_allocator()` are the whole API and you can skip ahead.

A trait, here, is a struct of function pointers used as an interface - C's version of a virtual table, written out by hand. `proven_allocator_t` is the library's most important one; `stream.h` and `random.h` use the same shape.

### Function pointer types

```

typedef proven_result_mem_mut_t (*proven_alloc_fn_t)(
    void *ctx,
    proven_size_t size,
    proven_size_t align
);

```

```

typedef proven_result_mem_mut_t (*proven_realloc_fn_t)(
    void *ctx,
    void *old_ptr,
    proven_size_t old_size,
    proven_size_t new_size,
    proven_size_t align
);

```

```

typedef void (*proven_free_fn_t)(void *ctx, void *ptr);

```

Intent:

- `alloc_fn` allocates a new byte slice.
- `realloc_fn` changes allocation size and must be failure-atomic: on failure the old block is still valid and unmodified, so the caller has lost nothing.
- A block must be reallocated and freed with the **same** `align` it was allocated with. An allocator may pick a different underlying mechanism for over-aligned requests than for ordinary ones, and the heap allocator does: `align <= alignof(max_align_t)` - which is every string, buffer and byte array in this library - goes through `malloc/realloc` so that growth can happen in place, and

anything more strictly aligned goes through an aligned allocator. Handing a block back under a different alignment class is undefined.

- `free_fn` releases memory. If an allocator needs size metadata, it must track that internally.

```
proven_allocator_t
typedef struct {
    void *ctx;
    proven_alloc_fn_t alloc_fn;
    proven_realloc_fn_t realloc_fn;
    proven_free_fn_t free_fn;
} proven_allocator_t;
```

Fields:

- `ctx`: allocator-specific state.
- `alloc_fn`: allocation function.
- `realloc_fn`: reallocation function.
- `free_fn`: deallocation function.

## Reference

| Member / API                              | Shape                            | Contract                                                                               |
|-------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------|
| <code>ctx</code>                          | <code>void *</code>              | The allocator's own state. Opaque to callers; passed back as the first argument.       |
| <code>alloc_fn</code>                     | <code>proven_alloc_fn_t</code>   | Allocate size bytes at align. Returns <code>proven_result_mem_mut_t</code> .           |
| <code>realloc_fn</code>                   | <code>proven_realloc_fn_t</code> | Resize. <b>Failure-atomic</b> : on failure the old block is untouched and still valid. |
| <code>free_fn</code>                      | <code>proven_free_fn_t</code>    | Release. Must be given the same align class the block was allocated with.              |
| <code>proven_alloc_is_valid(alloc)</code> | <code>static inline bool</code>  | True when every function pointer is non-null - i.e. the trait can be called.           |

The three rules that every allocator in this library obeys, and that yours must:

1. **Same allocator, whole lifetime.** Allocate, reallocate and free a block with one allocator.
2. **Same alignment class.** A block allocated at one alignment must be reallocated and freed at that alignment; allocators may use a different mechanism for over-aligned requests.
3. **Failure changes nothing.** A failed `realloc_fn` leaves the old block valid and unmodified.

```
static inline bool proven_alloc_is_valid(proven_allocator_t alloc);
```

Purpose: check that all function pointers are non-null.

Return: true if the trait can be called.

Correct:

```
proven_allocator_t heap = proven_heap_allocator();
proven_err_t err = proven_alloc_is_valid(heap) ? PROVEN_OK : PROVEN_ERR_UNSUPPORTED;
(void)err; /* a freestanding build hands back a zero allocator, not a valid one */
```

Wrong:

```
proven_allocator_t alloc = {0};
proven_result_mem_mut_t r = alloc.alloc_fn(alloc.ctx, 64, 8); /* wrong: null call */
```

## 5. Raw byte buffer

### What it is for

`proven_buf_t` is the plainest thing in this chapter: a pointer, a length and a capacity. It owns its bytes, it never grows, and it is what `proven_u8str_t` and `proven_u16str_t` are built out of.

Reach for it directly when you want *bytes* rather than *text* - a record you are assembling, a frame you are about to write to a socket - and you know the maximum size in advance. When the content is text, use the string types in [Chapter 3](#) instead: they give you the same storage plus NUL-termination, views, searching and formatting.

Like everything else here it does **not** store its allocator, so `proven_buf_destroy` takes the same one `proven_buf_create` was given.

```
proven_buf_t
typedef struct {
    proven_byte_t *ptr;
    proven_size_t len;
    proven_size_t cap;
} proven_buf_t;
```

Fields:

- `ptr`: storage pointer.
- `len`: bytes currently used.
- `cap`: bytes allocated.

```
proven_result_buf_t
typedef struct {
    proven_err_t err;
    proven_buf_t value;
} proven_result_buf_t;
```

### Buffer functions

| API                                         | Intent                                           | Return                                                                                                         |
|---------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>proven_buf_create(alloc, cap)</code>  | Allocate a non-zero-capacity buffer.             | <code>proven_result_buf_t</code> ; invalid allocator or zero cap returns <code>PROVEN_ERR_INVALID_ARG</code> . |
| <code>proven_buf_append(buf, data)</code>   | Append raw bytes if they fit.                    | <code>PROVEN_OK</code> or error.                                                                               |
| <code>proven_buf_destroy(alloc, buf)</code> | Free buffer storage with the matching allocator. | void.                                                                                                          |

Example:

```
proven_result_buf_t r = proven_buf_create(alloc, 64);
if (!proven_is_ok(r.err)) {
    return;
}
proven_buf_t buf = r.value;

proven_err_t e = proven_buf_append(&buf, (proven_mem_view_t){
    .ptr = (const proven_byte_t *)"abc",
    .size = 3,
});
(void)e; /* PROVEN_ERR_OUT_OF_BOUNDS if it would not fit: the buffer never grows */

proven_buf_destroy(alloc, &buf);
```

## 6. Examples and misuse cases

### Match allocator on destroy

Correct:

```
proven_allocator_t heap = proven_heap_allocator();

proven_result_buf_t r = proven_buf_create(heap, 128);
if (!proven_is_ok(r.err)) {
    return;
}
proven_buf_destroy(heap, &r.value); /* the same allocator that created it */
```

Wrong:

```
proven_result_buf_t r = proven_buf_create(heap, 128);
proven_buf_destroy(other_alloc, &r.value); /* wrong: allocator mismatch */
```

Nothing reports this at the call. To catch it in a test, put the allocators behind `alloc_check.h` ([section 7](#)).

### Arena free is a no-op

```
alignas(max_align_t) proven_byte_t storage[1024];
proven_arena_t arena = proven_arena_create((proven_mem_mut_t){
    .ptr = storage,
    .size = sizeof storage,
});
proven_allocator_t a = proven_arena_as_allocator(&arena);

proven_result_mem_mut_t r = proven_arena_alloc(&arena, 32);
if (proven_is_ok(r.err)) {
    /* Legal, and correct to write - but it intentionally reclaims nothing. */
    a.free_fn(a.ctx, r.value.ptr);
}
proven_arena_reset(&arena); /* only this gives the 32 bytes back */
```

### Pool size contract

Correct:

```
typedef struct Node { int value; } Node;

proven_pool_t pool = {0};
if (!proven_is_ok(proven_pool_init(&pool, alloc, sizeof(Node), alignof(Node), 8))) {
    return;
}
proven_allocator_t node_alloc = proven_pool_as_allocator(&pool);

/* The pool serves exactly the size and alignment it was configured with. */
proven_result_mem_mut_t n = node_alloc.alloc_fn(node_alloc.ctx, sizeof(Node), alignof(Node));
if (proven_is_ok(n.err)) {
    node_alloc.free_fn(node_alloc.ctx, n.value.ptr);
}

/* Anything else is refused with PROVEN_ERR_INVALID_ARG rather than served. */
proven_result_mem_mut_t wrong = node_alloc.alloc_fn(node_alloc.ctx, sizeof(Node) * 2, alignof(Node));
(void)wrong;

proven_pool_destroy(&pool);
```

Wrong:

```
node_alloc.alloc_fn(node_alloc.ctx, sizeof(Other), alignof(Other));
/* wrong: PROVEN_ERR_INVALID_ARG - a pool is for one size and one alignment */
```

## Buffer append does not grow

Wrong:

```
proven_buf_append(&buf, huge_data); /* returns out-of-bounds; no automatic growth */
```

Use `proven_u8str_append_grow()` or an array if you need growth.

## Buffer append tolerates overlap

Correct:

```
proven_result_buf_t r = proven_buf_create(alloc, 64);
if (!proven_is_ok(r.err)) {
    return;
}
proven_buf_t buf = r.value;

proven_err_t e = proven_buf_append(&buf, (proven_mem_view_t){
    .ptr = (const proven_byte_t *)"abcdefgh",
    .size = 8,
});
if (proven_is_ok(e)) {
    /* The source view points back into the buffer's own bytes. That is allowed:
     * append moves rather than copies, so the overlap is well defined. */
    e = proven_buf_append(&buf, (proven_mem_view_t){
        .ptr = buf.ptr + 2,
        .size = 4,
    });
}
(void)e;

proven_buf_destroy(alloc, &buf);
```

Wrong:

```
proven_sys_mem_copy(buf.ptr + buf.len, buf.ptr + 2, 4); /* copy is not the public contract here */
```

## Worked example: an arena over caller-supplied memory

Compiled and run by the test suite. It shows the bump-and-drop lifetime that makes an arena worth reaching for: allocations are nearly free, nothing is individually freed, and `proven_arena_reset` reclaims the whole region at once.

```
/*
 * An arena does not own memory: it bumps a pointer through memory YOU own. That
 * is the whole trade. Allocation is an add, individual frees do not exist, and
 * you get everything back at once with a reset.
 *
 * The shape that makes it worth using is bump-then-drop: a phase allocates
 * freely, the phase ends, one reset reclaims the lot. No per-object bookkeeping
 * to get wrong, and nothing to leak - the backing storage below is a plain
 * array with automatic storage duration.
 */

int main(void) {
    /* The backing store is the caller's. Over-align it so the arena can satisfy
     * any alignment a caller asks for out of the first byte. */
    alignas(max_align_t) static proven_byte_t storage[4096];

    proven_arena_t arena = proven_arena_create((proven_mem_mut_t){
        .ptr = storage,
        .size = sizeof storage,
    });
```

```

/* --- bump ----- */
proven_result_mem_mut_t a = proven_arena_alloc(&arena, 64);
EXAMPLE_REQUIRE(proven_is_ok(a.err), "64 bytes must fit in a 4 KiB arena");
EXAMPLE_REQUIRE(a.value.ptr == storage, "the first allocation starts at the backing store");

/* Explicit alignment when the type demands more than PROVEN_DEFAULT_ALIGNMENT.
 * The arena pads to reach it, so the bytes it skips are simply gone until reset. */
proven_result_mem_mut_t b = proven_arena_alloc_aligned(&arena, 32, 64);
EXAMPLE_REQUIRE(proven_is_ok(b.err), "an over-aligned block must still fit");
EXAMPLE_REQUIRE(((uintptr_t)b.value.ptr % 64) == 0, "the block must honour the requested alignment");

/* --- the arena as an allocator for another API ----- */
/* Anything in proven that takes a proven_allocator_t can be driven by the
 * arena. The string below therefore lives inside `storage`. */
proven_allocator_t arena_alloc = proven_arena_as_allocator(&arena);
EXAMPLE_REQUIRE(proven_alloc_is_valid(arena_alloc), "the arena must expose a usable allocator");

proven_result_u8str_t s = proven_u8str_create(arena_alloc, 32);
EXAMPLE_REQUIRE(proven_is_ok(s.err), "the arena should be able to back a 32-byte string");

proven_err_t err = proven_u8str_append_grow(arena_alloc, &s.value, PROVEN_LIT("scratch line"));
EXAMPLE_REQUIRE(proven_is_ok(err), "appending into an arena-backed string must succeed");

/* Destroying it is still correct and still required by the ownership rules -
 * but arena free is a no-op, so it reclaims nothing. That is not a leak: the
 * bytes belong to `storage`, and the reset below is what returns them. */
proven_u8str_destroy(arena_alloc, &s.value);

proven_size_t used = arena.offset;
EXAMPLE_REQUIRE(used > 64, "every allocation above came out of the same backing store");

/* --- drop ----- */
/* One statement frees the 64-byte block, the aligned block and the string.
 * Reset costs the same whether ten objects were allocated or ten thousand. */
proven_arena_reset(&arena);
EXAMPLE_REQUIRE(arena.offset == 0, "reset must reclaim every allocation at once");

/* Proof that the storage really is reusable: the next allocation lands back
 * at the start. Every pointer handed out before the reset is dangling now -
 * that is the price of the reset being free. */
proven_result_mem_mut_t c = proven_arena_alloc(&arena, 64);
EXAMPLE_REQUIRE(proven_is_ok(c.err), "allocation after reset must succeed");
EXAMPLE_REQUIRE(c.value.ptr == storage, "after reset the arena bumps from the beginning again");

/* --- exhaustion is an error, not a crash ----- */
proven_result_mem_mut_t too_big = proven_arena_alloc(&arena, sizeof storage);
EXAMPLE_REQUIRE(too_big.err == PROVEN_ERR_NOMEM, "an arena cannot grow: it reports NOMEM instead");

printf("arena: %zu bytes used before reset, %zu in use now\n",
       (size_t)used, (size_t)arena.offset);

/* Formal cleanup. A no-op for a caller-backed arena, but writing it keeps
 * the lifetime obvious if the backing store later becomes heap memory. */
proven_arena_destroy(&arena);
return EXAMPLE_OK();
}

```

### Worked example: a pool that recycles fixed-size blocks

Compiled and run by the test suite. It shows a freed block coming straight back out of the recycle bin, and what the pool refuses.

```

/*
 * A pool is a churn optimizer, not a region. It is for one type: allocate and
 * free the same fixed-size block over and over - list nodes, events, particles -
 * without paying malloc every time.
 *
 * It keeps a small stack of freed blocks (the "bin"). Freeing pushes a block
 * onto the bin instead of returning it to the base allocator; allocating pops
 * one back off. Both are O(1) and neither touches the heap. That recycling is
 * the entire point, and the check below proves it happens.
 *
 * Ownership: the pool caches freed blocks, but it does NOT track the blocks it
 * has handed out. Every block you take, you must give back before destroy - the
 * pool cannot free what it does not know about.
 */

typedef struct {
    int id;
    int score;
} node_t;

int main(void) {
    proven_allocator_t heap = proven_heap_allocator();
    EXAMPLE_REQUIRE(proven_alloc_is_valid(heap), "hosted builds have a heap allocator");

    /* The pool takes a base allocator for the blocks it cannot serve from the
     * bin, plus the exact size and alignment of the one type it manages. The
     * last argument caps how many freed blocks are parked for reuse. */
    proven_pool_t pool = {0};
    proven_err_t err = proven_pool_init(&pool, heap, sizeof(node_t), alignof(node_t), 4);
    EXAMPLE_REQUIRE(proven_is_ok(err), "initializing a pool of node_t must succeed");
    if (!proven_is_ok(err)) {
        return 1;
    }

    proven_allocator_t nodes = proven_pool_as_allocator(&pool);

    /* --- first block: nothing in the bin, so it comes from the heap ----- */
    proven_result_mem_mut_t first = nodes.alloc_fn(nodes.ctx, sizeof(node_t), alignof(node_t));
    EXAMPLE_REQUIRE(proven_is_ok(first.err), "the pool must be able to serve its own item type");
    if (!proven_is_ok(first.err)) {
        proven_pool_destroy(&pool);
        return 1;
    }

    node_t *n = (node_t *)first.value.ptr;
    *n = (node_t){ .id = 1, .score = 100 };
    void *first_addr = n;

    /* --- hand it back: it lands in the bin, not back on the heap ----- */
    nodes.free_fn(nodes.ctx, n);
    EXAMPLE_REQUIRE(pool.bin_len == 1, "a freed block is cached for reuse, not returned to the heap");
    /* `n` is dangling from here on. The pool owns those bytes again. */

    /* --- second block: the freed one is handed straight back ----- */
    proven_result_mem_mut_t second = nodes.alloc_fn(nodes.ctx, sizeof(node_t), alignof(node_t));
    EXAMPLE_REQUIRE(proven_is_ok(second.err), "allocating from a non-empty bin must succeed");
    EXAMPLE_REQUIRE(second.value.ptr == first_addr, "the recycled block is the one that was freed");
    EXAMPLE_REQUIRE(pool.bin_len == 0, "taking it back out empties the bin");

    /* Recycled memory is NOT zeroed for you - it is whatever the pool left there.

```

```

    * Initialize every field, exactly as you would for a fresh malloc. */
node_t *m = (node_t *)second.value.ptr;
*m = (node_t){ .id = 2, .score = 50 };
EXAMPLE_REQUIRE(m->id == 2, "the recycled block is ours to overwrite");

/* --- one pool serves one size and one alignment ----- */
/* A request for anything else is refused: this is not a general allocator, and it will
 * not silently hand you a block of the wrong size. The code is PROVEN_ERR_UNSUPPORTED -
 * "not my job" - and not INVALID_ARG, which would read as "you passed me garbage" and
 * send you hunting for a bug in your own code. */
proven_result_mem_mut_t wrong = nodes.alloc_fn(nodes.ctx, sizeof(node_t) * 2, alignof(node_t));
EXAMPLE_REQUIRE(wrong.err == PROVEN_ERR_UNSUPPORTED, "the pool only serves its configured item size");

/* --- return every live block before destroying ----- */
/* proven_pool_destroy frees what is in the bin and the bin itself. `m` is
 * still handed out, so if we skipped this free it would leak. */
nodes.free_fn(nodes.ctx, m);

printf("pool: %zu block(s) cached for reuse at teardown\n", (size_t)pool.bin_len);

proven_pool_destroy(&pool);
return EXAMPLE_OK();
}

```

### Worked example: start-up allocation, growing in place, and wrapping an allocator

The two examples above use an arena and a pool the way most code does. This one covers the three things you reach for once the program is real:

1. **Allocation during start-up.** Some allocations have no sensible failure path: if the program cannot get its own configuration table, there is nothing left for it to do.  
`proven_arena_alloc_or_panic()` and `proven_arena_alloc_aligned_or_panic()` return the block itself instead of a result, and hand the failure to the panic handler ([Chapter 1 section 6](#)).  
`proven_set_panic_handler()` is how you choose what happens then - the example installs one that records the message so the failure can be shown here, and puts the default back afterwards.
2. **Growing the block you took last.** `proven_arena_realloc_aligned()` extends it where it stands, because the most recent block is the one at the end of the used region. Any earlier block is copied to the end instead - still correct, no longer free.
3. **Wrapping an allocator.** An allocator is three function pointers and a context pointer, so a wrapper that counts, logs, or fails the tenth call on purpose is three forwarding functions. The arena's own three -- `proven_arena_alloc_trait()`, `proven_arena_realloc_trait()` and `proven_arena_free_trait()` - are public so your wrapper can call them instead of re-implementing the arena. Everything in the library that takes a `proven_allocator_t` then runs through your wrapper without knowing it.

```

/*
 * Three things a program that owns its own memory eventually has to do, and the
 * calls that do them:
 *
 * - Allocate during start-up, where running out of memory is not a condition
 *   the program can carry on from. That is what the `_or_panic` calls are
 *   for, and installing a panic handler is how you decide what "cannot carry
 *   on" means for your program - a log line and an exit, rather than a trap.
 *
 * - Grow the block you allocated last, without copying it. An arena can do
 *   that in place, because the block that was allocated last is the one
 *   sitting at the end of the used region.
 *
 * - Instrument the allocator - count allocations, or fail the tenth one on

```

```

*   purpose in a test - without changing the code being measured. An
*   allocator here is three function pointers and a context pointer, so
*   wrapping one is writing three forwarding functions. The arena's own three
*   are public for exactly this reason: your wrapper forwards to them.
*/

/* --- what a panic handler is for ----- */

static int g_panics = 0;
static char g_last_panic[128];

/* A panic handler receives the message and decides the program's fate. The
* default one traps immediately, which is right in production and useless in a
* test - so this one records the message and returns. Returning is allowed ONLY
* when you are deliberately testing the panic path, and the memory block the
* panicking call returns must then not be used. */
static void record_panic(const char *msg) {
    ++g_panics;
    snprintf(g_last_panic, sizeof g_last_panic, "%s", msg);
}

/* --- an allocator that counts what passes through it ----- */

typedef struct {
    proven_arena_t *arena;
    proven_size_t   live_bytes;
    proven_size_t   alloc_calls;
    proven_size_t   free_calls;
} counting_ctx_t;

/* Each of the three matches one field of proven_allocator_t. Each does its own
* bookkeeping and then forwards to the arena's public trait function, so the
* behaviour being measured is exactly the arena's behaviour and not a
* re-implementation of it. */
static proven_result_mem_mut_t counting_alloc(void *ctx, proven_size_t size, proven_size_t align) {
    counting_ctx_t *c = (counting_ctx_t *)ctx;
    proven_result_mem_mut_t r = proven_arena_alloc_trait(c->arena, size, align);
    if (proven_is_ok(r.err)) {
        c->live_bytes += size;
        ++c->alloc_calls;
    }
    return r;
}

static proven_result_mem_mut_t counting_realloc(void *ctx, void *old_ptr, proven_size_t old_size,
                                              proven_size_t new_size, proven_size_t align) {
    counting_ctx_t *c = (counting_ctx_t *)ctx;
    proven_result_mem_mut_t r = proven_arena_realloc_trait(c->arena, old_ptr, old_size, new_size, align);
    if (proven_is_ok(r.err)) {
        c->live_bytes = c->live_bytes - old_size + new_size;
    }
    return r;
}

static void counting_free(void *ctx, void *ptr) {
    counting_ctx_t *c = (counting_ctx_t *)ctx;
    ++c->free_calls;
    proven_arena_free_trait(c->arena, ptr); /* an arena free is a no-op; the count is the point */
}

```

```

int main(void) {
    alignas(max_align_t) static proven_byte_t storage[1024];
    proven_arena_t arena = proven_arena_create((proven_mem_mut_t){ .ptr = storage, .size = sizeof storage });

    /* --- 1. start-up allocation that must not fail ----- */

    proven_set_panic_handler(record_panic);

    /* No result to unwrap: these return the block directly, because there is no
     * error the caller could act on. That is the entire difference. */
    proven_mem_mut_t table = proven_arena_alloc_or_panic(&arena, 256);
    EXAMPLE_REQUIRE(table.ptr != NULL, "a 256-byte start-up allocation must succeed");
    EXAMPLE_REQUIRE(g_panic == 0, "a successful allocation must not panic");

    /* The aligned form, for a type that needs more than the default boundary -
     * a 64-byte cache line here, the usual reason. */
    proven_mem_mut_t cache_line = proven_arena_alloc_aligned_or_panic(&arena, 64, 64);
    EXAMPLE_REQUIRE(((proven_uintptr_t)cache_line.ptr % 64) == 0,
        "the block must start on the boundary that was asked for");
    EXAMPLE_REQUIRE(g_panic == 0, "an over-aligned allocation that fits must not panic either");

    /* Now the failing case, on purpose: more than the arena could ever hold.
     * With the recording handler installed we can observe it; with the default
     * handler the program would stop here, which is what it is for. */
    proven_mem_mut_t impossible = proven_arena_alloc_or_panic(&arena, sizeof storage * 2);
    (void)impossible; /* after a handler returns, this block means nothing */
    EXAMPLE_REQUIRE(g_panic == 1, "exhausting the arena through _or_panic must panic");
    EXAMPLE_REQUIRE(g_last_panic[0] != '\0', "the handler receives a message naming the call");
    printf("panic handler saw: %s\n", g_last_panic);

    /* Passing NULL puts the default trapping handler back. Leaving a test
     * handler installed turns a real failure into silent corruption. */
    proven_set_panic_handler(NULL);

    /* --- 2. growing the most recent block in place ----- */

    /* A parser that reads a header, then discovers the body is longer than it
     * guessed, wants to extend the buffer it just took - not copy it. */
    proven_result_mem_mut_t buf = proven_arena_alloc_aligned(&arena, 32, alignof(proven_u32));
    EXAMPLE_REQUIRE(proven_is_ok(buf.err), "the initial 32-byte buffer must fit");

    proven_size_t before = arena.offset;
    proven_result_mem_mut_t grown = proven_arena_realloc_aligned(&arena, buf.value.ptr, 32, 96,
        alignof(proven_u32));
    EXAMPLE_REQUIRE(proven_is_ok(grown.err), "growing the most recent block must succeed");
    EXAMPLE_REQUIRE(grown.value.ptr == buf.value.ptr,
        "the most recent block grows in place: same address, no copy");
    EXAMPLE_REQUIRE(arena.offset == before + 64, "only the extra 64 bytes were taken");

    /* The in-place path is available only for the block allocated LAST. Take
     * another block, and the earlier one can no longer be extended where it
     * stands - the arena copies it to the end instead, and the old bytes are
     * dead until the next reset. Correct either way; just not free. */
    proven_result_mem_mut_t other = proven_arena_alloc(&arena, 16);
    EXAMPLE_REQUIRE(proven_is_ok(other.err), "a second block must fit");
    proven_result_mem_mut_t moved = proven_arena_realloc_aligned(&arena, grown.value.ptr, 96, 128,
        alignof(proven_u32));
    EXAMPLE_REQUIRE(proven_is_ok(moved.err), "growing an older block must still succeed");
    EXAMPLE_REQUIRE(moved.value.ptr != grown.value.ptr, "but it is relocated, not extended");

```

```

/* --- 3. the arena behind a counting wrapper ----- */

proven_arena_reset(&arena);

counting_ctx_t counted = { .arena = &arena };
proven_allocator_t alloc = {
    .ctx      = &counted,
    .alloc_fn  = counting_alloc,
    .realloc_fn = counting_realloc,
    .free_fn   = counting_free,
};
EXAMPLE_REQUIRE(proven_alloc_is_valid(alloc), "all three function pointers must be present");

/* Any part of the library that takes an allocator now runs through the
 * wrapper, unchanged and unaware. */
proven_result_u8str_t s = proven_u8str_create(alloc, 16);
EXAMPLE_REQUIRE(proven_is_ok(s.err), "creating a string through the wrapper must succeed");

proven_err_t err = proven_u8str_append_grow(alloc, &s.value, PROVEN_LIT("a line long enough to need more
room"));
EXAMPLE_REQUIRE(proven_is_ok(err), "appending past the initial capacity must succeed");

proven_u8str_destroy(alloc, &s.value);

EXAMPLE_REQUIRE(counted.alloc_calls >= 1, "the wrapper saw the string being created");
EXAMPLE_REQUIRE(counted.free_calls >= 1, "and saw it being destroyed");
printf("counting allocator: %zu alloc call(s), %zu free call(s), %zu bytes handed out\n",
       (size_t)counted.alloc_calls, (size_t)counted.free_calls, (size_t)counted.live_bytes);

proven_arena_destroy(&arena);
return EXAMPLE_OK();
}

```

Wrong - leaving a test panic handler installed:

```

proven_set_panic_handler(record_panic); /* returns instead of stopping */
/* ... the rest of the program ... */ /* wrong: no restore */

```

A handler that returns turns every later out-of-memory panic into a call that hands back a null block and carries on. Install one only around the code you are testing, and restore the default with `proven_set_panic_handler(NULL)`.

Wrong - using the block after a handler that returned:

```

proven_mem_mut_t m = proven_arena_alloc_or_panic(&arena, huge);
memset(m.ptr, 0, huge); /* wrong: m.ptr is null if the handler returned */

```

Wrong - a wrapper that forgets one of the three:

```

proven_allocator_t alloc = { .ctx = &counted, .alloc_fn = counting_alloc }; /* wrong */

```

`proven_alloc_is_valid()` returns false for this, and the first `realloc` or `destroy` through it calls a null pointer. Fill in all three, even when two of them only forward.

## 7. Catching the wrong allocator: `alloc_check.h`

### The problem: a mistake that fails somewhere else

Every owner in this library - a string, a buffer, an array - is freed through an allocator the caller passes, and nothing records which allocator the memory came from. Destroying an arena string through the heap, freeing a block twice, or growing a block with an old size it was not allocated with does not fail where it happens. The heap is handed memory it never gave out, its bookkeeping is corrupted, and the crash comes later, in code that did nothing wrong. Storing the

allocator in every owner would double the size of every string for a check most programs never need, so the library does not (B-023). The check belongs on the other side: in the allocator, which can know exactly which blocks are its own.

### What `alloc_check.h` does

`proven_alloc_check_wrap` puts a checker in front of any allocator - heap, arena, pool, or one of your own - and returns an allocator you use in its place. The checker keeps a record of every block it hands out, in memory you supply, and refuses anything else: a free or realloc of a block that is not in the record (another allocator's block, or one already freed), and a realloc whose old size or alignment is not the one the block was allocated with. A refusal is a `proven_panic` at the call that made the mistake, and the refused pointer is never passed on, so the underlying allocator is not corrupted even when a test's panic handler returns. At the end of a test, `proven_alloc_check_live` is a leak check.

**It is for tests and bug hunts, not for production.** Every free and realloc searches the record, linearly. Write `proven_alloc_checked` where you create the allocator and leave it there: it wraps only when `PROVEN_ALLOC_CHECK` is defined, and otherwise returns the allocator it was given, at no cost. Define the macro on the build (`-DPROVEN_ALLOC_CHECK`) for tests and debug runs. It must be defined before the first proven header a file includes - the switch is read with the header, so a `#define` after `#include "proven.h"` leaves the checks silently off.

```
proven_alloc_check_t heap_chk;
proven_alloc_check_entry_t heap_rec[256]; /* up to 256 live blocks */
proven_allocator_t checked = proven_alloc_checked(&heap_chk, proven_heap_allocator(), heap_rec, 256);
/* ... the whole program or test uses `checked` ... */
(void)checked;
if (proven_alloc_check_live(&heap_chk) != 0) { /* a leak, when the checks are on */ }
```

| API                                                                | Intent                                                                               | Return                                                     |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------|
| <code>proven_alloc_check_wrap(&amp;st, inner, entries, cap)</code> | Wrap inner, recording up to <code>cap</code> live blocks. Always checks.             | <code>proven_allocator_t</code> (invalid on bad arguments) |
| <code>proven_alloc_checked(&amp;st, inner, entries, cap)</code>    | The same when <code>PROVEN_ALLOC_CHECK</code> is defined; otherwise inner unchanged. | <code>proven_allocator_t</code>                            |
| <code>proven_alloc_check_owns(&amp;st, ptr)</code>                 | Whether <code>ptr</code> is a live block from this checker.                          | <code>bool</code>                                          |
| <code>proven_alloc_check_live(&amp;st)</code>                      | Live blocks: 0 at the end of a test means no leak.                                   | <code>proven_size_t</code>                                 |

### The structures you hold

```
typedef struct { void *ptr; proven_size_t size; proven_size_t align; } proven_alloc_check_entry_t;
```

```
typedef struct {
    proven_allocator_t    inner;
    proven_alloc_check_entry_t *entries; /* your record storage */
    proven_size_t         cap, live, live_bytes, peak_live;
    proven_size_t         faults; /* refused operations, each also a panic */
} proven_alloc_check_t; /* caller-owned; outlives the allocator; do not copy once wrapped */
```

### Cautions, and what goes wrong

**It only checks the allocators you wrap.** Destroying an arena string through an unwrapped heap is still unchecked: the heap is not asked. Wrap every allocator the test uses.

**A full record is a panic, not a pass.** A record that quietly stopped recording would report the next free of an unrecorded block as foreign. Size cap for the most blocks the test holds at once; peak\_live tells you what that was. Wrong - a record too small for the test:

```
proven_alloc_check_entry_t rec[4];
proven_allocator_t a = proven_alloc_check_wrap(&st, heap, rec, 4);
/* wrong: a fifth live block panics "the record is full" - enlarge rec, do not ignore it */
```

Wrong - switching it on after the include. The switch is read with the header, so this file runs with the checks off and nothing says so:

```
#include "proven.h"
#define PROVEN_ALLOC_CHECK /* wrong: too late - use -DPROVEN_ALLOC_CHECK */
```

**Leaving it on in a hot path is a slowdown you chose.** The search is linear in the live blocks.

### Worked example: a string destroyed through the wrong allocator

```
#include <string.h>

/*
 * Catching the wrong allocator where the mistake is made.
 *
 * Nothing in this library remembers which allocator a block came from: the caller
 * passes one to create and one to destroy, and if they differ the allocator's
 * bookkeeping is corrupted and the program breaks later, somewhere else. The
 * alloc_check wrapper goes in front of any allocator, keeps a record of the
 * blocks it handed out, and refuses - with a panic, at the call - anything else.
 *
 * It is for tests and bug hunts. In your own code, write proven_alloc_checked()
 * and switch it with -DPROVEN_ALLOC_CHECK: without the macro it returns the
 * allocator it was given and costs nothing. (A #define after the first proven
 * #include is too late - the switch is read with the header.) This example
 * calls proven_alloc_check_wrap(), which checks unconditionally, so it shows
 * the checks whatever the build flags.
 */

/* A panic handler that returns, so this example can show a refusal and go on. A
 * real program keeps the default, which stops at the mistake. */
static int g_refused;
static void note_panic(const char *msg) {
    ++g_refused;
    printf("refused: %s\n", msg);
}

int main(void) {
    proven_set_panic_handler(note_panic);

    /* Two allocators, each behind its own checker with room to record 16 blocks. */
    static proven_byte_t arena_mem[2048];
    proven_arena_t arena = proven_arena_create((proven_mem_mut_t){ arena_mem, sizeof arena_mem });
    proven_alloc_check_t arena_chk, heap_chk;
    proven_alloc_check_entry_t arena_rec[16], heap_rec[16];
    proven_allocator_t scratch = proven_alloc_check_wrap(&arena_chk, proven_arena_as_allocator(&arena),
    arena_rec, 16);
    proven_allocator_t heap = proven_alloc_check_wrap(&heap_chk, proven_heap_allocator(), heap_rec, 16);

    /* --- correct use passes straight through ----- */

    proven_result_u8str_t name = proven_u8str_create_from_view(heap, PROVEN_LIT("report"));
    EXAMPLE_REQUIRE(proven_is_ok(name.err), "a string from the checked heap");
    EXAMPLE_REQUIRE(proven_is_ok(proven_u8str_append_grow(heap, &name.value, PROVEN_LIT(".txt"))), "grown by
```

```

the same allocator");
    EXAMPLE_REQUIRE(proven_alloc_check_owns(&heap_chk, name.value.internal.ptr), "the heap checker owns its
block");

    /* --- the mistake, caught at the call ----- */

    /* A temporary built in the scratch arena... */
    proven_result_u8str_t tmp = proven_u8str_create(scratch, 64);
    EXAMPLE_REQUIRE(proven_is_ok(tmp.err), "a temporary in the arena");
    proven_u8str_t t = tmp.value;
    /* ...and destroyed through the heap. Unchecked, the heap would free arena memory. */
    proven_u8str_destroy(heap, &t);
    EXAMPLE_REQUIRE(g_refused == 1 && heap_chk.faults == 1, "the heap refuses a block it never gave out");
    EXAMPLE_REQUIRE(proven_alloc_check_live(&arena_chk) == 1, "and the arena block is untouched");
    t = tmp.value;
    proven_u8str_destroy(scratch, &t); /* the right one */

    /* --- a leak check at the end of a test ----- */

    proven_u8str_destroy(heap, &name.value);
    EXAMPLE_REQUIRE(proven_alloc_check_live(&heap_chk) == 0 && proven_alloc_check_live(&arena_chk) == 0,
        "nothing is left live");
    printf("peak %zu live block(s) on the heap, %zu refusal(s)\n", (size_t)heap_chk.peak_live,
(size_t)heap_chk.faults);

    proven_set_panic_handler(NULL);
    return EXAMPLE_OK();
}

```

## Chapter 3: Strings, Formatting, and Scanning

**Part II - The vocabulary every program uses. Prerequisites: [Chapter 1](#) and [Chapter 2](#).** After this chapter you can hold text without a NUL terminator deciding your program's fate, build strings that refuse to overflow, and format and parse the everyday cases.

This chapter covers `u8str.h`, `u16str.h`, `utf.h`, `fmt.h`, and `scan.h`. It is the **tutorial half** of the text material: it introduces the formatter and the scanner with the cases you meet daily. [Chapter 8](#) is the reference half - the complete grammar, every argument constructor, and the scanner's error and recovery rules. Read this one first.

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### 1. U8 strings and views

#### The problem: a C string does not know how long it is

A C string is a pointer, and where it ends is decided by a zero byte somewhere in memory. That one decision - made in 1972 to save a byte per string - is behind a remarkable amount of damage:

- **Length is a search.** `strlen` walks the string. A loop that checks `strlen(s)` on each iteration is quadratic, and it looks like ordinary code.
- **Text cannot contain a zero byte.** So a C string cannot hold a UTF-16 buffer, a protocol frame, a slice of a file, or any binary data - the string type and the byte type are different things and the language pretends otherwise.
- **A missing terminator is not detectable.** `strcpy` into a buffer with no room writes until it finds a zero somewhere in your stack frame. Nothing reports it. This is the single most exploited bug class in the language's history.
- **You cannot cheaply refer to part of a string.** "The third field of this line" means either copying those bytes out or writing a zero over the original, destroying it. `strtok` chose the second, which is why it mutates its input and cannot be nested.

`strncpy`, the traditional patch, does not always NUL-terminate - so the "safe" function can produce a string that is not a string.

#### What this library does instead

Two types, and the difference between them is ownership:

- `proven_u8str_view_t` - **borrowed**. A pointer and a size, together, pointing at bytes someone else owns. Copying it is free. It allocates nothing, destroys nothing, and stops being valid when its owner does. This is what you pass to functions.
- `proven_u8str_t` - **owned**. It has its own storage, a capacity, and a NUL terminator kept for you so `proven_u8str_as_cstr` can hand the bytes to a libc function that still wants one. You created it with an allocator; you destroy it with the same one.

**The owned string does not remember that allocator, and nothing checks it.** Every call that may allocate or free - `_create`, `_reserve`, the `_grow` family, `_destroy` - takes the allocator as an argument, and it must be the one the string was created with. Pass a different one and there is no error to return: the other allocator is handed memory it never gave out, its bookkeeping is corrupted, and the crash comes later and somewhere else. Keeping the allocator in every string would make each one

twice the size and would mean nothing for a borrowed string, so the library does not; keep the pairing in your own code - one allocator per owner, passed from the same place.

Wrong - a string made in an arena and destroyed through the heap:

```
proven_u8str_t s = proven_u8str_create(arena, 64).value;
proven_u8str_destroy(proven_heap_allocator(), &s); /* wrong: the heap frees arena memory */
```

Both count **bytes**, not characters. "한" is three bytes in UTF-8 and one character, and this library will tell you three, because that is what it knows. Text is a sequence of bytes here; interpreting those bytes as characters is a job for a Unicode layer this library does not have.

Because a view carries its length, everything the NUL terminator made hard becomes ordinary: length is a field, text may contain zero bytes, a sub-range is a view into the same memory with no copy, and a write that would not fit is refused rather than performed.

Wrong - treating a view as a C string:

```
proven_u8str_view_t v = proven_u8str_view_slice(line, 4, 8); /* a field inside a line */
printf("%s\n", (const char *)v.ptr); /* wrong: no terminator, prints until it finds a zero */
```

A view is deliberately *not* NUL-terminated: it usually points into the middle of someone else's buffer, where writing a terminator would corrupt the next field. Use `proven_println("{}"`, `PROVEN_ARG(v)`), which takes the size, or copy it into an owned string first.

## Structures

```
typedef struct {
    const proven_byte_t *ptr;
    proven_size_t size;
} proven_u8str_view_t;
```

```
typedef struct {
    proven_byte_t *ptr;
    proven_size_t size;
} proven_u8str_mut_t;
```

```
typedef struct {
    proven_buf_t internal;
    bool        borrowed;
} proven_u8str_t;
```

```
typedef struct {
    proven_err_t err;
    proven_u8str_t value;
} proven_result_u8str_t;
```

```
typedef struct {
    proven_err_t err;
    const char *value;
} proven_result_cstr_t;
```

Intent:

- `proven_u8str_view_t`: borrowed read-only byte string view.
- `proven_u8str_mut_t`: borrowed mutable byte string view.
- `proven_u8str_t`: owned string with NUL termination.
- `proven_result_u8str_t`: result wrapper for owned strings.
- `proven_result_cstr_t`: result wrapper for an allocated NUL-terminated C string.

## Internal layout

The views are exactly what they look like - a pointer and a byte count:

```
typedef struct { const proven_byte_t *ptr; proven_size_t size; } proven_u8str_view_t;
typedef struct { proven_byte_t *ptr;          proven_size_t size; } proven_u8str_mut_t;
```

The owned string wraps a `proven_buf_t` (the fixed-capacity byte buffer from Chapter 2) plus one flag:

```
typedef struct {
    proven_buf_t internal; /* the bytes + length + capacity; always keeps room for a NUL */
    bool         borrowed; /* false = allocator-owned (default); true = wraps caller memory */
} proven_u8str_t;
```

- `internal.len` is the byte length (`proven_buf_t` is `ptr / len / cap` - there is no `size` member). The byte at `internal.len` is always the NUL terminator while the string is valid, so `proven_u8str_as_cstr()` is  $O(1)$ .
- `borrowed` is false for a zero-initialized handle, so an allocator-owned string is the safe default. It is set true only by `proven_u8str_borrow`, which wraps `[buf, buf+cap)` you own: growing operations then refuse to reallocate (`PROVEN_ERR_OUT_OF_BOUNDS`) and `proven_u8str_destroy` is a no-op.
- **Do not** read or write these fields directly to change the string; use the functions below. Reading `internal.len` for length is fine, but prefer `proven_u8str_as_view()`.

Counter-example - treating a borrowed string like an owned one:

```
proven_byte_t stack[8];
proven_u8str_t s = proven_u8str_borrow(stack, sizeof stack); /* borrowed */

/* This would have to reallocate caller memory, so it refuses: the call returns
   PROVEN_ERR_OUT_OF_BOUNDS and `stack` is left exactly as it was. A borrowed
   string never silently escapes to the heap. */
proven_err_t e = proven_u8str_append_grow(&s, PROVEN_LIT("far too long for eight bytes"));
(void)e; /* == PROVEN_ERR_OUT_OF_BOUNDS */

/* destroy is a no-op here: `stack` is yours, and the library will not free it.
   Writing it anyway is correct, and it keeps teardown code uniform. */
proven_u8str_destroy(&s);
```

## Macros

| Macro                               | Intent                                                         | Notes                                   |
|-------------------------------------|----------------------------------------------------------------|-----------------------------------------|
| <code>PROVEN_LIT(s)</code>          | Make a <code>proven_u8str_view_t</code> from a string literal. | Use only with string literals.          |
| <code>PROVEN_LIT_INIT(s)</code>     | Initializer form for literal views.                            | Use in aggregate initialization.        |
| <code>PROVEN_INDEX_NOT_FOUND</code> | Sentinel returned by find functions.                           | Equal to $(\text{proven\_size\_t})-1$ . |

## U8 functions

| API                                                     | Intent                                                                                                                                                                                                                                                             | Return                               |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <code>proven_u8str_create(alloc, limit)</code>          | Create an empty owned string with capacity for <code>limit</code> bytes plus NUL.                                                                                                                                                                                  | <code>proven_result_u8str_t</code> . |
| <code>proven_u8str_create_from_view(alloc, view)</code> | Copy a view into a new owned string.                                                                                                                                                                                                                               | <code>proven_result_u8str_t</code> . |
| <code>proven_u8str_borrow(buf, cap)</code>              | Wrap caller-owned <code>[buf, buf+cap)</code> as a fixed-capacity string (no allocation). Fixed-capacity ops and <code>append_fmt</code> work; growing ops refuse to reallocate caller memory; <code>destroy</code> is a no-op. <code>cap</code> includes the NUL. | <code>proven_u8str_t</code> .        |

|                                                                    |                                                                                                        |                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| proven_u8str_reset(str)                                            | Truncate to empty, keeping the buffer/capacity for reuse (owned or borrowed).                          | proven_err_t.                                      |
| proven_u8str_is_valid(str)                                         | Validate public string invariants.                                                                     | bool.                                              |
| proven_u8str_reserve(alloc, str, new_cap)                          | Ensure at least new_cap bytes of internal capacity. (Borrowed strings cannot grow.)                    | proven_err_t.                                      |
| proven_u8str_append(str, data)                                     | Atomic fixed-capacity append.                                                                          | PROVEN_OK or PROVEN_ERR_OUT_OF_BOUNDS.             |
| proven_u8str_append_partial(str, data)                             | Truncating append.                                                                                     | proven_result_size_t; value is bytes written.      |
| proven_u8str_append_grow(alloc, str, data)                         | Atomic growable append.                                                                                | proven_err_t.                                      |
| proven_u8str_append_byte(alloc, str, b)                            | Append one byte, growing if needed.                                                                    | proven_err_t.                                      |
| proven_u8str_replace_at(str, index, old_len, data)                 | Replace an exact byte range with fixed-capacity semantics.                                             | proven_err_t.                                      |
| proven_u8str_replace_at_grow(alloc, str, index, old_len, data)     | Like replace_at, but grows the buffer (doubling) when the edit does not fit instead of failing.        | proven_err_t; string unchanged on alloc failure.   |
| proven_u8str_insert(str, index, data)                              | Insert bytes at index with fixed-capacity semantics.                                                   | proven_err_t.                                      |
| proven_u8str_insert_grow(alloc, str, index, data)                  | Like insert, but grows the buffer when needed. No manual reserve required.                             | proven_err_t; string unchanged on alloc failure.   |
| proven_u8str_remove(str, index, len)                               | Remove a byte range.                                                                                   | proven_err_t.                                      |
| proven_u8str_replace_first(str, start_offset, target, replacement) | Replace first matching target at or after start.                                                       | PROVEN_OK even when not found.                     |
| proven_u8str_view_find(haystack, start_offset, needle)             | Find the first occurrence of a byte substring (correctly handles any byte values; not NUL-terminated). | index or PROVEN_INDEX_NOT_FOUND.                   |
| proven_u8str_view_starts_with(str, prefix)                         | Prefix test.                                                                                           | int truth value.                                   |
| proven_u8str_view_ends_with(str, suffix)                           | Suffix test.                                                                                           | int truth value.                                   |
| proven_u8str_view_slice(str, index, len)                           | Return a clamped subview.                                                                              | proven_u8str_view_t.                               |
| proven_u8str_as_cstr(str)                                          | Return internal NUL-terminated pointer.                                                                | const char *; invalidated by growth/destroy.       |
| proven_mem_view_from_u8(view)                                      | Convert U8 view to byte view.                                                                          | proven_mem_view_t.                                 |
| proven_u8str_view_to_cstr(view, alloc)                             | Allocate a NUL-terminated C string from any view.                                                      | proven_result_cstr_t; caller frees with allocator. |
| proven_cstr_len(s)                                                 | Count bytes until NUL.                                                                                 | proven_size_t.                                     |

|                                  |                                                   |                              |
|----------------------------------|---------------------------------------------------|------------------------------|
| proven_u8str_view_from_cstr(s)   | Make a view from a trusted NUL-terminated string. | Empty view for null pointer. |
| proven_u8str_view_eq(a, b)       | Byte equality test.                               | int truth value.             |
| proven_u8str_destroy(alloc, str) | Free owned storage with matching allocator.       | void.                        |
| proven_u8str_as_view(str)        | Borrow current contents.                          | proven_u8str_view_t.         |

### Basic U8 example

```

proven_result_u8str_t r = proven_u8str_create_from_view(alloc, PROVEN_LIT("log"));
if (!proven_is_ok(r.err)) {
    return;
}
proven_u8str_t s = r.value;

proven_err_t e = proven_u8str_append_grow(alloc, &s, PROVEN_LIT(": ready"));
if (!proven_is_ok(e)) {
    proven_u8str_destroy(alloc, &s);
    return;
}

/* Valid until the next growing call: as_cstr points into the string's storage. */
const char *cstr = proven_u8str_as_cstr(&s);
(void)cstr;

proven_u8str_destroy(alloc, &s);

```

### Splitting, trimming, searching from the end, and ordering views

Views could be found in, sliced, and compared for equality, and that was all. Everything else a program does with a line of text - cut it into fields, trim the fields, find the last dot, sort the results - every caller wrote by hand, and the hand-written versions were wrong in the same places. Splitting is the clearest case. The loop a competent person writes first is *"keep going while a separator is found"*:

```

proven_size_t off = 0, at;
while ((at = proven_u8str_view_find(s, off, sep)) != PROVEN_INDEX_NOT_FOUND) {
    emit(proven_u8str_view_slice(s, off, at - off)); /* wrong: the tail is never emitted */
    off = at + sep.size;
}

```

It loses the text after the last separator, and that is wrong on the *common* case: "a,b,c" gives two fields, "a" gives none, "" gives none. Getting it right means hoisting the last field out of the loop, which is exactly the step that gets skipped under a deadline. The alternative callers actually reached for - one owned string per field - measured 3.4x slower and one allocation per field (RFC-0004 section 2.2).

So the view vocabulary now has the rest of the words, all pure functions over views with no allocation. Two rules hold for every one of them. **An ill-formed view** - ptr == NULL with size > 0 - **is treated as empty**, by a guard in each function. And **every empty result is {NULL, 0}**, the spelling proven\_u8str\_view\_slice already used, so an empty result carries no position: test a view by its size, never by its pointer.

**Splitting keeps a permanent contract: n separators yield n + 1 fields**, counting non-overlapping occurrences from the left. "a," is "a" and an empty field; ",a" is an empty field and "a"; "a,,b" keeps its empty middle field, unlike strtok; "" is one empty field, not zero. An empty separator yields the whole input once - never an endless run of empty fields, which is what the obvious

implementation does, because a search for an empty needle matches where it starts. The iterator points into your source, never into itself, so it can be copied, and the copy continues on its own.

**Trimming knows exactly six bytes:** space, \t, \n, \v, \f, \r. Not the locale, and not Unicode - a no-break space is not whitespace here. Prefix and suffix removal leave the view unchanged when the affix is absent; that is not an error, and if you need to know whether it fired, ask

`proven_u8str_view_starts_with` first.

`proven_u8str_view_find_last` **returns a position**, the start of the last occurrence, and occurrences may overlap: `find_last("aaa", "aa")` is 1. For an empty needle the position is `size` - the same answer `_find` gives - which is the one result that is not a valid byte index. It makes the same choices the forward search makes, from the same sample of the haystack: on ordinary text it finds the rarest needle byte from the end and verifies around it, and on a low-entropy haystack - long runs, a small alphabet - it switches to an algorithm that is linear whatever the input (backward Shift-Or up to 64 bytes, a reverse Two-Way beyond).

`proven_u8str_view_cmp` **orders bytes unsigned, with a prefix first**, so `"\xFF"` sorts after `"a"` and `"app"` before `"apple"`; a NUL inside a view is data. It answers by sign - not necessarily -1, 0 or 1.

`proven_u8str_view_cmp_ptr` is the same comparison shaped for `proven_array_sort`, which passes pointers to the elements.

| API                                                                             | Intent                                                                 | Return                                 |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------|
| <code>proven_u8str_view_split(src, sep)</code>                                  | Start splitting; the fields are views into <code>src</code> .          | <code>proven_u8str_view_split_t</code> |
| <code>proven_u8str_view_split_next(&amp;it, &amp;field)</code>                  | The next field, or false when there are none.                          | bool                                   |
| <code>proven_u8str_view_trim(s)</code>                                          | Drop the six whitespace bytes from both ends.                          | <code>proven_u8str_view_t</code>       |
| <code>proven_u8str_view_trim_start(s),<br/>proven_u8str_view_trim_end(s)</code> | One end only.                                                          | <code>proven_u8str_view_t</code>       |
| <code>proven_u8str_view_remove_prefix(s,<br/>prefix)</code>                     | <code>s</code> without prefix, or <code>s</code> unchanged.            | <code>proven_u8str_view_t</code>       |
| <code>proven_u8str_view_remove_suffix(s,<br/>suffix)</code>                     | <code>s</code> without suffix, or <code>s</code> unchanged.            | <code>proven_u8str_view_t</code>       |
| <code>proven_u8str_view_find_last(haystack,<br/>needle)</code>                  | Start of the last occurrence, or <code>PROVEN_INDEX_NOT_FOUND</code> . | <code>proven_size_t</code>             |
| <code>proven_u8str_view_contains(haystack,<br/>needle)</code>                   | find from 0 is not <code>NOT_FOUND</code> .                            | bool                                   |
| <code>proven_u8str_view_cmp(a, b)</code>                                        | Bytewise unsigned order, prefix first; sign only.                      | int                                    |
| <code>proven_u8str_view_cmp_ptr(pa, pb)</code>                                  | cmp for <code>proven_array_sort</code> : pointers to views.            | int                                    |
| <code>proven_u8str_view_is_well_formed(s)</code>                                | False only for <code>{NULL, n &gt; 0}</code> .                         | bool                                   |

```
typedef struct {
    proven_u8str_view_t rest; /* not yet yielded; points into the caller's source bytes */
    proven_u8str_view_t sep;
    bool done; /* set once the final field has been yielded */
} proven_u8str_view_split_t; /* copyable: a copy continues independently */
```

Wrong - ending a loop on the field, or on `_is_well_formed`. An empty field is `{NULL, 0}`, and so is a slice past the end of the text; both are well formed. `_is_well_formed` says only that a view is safe to read. End a split on `split_next`'s return value and a search on `PROVEN_INDEX_NOT_FOUND`:

```
while (proven_u8str_view_split_next(&it, &f) && f.ptr != NULL) { ... } /* wrong: stops at the first empty field */
while (proven_u8str_view_is_well_formed(rest)) { ... } /* wrong: never false for a slice */
```

Wrong - reading the byte at `find_last`'s answer without checking the needle. For an empty needle the answer is size:

```
proven_size_t at = proven_u8str_view_find_last(s, needle);
char c = (char)s.ptr[at]; /* wrong when needle is empty: at == s.size, one past the end */
```

The worked example splits a record with trailing and doubled separators, trims each field, finds the last dot, strips a double suffix, and sorts the names:

```
#include <string.h>

/*
 * The everyday text jobs, done on views: split a line into fields, trim them,
 * strip a known prefix or suffix, find the last dot, and sort what you got.
 * Nothing here allocates except the array the sort works on; every result
 * points into the text you started with.
 *
 * Two rules hold throughout. An empty result is {NULL, 0}, so test a view by
 * its size, never its pointer. And a split loop ends on the return value of
 * split_next - n separators always give n + 1 fields, empty ones included.
 */

static bool is(proven_u8str_view_t v, const char *want) {
    return v.size == strlen(want) && (v.size == 0 || memcmp(v.ptr, want, v.size) == 0);
}

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- split, then trim each field ----- */

    /* A record with a trailing separator and an empty field: five separators,
     * so six fields - the last one empty. The loop a person writes first
     * ("while a separator is found") would give five and lose the tail. */
    proven_u8str_view_t record = PROVEN_LIT(" report.tar.gz , draft.txt,,notes.md ,\tREADME ,");
    proven_u8str_view_t fields[8];
    int n = 0;
    proven_u8str_view_split_t it = proven_u8str_view_split(record, PROVEN_LIT(","));
    proven_u8str_view_t f;
    while (n < 8 && proven_u8str_view_split_next(&it, &f)) {
        fields[n++] = proven_u8str_view_trim(f); /* ' ', \t, \n, \v, \f, \r - nothing else */
    }
    EXAMPLE_REQUIRE(n == 6, "five separators give six fields");
    EXAMPLE_REQUIRE(is(fields[0], "report.tar.gz") && is(fields[3], "notes.md") && is(fields[4], "README"),
        "each field is trimmed on both ends");
    EXAMPLE_REQUIRE(fields[2].size == 0 && fields[5].size == 0, "the empty fields are kept, and tested by
size");

    /* One-sided trims, for text where one end is significant. */
    EXAMPLE_REQUIRE(is(proven_u8str_view_trim_start(PROVEN_LIT(" indented ")), "indented "), "trim_start");
    EXAMPLE_REQUIRE(is(proven_u8str_view_trim_end(PROVEN_LIT(" indented ")), " indented"), "trim_end");

    /* --- prefixes, suffixes, and the last dot ----- */

    proven_u8str_view_t name = fields[0];
    /* The extension is after the LAST dot: find_last, not find. */
    proven_size_t dot = proven_u8str_view_find_last(name, PROVEN_LIT("."));
```

```

EXAMPLE_REQUIRE(dot == 10, "the last dot in report.tar.gz is at 10");
EXAMPLE_REQUIRE(is(proven_u8str_view_slice(name, dot + 1, name.size), "gz"), "so the extension is gz");

/* remove_suffix leaves the view unchanged when the suffix is absent - no
 * error, so ask starts_with/ends_with first when you need to know. */
EXAMPLE_REQUIRE(is(proven_u8str_view_remove_suffix(name, PROVEN_LIT(".tar.gz")), "report"), "the double
suffix goes");
EXAMPLE_REQUIRE(is(proven_u8str_view_remove_suffix(name, PROVEN_LIT(".zip")), "report.tar.gz"),
    "an absent suffix changes nothing");
EXAMPLE_REQUIRE(is(proven_u8str_view_remove_prefix(PROVEN_LIT("# heading"), PROVEN_LIT("# ")), "heading"),
    "a known prefix goes");
EXAMPLE_REQUIRE(proven_u8str_view_contains(fields[1], PROVEN_LIT("draft")), "contains is find !=
NOT_FOUND");

/* --- sort the non-empty fields ----- */

proven_result_array_t ra = proven_array_create(alloc, 8, sizeof(proven_u8str_view_t),
alignof(proven_u8str_view_t));
EXAMPLE_REQUIRE(proven_is_ok(ra.err), "creating the array must succeed");
if (!proven_is_ok(ra.err)) return EXAMPLE_OK();
proven_array_t names = ra.value;
for (int i = 0; i < n; ++i) {
    if (fields[i].size > 0) (void)proven_array_push(&names, &fields[i]);
}
/* cmp_ptr is cmp shaped for a sort: it receives pointers to the views. Bytes
 * compare unsigned, and a prefix sorts first; "README" sorts before
 * lowercase names because 'R' is below 'd' in ASCII. */
proven_array_sort(&names, proven_u8str_view_cmp_ptr);
const proven_u8str_view_t *first = proven_array_get(&names, 0);
const proven_u8str_view_t *last = proven_array_get(&names, 3);
EXAMPLE_REQUIRE(first && is(*first, "README") && last && is(*last, "report.tar.gz"), "sorted bytewise");
EXAMPLE_REQUIRE(proven_u8str_view_cmp(PROVEN_LIT("app"), PROVEN_LIT("apple")) < 0,
    "cmp answers by sign - never compare it with -1");

/* --- well-formed is not "found" or "not empty" ----- */

/* A slice past the end is {NULL, 0}, which is well formed - so this
 * predicate cannot end a loop or tell "empty" from "past the end". It only
 * says the view is safe to read. */
proven_u8str_view_t past = proven_u8str_view_slice(name, 100, 5);
EXAMPLE_REQUIRE(proven_u8str_view_is_well_formed(past) && past.size == 0, "past the end: empty and well
formed");
EXAMPLE_REQUIRE(!proven_u8str_view_is_well_formed((proven_u8str_view_t){ NULL, 3 }), "only {NULL, n > 0}
is not");

printf("%d fields, %zu names sorted\n", n, (size_t)names.len);
proven_array_destroy(&names);
return EXAMPLE_OK();
}

```

## 2. U16 strings and views

### Why a second string type exists at all

UTF-8 is the right default and this library commits to it. `proven_u16str_t` exists for one reason: **the Windows API is UTF-16**. Every "wide" entry point - `CreateFileW`, `GetEnvironmentVariableW`, the whole `W` family - takes `wchar_t *`, which on Windows is 16 bits. A library that talks to those APIs needs a type that holds their code units without pretending they are bytes.

So this is a boundary type. You use it where you touch a UTF-16 API, and you use `proven_u8str_t` everywhere else. It is deliberately small - create, destroy, append, and length - because it is not meant to be the type your program thinks in.

Two things to hold on to, because they are the source of every UTF-16 bug:

- **A code unit is not a character.** UTF-16 encodes anything outside the Basic Multilingual Plane as a *surrogate pair* - two `proven_u16` values that mean one character. An emoji is two code units. Slicing between them produces an unpaired surrogate, which is not valid UTF-16.
- **size here counts code units, not bytes.** `proven_u16str_view_t.size` is a count of `proven_u16` values; the `proven_buf_t` underneath tracks bytes, which is why `proven_u16str_len` divides. Mixing the two units is the most common mistake with this type.

You build a `proven_u16str_t` from a `u"..."` literal, from code units you already have, or - the usual case - by converting the UTF-8 your program holds, with `utf.h` (next subsection).

U16 APIs are excluded when `PROVEN_NO_U16STR` is defined, which is the default in freestanding builds - a bare-metal target has no Windows API to talk to.

Wrong - assuming one code unit is one character:

```
proven_u16str_view_t v = ...;           /* text containing an emoji */
proven_size_t chars = proven_u16str_len(&s); /* wrong: that is code units */
```

## Structures

```
typedef struct {
    const proven_u16 *ptr;
    proven_size_t size;
} proven_u16str_view_t;
```

```
typedef struct {
    proven_buf_t internal;
} proven_u16str_t;
```

```
typedef struct {
    proven_err_t err;
    proven_u16str_t value;
} proven_result_u16str_t;
```

## Macro

| Macro                          | Intent                                                                                          |
|--------------------------------|-------------------------------------------------------------------------------------------------|
| <code>PROVEN_U16_LIT(s)</code> | Make a <code>proven_u16str_view_t</code> from a UTF-16 literal expression <code>u"..."</code> . |

## U16 functions

| API                                                      | Intent                                           | Return                                |
|----------------------------------------------------------|--------------------------------------------------|---------------------------------------|
| <code>proven_u16str_create(alloc, unit_limit)</code>     | Create empty owned U16 string.                   | <code>proven_result_u16str_t</code> . |
| <code>proven_u16str_create_from_view(alloc, view)</code> | Copy U16 view into owned string.                 | <code>proven_result_u16str_t</code> . |
| <code>proven_u16str_destroy(alloc, str)</code>           | Free owned U16 storage.                          | <code>void</code> .                   |
| <code>proven_u16str_append(str, data)</code>             | Atomic fixed-capacity append.                    | <code>proven_err_t</code> .           |
| <code>proven_u16str_append_partial(str, data)</code>     | Truncating append.                               | <code>proven_result_size_t</code> .   |
| <code>proven_u16str_append_grow(alloc, str, data)</code> | Atomic growable append.                          | <code>proven_err_t</code> .           |
| <code>proven_u16str_as_ptr(str)</code>                   | Return internal <code>proven_u16</code> pointer. | <code>const proven_u16 *</code> .     |
| <code>proven_u16str_len(str)</code>                      | Return length in code units.                     | <code>proven_size_t</code> .          |

Example:

```

#ifdef PROVEN_NO_U16STR
proven_result_u16str_t r =
    proven_u16str_create_from_view(alloc, PROVEN_U16_LIT("hello"));
if (!proven_is_ok(r.err)) {
    return;
}
proven_u16str_t s = r.value;

(void)proven_u16str_append_grow(alloc, &s, PROVEN_U16_LIT(" world"));

/* Length is in code units, not characters: "hello world" is 11 units. */
proven_size_t units = proven_u16str_len(&s);
(void)units;

proven_u16str_destroy(alloc, &s);
#endif

```

Note: proven\_u16 is a code unit, not necessarily one full Unicode character. UTF-16 surrogate pairs use two code units.

### Converting between UTF-8 and UTF-16 (utf.h)

The program holds UTF-8; the Windows wide API wants UTF-16; the names and text that API hands back are UTF-16 and the rest of the program wants them as UTF-8. Every program that crosses that line needs a converter, and a converter has to decide what to do with text that is not valid: a stray continuation byte, an overlong encoding, an encoded surrogate, a lone surrogate on the UTF-16 side. The tempting answer is to repair it - swap in U+FFFD and carry on - and it is the wrong one for a library that hands file names to the operating system: a repaired name is a different name, and the program opens a file it was never asked to open.

**So utf.h is strict everywhere.** Malformed input is PROVEN\_ERR\_INVALID\_ENCODING, and the forms that write into your memory write nothing when they refuse. There is no lossy mode.

**Malformed and incomplete are different answers.** Text read in pieces - from a pipe, a buffered writer, a console - can end in the middle of a character that the next piece completes. The partial forms stop there with PROVEN\_ERR\_NEED\_MORE and report how much input they used, so you keep the tail and try again with more. The whole-text forms have no "more" to wait for, so for them a text that ends mid-character is malformed.

The three shapes are the ones the rest of the library uses: measure then convert into a buffer you own, all or nothing; grow an owned string, all or nothing; and a partial form that converts whole characters until the input, the output or the valid text runs out. A partial conversion never writes half a surrogate pair or half a UTF-8 sequence.

| API                                                  | Intent                                                           | Return               |
|------------------------------------------------------|------------------------------------------------------------------|----------------------|
| proven_utf8_to_utf16_size(src)                       | Code units src converts to; validates.                           | proven_result_size_t |
| proven_utf16_to_utf8_size(src, n)                    | UTF-8 bytes n code units convert to; validates.                  | proven_result_size_t |
| proven_utf8_to_utf16(src, out, out_cap, &written)    | All of src into out, or nothing. No terminator.                  | proven_err_t         |
| proven_utf16_to_utf8(src, n, out, out_cap, &written) | The same, the other way.                                         | proven_err_t         |
| proven_utf8_to_utf16_partial(src, out, out_cap)      | Whole characters until something stops it.                       | proven_utf_step_t    |
| proven_utf16_to_utf8_partial(src, n, out, out_cap)   | The same, the other way; a trailing high surrogate is NEED_MORE. | proven_utf_step_t    |

|                                                  |                                                                            |                    |
|--------------------------------------------------|----------------------------------------------------------------------------|--------------------|
| proven_utf8_append_to_u16str(alloc, dst, src)    | Append the UTF-16 of src, growing dst; all or nothing.                     | proven_err_t       |
| proven_utf16_append_to_u8str(alloc, dst, src, n) | Append the UTF-8 of n code units, growing dst; all or nothing.             | proven_err_t       |
| proven_utf8_decode_next(s, pos)                  | The one character at byte pos, or why not; len is always the step to take. | proven_utf8_char_t |

```
typedef struct {
    proven_err_t err;          /* OK, OUT_OF_BOUNDS (output full), NEED_MORE (input ends
                               mid-character), INVALID_ENCODING (malformed at `consumed`) */
    proven_size_t consumed;    /* input units used: bytes for UTF-8, code units for UTF-16 */
    proven_size_t written;     /* output units written */
} proven_utf_step_t;

typedef enum {
    PROVEN_TEXT_UTF8, PROVEN_TEXT_UTF16LE, PROVEN_TEXT_UTF16BE,
    PROVEN_TEXT_AUTO        /* readers only: decided by a byte order mark */
} proven_text_encoding_t;    /* for the u16 writers and readers in stream.h (chapter 5) */

typedef struct {
    proven_err_t err;          /* OK, INVALID_ENCODING, NEED_MORE, EOF, OUT_OF_BOUNDS, INVALID_ARG */
    proven_u32 cp;             /* the scalar value when err is OK; 0 otherwise */
    proven_size_t len;         /* bytes this step covers: the character, the maximal subpart of
                               malformed input, or the bytes left when the text ends mid-character */
} proven_utf8_char_t;
```

Wrong - widening each byte into a code unit. It is right for ASCII, which is why it survives testing, and it turns every other character into two or three meaningless units. The library's own `proven_time_u16_fmt` did exactly this with a caller's non-ASCII day names until `utf.h` existed:

```
for (size_t i = 0; i < text.size; ++i) {
    proven_u16 unit = (proven_u16)text.ptr[i];          /* wrong: a byte is not a code unit */
    (void)proven_u16str_append_grow(alloc, &wide, (proven_u16str_view_t){ &unit, 1 });
}
```

Wrong - converting pieces with the whole-text form. The piece that ends inside a character is refused as malformed, although the text is fine; use the partial form and carry the tail:

```
proven_utf8_to_utf16(piece, out, cap, &n);    /* wrong for pieces: a split character is INVALID */
```

**One character at a time.** A segmenter, a shaper or a lexer walks text a character at a time, and wants a malformed byte reported where it is rather than the whole text refused.

`proven_utf8_decode_next` decodes the one character at a byte offset under the same strict rules, and its `len` is always the step to take: the character's length, or for malformed input the *maximal subpart* - the longest prefix that was still a valid start - which is where Unicode says to resynchronise. Nothing is substituted. A program that draws text puts one U+FFFD per malformed step, which is Unicode's recommended practice; one that validates stops at the first.

```
proven_u8str_view_t text = PROVEN_LIT("a\xC3\xA9\xFF" "z\xE2\x82");
proven_size_t chars = 0, malformed = 0;
for (proven_size_t pos = 0; pos < text.size;) {
    proven_utf8_char_t c = proven_utf8_decode_next(text, pos);
    if (c.err == PROVEN_OK) ++chars;          /* c.cp: 'a', U+00E9, 'z' */
    else if (c.err == PROVEN_ERR_INVALID_ENCODING) ++malformed; /* the FF: one step */
    /* PROVEN_ERR_NEED_MORE: E2 82 start a character this text never finished */
    pos += c.len;                             /* at least 1 inside the text */
}
(void)chars; (void)malformed;
```

The worked example converts a Korean file name for a wide API and back, shows the refusal that writes nothing, grows owned strings both ways, and carries a character split between two pieces:

```
#include <string.h>

/*
 * The program thinks in UTF-8; a Windows "wide" call wants UTF-16. utf.h is the
 * crossing, in both directions, and it is strict: text that is not valid UTF-8
 * or UTF-16 is refused, never repaired. A converter that quietly swaps a bad
 * byte for a question mark has changed a file name you were about to open.
 *
 * Three shapes, as everywhere in this library: measure-then-convert into a
 * buffer you own (all or nothing), grow an owned string (all or nothing), and a
 * partial form for text that arrives in pieces.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* A Korean file name ("report.txt": three Hangul syllables and ".txt"), as the rest
     * of the program holds it: UTF-8. */
    proven_u8str_view_t name = PROVEN_LIT("\xEB\xB3\xB4\xEA\xB3\xA0\xEC\x84\x9C.txt");

    /* --- measure, then convert into your own buffer ----- */

    /* Ask first: the answer is in code units, and it also validates. Nine bytes
     * of Hangul are three code units here, not nine. */
    proven_result_size_t need = proven_utf8_to_utf16_size(name);
    EXAMPLE_REQUIRE(proven_is_ok(need.err) && need.value == 7, "3 syllables + \".txt\" are 7 code units");

    /* One more unit for the NUL a wide API expects; the converter writes none. */
    proven_u16 wide[16];
    proven_size_t units = 0;
    proven_err_t err = proven_utf8_to_utf16(name, wide, 15, &units);
    EXAMPLE_REQUIRE(proven_is_ok(err) && units == 7, "the name converts whole");
    wide[units] = 0;
    EXAMPLE_REQUIRE(wide[0] == 0xBCF4, "the first code unit is the syllable, not its first byte");
    /* On Windows: CreateFileW((LPCWSTR)wide, ...). Not called here, so this runs everywhere. */

    /* And back: a name a wide API returned, as UTF-8 for everything else. */
    proven_result_size_t back = proven_utf16_to_utf8_size(wide, units);
    EXAMPLE_REQUIRE(proven_is_ok(back.err) && back.value == name.size, "the reverse size is the original byte
count");
    proven_byte_t bytes[32];
    proven_size_t nbytes = 0;
    err = proven_utf16_to_utf8(wide, units, bytes, sizeof bytes, &nbytes);
    EXAMPLE_REQUIRE(proven_is_ok(err) && nbytes == name.size && memcmp(bytes, name.ptr, nbytes) == 0,
        "the round trip gives the same bytes");

    /* --- strict: malformed text is refused, and nothing is written ----- */

    /* A lone continuation byte, as a truncated or mis-decoded name would have. */
    proven_u8str_view_t broken = PROVEN_LIT("bad\x80.txt");
    wide[0] = 0x1234;
    err = proven_utf8_to_utf16(broken, wide, 15, &units);
    EXAMPLE_REQUIRE(err == PROVEN_ERR_INVALID_ENCODING, "a stray continuation byte is refused");
    EXAMPLE_REQUIRE(units == 0 && wide[0] == 0x1234, "and the output is untouched");

    /* --- grow an owned string ----- */
}
```

```

proven_result_u16str_t r = proven_u16str_create(alloc, 4);
EXAMPLE_REQUIRE(proven_is_ok(r.err), "creating the wide path must succeed");
proven_u16str_t path = r.value;
err = proven_utf8_append_to_u16str(alloc, &path, PROVEN_LIT("C:\\reports\\"));
EXAMPLE_REQUIRE(proven_is_ok(err), "the directory appends");
err = proven_utf8_append_to_u16str(alloc, &path, name);
EXAMPLE_REQUIRE(proven_is_ok(err) && proven_u16str_len(&path) == 11 + 7, "and the name, growing the
string");
EXAMPLE_REQUIRE(proven_u16str_as_ptr(&path)[18] == 0, "still NUL-terminated for the system call");

/* A failed append changes nothing: no half-converted tail. */
err = proven_utf8_append_to_u16str(alloc, &path, broken);
EXAMPLE_REQUIRE(err == PROVEN_ERR_INVALID_ENCODING && proven_u16str_len(&path) == 18, "a refused append
leaves the path as it was");

proven_result_u8str_t r8 = proven_u8str_create(alloc, 8);
EXAMPLE_REQUIRE(proven_is_ok(r8.err), "creating the UTF-8 copy must succeed");
proven_u8str_t copy = r8.value;
err = proven_utf16_append_to_u8str(alloc, &copy, proven_u16str_as_ptr(&path), proven_u16str_len(&path));
EXAMPLE_REQUIRE(proven_is_ok(err) && copy.internal.len == 11 + name.size, "the whole path comes back as
UTF-8");

/* --- partial: text that arrives in pieces ----- */

/* A reader hands over 4 bytes at a time, and the second syllable is split after its first
 * byte. The partial form converts what is whole, stops with NEED_MORE, and
 * says how much it used; the caller keeps the rest and adds the next piece. */
proven_u16 out[8];
proven_utf_step_t st = proven_utf8_to_utf16_partial((proven_u8str_view_t){ name.ptr, 4 }, out, 8);
EXAMPLE_REQUIRE(st.err == PROVEN_ERR_NEED_MORE && st.consumed == 3 && st.written == 1,
    "one whole syllable, and the start of the next kept back");
st = proven_utf8_to_utf16_partial((proven_u8str_view_t){ name.ptr + st.consumed, 6 }, out + 1, 7);
EXAMPLE_REQUIRE(proven_is_ok(st.err) && st.written == 2 && out[1] == 0xACE0, "with the next piece it
completes");

/* The same distinction in the other direction: a high surrogate at the end
 * may be half of a pair whose other half is in the next piece. */
const proven_u16 emoji_half[] = { 'o', 'k', 0xD83D };
proven_byte_t u8out[16];
st = proven_utf16_to_utf8_partial(emoji_half, 3, u8out, sizeof u8out);
EXAMPLE_REQUIRE(st.err == PROVEN_ERR_NEED_MORE && st.consumed == 2, "a trailing high surrogate waits");

printf("converted a %zu-byte name to %zu code units and back\n", (size_t)name.size, (size_t)need.value);

proven_u8str_destroy(alloc, &copy);
proven_u16str_destroy(alloc, &path);
return EXAMPLE_OK();
}

```

### 3. Formatting

#### The problem: printf is told the types twice

printf("%d", x) states the type of x twice - once in the format string and once by passing x - and nothing checks that the two agree. Varargs erases the type, so the function reads whatever bytes the calling convention left, in whatever shape the format demanded:

```

printf("%d\n", 3.0);    /* wrong: reads a double's bytes as an int */
printf("%s\n", 42);     /* wrong: dereferences 42 as a pointer */
printf("%d %d\n", 1);   /* wrong: reads an argument that was never passed */

```

All three compile. Modern compilers warn when the format is a literal, which helps until the format is a variable - and then you have a function that will read arbitrary stack memory on demand, which is a class of vulnerability with its own name.

The second problem is where the output goes. `sprintf` writes to a buffer whose size it does not know. `snprintf` takes a size and then **truncates**, returning the length it *would* have written - so the caller who forgets to compare gets a silently shortened path, command, or identifier.

### What this library does instead

**The placeholder has no type in it.** `{}` says "a value goes here"; the type comes from the argument, resolved at compile time:

```
proven_println("{} scored {}", PROVEN_ARG(name), PROVEN_ARG(score));
```

`PROVEN_ARG` is a `_Generic` dispatch - the compiler picks the right constructor for the argument's static type. A mismatch between the format and the argument is not possible, because the format never states a type. What the spec after `:` controls is *presentation* - width, fill, alignment, precision, base - never interpretation.

**The destination is a sized object**, and the fixed-capacity form refuses rather than truncates:

`proven_u8str_append_fmt` fails with `PROVEN_ERR_OUT_OF_BOUNDS` and writes nothing, while `proven_u8str_append_fmt_grow` takes an allocator and grows. Which one you called is visible in the call itself.

**Your own types can join in.** `PROVEN_ARG_OF(&obj, render_fn)` lets a type you defined print with `{}` like everything else - the extension point is compile-time and typed, not a registry of names. [Chapter 8 section 5.1](#) shows how.

The cost, stated plainly: `PROVEN_ARG` around each argument is more typing than `%d`, and the format language is not the one in your fingers. What you buy is that the class of bug at the top of this section cannot be written.

The formatter writes into `proven_u8str_t` or PAL-backed streams. It uses a small structural format language with `{}` placeholders, explicit indexes like `{1}`, escaped braces `{{` and `}}`, and width/alignment specs such as `{:0>5}`, `{:*^10}`, and `{:.<10}`.

Wrong - assuming the spec chooses the type:

```
proven_println("{:d}", PROVEN_ARG(3.5)); /* wrong: the spec formats, it does not convert */
```

### Structures and enums

```
typedef struct {
    proven_err_t err;
    proven_size_t written;
    proven_size_t required;
} proven_fmt_result_t;
```

Fields:

- `err`: result code.
- `written`: bytes actually written.
- `required`: bytes required for full output.

`proven_arg_type_t` variants:

- `PROVEN_ARG_NONE`
- `PROVEN_ARG_I32`
- `PROVEN_ARG_U32`
- `PROVEN_ARG_I64`
- `PROVEN_ARG_U64`
- `PROVEN_ARG_F64` unless `PROVEN_FMT_NO_FLOAT` is defined

- PROVEN\_ARG\_CSTR
- PROVEN\_ARG\_STR\_VIEW
- PROVEN\_ARG\_DATETIME
- PROVEN\_ARG\_PTR
- PROVEN\_ARG\_FN

```
typedef struct {
    proven_arg_type_t type;
    union {
        proven_i32 i32;
        proven_u32 u32;
        proven_i64 i64;
        proven_u64 u64;
        double f64;
        const char *cstr;
        proven_u8str_view_t str_view;
        proven_datetime_t datetime;
        const void *ptr;
        void (*fn)(void);
    } value;
} proven_arg_t;
```

### Format argument constructors

| API                                               | Intent                                                                                                                                                                                                          |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_arg_none()</code>                    | Internal sentinel value.                                                                                                                                                                                        |
| <code>proven_arg_i32(v), proven_arg_u32(v)</code> | Integer arguments.                                                                                                                                                                                              |
| <code>proven_arg_i64(v), proven_arg_u64(v)</code> | Wide integer arguments.                                                                                                                                                                                         |
| <code>proven_arg_f64(v)</code>                    | Floating-point argument unless float formatting is disabled.                                                                                                                                                    |
| <code>proven_arg_cstr(v)</code>                   | Trusted live NUL-terminated C string.                                                                                                                                                                           |
| <code>proven_arg_cstr_n(v, max_len)</code>        | Bounded C-string argument; scans for NUL only up to <code>max_len</code> .                                                                                                                                      |
| <code>proven_arg_str_view(v)</code>               | Borrowed string view argument.                                                                                                                                                                                  |
| <code>proven_arg_u16(v)</code>                    | Borrowed UTF-16 view, rendered as UTF-8. Strict: an unpaired surrogate fails the format. Width counts UTF-8 bytes, as for a u8 view. <code>PROVEN_ARG</code> picks it for a <code>proven_u16str_view_t</code> . |
| <code>proven_arg_datetime(v)</code>               | Datetime argument.                                                                                                                                                                                              |
| <code>proven_arg_ptr(v)</code>                    | Object pointer argument.                                                                                                                                                                                        |
| <code>proven_arg_fn(v)</code>                     | Function pointer argument.                                                                                                                                                                                      |
| <code>proven_arg_ucstr(v)</code>                  | Unsigned-char C string helper.                                                                                                                                                                                  |
| <code>proven_arg_identity(v)</code>               | Pass-through for existing <code>proven_arg_t</code> .                                                                                                                                                           |

### Format macros

| Macro                                                           | Intent                                                       |
|-----------------------------------------------------------------|--------------------------------------------------------------|
| <code>PROVEN_ARG(x)</code>                                      | <code>_Generic</code> selector for supported argument types. |
| <code>PROVEN_ARG_FN(f)</code>                                   | Function pointer formatting helper.                          |
| <code>PROVEN_ARG_CSTR_N(v, max_len)</code>                      | Bounded C-string helper.                                     |
| <code>proven_u8str_append_fmt(str, fmt, ...)</code>             | Atomic fixed-capacity formatting.                            |
| <code>proven_u8str_append_fmt_trunc(str, fmt, ...)</code>       | Best-effort truncating formatting.                           |
| <code>proven_u8str_append_fmt_grow(alloc, str, fmt, ...)</code> | Atomic growable formatting.                                  |

|                                                                     |                                                                             |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------|
| proven_u8str_append_fmt_with_scratch(alloc, str, fmt, scratch, ...) | Growable formatting with scratch allocator for temporary patch allocations. |
| PROVEN_FMT_IS_OK(res)                                               | Check proven_fmt_result_t.                                                  |

## Formatting engine

```
proven_fmt_result_t proven_u8str_fmt_internal(
    proven_allocator_t alloc,
    proven_u8str_t *str,
    bool trunc,
    const char *fmt,
    proven_allocator_t scratch,
    const proven_arg_t *args,
    proven_size_t args_count
);
```

Normal user code should call the macros instead of this internal engine. The engine expects a leading `proven_arg_none()` sentinel at index 0.

Extra unused format arguments return `PROVEN_ERR_INVALID_ARG`.

Example:

```
proven_result_u8str_t r = proven_u8str_create(alloc, 8);
if (!proven_is_ok(r.err)) {
    return;
}
proven_u8str_t s = r.value;

/* The target is only 8 bytes; the _grow form reallocates rather than truncate. */
proven_fmt_result_t fr = proven_u8str_append_fmt_grow(
    alloc,
    &s,
    "name={ } score={:0>4}",
    PROVEN_ARG(PROVEN_LIT("ada")),
    PROVEN_ARG(42)
);
if (!PROVEN_FMT_IS_OK(fr)) {
    proven_u8str_destroy(alloc, &s);
    return;
}
/* s == "name=ada score=0042" */

proven_u8str_destroy(alloc, &s);
```

## 4. Scanning

### The problem: scanf will not tell you where it stopped

Parsing input is the mirror of formatting, and libc's answer is worse. `sscanf` returns *how many fields it filled* and nothing else - not which one failed, not how far it got, not why:

```
int n = sscanf(line, "%d %d %d", &a, &b, &c);
if (n != 3) { /* wrong: which field? at what offset? was it malformed or missing? */ }
```

If the third field is malformed, you know only that you got two. You cannot report the position to the user, cannot skip the bad record and resume, and cannot tell "ran out of input" from "found something that is not a number". And `%s` into a `char *` has the same unbounded-write problem as `strcpy`, with the input now coming from outside your program.

Then there is `strtol`, whose contract requires you to clear `errno` first, check it after, and compare `endptr` against the input to detect "no digits at all" - three separate things to get right for one conversion.

### What this library does instead

**The cursor is yours.** A `proven_scan_t` holds the view being parsed and an offset into it. Each scan reads from the cursor and moves it forward on success. Because the cursor is a field you can read, you always know exactly where parsing stopped - which is the position you show the user.

**Each scan returns a result.** `proven_scan_i64` hands back `{err, val}`, so "not a number", "out of range" and "end of input" are different errors rather than one missing field.

**Failure restores the cursor** for the primitive scanners: a failed `proven_scan_i64` leaves the cursor where it was, so you can try something else at the same position. That is what makes recovery possible instead of guesswork.

One thing to know before you rely on it: the *structural* scan (`proven_scan_fmt`, the `{}` form) is **not** transactional across fields. If the third placeholder fails, the first two destinations have already been written. [Chapter 8 section 11.1](#) covers the error codes and the recovery patterns in full.

The scanner parses from a borrowed `proven_u8str_view_t`. A cursor tracks progress.

Wrong - treating a partial structural scan as if nothing happened:

```
proven_err_t e = proven_scan_fmt(&sc, "{} {} {}", ...);
if (!proven_is_ok(e)) {
    /* wrong: destinations for the fields that DID parse have already been written */
}
```

### Result structs

```
typedef struct { proven_err_t err; proven_i64 val; } proven_result_i64_t;
typedef struct { proven_err_t err; proven_u64 val; } proven_result_u64_t;
typedef struct { proven_err_t err; double val; } proven_result_f64_t;
typedef struct { proven_err_t err; proven_u8str_view_t val; } proven_result_u8str_view_t;
```

```
proven_scan_t
typedef struct {
    proven_u8str_view_t view;
    proven_size_t cursor;
} proven_scan_t;
```

Purpose: hold input and current parse position. `proven_scan_init()` normalizes invalid non-empty null views to empty views.

### Scanner functions

| API                                            | Intent                                                               | Return                                      |
|------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------|
| <code>proven_scan_init(view)</code>            | Create scanner from view.                                            | <code>proven_scan_t</code> .                |
| <code>proven_scan_skip_whitespace(scan)</code> | Advance past whitespace.                                             | <code>void</code> .                         |
| <code>proven_scan_i64(scan)</code>             | Parse signed 64-bit integer.                                         | <code>proven_result_i64_t</code> .          |
| <code>proven_scan_u64(scan)</code>             | Parse unsigned 64-bit integer.                                       | <code>proven_result_u64_t</code> .          |
| <code>proven_scan_f64(scan)</code>             | Parse floating-point value.                                          | <code>proven_result_f64_t</code> .          |
| <code>proven_parse_double_ascii(view)</code>   | Parse one locale-free ASCII float token without skipping whitespace. | <code>proven_parse_double_result_t</code> . |
| <code>proven_parse_f64_ascii(view)</code>      | Compatibility alias for the same locale-free binary64 parser.        | <code>proven_parse_f64_result_t</code> .    |

|                                      |                                                                             |                             |
|--------------------------------------|-----------------------------------------------------------------------------|-----------------------------|
| proven_strtod(nptr, endptr)          | Parse one strtod-style token with whitespace skipping and endptr reporting. | double.                     |
| proven_scan_str(scan)                | Parse a whitespace-delimited token as view into input.                      | proven_result_u8str_view_t. |
| proven_scan_skip_until(scan, target) | Move cursor to target if found.                                             | proven_err_t.               |
| proven_scan_skip_until_number(scan)  | Move cursor to next number-looking position.                                | void.                       |

## Scan argument types

proven\_scan\_arg\_type\_t identifies the destination kind stored in a proven\_scan\_arg\_t. PROVEN\_SCAN\_ARG(&x) supports pointers to:

- short, unsigned short
- int, unsigned int
- long, unsigned long
- long long, unsigned long long
- double
- proven\_u8str\_view\_t

Native destination constructors:

- proven\_scan\_arg\_short, proven\_scan\_arg\_ushort
- proven\_scan\_arg\_int, proven\_scan\_arg\_uint
- proven\_scan\_arg\_long, proven\_scan\_arg\_ulong
- proven\_scan\_arg\_llong, proven\_scan\_arg\_ullong

Explicit fixed-width and utility helpers:

- proven\_scan\_arg\_i32, proven\_scan\_arg\_u32
- proven\_scan\_arg\_i64, proven\_scan\_arg\_u64
- proven\_scan\_arg\_f64
- proven\_scan\_arg\_str\_view
- proven\_scan\_arg\_none
- proven\_scan\_arg\_identity

Long aliases:

```
#define PROVEN_SCAN_ARG_LONG(ptr) proven_scan_arg_long(ptr)
#define PROVEN_SCAN_ARG_ULONG(ptr) proven_scan_arg_ulong(ptr)
```

## Format scanning macros

| Macro                                      | Intent                                    |
|--------------------------------------------|-------------------------------------------|
| PROVEN_SCAN_ARG(x)                         | _Generic destination selector.            |
| proven_scan_fmt_cursor(scan_ptr, fmt, ...) | Scan from an existing cursor.             |
| proven_scan_fmt(view, fmt, ...)            | Scan from a view with a temporary cursor. |

## Scan engine

```
proven_err_t proven_scan_fmt_internal(
    proven_scan_t *scan,
    const char *fmt,
    const proven_scan_arg_t *args,
    proven_size_t args_count
);
```

| API                                                                | Intent                                                            | Return                     |
|--------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------|
| <code>proven_scan_fmt_internal(scan, fmt, args, args_count)</code> | Structural scanner engine over an existing cursor.                | <code>proven_err_t.</code> |
| <code>proven_scan_fmt_internal_view(view, fmt, args, count)</code> | Convenience wrapper that creates a temporary scanner over a view. | <code>proven_err_t.</code> |

Normal user code should call the macros. The engine expects a leading sentinel argument.

Important behavior: `proven_scan_fmt_internal()` may advance the cursor and write earlier destinations before returning an error if a later literal mismatch occurs. Save the cursor and destination values first if you need transaction-like parsing.

Example:

```
proven_scan_t scan = proven_scan_init(PROVEN_LIT("ID: 402 SCORE: 99.5 ada"));
int id = 0;
double score = 0.0;
proven_u8str_view_t user = {0};

proven_err_t e = proven_scan_fmt_cursor(
    &scan,
    "ID: {} SCORE: {} {}",
    PROVEN_SCAN_ARG(&id),
    PROVEN_SCAN_ARG(&score),
    PROVEN_SCAN_ARG(&user)
);
if (!proven_is_ok(e)) {
    return;
}
/* id == 402, score == 99.5, user borrows "ada" out of the input - it is not a
 * copy, so it is only valid while the scanned bytes are. */
```

## Float parsing notes

- `proven_scan_fmt64()` and `proven_parse_double_ascii()` route through the shared decimal-to-binary64 backend.
- Finite decimal inputs are rounded to IEEE-754 binary64 with round-to-nearest, ties-to-even behavior.
- The current conversion stack is Clinger fast path → direct Eisel-Lemire → staged Eisel-Lemire layer → exact bigint fallback, with internal counters used by tests to confirm which path took a representative input.
- The direct Eisel-Lemire layer multiplies the significand by a truncated 128-bit power of five and proves the rounding from the product's low bits; it declines anything it cannot prove (a possible tie, a subnormal or overflowing result). A significand longer than 19 digits is bounded by its first 19 digits and the same plus one, and taken when both bounds round alike.
- The staged Eisel-Lemire layer currently accepts generated- $5^q$  positive exponent cases and exact negative-exponent cases where the decimal significand cleanly cancels the required  $5^q$ .
- On every target (it needs no 128-bit integer type), the same layer also accepts a conservative rounded negative-exponent ratio subset in normal-range cases, including wider left-shift normalization than the original narrow prototype allowed.
- The widened cached-power product path now also stages some subnormal cases, including  $5e-324$ , while values below the half-threshold to true-min still defer to the exact bigint fallback.
- The checked-in cached  $5^q$  constants are generated locally by `scripts/generate_float_decimal_tables.py`.
- `proven_parse_double_ascii()` does not skip leading whitespace.

- `proven_strtod()` skips leading ASCII whitespace, updates `endptr`, returns signed infinity on overflow, and preserves signed zero on underflow.

## 5. Examples and misuse cases

### Views are not C strings

Wrong:

```
proven_u8str_view_t view = get_view();
printf("%s\n", (const char *)view.ptr); /* wrong: view may not be NUL-terminated */
```

Correct:

```
proven_u8str_view_t view = proven_u8str_view_slice(PROVEN_LIT("/etc/hosts"), 5, 5);

/* A view is a pointer and a length into somebody else's bytes. To hand it to a
 * C API that wants a NUL, allocate a real C string from it. */
proven_result_cstr_t c = proven_u8str_view_to_cstr(view, alloc);
if (!proven_is_ok(c.err)) {
    return;
}
/* c.value is "hosts", NUL-terminated. It is yours, so free it. */
alloc.free_fn(alloc.ctx, (void *)c.value);
```

### PROVEN\_LIT is for literals

Correct:

```
proven_u8str_view_t a = PROVEN_LIT("abc");
(void)a;
```

Wrong:

```
const char *runtime = getenv("NAME");
proven_u8str_view_t a = PROVEN_LIT(runtime); /* wrong: macro requires literal syntax */
```

Use:

```
const char *runtime = "NAME=value"; /* any trusted NUL-terminated string */
proven_u8str_view_t a = proven_u8str_view_from_cstr(runtime);
(void)a;
```

### Distinguish not found from replaced

`proven_u8str_replace_first()` returns `PROVEN_OK` when the target is not found. Search first if that matters.

```
proven_result_u8str_t r = proven_u8str_create_from_view(alloc, PROVEN_LIT("the old way"));
if (!proven_is_ok(r.err)) {
    return;
}
proven_u8str_t s = r.value;

proven_size_t at = proven_u8str_view_find(proven_u8str_as_view(&s), 0, PROVEN_LIT("old"));
if (at != PROVEN_INDEX_NOT_FOUND) {
    (void)proven_u8str_replace_first(&s, 0, PROVEN_LIT("old"), PROVEN_LIT("new"));
}

proven_u8str_destroy(alloc, &s);
```

### Bounded format input

Wrong:

```
char *untrusted = get_untrusted_pointer();
proven_println("{", PROVEN_ARG(untrusted));
/* wrong: C-string formatting scans until NUL */
```

Correct:

```
char untrusted[16] = { 'n', 'o', ' ', 'n', 'u', 'l', ' ', 'h', 'e', 'r', 'e', '!', '!', '!', '!', '!' };
proven_size_t max_len = sizeof untrusted;

/* The bounded form stops looking for a NUL after max_len bytes. */
proven_println("{", PROVEN_ARG_CSTR_N(untrusted, max_len));
```

## Borrowed fixed-capacity strings

Use `proven_u8str_borrow` to format into a stack or static buffer without any allocation - useful in allocator-free code and on hot paths. Use the fixed-capacity operations (and `proven_u8str_append_fmt`), reuse with `proven_u8str_reset`, and do not call growing operations or `proven_u8str_destroy` on it (the caller owns the memory).

```
int cur = 3;
int total = 10;

proven_byte_t line[64];
proven_u8str_t s = proven_u8str_borrow(line, sizeof line); /* cap includes NUL */

proven_fmt_result_t fr = proven_u8str_append_fmt(&s, "L{}/{}", PROVEN_ARG(cur), PROVEN_ARG(total));
if (PROVEN_FMT_IS_OK(fr)) {
    proven_println("{", PROVEN_ARG(proven_u8str_as_view(&s)));
}

(void)proven_u8str_reset(&s); /* reuse next frame, no allocation */
```

Misuse: a growing call that would exceed the borrowed capacity returns `PROVEN_ERR_OUT_OF_BOUNDS` rather than reallocating caller memory.

```
proven_byte_t small[4];
proven_u8str_t t = proven_u8str_borrow(small, sizeof small);
proven_err_t e = proven_u8str_append_grow(&t, PROVEN_LIT("toolong"));
(void)e; /* e == PROVEN_ERR_OUT_OF_BOUNDS; small[] is untouched */
```

## Self-referential formatting

Wrong:

```
const char *inside = proven_u8str_as_cstr(&s);
proven_u8str_append_fmt_grow(&s, "{}", PROVEN_ARG(inside));
/* wrong: self-aliasing C-string arguments are rejected */
```

Correct:

```
proven_result_u8str_t r = proven_u8str_create_from_view(&s, PROVEN_LIT("ab"));
if (!proven_is_ok(r.err)) {
    return;
}
proven_u8str_t s = r.value;

/* A view carries a length, so the formatter knows exactly which bytes to snapshot. */
proven_u8str_view_t before = proven_u8str_as_view(&s);
proven_fmt_result_t fr = proven_u8str_append_fmt_grow(&s, "{}", PROVEN_ARG(before));
(void)fr; /* s == "abab" */

proven_u8str_destroy(&s);
```

## Transactional scanning

Wrong assumption:

```
proven_err_t e = proven_scan_fmt_cursor(&scan, "{} suffix", PROVEN_SCAN_ARG(&x));
/* if suffix mismatches, x and scan.cursor may already have changed */
```

Correct pattern:

```

proven_scan_t scan = proven_scan_init(PROVEN_LIT("42 prefix"));
int x = -1;

proven_size_t old_cursor = scan.cursor;
int old_x = x;

proven_err_t e = proven_scan_fmt_cursor(&scan, "{} suffix", PROVEN_SCAN_ARG(&x));
if (!proven_is_ok(e)) {
    /* The literal "suffix" did not match, but 42 was already written into x and
     * the cursor already moved. Put both back yourself. */
    scan.cursor = old_cursor;
    x = old_x;
}

```

### Worked example: owned strings and borrowed strings

Compiled and run by the test suite. The distinction that matters: an owned string may reallocate and must be destroyed; a borrowed one wraps caller memory, never reallocates, and refuses to grow past the buffer you gave it rather than quietly moving it.

```

/*
 * There are two string handles here and the difference is ownership, not size:
 *
 * proven_u8str_t      - a byte string you can edit. It either owns an
 *                      allocation (create) or borrows one of yours (borrow).
 * proven_u8str_view_t - a pointer and a length into somebody else's bytes.
 *                      It owns nothing, it is not NUL-terminated, and it is
 *                      only valid while those bytes are.
 *
 * A view is what you pass to a function that reads. A u8str is what you keep.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- an OWNED string: the allocator's memory, yours to destroy ----- */
    /* The capacity argument is content bytes; the NUL is extra, so as_cstr is
     * always 0(1) and always safe. */
    proven_result_u8str_t r = proven_u8str_create(alloc, 16);
    EXAMPLE_REQUIRE(proven_is_ok(r.err), "creating a 16-byte string must succeed");
    if (!proven_is_ok(r.err)) {
        return 1;
    }
    proven_u8str_t path = r.value;

    /* append is fixed-capacity: it fits or it fails, and on failure it has not
     * touched the string. It never reallocates, so it needs no allocator. */
    proven_err_t err = proven_u8str_append(&path, PROVEN_LIT("/etc/hosts"));
    EXAMPLE_REQUIRE(proven_is_ok(err), "10 bytes fit in a 16-byte string");

    /* append_grow is the growable twin: give it the allocator the string was
     * created with and it reallocates when needed. Still failure-atomic - if the
     * allocation fails, the string is exactly as it was. */
    err = proven_u8str_append_grow(alloc, &path, PROVEN_LIT(".backup.original"));
    EXAMPLE_REQUIRE(proven_is_ok(err), "append_grow must reallocate rather than fail");

    /* Edits in the middle. insert shifts the tail right; remove shifts it left. */
    err = proven_u8str_insert_grow(alloc, &path, 0, PROVEN_LIT("/srv"));
    EXAMPLE_REQUIRE(proven_is_ok(err), "inserting a prefix must succeed");

    err = proven_u8str_remove(&path, proven_u8str_as_view(&path).size - 9, 9); /* drop ".original" */
}

```

```

EXAMPLE_REQUIRE(proven_is_ok(err), "removing the trailing suffix must succeed");

/* replace_first returns PROVEN_OK when the target is absent - "nothing to do"
 * is not an error. Search first when the difference matters to you. */
err = proven_u8str_replace_first(&path, 0, PROVEN_LIT("hosts"), PROVEN_LIT("fstab"));
EXAMPLE_REQUIRE(proven_is_ok(err), "replacing an existing substring must succeed");

/* --- reading it: borrow a view, do not copy ----- */
/* as_view is free. The view is only good until the next edit: any growing
 * call may reallocate and leave the view (and any cstr) dangling. */
proven_u8str_view_t v = proven_u8str_as_view(&path);

EXAMPLE_REQUIRE(proven_u8str_view_eq(v, PROVEN_LIT("/srv/etc/fstab.backup")),
    "the edits above should have produced /srv/etc/fstab.backup");
EXAMPLE_REQUIRE(proven_u8str_view_starts_with(v, PROVEN_LIT("/srv")),
    "the inserted prefix is at the front");

proven_size_t dot = proven_u8str_view_find(v, 0, PROVEN_LIT(".backup"));
EXAMPLE_REQUIRE(dot != PROVEN_INDEX_NOT_FOUND, "the suffix must be found");

/* A slice is a view into the SAME bytes - no allocation, no copy. */
proven_u8str_view_t stem = proven_u8str_view_slice(v, 0, dot);
EXAMPLE_REQUIRE(proven_u8str_view_eq(stem, PROVEN_LIT("/srv/etc/fstab")),
    "slicing at the suffix leaves the stem");

/* as_cstr is the escape hatch to C APIs, and it is only valid because the
 * owned string keeps a NUL past its length. Do NOT do this with a view:
 * `stem.ptr` is not NUL-terminated - it just points into `path`. */
printf("owned: %s\n", proven_u8str_as_cstr(&path));

/* --- a BORROWED string: your memory, no allocation at all ----- */
/* Same type, same operations - but the bytes are this stack buffer. `cap`
 * includes the NUL, so this holds 31 content bytes. */
proven_byte_t line[32];
proven_u8str_t status = proven_u8str_borrow(line, sizeof line);

err = proven_u8str_append(&status, PROVEN_LIT("mounted "));
EXAMPLE_REQUIRE(proven_is_ok(err), "appending into a borrowed buffer needs no allocator");
err = proven_u8str_append(&status, stem);
EXAMPLE_REQUIRE(proven_is_ok(err), "a view can be appended just like a literal");

/* The growing calls exist for a borrowed string, but they refuse to
 * reallocate memory they do not own: too much data is OUT_OF_BOUNDS, and
 * `line` is left untouched. A borrowed string cannot silently escape to the
 * heap behind your back. */
err = proven_u8str_append_grow(alloc, &status,
    PROVEN_LIT(" ...and a great deal more text than fits"));
EXAMPLE_REQUIRE(err == PROVEN_ERR_OUT_OF_BOUNDS,
    "a borrowed string reports overflow instead of reallocating caller memory");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&status), PROVEN_LIT("mounted /srv/etc/fstab")),
    "the failed append must have left the string unchanged");

printf("borrowed: %s\n", proven_u8str_as_cstr(&status));

/* reset truncates to empty and keeps the buffer, so the next frame reuses
 * the same 32 bytes with no allocation. */
err = proven_u8str_reset(&status);
EXAMPLE_REQUIRE(proven_is_ok(err), "reset must succeed on a borrowed string");
EXAMPLE_REQUIRE(proven_u8str_as_view(&status).size == 0, "reset empties the string");

```

```

/* --- destroy: the ownership rule, spelled out ----- */
/* destroy on the borrowed string is a no-op - `line` is not the library's to
 * free. Calling it anyway is correct and costs nothing, and it means the
 * teardown code does not have to know which kind of string it holds. */
proven_u8str_destroy(alloc, &status);

/* destroy on the owned string frees the allocation, and it must be given the
 * allocator the string was created with. */
proven_u8str_destroy(alloc, &path);
return EXAMPLE_OK();
}

```

### Worked example: when the buffer is fixed and the data is not

The example above grows the string whenever it runs out of room. A great deal of real code cannot do that: a record has a fixed field width, a log line has a hard limit, a string lives in an arena where every reallocation leaves the old copy behind until the next reset. Those callers need to know which of three answers a call gives when the data does not fit.

| Kind of call                    | What it does when the data does not fit                                                             | Calls                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Atomic, fixed capacity</b>   | Refuses with <code>PROVEN_ERR_OUT_OF_BOUNDS</code> and changes nothing.                             | <code>proven_u8str_append</code> ,<br><code>proven_u8str_insert</code> ,<br><code>proven_u8str_replace_at</code>                |
| <b>Best-effort (truncating)</b> | Writes what fits, returns <code>PROVEN_ERR_OUT_OF_BOUNDS</code> <b>and</b> the byte count it wrote. | <code>proven_u8str_append_partial</code>                                                                                        |
| <b>Atomic, growable</b>         | Reallocates through the allocator; on allocation failure changes nothing.                           | <code>proven_u8str_append_grow</code> ,<br><code>proven_u8str_replace_at_grow</code> ,<br><code>proven_u8str_append_byte</code> |

Two supporting calls appear here as well. `proven_u8str_reserve()` raises the capacity once, up front, so later growth does not reallocate - on the heap that saves copying, and in an arena it saves the dead storage every reallocation leaves behind. `proven_u8str_is_valid()` checks that a string handle's own fields are consistent; it is worth asserting where a string arrives from other code, not after every edit.

The program below writes a bounded log line, refuses an oversized append, truncates deliberately with the best-effort call, and edits a path in the middle with `proven_u8str_replace_at()` - first shrinking (which always fits), then growing (which does not, and is refused), then the same edit again through `proven_u8str_replace_at_grow()`. It ends by checking the file extension with `proven_u8str_view_ends_with()`.

```

/*
 * The previous example grew a string whenever it ran out of room. This one is
 * about the case where growing is not allowed - a fixed-size record, a log line
 * with a hard length limit, a buffer in an arena that must not be reallocated -
 * and about the two honest answers a call can give when the data does not fit:
 *
 * "no, and I changed nothing"      - the atomic calls: append, insert,
 *                                replace_at. They check the capacity
 *                                first, so a refusal leaves the string
 *                                exactly as it was.
 * "some of it, and here is how much" - the best-effort call:
 *                                append_partial. It fills what it can and
 *                                tells you the byte count it wrote.
 */

```

```

* Both are useful; picking the wrong one silently truncates a record or
* silently drops one. The `_grow` variants are the third answer - "yes, I found
* more room" - and appear at the end for contrast.
*/

#define FIELD_CAP 32u

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    proven_result_u8str_t r = proven_u8str_create(alloc, FIELD_CAP);
    EXAMPLE_REQUIRE(proven_is_ok(r.err), "creating the field buffer must succeed");
    if (!proven_is_ok(r.err)) {
        return 1;
    }
    proven_u8str_t field = r.value;

    /* is_valid checks the handle's own structure - a pointer with a capacity and
    * a length that do not contradict each other. Worth asserting once at the
    * boundary of your code when a string arrives from somewhere else; it is not
    * a check you need after every edit, because every edit maintains it. */
    EXAMPLE_REQUIRE(proven_u8str_is_valid(&field), "a freshly created string must be structurally valid");

    /* --- reserving room up front ----- */

    /* reserve raises the capacity now, so later growth does not reallocate. On
    * the heap that saves copies; in an arena it saves something worse, because
    * every reallocation there leaks the old block until the next reset. Ask for
    * what you expect to need, once. */
    proven_err_t err = proven_u8str_reserve(alloc, &field, 64);
    EXAMPLE_REQUIRE(proven_is_ok(err), "reserving 64 bytes must succeed");
    EXAMPLE_REQUIRE(field.internal.cap >= 64, "the capacity must actually be at least what was asked for");

    /* --- the atomic calls: fit, or change nothing ----- */

    err = proven_u8str_append(&field, PROVEN_LIT("2026-01-01 level=info "));
    EXAMPLE_REQUIRE(proven_is_ok(err), "the prefix fits in the reserved capacity");

    /* append_byte adds one byte, which is what separators, terminators and
    * escape characters are. It takes the allocator because it is a growing
    * call - one byte is exactly the case where a capacity check would fail on
    * a boundary you did not think about. */
    err = proven_u8str_append_byte(alloc, &field, (proven_u8)'\n');
    EXAMPLE_REQUIRE(proven_is_ok(err), "appending a single separator byte must succeed");

    err = proven_u8str_append(&field, PROVEN_LIT("disk full"));
    EXAMPLE_REQUIRE(proven_is_ok(err), "the message fits");

    err = proven_u8str_append_byte(alloc, &field, (proven_u8)']');
    EXAMPLE_REQUIRE(proven_is_ok(err), "closing the bracket must succeed");

    /* Now ask for more than the capacity can hold. The atomic append refuses and
    * - this is the property worth relying on - the string still holds exactly
    * what it held before the call. */
    proven_size_t before = proven_u8str_as_view(&field).size;
    err = proven_u8str_append(&field, PROVEN_LIT(" and a very long trailing explanation that certainly does
not fit"));
    EXAMPLE_REQUIRE(err == PROVEN_ERR_OUT_OF_BOUNDS, "an oversized atomic append must be refused");
    EXAMPLE_REQUIRE(proven_u8str_as_view(&field).size == before, "and must leave the string untouched");

```

```

/* --- the best-effort call: as much as fits, and the count ----- */

/* A fixed-width column in a report is the case for this one: write what
 * fits, and know how much was written so the caller can mark the value as
 * truncated instead of pretending it is complete. */
proven_result_size_t part = proven_u8str_append_partial(&field, PROVEN_LIT(" ...more text than there is
room for"));
EXAMPLE_REQUIRE(part.err == PROVEN_ERR_OUT_OF_BOUNDS, "a partial append that truncates still reports the
truncation");
EXAMPLE_REQUIRE(part.value > 0, "but it wrote what it could");
EXAMPLE_REQUIRE(proven_u8str_as_view(&field).size == before + part.value,
                "and the string grew by exactly the number of bytes it reports");
printf("partial append wrote %zu byte(s) before the buffer was full\n", (size_t)part.value);

/* --- editing in the middle without growing ----- */

proven_result_u8str_t r2 = proven_u8str_create(alloc, FIELD_CAP);
EXAMPLE_REQUIRE(proven_is_ok(r2.err), "creating the second buffer must succeed");
proven_u8str_t path = r2.value;
err = proven_u8str_append(&path, PROVEN_LIT("var/log/service.log"));
EXAMPLE_REQUIRE(proven_is_ok(err), "the path fits");

/* insert shifts the tail right. Fixed-capacity: it fits or it refuses. */
err = proven_u8str_insert(&path, 0, PROVEN_LIT("/"));
EXAMPLE_REQUIRE(proven_is_ok(err), "inserting a leading slash must succeed");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&path), PROVEN_LIT("/var/log/service.log")),
                "the insert lands at index 0");

/* replace_at replaces old_len bytes at an index with data of any length, as
 * long as the result still fits. Replacing "service" (7) with "daemon" (6)
 * shrinks the string, so this cannot fail on capacity. */
proven_size_t at = proven_u8str_view_find(proven_u8str_as_view(&path), 0, PROVEN_LIT("service"));
EXAMPLE_REQUIRE(at != PROVEN_SIZE_MAX, "the substring must be found before it can be replaced");
err = proven_u8str_replace_at(&path, at, 7, PROVEN_LIT("daemon"));
EXAMPLE_REQUIRE(proven_is_ok(err), "a shortening replacement must succeed");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&path), PROVEN_LIT("/var/log/daemon.log")),
                "and produce the expected path");

/* The same edit the other way round overflows a 32-byte buffer, and the
 * fixed-capacity call refuses it rather than truncating a path - which is
 * the failure that silently writes to the wrong file. */
before = proven_u8str_as_view(&path).size;
err = proven_u8str_replace_at(&path, at, 6, PROVEN_LIT("a-replacement-name-far-too-long-for-this-
buffer"));
EXAMPLE_REQUIRE(err == PROVEN_ERR_OUT_OF_BOUNDS, "a replacement that does not fit must be refused");
EXAMPLE_REQUIRE(proven_u8str_as_view(&path).size == before, "and must leave the path unchanged");

/* replace_at_grow is the same edit with permission to reallocate. Use it
 * when the buffer is heap-backed and the length is genuinely unbounded;
 * prefer the fixed-capacity call when the limit is part of the format. */
err = proven_u8str_replace_at_grow(alloc, &path, at, 6, PROVEN_LIT("a-replacement-name-far-too-long-for-
this-buffer"));
EXAMPLE_REQUIRE(proven_is_ok(err), "the growing variant makes room instead of refusing");
EXAMPLE_REQUIRE(proven_u8str_view_ends_with(proven_u8str_as_view(&path), PROVEN_LIT(".log")),
                "the extension is still at the end after the edit");

/* ends_with answers the question an extension check actually asks. Doing it
 * with an index computed by hand is where the off-by-one lives; doing it
 * with strcmp on a pointer requires a NUL that a view does not have. */
EXAMPLE_REQUIRE(!proven_u8str_view_ends_with(proven_u8str_as_view(&path), PROVEN_LIT(".txt")),

```

```

        "and it is not a .txt file");

/* An empty suffix is a suffix of everything, which is the answer that keeps
 * loops over a list of suffixes from needing a special case. */
EXAMPLE_REQUIRE(proven_u8str_view_ends_with(proven_u8str_as_view(&path), PROVEN_LIT("")),
    "every string ends with the empty suffix");

printf("log line: %s\n", proven_u8str_as_cstr(&field));
printf("path: %s\n", proven_u8str_as_cstr(&path));

proven_u8str_destroy(alloc, &path);
proven_u8str_destroy(alloc, &field);
return EXAMPLE_OK();
}

```

Wrong - treating the best-effort call as if it were atomic:

```
(void)proven_u8str_append_partial(&line, field); /* wrong: ignores the count */
```

The return value is the only place the truncation is reported. Discarding it turns "the record was cut short" into "the record looked fine".

Wrong - reading the byte count as a character count:

```
if (part.value == field.size) { /* all of it was written */ }
```

That comparison is correct, and it is correct in **bytes**. A UTF-8 character can be up to four of them, so a partial append can stop in the middle of one. When the tail must stay valid UTF-8, decide the cut point yourself rather than letting the capacity decide it.

### Worked example: assembling a UTF-16 string for a system call

`proven_u16str_t` earns its place only at the boundary where an operating system call demands UTF-16 - the Windows wide API being the usual reason. The pattern is always the same: assemble the code units, then hand `proven_u16str_as_ptr()` to the call.

Three points decide whether this code is right:

- **The unit is a code unit, not a byte and not a character.** A capacity of 32 is 32 code units, which is 64 bytes; a character outside the Basic Multilingual Plane (BMP) - an emoji, many rarer CJK characters - occupies two of them.
- `proven_u16str_as_ptr()` **does not copy**, and the pointer it returns is good only until the next append that grows the string.
- **The result is NUL-terminated**, including after a deliberate truncation, so it is safe to pass to a system call that expects a terminator.

```

/*
 * UTF-16 exists in this library for one reason: some operating system calls
 * take it and nothing else. The Windows "wide" API is the usual case - the file
 * name you hand to CreateFileW is a NUL-terminated run of 16-bit code units,
 * not bytes.
 *
 * So the job this type does is narrow: assemble the code units, keep the count
 * right, and produce the pointer the system call wants. Everything else in your
 * program should stay UTF-8.
 *
 * The one thing to keep straight is the unit. A capacity of 32 here means 32
 * CODE UNITS, which is 64 bytes, and a character outside the Basic Multilingual
 * Plane - an emoji, most of the rarer CJK characters - costs two of them. A
 * count of code units is not a count of characters and never has been.
 */

```

```
int main(void) {
```

```

proven_allocator_t alloc = proven_heap_allocator();

/* The argument is a code-unit limit, not a byte limit. */
proven_result_u16str_t r = proven_u16str_create(alloc, 32);
EXAMPLE_REQUIRE(proven_is_ok(r.err), "creating a 32-code-unit string must succeed");
if (!proven_is_ok(r.err)) {
    return 1;
}
proven_u16str_t name = r.value;
EXAMPLE_REQUIRE(proven_u16str_len(&name) == 0, "a new string is empty");

/* PROVEN_U16_LIT builds a view from a u"..." literal and computes the unit
 * count from the literal itself, so the count cannot disagree with the text. */
proven_err_t err = proven_u16str_append(&name, PROVEN_U16_LIT("C:\\logs\\"));
EXAMPLE_REQUIRE(proven_is_ok(err), "the directory prefix fits in 32 code units");

err = proven_u16str_append(&name, PROVEN_U16_LIT("service.log"));
EXAMPLE_REQUIRE(proven_is_ok(err), "the file name fits too");
EXAMPLE_REQUIRE(proven_u16str_len(&name) == 8 + 11, "the length is a count of code units");

/* Atomic, like its byte-string twin: too much data is refused and the string
 * is left exactly as it was, so a path is never half-written. */
proven_size_t before = proven_u16str_len(&name);
err = proven_u16str_append(&name, PROVEN_U16_LIT(".a-suffix-long-enough-to-overflow-the-capacity"));
EXAMPLE_REQUIRE(err == PROVEN_ERR_OUT_OF_BOUNDS, "an oversized append must be refused");
EXAMPLE_REQUIRE(proven_u16str_len(&name) == before, "and must not truncate the path");

/* --- the pointer the system call wants ----- */

/* as_ptr hands back the internal code units, NUL-terminated, without
 * copying. It is the last step before the call, and the pointer is only
 * valid until the next append: a growing append may move the storage. */
const proven_u16 *wide = proven_u16str_as_ptr(&name);
EXAMPLE_REQUIRE(wide != NULL, "an assembled string must yield a pointer");
EXAMPLE_REQUIRE(wide[0] == (proven_u16)'C', "the first code unit is the drive letter");
EXAMPLE_REQUIRE(wide[proven_u16str_len(&name)] == 0, "the sequence is NUL-terminated for the system
call");
/* On Windows this is the whole point of the type:
 *     HANDLE h = CreateFileW((LPCWSTR)wide, ...);
 * Nothing here calls it, because this example must also run everywhere else. */

/* --- when truncation is the correct answer ----- */

/* Some system structures have a fixed-width field - a 16-unit label, say -
 * where a name that does not fit is meant to be cut, not rejected. That is
 * what the partial append is for: it fills what it can and reports the unit
 * count it wrote, so the caller can mark the value as truncated. */
proven_result_u16str_t r2 = proven_u16str_create(alloc, 16);
EXAMPLE_REQUIRE(proven_is_ok(r2.err), "creating the fixed-width label must succeed");
proven_u16str_t label = r2.value;

proven_result_size_t wrote = proven_u16str_append_partial(&label, PROVEN_U16_LIT("a-label-that-is-longer-
than-the-field"));
EXAMPLE_REQUIRE(wrote.err == PROVEN_ERR_OUT_OF_BOUNDS, "the truncation is reported, not hidden");
EXAMPLE_REQUIRE(wrote.value == 16, "it filled the field exactly");
EXAMPLE_REQUIRE(proven_u16str_len(&label) == wrote.value, "and the length matches what it says it wrote");
EXAMPLE_REQUIRE(proven_u16str_as_ptr(&label)[wrote.value] == 0,
    "a truncated string is still NUL-terminated, so it is still safe to pass on");

printf("assembled %zu code unit(s); label truncated to %zu\n",

```

```

        (size_t)proven_u16str_len(&name), (size_t)wrote.value);

    proven_u16str_destroy(alloc, &label);
    proven_u16str_destroy(alloc, &name);
    return EXAMPLE_OK();
}

```

Wrong - sizing a UTF-16 buffer in bytes:

```
proven_result_u16str_t r = proven_u16str_create(alloc, sizeof(buf)); /* wrong */
```

The argument is a count of code units. Passing a byte count asks for twice the storage you meant, or - when the byte count came from a UTF-8 string - for a buffer that cannot hold the conversion at all.

Wrong - keeping the pointer across an append:

```

const proven_u16 *w = proven_u16str_as_ptr(&name);
proven_err_t e = proven_u16str_append_grow(alloc, &name, more); /* may reallocate */
use_wide(w); /* wrong: w may dangle */

```

Take the pointer immediately before the call that consumes it.

## Chapter 4: Containers and Algorithms

**Part III - Data structures. Prerequisite: Part II (1, 2, 3).** After this chapter you can pick the right container for a job, sort and search with a guaranteed bound, hash for the right reason, and turn bytes into text and back.

This chapter covers `array.h`, `list.h`, `ring.h`, `map.h`, `algorithm.h`, `hash.h`, and `encode.h`. Every container here takes an allocator, which is why Chapter 2 comes first.

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### 1. Dynamic array

#### The problem: the array you write by hand, every time

A growable array is the data structure every C program eventually writes, and writes slightly differently:

```
int *items = malloc(cap * sizeof(int));
/* ... later ... */
if (n == cap) {
    cap *= 2;
    items = realloc(items, cap * sizeof(int)); /* wrong on two counts */
    items[n++] = x;
}
```

Two bugs in one line, and both are classics. `cap * sizeof(int)` can **wrap** for a large `cap`, producing a small allocation for a huge count - see [Chapter 1 section 4](#). And assigning `realloc`'s result straight back to `items` **leaks the old block when it returns NULL**, because the original pointer is gone and the memory it named is still allocated.

Beyond the bugs, the hand-written version has to be rewritten for every element type, or made generic with `void *` and lose type checking.

#### What this library does instead

`proven_array_t` is a growable vector that keeps the element size and alignment it was created with, so it works for any type without a template and without `void *` at the call site - the `PROVEN_ARRAY_*` macros take the type and do the casting where the compiler can still check it.

- Growth uses checked arithmetic, so the overflow above is `PROVEN_ERR_OVERFLOW`, not a small allocation.
- Growth is **failure-atomic**: if it cannot grow, your existing elements are untouched and still valid. Nothing is leaked and nothing is lost.
- It **stores its allocator internally**, unlike the string types - which is why `PROVEN_ARRAY_DESTROY` takes only the array.

`proven_array_t` owns contiguous element storage, so it is also what you hand to `proven_array_sort` and the searches in section 5.

Wrong - holding a pointer across a push:

```

int *first = PROVEN_ARRAY_GET_MUT(&arr, int, 0);
(void)PROVEN_ARRAY_PUSH(&arr, int, 42); /* may reallocate */
*first = 7; /* wrong: `first` may point at freed memory */

```

This is the one hazard the type cannot remove. Growth moves the storage, so **any pointer or view into the array is invalidated by any operation that can grow it**. Index instead of pointing, or re-fetch the pointer after the push.

## Structures

```

typedef struct {
    proven_allocator_t alloc;
    proven_byte_t *data;
    proven_size_t len;
    proven_size_t cap;
    proven_size_t elem_size;
    proven_size_t align;
} proven_array_t;

```

```

typedef struct {
    proven_err_t err;
    proven_array_t value;
} proven_result_array_t;

```

Fields:

- `alloc`: allocator used for growth and destruction.
- `data`: byte storage for elements.
- `len`: current element count.
- `cap`: capacity in elements.
- `elem_size`: size of each element.
- `align`: alignment of each element.

## Functions

| API                                                                 | Intent                                                             | Return                                                            |
|---------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------|
| <code>proven_array_create(alloc, init_cap, elem_size, align)</code> | Create a generic array.                                            | <code>proven_result_array_t</code> .                              |
| <code>proven_array_is_valid(arr)</code>                             | Validate public array invariants.                                  | <code>bool</code> .                                               |
| <code>proven_array_reserve(arr, new_cap)</code>                     | Ensure at least <code>new_cap</code> elements.                     | <code>proven_err_t</code> .                                       |
| <code>proven_array_push(arr, element)</code>                        | Append one element by copying from pointer.                        | <code>proven_err_t</code> .                                       |
| <code>proven_array_pop(arr, out_element)</code>                     | Pop last element; <code>out_element</code> may be null to discard. | <code>proven_err_t</code> .                                       |
| <code>proven_array_get_mut(arr, index)</code>                       | Get mutable element pointer.                                       | pointer or null.                                                  |
| <code>proven_array_get(arr, index)</code>                           | Get const element pointer.                                         | pointer or null.                                                  |
| <code>proven_array_destroy(arr)</code>                              | Free storage and clear state.                                      | <code>void</code> .                                               |
| <code>proven_array_clear(arr)</code>                                | Remove every element; capacity stays.                              | <code>void</code> .                                               |
| <code>proven_array_truncate(arr, new_len)</code>                    | Shorten to <code>new_len</code> .                                  | <code>proven_err_t</code> ; <code>OUT_OF_BOUNDS</code> if longer. |

|                                                   |                                                                                   |               |
|---------------------------------------------------|-----------------------------------------------------------------------------------|---------------|
| proven_array_insert(arr, index, element)          | Insert at index (0..len), moving the rest up. element may point into the array.   | proven_err_t. |
| proven_array_remove_at(arr, index, out_element)   | Remove at index, keeping order.                                                   | proven_err_t. |
| proven_array_swap_remove(arr, index, out_element) | Remove at index in O(1) by moving the last element in; order is not kept.         | proven_err_t. |
| proven_array_extend(arr, elements, count)         | Append count elements, one reallocation at most; may be the array's own elements. | proven_err_t. |

The editing operations change nothing when they fail, refuse an index past the end with `PROVEN_ERR_OUT_OF_BOUNDS`, and - for the two that can grow the array - invalidate pointers into it as push does.

```

/* A to-do list: insert at the front, take one out from the middle keeping order, drop
 * finished items in O(1) when order does not matter, and append a batch at once. */
proven_result_array_t todo = PROVEN_ARRAY_INIT(alloc, int, 4);
if (proven_is_ok(todo.err)) {
    const int batch[] = { 10, 20, 30, 40 };
    int first = 5, taken = 0;
    proven_err_t e = proven_array_extend(&todo.value, batch, 4);           /* 10 20 30 40 */
    if (proven_is_ok(e)) e = proven_array_insert(&todo.value, 0, &first); /* 5 10 20 30 40 */
    if (proven_is_ok(e)) e = proven_array_remove_at(&todo.value, 2, &taken); /* 5 10 30 40, took 20 */
    if (proven_is_ok(e)) e = proven_array_swap_remove(&todo.value, 0, NULL); /* 40 10 30 */
    if (proven_is_ok(e)) e = proven_array_truncate(&todo.value, 2);        /* 40 10 */
    proven_println("{} items, took {}", PROVEN_ARG(todo.value.len), PROVEN_ARG(taken));
    proven_array_clear(&todo.value);                                       /* empty, capacity kept */
    proven_array_destroy(&todo.value);
}

```

## Macros

| Macro                                      | Intent                                                 |
|--------------------------------------------|--------------------------------------------------------|
| PROVEN_ARRAY_INIT(alloc, type, init_cap)   | Type-safe create using sizeof(type) and alignof(type). |
| PROVEN_ARRAY_PUSH(arr_ptr, type, value)    | Push rvalue or lvalue by temporary compound literal.   |
| PROVEN_ARRAY_POP(arr_ptr, type, out_ptr)   | Pop into output pointer or discard with null.          |
| PROVEN_ARRAY_GET(arr_ptr, type, index)     | Typed const element pointer.                           |
| PROVEN_ARRAY_GET_MUT(arr_ptr, type, index) | Typed mutable element pointer.                         |
| PROVEN_ARRAY_DESTROY(arr_ptr)              | Destroy array.                                         |

## Example:

```

proven_result_array_t r = PROVEN_ARRAY_INIT(alloc, int, 4);
if (!proven_is_ok(r.err)) {
    return;
}
proven_array_t nums = r.value;

(void)PROVEN_ARRAY_PUSH(&nums, int, 10);
(void)PROVEN_ARRAY_PUSH(&nums, int, 20);

/* GET points into the array's own storage, and returns NULL out of range. */
const int *first = PROVEN_ARRAY_GET(&nums, int, 0);

```

```

if (first) {
    proven_println("first={}", PROVEN_ARG(*first));
}

/* The array stores its allocator, so destroy needs nothing but the array. */
PROVEN_ARRAY_DESTROY(&nums);

```

## 2. Intrusive list

### The problem: the list you were taught allocates a node per element

```

struct node { struct node *next; void *data; };

```

That is the textbook linked list, and it has two costs the textbook does not mention. Every insertion **allocates** - a thousand items means a thousand allocations that exist only to hold pointers, each one able to fail and each one needing a matching free. And data is a void \*, so the list has no idea what it holds: every read back out is a cast the compiler cannot check.

### What an intrusive list does instead

It turns the relationship inside out: **you own the memory, and the link lives inside it.**

```

typedef struct {
    int            id;
    proven_list_node_t link; /* the list's hook, inside your struct */
} task_t;

```

Now inserting is writing two pointers. It allocates nothing, so it **cannot fail** - notice that proven\_list\_push\_back returns void, which is unusual in this library and is the point. Removing is the same. The objects can live on the stack, in an array, in an arena, anywhere; the list only rearranges pointers that are already inside them.

Getting from a link back to your object is PROVEN\_LIST\_ENTRY(node, task\_t, link), which subtracts the member's offset from the node's address. It is arithmetic the compiler checks the types of, not a cast you hope is right.

What you give up: **an object can be in only as many lists as it has link members**, and you decide that when you declare the struct. If an item needs to be in a queue and an index at the same time, it needs two links.

proven\_list\_t is an intrusive doubly-linked **circular** list - the head is a sentinel node, so insertion and removal never special-case the ends. It allocates no nodes.

Wrong - removing while walking with the plain iterator:

```

PROVEN_LIST_FOR_EACH(it, &queue) {
    if (should_drop(it)) proven_list_remove(it); /* wrong: it->next is read after unlink */
}

```

proven\_list\_remove writes through the node's own pointers, so the loop then reads next from a node that has just been detached. PROVEN\_LIST\_FOR\_EACH\_SAFE takes a second variable and reads the next pointer *before* the body runs, which is exactly why it exists.

### Worked example: a queue whose links live in the caller's structs

Compiled and run by the test suite. It builds a queue of stack-allocated tasks, walks it, removes one from inside a safe walk, and inserts in the middle - with no allocator anywhere in the program.

```

/*
 * An INTRUSIVE list puts the links inside your struct instead of allocating a
 * node to hold your data.
 *
 * The list you were taught looks like this:
 *
 * struct node { struct node *next; void *data; };
 */

```

```

*
* Every insertion allocates a node, so a list of a thousand items costs a
* thousand allocations that exist only to hold pointers, each one a chance to
* fail and a thing to free. Worse, `data` is a void* - the list has no idea what
* it holds, so every read is a cast the compiler cannot check.
*
* Intrusive lists invert it: YOU own the memory, and the link lives in it.
* Inserting allocates nothing and cannot fail. Removing allocates nothing and
* cannot fail. And because the link is a member of a known type, getting back
* from a link to the object is arithmetic the compiler does for you, not a cast.
*
* The trade is that an object can only be in as many lists as it has link
* members, and that is a decision you make when you declare the struct.
*/

/* The link is a member. This task can be in exactly one list at a time. */
typedef struct {
    int id;
    proven_list_node_t link;
} task_t;

int main(void) {
    /* No allocator anywhere in this program: the tasks are on the stack, and the
     * list only ever rearranges pointers that live inside them. */
    task_t a = { .id = 1 };
    task_t b = { .id = 2 };
    task_t c = { .id = 3 };

    proven_list_t queue;
    proven_list_init(&queue);
    EXAMPLE_REQUIRE(proven_list_is_empty(&queue), "a freshly initialised list is empty");

    /* --- pushing cannot fail, because nothing is allocated ----- */
    proven_list_push_back(&queue, &a.link);
    proven_list_push_back(&queue, &b.link);
    proven_list_push_back(&queue, &c.link);
    EXAMPLE_REQUIRE(!proven_list_is_empty(&queue), "three tasks are queued");

    /* --- walking: PROVEN_LIST_ENTRY gets the object back from the link -- */
    proven_list_node_t *it = NULL;
    int seen[3] = {0}, n = 0;
    PROVEN_LIST_FOR_EACH(it, &queue) {
        task_t *t = PROVEN_LIST_ENTRY(it, task_t, link);
        if (n < 3) seen[n++] = t->id;
    }
    EXAMPLE_REQUIRE(n == 3 && seen[0] == 1 && seen[1] == 2 && seen[2] == 3,
        "the walk visits every task, in insertion order");

    /* --- removing while walking needs the SAFE form ----- */
    /* proven_list_remove writes through the node's own next/prev pointers, so a
     * plain FOR_EACH would read `it->next` from a node that has just been
     * unlinked. The _SAFE form reads the next pointer BEFORE the body runs. */
    proven_list_node_t *safe = NULL;
    PROVEN_LIST_FOR_EACH_SAFE(it, safe, &queue) {
        task_t *t = PROVEN_LIST_ENTRY(it, task_t, link);
        if (t->id == 2) proven_list_remove(it);
    }

    n = 0;
    PROVEN_LIST_FOR_EACH(it, &queue) {

```

```

        task_t *t = PROVEN_LIST_ENTRY(it, task_t, link);
        if (n < 3) seen[n++] = t->id;
    }
    EXAMPLE_REQUIRE(n == 2 && seen[0] == 1 && seen[1] == 3, "task 2 was unlinked");

    /* --- inserting in the middle is a pointer swap ----- */
    task_t d = { .id = 4 };
    proven_list_insert_after(&a.link, &d.link);

    n = 0;
    PROVEN_LIST_FOR_EACH(it, &queue) {
        task_t *t = PROVEN_LIST_ENTRY(it, task_t, link);
        if (n < 3) seen[n++] = t->id;
    }
    EXAMPLE_REQUIRE(n == 3 && seen[0] == 1 && seen[1] == 4 && seen[2] == 3,
        "task 4 sits directly after task 1");

    /* There is nothing to destroy. The list never owned anything: `a`, `c` and
     * `d` are still perfectly good local variables, and their lifetime is the
     * function's, exactly as it would be without the list. */
    return EXAMPLE_OK();
}

```

## Structures

```

typedef struct proven_list_node_t {
    struct proven_list_node_t *next;
    struct proven_list_node_t *prev;
} proven_list_node_t;

```

```

typedef struct {
    proven_list_node_t head;
} proven_list_t;

```

## Functions and macros

| API                                              | Intent                                           | Return           |
|--------------------------------------------------|--------------------------------------------------|------------------|
| proven_list_init(list)                           | Initialize circular sentinel.                    | void.            |
| proven_list_insert_after(target, node)           | Insert node after target.                        | void.            |
| proven_list_push_back(list, node)                | Insert at tail.                                  | void.            |
| proven_list_remove(node)                         | Detach node and poison links to null.            | void.            |
| proven_list_is_empty(list)                       | Test empty or null list.                         | int truth value. |
| PROVEN_CONTAINER_OF(ptr, type, member)           | Convert member pointer to parent object pointer. | pointer.         |
| PROVEN_LIST_FOR_EACH(iter, list)                 | Iterate nodes.                                   | loop macro.      |
| PROVEN_LIST_FOR_EACH_SAFE(iter, safe_next, list) | Iterate safely while removing current node.      | loop macro.      |
| PROVEN_LIST_ENTRY(ptr, type, member)             | Convert list node to containing object.          | pointer.         |

Example:

```

typedef struct Item {
    int value;
    proven_list_node_t link; /* the list lives inside the object; it allocates nothing */
} Item;

proven_list_t list;
proven_list_init(&list);

```

```

Item a = { .value = 1 };
Item b = { .value = 2 };
proven_list_push_back(&list, &a.link);
proven_list_push_back(&list, &b.link);

int total = 0;
proven_list_node_t *it = NULL;
PROVEN_LIST_FOR_EACH(it, &list) {
    /* The iterator walks nodes; ENTRY converts a node back to its owner. */
    Item *item = PROVEN_LIST_ENTRY(it, Item, link);
    total += item->value;
}
proven_println("total={}", PROVEN_ARG(total)); /* 3 */

```

### 3. Ring buffer

#### The problem: a queue that must not grow

A queue between a fast producer and a slow consumer has to answer one question: what happens when the consumer falls behind? A growable queue answers "allocate more", which turns a temporary slowdown into unbounded memory growth and, eventually, into something worse than dropped work.

The hand-written alternative is an array plus a head index plus a tail index plus modulo arithmetic - and it has one famous bug. When `head == tail`, is the buffer empty or full? Both states look identical unless you keep a separate count or deliberately waste one slot, and every generation of programmers rediscovers this.

#### What this library does instead

`proven_ring_t` is a fixed-capacity FIFO that keeps the count, so empty and full are distinct, and that **refuses** rather than grows:

- `proven_ring_push` on a full ring returns `PROVEN_ERR_OUT_OF_BOUNDS`. It does not overwrite the oldest entry and it does not reallocate. The caller decides - wait, drop the new item, or report backpressure - because only the caller knows which is right.
- `proven_ring_pop` on an empty ring fails rather than handing back a stale slot.

The capacity is chosen once, at creation, and that number is the policy: it says how far ahead the producer may get. Reach for a ring for an event queue, a log of recent items, an audio or sensor buffer - anywhere a bound is part of the design rather than a limitation.

Unlike most types in this library, `proven_ring_t` **stores its allocator internally**, which is why `PROVEN_RING_DESTROY` takes only the ring.

Wrong - treating a full ring as an error to retry immediately:

```
while (PROVEN_RING_PUSH(&ring, event_t, e) != PROVEN_OK) { } /* wrong: spins forever */
```

Nothing in that loop consumes anything, so the ring stays full. A refusal from a bounded queue is information about the *consumer*, and the loop that ignores it is a hang.

#### Worked example: pushing until full, and what refusal looks like

Compiled and run by the test suite.

```

/*
 * A ring buffer is a fixed-size queue that never moves its contents and never
 * grows. You give it a capacity once; push adds at the tail, pop removes from
 * the head, and when it is full, push REFUSES.
 *
 * The C you would otherwise write is an array plus two indices plus the modulo
 * arithmetic to wrap them, and the bug is always the same: "is it full or is it

```

```

* empty?" - both states have head == tail unless you keep a count or waste a
* slot. This ring keeps the count, so the question does not arise.
*
* Use one when a producer and a consumer run at different speeds and you want a
* hard bound on how far ahead the producer may get: an event queue, a log ring,
* an audio buffer. The refusal on a full ring is the feature - it is
* backpressure. A growable queue would answer a burst by eating memory until
* something worse happens.
*/

```

```

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* Capacity of 4 events. It will never be 5. */
    proven_result_ring_t r = PROVEN_RING_INIT(alloc, int, 4);
    EXAMPLE_REQUIRE(proven_is_ok(r.err), "creating a 4-slot ring should succeed");
    proven_ring_t ring = r.value;

    /* --- push until full ----- */
    for (int i = 1; i <= 4; ++i) {
        proven_err_t err = PROVEN_RING_PUSH(&ring, int, i);
        EXAMPLE_REQUIRE(proven_is_ok(err), "the first four pushes fit");
    }

    /* The fifth push is refused. It does not overwrite the oldest entry, and it
     * does not grow: a full ring is a full ring, and the caller decides what to
     * do about it - wait, drop the new event, or report backpressure. */
    proven_err_t full = PROVEN_RING_PUSH(&ring, int, 5);
    EXAMPLE_REQUIRE(full == PROVEN_ERR_OUT_OF_BOUNDS,
        "pushing into a full ring must be refused, not silently absorbed");

    /* --- pop in the order they were pushed ----- */
    int out = 0;
    proven_err_t err = PROVEN_RING_POP(&ring, int, &out);
    EXAMPLE_REQUIRE(proven_is_ok(err) && out == 1, "pop returns the oldest entry first");

    /* Now there is room again, and the ring wraps around its storage without
     * moving anything. */
    err = PROVEN_RING_PUSH(&ring, int, 5);
    EXAMPLE_REQUIRE(proven_is_ok(err), "after a pop there is room for one more");

    int expected[] = { 2, 3, 4, 5 };
    for (int i = 0; i < 4; ++i) {
        err = PROVEN_RING_POP(&ring, int, &out);
        EXAMPLE_REQUIRE(proven_is_ok(err) && out == expected[i],
            "entries come out in the order they went in");
    }

    /* --- empty behaves like full: it refuses, it does not invent data --- */
    err = PROVEN_RING_POP(&ring, int, &out);
    EXAMPLE_REQUIRE(!proven_is_ok(err), "popping an empty ring must fail rather than return junk");

    PROVEN_RING_DESTROY(&ring);
    return EXAMPLE_OK();
}

```

## Structures

```

typedef struct {
    proven_allocator_t alloc;
    proven_mem_mut_t internal;
}

```

```

    proven_size_t head;
    proven_size_t tail;
    proven_size_t len;
    proven_size_t cap;
    proven_size_t elem_size;
    proven_size_t align;
} proven_ring_t;

```

```

typedef struct {
    proven_err_t err;
    proven_ring_t value;
} proven_result_ring_t;

```

## Functions and macros

| API                                                           | Intent                                 | Return                              |
|---------------------------------------------------------------|----------------------------------------|-------------------------------------|
| <code>proven_ring_create(alloc, cap, elem_size, align)</code> | Create fixed-capacity ring.            | <code>proven_result_ring_t</code> . |
| <code>proven_ring_is_valid(ring)</code>                       | Validate ring invariants.              | <code>bool</code> .                 |
| <code>proven_ring_push(ring, element)</code>                  | Push one element; fails when full.     | <code>proven_err_t</code> .         |
| <code>proven_ring_pop(ring, out_element)</code>               | Pop one element; null output discards. | <code>proven_err_t</code> .         |
| <code>proven_ring_destroy(ring)</code>                        | Free storage.                          | <code>void</code> .                 |
| <code>PROVEN_RING_INIT(alloc, type, cap)</code>               | Type-safe create.                      | <code>proven_result_ring_t</code> . |
| <code>PROVEN_RING_PUSH(ring_ptr, type, value)</code>          | Type-safe push.                        | <code>proven_err_t</code> .         |
| <code>PROVEN_RING_POP(ring_ptr, type, out_ptr)</code>         | Type-safe pop.                         | <code>proven_err_t</code> .         |
| <code>PROVEN_RING_DESTROY(ring_ptr)</code>                    | Destroy ring.                          | <code>void</code> .                 |

Example:

```

proven_result_ring_t r = PROVEN_RING_INIT(alloc, int, 8);
if (!proven_is_ok(r.err)) {
    return;
}
proven_ring_t q = r.value;

proven_err_t e = PROVEN_RING_PUSH(&q, int, 7);
if (!proven_is_ok(e)) {
    /* The ring is fixed-capacity: a full ring is PROVEN_ERR_OUT_OF_BOUNDS,
     * never a silent grow. */
    PROVEN_RING_DESTROY(&q);
    return;
}

int out = 0;
e = PROVEN_RING_POP(&q, int, &out); /* FIFO: out == 7. Empty ring is OUT_OF_BOUNDS. */
(void)e;

PROVEN_RING_DESTROY(&q);

```

## 4. Hash map

`proven_map_t` is an open-addressing hash map with tombstones. It supports integer keys and borrowed or owned U8 string keys. It stores its allocator internally.

## Structures

```
typedef enum {
    PROVEN_KEY_TYPE_INT,          /* keys are proven_size_t integers */
    PROVEN_KEY_TYPE_U8_BORROWED, /* keys are u8 views; caller keeps the bytes alive */
    PROVEN_KEY_TYPE_U8_OWNED     /* keys are u8 views; the map copies and frees the bytes */
} proven_key_type_t;

typedef union {
    proven_size_t id;             /* used when key_type == PROVEN_KEY_TYPE_INT */
    proven_u8str_view_t str;      /* used for the two U8 key modes */
} proven_map_key_t;

typedef struct {
    proven_allocator_t alloc;     /* allocator for the bucket array and owned keys */
    proven_mem_mut_t internal;    /* the single contiguous bucket array (ptr + size) */
    proven_size_t len;           /* live entries */
    proven_size_t used;          /* live entries + tombstones (drives the load factor) */
    proven_size_t cap;           /* number of buckets, always a power of two */
    proven_size_t elem_size;     /* size of one stored value */
    proven_size_t align;         /* alignment requested for the value */
    proven_size_t bucket_stride; /* bytes per bucket: align_up(header + elem_size) */
    proven_size_t payload_offset; /* bytes from a bucket start to its value payload */
    proven_key_type_t key_type;  /* key mode chosen at create time */
} proven_map_t;

typedef struct {
    proven_err_t err;
    proven_map_t value;
} proven_result_map_t;
```

## How it works internally

The map is a single flat array of `cap` buckets - there are no per-entry allocations for values (and none for keys in INT/BORROWED mode), so lookups stay cache-friendly. Each bucket is laid out as:

```
[ header: state + key ][ padding to the value's alignment ][ value payload ]
^ bucket start                                     ^ payload_offset
|<----- bucket_stride ----->|
```

The header's state is one of **EMPTY** (never used), **OCCUPIED** (holds a live key+value), or **TOMBSTONE** (a removed entry). `payload_offset` and `bucket_stride` are computed once at create time from `elem_size/align`, so addressing bucket `i`'s value is just `internal.ptr + i*bucket_stride + payload_offset`.

- **Hashing, and why the default is the safe one.** Keys are hashed under a per-process secret drawn once from the OS CSPRNG - because a map that hashes *untrusted* keys with a predictable function is a denial of service waiting to happen: an attacker who controls the keys computes collisions offline, floods them all into one bucket, and turns every lookup into a linear scan. Keying the hash with a secret they cannot see is what closes that, and it is the same choice Python, Rust, and the Linux kernel made for their own tables. String keys use SipHash-2-4; integer keys use SipHash-1-3 over the key's 8 bytes - an integer is no safer: user ids and record numbers from a request are chosen by whoever sent it. If your keys all come from your own program, `proven_map_create_trusted` opts into the fast unkeyed functions instead - FNV-1a for strings, a SplitMix-style bit-mix finaliser for integers (about 7 ns less per lookup on a small table); `proven_map_hash` shows you which function a given map actually uses. (On a freestanding target, which has no CSPRNG and no attacker model, keys fall back to the unkeyed functions.)
- **Probing.** Linear open addressing: the start bucket is `hash & (cap - 1)` (cheap because `cap` is a power of two), then the search walks forward one bucket at a time, wrapping around, until it finds the

key (OCCUPIED with an equal key) or an EMPTY bucket (which proves the key is absent). Linear probing keeps the walked buckets contiguous in memory.

- **Removal and tombstones.** `proven_map_remove` cannot just blank a bucket -- that would cut a probe chain and hide later keys. It marks the bucket TOMBSTONE instead. Tombstones are skipped by lookups but still consume a slot, which is why `used (live + tombstones)` - not `len` - drives growth.
- **Load factor and resize.** When used  $\geq \text{cap} * 3/4$ , the map allocates a new bucket array of the next power of two and **rehashes** every OCCUPIED entry into it, dropping all tombstones in the process. Capacity only grows; it never shrinks. Reserve ahead with `proven_map_reserve` (especially with arena allocators, to avoid leaving dead arrays behind).

### Key modes - choosing one

- `PROVEN_KEY_TYPE_INT`: keys are `proven_size_t`. No key storage.
- `PROVEN_KEY_TYPE_U8_BORROWED`: the bucket stores the *view* (pointer + length). **The map never copies the bytes**, so the caller must keep the exact bytes alive and unmoved for the whole time the entry exists. Cheapest for keys that already live somewhere stable (string literals, interned strings).
- `PROVEN_KEY_TYPE_U8_OWNED`: on insert (`proven_map_set_u8_owned` / `PROVEN_MAP_SET_U8_OWNED`) the map **duplicates** the key bytes into its own storage and frees them on remove/destroy, so the source buffer may be released or reused immediately after the call.

### The `set_with_scratch` / `alias` case

If the element you pass to `proven_map_set` points *into the map's own bucket array* (for example, copying a value from one key to another via a pointer returned by `proven_map_get`), a rehash triggered by that same insert would move or free the bytes you are reading. `proven_map_set_with_scratch` (and the `*_WITH_SCRATCH*` macros) capture the source bytes into a temporary buffer from the scratch allocator first, so the insert is alias-safe. Persistent storage still uses `map->alloc`; the scratch allocator is only for that transient copy.

### Functions

| API                                                                         | Intent                                                                                                                                       | Return                             |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <code>proven_map_create(alloc, init_cap, key_type, elem_size, align)</code> | Create a map.                                                                                                                                | <code>proven_result_map_t</code> . |
| <code>proven_map_is_valid(map)</code>                                       | Validate map invariants.                                                                                                                     | <code>bool</code> .                |
| <code>proven_map_reserve(map, new_cap)</code>                               | Ensure capacity.                                                                                                                             | <code>proven_err_t</code> .        |
| <code>proven_map_set_with_scratch(map, key, element, scratch)</code>        | Insert/update using scratch for temporary alias-safe copies.                                                                                 | <code>proven_err_t</code> .        |
| <code>proven_map_set(map, key, element)</code>                              | Insert/update.                                                                                                                               | <code>proven_err_t</code> .        |
| <code>proven_map_set_u8_owned(map, key, element)</code>                     | Insert/update with map-owned U8 key storage.                                                                                                 | <code>proven_err_t</code> .        |
| <code>proven_map_get_mut(map, key)</code>                                   | Lookup mutable value.                                                                                                                        | pointer or null.                   |
| <code>proven_map_get(map, key)</code>                                       | Lookup const value.                                                                                                                          | pointer or null.                   |
| <code>proven_map_remove(map, key)</code>                                    | Remove key if present.                                                                                                                       | <code>proven_err_t</code> .        |
| <code>proven_map_destroy(map)</code>                                        | Free map storage.                                                                                                                            | <code>void</code> .                |
| <code>proven_map_len(map)</code>                                            | Number of entries; 0 for NULL.                                                                                                               | <code>proven_size_t</code> .       |
| <code>proven_map_iter_init(map)</code>                                      | Start a walk over every entry, in bucket order.                                                                                              | <code>proven_map_iter_t</code> .   |
| <code>proven_map_iter_next(&amp;it, &amp;key, &amp;value)</code>            | The next entry. <code>PROVEN_ERR_EOF</code> after the last; <code>PROVEN_ERR_INVALID_STATE</code> if a new key or a reserve rehashed the map | <code>proven_err_t</code> .        |

|  |                                                                   |  |
|--|-------------------------------------------------------------------|--|
|  | during the walk. Removing entries and updating values is allowed. |  |
|--|-------------------------------------------------------------------|--|

**Visiting every entry.** The map has no order a caller can rely on - bucket order, and a keyed map's differs per process - but it can be walked, to list what it holds, serialise it, or release what its values own before `proven_map_destroy`. Removing the entry just returned is allowed:

```
/* Count words, then drop the rare ones while walking the map. */
proven_result_map_t words = proven_map_create(alloc, 16, PROVEN_KEY_TYPE_U8_BORROWED, sizeof(int),
_Alignof(int));
if (proven_is_ok(words.err)) {
    int one = 1, three = 3;
    (void)proven_map_set(&words.value, (proven_map_key_t){ .str = PROVEN_LIT("rare") }, &one);
    (void)proven_map_set(&words.value, (proven_map_key_t){ .str = PROVEN_LIT("common") }, &three);
    proven_map_iter_t it = proven_map_iter_init(&words.value);
    proven_map_key_t key;
    void *value;
    while (proven_map_iter_next(&it, &key, &value) == PROVEN_OK) {
        if (*(int *)value < 2) (void)proven_map_remove(&words.value, key);
    }
    proven_println("{ } word(s) left", PROVEN_ARG(proven_map_len(&words.value)));
    proven_map_destroy(&words.value);
}
```

## Macros

| Macro                                                                                        | Intent                                       |
|----------------------------------------------------------------------------------------------|----------------------------------------------|
| <code>proven_map_create_with_capacity(...)</code>                                            | Alias emphasizing capacity allocation.       |
| <code>PROVEN_MAP_INIT_INT(alloc, type, init_cap)</code>                                      | Create integer-key typed map.                |
| <code>PROVEN_MAP_INIT_U8_BORROWED(alloc, type, init_cap)</code>                              | Create borrowed-string-key typed map.        |
| <code>PROVEN_MAP_INIT_U8_OWNED(alloc, type, init_cap)</code>                                 | Create owned-string-key typed map.           |
| <code>PROVEN_MAP_SET_INT(map_ptr, int_key, type, value)</code>                               | Set integer key.                             |
| <code>PROVEN_MAP_SET_WITH_SCRATCH_INT(map_ptr, int_key, type, value, scratch)</code>         | Set integer key using scratch allocator.     |
| <code>PROVEN_MAP_SET_U8_BORROWED(map_ptr, u8_view, type, value)</code>                       | Set borrowed U8 key.                         |
| <code>PROVEN_MAP_SET_U8_OWNED(map_ptr, u8_view, type, value)</code>                          | Set owned U8 key.                            |
| <code>PROVEN_MAP_SET_WITH_SCRATCH_U8_BORROWED(map_ptr, u8_view, type, value, scratch)</code> | Set borrowed U8 key using scratch allocator. |
| <code>PROVEN_MAP_GET_INT(map_ptr, type, int_key)</code>                                      | Get const value by integer key.              |
| <code>PROVEN_MAP_GET_U8_BORROWED(map_ptr, type, u8_view)</code>                              | Get const value by borrowed U8 key.          |
| <code>PROVEN_MAP_GET_U8_OWNED(map_ptr, type, u8_view)</code>                                 | Get const value by owned U8 key.             |
| <code>PROVEN_MAP_GET_MUT_INT(map_ptr, type, int_key)</code>                                  | Get mutable value by integer key.            |
| <code>PROVEN_MAP_GET_MUT_U8_BORROWED(map_ptr, type, u8_view)</code>                          | Get mutable value by borrowed U8 key.        |
| <code>PROVEN_MAP_GET_MUT_U8_OWNED(map_ptr, type, u8_view)</code>                             | Get mutable value by owned U8 key.           |
| <code>PROVEN_MAP_REMOVE_INT(map_ptr, int_key)</code>                                         | Remove integer key.                          |
| <code>PROVEN_MAP_REMOVE_U8_BORROWED(map_ptr, u8_view)</code>                                 | Remove U8 key.                               |
| <code>PROVEN_MAP_REMOVE_U8_OWNED(map_ptr, u8_view)</code>                                    | Remove owned U8 key.                         |
| <code>PROVEN_MAP_DESTROY(map_ptr)</code>                                                     | Destroy map.                                 |

Owned key path:

PROVEN\_KEY\_TYPE\_U8\_OWNED stores a duplicate of the key bytes inside the map. Use it when the source buffer may move or be freed after insertion. The owned bytes are released on remove and destroy, and they move with the entry during rehash.

PROVEN\_HARDENED and debug validation can reject some borrowed-key pointers that fall inside the map's own internal storage. That check is defensive only; it does not make borrowed keys self-owning.

Example:

```
typedef struct UserInfo {
    int level;
    double budget;
} UserInfo;

proven_result_map_t r = PROVEN_MAP_INIT_INT(alloc, UserInfo, 8);
if (!proven_is_ok(r.err)) {
    return;
}
proven_map_t users = r.value;

UserInfo u = { .level = 3, .budget = 99.0 };
(void)PROVEN_MAP_SET_INT(&users, 404, UserInfo, u);

/* GET returns a pointer into the bucket array, or NULL when absent. Any insert
 * that rehashes invalidates it: look it up, use it, drop it. */
const UserInfo *found = PROVEN_MAP_GET_INT(&users, UserInfo, 404);
if (found) {
    proven_println("level={}", PROVEN_ARG(found->level));
}

PROVEN_MAP_DESTROY(&users);
```

## 5. Algorithms

Array algorithms operate on `proven_array_t` and a comparison callback.

### Comparison function

```
typedef int (*proven_compare_fn_t)(const void *a, const void *b);
```

Return negative if  $a < b$ , zero if equal, positive if  $a > b$ .

### Functions

| API                                                    | Intent                        | Return                      |
|--------------------------------------------------------|-------------------------------|-----------------------------|
| <code>proven_array_sort(arr, cmp)</code>               | Sort array in place.          | void.                       |
| <code>proven_array_binary_search(arr, key, cmp)</code> | Search sorted array.          | pointer to element or null. |
| <code>proven_array_linear_search(arr, key, cmp)</code> | Search any array by scanning. | pointer to element or null. |

`proven_array_sort` is an introsort: a Bentley-McIlroy three-way partition, an insertion-sort cutoff for small ranges, and a heapsort fallback once the recursion exceeds  $2 \cdot \log_2(n)$  levels.

Two properties are worth stating, because they are the ones that bite:

- **$O(n \log n)$  is a guarantee, not a typical case.** The heapsort fallback is what makes it one. A sort whose worst case can be reached by an adversarial ordering is a denial of service in any program that sorts data it did not author.

- **Duplicate keys are the fast case, not the slow one.** Elements equal to the pivot are collected into a run that is final and never recursed into, so all-equal input costs a single pass. This matters because low-cardinality keys
  - a status column, an enum, a bucket id - are what callers actually sort by, and they are exactly what a naive two-way partition degrades on.

The sort is not stable: equal elements may be reordered.

The comparator is an ordinary file-scope function, so the shape is:

```
static int cmp_int(const void *a, const void *b) {
    int x = *(const int *)a;
    int y = *(const int *)b;
    return (x > y) - (x < y);    /* not (x - y): that overflows */
}
```

```
proven_array_sort(&nums, cmp_int);
int key = 20;
int *hit = proven_array_binary_search(&nums, &key, cmp_int);
```

The worked example at the end of this chapter (`manual/examples/ex_04_array.c`) is the compiled-and-run version of exactly this.

## 6. Hashing, by use case

There is no single "hash". Which function is correct depends entirely on what you are doing with the result, and reaching for the wrong one gives you either a program that is needlessly slow or one that is quietly insecure. `proven/hash.h` offers exactly one primitive per job so the choice is made once you name the job:

| You are...                                                             | Use                                          | And crucially                                                                                   |
|------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------|
| hashing keys into <b>your own</b> table, <b>trusted</b> input          | <code>proven_hash_bytes</code> (FNV-1a)      | fast; a crypto hash here is ~50x slower for no gain                                             |
| hashing keys from <b>untrusted</b> input into a table                  | <code>proven_hash_keyed</code> (SipHash-2-4) | FNV lets an attacker collide every key into one bucket and turn your $O(1)$ table into $O(n^2)$ |
| checking data was not <b>corrupted</b> in transit or on disk           | <code>proven_crc32</code>                    | a checksum; interoperates with gzip/zlib/PNG                                                    |
| <b>fingerprinting</b> content: dedup, content-addressing, "same file?" | <code>proven_sha256</code>                   | the only one safe against a <i>deliberately</i> forged match                                    |

The one line to remember: **CRC-32 and FNV detect accident, not attack.** Do not use them to decide whether two things are "the same" when someone might benefit from fooling you – that is what `proven_sha256` is for. And a keyed hash is only safe if the key is a real secret chosen once from real randomness; a fixed key is no key at all.

Every function is byte-exact and endianness-independent: the same input gives the same output on any target, because a fingerprint that changed with the machine would be a fingerprint of the machine, not the content. All four are royalty-free algorithms (FNV, CRC-32: public domain; SipHash: CC0; SHA-256: FIPS 180-4, unpatented), implemented from their specifications and checked against each one's official known-answer vectors.

### Reference

| API | Intent | Return |
|-----|--------|--------|
|-----|--------|--------|

|                                       |                                                                                                                                                |                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| proven_hash_bytes(view)               | FNv-1a 64. A hash-table hash for keys you chose yourself.                                                                                      | proven_u64.                            |
| proven_hash_keyed(view, key[16])      | SipHash-2-4 under a 16-byte secret. The same job, for keys an attacker supplies.                                                               | proven_u64.                            |
| proven_crc32(view)                    | One-shot CRC-32 (IEEE, reflected) - the one gzip/zlib/PNG carry.                                                                               | proven_u32.                            |
| proven_crc32_update(crc, view)        | The same CRC over a stream of chunks. Start from 0; the value you hold between calls is the real CRC, so you can store it, log it, and resume. | proven_u32.                            |
| proven_sha256(view, out[32])          | One-shot SHA-256.                                                                                                                              | void; writes PROVEN_SHA256_SIZE bytes. |
| proven_sha256_init/_update/_final     | The same digest over content you cannot hold in memory at once. The digest depends only on the bytes, never on how they were chunked.          | void.                                  |
| proven_sha256_to_hex(digest, out[65]) | The 64-character lowercase spelling sha256sum and git print. NUL-terminated.                                                                   | void.                                  |

The one structure you hold:

```
typedef struct {
    /* Opaque. A running SHA-256: the 8-word chain value, a 64-byte block being
       filled, and the message length. Declare one, init it, update it, finalise it. */
} proven_sha256_t;
```

```
#define PROVEN_SHA256_SIZE 32 /* the digest; size your output buffer with this */
```

proven\_sha256\_t allocates nothing - it is [caller-owned state](#), so there is nothing to destroy.

### Cautions, and what goes wrong

**A keyed hash with a fixed key is not a keyed hash.** The security of proven\_hash\_keyed rests entirely on the key being a secret drawn once from real randomness. This is why proven\_map\_create does it for you and why you should almost never call proven\_hash\_keyed yourself for a table.

Wrong:

```
proven_byte_t key[16] = { 0 }; /* wrong: a "secret" everyone knows */
proven_u64 h = proven_hash_keyed(user_input, key);

proven_byte_t key[16];
memcpy(key, &timestamp, sizeof timestamp); /* wrong: guessable, and mostly zero */
```

Correct - draw it once, at startup, from the OS:

```
proven_byte_t key[16];
if (proven_random_bytes(key, sizeof key)) {
    proven_u64 h = proven_hash_keyed(
        proven_mem_view_from_u8(PROVEN_LIT("session-token")), key);
    (void)h;
}
```

**Do not use CRC-32 or FNV to decide two things are "the same"** when someone might gain by fooling you. Both are trivially forgeable: producing a second input with the same CRC-32 is schoolbook arithmetic.

Wrong - a content-addressed store an attacker can poison:

```
if (proven_crc32(incoming) == stored_crc) {
    accept_as_identical(incoming); /* wrong: a forged collision costs seconds */
}
```

Correct - a fingerprint that must not be foolable is proven\_sha256, compared over all 32 bytes.

**A digest buffer that is not PROVEN\_SHA256\_SIZE is a buffer overflow.** proven\_sha256 writes exactly 32 bytes and does not know how big your array is.

```
proven_byte_t digest[16]; /* wrong: SHA-256 is 32 bytes */
proven_sha256(data, digest); /* writes 32 into a 16-byte buffer */
```

Compiled and run by the test suite:

```
/*
 * Hashing, by use case. The module gives you exactly one function per job, so the only
 * decision is which job you have - and getting THAT wrong is the whole danger.
 */

int main(void) {
    proven_mem_view_t data = proven_mem_view_from_u8(PROVEN_LIT("the quick brown fox"));

    /* Job 1: hash a key into your own table, trusted input. Fast, non-cryptographic. */
    proven_u64 table_hash = proven_hash_bytes(data);
    EXAMPLE_REQUIRE(table_hash != 0, "FNV-1a produces a spread-out 64-bit value");

    /* Job 2: hash a key from UNTRUSTED input. Same purpose, but an attacker who picks the
     * input still cannot make everything collide, because they do not have the key. Pick
     * the key once at startup from real randomness; a fixed key defeats the point. */
    proven_byte_t key[16] = { 0 }; /* in real code: fill from a random source, once */
    proven_u64 safe_hash = proven_hash_keyed(data, key);
    EXAMPLE_REQUIRE(safe_hash != table_hash, "a keyed hash is a different function");

    /* Job 3: did these bytes get corrupted? A checksum, not a hash. Interoperates with
     * gzip/zlib/PNG, which all carry this exact CRC-32. */
    proven_u32 checksum = proven_crc32(data);
    /* The canonical CRC-32 sanity value, so you can see it is the real one: */
    EXAMPLE_REQUIRE(proven_crc32(proven_mem_view_from_u8(PROVEN_LIT("123456789"))) == 0xcbf43926u,
        "CRC-32 of \"123456789\" is the shared check value");
    (void)checksum;

    /* Job 4: fingerprint content - dedup, content-addressing, "are these the same file",
     * answered safely even against someone trying to forge a match. This is the one you
     * reach for when the answer must not be foolable. */
    proven_byte_t digest[PROVEN_SHA256_SIZE];
    proven_sha256(data, digest);

    char hex[65];
    proven_sha256_to_hex(digest, hex);
    /* The same spelling sha256sum and git print, so it interoperates: */
    EXAMPLE_REQUIRE(hex[64] == '\0' && proven_cstr_len(hex) == 64,
        "a SHA-256 fingerprint is 64 lowercase hex characters");

    /* SHA-256 streams, for content you cannot hold in memory at once - the digest depends
     * only on the bytes, never on how they were chunked. */
    proven_sha256_t ctx;
```

```

proven_sha256_init(&ctx);
proven_sha256_update(&ctx, proven_mem_view_from_u8(PROVEN_LIT("the quick ")));
proven_sha256_update(&ctx, proven_mem_view_from_u8(PROVEN_LIT("brown fox")));
proven_byte_t streamed[PROVEN_SHA256_SIZE];
proven_sha256_final(&ctx, streamed);

bool same = true;
for (proven_size_t i = 0; i < PROVEN_SHA256_SIZE; ++i) {
    if (streamed[i] != digest[i]) same = false;
}
EXAMPLE_REQUIRE(same, "two updates of the halves equal one hash of the whole");

return EXAMPLE_OK();
}

```

## Legacy digests: SHA-1 and MD5

proven/hash\_legacy.h holds two digests that are broken and that other people's formats still name. Someone who chooses the input can build two different messages with the same MD5 in seconds, and with the same SHA-1 for a known and affordable cost. Neither may decide whether two things are "the same" when someone gains by fooling you - that stays proven\_sha256.

They are here for the cases where you do not get to choose:

| The format says... | Use         | Where you meet it                                                         |
|--------------------|-------------|---------------------------------------------------------------------------|
| SHA-1              | proven_sha1 | the WebSocket accept key (RFC 6455); sha1sum files; Git object names      |
| MD5                | proven_md5  | HTTP Digest authentication against old servers; md5sum files; Content-MD5 |

| API                                 | Intent                                                                     | Return                               |
|-------------------------------------|----------------------------------------------------------------------------|--------------------------------------|
| proven_sha1(view, out[20])          | One-shot SHA-1.                                                            | void; writes PROVEN_SHA1_SIZE bytes. |
| proven_sha1_init/_update/_final     | The same digest over a stream of chunks.                                   | void.                                |
| proven_sha1_to_hex(digest, out[41]) | The 40-character lowercase spelling sha1sum and git print. NUL-terminated. | void.                                |
| proven_md5(view, out[16])           | One-shot MD5.                                                              | void; writes PROVEN_MD5_SIZE bytes.  |
| proven_md5_init/_update/_final      | The same digest over a stream of chunks.                                   | void.                                |
| proven_md5_to_hex(digest, out[33])  | The 32-character lowercase spelling md5sum prints. NUL-terminated.         | void.                                |

proven\_sha1\_t and proven\_md5\_t are caller-owned state like proven\_sha256\_t: nothing is allocated and there is nothing to destroy. Both digests are implemented from their specifications (FIPS 180-4, RFC 1321) and checked against the vectors those documents print; the header is available in freestanding builds.

Wrong - a new design that picks a broken digest because it is shorter:

```

proven_byte_t id[PROVEN_MD5_SIZE];
proven_md5(upload, id);          /* wrong: two different uploads can share this id */
store_under(id, upload);

```

Correct - when the choice is yours it is proven\_sha256; truncate its output if 32 bytes is too long for the place it goes.

Compiled and run by the test suite:

```

/*
 * SHA-1 and MD5. Both are broken for anything an adversary can influence, and both are still
 * written into formats nobody can change - which is the only reason to call them.
 */

int main(void) {
    /* The WebSocket handshake (RFC 6455): the server proves it read the client's key by
     * returning SHA-1(key + a fixed GUID), in Base64. The format says SHA-1, so SHA-1 it is. */
    proven_sha1_t ctx;
    proven_sha1_init(&ctx);
    proven_sha1_update(&ctx, proven_mem_view_from_u8(PROVEN_LIT("dGhlIHNhbXBsZSBub25jZQ==")));
    proven_sha1_update(&ctx, proven_mem_view_from_u8(PROVEN_LIT("258EAF45-E914-47DA-95CA-C5AB0DC85B11")));
    proven_byte_t sha1[PROVEN_SHA1_SIZE];
    proven_sha1_final(&ctx, sha1);

    proven_byte_t accept[32];
    proven_size_t accept_len = 0;
    proven_err_t err = proven_base64_encode((proven_mem_view_t){ .ptr = sha1, .size = sizeof sha1 },
                                           accept, sizeof accept, &accept_len);
    EXAMPLE_REQUIRE(proven_is_ok(err), "20 bytes encode to 28 Base64 characters");
    EXAMPLE_REQUIRE(proven_u8str_view_eq((proven_u8str_view_t){ .ptr = accept, .size = accept_len },
                                         PROVEN_LIT("s3pPLMBiTxaQ9kYGzzhZRbK+x0o=")),
                    "the accept key printed in RFC 6455 section 1.3");

    /* One-shot, and the hex spelling shasum and git print. */
    proven_mem_view_t data = proven_mem_view_from_u8(PROVEN_LIT("abc"));
    proven_sha1(data, sha1);
    char sha1_hex[41];
    proven_sha1_to_hex(sha1, sha1_hex);
    EXAMPLE_REQUIRE(proven_cstr_len(sha1_hex) == 40, "a SHA-1 digest is 40 hex characters");

    /* MD5: checking a download against a checksum file published as md5sum output. That
     * catches a damaged download. It does not catch a forged one - anyone can build a second
     * file with the same MD5. */
    proven_byte_t md5[PROVEN_MD5_SIZE];
    proven_md5(data, md5);
    char md5_hex[33];
    proven_md5_to_hex(md5, md5_hex);
    EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_view_from_cstr(md5_hex),
                                         PROVEN_LIT("900150983cd24fb0d6963f7d28e17f72")),
                    "the digest md5sum prints for \"abc\"");

    /* MD5 streams too; the digest depends only on the bytes. */
    proven_md5_t mctx;
    proven_md5_init(&mctx);
    proven_md5_update(&mctx, proven_mem_view_from_u8(PROVEN_LIT("a")));
    proven_md5_update(&mctx, proven_mem_view_from_u8(PROVEN_LIT("bc")));
    proven_byte_t streamed[PROVEN_MD5_SIZE];
    proven_md5_final(&mctx, streamed);

    bool same = true;
    for (proven_size_t i = 0; i < PROVEN_MD5_SIZE; ++i) {

```

```

    if (streamed[i] != md5[i]) same = false;
}
EXAMPLE_REQUIRE(same, "two updates of the pieces equal one hash of the whole");

return EXAMPLE_OK();
}

```

## 7. Bytes to text: hex and Base64

Once you can hash a thing (above) and draw a random token (`random.h`), you need to write those bytes somewhere that only holds text - a URL, an HTTP header, a log line, a JSON string. That is `encode.h`. No cryptography, no compression; the two encodings everything already agrees on, done without hidden allocation and without the two ways they are usually got wrong.

| You want                              | Use                                  | Alphabet                                                                           |
|---------------------------------------|--------------------------------------|------------------------------------------------------------------------------------|
| A digest or a few bytes a human reads | <code>proven_hex_encode</code>       | lowercase hex, what <code>sha256sum</code> and <code>git print</code>              |
| Bytes in a URL, a cookie, a filename  | <code>proven_base64url_encode</code> | - <code>_</code> , <b>no</b> padding - nothing to escape, <code>=</code> to mangle |
| Bytes in an HTTP header, MIME, JSON   | <code>proven_base64_encode</code>    | standard <code>+</code> <code>/</code> , <code>=</code> -padded                    |

Two refusals are the point:

- **A decoder validates its whole input before writing a byte.** Text from outside the program is not guaranteed to be valid; a stray character, a bad length, bad padding, or embedded whitespace is `PROVEN_ERR_INVALID_ENCODING` with nothing committed - not a read past the end, and not a silently short result one byte into which the caller finds the corruption. `proven_base64_decode` accepts **both** alphabets and padded-or-not, because a decoder that only takes what it emits rejects half the Base64 in the world.
- **The output size is a call, not a guess** - `proven_hex_encoded_size`, `proven_base64_encoded_size`, and their decode counterparts. A buffer one byte too small is `PROVEN_ERR_OUT_OF_BOUNDS` with nothing written, never a truncated prefix.

It is pure computation - no allocation, no OS - and available freestanding.

### Reference

Every call takes the input, a caller-owned output buffer and its capacity, and an optional `written_out`. None of them allocate.

| API                                                    | Intent                                                                                               | Return                                                                                                                                                                                                                                                   |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_hex_encoded_size(n)</code>                | The characters <code>proven_hex_encode</code> will write for <code>n</code> bytes: $n * 2$ , no NUL. | <code>proven_size_t</code> ; <code>PROVEN_SIZE_MAX</code> when $n * 2$ does not fit.                                                                                                                                                                     |
| <code>proven_hex_decoded_size(n)</code>                | The bytes <code>proven_hex_decode</code> will write for <code>n</code> characters: $n / 2$ .         | <code>proven_size_t</code> .                                                                                                                                                                                                                             |
| <code>proven_hex_encode(data, out, cap, &amp;w)</code> | Lowercase hex.                                                                                       | <code>PROVEN_OK</code> ; <code>OUT_OF_BOUNDS</code> if <code>cap</code> is short ( <b>nothing written</b> ); <code>OVERFLOW</code> if the output size does not fit at all; <code>INVALID_ARG</code> for a NULL out or a <code>{NULL, &gt;0}</code> view. |

|                                             |                                                                              |                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| proven_hex_decode(text, out, cap, &w)       | Decode hex; upper and lower case both accepted.                              | INVALID_ENCODING for an odd length or any non-hex byte ( <b>nothing committed</b> ). |
| proven_base64_encoded_size(n)               | An upper bound for both Base64 forms: $4 * \text{ceil}(n/3)$ .               | proven_size_t; PROVEN_SIZE_MAX when that does not fit.                               |
| proven_base64_decoded_size(n)               | An upper bound for <b>padded and unpadded</b> text: $3 * \text{ceil}(n/4)$ . | proven_size_t.                                                                       |
| proven_base64_encode(data, out, cap, &w)    | Standard alphabet (+ /), <b>no</b> padded.                                   | as above.                                                                            |
| proven_base64url_encode(data, out, cap, &w) | URL-safe alphabet (- _), <b>no padding</b> .                                 | as above.                                                                            |
| proven_base64_decode(text, out, cap, &w)    | Decode. Accepts <b>both</b> alphabets, padded or not.                        | INVALID_ENCODING for a stray byte, a bad length, or bad padding.                     |

written\_out may be NULL. It is set to 0 on entry and on every error path, so it is never stale.

### Cautions, and what goes wrong

**Size the output with the size function, not by eye.** The encoders refuse a short buffer rather than truncating - which means the failure you get from a hand-computed size is an error you have to handle, not a silent corruption. That is the good outcome; the point is you will still have to handle it.

Wrong - the classic off-by-one, and it now *fails* instead of overflowing:

```
proven_byte_t out[16];                /* wrong: 12 bytes of hex needs 24 chars */
proven_hex_encode(twelve_bytes, out, sizeof out, &w); /* OUT_OF_BOUNDS, nothing written */
```

Correct:

```
proven_mem_view_t data = proven_mem_view_from_u8(PROVEN_LIT("twelve bytes"));
proven_byte_t out[64];
proven_size_t w = 0;
if (proven_hex_encoded_size(data.size) <= sizeof out &&
    proven_is_ok(proven_hex_encode(data, out, sizeof out, &w))) {
    /* `w` characters of lowercase hex in `out` - not NUL-terminated */
}
```

**Base64URL emits no padding, and the decoder accepts that.** proven\_base64\_decoded\_size rounds *up* so it is a correct upper bound for unpadded text too. Do not "improve" on it with  $3 * (n / 4)$  - that floors away the 1-2 bytes an unpadded tail carries, and you will fail to decode this library's own URL-safe output. (It did exactly that, once, and an audit caught it.)

```
proven_size_t cap = 3 * (text_len / 4); /* wrong: 0 for "QQ", which decodes to 1 byte */
```

**Never hand-roll the decoder.** The whole reason these exist is that a decode reads text from outside your program, and the two-line loop everyone writes trusts it.

Wrong - reads past the end on odd input, and accepts junk as data:

```
for (size_t i = 0; i < len; i += 2) /* wrong: no length check */
    out[i/2] = (hexval(text[i]) << 4) | hexval(text[i+1]); /* wrong: no validation */
```

proven\_hex\_decode validates the **whole** input before writing a single byte, so a stray character near the end cannot leave you holding a half-decoded prefix you believe is complete.

**Whitespace is not skipped, on purpose.** A pasted, line-wrapped Base64 blob is INVALID\_ENCODING, not a silently different result. If you *want* to accept wrapped input, strip the whitespace yourself - deliberately, where you can see it.

**Size the destination with the size functions, never by hand.** Each encoding has a pair - `proven_hex_encoded_size()` / `proven_hex_decoded_size()` and `proven_base64_encoded_size()` / `proven_base64_decoded_size()` - and they differ in what they promise:

| Direction | What the number is                                                                                                                        | How to use it                                                                  |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Encoding  | <b>Exact</b> for hex and standard Base64. Base64URL omits padding, so its output can be shorter - the number is still a safe upper bound. | Allocate this many bytes; the call reports what it wrote.                      |
| Decoding  | An <b>upper bound</b> , because padding and the alphabet in use are not known until the text is read.                                     | Allocate the bound, then use the reported count, not the bound, as the length. |

The example below ends with that: it allocates a destination sized by `proven_base64_encoded_size()`, encodes into it, sizes the reverse buffer with `proven_base64_decoded_size()`, decodes, and round-trips the same bytes through `proven_hex_decode()` - including the uppercase input real tools produce, and the odd-length input that cannot be a whole number of bytes.

Compiled and run by the test suite:

```
/*
 * Bytes to text, by use case. The rule is the same one hashing follows: one function per job,
 * and the danger is picking the wrong job. Hex for something a human reads; Base64URL for
 * something that goes in a URL; standard Base64 for something that goes on the wire.
 */

int main(void) {
    proven_mem_view_t data = proven_mem_view_from_u8(PROVEN_LIT("the quick brown fox"));

    /* Job 1: a digest a human will read or paste - hex, the spelling sha256sum and git use. */
    proven_byte_t hex[64]; /* proven_hex_encoded_size(19) = 38 */
    proven_size_t hn = 0;
    EXAMPLE_REQUIRE(proven_is_ok(proven_hex_encode(data, hex, sizeof hex, &hn)),
        "hex encode into a buffer sized by proven_hex_encoded_size");
    EXAMPLE_REQUIRE(hn == proven_hex_encoded_size(data.size), "two hex chars per byte");

    /* Job 2: a token that goes in a URL - Base64URL, so nothing needs percent-escaping and
     * there is no '=' padding for a parser to trip over. */
    proven_byte_t token_bytes[16] = { 0 }; /* in real code: proven_random_bytes(token_bytes, 16) */
    proven_byte_t url[32];
    proven_size_t un = 0;
    EXAMPLE_REQUIRE(proven_is_ok(proven_base64url_encode(
        (proven_mem_view_t){ token_bytes, sizeof token_bytes }, url, sizeof url, &un)),
        "base64url encode a token");
    /* No '=' in a URL-safe token. */
    bool has_pad = false;
    for (proven_size_t i = 0; i < un; ++i) if (url[i] == '=') has_pad = true;
    EXAMPLE_REQUIRE(!has_pad, "the URL form emits no padding");

    /* Job 3: bytes on the wire - standard Base64, the +/= alphabet HTTP and MIME expect. */
    proven_byte_t b64[64];
    proven_size_t bn = 0;
    EXAMPLE_REQUIRE(proven_is_ok(proven_base64_encode(data, b64, sizeof b64, &bn)),
        "standard base64 encode");

    /* And it round-trips: decode gives back exactly the bytes. A decoder that accepts both
     * alphabets and padded-or-not is deliberate - real input comes in every shape. */
    proven_byte_t back[32];
```

```

proven_size_t dn = 0;
EXAMPLE_REQUIRE(proven_is_ok(proven_base64_decode(
    (proven_mem_view_t){ b64, bn }, back, sizeof back, &dn)),
    "decode the base64 back");
EXAMPLE_REQUIRE(dn == data.size && proven_memcmp(back, data.ptr, dn) == 0,
    "what comes back is exactly what went in");

/* The point of a validating decoder: junk is refused, not guessed. A caller who fed this
 * to a two-line loop would read past the end or get a silently short result. */
proven_err_t bad = proven_base64_decode(
    proven_mem_view_from_u8(PROVEN_LIT("not valid base64!!")), back, sizeof back, &dn);
EXAMPLE_REQUIRE(bad == PROVEN_ERR_INVALID_ENCODING,
    "a stray character is INVALID_ENCODING, with nothing committed");

/* And a buffer one byte too small is refused, never truncated. */
proven_byte_t tiny[4];
EXAMPLE_REQUIRE(proven_hex_encode(data, tiny, sizeof tiny, &hn) == PROVEN_ERR_OUT_OF_BOUNDS,
    "a too-small output buffer is OUT_OF_BOUNDS, not a truncated prefix");

/* --- sizing the destination buffer, instead of guessing ----- */

/* Every encode and decode above wrote into a fixed array that was obviously
 * big enough. Real code allocates the destination, and then the size has to
 * be computed rather than remembered. That is what the four size functions
 * are for - one per direction, per encoding. */
proven_allocator_t alloc = proven_heap_allocator();

proven_size_t need = proven_base64_encoded_size(data.size);
EXAMPLE_REQUIRE(need > 0, "an encoded size must be computed, not guessed");

proven_result_mem_mut_t out = alloc.alloc_fn(alloc.ctx, need, 1);
EXAMPLE_REQUIRE(proven_is_ok(out.err), "allocating exactly the encoded size must succeed");

proven_size_t wrote = 0;
EXAMPLE_REQUIRE(proven_is_ok(proven_base64_encode(data, out.value.ptr, out.value.size, &wrote)),
    "encoding into a buffer sized by the size function must fit exactly");
EXAMPLE_REQUIRE(wrote == need, "and use all of it: this size is exact for standard base64");

/* The decode direction is an UPPER BOUND, not an exact count: base64 text
 * may carry padding, so the decoder reports how many bytes it really wrote.
 * Size the buffer with the bound; use the reported count afterwards. */
proven_size_t bound = proven_base64_decoded_size(wrote);
EXAMPLE_REQUIRE(bound >= data.size, "the decoded bound must cover the real output");

proven_result_mem_mut_t plain = alloc.alloc_fn(alloc.ctx, bound, 1);
EXAMPLE_REQUIRE(proven_is_ok(plain.err), "allocating the decode bound must succeed");

proven_size_t got = 0;
EXAMPLE_REQUIRE(proven_is_ok(proven_base64_decode(
    (proven_mem_view_t){ out.value.ptr, wrote }, plain.value.ptr, plain.value.size,
&got)),
    "decoding back must succeed");
EXAMPLE_REQUIRE(got == data.size && proven_memcmp(plain.value.ptr, data.ptr, got) == 0,
    "and reproduce the original bytes exactly");

/* Hex has the same pair. Its decoded bound is exact when the input length is
 * even, which valid hex always is - an odd length is malformed input, and
 * the decoder says so rather than dropping the last digit. */
proven_size_t hex_len = 0;
EXAMPLE_REQUIRE(proven_is_ok(proven_hex_encode(data, hex, sizeof hex, &hex_len)),

```

```

        "re-encode to hex, since the refused call above left its count unusable");

proven_size_t hex_bound = proven_hex_decoded_size(hex_len);
EXAMPLE_REQUIRE(hex_bound == data.size, "two hex characters decode to one byte");

proven_byte_t from_hex[32];
proven_size_t hex_got = 0;
EXAMPLE_REQUIRE(proven_is_ok(proven_hex_decode(
    (proven_mem_view_t){ hex, hex_len }, from_hex, sizeof from_hex, &hex_got)),
    "hex decodes back to the original bytes");
EXAMPLE_REQUIRE(hex_got == data.size && proven_memcmp(from_hex, data.ptr, hex_got) == 0,
    "and the round trip is exact");

/* Hex is case-insensitive on the way in, which matters because different
 * tools print different cases of the same digest. */
EXAMPLE_REQUIRE(proven_is_ok(proven_hex_decode(
    proven_mem_view_from_u8(PROVEN_LIT("DEADBEEF")), from_hex, sizeof from_hex,
    &hex_got)),
    "uppercase hex decodes too");
EXAMPLE_REQUIRE(hex_got == 4, "and yields one byte per two characters");

/* An odd number of characters cannot be a whole number of bytes. */
EXAMPLE_REQUIRE(proven_hex_decode(proven_mem_view_from_u8(PROVEN_LIT("abc")),
    from_hex, sizeof from_hex, &hex_got) == PROVEN_ERR_INVALID_ENCODING,
    "an odd-length hex string is malformed, not silently truncated");

alloc.free_fn(alloc.ctx, plain.value.ptr);
alloc.free_fn(alloc.ctx, out.value.ptr);

(void)url; (void)un;
return EXAMPLE_OK();
}

```

## 8. Examples and misuse cases

### Pointers into arrays can become stale

Wrong:

```

int *p = PROVEN_ARRAY_GET_MUT(&nums, int, 0);
PROVEN_ARRAY_PUSH(&nums, int, 30);
*p = 99; /* wrong: push may have reallocated the array */

```

Correct:

```

proven_result_array_t r = PROVEN_ARRAY_INIT(alloc, int, 2);
if (!proven_is_ok(r.err)) {
    return;
}
proven_array_t nums = r.value;
(void)PROVEN_ARRAY_PUSH(&nums, int, 10);

/* Push first, then take the pointer. A pointer fetched before a push may point
 * at a block the array has already reallocated away. */
(void)PROVEN_ARRAY_PUSH(&nums, int, 30);
int *p = PROVEN_ARRAY_GET_MUT(&nums, int, 0);
if (p) {
    *p = 99;
}

PROVEN_ARRAY_DESTROY(&nums);

```

## One intrusive node belongs to one list at a time

Wrong:

```
proven_list_push_back(&list_a, &item.link);
proven_list_push_back(&list_b, &item.link); /* wrong */
```

Use one embedded node per membership.

## Remove while iterating with the safe macro

Correct:

```
typedef struct Item {
    int value;
    proven_list_node_t link;
} Item;

proven_list_t list;
proven_list_init(&list);
Item a = { .value = 1 };
Item b = { .value = 2 };
proven_list_push_back(&list, &a.link);
proven_list_push_back(&list, &b.link);

/* The SAFE form reads `it->next` into `next` before the body runs, so removing
 * `it` (which nulls its links) cannot strand the loop. */
proven_list_node_t *it = NULL;
proven_list_node_t *next = NULL;
PROVEN_LIST_FOR_EACH_SAFE(it, next, &list) {
    Item *item = PROVEN_LIST_ENTRY(it, Item, link);
    if (item->value % 2 == 0) {
        proven_list_remove(it);
    }
}
```

## Ring push does not grow

Wrong:

```
PROVEN_RING_PUSH(&q, int, value); /* wrong if you ignore full-ring errors */
```

Correct:

```
proven_result_ring_t r = PROVEN_RING_INIT(alloc, int, 2);
if (!proven_is_ok(r.err)) {
    return;
}
proven_ring_t q = r.value;

int value = 7;
for (int i = 0; i < 3; ++i) {
    proven_err_t e = PROVEN_RING_PUSH(&q, int, value);
    if (e == PROVEN_ERR_OUT_OF_BOUNDS) {
        /* The ring is full. Drop the item, or drain one first - but decide. */
        int drop = 0;
        (void)PROVEN_RING_POP(&q, int, &drop);
        e = PROVEN_RING_PUSH(&q, int, value);
    }
    (void)e;
}

PROVEN_RING_DESTROY(&q);
```

## Borrowed map keys must outlive entries

Wrong:

```
proven_u8str_t key = make_key(alloc);
PROVEN_MAP_SET_U8_BORROWED(&m, proven_u8str_as_view(&key), int, 1);
proven_u8str_destroy(alloc, &key);
/* wrong: map still points at freed key bytes */
```

Correct options:

- Use string literals or other stable storage for borrowed keys.
- Store owned key objects elsewhere and destroy them after removing map entries.
- Use `PROVEN_KEY_TYPE_U8_OWNED` and `proven_map_set_u8_owned()` when the map should own key storage.

## Use scratch when inserting a value borrowed from the same map

If `element` points into map storage and insertion can rehash, use `proven_map_set_with_scratch()` or the `scratch` macros.

```
proven_result_map_t r = PROVEN_MAP_INIT_INT(alloc, int, 4);
if (!proven_is_ok(r.err)) {
    return;
}
proven_map_t m = r.value;
(void)PROVEN_MAP_SET_INT(&m, 1, int, 42);

/* `src` points into the map's own bucket array. Passing it straight to
 * proven_map_set would be an alias: a rehash triggered by that same insert can
 * free the bytes it is still reading from. The scratch form captures the source
 * bytes into a temporary from `scratch` first, so the insert is alias-safe. */
const int *src = PROVEN_MAP_GET_INT(&m, int, 1);
if (src) {
    (void)proven_map_set_with_scratch(&m, (proven_map_key_t){ .id = 2 }, src, scratch);
}

PROVEN_MAP_DESTROY(&m);
```

## Binary search requires sorted input

Wrong:

```
int *hit = proven_array_binary_search(&nums, &key, cmp_int);
/* wrong if nums was not sorted by cmp_int */
```

## Worked example: array, sort, binary search

Compiled and run by the test suite. Note the sort's behaviour on duplicate keys: they are the *fast* case, not the quadratic one.

```
/*
 * proven_array_t is a growable vector of one element type. It stores the
 * allocator you created it with, so push can grow and destroy can free without
 * you passing the allocator back in - which also means the array must be
 * destroyed with nothing but itself.
 *
 * The PROVEN_ARRAY_* macros are the type-safe face of the void* core: they
 * derive sizeof/alignof from the type and hand the element to the array by
 * pointer, so pushing an int into an array of records will not compile.
 */

/* A task queue keyed by priority. Real data is low-cardinality: three
 * priorities, many tasks. That matters for the sort - see below. */
typedef struct {
    int priority;
    int id;
```

```

} task_t;

/* The comparator is the array's ordering. The same one MUST be used for the
 * sort and for the binary search - a search under a different order is a bug
 * the compiler cannot see.
 *
 * The (x > y) - (x < y) form is deliberate: subtracting the two ints would
 * overflow for large values and hand back a nonsense sign. */
static int cmp_priority(const void *a, const void *b) {
    int x = ((const task_t *)a)->priority;
    int y = ((const task_t *)b)->priority;
    return (x > y) - (x < y);
}

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- push / get / pop ----- */
    /* init_cap is a hint, not a limit: push grows past it. Sizing it right just
     * avoids the reallocations. */
    proven_result_array_t r = PROVEN_ARRAY_INIT(alloc, task_t, 4);
    EXAMPLE_REQUIRE(proven_is_ok(r.err), "creating an array of task_t must succeed");
    if (!proven_is_ok(r.err)) {
        return 1;
    }
    proven_array_t tasks = r.value;

    /* Deliberately duplicate-heavy, because that is what real keys look like:
     * a status column, an enum, a bucket id. */
    static const task_t seed[] = {
        { .priority = 2, .id = 10 }, { .priority = 0, .id = 11 },
        { .priority = 2, .id = 12 }, { .priority = 1, .id = 13 },
        { .priority = 2, .id = 14 }, { .priority = 0, .id = 15 },
        { .priority = 1, .id = 16 }, { .priority = 2, .id = 17 },
    };

    for (proven_size_t i = 0; i < sizeof seed / sizeof seed[0]; ++i) {
        proven_err_t err = PROVEN_ARRAY_PUSH(&tasks, task_t, seed[i]);
        EXAMPLE_REQUIRE(proven_is_ok(err), "pushing into a growable array must succeed");
        if (!proven_is_ok(err)) {
            PROVEN_ARRAY_DESTROY(&tasks);
            return 1;
        }
    }
    EXAMPLE_REQUIRE(tasks.len == 8, "eight pushes give eight elements");

    /* get returns a pointer INTO the array's storage - it is not a copy, and the
     * next push may reallocate and leave it dangling. Fetch it after the pushes,
     * use it, do not store it. */
    const task_t *front = PROVEN_ARRAY_GET(&tasks, task_t, 0);
    EXAMPLE_REQUIRE(front && front->id == 10, "element 0 is the first one pushed");

    /* Out of range is a null pointer, not a crash and not UB. */
    EXAMPLE_REQUIRE(PROVEN_ARRAY_GET(&tasks, task_t, 99) == NULL, "an out-of-range index yields NULL");

    /* pop copies the last element out (pass NULL to just discard it). */
    task_t last = {0};
    proven_err_t err = PROVEN_ARRAY_POP(&tasks, task_t, &last);
    EXAMPLE_REQUIRE(proven_is_ok(err), "popping a non-empty array must succeed");
    EXAMPLE_REQUIRE(last.id == 17 && tasks.len == 7, "pop returns the last element and shrinks the array");

```

```

/* Put it back: the rest of the example wants all eight. */
err = PROVEN_ARRAY_PUSH(&tasks, task_t, last);
EXAMPLE_REQUIRE(proven_is_ok(err), "pushing the popped element back must succeed");

/* --- sort ----- */
/* An introsort: O(n log n) is a guarantee, not an average - the heapsort
 * fallback rules out the quadratic case an adversary could otherwise steer
 * you into. And equal keys are the FAST path here: everything equal to the
 * pivot is partitioned into a run that is never recursed into, so the
 * duplicate priorities above cost less than distinct ones would, not more.
 *
 * It is not stable: task 10 and task 12 may come out in either order. */
proven_array_sort(&tasks, cmp_priority);

for (proven_size_t i = 1; i < tasks.len; ++i) {
    const task_t *prev = PROVEN_ARRAY_GET(&tasks, task_t, i - 1);
    const task_t *cur  = PROVEN_ARRAY_GET(&tasks, task_t, i);
    EXAMPLE_REQUIRE(prev && cur && prev->priority <= cur->priority,
        "after sorting, priorities must be non-decreasing");
}

/* --- binary search ----- */
/* Only legal because the array is sorted by this exact comparator. The key
 * is a whole element, but only the fields the comparator reads matter. */
task_t key = { .priority = 1, .id = 0 };
const task_t *hit = (const task_t *)proven_array_binary_search(&tasks, &key, cmp_priority);
EXAMPLE_REQUIRE(hit != NULL, "priority 1 is present, so the search must find it");
EXAMPLE_REQUIRE(hit && hit->priority == 1, "the hit must be an element with the searched key");
/* With duplicate keys it finds SOME element with that key, not the first one.
 * If you need the first, scan backwards from the hit. */

task_t absent = { .priority = 9, .id = 0 };
EXAMPLE_REQUIRE(proven_array_binary_search(&tasks, &absent, cmp_priority) == NULL,
    "a key that is not there must return NULL");

printf("tasks: %zu sorted, found priority %d (id %d)\n",
    (size_t)tasks.len, hit ? hit->priority : -1, hit ? hit->id : -1);

/* The array owns its element storage; destroy returns it through the
 * allocator the array was created with. Elements are plain data here - if
 * they owned anything, you would have to destroy each one first. */
PROVEN_ARRAY_DESTROY(&tasks);
return EXAMPLE_OK();
}

```

### Worked example: a map with integer keys and with owned string keys

Compiled and run by the test suite. The owned-key half proves the point that is easy to get wrong: the map copies the key, so the buffer you built it in is yours to reuse immediately.

```

/*
 * proven_map_t is a flat open-addressing hash map. The value type is fixed at
 * create time and stored inline in the bucket array - there is no per-entry
 * allocation for values, and get hands back a pointer straight into that array.
 *
 * The interesting decision is the KEY:
 *
 * PROVEN_KEY_TYPE_INT          - the key is a proven_size_t. Nothing to own.
 * PROVEN_KEY_TYPE_U8_BORROWED - the bucket stores your pointer and length.
 *                               The map never copies the bytes, so YOU must

```

```

*           keep them alive and unmoved for as long as
*           the entry exists. Right for string literals.
*   PROVEN_KEY_TYPE_U8_OWNED   - the map copies the key bytes into its own
*                               storage and frees them again. Right for keys
*                               built at runtime, which is most of them.
*
* The second half of this example is the reason OWNED exists.
*/

typedef struct {
    int level;
    long score;
} player_t;

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- integer keys ----- */
    proven_result_map_t r = PROVEN_MAP_INIT_INT(alloc, player_t, 8);
    EXAMPLE_REQUIRE(proven_is_ok(r.err), "creating an int-keyed map must succeed");
    if (!proven_is_ok(r.err)) {
        return 1;
    }
    proven_map_t by_id = r.value;

    proven_err_t err = PROVEN_MAP_SET_INT(&by_id, 404, player_t, ((player_t){ .level = 3, .score = 990 }));
    EXAMPLE_REQUIRE(proven_is_ok(err), "inserting into the map must succeed");
    err = PROVEN_MAP_SET_INT(&by_id, 7, player_t, ((player_t){ .level = 1, .score = 10 }));
    EXAMPLE_REQUIRE(proven_is_ok(err), "inserting a second key must succeed");

    /* set on an existing key replaces the value; it does not add an entry. */
    err = PROVEN_MAP_SET_INT(&by_id, 7, player_t, ((player_t){ .level = 2, .score = 40 }));
    EXAMPLE_REQUIRE(proven_is_ok(err), "re-setting an existing key must succeed");
    EXAMPLE_REQUIRE(by_id.len == 2, "re-setting a key replaces its value rather than adding an entry");

    /* get returns a pointer into the bucket array, or NULL when absent. It is
     * invalidated by any insert that rehashes - look it up, use it, drop it. */
    const player_t *p = PROVEN_MAP_GET_INT(&by_id, player_t, 7);
    EXAMPLE_REQUIRE(p && p->level == 2 && p->score == 40, "get must see the replaced value");
    EXAMPLE_REQUIRE(PROVEN_MAP_GET_INT(&by_id, player_t, 999) == NULL, "a missing key yields NULL");

    err = PROVEN_MAP_REMOVE_INT(&by_id, 7);
    EXAMPLE_REQUIRE(proven_is_ok(err), "removing a present key must succeed");
    EXAMPLE_REQUIRE(PROVEN_MAP_GET_INT(&by_id, player_t, 7) == NULL, "a removed key is gone");
    EXAMPLE_REQUIRE(by_id.len == 1, "removal decrements the live entry count");

    PROVEN_MAP_DESTROY(&by_id);

    /* --- owned string keys ----- */
    /* Same map, keyed by a name that we build at runtime - the case where a
     * borrowed key would be a dangling pointer waiting to happen. */
    proven_result_map_t rm = PROVEN_MAP_INIT_U8_OWNED(alloc, player_t, 8);
    EXAMPLE_REQUIRE(proven_is_ok(rm.err), "creating an owned-string-keyed map must succeed");
    if (!proven_is_ok(rm.err)) {
        return 1;
    }
    proven_map_t by_name = rm.value;

    /* A scratch buffer we intend to reuse for every key. With a BORROWED map
     * that plan is fatal: every entry would point at these same bytes. */

```

```

proven_byte_t scratch[32];
proven_u8str_t name = proven_u8str_borrow(scratch, sizeof scratch);

err = proven_u8str_append(&name, PROVEN_LIT("ada"));
EXAMPLE_REQUIRE(proven_is_ok(err), "building the first key must succeed");

/* set_u8_owned COPIES the key bytes into map storage. After it returns, the
 * map's key no longer has anything to do with `scratch`. */
err = PROVEN_MAP_SET_U8_OWNED(&by_name, proven_u8str_as_view(&name), player_t,
                              ((player_t){ .level = 9, .score = 5000 }));
EXAMPLE_REQUIRE(proven_is_ok(err), "inserting with an owned key must succeed");

/* So the buffer is immediately free to be reused for the next key... */
err = proven_u8str_reset(&name);
EXAMPLE_REQUIRE(proven_is_ok(err), "the key buffer is ours again the moment set returns");
err = proven_u8str_append(&name, PROVEN_LIT("grace"));
EXAMPLE_REQUIRE(proven_is_ok(err), "overwriting the buffer with the next key must succeed");

err = PROVEN_MAP_SET_U8_OWNED(&by_name, proven_u8str_as_view(&name), player_t,
                              ((player_t){ .level = 4, .score = 700 }));
EXAMPLE_REQUIRE(proven_is_ok(err), "inserting the second owned key must succeed");

/* ...and the first entry is untouched by that overwrite. This is the whole
 * point: the map holds its own copy of "ada", not a view of a buffer that
 * now says "grace". A BORROWED map would report two entries both keyed
 * "grace" - or worse, one keyed by freed memory. */
const player_t *ada = PROVEN_MAP_GET_U8_OWNED(&by_name, player_t, PROVEN_LIT("ada"));
EXAMPLE_REQUIRE(ada && ada->score == 5000, "the copied key survives the caller reusing its buffer");

const player_t *grace = PROVEN_MAP_GET_U8_OWNED(&by_name, player_t, PROVEN_LIT("grace"));
EXAMPLE_REQUIRE(grace && grace->score == 700, "the second key is a separate entry");
EXAMPLE_REQUIRE(by_name.len == 2, "two distinct keys means two entries");

/* Remove frees the key copy the map made - you never free it yourself. */
err = PROVEN_MAP_REMOVE_U8_OWNED(&by_name, PROVEN_LIT("ada"));
EXAMPLE_REQUIRE(proven_is_ok(err), "removing an owned key must succeed");
EXAMPLE_REQUIRE(PROVEN_MAP_GET_U8_OWNED(&by_name, player_t, PROVEN_LIT("ada")) == NULL,
                "the removed entry is gone");

printf("map: %zu name(s) left, grace at level %d\n",
       (size_t)by_name.len, grace ? grace->level : -1);

/* destroy frees the bucket array AND every key copy still in it ("grace"
 * here). `scratch` is ours and outlives the map; the borrowed `name` handle
 * has nothing to free. */
PROVEN_MAP_DESTROY(&by_name);
proven_u8str_destroy(&name);
return EXAMPLE_OK();
}

```

### Worked example: sizing containers, searching unsorted data, and streaming a checksum

The worked examples above take one container at a time. This one is the program those pieces end up inside - a small event intake - and it answers the questions that only come up once several containers are in play:

- **Stop a container reallocating while it fills.** `proven_array_reserve()` and `proven_map_reserve()` set the capacity once. On the heap that saves copying; behind an arena it saves the dead storage every reallocation leaves behind until the next reset. For a map it also matters for correctness of your pointers: a rehash invalidates every pointer a previous `get_mut` returned.

- **Check a handle you did not create.** `proven_array_is_valid()`, `proven_ring_is_valid()` and `proven_map_is_valid()` ask whether the handle's own fields are consistent - the check that catches a zero-initialised or never-created container at the boundary of your code, rather than at the first push.
- **Search data that is not sorted.** The event log is kept in arrival order, because that order is the thing being recorded, so binary search may not be used on it - on unsorted input it does not return "not found", it returns a wrong answer. `proven_array_linear_search()` is the correct call, and it returns the *first* match in order.
- **Update a value already in a map, with one lookup.** `proven_map_get_mut()` returns a pointer into the map's own storage, so a counter is incremented where it lives instead of being read, changed and written back.
- **Decide which hash the keys deserve.** `proven_map_create()` hashes string keys with keyed SipHash, so an attacker who chooses the keys cannot collide them all into one bucket. `proven_map_create_trusted()` uses fast FNV-1a and is right only when your own code chooses every key. `proven_map_hash()` returns the value the map actually places a key by, so the difference between the two is observable rather than a claim.
- **Checksum data that arrives in pieces.** `proven_crc32_update()` folds each chunk into a running value; the result equals `proven_crc32()` over the whole, whatever the chunk boundaries were.

```
/*
 * The container chapters show each structure on its own. A real program uses
 * several at once, and the questions that come up then are not about any single
 * container:
 *
 * - How do I stop a container reallocating while it fills?  reserve.
 * - How do I check a container somebody handed me is usable?  is_valid.
 * - How do I search when the data is NOT sorted?  linear search - and knowing
 *   why binary search would give a wrong answer here.
 * - How do I update a value already in a map without looking it up twice?
 *   get_mut.
 * - Do I need the attack-resistant hash, or the fast one?  it depends on who
 *   chooses the keys, and map_hash lets you see the difference.
 * - How do I checksum data that arrives in pieces?  crc32_update.
 *
 * The program is a small event intake: events arrive, are counted per client,
 * the most recent few are kept for a diagnostic dump, and the batch is
 * checksummed as it streams past.
 */
```

```
typedef struct {
    proven_u32 client;
    proven_u32 code;      /* an event code; 0 means "connection closed" */
} event_t;

/* Search by client id: this is the comparison for finding an event, not for
 * sorting them. The array below is in ARRIVAL order and stays that way. */
static int by_client(const void *a, const void *b) {
    const event_t *x = (const event_t *)a;
    const event_t *y = (const event_t *)b;
    return (x->client > y->client) - (x->client < y->client);
}

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- 1. reserve: decide the capacity once ----- */
```

```

proven_result_array_t ar = PROVEN_ARRAY_INIT(alloc, event_t, 4);
EXAMPLE_REQUIRE(proven_is_ok(ar.err), "creating the event log must succeed");
if (!proven_is_ok(ar.err)) {
    return 1;
}
proven_array_t events = ar.value;

/* We know the batch size before we start, so ask for the room once. Without
 * this the array doubles as it fills, copying its contents each time; behind
 * an arena allocator each of those copies also leaves the old block behind
 * until the arena is reset. */
proven_err_t err = proven_array_reserve(&events, 16);
EXAMPLE_REQUIRE(proven_is_ok(err), "reserving room for the whole batch must succeed");
EXAMPLE_REQUIRE(events.cap >= 16, "the capacity is now at least what was asked for");

/* is_valid asks whether the handle itself is structurally sound - a pointer,
 * a length and a capacity that agree. Assert it where a container arrives
 * from other code, or after a zero-initialised handle might have escaped;
 * it is not something to repeat after every push. */
EXAMPLE_REQUIRE(proven_array_is_valid(&events), "a created array must be structurally valid");

proven_array_t never_created = {0};
EXAMPLE_REQUIRE(!proven_array_is_valid(&never_created),
    "a zero-initialised handle is not a usable array");

static const event_t batch[] = {
    { .client = 7, .code = 200 }, { .client = 3, .code = 200 },
    { .client = 7, .code = 404 }, { .client = 9, .code = 200 },
    { .client = 3, .code = 500 }, { .client = 7, .code = 0 },
};
proven_size_t batch_len = sizeof batch / sizeof batch[0];

proven_size_t cap_before = events.cap;
for (proven_size_t i = 0; i < batch_len; ++i) {
    err = PROVEN_ARRAY_PUSH(&events, event_t, batch[i]);
    EXAMPLE_REQUIRE(proven_is_ok(err), "pushing an event must succeed");
}
EXAMPLE_REQUIRE(events.cap == cap_before, "the reserve was enough: nothing reallocated while filling");

/* --- 2. searching data that is not sorted ----- */

/* The log is in arrival order, which is the order we want to keep: it is the
 * thing being recorded. Binary search would be faster and WRONG here, since
 * it may only be used on a sorted range - on unsorted data it does not
 * return "not found", it returns nonsense. Linear search is the correct
 * tool, and O(n) over a batch this size is nothing. */
event_t key = { .client = 9, .code = 0 };
const event_t *found = (const event_t *)proven_array_linear_search(&events, &key, by_client);
EXAMPLE_REQUIRE(found != NULL, "client 9 appears in the batch");
EXAMPLE_REQUIRE(found->code == 200, "and linear search returns the FIRST match in order");

event_t absent = { .client = 42, .code = 0 };
EXAMPLE_REQUIRE(proven_array_linear_search(&events, &absent, by_client) == NULL,
    "a client that never appeared is reported as not found");

/* --- 3. a ring for the most recent events ----- */

proven_result_ring_t rr = PROVEN_RING_INIT(alloc, event_t, 4);
EXAMPLE_REQUIRE(proven_is_ok(rr.err), "creating the recent-events ring must succeed");
proven_ring_t recent = rr.value;

```

```

EXAMPLE_REQUIRE(proven_ring_is_valid(&recent), "a created ring must be structurally valid");

proven_ring_t unset = {0};
EXAMPLE_REQUIRE(!proven_ring_is_valid(&unset), "a zero-initialised ring handle is not usable");

/* This ring refuses when full rather than overwriting, so "keep the most
 * recent four" means dropping the oldest ourselves before pushing. */
for (proven_size_t i = 0; i < batch_len; ++i) {
    if (recent.len == recent.cap) {
        event_t dropped;
        err = PROVEN_RING_POP(&recent, event_t, &dropped);
        EXAMPLE_REQUIRE(proven_is_ok(err), "a full ring must yield its oldest element");
    }
    err = PROVEN_RING_PUSH(&recent, event_t, batch[i]);
    EXAMPLE_REQUIRE(proven_is_ok(err), "pushing after making room must succeed");
}
EXAMPLE_REQUIRE(recent.len == 4, "the ring holds the four most recent events");

/* --- 4. counting per client, with one lookup per update ----- */

/* create_with_capacity is proven_map_create under a name that says why the
 * capacity argument is there: sizing it now avoids rehashing later, and a
 * rehash both copies every bucket and invalidates every pointer previously
 * returned by get_mut. */
proven_result_map_t mr = proven_map_create_with_capacity(alloc, 8, PROVEN_KEY_TYPE_INT,
                                                         sizeof(proven_u32), alignof(proven_u32));
EXAMPLE_REQUIRE(proven_is_ok(mr.err), "creating the counter map must succeed");
proven_map_t counts = mr.value;
EXAMPLE_REQUIRE(proven_map_is_valid(&counts), "a created map must be structurally valid");

err = proven_map_reserve(&counts, 32);
EXAMPLE_REQUIRE(proven_is_ok(err), "reserving map capacity must succeed");

for (proven_size_t i = 0; i < batch_len; ++i) {
    proven_map_key_t k = { .id = batch[i].client };

    /* get_mut returns a pointer INTO the map's storage, so the counter is
     * incremented where it lives: one lookup, no copy back. The pointer is
     * good only until the next insert - which is another reason the capacity
     * was reserved above. */
    proven_u32 *seen = (proven_u32 *)proven_map_get_mut(&counts, k);
    if (seen) {
        *seen += 1;
    } else {
        proven_u32 one = 1;
        err = proven_map_set(&counts, k, &one);
        EXAMPLE_REQUIRE(proven_is_ok(err), "inserting a new client must succeed");
    }
}

const proven_u32 *seven = PROVEN_MAP_GET_INT(&counts, proven_u32, 7);
EXAMPLE_REQUIRE(seven != NULL && *seven == 3, "client 7 sent three events");

/* A closing event (code 0) means the client is gone: drop its counter. */
for (proven_size_t i = 0; i < batch_len; ++i) {
    if (batch[i].code == 0) {
        err = proven_map_remove(&counts, (proven_map_key_t){ .id = batch[i].client });
        EXAMPLE_REQUIRE(proven_is_ok(err), "removing a present key must succeed");
    }
}

```

```

EXAMPLE_REQUIRE(PROVEN_MAP_GET_INT(&counts, proven_u32, 7) == NULL, "the closed client is gone");

/* --- 5. which hash, and how to see the difference ----- */

/* Two string-key maps over the same keys. The default one hashes with keyed
 * SipHash, so an attacker who chooses the keys cannot force them all into
 * one bucket. The trusted one uses fast FNV-1a and is the right choice ONLY
 * when your own code chooses every key. */
proven_result_map_t untrusted = PROVEN_MAP_INIT_U8_BORROWED(alloc, proven_u32, 8);
EXAMPLE_REQUIRE(proven_is_ok(untrusted.err), "creating the default string-key map must succeed");
proven_map_t from_network = untrusted.value;

proven_result_map_t tr = proven_map_create_trusted(alloc, 8, PROVEN_KEY_TYPE_U8_BORROWED,
                                                    sizeof(proven_u32), alignof(proven_u32));
EXAMPLE_REQUIRE(proven_is_ok(tr.err), "creating the trusted-key map must succeed");
proven_map_t internal = tr.value;

EXAMPLE_REQUIRE(from_network.trusted_keys == false, "the default map defends against chosen keys");
EXAMPLE_REQUIRE(internal.trusted_keys == true, "the trusted map opts out of that defence");

/* map_hash exposes the value the map actually places a key by, so the
 * choice is observable rather than something you take on faith. */
proven_map_key_t name = { .str = PROVEN_LIT("user-agent") };
proven_u64 keyed = proven_map_hash(&from_network, name);
proven_u64 fast = proven_map_hash(&internal, name);
EXAMPLE_REQUIRE(keyed != fast, "the same key hashes differently under the two functions");
EXAMPLE_REQUIRE(proven_map_hash(&internal, name) == fast, "and each function is deterministic");

/* Keys that live in memory the map does not own are BORROWED: the bytes must
 * outlive the map. These are string literals, so they do. When the key comes
 * from a buffer you are about to reuse, use an owned-key map instead
 * (PROVEN_MAP_INIT_U8_OWNED / proven_map_set_u8_owned), which copies. */
proven_u32 hits = 1;
err = proven_map_set(&from_network, name, &hits);
EXAMPLE_REQUIRE(proven_is_ok(err), "inserting a borrowed string key must succeed");
EXAMPLE_REQUIRE(PROVEN_MAP_GET_U8_BORROWED(&from_network, proven_u32, PROVEN_LIT("user-agent")) != NULL,
                "and it can be looked up by an equal view, not the same pointer");

/* --- 6. checksumming a stream in chunks ----- */

/* The batch is checksummed as it goes past, which is what a program reading
 * a file or a socket has to do: it never holds the whole thing. Start the
 * running value at 0, feed each chunk in, and the final value is the same
 * one a single call over the concatenation would produce. */
proven_u32 running = 0;
for (proven_size_t i = 0; i < batch_len; ++i) {
    proven_mem_view_t chunk = { .ptr = (const proven_byte_t *)&batch[i], .size = sizeof batch[i] };
    running = proven_crc32_update(running, chunk);
}
proven_mem_view_t whole = { .ptr = (const proven_byte_t *)batch, .size = sizeof batch };
EXAMPLE_REQUIRE(running == proven_crc32(whole),
                "chunked and whole-buffer CRC-32 must agree, whatever the chunking");

printf("events=%zu recent=%zu crc32=%08x\n",
        (size_t)events.len, (size_t)recent.len, (unsigned)running);

proven_map_destroy(&internal);
proven_map_destroy(&from_network);
proven_map_destroy(&counts);
PROVEN_RING_DESTROY(&recent);

```

```

    PROVEN_ARRAY_DESTROY(&events);
    return EXAMPLE_OK();
}

```

Wrong - binary search on the arrival-ordered log:

```

const event_t *e = proven_array_binary_search(&events, &key, by_client);    /* wrong */

```

`proven_array_binary_search()` may only be used on a range sorted by the same comparison. On unsorted input it does not fail; it halves its way to whatever element happens to be there and reports it, or reports nothing while the element is present two slots away.

Wrong - holding a `get_mut` pointer across an insert:

```

proven_u32 *seen = proven_map_get_mut(&counts, k);
proven_err_t e = proven_map_set(&counts, other_key, &one);    /* may rehash */
*seen += 1;                                                    /* wrong: may dangle */

```

Hold the key, not the pointer, across anything that can insert.

Wrong - the fast hash on keys from outside the program:

```

proven_result_map_t m = proven_map_create_trusted(alloc, 64, PROVEN_KEY_TYPE_U8_BORROWED,
                                                    sizeof(session_t), alignof(session_t));
/* keys are HTTP header names taken from the request */    /* wrong */

```

That is exactly the case the keyed hash defends against. Use `proven_map_create()` whenever the keys are chosen by anyone other than you.

## Chapter 8: Formatting and Scanning (v0.11.0)

**Part IV - Text in and out. Prerequisite:** [Chapter 3](#) section 3-section 4. After this chapter you can format any value, teach the formatter a type of your own, parse input with the cursor under your control, and recover from a partial parse.

This chapter is the **reference half** of the text material; [Chapter 3](#) is the tutorial half. Chapter 3 introduces the formatter and the scanner beside the string types and gives you enough to be productive. This chapter gives the exact syntax, every parameter shape, every return value, and the places where callers usually go wrong. If you are meeting {} for the first time, read Chapter 3 first - this one assumes you have.

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### 1. Design model

#### Why the type is never written twice

Both halves of this chapter exist to eliminate one thing: a place where the programmer states a type that the compiler cannot check against the value.

`printf("%d", x)` states it twice - once as `%d`, once by passing `x` - and `varargs` erases the second, so nothing can compare them. `scanf("%d", &x)` is worse: the format decides both how to parse *and* what to write through the pointer, so a mismatch corrupts memory rather than printing nonsense. Both are the same defect from opposite directions, and both compile silently.

Here the placeholder carries **no type at all**. {} marks a position; the type comes from `PROVEN_ARG(x)` on the formatting side and `PROVEN_SCAN_ARG(&x)` on the scanning side, both resolved by `_Generic` at compile time against the argument's static type. The spec after `:` controls presentation only - width, fill, alignment, precision, base - never interpretation. There is no `%d`-versus-`double` to get wrong because you never wrote a type in the string.

The second design decision is that **neither side owns a buffer**. Formatting appends into a destination you supply and refuses when it does not fit; scanning reads from a view you supply and moves a cursor you can read. Nothing here allocates unless you hand it an allocator, which is what lets the whole chapter work in a freestanding build (section 13).

The formatting side and the scanning side solve opposite problems.

- Formatting takes typed values and renders text.

- Scanning takes text and writes typed values.

The project keeps both sides intentionally small:

- formatting supports a compact placeholder language, positional reuse, simple alignment, width, and hex rendering for numeric values;
- scanning supports typed destination pointers, strict placeholder counting, `{:x}` for hexadecimal integers, and literal matching with whitespace collapsing;
- neither side tries to become a full `printf` or `scanf` clone.

The practical result is that the APIs are easier to reason about than large general-purpose format engines, but the syntax is still expressive enough for common systems-code tasks.

## 2. Formatter data model

```
proven_fmt_result_t
typedef struct {
    proven_err_t err;
    proven_size_t written;
    proven_size_t required;
} proven_fmt_result_t;
```

Meaning:

- `err`: the status code for the operation.
- `written`: bytes actually written into the destination.
- `required`: total bytes needed for the full formatted output.

Use `err` first. The other fields are most useful for truncating or partially successful operations.

A successful result looks like this:

```
proven_result_u8str_t rs = proven_u8str_create(alloc, 64);
if (proven_is_ok(rs.err)) {
    proven_u8str_t s = rs.value;

    proven_fmt_result_t r = proven_u8str_append_fmt_trunc(
        &s,
        "hello {}",
        PROVEN_ARG("world")
    );
    if (proven_is_ok(r.err)) {
        proven_println("{", PROVEN_ARG(proven_u8str_as_view(&s)));
    }

    proven_u8str_destroy(alloc, &s);
}
```

A truncating result can still tell you how much more space you would have needed. Here the destination is deliberately too small, so `written` stops short of `required` - and the string is still valid:

```
proven_result_u8str_t rs = proven_u8str_create(alloc, 8); /* on purpose: too small */
if (proven_is_ok(rs.err)) {
    proven_u8str_t s = rs.value;

    proven_fmt_result_t r = proven_u8str_append_fmt_trunc(
        &s,
        "name={} score={}",
        PROVEN_ARG("ada"),
        PROVEN_ARG(42)
    );
    /* r.written is what fit; r.required is what the whole output needed. */
}
```

```

    proven_println("wrote {} of {} bytes",
        PROVEN_ARG(r.written), PROVEN_ARG(r.required));

    proven_u8str_destroy(alloc, &s);
}

proven_arg_type_t
typedef enum {
    PROVEN_ARG_NONE,
    PROVEN_ARG_I32,
    PROVEN_ARG_U32,
    PROVEN_ARG_I64,
    PROVEN_ARG_U64,
#ifdef PROVEN_FMT_NO_FLOAT
    PROVEN_ARG_F64,
#endif
    PROVEN_ARG_CSTR,
    PROVEN_ARG_STR_VIEW,
    PROVEN_ARG_DATETIME,
    PROVEN_ARG_PTR,
    PROVEN_ARG_FN,
} proven_arg_type_t;

```

The formatter currently recognizes these value classes:

- signed 32-bit integers
- unsigned 32-bit integers
- signed 64-bit integers
- unsigned 64-bit integers
- floating-point values, unless `PROVEN_FMT_NO_FLOAT` is defined
- trusted C strings
- borrowed U8 string views
- datetimes
- object pointers
- function pointers

```

proven_arg_t
typedef struct {
    proven_arg_type_t type;
    union {
        proven_i32 i32;
        proven_u32 u32;
        proven_i64 i64;
        proven_u64 u64;
        double f64;
        const char *cstr;
        proven_u8str_view_t str_view;
        proven_datetime_t datetime;
        const void *ptr;
        void (*fn)(void);
    } value;
} proven_arg_t;

```

The union field must match the selected type. Do not manufacture a `proven_arg_t` by writing a mismatched union field and hoping the formatter will guess.

Wrong:

```

proven_arg_t arg = {0};
arg.type = PROVEN_ARG_I64;
arg.value.u64 = 123; /* wrong: type and union field do not match */

```

Correct:

```

proven_arg_t arg = proven_arg_i64(123);
(void)arg; /* pass it to a formatting macro; PROVEN_ARG(arg) accepts it as-is */

```

### 3. Formatter constructors and selectors

#### Constructor summary

| API                                                                  | Parameters                | Returns                   | Intent                                                                                                                                                                  |
|----------------------------------------------------------------------|---------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_arg_none(void)</code>                                   | none                      | <code>proven_arg_t</code> | Internal sentinel value.                                                                                                                                                |
| <code>proven_arg_i32(int v)</code>                                   | signed integer            | <code>proven_arg_t</code> | Render as 32-bit signed integer.                                                                                                                                        |
| <code>proven_arg_u32(unsigned int v)</code>                          | unsigned integer          | <code>proven_arg_t</code> | Render as 32-bit unsigned integer.                                                                                                                                      |
| <code>proven_arg_i64(long long v)</code>                             | wide signed integer       | <code>proven_arg_t</code> | Render as 64-bit signed integer.                                                                                                                                        |
| <code>proven_arg_u64(unsigned long long v)</code>                    | wide unsigned integer     | <code>proven_arg_t</code> | Render as 64-bit unsigned integer.                                                                                                                                      |
| <code>proven_arg_f64(double v)</code>                                | floating-point value      | <code>proven_arg_t</code> | Render as floating-point text, unless float formatting is disabled.                                                                                                     |
| <code>proven_arg_cstr(const char *v)</code>                          | trusted live C string     | <code>proven_arg_t</code> | Render a NUL-terminated C string.                                                                                                                                       |
| <code>proven_arg_cstr_n(const char *v, proven_size_t max_len)</code> | possibly bounded C string | <code>proven_arg_t</code> | Render only up to <code>max_len</code> while searching for NUL.                                                                                                         |
| <code>proven_arg_str_view(proven_u8str v)</code>                     | borrowed U8 view          | <code>proven_arg_t</code> | Render a borrowed view without assuming NUL termination.                                                                                                                |
| <code>proven_arg_datetime(proven_datetime v)</code>                  | datetime value            | <code>proven_arg_t</code> | Render a datetime using the formatter's datetime rules.                                                                                                                 |
| <code>proven_arg_ptr(const void *v)</code>                           | object pointer            | <code>proven_arg_t</code> | Render the pointer value.                                                                                                                                               |
| <code>proven_arg_fn(void (*v)(void))</code>                          | function pointer          | <code>proven_arg_t</code> | Render the raw function-pointer representation.                                                                                                                         |
| <code>proven_arg_ucstr(const unsigned char *v)</code>                | unsigned-char string      | <code>proven_arg_t</code> | Convenience wrapper around <code>proven_arg_cstr</code> .                                                                                                               |
| <code>proven_arg_identity(proven_arg_t v)</code>                     | existing argument object  | <code>proven_arg_t</code> | Pass-through helper.                                                                                                                                                    |
| <code>proven_arg_bool(bool v)</code>                                 | boolean                   | <code>proven_arg_t</code> | Render true / false as words, not as 1 / 0.                                                                                                                             |
| <code>proven_arg_char(char v)</code>                                 | a character               | <code>proven_arg_t</code> | Render the <b>character</b> . This is why a <code>char</code> <code>VARIABLE</code> renders as a character while the literal <code>PROVEN_ARG('Z')</code> still renders |

|                                                           |                 |              |                                                                                                                                   |
|-----------------------------------------------------------|-----------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
|                                                           |                 |              | 90: in C, 'z' has type int, and no amount of _Generic can tell it apart from the number 90.                                       |
| proven_arg_custom(const void *v, proven_fmt_custom_fn fn) | any type at all | proven_arg_t | Render a type the library has never heard of, through a function you supply. See <a href="#">Formatting a user-defined type</a> . |

#### PROVEN\_ARG(x)

PROVEN\_ARG(x) is the usual entry point. It uses \_Generic so the compiler chooses a constructor from the type of x.

The current mapping is:

- \_Bool, char, signed char, short, int -> proven\_arg\_i32
- unsigned char, unsigned short, unsigned int -> proven\_arg\_u32
- long, long long -> proven\_arg\_i64
- unsigned long, unsigned long long -> proven\_arg\_u64
- double, float -> proven\_arg\_f64, unless PROVEN\_FMT\_NO\_FLOAT is defined
- const char \*, char \* -> proven\_arg\_cstr
- unsigned char \*, const unsigned char \* -> proven\_arg\_ucstr
- void \*, const void \* -> proven\_arg\_ptr
- proven\_u8str\_view\_t -> proven\_arg\_str\_view
- proven\_datetime\_t -> proven\_arg\_datetime
- proven\_arg\_t -> proven\_arg\_identity

Important consequence:

- PROVEN\_ARG does not select function pointers.
- Use PROVEN\_ARG\_FN(f) for function pointers.

Wrong:

```
void helper(void) {}
proven_u8str_append_fmt_grow(alloc, &s, "{}", PROVEN_ARG(helper)); /* wrong */
```

Correct:

```
void helper(void); /* whatever function you want to print the address of */

proven_result_u8str_t rs = proven_u8str_create(alloc, 64);
if (proven_is_ok(rs.err)) {
    proven_fmt_result_t r = proven_u8str_append_fmt_grow(alloc, &rs.value, "{}",
                                                         PROVEN_ARG_FN(helper));
    (void)r;
    proven_u8str_destroy(alloc, &rs.value);
}
```

#### PROVEN\_ARG\_FN(f)

This macro exists so callers can pass function pointers without casting them through void \*. It is a small safety wrapper around proven\_arg\_fn.

Example:

```
void helper(void);

proven_result_u8str_t rs = proven_u8str_create(alloc, 64);
```

```

if (proven_is_ok(rs.err)) {
    proven_u8str_t s = rs.value;

    proven_fmt_result_t r = proven_u8str_append_fmt_grow(
        alloc,
        &s,
        "callback = {}",
        PROVEN_ARG_FN(helper)
    );
    if (!PROVEN_FMT_IS_OK(r)) {
        proven_eprintln("formatting the callback failed");
    }

    proven_u8str_destroy(alloc, &s);
}

```

PROVEN\_ARG\_CSTR\_N(v, max\_len)

This macro is the bounded-string helper. Use it when the source may not be a fully trusted C string, but you still want C-string-like input handling.

It scans for NUL only up to max\_len and then formats the bounded prefix as a view.

Good use case - buf came off the wire, so nothing promises it is NUL-terminated:

```

char buf[128]; /* filled from an untrusted source */
(void)proven_mem_copy(buf, sizeof buf, proven_mem_view_from_u8(PROVEN_LIT("payload")));

proven_result_u8str_t rs = proven_u8str_create(alloc, 64);
if (proven_is_ok(rs.err)) {
    proven_fmt_result_t r = proven_u8str_append_fmt_grow(
        alloc,
        &rs.value,
        "payload={}",
        PROVEN_ARG_CSTR_N(buf, sizeof buf) /* looks for NUL only within 128 bytes */
    );
    (void)r;
    proven_u8str_destroy(alloc, &rs.value);
}

```

Bad use case:

```

const char *buf = get_network_buffer();
proven_u8str_append_fmt_grow(alloc, &s, "{}", PROVEN_ARG(buf)); /* wrong if buf is not trusted */

```

## Float formatting note

If PROVEN\_FMT\_NO\_FLOAT is defined, float support is removed from the generic selector and the float constructor is not available. That is a compile-time configuration choice, not a runtime toggle.

The default {} rendering of a double produces a fixed six-digit fractional form for finite values and switches to scientific notation when the magnitude is too large or too small for the compact form. Unlike earlier versions, this output is now computed by an **exact, integer-only** engine (no double/long double approximation): the digits are correctly rounded (round-half-to-even), so {} matches what printf("%.6f") / %.6e would print on the same value. For a shortest round-trippable form, or for a precision other than six, use the proven\_float\_format\_\* policy API described below.

## Accuracy and limits

- The default {} float output uses six fractional digits, correctly rounded to nearest with ties to even (matching glibc %.6f/%.6e). It is exact at any magnitude - there is no long double and no precision/magnitude ceiling beyond the configurable big-integer capacity.
- For round-trip serialization use the **shortest** policy (proven\_float\_format\_options\_shortest()), which emits the shortest decimal string that parses back to the exact same value.

- Decimal-to-double scanning is correctly rounded to IEEE-754 binary64 with round-to-nearest, ties-to-even, matching the host strtod bit for bit. Values below the half-way threshold to the smallest subnormal round to signed zero with the input sign preserved.
- The parser is layered Clinger fast path -> Eisel-Lemire -> exact big-integer fallback; every tier is exact and the fallback is the arbiter, so the result is correctly rounded for every input. The cached power tables are generated source (scripts/generate\_float\_decimal\_tables.py).
- The exact-fallback big-integer capacity is tunable with PROVEN\_FLOAT\_BIGINT\_LIMBS (see include/proven/float\_config.h) for embedded targets; the fast paths never touch the big integer.
- Validation: the formatter and parser are checked exhaustively over all 4.28 billion finite binary32 values and over 2.56 billion random binary64 values against the host C library, with zero mismatches. See docs/float-correctness-and-performance.md for algorithms, methodology, and a benchmark against glibc.

### Inside the engine (conceptual)

You do not need any of this to use the API - it is here for readers who want to know why the output is trustworthy. Full detail is in docs/float-correctness-and-performance.md.

**Parsing (decimal -> binary64), three tiers, fastest first.** The result is always correctly rounded; the tiers are purely a speed staircase, each one only taken when it can guarantee the exact answer:

1. *Clinger fast path.* When the value has few significant digits and a small exponent, both the significand and  $10^{\text{exp}}$  are exactly representable as double, so a single rounded multiply/divide is provably correct. Covers most everyday numbers.
2. *Eisel-Lemire.* A 64x128-bit fixed-point multiply by a cached power of ten, with a check that the result is far enough from a rounding boundary to be certain. If the check is inconclusive (the value sits on a halfway tie), it falls through.
3. *Exact big-integer fallback.* Builds the value as a ratio of big integers (significand and  $5^q/2^q$ ) and compares against the candidate double and its neighbor exactly - this is the arbiter that makes ties and subnormals correct. A seeded +/-16-ULP window keeps the search to a few comparisons. The big-integer capacity is bounded by PROVEN\_FLOAT\_BIGINT\_LIMBS; the tier is the only one that allocates limbs (on the stack), and the fast paths never reach it.

**Formatting (binary64/32 -> decimal).** Two engines, no long double anywhere:

- *Shortest* (proven\_float\_format\_options\_shortest()): a **Grisu3** fast path produces the minimal round-trippable digits for almost all inputs in ~90 ns; the rare cases where Grisu3 cannot prove minimality fall back to an exact **Dragon4** (Burger-Dybvig, round-to-even) core. The result is the unique shortest decimal that parses back to the same bits.
- *Fixed %f / scientific %e* (the default {} and the fixed options): an exact integer engine scales the value by  $10^{\text{precision}}$  with big-integer mul\_pow5/shift, does an integer divmod, and rounds half-to-even - so it matches glibc at any precision and magnitude, with no  $2^{64}/\text{precision}$  ceiling. Extreme exponents do real arbitrary-precision work and are correspondingly slower (rare in practice).

### Public float parsing APIs

Three entry points, sharing one correctly-rounded backend:

- proven\_scan\_f64(scan) - parse from a proven\_scan\_t cursor; restores the cursor on failure. The native, length-bounded path (no NUL terminator needed).
- proven\_parse\_double\_ascii(view) - parse one locale-free ASCII token from a proven\_u8str\_view\_t and report the consumed length.
- proven\_strtod(nptr, endptr) - a strtod-style convenience wrapper over a C string: skips leading ASCII whitespace and reports endptr.

### Worked example: parsing

```
#include "proven/scan.h"
#include "proven/float_parse.h"

/* (1) Native, view-based parsing through a scanner cursor. */
proven_scan_t sc = proven_scan_init(proven_u8str_view_from_cstr("3.14159e2 rest"));
proven_result_f64_t r = proven_scan_f64(&sc);
if (r.err == PROVEN_OK) {
    /* r.val == 314.159; the cursor now sits at " rest". */
    proven_println("parsed {}", PROVEN_ARG(r.val));
}

/* (2) strtod-style wrapper for C strings. endptr reports where parsing stopped. */
char *end = NULL;
double v = proven_strtod(" -0.5\t", &end); /* v == -0.5, *end == '\t' */
(void)v;

/* A trailing exponent marker with no digits stops like strtod: "1e" parses 1,
   leaving endptr at 'e'. Inputs with hundreds of significant digits and extreme
   exponents are still rounded correctly via the exact fallback. */
```

### Worked example: formatting with the policy API

proven\_float\_format\_f64\_policy / \_f32\_policy write directly into a caller buffer and report the number of bytes written. They never allocate.

```
#include "proven/float_format.h"

char buf[64];
proven_size_t n = 0;

/* Shortest round-trippable form: 0.1 -> "0.1" (not "0.10000000000000001"). */
(void)proven_float_format_f64_policy(buf, sizeof buf, 0.1,
    PROVEN_FLOAT_FORMAT_POLICY_RYU,
    proven_float_format_options_shortest(), &n);
/* buf == "0.1", n == 3 */

/* Fixed precision (correctly rounded, round-half-to-even). */
proven_float_format_options_t opt = proven_float_format_options_fixed_default();
opt.precision = 2;
(void)proven_float_format_f64_policy(buf, sizeof buf, 3.14159,
    PROVEN_FLOAT_FORMAT_POLICY_DEFAULT, opt, &n);
/* buf == "3.14" */

/* Always scientific - this is what {:e} selects. Six fractional digits by default,
   a signed two-digit-minimum exponent: exactly what printf's %e prints. */
proven_float_format_options_t sci = proven_float_format_options_scientific();
sci.precision = 2;
(void)proven_float_format_f64_policy(buf, sizeof buf, 42.0,
    PROVEN_FLOAT_FORMAT_POLICY_DEFAULT, sci, &n);
/* buf == "4.20e+01" - where fixed would give "42.00" and shortest "42" */
```

- PROVEN\_FLOAT\_FORMAT\_POLICY\_RYU selects shortest output; DEFAULT/SIMPLE select the exact fixed-precision path (%f, switching to %e for very large or very small magnitudes).
- Returns PROVEN\_ERR\_OUT\_OF\_BOUNDS if the buffer is too small (the value is never truncated silently) and PROVEN\_ERR\_INVALID\_ARG for an unsupported policy.
- The generic {} formatter (proven\_u8str\_append\_fmt\*) uses the DEFAULT policy internally, so everyday logging needs no direct call to this API.

## 4. Format string grammar

The formatter accepts a deliberately small grammar.

### Replacement fields

Supported forms:

- {}: next positional argument
- {0}: first user argument
- {1}: second user argument
- {2}: third user argument
- and so on

The numbering is user-facing and zero-based. The implementation stores a hidden sentinel at index 0 and maps user index 0 to internal argument slot 1.

### Escaped braces

- {{ becomes a literal {
- }} becomes a literal }

### The layout spec

{:[fill]align}[sign][#][0][width][.precision][type]}

Every part is optional, and the order is the one the rest of the world uses, so a spec copied from Python or Rust means here what it means there.

| Part       | Values                         | What it does                                                             |
|------------|--------------------------------|--------------------------------------------------------------------------|
| align      | < > ^                          | left, right, centre. Default >.                                          |
| fill       | any character, before an align | the pad character. Default space.                                        |
| sign       | + or a space                   | force a sign on a non-negative number, or reserve a space for one.       |
| #          |                                | alternate form: 0x, 0X, 0o, 0b prefixes. Integers only.                  |
| 0          |                                | zero-fill. {:08} on 42 is 00000042.                                      |
| width      | digits, up to 10000            | minimum field width.                                                     |
| .precision | .N, up to 60                   | decimals. <b>Floats only.</b>                                            |
| type       | x X o b d (int), f g e (float) | base and case; f fixed, g shortest round-trip, e scientific (printf %e). |

Two things are worth knowing because they used to be false:

- **A leading 0 is zero-fill, not the first digit of the width.** {:08} produced " 42" - space-padded, no error - until v26.07.12f. An explicit fill still wins: {:\*>08} pads with \*.
- **Zero-padding goes between the sign and the digits.** {:+08} on 42 is +0000042, never 0000+42. Padding is part of the number, and a number's sign comes first.

### Floats

{:} gives six decimals, correctly rounded. {:.3} gives three, {:.0} gives none, {:e} forces scientific notation - mantissa, precision fractional digits (six by default), a signed two-digit-minimum exponent, correctly rounded, exactly as printf's %e - which is the form {:f} and {:g} do not reach: {:f} never shows an exponent, and {:g} uses one only when it is shorter. {:f} forces the fixed form, and {:g} gives the shortest representation that round-trips.

Until v26.07.12i **none of these existed**: every float came out with exactly six decimals, forever. The exact engine could always do all of it - the {} grammar simply could not reach it. The visible cost

was that a float column could not be aligned, because 12.5 rendered nine characters wide and 100.0 rendered ten:

```
proven_byte_t buf[64];
proven_u8str_t line = proven_u8str_borrow(buf, sizeof buf);
(void)proven_u8str_append_fmt(&line, "{:>9.2}", PROVEN_ARG(12.5));    /* "    12.50" */
```

### A spec the argument cannot honour is an error

{:x} on a double, {:.2} on an integer, {:f} on an integer, {:#} on a string: all `PROVEN_ERR_INVALID_FORMAT`.

They used to be *ignored* - {:x} on a double printed 3.500000 and reported success. The caller asked for something, got something else, and was told it had worked. That is the worst available outcome, and it is worse than refusing.

char **and** bool

`PROVEN_ARG('Z')` renders Z, and a bool renders true or false. Both used to go through the integer path, so a character printed as 90 and there was no way to emit one at all - the ASCII column of a hex dump had to be built by hand in a separate buffer and passed as a string. An uppercase hex dump is now one loop:

```
proven_byte_t hexbuf[64];
proven_u8str_t hexline = proven_u8str_borrow(hexbuf, sizeof hexbuf);
unsigned char byte = 0xde;
(void)proven_u8str_append_fmt(&hexline, "{:02X} ", PROVEN_ARG((unsigned)byte));
(void)proven_u8str_append_fmt(&hexline, "{}", PROVEN_ARG((char)'.'));
```

## 5. Formatting APIs

```
proven_u8str_fmt_internal(...)
proven_fmt_result_t proven_u8str_fmt_internal(
    proven_allocator_t alloc,
    proven_u8str_t *str,
    bool trunc,
    const char *fmt,
    proven_allocator_t scratch,
    const proven_arg_t *args,
    proven_size_t args_count
);
```

This is the internal formatting engine. User code should normally call the public macros instead.

Parameters:

- `alloc`: allocator used when the string must grow
- `str`: destination U8 string
- `trunc`: if true, allow best-effort truncation; if false, keep atomic behavior
- `fmt`: format text
- `scratch`: allocator used for temporary alias-patching when needed
- `args`: array of format arguments, including the hidden sentinel at index 0
- `args_count`: total length of args, including the sentinel

Return value:

- a `proven_fmt_result_t`

Important rules:

- `args_count` must match the number of placeholders plus the hidden sentinel
- extra unused arguments are an error
- missing arguments are an error

- if the engine detects aliasing between the destination string and a borrowed view argument, it may use the scratch allocator to preserve failure atomicity

`proven_u8str_append_fmt(str, fmt, ...)`

Atomic formatting into a fixed-capacity string. If the result does not fit, the function reports failure and leaves the destination unchanged.

Use this when you want all-or-nothing behavior.

`proven_u8str_append_fmt_trunc(str, fmt, ...)`

Best-effort formatting. It writes as much as fits and reports how much was written and how much was required.

Use this when partial output is acceptable.

`proven_u8str_append_fmt_grow(alloc, str, fmt, ...)`

Growable formatting. It may reallocate the destination string through the supplied allocator. On allocation failure, the old string remains valid.

Use this when you want the output to fit without manual capacity planning.

`proven_u8str_append_fmt_with_scratch(alloc, str, fmt, scratch, ...)`

Growable formatting with a separate scratch allocator. This is useful when the argument list contains string views that may alias the destination buffer and temporary patching is needed.

Use a real allocator for both `alloc` and `scratch`. Do not pass a dead arena or a one-shot temporary buffer unless its lifetime is long enough for the call.

### Float format policy seam

The public float policy header provides an explicit policy layer for float formatting. It is intentionally small and keeps the exact fixed-precision formatter as the default path.

The main entry points are:

- `proven_float_format_f64_policy(...)`
- `proven_float_format_f32_policy(...)`
- `proven_float_format_options_fixed_default()`
- `proven_float_format_options_shortest()`

Policy notes:

- `PROVEN_FLOAT_FORMAT_POLICY_DEFAULT` and `PROVEN_FLOAT_FORMAT_POLICY_SIMPLE` select the exact fixed-precision output (correctly rounded, round-half-to-even).
- `PROVEN_FLOAT_FORMAT_POLICY_RYU` is the shortest-output policy branch.
- The policy API returns `PROVEN_ERR_INVALID_ARG` for unsupported enum values.
- The policy API returns `PROVEN_ERR_OUT_OF_BOUNDS` when the caller-provided buffer is too small.

Example:

```
char buf[128];
proven_size_t written = 0;
proven_err_t err = proven_float_format_f64_policy(
    buf,
    sizeof buf,
    0.1,
    PROVEN_FLOAT_FORMAT_POLICY_RYU,
    proven_float_format_options_shortest(),
    &written
);
if (proven_is_ok(err)) {
    /* buf holds the shortest form that parses back to exactly 0.1 */
}
```

```
    proven_println("{} ", PROVEN_ARG_CSTR_N(buf, written));
}
```

PROVEN\_FMT\_IS\_OK(res)

A small helper macro for checking proven\_fmt\_result\_t. Use it when you want the intent to stay compact.

Example:

```
proven_result_u8str_t rs = proven_u8str_create(alloc, 32);
if (proven_is_ok(rs.err)) {
    proven_u8str_t s = rs.value;

    proven_fmt_result_t r = proven_u8str_append_fmt_grow(
        alloc,
        &s,
        "name={} score={:0>4}",
        PROVEN_ARG("ada"),
        PROVEN_ARG(42)
    );
    if (!PROVEN_FMT_IS_OK(r)) {
        /* the string is untouched: grow-mode formatting is failure-atomic */
        proven_eprintln("formatting failed");
    }

    proven_u8str_destroy(alloc, &s);
}
```

## Console-style helpers

The sysio layer provides print helpers that use the same formatter machinery:

- proven\_print(fmt, ...)
- proven\_println(fmt, ...)
- proven\_eprint(fmt, ...)
- proven\_eprintln(fmt, ...)

They are convenient when you want formatted output directly to stdout or stderr. They still return proven\_err\_t, so check the result when the output matters.

Example:

```
if (!proven_is_ok(proven_println("hello {}", PROVEN_ARG("world")))) {
    /* the write to stdout failed - a closed pipe, a full disk */
    proven_eprintln("stdout is not writable");
}
```

### 5.1. Formatting a type of your own

PROVEN\_ARG is built on \_Generic, and \_Generic can only dispatch on types it was told about at compile time. It cannot be told about yours. So the formatter's argument set - integers, floats, strings, pointers, datetimes - was a **closed** one: a rect\_t, a uuid\_t, a vec3\_t could not be passed to {} at all.

The two ways around it were both bad. Pre-format the value into a scratch string and pass *that*: an allocation and a copy for every value, in the logging path, which is the one path that must keep working when allocation is exactly what has failed. Or print the fields one at a time and give up on ever aligning the column.

PROVEN\_ARG\_OF(&obj, render) is the door:

```
proven_err_t render(proven_fmt_sink_t out, const void *obj);
```

Three things follow from the shape of that signature:

- **The renderer gets a sink, not a buffer.** It does not need to know how much room there is, and it cannot overflow anything. It emits with `proven_fmt_put`.
- **It composes.** The renderer may call the formatter again - into a stack buffer, with no allocator - and hand the result to the sink. A type whose fields are themselves user types nests naturally.
- **Width, fill and alignment work.** The formatter runs the renderer **twice**: once against a counting sink to learn how wide the output is, then once for real, with the padding applied around it. This is why `{:>10}` lines up a column of your type exactly as it lines up a column of ints - and it is why the renderer must be **deterministic** and must not mutate `obj`. If the two passes disagree, the formatter returns `PROVEN_ERR_INVALID_ARG` rather than emit a field of the wrong width into an aligned column and let you find out later.

What the formatter will **not** do is guess. `{:x}` on a rectangle, `{:.2}` on a UUID, `{:~}` on a matrix - the library has no idea what those would mean for your type, and so it refuses them with `PROVEN_ERR_INVALID_FORMAT`. Inventing a plausible answer and reporting success is how a formatter starts lying; a type letter you did not ask for is not better than an error.

Compiled and run by the test suite:

```
/*
 * Formatting a type the library has never heard of.
 *
 * `PROVEN_ARG` is built on `_Generic`, which can only dispatch on types it was told
 * about - and it cannot be told about yours. So before `PROVEN_ARG_OF` existed, a
 * `rect_t` simply could not be printed. You either pre-formatted it into a scratch
 * string and passed that (an allocation and a copy per value, in the logging path,
 * which is the one path that must not allocate), or you printed the fields one at a
 * time and gave up on ever aligning the column.
 *
 * A renderer receives a *sink*, not a buffer. That is what makes it compose: it can
 * call the formatter again, and its output is just bytes going somewhere. And because
 * the formatter measures the renderer's output before emitting it - by running it once
 * against a counting sink - width, fill and alignment work on a user type exactly as
 * they do on an int.
 */

typedef struct { int w, h; } rect_t;

static proven_err_t render_rect(proven_fmt_sink_t out, const void *obj) {
    const rect_t *r = (const rect_t *)obj;

    /* Compose: the formatter, into a stack buffer, no allocator anywhere. */
    proven_byte_t tmp[64];
    proven_u8str_t s = proven_u8str_borrow(tmp, sizeof tmp);
    proven_fmt_result_t f = proven_u8str_append_fmt(&s, "{}x{}",
                                                    PROVEN_ARG(r->w), PROVEN_ARG(r->h));
    if (!PROVEN_FMT_IS_OK(f)) return f.err;

    return proven_fmt_put(out, proven_u8str_as_view(&s));
}

int main(void) {
    rect_t a = { .w = 1920, .h = 1080 };
    rect_t b = { .w = 640, .h = 480 };

    proven_byte_t buf[128];
    proven_u8str_t line = proven_u8str_borrow(buf, sizeof buf);

    /* Just like any other argument. */
    proven_fmt_result_t r = proven_u8str_append_fmt(&line, "mode={}", PROVEN_ARG_OF(&a, render_rect));
}
```

```

EXAMPLE_REQUIRE(PROVEN_FMT_IS_OK(r), "a user type should format");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&line), PROVEN_LIT("mode=1920x1080")),
    "the renderer's bytes should be what came out");

/* And it aligns, which is the whole reason the formatter measures it first: a
 * column of user-defined values lines up like a column of anything else. */
(void)proven_u8str_reset(&line);
r = proven_u8str_append_fmt(&line, "[{:>10}]\n[{:>10}]",
    PROVEN_ARG_OF(&a, render_rect),
    PROVEN_ARG_OF(&b, render_rect));
EXAMPLE_REQUIRE(PROVEN_FMT_IS_OK(r), "two user types should format");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&line),
    PROVEN_LIT("[ 1920x1080]\n[ 640x480]")),
    "both rows should be right-aligned to the same width");

/* A spec the library cannot interpret for your type is refused, not guessed at.
 * `{:x}` on a rectangle has no meaning, and answering it with something plausible
 * while reporting success is how a formatter starts lying to you. */
(void)proven_u8str_reset(&line);
r = proven_u8str_append_fmt(&line, "{:x}", PROVEN_ARG_OF(&a, render_rect));
EXAMPLE_REQUIRE(r.err == PROVEN_ERR_INVALID_FORMAT,
    "a type letter on a user type should be an error");

return EXAMPLE_OK();
}

```

## 6. Console print helpers

### What proven\_println is, and what it costs

`proven_println("{ }", PROVEN_ARG(x))` is the shortest way to get text out of a program, and it is the right tool for a diagnostic, a one-off tool, or a program whose output is a few lines.

It is the wrong tool for a loop, and the reason is worth stating because it is invisible at the call site: **each call is its own write syscall**. Ten thousand `proven_println` calls are ten thousand syscalls, which is roughly two orders of magnitude more expensive than the formatting itself. `printf` hides this behind a buffer that `libc` flushes for you; this library does not buffer behind your back, because a buffer you did not ask for is a buffer that surprises you at exit, on a crash, or when two writers interleave.

**One allocation caveat.** The line is formatted into a 512-byte stack buffer, so a typical line costs zero allocations. A line that does *not* fit falls back to the global heap for that call rather than being refused - refusing to print something because it is long would be worse. This is the one place in the library where a call with no allocator parameter can still allocate, and it is bounded to the over-long case. If you need the guarantee that nothing allocates, format into your own buffer and write that.

When output volume matters, take a buffered writer from `proven_sysio_stdout_buffered` and format into that - same argument rules, one syscall per flush instead of one per line. [Chapter 5](#) covers the stream layer; this section is short because that is where the I/O API is documented.

Wrong - reaching for the scanning argument constructor when printing:

```
proven_println("{ }", PROVEN_SCAN_ARG(&x)); /* wrong: that builds a scan destination */
```

The important point for formatter users is that the console helpers share the same argument rules as the string append APIs.

Common mistakes:

- using `PROVEN_SCAN_ARG` with `proven_println`

- assuming `PROVEN_LIT` is needed for every format string
- forgetting that output functions can still fail

## 7. Scanner data model

The scanner is a cursor over bytes you already have. It does not own the text, it does not copy it, and it does not read from anywhere - `proven_scan_t` is a view plus an offset, and that is the whole of it.

```
typedef struct {
    proven_u8str_view_t view;    /* the bytes being read; not owned */
    proven_size_t cursor; /* how far in we are */
} proven_scan_t;
```

Two consequences worth stating, because they are what make this different from `scanf`:

- **The scanner never allocates and never writes into your input.** A scanned word comes back as a `proven_u8str_view_t` pointing *into* the original bytes. It is valid exactly as long as those bytes are, and no longer. If it must outlive them, copy it with `proven_u8str_create_from_view()`.
- **The cursor is yours.** It is a plain field. You may save it, restore it, or step it by hand (section 12 does exactly that after `proven_scan_skip_until`). Nothing in the scanner is hidden from you, so nothing has to be undone for you.

`proven_scan_init()` normalizes a malformed view (`size > 0` with a null pointer) to an empty one rather than trusting it, so a scanner built from garbage reads as end-of-input instead of dereferencing.

Each primitive returns a result struct pairing the value with the error that guards it:

`proven_result_i64_t`, `proven_result_u64_t`, `proven_result_f64_t`, `proven_result_u8str_view_t`. **The value is meaningless unless the error is `PROVEN_OK`** - the scanner does not use a sentinel value to mean failure, because every sentinel is also a legitimate input.

## 8. Scanner primitive APIs

```
void proven_scan_skip_whitespace(proven_scan_t *scan);
proven_result_i64_t proven_scan_i64(proven_scan_t *scan);
proven_result_u64_t proven_scan_u64(proven_scan_t *scan);
proven_result_i64_t proven_scan_i64_hex(proven_scan_t *scan);
proven_result_u64_t proven_scan_u64_hex(proven_scan_t *scan);
proven_result_f64_t proven_scan_f64(proven_scan_t *scan);
proven_result_u8str_view_t proven_scan_str(proven_scan_t *scan);
proven_err_t proven_scan_skip_until(proven_scan_t *scan, proven_u8str_view_t target);
void proven_scan_skip_until_number(proven_scan_t *scan);
```

The value-returning ones are `[[nodiscard]]`: a scan whose result you throw away is a scan you did not need to make.

### Shared behaviour

- **Leading whitespace is skipped** by every value scanner. You do not need to call `proven_scan_skip_whitespace()` first; it exists for when you want to position the cursor yourself.
- **Scanning stops at the first byte that cannot belong to the value.** `"12abc"` yields 12 and leaves the cursor on the `a`. That is not an error - the scanner answered the question you asked and left the rest for whoever asks next.
- **On failure the cursor is restored**, so a failed scan is a non-event: you can turn around and parse the same position as something else. section 12 does this.

### The integer scanners

| Input                                                         | <code>proven_scan_i64</code>                              | Why                          |
|---------------------------------------------------------------|-----------------------------------------------------------|------------------------------|
| <code>"42"</code> , <code>" +42"</code> , <code>" -42"</code> | <code>OK - 42</code> , <code>42</code> , <code>-42</code> | a sign is part of the number |

|                       |                        |                                               |
|-----------------------|------------------------|-----------------------------------------------|
| "9223372036854775808" | PROVEN_ERR_OVERFLOW    | one past INT64_MAX; it does <b>not</b> wrap   |
| "abc", ""             | PROVEN_ERR_INVALID_ARG | there is no number here                       |
| "0x10"                | OK - 0, cursor at 1    | <b>decimal only:</b> a zero, followed by text |

That last row is the one that surprises people. `proven_scan_i64` and `proven_scan_u64` read decimal: no octal, no base prefix. `0x10` is the integer zero, and `x10` is still in the input.

Hexadecimal is a separate, explicit request. `proven_scan_u64_hex` reads hex digits in either case, with an optional `0x` taken only when a digit follows it (as `strtoul` takes it, so `"0xg"` is zero followed by `"xg"`); `proven_scan_i64_hex` adds a sign. In a format string, `{:x}` (or `{:X}`) reads any integer argument the same way - the formatter's spec for writing hex, read backwards. Overflow, cursor restore and the stream signal behave as for decimal.

```
/* A UnicodeData-style range, "0041..005A", read as two hexadecimal numbers. */
proven_u32 first = 0, last = 0;
proven_err_t e = proven_scan_fmt(PROVEN_LIT("0041..005A"), "{:x}..{:x}", PROVEN_SCAN_ARG(&first),
PROVEN_SCAN_ARG(&last));
/* e == PROVEN_OK, first == 0x41, last == 0x5A */

/* The primitive, for a hand-written loop: */
proven_scan_t s = proven_scan_init(PROVEN_LIT("0x1F rest"));
proven_result_u64_t h = proven_scan_u64_hex(&s); /* h.val == 0x1F, cursor on " rest" */
proven_scan_t t = proven_scan_init(PROVEN_LIT("-7f"));
proven_result_i64_t g = proven_scan_i64_hex(&t); /* g.val == -0x7F: the signed form takes a sign */
(void)e; (void)h; (void)g;
```

`proven_scan_u64` means unsigned: `"-1"` is `PROVEN_ERR_INVALID_ARG`, not a wrap to 18446744073709551615. A scanner that quietly turns a negative number into a huge positive one is how a bounds check gets defeated.

## The float scanner

`proven_scan_f64` routes through the same correctly-rounded decimal engine as the rest of the library: round-to-nearest, ties-to-even, no long double anywhere. It accepts `nan` and `inf`.

Its two boundary behaviours are **deliberately asymmetric**, and the asymmetry is the point:

- `"1e309"` gives `PROVEN_ERR_OVERFLOW`. There is no correct finite answer, so it refuses rather than handing you an infinity you did not ask for.
- `"1e-400"` gives `PROVEN_OK` and `0.0`, with the sign preserved. Underflow to zero is the correctly rounded answer. Reporting it as an error would mean reporting correct arithmetic as a failure.

## Words and navigation

`proven_scan_str` returns the next whitespace-delimited run as a view into the input. Nothing left but whitespace is `PROVEN_ERR_INVALID_ARG`.

On a **complete view**, "the input ran out" and "the input is wrong" are the same fact - there is no more input, so a number cut off at the end really is malformed. Over a **stream** they are opposite facts, and the difference decides whether you wait or report an error. The scanner sets `proven_scan_t::needs_more` when the parse ran off the end of what it had, and the buffered scanner uses exactly that to refill and retry: a pipe that delivers - and then, a moment later, 12 scans as -12. Before, it was a malformed number. A wrong byte that is actually *present* - a letter where a digit belongs - is still an error, and no amount of further input will change that; the scanner does not wait for it.

`proven_scan_skip_until(scan, target)` moves the cursor **to** the target, not past it - you decide how much of it to consume. If the target is not there the result is `PROVEN_ERR_NOT_FOUND` **and the cursor does not move**: the scanner does not consume input it failed to navigate.

proven\_scan\_skip\_until\_number stops at the first digit, or at a sign immediately followed by a digit. If there is no number it runs the cursor to the end of the input - so check `scan.cursor < scan.view.size` before assuming there is something to read.

## 9. Scan argument model

A scan argument is a **typed pointer to your destination**, selected by the compiler:

```
PROVEN_SCAN_ARG(&x)    /* _Generic on the pointer type */
```

This is where the scanner differs most sharply from `scanf`. There is no format letter to get wrong, because there is no format letter at all. `%d` against a `long`, or `%s` against a buffer that is too small, are not mistakes available here: the destination's type is the specification, and a mismatch is a compile error rather than a corrupted stack.

Supported destinations: `short`, `unsigned short`, `int`, `unsigned int`, `long`, `unsigned long`, `long long`, `unsigned long long`, `double`, and `proven_u8str_view_t`.

`PROVEN_SCAN_ARG_LONG(&x)` and `PROVEN_SCAN_ARG_ULONG(&x)` exist for callers who want to be explicit at the call site; `PROVEN_SCAN_ARG` already handles `long*` and `unsigned long*`.

**Narrow destinations are range-checked.** Scanning `"70000"` into a `short` is `PROVEN_ERR_OVERFLOW`, not a truncated 4464. The value is parsed at 64 bits and checked against the destination's range before anything is stored.

The `proven_scan_arg.*` constructors are public if you need to build an argument array by hand, but the macros are what callers use.

## 10. Structural scan grammar

**Why parsing with a format string is more dangerous than printing with one**

Formatting with a bad format string produces wrong output. **Parsing with one corrupts memory**, because the format decides what to write through the pointers you passed. `sscanf("%d", &c)` where `c` is a `char` writes four bytes into a one-byte object, and nothing in the call says so.

That asymmetry shapes this section. The structural scan uses the same `{}` grammar as the formatter, but the destination type comes from `PROVEN_SCAN_ARG(&x)` - the same `_Generic` dispatch, so the width written is the width of the object, and a value too large for it is `PROVEN_ERR_OVERFLOW` rather than three neighbouring bytes.

The one property to internalise before using it: **the structural scan is not transactional across placeholders**. If the third `{}` fails, the first two destinations have already been written. That is a deliberate trade - buffering every destination until the whole line parsed would need allocation, and this scanner allocates nothing - but it means a failed `proven_scan_fmt` leaves your variables in a partly-updated state. Either treat them as garbage on failure, or scan into locals and copy out only on success. section 11.1 shows both patterns.

Literals in the pattern are the other half of the grammar and the part people underuse: anything that is not a placeholder must match the input exactly, so `"{:}"` on `"12-34"` fails at the literal rather than quietly returning one field.

The scan format string is the formatter's, read backwards:

- a placeholder consumes one argument, in order;
- anything else is a **literal that must match the input exactly**.

One spec exists on the scanning side: `{:x}` (or `{:X}`) reads an integer argument as hexadecimal (section 8); on a float or string argument it is `PROVEN_ERR_INVALID_FORMAT`, as in the formatter. Width, fill and alignment are formatting concerns; the scanner reads what is there.

Whitespace in the format is not special. The value scanners skip leading whitespace themselves, so a format with a space between two placeholders and one without parse "7 8" identically - the space in the format matches the space in the input, and had it not been there, the second scanner would have skipped it anyway.

The number of placeholders must equal the number of arguments. Too few values in the input is an error; **too many is not** (section 11.1).

## 11. Scan formatting APIs

```
proven_scan_fmt(view, fmt, ...) /* scan a view from the beginning */
proven_scan_fmt_cursor(&scan, fmt, ...) /* continue from an existing cursor */
proven_err_t proven_scan_fmt_internal(...) /* what the macros expand to */
```

Use `proven_scan_fmt` for a self-contained line. Use `proven_scan_fmt_cursor` when the scan is one step in a longer walk over the same input: it advances the cursor you own, so it mixes freely with the primitives of section 8.

### 11.1. Scan error code guide and recovery

| Code                      | What actually happened                                                                                                                                                                                                                                        | What to do                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PROVEN_OK                 | Every placeholder was filled and every literal matched.                                                                                                                                                                                                       | Publish the values.                                                                                                                                                  |
| PROVEN_ERR_INVALID_ARG    | The input is not the shape you asked for - a placeholder had no value to read, or the input ran out.                                                                                                                                                          | The line does not match. Report it; do not retry the same shape.                                                                                                     |
| PROVEN_ERR_NEED_MORE      | <b>Buffered scanner only.</b> The token is cut in half by the read boundary: the rest of it has not arrived yet. You will not normally see this - <code>proven_sysio_scanner_scan</code> refills and retries for you - it is what the scanner says to itself. | Nothing. It is handled.                                                                                                                                              |
| PROVEN_ERR_NOT_FOUND      | A <b>literal</b> in the format did not match.                                                                                                                                                                                                                 | The line has a different shape than expected. Try another format, from a saved cursor.                                                                               |
| PROVEN_ERR_OVERFLOW       | A number was well-formed but does not fit the destination.                                                                                                                                                                                                    | The input may be valid and your destination too narrow, or the input may be hostile. Those are very different situations - tell them apart before widening the type. |
| PROVEN_ERR_INVALID_FORMAT | The format string itself is malformed.                                                                                                                                                                                                                        | A bug in your code, not in the input.                                                                                                                                |

**The structural scanner is not transactional, and this is the one that bites.**

When a literal fails to match, the placeholders *before* the mismatch have already been written through. The call returns `PROVEN_ERR_NOT_FOUND` and your destination holds a value anyway - `id` below is 7, from a call that failed:

```
int id = -1;
double ratio = -1.0;
proven_err_t err = proven_scan_fmt(line, "id={ } XXX={ }",
                                   PROVEN_SCAN_ARG(&id), PROVEN_SCAN_ARG(&ratio));
/* err == PROVEN_ERR_NOT_FOUND, and id == 7: it was written before the failure. */
```

So: **on failure, treat every destination as clobbered**. If you need all-or-nothing, scan into locals and publish them only once the call has succeeded - the worked example in section 12 shows the shape. Alternatively, save `scan.cursor` before the call and restore it afterwards. The cursor is a plain field, and that is deliberate.

**Trailing input is not an error**. Scanning one placeholder against "7 8" succeeds with the value 7 and leaves 8 unconsumed. The scanner matched what you asked for and stopped; it does not police what you did not ask about. If the whole line must be consumed, check the cursor yourself:

```
if (scan.cursor != scan.view.size) { /* there is unparsed input left */ }
```

## 12. Examples and misuse cases

### Worked example: formatting a line and scanning it back

Compiled and run by the test suite. It formats into a stack-borrowed string (atomic on overflow) and into an allocator-backed one (grows), uses a bounded argument for untrusted bytes, and then parses the line back - with the float round-tripping exactly.

```
/*
 * Formatting and scanning are the two halves of the same idea: `{}` renders typed
 * values into text, and the scanner reads text back into typed destinations. Both
 * are type-checked at the call site (_Generic picks the constructor), so there is
 * no format-string/argument mismatch to get wrong at runtime.
 *
 * The choice that matters is *where the bytes go*:
 *
 * append_fmt      - fixed capacity, atomic. Too long? Nothing is written and
 *                   you get PROVEN_ERR_OUT_OF_BOUNDS. No allocator involved,
 *                   so it works on a stack buffer.
 * append_fmt_grow - allocator-backed. Grows to fit; on allocation failure the
 *                   string is left exactly as it was.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- fixed capacity: no allocator, no allocation ----- */
    /* borrow wraps caller memory, so this string lives entirely on the stack. `cap`
     * includes the NUL, so 32 bytes hold 31 of content. Nothing to destroy. */
    proven_byte_t stack_buf[32];
    proven_u8str_t fixed = proven_u8str_borrow(stack_buf, sizeof stack_buf);

    proven_fmt_result_t r = proven_u8str_append_fmt(&fixed, "port={}", PROVEN_ARG(8080));
    EXAMPLE_REQUIRE(PROVEN_FMT_IS_OK(r), "a short line should fit in 32 bytes");
    EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&fixed), PROVEN_LIT("port=8080")),
        "the fixed-capacity append should have rendered the port");

    /* Atomic means atomic: an append that does not fit changes nothing. The string
     * is still valid and still holds what it held before - no truncated tail to
     * clean up. (Use append_fmt_trunc if a truncated tail is what you want.) */
    proven_fmt_result_t too_long = proven_u8str_append_fmt(
        &fixed, " and a great deal more text than will ever fit here {}", PROVEN_ARG(1));
    EXAMPLE_REQUIRE(too_long.err == PROVEN_ERR_OUT_OF_BOUNDS, "the overlong append must fail");
    EXAMPLE_REQUIRE(too_long.required > too_long.written, "it reports what it would have needed");
    EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&fixed), PROVEN_LIT("port=8080")),
        "a failed atomic append must leave the string untouched");

    /* --- specs: fill, align, width, hex ----- */
    proven_result_u8str_t created = proven_u8str_create(alloc, 8); /* deliberately small */
    EXAMPLE_REQUIRE(proven_is_ok(created.err), "creating the output string should succeed");
```

```

if (!proven_is_ok(created.err)) return 1;
proven_u8str_t out = created.value;

/* grow reallocates as needed, so the initial capacity is a hint, not a limit.
 * `{:0>4}` = fill '0', align right, width 4. `{:x}` = lowercase hex, no 0x. */
r = proven_u8str_append_fmt_grow(alloc, &out, "id={:0>4} tag={:*^9} addr=0x{:x}",
    PROVEN_ARG(7),
    PROVEN_ARG(PROVEN_LIT("ok")),
    PROVEN_ARG(48879));
EXAMPLE_REQUIRE(PROVEN_FMT_IS_OK(r), "the growing append should succeed");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&out),
    PROVEN_LIT("id=0007 tag=***ok*** addr=0xbeef")),
    "fill/align/width/hex should render exactly this");
printf("%s\n", proven_u8str_as_cstr(&out));

/* --- untrusted text is bounded, never trusted to be NUL-terminated ----- */
/* PROVEN_ARG on a char* means "walk it until a NUL turns up" - fine for a
 * literal, a buffer-overread for anything that came off a socket. This buffer
 * has no NUL at all; PROVEN_ARG_CSTR_N stops at the length instead, so it reads
 * only what actually exists. Use it for anything you did not create yourself. */
const char untrusted[4] = {'a', 'b', 'c', 'd'}; /* no terminator, on purpose */
EXAMPLE_REQUIRE(proven_is_ok(proven_u8str_reset(&out)), "reset should keep the buffer");
r = proven_u8str_append_fmt_grow(alloc, &out, "payload={}",
    PROVEN_ARG_CSTR_N(untrusted, sizeof untrusted));
EXAMPLE_REQUIRE(PROVEN_FMT_IS_OK(r), "the bounded append should succeed");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&out), PROVEN_LIT("payload=abcd")),
    "the bounded argument should render its whole 4 bytes and stop");

/* --- format a record, then scan it back ----- */
proven_i64 sensor_id = 42;
double reading = 3.14159;

EXAMPLE_REQUIRE(proven_is_ok(proven_u8str_reset(&out)), "reset should keep the buffer");
r = proven_u8str_append_fmt_grow(alloc, &out, "{ } { }",
    PROVEN_ARG(sensor_id),
    PROVEN_ARG(PROVEN_LIT("boiler")),
    PROVEN_ARG(reading));
EXAMPLE_REQUIRE(PROVEN_FMT_IS_OK(r), "formatting the record should succeed");
printf("record: %s\n", proven_u8str_as_cstr(&out));

/* One scanner over one view. Each call advances the cursor past what it
 * consumed, so the calls compose left to right - and each one can fail
 * independently, which is the difference between a parser and a guess. */
proven_scan_t sc = proven_scan_init(proven_u8str_as_view(&out));

proven_result_i64_t id = proven_scan_i64(&sc);
EXAMPLE_REQUIRE(proven_is_ok(id.err), "the first field should parse as an integer");
EXAMPLE_REQUIRE(id.val == sensor_id, "the integer should round-trip");

/* scan_str returns a view *into the scanned string* - it copies nothing and
 * owns nothing, so it is only valid while `out` is. */
proven_result_u8str_view_t name = proven_scan_str(&sc);
EXAMPLE_REQUIRE(proven_is_ok(name.err), "the second field should parse as a word");
EXAMPLE_REQUIRE(proven_u8str_view_eq(name.val, PROVEN_LIT("boiler")), "the name should round-trip");

proven_result_f64_t temp = proven_scan_f64(&sc);
EXAMPLE_REQUIRE(proven_is_ok(temp.err), "the third field should parse as a float");

/* Exactly equal, not approximately: the scanner is correctly rounded, so it
 * returns the nearest double to the text - and the text the formatter produced

```

```

    * (six fractional digits) names this value unambiguously. Bit-for-bit, this is
    * the same double we started with. For a value that needs more than six
    * fractional digits, format it with the shortest policy
    * (proven_float_format_options_shortest) and the same round-trip holds. */
EXAMPLE_REQUIRE(temp.val == reading, "the float must round-trip exactly, not approximately");

/* The input is fully consumed: nothing was silently left on the table. */
proven_result_i64_t extra = proven_scan_i64(&sc);
EXAMPLE_REQUIRE(!proven_is_ok(extra.err), "there should be nothing left to scan");

proven_u8str_destroy(alloc, &out);
return EXAMPLE_OK();
}

```

### Worked example: the scanner's error codes, and recovering from them

Compiled and run by the test suite. Every code in the table above is provoked on purpose here, including the non-transactional failure - because a contract you have only read is a contract you have not learned.

```

/*
 * The scanner's error codes, and how to recover from them.
 *
 * The scanner is not scanf. It never writes through a pointer it was not given,
 * it never guesses a width, and it tells you which of several different things
 * went wrong. That last part only helps if you know what the codes mean - so
 * this program provokes each one on purpose.
 */

static proven_u8str_view_t v(const char *s) {
    return proven_u8str_view_from_cstr(s);
}

int main(void) {
    /* --- the primitives restore the cursor when they fail ----- */
    /* A failed scan is a non-event: the cursor is where it was, so you can try
     * to parse the same position as something else. */
    {
        proven_scan_t sc = proven_scan_init(v("abc"));
        proven_result_i64_t n = proven_scan_i64(&sc);
        EXAMPLE_REQUIRE(n.err == PROVEN_ERR_INVALID_ARG, "'abc' is not an integer");
        EXAMPLE_REQUIRE(sc.cursor == 0, "a failed integer scan leaves the cursor alone");

        /* So the same position can be read as a word instead. */
        proven_result_u8str_view_t w = proven_scan_str(&sc);
        EXAMPLE_REQUIRE(proven_is_ok(w.err) && proven_u8str_view_eq(w.val, PROVEN_LIT("abc")),
            "the same bytes parse fine as a word");
    }

    /* --- a number that does not fit is OVERFLOW, not a wrapped value ----- */
    {
        proven_scan_t sc = proven_scan_init(v("9223372036854775808")); /* INT64_MAX + 1 */
        proven_result_i64_t n = proven_scan_i64(&sc);
        EXAMPLE_REQUIRE(n.err == PROVEN_ERR_OVERFLOW, "one past INT64_MAX must not wrap");
        EXAMPLE_REQUIRE(sc.cursor == 0, "the cursor is restored on overflow too");
    }

    /* --- but a float that underflows is NOT an error ----- */
    /* Too large is OVERFLOW; too small is zero, with the sign kept. That
     * asymmetry is deliberate - underflow to zero is the correctly rounded
     * answer, while overflow has no correct finite answer at all. */

```

```

{
    proven_scan_t big = proven_scan_init(v("1e309"));
    proven_result_f64_t b = proven_scan_f64(&big);
    EXAMPLE_REQUIRE(b.err == PROVEN_ERR_OVERFLOW, "1e309 does not fit a double");

    proven_scan_t tiny = proven_scan_init(v("-1e-400"));
    proven_result_f64_t t = proven_scan_f64(&tiny);
    EXAMPLE_REQUIRE(proven_is_ok(t.err), "1e-400 underflows, which is not an error");
    EXAMPLE_REQUIRE(t.val == 0.0, "it rounds to zero");
}

/* --- the integer scanners are decimal only ----- */
/* "0x10" is not sixteen. It is a zero, followed by text the scanner has not
 * been asked to look at. This surprises people, so it is worth knowing. */
{
    proven_scan_t sc = proven_scan_init(v("0x10"));
    proven_result_i64_t n = proven_scan_i64(&sc);
    EXAMPLE_REQUIRE(proven_is_ok(n.err) && n.val == 0, "0x10 scans as the integer 0");
    EXAMPLE_REQUIRE(sc.cursor == 1, "and the cursor stops before the 'x'");
}

/* --- scanning stops at the first byte that cannot belong to the value -- */
{
    proven_scan_t sc = proven_scan_init(v("12abc"));
    proven_result_i64_t n = proven_scan_i64(&sc);
    EXAMPLE_REQUIRE(proven_is_ok(n.err) && n.val == 12, "12abc yields 12");
    EXAMPLE_REQUIRE(sc.cursor == 2, "and leaves 'abc' for whoever asks next");
}

/* --- unsigned means unsigned ----- */
{
    proven_scan_t sc = proven_scan_init(v("-1"));
    proven_result_u64_t n = proven_scan_u64(&sc);
    EXAMPLE_REQUIRE(n.err == PROVEN_ERR_INVALID_ARG,
        "-1 is rejected rather than wrapping to a huge unsigned value");
}

/* --- navigating to a value: skip_until ----- */
/* skip_until leaves the cursor ON the target, not past it, so you decide
 * how much of it to consume. */
{
    proven_scan_t sc = proven_scan_init(v("port=8080"));
    proven_err_t err = proven_scan_skip_until(&sc, PROVEN_LIT("="));
    EXAMPLE_REQUIRE(proven_is_ok(err), "the '=' is there");
    EXAMPLE_REQUIRE(sc.cursor == 4, "the cursor sits on the '=' itself");

    ++sc.cursor; /* step over it */
    proven_result_i64_t port = proven_scan_i64(&sc);
    EXAMPLE_REQUIRE(proven_is_ok(port.err) && port.val == 8080, "the port parses");

    /* Not finding it is NOT_FOUND, and the cursor does not move - the
     * scanner does not consume the input it failed to navigate. */
    proven_scan_t sc2 = proven_scan_init(v("port=8080"));
    proven_err_t missing = proven_scan_skip_until(&sc2, PROVEN_LIT("#"));
    EXAMPLE_REQUIRE(missing == PROVEN_ERR_NOT_FOUND, "there is no '#'");
    EXAMPLE_REQUIRE(sc2.cursor == 0, "and the cursor stayed put");
}

/* --- the structural scanner ----- */
{

```

```

int id = 0;
double ratio = 0.0;
proven_u8str_view_t name = {};

proven_err_t err = proven_scan_fmt(v("id=7 ratio=0.5 name=ada"),
    "id={} ratio={} name={}",
    PROVEN_SCAN_ARG(&id),
    PROVEN_SCAN_ARG(&ratio),
    PROVEN_SCAN_ARG(&name));
EXAMPLE_REQUIRE(proven_is_ok(err), "the line matches the shape");
EXAMPLE_REQUIRE(id == 7 && ratio == 0.5, "the values land in the right places");
EXAMPLE_REQUIRE(proven_u8str_view_eq(name, PROVEN_LIT("ada")), "including the word");
}

/* --- the structural scanner is NOT transactional ----- */
/*
 * This is the one that bites. When a literal fails to match, the scan
 * returns an error - but the placeholders BEFORE the mismatch have already
 * been written through. `id` is 7 even though the call failed.
 *
 * So: on failure, treat every destination as clobbered. If you need
 * all-or-nothing, scan into locals and only publish them once the call
 * succeeded, which is what the code below does.
 */
{
    int id = -1;
    double ratio = -1.0;
    proven_err_t err = proven_scan_fmt(v("id=7 ratio=0.5"),
        "id={} XXX={}", /* the literal is wrong */
        PROVEN_SCAN_ARG(&id),
        PROVEN_SCAN_ARG(&ratio));
    EXAMPLE_REQUIRE(err == PROVEN_ERR_NOT_FOUND, "the literal 'XXX=' is not in the input");
    EXAMPLE_REQUIRE(id == 7, "and yet id was already written: the scan is not atomic");

    /* The safe shape: scan into locals, publish on success. */
    int good_id = 0;
    double good_ratio = 0.0;
    int published_id = -1;
    proven_err_t ok = proven_scan_fmt(v("id=7 ratio=0.5"), "id={} ratio={}",
        PROVEN_SCAN_ARG(&good_id), PROVEN_SCAN_ARG(&good_ratio));
    if (proven_is_ok(ok)) published_id = good_id;
    EXAMPLE_REQUIRE(published_id == 7, "publish only what a successful scan produced");
}

/* --- running out of input, and having input left over ----- */
{
    int a = 0, b = 0;
    proven_err_t short_input = proven_scan_fmt(v("5"), "{} {}",
        PROVEN_SCAN_ARG(&a), PROVEN_SCAN_ARG(&b));
    EXAMPLE_REQUIRE(!proven_is_ok(short_input), "two placeholders, one value: that fails");

    /* Trailing input is NOT an error. The scanner matched what you asked for
     * and stopped; it does not police what you did not ask about. If the
     * whole line must be consumed, check that yourself. */
    int only = 0;
    proven_scan_t sc = proven_scan_init(v("7 8"));
    proven_err_t err = proven_scan_fmt_cursor(&sc, "{}", PROVEN_SCAN_ARG(&only));
    EXAMPLE_REQUIRE(proven_is_ok(err) && only == 7, "the first value scans");
    EXAMPLE_REQUIRE(sc.cursor < sc.view.size, "and '8' is still sitting there, unconsumed");
}

```

```

/* --- narrow destinations are range-checked ----- */
{
    short small = 0;
    proven_err_t err = proven_scan_fmt(v("70000"), "{}", PROVEN_SCAN_ARG(&small));
    EXAMPLE_REQUIRE(err == PROVEN_ERR_OVERFLOW,
        "70000 does not fit a short, and the scanner says so rather than truncating");
}

return EXAMPLE_OK();
}

```

#### **Misuse: assuming 0x10 is sixteen**

It is zero. The decimal scanners are decimal only, and x10 is still in the input. If the input is hex, say so: `proven_scan_u64_hex`, or `{:x}` in the format.

#### **Misuse: treating trailing input as an error**

It is not one. One placeholder against "7 8" succeeds. Check the cursor if you care.

#### **Misuse: trusting destinations after a failed scan**

They are clobbered. See section 11.1.

#### **Misuse: keeping a scanned word after its input is gone**

`proven_scan_str` returns a **view into the input**, not a copy. When the buffer goes, so does the word. Copy it with `proven_u8str_create_from_view()` if it has to outlive the bytes it came from.

## **13. Freestanding and build-mode notes**

### **Why float is the one thing that gets compiled out**

Everything in this chapter is portable computation except one part, and that part is unusually expensive.

Formatting a double correctly - so that the shortest decimal that round-trips is what you get, on every input including subnormals - needs big-integer arithmetic and lookup tables. It is the largest single piece of code in the formatter, and most firmware never prints a double at all. So the freestanding profile sets `PROVEN_FMT_NO_FLOAT` and drops it, and a build for a microcontroller does not carry kilobytes of decimal-conversion tables it will never call.

This is a build-time decision rather than a run-time one on purpose: the code is *absent*, not merely unreachable, so the linker cannot be talked into keeping it. section 8a of the [freestanding guide](#) covers the related knob - the big-integer capacity - for builds that do want floats on a small target.

The scanner is core: it does no I/O, allocates nothing, and touches no platform layer, so it is available in a freestanding build exactly as it is in a hosted one.

The one build-mode dependency is float. The freestanding build sets `PROVEN_FMT_NO_FLOAT` (which compiles out the float *formatter*) along with `PROVEN_NO_U16STR`. `proven_scan_f64` pulls in the decimal parsing engine (`float_parse.c` and `float_decimal.c`), which is integer-only - no long double, no libm, no soft-float helper calls - but it is not free in code size. On a target where that matters and you do not need to read floats, simply do not call it: nothing else in the scanner references the float path.

`proven_sysio_scanner_*` (Chapter 5) is a **different** thing: a buffered scanner over a file, hosted-only, because it does I/O. The scanner described in this chapter reads bytes you already have.

### **Worked example: wider numbers, loose text, and floating point**

The examples above scan rigid formats into `proven_i32`. Three things come up as soon as the input is real, and this example covers them together.

**The destination type is the contract.** `proven_scan_arg_i64()`, `proven_scan_arg_u32()` and `proven_scan_arg_u64()` name the type the value is written into, and the scanner reports an overflow rather than wrapping. A byte count that needs more than 32 bits, and a value that must never be negative, are the two cases where the wrong destination type is a bug nobody notices until the numbers get big. The generic `PROVEN_SCAN_ARG()` macro chooses these from the pointer you hand it; the named forms are what it chooses, written out.

**Not every input is a rigid format.** `proven_scan_skip_whitespace()` advances past spaces, tabs and newlines, and `proven_scan_skip_until_number()` runs the cursor forward to the next digit - or to a sign immediately followed by one, so a negative number keeps its sign. Neither can fail: "there was nothing to skip" is not an error, and at the end of the input the cursor stops instead of running off.

**Floating point has a locale problem, and this library does not have it.** `strtod()` reads "3,5" as 3.5 under one locale and as 3 under another, so the same program disagrees with itself across machines. `proven_parse_double_ascii()` is locale-free by construction: a comma is never a decimal point. It reports how many bytes the number used, so a caller can carry on from there, and it reports unparsable text as an error rather than as a 0 that cannot be told from a real one. `proven_parse_f64_ascii()` is the same function under its earlier name, kept so existing call sites still read correctly.

On the way out, `proven_arg_f64()` is how a floating-point value enters the formatter - both `float` and `double` go through it, so what a program prints does not depend on the width of the variable it was kept in. When the exact spelling matters, `proven_float_format_f32_policy()` and its `f64` sibling take the policy and options explicitly. Shortest mode asks for the fewest digits that read back as exactly this value, which is what a serialiser wants: an exact round trip without printing seventeen digits for numbers that do not need them.

```
#include <string.h>
```

```
/*
 * Reading numbers out of text, and writing them back.
 *
 * The text here is the sort a program actually meets: a log line with mixed
 * widths and signs, and a measurements file where the interesting number is
 * buried in prose. That means three jobs the earlier chapter-8 examples do not
 * cover:
 *
 * - Scanning into types WIDER than int, and into unsigned ones, where the
 *   destination type is the whole question - a byte count that does not fit in
 *   32 bits is the classic overflow nobody notices until the file is 5 GB.
 * - Positioning the cursor by hand - skipping whitespace, or skipping ahead to
 *   wherever the next number starts - when the input is not a rigid format.
 * - Converting a decimal string to a double and back, exactly, without going
 *   through the C library's locale-dependent conversions.
 *
 * The float part matters more than it sounds. strtod reads "3,5" as 3.5 in one
 * locale and as 3 in another, and the same program then disagrees with itself
 * across machines. The parser used here is locale-free by construction: a comma
 * is never a decimal point, whatever the environment says.
 */

int main(void) {
    /* --- 1. scanning into the type the value needs ----- */

    /* A log line: a request id that needs 64 bits, a byte count that is never
     * negative and can exceed 4 GB, a small status code, and a negative offset. */
    proven_scan_t scan = proven_scan_init(
        PROVEN_LIT("id=9007199254740993 bytes=5368709120 status=404 delta=-17"));
```

```

proven_i64 id = 0;
proven_u64 bytes = 0;
proven_u32 status = 0;
proven_i32 delta = 0;

/* Each argument constructor names the destination type, so the scanner
 * writes the right width and reports an overflow instead of wrapping. The
 * generic PROVEN_SCAN_ARG() macro picks these for you from the pointer's
 * type; the named forms are what it picks, and what you write when you want
 * the choice visible. */
proven_err_t err = proven_scan_fmt_cursor(&scan, "id={} bytes={} status={} delta={}",
    proven_scan_arg_i64(&id),
    proven_scan_arg_u64(&bytes),
    proven_scan_arg_u32(&status),
    proven_scan_arg_i32(&delta));
EXAMPLE_REQUIRE(proven_is_ok(err), "scanning the log line must succeed");
EXAMPLE_REQUIRE(id == 9007199254740993LL, "a 64-bit id survives, which a 32-bit destination could not");
EXAMPLE_REQUIRE(bytes == 5368709120ULL, "and so does a byte count larger than 4 GiB");
EXAMPLE_REQUIRE(status == 404u, "the small unsigned value reads normally");
EXAMPLE_REQUIRE(delta == -17, "and a signed destination accepts the minus sign");

/* The destination type is a contract, not a hint. A negative number has no
 * unsigned representation, and the scanner refuses rather than wrapping it
 * to something enormous. */
proven_scan_t neg = proven_scan_init(PROVEN_LIT("-17"));
proven_u32 nowhere = 12345;
err = proven_scan_fmt_cursor(&neg, "{}", proven_scan_arg_u32(&nowhere));
EXAMPLE_REQUIRE(err != PROVEN_OK, "a negative value cannot be scanned into an unsigned destination");
EXAMPLE_REQUIRE(nowhere == 12345, "and the destination is left as it was");

/* --- 2. moving the cursor when the input is not rigid ----- */

/* Free-form text: the numbers matter, the words between them do not. */
proven_scan_t notes = proven_scan_init(
    PROVEN_LIT(" sample A measured 42 units; sample B measured -8 units"));

/* skip_whitespace advances past spaces, tabs and newlines. It is what you
 * call between fields of a format you are driving by hand - it never fails,
 * because "there was no whitespace" is not an error. */
proven_scan_skip_whitespace(&notes);
EXAMPLE_REQUIRE(notes.cursor == 3, "the three leading spaces are consumed");

/* skip_until_number runs the cursor forward to the first digit, or to a sign
 * immediately followed by one. It is the call that turns "find the number in
 * this line" from a hand-written loop into one statement. */
proven_scan_skip_until_number(&notes);
proven_i32 first = 0;
err = proven_scan_fmt_cursor(&notes, "{}", proven_scan_arg_i32(&first));
EXAMPLE_REQUIRE(proven_is_ok(err) && first == 42, "the first number in the line is 42");

proven_scan_skip_until_number(&notes);
proven_i32 second = 0;
err = proven_scan_fmt_cursor(&notes, "{}", proven_scan_arg_i32(&second));
EXAMPLE_REQUIRE(proven_is_ok(err) && second == -8,
    "and the next one keeps its sign, because the sign is part of the number");

/* At the end there is nothing left to find, and the cursor stops rather than
 * running off: the scan is over, not broken. */
proven_scan_skip_until_number(&notes);

```

```

EXAMPLE_REQUIRE(notes.cursor == notes.view.size, "with no number left, the cursor lands at the end");

/* --- 3. decimal text to double, without a locale ----- */

proven_parse_double_result_t d = proven_parse_double_ascii(PROVEN_LIT("3.14159 rest"));
EXAMPLE_REQUIRE(proven_is_ok(d.err), "parsing a decimal number must succeed");
EXAMPLE_REQUIRE(d.val > 3.14158 && d.val < 3.14160, "and produce the value the digits spell");

/* consumed says where the number ended, so the caller can carry on from
 * there - which is how you parse a list without copying it into pieces. */
EXAMPLE_REQUIRE(d.consumed == 7, "consumed reports exactly the bytes the number used");

/* A comma is not a decimal point here and never will be, whatever locale the
 * program is running under. The number ends at the comma. */
proven_parse_double_result_t comma = proven_parse_double_ascii(PROVEN_LIT("3,5"));
EXAMPLE_REQUIRE(proven_is_ok(comma.err) && comma.consumed == 1 && comma.val == 3.0,
    "a comma ends the number: the parser is locale-free by construction");

/* proven_parse_f64_ascii is the same function under its earlier name, kept
 * for code written before the current one existed. New code should use
 * proven_parse_double_ascii; both are here so an old call site still reads
 * correctly. */
proven_parse_f64_result_t same = proven_parse_f64_ascii(PROVEN_LIT("3.14159 rest"));
EXAMPLE_REQUIRE(same.val == d.val && same.consumed == d.consumed,
    "the compatibility name is the same parser");

/* Text that is not a number at all is an error, not a silent zero - which is
 * what atof() would have handed back with nothing to distinguish it from a
 * genuine "0". */
proven_parse_double_result_t junk = proven_parse_double_ascii(PROVEN_LIT("not a number"));
EXAMPLE_REQUIRE(!proven_is_ok(junk.err), "unparsable text is reported, not turned into 0");
EXAMPLE_REQUIRE(junk.consumed == 0, "and nothing was consumed");

/* --- 4. writing a float back ----- */

/* proven_arg_f64 is how a floating-point value enters the formatter. Both
 * float and double go through it, so the digits a program prints do not
 * depend on the width of the variable it happened to be kept in. */
proven_allocator_t alloc = proven_heap_allocator();
proven_result_u8str_t line = proven_u8str_create(alloc, 64);
EXAMPLE_REQUIRE(proven_is_ok(line.err), "creating the output string must succeed");

proven_fmt_result_t out = proven_u8str_append_fmt(&line.value, "measured {} units",
    proven_arg_f64(d.val));
EXAMPLE_REQUIRE(proven_is_ok(out.err), "formatting the parsed value must succeed");
EXAMPLE_REQUIRE(proven_u8str_view_starts_with(proven_u8str_as_view(&line.value),
    PROVEN_LIT("measured 3.14159")),
    "and spell the number it was given");

/* The policy form is the explicit one, for when the caller needs a
 * particular spelling rather than the default. SHORTEST asks for the fewest
 * digits that read back as exactly this value - which is the property a
 * serialiser wants, because it makes the round trip exact without printing
 * seventeen digits for numbers that do not need them. */
char shortest[64];
proven_size_t wrote = 0;
float measured = 0.1f;
proven_err_t ferr = proven_float_format_f32_policy(shortest, sizeof shortest, measured,
    PROVEN_FLOAT_FORMAT_POLICY_RYU,
    proven_float_format_options_shortest(),

```

```

                                &wrote);
EXAMPLE_REQUIRE(proven_is_ok(ferr), "formatting a float in shortest mode must succeed");
EXAMPLE_REQUIRE(wrote > 0 && shortest[wrote] == '\0', "the result is written and terminated");
EXAMPLE_REQUIRE(strcmp(shortest, "0.1") == 0,
    "0.1f prints as 0.1: the shortest text that reads back as the same float");

/* And it round-trips: the text reads back as the value it came from. That is
 * the guarantee shortest mode exists to provide. */
proven_parse_double_result_t roundtrip = proven_parse_double_ascii(
    (proven_u8str_view_t){ .ptr = (const proven_byte_t *)shortest, .size = wrote });
EXAMPLE_REQUIRE(proven_is_ok(roundtrip.err), "the shortest form parses back");
EXAMPLE_REQUIRE((float)roundtrip.val == measured, "as exactly the float it was printed from");

printf("scanned id=%lld bytes=%llu; formatted %s and %s\n",
    (long long)id, (unsigned long long)bytes, proven_u8str_as_cstr(&line.value), shortest);

proven_u8str_destroy(alloc, &line.value);
return EXAMPLE_OK();
}

```

Wrong - scanning a large value into a 32-bit destination:

```

proven_i32 bytes = 0;
proven_err_t e = proven_scan_fmt_cursor(&scan, "bytes={}", proven_scan_arg_i32(&bytes));
/* input says 5368709120 */                                /* wrong type */

```

The value does not fit. Choosing the destination type by what is convenient, rather than by the range the field can hold, is the same mistake as declaring a file offset int.

Wrong - using atof or strtod for data that crosses machines:

```

double v = strtod(text, NULL); /* wrong: the decimal point depends on the locale */

```

And atof cannot report failure at all: unparsable text becomes 0, which is indistinguishable from a genuine zero in the data.

Wrong - assuming skip\_until\_number reports "not found":

```

proven_scan_skip_until_number(&scan);
proven_i32 n = 0;
proven_err_t e = proven_scan_fmt_cursor(&scan, "{}", proven_scan_arg_i32(&n)); /* may fail */

```

It moves the cursor and returns nothing. When there is no number left the cursor lands at the end of the input, and it is the scan afterwards that tells you so -- check that error rather than assuming a number was found.

### Worked example: naming the argument type yourself

PROVEN\_ARG(x) and PROVEN\_SCAN\_ARG(&x) choose an argument constructor from the type of what you hand them, and most code never needs to know which one they chose. Two situations take the choice back:

1. **The macro has no type to dispatch on that means what you mean.** A proven\_u8str\_view\_t, a broken-down date, a raw address printed for a diagnostic - proven\_arg\_str\_view(), proven\_arg\_datetime() and proven\_arg\_ptr() exist because those are deliberate choices, not defaults to fall into.
2. **The argument list is built at run time.** A logging helper whose parameters are already proven\_arg\_t values cannot re-wrap them - a macro that turns a value into an argument has nothing to do with something that is already one. proven\_arg\_identity() and proven\_scan\_arg\_identity() are what let the same macro-driven code path accept an argument built earlier.

The example writes both sides out by name: the formatting constructors for a character, a boolean, unsigned 32- and 64-bit values, the three ways of passing text (`proven_arg_str_view()`, `proven_arg_cstr()`, `proven_arg_ucstr()`), a date and a pointer; and the scanning constructors for every plain C integer width (short, int, long, long long and their unsigned forms), a double, and a string view captured without copying.

The three text constructors differ in how much they trust the caller, and the order is worth remembering:

| Constructor                      | What it is given              | What can go wrong                                                                              |
|----------------------------------|-------------------------------|------------------------------------------------------------------------------------------------|
| <code>proven_arg_str_view</code> | a pointer <b>and</b> a length | nothing is scanned for a terminator, so there is none to be missing - prefer this              |
| <code>proven_arg_cstr</code>     | a NUL-terminated C string     | the terminator must be there and the memory must be alive, or the formatter reads past the end |
| <code>proven_arg_ucstr</code>    | the same, as unsigned char *  | exists so a byte buffer does not need a cast that silences a real warning                      |

```

/*
 * Every value handed to the formatter arrives as a proven_arg_t, and every value
 * the scanner writes into arrives as a proven_scan_arg_t. Most of the time you
 * never see either: PROVEN_ARG(x) and PROVEN_SCAN_ARG(&x) pick the right
 * constructor from the type of what you passed, and that is the whole point of
 * them.
 *
 * Two situations take the choice back off the macro, and both are ordinary:
 *
 * 1. The macro cannot see a type it recognises. A `proven_u8str_view_t`, a
 *    broken-down date, a raw address printed for a diagnostic - these need the
 *    constructor named, because there is no plain C type for the macro to
 *    dispatch on that means what you mean.
 *
 * 2. You are building the argument list at RUNTIME. A logging helper that
 *    takes "whatever the caller already assembled" receives proven_arg_t
 *    values, not ints and strings - and a macro that turns a value into an
 *    argument cannot be handed something that is already an argument. That is
 *    what the identity constructors are for: they let the same macro-driven
 *    code path accept an argument that was built earlier.
 *
 * So this example writes both sides out by name. The width and signedness of a
 * destination is not decoration - it is what decides whether 70000 arrives as
 * 70000 or as 4464.
 */

/* A logging helper that takes arguments the caller has already built. Because
 * its parameters are proven_arg_t, PROVEN_ARG would be the wrong tool inside it
 * - the value is already an argument, and proven_arg_identity is what says so. */
static proven_fmt_result_t log_pair(proven_u8str_t *out, const char *fmt,
                                   proven_arg_t a, proven_arg_t b) {
    return proven_u8str_append_fmt(out, fmt, proven_arg_identity(a), proven_arg_identity(b));
}

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    proven_result_u8str_t line = proven_u8str_create(alloc, 256);
    EXAMPLE_REQUIRE(proven_is_ok(line.err), "creating the output string must succeed");

```

```

if (!proven_is_ok(line.err)) {
    return 1;
}
proven_u8str_t out = line.value;

/* --- naming the formatting argument yourself ----- */

/* A single character and a flag. Written by name here so it is visible that
 * a bool prints as true/false rather than as 1/0. */
proven_fmt_result_t r = proven_u8str_append_fmt(&out, "flag={} mark={}",
                                                proven_arg_bool(true),
                                                proven_arg_char('!'));
EXAMPLE_REQUIRE(proven_is_ok(r.err), "formatting a bool and a char must succeed");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&out), PROVEN_LIT("flag=true mark=!")),
                "a bool renders as a word, not as a digit");

EXAMPLE_REQUIRE(proven_is_ok(proven_u8str_reset(&out)), "clearing the line must succeed");

/* Unsigned widths. The constructor is the declaration of what the value is:
 * an unsigned 32-bit count and an unsigned 64-bit byte total are different
 * facts, and writing them out says which one you have. */
r = proven_u8str_append_fmt(&out, "files={} bytes={}",
                            proven_arg_u32(1200u),
                            proven_arg_u64(5368709120ULL));
EXAMPLE_REQUIRE(proven_is_ok(r.err), "formatting unsigned values must succeed");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&out), PROVEN_LIT("files=1200
bytes=5368709120")),
                "a 64-bit total is printed in full, not truncated to 32 bits");

EXAMPLE_REQUIRE(proven_is_ok(proven_u8str_reset(&out)), "clearing the line must succeed");

/* Three ways to give the formatter text, in order of how much they trust
 * the caller:
 *
 * proven_arg_str_view - a pointer AND a length. Nothing is scanned for a
 * terminator, so there is no terminator to be
 * missing. Prefer this.
 * proven_arg_cstr - a NUL-terminated C string. The formatter walks it
 * looking for the terminator, so it must be there,
 * and the memory must still be alive.
 * proven_arg_ucstr - the same, for `unsigned char *`, which is what a
 * byte buffer is usually typed as. It exists so the
 * caller does not have to write a cast that
 * silences a real warning.
 */
const char *name = "report.txt";
const unsigned char *tag = (const unsigned char *)"draft";
proven_u8str_view_t note = PROVEN_LIT("first pass");

r = proven_u8str_append_fmt(&out, "{} [{}] {}",
                            proven_arg_cstr(name),
                            proven_arg_ucstr(tag),
                            proven_arg_str_view(note));
EXAMPLE_REQUIRE(proven_is_ok(r.err), "formatting the three text forms must succeed");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&out),
                                    PROVEN_LIT("report.txt [draft] first pass")),
                "all three produce the same kind of output from different kinds of pointer");

EXAMPLE_REQUIRE(proven_is_ok(proven_u8str_reset(&out)), "clearing the line must succeed");

```

```

/* A date, and an address. Neither has a plain C type that means what it
 * means: a date is a struct, and an address printed for a diagnostic is a
 * deliberate act rather than something to fall into. */
proven_datetime_t when = proven_time_breakdown(0); /* the epoch: a fixed, checkable value */
int local = 0;

r = proven_u8str_append_fmt(&out, "at {} object {}",
                           proven_arg_datetime(when),
                           proven_arg_ptr(&local));
EXAMPLE_REQUIRE(proven_is_ok(r.err), "formatting a date and a pointer must succeed");
EXAMPLE_REQUIRE(proven_u8str_view_starts_with(proven_u8str_as_view(&out), PROVEN_LIT("at 1970-01-01")),
                "the epoch breaks down to the first of January 1970");
EXAMPLE_REQUIRE(proven_u8str_view_find(proven_u8str_as_view(&out), 0, PROVEN_LIT("0x")) !=
PROVEN_SIZE_MAX,
                "and an address is rendered in hexadecimal");

EXAMPLE_REQUIRE(proven_is_ok(proven_u8str_reset(&out)), "clearing the line must succeed");

/* Arguments built here, passed on, and formatted there. */
r = log_pair(&out, "status={} retries={}", proven_arg_cstr("ok"), proven_arg_u32(3u));
EXAMPLE_REQUIRE(proven_is_ok(r.err), "formatting pre-built arguments must succeed");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&out), PROVEN_LIT("status=ok retries=3")),
                "an argument built by the caller formats exactly as one built in place");

/* --- naming the scanning destination yourself ----- */

/* On the way in, the constructor names the destination, and the destination
 * decides what "too big" means. These are the plain C types - short, int,
 * long, long long and their unsigned forms - for the very common case where
 * the variable you already have is one of them rather than a fixed-width
 * type. */
proven_scan_t scan = proven_scan_init(
    PROVEN_LIT("h=-32000 uh=65000 i=-2000000 ui=4000000000 l=-9000000 ul=9000000 "
              "ll=-9007199254740993 ull=18446744073709551615 f=2.5 word=alpha"));

short h = 0;
unsigned short uh = 0;
int i = 0;
unsigned int ui = 0;
long l = 0;
unsigned long ul = 0;
long long ll = 0;
unsigned long long ull = 0;
double f = 0.0;
proven_u8str_view_t word = {0};

proven_err_t err = proven_scan_fmt_cursor(
    &scan, "h={} uh={} i={} ui={} l={} ul={} ll={} ull={} f={} word={}",
    proven_scan_arg_short(&h),
    proven_scan_arg_ushort(&uh),
    proven_scan_arg_int(&i),
    proven_scan_arg_uint(&ui),
    proven_scan_arg_long(&l),
    proven_scan_arg_ulong(&ul),
    proven_scan_arg_llong(&ll),
    proven_scan_arg_ullong(&ull),
    proven_scan_arg_f64(&f),
    proven_scan_arg_str_view(&word));
EXAMPLE_REQUIRE(proven_is_ok(err), "scanning every plain integer width must succeed");
EXAMPLE_REQUIRE(h == -32000 && uh == 65000u, "the short forms hold their values");

```

```

EXAMPLE_REQUIRE(i == -2000000 && ui == 4000000000u, "and so do the int forms");
EXAMPLE_REQUIRE(l == -9000000L && ul == 9000000UL, "and the long forms");
EXAMPLE_REQUIRE(ll == -9007199254740993LL, "a value needing 64 bits arrives whole");
EXAMPLE_REQUIRE(ull == 18446744073709551615ULL, "including the largest unsigned 64-bit value");
EXAMPLE_REQUIRE(f > 2.4999 && f < 2.5001, "the floating-point destination reads a decimal");

/* A scanned string view points INTO the text being scanned. Nothing was
 * copied and nothing was allocated, so it is valid exactly as long as that
 * text is - copy it into a proven_u8str_t if it must outlive the scan. */
EXAMPLE_REQUIRE(proven_u8str_view_eq(word, PROVEN_LIT("alpha")), "the word is captured as a view");

/* The destination's width is a promise the scanner keeps. 70000 does not fit
 * in a short, so it is refused rather than wrapped to 4464 - which is what a
 * cast would have produced, silently, in a file nobody looks at again. */
proven_scan_t narrow = proven_scan_init(PROVEN_LIT("70000"));
short too_small = 7;
err = proven_scan_fmt_cursor(&narrow, "{}", proven_scan_arg_short(&too_small));
EXAMPLE_REQUIRE(err != PROVEN_OK, "a value that does not fit the destination is refused");
EXAMPLE_REQUIRE(too_small == 7, "and the destination keeps the value it had");

/* The scanning identity constructor, for the same reason as the formatting
 * one: a helper that receives ready-made scan arguments cannot re-wrap them. */
proven_scan_t again = proven_scan_init(PROVEN_LIT("41"));
proven_i32 answer = 0;
proven_scan_arg_t prebuilt = proven_scan_arg_i32(&answer);
err = proven_scan_fmt_cursor(&again, "{}", proven_scan_arg_identity(prebuilt));
EXAMPLE_REQUIRE(proven_is_ok(err) && answer == 41, "a pre-built scan argument works unchanged");

printf("arguments: %s\n", proven_u8str_as_cstr(&out));

proven_u8str_destroy(alloc, &out);
return EXAMPLE_OK();
}

```

Wrong - wrapping an argument that is already an argument:

```

static proven_fmt_result_t log_one(proven_u8str_t *out, proven_arg_t a) {
    return proven_u8str_append_fmt(out, "{}", PROVEN_ARG(a)); /* wrong */
}

```

Use proven\_arg\_identity(a). PROVEN\_ARG is for values; a is already an argument.

Wrong - keeping a scanned string view after the text is gone:

```

proven_u8str_view_t word = {};
{
    proven_u8str_t line = read_a_line(alloc);
    proven_scan_t s = proven_scan_init(proven_u8str_as_view(&line));
    proven_err_t e = proven_scan_fmt_cursor(&s, "word={}", proven_scan_arg_str_view(&word));
    proven_u8str_destroy(alloc, &line); /* wrong: `word` pointed into `line` */
}
use(word);

```

A scanned view points into the text being scanned. Copy it into a proven\_u8str\_t if it has to outlive that text.

## Chapter 5: Hosted Services

**Part V - Talking to the operating system. Prerequisite: Part II (1, 2, 3).** After this chapter you can read and write files without losing data on a crash, read input a line at a time, generate randomness that suits the job, and tell wall-clock time from elapsed time.

This chapter covers `fs.h`, `sysio.h`, `mmap.h`, `time.h`, `stream.h`, and `random.h`. Everything in it needs an operating system: this is the only part of the manual that does not apply to a [freestanding](#) build.

These APIs require hosted platform support and are excluded from the current freestanding subset.

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### 1. Filesystem API

The filesystem layer wraps platform file handles, paths, directory listing, metadata, permissions, links, and locks.

Raw filesystem helpers do not sanitize untrusted paths, enforce root confinement, or defend against symlink-race TOCTOU. Callers that accept untrusted paths must validate them before using the API.

#### Structures and enums

```
typedef struct {
    union {
        void *ptr;
        int fd;
    } internal;
} proven_fs_handle_t;
```

```
typedef proven_fs_handle_t proven_file_t;
```

Intent: represent a platform file handle without exposing POSIX or Win32 details to higher layers.

```
typedef enum {
    PROVEN_FS_READ    = 1 << 0,
    PROVEN_FS_WRITE   = 1 << 1,
    PROVEN_FS_APPEND  = 1 << 2,
    PROVEN_FS_CREATE  = 1 << 3,
    PROVEN_FS_TRUNC   = 1 << 4,
    PROVEN_FS_CREATE_NEW = 1 << 5
} proven_fs_mode_t;
```

File mode flags can be combined. Examples:

- `PROVEN_FS_READ`
- `PROVEN_FS_WRITE | PROVEN_FS_CREATE | PROVEN_FS_TRUNC`
- `PROVEN_FS_WRITE | PROVEN_FS_APPEND | PROVEN_FS_CREATE`

```
typedef enum {
    PROVEN_FS_TYPE_FILE,
    PROVEN_FS_TYPE_DIR,
    PROVEN_FS_TYPE_OTHER
} proven_fs_type_t;
```

```
typedef struct {
    proven_u8str_t name;
    proven_fs_type_t type;
    proven_size_t size;
} proven_fs_entry_t;
```

Directory entries own name. Use `proven_fs_list_destroy()` for arrays returned by `proven_fs_list()`.

```
typedef struct {
    proven_err_t err;
    proven_file_t value;
} proven_result_file_t;
```

```
typedef enum {
    PROVEN_FS_PERM_OWNER_R = 1 << 8,
    PROVEN_FS_PERM_OWNER_W = 1 << 7,
    PROVEN_FS_PERM_OWNER_X = 1 << 6,
    PROVEN_FS_PERM_GROUP_R = 1 << 5,
    PROVEN_FS_PERM_GROUP_W = 1 << 4,
    PROVEN_FS_PERM_GROUP_X = 1 << 3,
    PROVEN_FS_PERM_OTHER_R = 1 << 2,
    PROVEN_FS_PERM_OTHER_W = 1 << 1,
    PROVEN_FS_PERM_OTHER_X = 1 << 0,
    PROVEN_FS_PERM_DEFAULT = PROVEN_FS_PERM_OWNER_R | PROVEN_FS_PERM_OWNER_W |
                             PROVEN_FS_PERM_GROUP_R | PROVEN_FS_PERM_OTHER_R
} proven_fs_perms_t;
```

```
typedef enum {
    PROVEN_FS_LOCK_SHARED,
    PROVEN_FS_LOCK_EXCLUSIVE,
    PROVEN_FS_LOCK_UNLOCK
} proven_fs_lock_type_t;
```

```
typedef struct {
    proven_size_t size;           /* size in bytes (0 for directories on some hosts) */
    proven_fs_type_t type;        /* PROVEN_FS_TYPE_FILE / _DIR - there is no symlink type */
    proven_fs_perms_t perms;      /* the nine permission bits only; the file type is in `type` */
    proven_i64 created_at;        /* always 0: no birth time is queried - see below */
    proven_i64 modified_at;       /* last-modification time, seconds since the Unix epoch */
    unsigned long long dev;       /* device id (POSIX st_dev; 0 where unavailable) */
    unsigned long long ino;       /* inode number (POSIX st_ino; 0 where unavailable) */
    unsigned long long uid;       /* owner id (POSIX st_uid; 0 on Windows - no uid/gid) */
    unsigned long long gid;       /* group id (POSIX st_gid; 0 on Windows) */
} proven_fs_stat_t;
```

`uid/gid` (added in v26.06.22a) hold the POSIX owner and group ids; both are 0 on Windows, which has no `uid/gid` concept. Resolve them to names with the host's `getpwuid/getgrgid` when displaying an `ls -l`-style owner/group column.

One field is narrower than it looks, and one has a sharp edge:

- `created_at` is **always 0**. Plain `stat()` has no portable birth time, and the PAL does not ask for one. Only `modified_at` carries a real timestamp.
- `type` is `FILE` for a regular file, `DIR` for a directory, and `PROVEN_FS_TYPE_OTHER` for everything else - a FIFO, a socket, a device, a dangling symlink. (Until v26.07.13a all of those `stat'd` as `FILE`, which told a caller it could open them and read bytes out of them. A dangling symlink cannot be opened at all.)

type **follows symlinks**, here and in the directory walk, which is what makes the two agree: a symlink to a regular file is FILE, and a symlink to a directory is DIR. The edge that follows is real - **a recursive walker can loop**, because a symlink pointing at an ancestor is a cycle and its type says DIR. Carry a depth limit, or remember (dev, ino) pairs and refuse to descend into one you have seen.

```

proven_fs_stat_t st;
if (proven_is_ok(proven_fs_stat(scratch, PROVEN_LIT("/etc/hosts"), &st))) {
    proven_println("size={} uid={} gid={}",
        PROVEN_ARG(st.size), PROVEN_ARG(st.uid), PROVEN_ARG(st.gid));
}

```

## Macro

| Macro              | Intent                                                      |
|--------------------|-------------------------------------------------------------|
| PROVEN_FS_PATH_SEP | Preferred library-level separator character, currently '/'. |

## File functions

| API                                   | Intent                                                                                                                                  | Return                                                                                                                                                                                                                       |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proven_fs_open(scratch, path, mode)   | Open a file path. scratch is used for path conversion.                                                                                  | proven_result_file_t.                                                                                                                                                                                                        |
| proven_fs_close(file)                 | Close file handle.                                                                                                                      | <b>Returns an error.</b> On a file you wrote to, the close is part of the write: NFS, CIFS and quota-enforcing filesystems report a failed write-back here and nowhere else. On a file you only read, (void)-ing it is fine. |
| proven_fs_read(file, dest)            | Single read attempt into mutable slice.                                                                                                 | proven_result_size_t.                                                                                                                                                                                                        |
| proven_fs_write(file, src)            | Single write attempt from byte view.                                                                                                    | proven_result_size_t.                                                                                                                                                                                                        |
| proven_fs_write_all(file, src)        | Retry until all bytes are written or an error occurs.                                                                                   | proven_err_t.                                                                                                                                                                                                                |
| proven_fs_size(file)                  | Query open file size.                                                                                                                   | proven_result_size_t.                                                                                                                                                                                                        |
| proven_fs_rename(scratch, src, dest)  | Rename or move path, replacing dest.<br>PROVEN_ERR_NOT_FOUND when src is not there.                                                     | proven_err_t.                                                                                                                                                                                                                |
| proven_fs_remove(scratch, path)       | Remove a file, or an empty directory (as POSIX remove(), on Windows too). A directory that is not empty is<br>PROVEN_ERR_INVALID_STATE. | proven_err_t.                                                                                                                                                                                                                |
| proven_fs_copy(temp_alloc, src, dest) | Copy file using temporary buffer allocation.                                                                                            | proven_err_t.                                                                                                                                                                                                                |
| proven_fs_mkdir(scratch, path)        | Create one directory; its parent must exist.<br>PROVEN_ERR_EXISTS when the name is taken,                                               | proven_err_t.                                                                                                                                                                                                                |

|                                         |                                                                                                                                                       |                        |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|                                         | PROVEN_ERR_NOT_FOUND when the parent is missing.                                                                                                      |                        |
| proven_fs_mkdir_all(scratch, path)      | Create a directory and every missing directory above it. A level that is already a directory is not an error; a file in the way is PROVEN_ERR_EXISTS. | proven_err_t.          |
| proven_fs_rmdir(scratch, path)          | Remove empty directory.<br>PROVEN_ERR_NOT_FOUND when it is not there;<br>PROVEN_ERR_INVALID_STATE when it is not empty.                               | proven_err_t.          |
| proven_fs_list(alloc, path)             | List directory into proven_array_t of proven_fs_entry_t.                                                                                              | proven_result_array_t. |
| proven_fs_list_destroy(alloc, list)     | Destroy directory listing and entry names.                                                                                                            | void.                  |
| proven_fs_chmod(scratch, path, perms)   | Set permissions.<br>PROVEN_ERR_NOT_FOUND when the name is not there.                                                                                  | proven_err_t.          |
| proven_fs_lock(file, type, wait)        | Acquire/release file lock. With wait false, a lock someone else holds is PROVEN_ERR_BUSY.                                                             | proven_err_t.          |
| proven_fs_stat(scratch, path, out_stat) | Fill metadata.                                                                                                                                        | proven_err_t.          |

proven\_fs\_stat() reports only the nine permission bits in perms, so a stat's perms can be handed straight back to proven\_fs\_chmod(). It used to carry the raw POSIX st\_mode, whose file-type bits chmod rejects - which made that round-trip, the obvious use of the field, fail with PROVEN\_ERR\_INVALID\_ARG for every real file. Read the file type from type.

### Position, and the difference between atomic and durable

**A handle that cannot seek says so.** A pipe, a FIFO or a terminal returns PROVEN\_ERR\_UNSUPPORTED from proven\_fs\_seek, not PROVEN\_ERR\_IO. Not being seekable is a property of the thing, not a failure of the call, and code that adapts to it - the scanner does - has to be able to tell them apart.

**pread and pwrite do not move the position.** That is what they are for: two readers sharing one handle cannot race on a cursor that neither of them moves.

### Atomic and durable are different promises, and conflating them is how data gets lost.

- proven\_fs\_write\_file\_atomic guarantees that a *reader* never sees a half-written file. It says nothing about a power cut: the kernel may still be holding your bytes, and the rename may reach the disk before the data it points at.
- proven\_fs\_write\_file\_durable closes that window, in the only order that works: fsync the temp file, **then** rename, **then** fsync the directory. Syncing the file but not the directory leaves a crash window in which the bytes are safe and the name that points at them is not - which is exactly the corruption an atomic write exists to prevent.

The durable form waits for the storage device twice. Use it when losing the write would be worse than the wait, and the atomic form when it would not.

There used to be a `proven_sysio_flush` here that was none of this: it claimed to flush a buffer that did not exist. It is **deleted**. Pushing a buffered writer's bytes to the OS is `proven_writer_flush`; pushing the OS's bytes to the disk is `proven_fs_sync`. Those are two different operations, and one word could not honestly mean both.

Important behavior:

- `proven_fs_read()` and `proven_fs_write()` are single-operation APIs and may process fewer bytes than requested.
- **A read at end-of-file returns `PROVEN_ERR_EOF`, not a zero-byte success.** A loop written the obvious way - `if (r.value == 0) break;` - never takes that branch, and treats the end of the file as an I/O failure instead. Check for `PROVEN_ERR_EOF` explicitly; the worked example at the end of this chapter shows the shape.
- Use `proven_fs_write_all()` when all bytes must be written.
- Zero-size read/write requests should succeed with zero bytes processed without requiring a non-null buffer.
- `proven_fs_is_absolute()` recognizes POSIX absolute paths, Windows drive-root paths, UNC paths, and extended Windows path forms.
- `proven_fs_read_all()` reads to EOF; it does not read to a pre-measured size. The file's reported size only seeds the initial capacity, so a regular file is still read in one allocation and one pass. This matters because `proven_fs_size()` reports 0 for anything that is not a regular file: a FIFO, a character device, or a `/proc` entry has no size that can be known up front, and reading to EOF is the only way to get their contents. It also means a file that grows while it is being read is not silently truncated. `value.size` is always the actual byte count, and an empty source yields `{ .ptr = NULL, .size = 0 }` with `PROVEN_OK`.
- `proven_fs_read_all()` and `proven_fs_read_all_u8str()` need an allocator with a `realloc_fn` if the source outgrows its reported size; a non-growing allocator returns `PROVEN_ERR_UNSUPPORTED` in that case.
- `proven_fs_read_all()` reads until the source ends, with no bound - use it on paths you trust. For a path from outside the program, `proven_fs_read_all_bounded()` refuses a source of more than `max_bytes` bytes with `PROVEN_ERR_OUT_OF_BOUNDS`; a file whose size says so before anything is allocated, and one whose size cannot be known (a FIFO, `/proc`, `/dev/zero`) as soon as the byte after the bound arrives.

```
/* An upload path comes from a request, so it may name /dev/zero, a FIFO that never
 * ends, or a 40 GB file. proven_fs_read_all would read until the allocator gives up;
 * the bounded form stops at the limit and allocates no more than it. */
proven_result_mem_mut_t body = proven_fs_read_all_bounded(alloc, PROVEN_LIT("upload.bin"), 1u << 20);
if (body.err == PROVEN_ERR_OUT_OF_BOUNDS) {
    proven_eprintln("upload.bin is larger than 1 MiB; refused");
} else if (proven_is_ok(body.err)) {
    proven_println("read {} bytes", PROVEN_ARG(body.value.size));
    if (body.value.ptr) alloc.free_fn(alloc.ctx, body.value.ptr);
}
```

- `proven_fs_read_all_u8str()` is the whole-file read most callers want: the result is NUL-terminated, so `proven_u8str_as_view()` and `proven_u8str_as_cstr()` work on it with no second copy. The terminator slot is reserved up front, so it costs no extra allocation. Contents are not validated as UTF-8. Release it with `proven_u8str_destroy()`.
- `proven_fs_write_file()` is not atomic: a reader can observe a partially written file, and a failure mid-write leaves the file truncated. `proven_fs_write_file_atomic()` writes a sibling temp file and renames it over the target, so a concurrent reader sees either the entire old file or the entire new one. It is atomic with respect to readers, not durable across power loss: the rename may reach the disk before the data. When you need durability, ask for it explicitly with `proven_fs_write_file_durable` (or `proven_fs_sync`), described above.

- The temp file is <path>.pvtmp plus 13 random characters, so it cannot be predicted: leftovers and names planted by another user never block the write, and a taken name is skipped (PROVEN\_ERR\_EXISTS only after sixteen in a row). A writer killed between creating and renaming it leaves the temp file behind, and proven never deletes one it did not just create - it cannot tell an abandoned file from a write in progress. A long-running service can list the directory and remove old entries for which proven\_fs\_is\_staging\_name() is true.

```
/* A cleanup pass a service might run at startup: count the staging files the atomic
 * writes of an earlier, killed run left in its state directory. Removing them is the
 * caller's decision - only the caller knows no writer is still filling one. */
proven_result_dir_t dir = proven_fs_dir_open(scratch, PROVEN_LIT("."));
if (proven_is_ok(dir.err)) {
    proven_size_t leftovers = 0;
    proven_fs_dir_entry_t entry;
    proven_err_t e;
    while ((e = proven_fs_dir_next(&dir.value, &entry)) != PROVEN_ERR_EOF) {
        if (proven_is_ok(e) && proven_fs_is_staging_name(entry.name)) ++leftovers;
        else if (e != PROVEN_OK && e != PROVEN_ERR_INVALID_ENCODING) break;
    }
    proven_fs_dir_close(&dir.value);
    proven_println("staging leftovers: {}", PROVEN_ARG(leftovers));
}
```

**A read-only destination is refused, by all of them.** proven\_fs\_write\_file, proven\_fs\_write\_file\_atomic, proven\_fs\_write\_file\_durable and proven\_fs\_copy all return PROVEN\_ERR\_PERMISSION when the destination's owner-write bit is clear, and leave the file exactly as it was. That bit is where both platforms record "do not write this file" - mode 0200 on POSIX, the READONLY attribute on Windows.

It is one rule because it used to be three answers to one question, on one platform: write\_file refused, because it opens the destination for writing; write\_file\_atomic succeeded, because rename asks the *directory* for permission and never consults the file's mode; and copy succeeded **and left the file writable afterwards**, because it carries the source's mode across. A protection the caller had set disappeared and nothing said so. Which function a caller reaches for is not a decision about permissions.

A caller who means to replace a protected file lifts the mark first - one line, and visible. The refusal is deliberately the recoverable direction; replacing a protected file by accident is not. It is a guard against accidents, not a security boundary: the mode is read before the work and acted on after it, and anyone who can chmod the file can lift the mark.

proven\_fs\_rename is on that list, and has to be: it replaces the destination, and it is what the atomic write is built on - without it, a caller refused by one function got the same result from the other. proven\_fs\_remove is deliberately **not** on it: deleting a name is a directory operation and POSIX has never let the file's own mode have a say. The platforms differ there - Windows will not delete a read-only file - so proven\_fs\_remove reports that as PROVEN\_ERR\_PERMISSION rather than hiding it behind an I/O error.

A refusal now says *which* refusal it is. proven\_fs\_open, proven\_fs\_rename and proven\_fs\_remove answer PROVEN\_ERR\_NOT\_FOUND, PROVEN\_ERR\_PERMISSION or PROVEN\_ERR\_BUSY where they used to answer PROVEN\_ERR\_IO for all of it. Asking the user, retrying, and giving up are three different answers, and one error code supports none of them.

The directory, mode, link and lock calls follow the same rule. proven\_fs\_mkdir and proven\_fs\_link answer PROVEN\_ERR\_EXISTS when the name they were asked to create is taken; proven\_fs\_mkdir, proven\_fs\_rmdir, proven\_fs\_chmod, proven\_fs\_link and proven\_fs\_rename answer PROVEN\_ERR\_NOT\_FOUND for a name, or a parent directory, that is not there, and PROVEN\_ERR\_PERMISSION when the platform refuses the caller; proven\_fs\_lock asked not to wait answers PROVEN\_ERR\_BUSY for a lock someone else holds. Two

refusals are named with codes that already had the right meaning. A directory that is not empty answers `PROVEN_ERR_INVALID_STATE` to `proven_fs_rmdir` and to `proven_fs_remove`: nothing failed, and the fix is the order of the calls - empty it first. A hard link or a rename that would have to cross from one file system to another answers `PROVEN_ERR_UNSUPPORTED`: that route does not exist, and the caller takes another - a copy, then a remove.

`proven_fs_mkdir` creates one level and reports a name that is taken, because for one level that is information. A caller that wants a whole path to exist, and does not care who made it, calls `proven_fs_mkdir_all`: it creates every missing level, treats a level that is already a directory as done, and still refuses - with `PROVEN_ERR_EXISTS` - when a level is something other than a directory. Directories it created before a failure stay.

Wrong - reading every refusal of `proven_fs_mkdir` as a failure:

```
proven_err_t e = proven_fs_mkdir(alloc, dir);
if (!proven_is_ok(e)) return e;           /* wrong: stops when dir is already there */
```

A program run twice fails the second time, at a directory it made itself. Either ask for what is meant - `proven_fs_mkdir_all`, "this path is a directory when I return" - or accept `PROVEN_ERR_EXISTS` on purpose.

Wrong - accepting `PROVEN_ERR_EXISTS` without asking what exists:

```
proven_err_t e = proven_fs_mkdir(alloc, dir);
if (e == PROVEN_ERR_EXISTS) e = PROVEN_OK;    /* wrong: a FILE named dir also answers this */
```

`proven_fs_mkdir` does not look at what holds the name. A file of that name gives the same answer, and the write into `dir/...` that follows fails somewhere less clear. `proven_fs_mkdir_all` asks, and returns `PROVEN_OK` only for a directory.

**A file someone is reading is still replaced.** When another process holds the destination open but allowed delete sharing - `proven_fs_open` does - the atomic write succeeds, and the reader keeps reading the old contents through the handle it already has. POSIX has always done this; Windows 10 1809 and later do the same through the POSIX-semantics rename. Where that rename is unknown - Windows before 1809, FAT32, exFAT, network drives - the library falls back to `MoveFileExW`, which refuses the replacement while anyone has the file open, and the answer is `PROVEN_ERR_BUSY`. A holder that did not allow delete sharing gets the same `PROVEN_ERR_BUSY` on every Windows. It is in use, not protected: a retry may succeed.

Example:

```
proven_result_file_t of = proven_fs_open(
    alloc,
    PROVEN_LIT("out.txt"),
    PROVEN_FS_WRITE | PROVEN_FS_CREATE | PROVEN_FS_TRUNC
);
if (proven_is_ok(of.err)) {
    proven_err_t e = proven_fs_write_all(
        of.value,
        proven_mem_view_from_u8(PROVEN_LIT("hello\n"))
    );
    /* The close is part of the write. Close on the failure path too - the handle is
     * ours either way - but do not throw the answer away: on a network filesystem or
     * over quota, close() is the only place the failure appears. */
    proven_err_t ce = proven_fs_close(of.value);
    if (proven_is_ok(e)) e = ce;
    if (!proven_is_ok(e)) {
        proven_eprintln("writing out.txt failed");
    }
}
```

## 2. System I/O and environment

### Standard streams

```
proven_file_t proven_sysio_stdin(void);
proven_file_t proven_sysio_stdout(void);
proven_file_t proven_sysio_stderr(void);
```

Purpose: expose the standard streams as `proven_file_t` handles. They are also writers and readers - see [The standard streams](#) below, which is what lets you read stdin a line at a time, buffer stdout, and format straight into either.

```
proven_sysio_scanner_t
typedef struct {
    proven_file_t file;
    proven_allocator_t alloc;
    proven_u8 *buffer;
    proven_size_t capacity;
    proven_size_t cursor;
    proven_size_t length;
    bool eof;
} proven_sysio_scanner_t;
```

Purpose: buffered scanner for bounded stream input, safe for pipes and stdin as well as seekable files. When a token reaches the end of the currently loaded fragment before EOF, the scanner refills the buffer and retries; only a token that cannot fit inside the whole buffer even after a refill is rejected, with `PROVEN_ERR_OUT_OF_BOUNDS`, instead of being accepted truncated.

### Sysio functions and macros

| API                                                                           | Intent                                          | Return                               |
|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------|
| <code>proven_sysio_scanner_init(scanner, file, alloc, buffer_capacity)</code> | Allocate scanner buffer and bind stream.        | <code>proven_err_t</code> .          |
| <code>proven_sysio_scanner_deinit(scanner)</code>                             | Free scanner buffer.                            | <code>void</code> .                  |
| <code>proven_sysio_scanner_scan_impl(scanner, fmt, args, args_count)</code>   | Internal scanner engine.                        | <code>proven_err_t</code> .          |
| <code>proven_sysio_scanner_scan(scanner, fmt, ...)</code>                     | Type-safe buffered scan macro.                  | <code>proven_err_t</code> .          |
| <code>proven_sysio_print_impl(handle, fmt, args, args_count)</code>           | Internal printing engine.                       | <code>proven_err_t</code> .          |
| <code>proven_sysio_scan_chunk_impl(handle, fmt, args, args_count)</code>      | One-chunk scan engine.                          | <code>proven_err_t</code> .          |
| <code>proven_print(fmt, ...)</code>                                           | Print to stdout.                                | <code>proven_err_t</code> .          |
| <code>proven_println(fmt, ...)</code>                                         | Print to stdout with newline.                   | <code>proven_err_t</code> .          |
| <code>proven_eprint(fmt, ...)</code>                                          | Print to stderr.                                | <code>proven_err_t</code> .          |
| <code>proven_eprintln(fmt, ...)</code>                                        | Print to stderr with newline.                   | <code>proven_err_t</code> .          |
| <code>proven_scan_fmt_from_file(file, fmt, ...)</code>                        | Scan from one fixed-size chunk of file.         | <code>proven_err_t</code> .          |
| <code>proven_scan_fmt_from_stdin(fmt, ...)</code>                             | Scan one fixed-size chunk from stdin.           | <code>proven_err_t</code> .          |
| <code>proven_env_get(alloc, key)</code>                                       | Read environment variable into owned U8 string. | <code>proven_result_u8str_t</code> . |

`proven_sysio_scan_chunk_impl()` is intended for seekable file inputs. It reads at most one fixed-size chunk. If the handle cannot be rewound, it returns `PROVEN_ERR_UNSUPPORTED` before reading. If the

chunk fills before a complete token is available, it returns `PROVEN_ERR_OUT_OF_BOUNDS` and rewinds the file cursor to the start of the chunk. It also refuses string-view destinations with `PROVEN_ERR_UNSUPPORTED` before reading: such a view would borrow the helper's local chunk buffer and dangle on return. Use the caller-owned `proven_sysio_scanner_t` for repeated buffered scanning or borrowed string results. A string view returned by that scanner remains valid only until the next scan call or scanner deinitialization, because either operation may refill, compact, or free the shared buffer.

Example:

```
proven_println("answer={}", PROVEN_ARG(42));
proven_eprintln("warning: {}", PROVEN_ARG(PROVEN_LIT("low memory")));
```

Environment example. `proven_env_get` hands back an owned string, so it has to be destroyed. A variable set to the empty string is an empty string, not an error, and a value that is not valid text (bytes that are not UTF-8 on POSIX, a lone surrogate on Windows) is `PROVEN_ERR_INVALID_ENCODING` - nothing is substituted. The same rule holds for directory entry names:

```
proven_result_u8str_t env = proven_env_get(alloc, PROVEN_LIT("PATH"));
if (proven_is_ok(env.err)) {
    proven_u8str_view_t path = proven_u8str_as_view(&env.value);
    proven_println("PATH is {} bytes", PROVEN_ARG(path.size));
    proven_u8str_destroy(alloc, &env.value);
}
```

### 3. Memory mapping

#### The problem: reading a file you do not want to copy

`proven_fs_read_all_u8str` reads a file into memory you own. For a configuration file that is exactly right. For a 4 GB database, a memory-mapped index, or a file two processes need to see at once, it is wrong in three ways: it needs 4 GB of RAM, it copies every byte whether or not you read them, and the copy is yours alone.

A memory mapping asks the operating system to make the file's contents *appear* at an address instead. Nothing is copied up front. The pages you touch are read from disk on demand; the ones you never touch are never read. Two processes mapping the same file with `PROVEN_MMAP_SHARED` see one set of pages, so a write in one is visible in the other.

#### What this costs, and when not to use it

This is the section of the manual where the trade is sharpest, because the failure modes do not look like errors:

- **A read can fault.** With a file in memory, an I/O error is a return value. With a mapping, touching a page whose disk read fails delivers a **signal** (`SIGBUS`), not an error code. There is no if you can write around it.
- **Truncation is a landmine.** If another process shortens the file after you map it, touching a page past the new end is also `SIGBUS`. Mapping a file that anything else may truncate is unsafe in a way no API can fix for you.
- **It is not free.** Setting up a mapping is a syscall and page-table work; each first touch is a page fault. For a small file, `proven_fs_read_all_u8str` is simply faster.
- `proven_mmap_sync` **is the durability point.** Writes to a shared mapping reach the file eventually; if you need them on disk *now*, ask.

Use a mapping for large files you read sparsely, for read-only data shared between processes, and for random access into a big file. Use an ordinary read for everything else.

The mapping is caller-owned state: `proven_mmap_as_view` hands you a view **into the mapping**, so that view is dead the moment `proven_mmap_destroy` runs.

Wrong - using the view after destroying the mapping:

```
proven_u8str_view_t data = proven_mmap_as_view(m);
proven_err_t e = proven_mmap_destroy(&m);
(void)e;
parse(data);          /* wrong: those addresses are no longer mapped - SIGSEGV */
```

Wrong - treating a mapping like a buffer you can grow:

```
/* wrong: a mapping is a window onto a file of a fixed size at map time.
   Appending means changing the file and mapping it again. */
```

## Structures and enums

```
typedef enum {
    PROVEN_MMAP_READ   = 0x01,
    PROVEN_MMAP_WRITE  = 0x02,
    PROVEN_MMAP_EXEC   = 0x04
} proven_mmap_prot_t;

typedef enum {
    PROVEN_MMAP_PRIVATE = 0x01,
    PROVEN_MMAP_SHARED  = 0x02
} proven_mmap_flags_t;

typedef struct {
    void *ptr;
    proven_size_t size;
    proven_fs_handle_t file;
    void *internal_handle;
} proven_mmap_t;

typedef struct {
    proven_err_t err;
    proven_mmap_t value;
} proven_result_mmap_t;
```

## Functions

| API                                                 | Intent                                                                                           | Return                |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------|
| proven_mmap_create(file, offset, size, prot, flags) | Map a file region. size == 0 maps until EOF. Offset must match the platform mapping granularity. | proven_result_mmap_t. |
| proven_mmap_destroy(mmap)                           | Unmap region and clear state.                                                                    | proven_err_t.         |
| proven_mmap_sync(mmap)                              | Flush shared writable changes to storage.                                                        | proven_err_t.         |
| proven_mmap_as_view(mmap)                           | Borrow mapped memory as U8 view.                                                                 | proven_u8str_view_t.  |

Example:

```
proven_result_file_t f = proven_fs_open(alloc, PROVEN_LIT("data.bin"), PROVEN_FS_READ);
if (proven_is_ok(f.err)) {
    proven_result_mmap_t mr = proven_mmap_create(
        f.value,
        0,
        0,
        PROVEN_MMAP_READ,
        PROVEN_MMAP_PRIVATE
    );
    if (proven_is_ok(mr.err)) {
        /* The view borrows the mapping: it dies when the mapping is destroyed. */
```

```

    proven_u8str_view_t bytes = proven_mmap_as_view(mr.value);
    proven_println("mapped {} bytes", PROVEN_ARG(bytes.size));
    (void)proven_mmap_destroy(&mr.value);
}
(void)proven_fs_close(f.value);
}

```

## 4. Time API

### The problem: two different questions, one word

"Time" means two things that look alike and behave nothing alike.

A **wall clock** answers *what time is it?* - the thing a user reads. It is allowed to jump: NTP corrects it, daylight saving shifts it, an administrator sets it. Measure a duration with it and you can get a negative answer, which is a bug that appears on leap-second days and on nobody's laptop during testing.

A **monotonic** clock answers *how long since?* It only moves forward and has no relationship to any calendar. It is what you time an operation with.

libc blurs the distinction. `time()` gives whole seconds of wall clock - too coarse to measure anything. `clock()` measures **CPU** time, so a program that waits on a socket appears to take no time at all. Neither name says which question it answers, and both are routinely used for the other one.

### What this library does instead

There are two clocks, and each has one job.

`proven_time_now()` is the **wall clock**: nanoseconds since the Unix epoch as a signed 64-bit value. It has a date, and it is set - by NTP, by an administrator - so it can step in either direction. Break it down or format it.

`proven_time_monotonic_now()` is the **monotonic clock**: nanoseconds from an origin fixed for the life of the process. It never goes backwards and nobody sets it, so the difference of two readings is an elapsed time. It has no date, and a reading means nothing in another process or a later run. Durations, timeouts and deadlines are measured with this one.

`proven_time_breakdown()` turns that number into `proven_datetime_t`, whose fields are the ones a human uses: month is **1-12**, not libc's 0-11, and year is the actual year, not years since

1. Those two off-by-N conventions in `struct tm` have caused enough bugs to be worth diverging from deliberately.

Formatting uses the same `{}` placeholders as [Chapter 3](#): the name picks the field and the spec pads it, so `{month:0>2}` is zero-filled to width two. Month and weekday names come from a `proven_time_locale_t`, so rendering another language is passing a different locale rather than setting a global.

Wrong - timing with a wall clock and trusting the sign:

```

proven_time_t t0 = proven_time_now();
do_work();
proven_time_t t1 = proven_time_now();
proven_u64 ns = (proven_u64)(t1 - t0); /* wrong: an NTP step back makes this enormous */

```

Correct - a duration comes from the monotonic clock, which cannot step:

```

proven_time_t t0 = proven_time_monotonic_now();
do_work();
proven_i64 elapsed_ns = proven_time_monotonic_now() - t0; /* never negative */

```

If two wall-clock readings are all you have - two timestamps from a log - keep the difference signed, and treat a negative one as "the clock moved", not as a duration.

Wrong - assuming sleep is precise:

```
proven_time_sleep(15);
/* wrong to assume exactly 15ms have passed: sleep guarantees AT LEAST that long,
   and the scheduler decides when you actually run again. */
```

### Worked example: a duration, a date, and a formatted timestamp

Compiled and run by the test suite. Note what it does *not* assert - an upper bound on the sleep -- because that would be a test that fails on a busy machine.

```
/*
 * Time comes in two flavours that look identical and are not, and picking the
 * wrong one is the classic timing bug.
 *
 * - A WALL CLOCK answers "what time is it?". It is what a user wants to see,
 *   and it is allowed to jump: NTP corrects it, daylight saving shifts it, an
 *   administrator sets it. Measuring a duration with it can produce a negative
 *   elapsed time, and did, famously, on leap-second days.
 *
 * - A MONOTONIC clock answers "how long since?". It only moves forward, at a
 *   steady rate, and has no relationship to any calendar. It is what you time
 *   an operation with.
 *
 * libc blurs this. time() is wall clock in whole seconds - useless for
 * measurement. clock() measures CPU time, not elapsed time, so a program that
 * sleeps looks instantaneous. Neither name tells you which of the two questions
 * it is answering.
 *
 * Here each has its own call. proven_time_now() is the wall clock, nanoseconds
 * since the Unix epoch: it formats as a date. proven_time_monotonic_now() is the
 * monotonic clock, nanoseconds from an origin that means nothing outside this
 * process: two readings subtract to a duration.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- as a duration ----- */
    proven_time_t start = proven_time_now();          /* the wall clock, for the date below */
    proven_time_t t0 = proven_time_monotonic_now();   /* the monotonic clock, for the duration */
    proven_time_sleep(15);                            /* milliseconds */
    proven_i64 elapsed_ns = proven_time_monotonic_now() - t0;

    EXAMPLE_REQUIRE(elapsed_ns > 0, "the monotonic clock moves forward across a sleep");
    /* Sleep guarantees AT LEAST the requested time, never at most: the scheduler
     * decides when you actually run again. Asserting an upper bound here would
     * be a test that fails on a busy machine, which is why this one does not. */
    EXAMPLE_REQUIRE(elapsed_ns >= 10 * 1000 * 1000,
        "sleeping 15ms must take at least ~10ms");

    /* --- as a date ----- */
    proven_datetime_t dt = proven_time_breakdown(start);
    EXAMPLE_REQUIRE(dt.year >= 2020 && dt.year < 3000, "the epoch breakdown gives a plausible year");
    EXAMPLE_REQUIRE(dt.month >= 1 && dt.month <= 12, "month is 1-12, not 0-11 as in libc's tm");
    EXAMPLE_REQUIRE(dt.day >= 1 && dt.day <= 31, "day is 1-31");
    EXAMPLE_REQUIRE(dt.hour <= 23 && dt.min <= 59 && dt.sec <= 60, "sec allows 60 for leap seconds");
    EXAMPLE_REQUIRE(dt.weekday <= 6, "weekday is 0-6 with 0 = Sunday");

    /* proven_time_now_datetime() is the two calls above in one, for when you
     * only want the calendar form. */
}
```

```

proven_datetime_t now = proven_time_now_datetime();
EXAMPLE_REQUIRE(now.year == dt.year, "both routes read the same clock");

/* --- formatting a timestamp ----- */
proven_result_u8str_t s = proven_u8str_create(alloc, 64);
EXAMPLE_REQUIRE(proven_is_ok(s.err), "a 64-byte string is enough for a timestamp");

/* The locale supplies month and weekday names; proven_time_locale_en is the
 * built-in English one. Pass your own to render other languages. */
proven_err_t err = proven_time_u8_fmt(alloc, &s.value, dt, &proven_time_locale_en,
                                     "{year}--{month:0>2}--{day:0>2} {hour:0>2}:{min:0>2}:{sec:0>2}");
EXAMPLE_REQUIRE(proven_is_ok(err), "formatting a datetime should succeed");

proven_u8str_view_t out = proven_u8str_as_view(&s.value);
EXAMPLE_REQUIRE(out.size == 19, "year-month-day hour:min:sec is exactly 19 characters");
EXAMPLE_REQUIRE(out.ptr[4] == '-' && out.ptr[7] == '-' && out.ptr[13] == ':',
               "the separators land where the pattern put them");

proven_println("formatted: {}", PROVEN_ARG(out));

proven_u8str_destroy(alloc, &s.value);
return EXAMPLE_OK();
}

```

## Types

typedef proven\_i64 proven\_time\_t;

Nanoseconds since UNIX epoch.

```

typedef struct {
    proven_i32 year;
    proven_u8 month;
    proven_u8 day;
    proven_u8 hour;
    proven_u8 min;
    proven_u8 sec;
    proven_u32 ms;
    proven_u8 weekday;
} proven_datetime_t;

```

Field ranges:

- month: 1 to 12.
- day: 1 to 31.
- hour: 0 to 23.
- min: 0 to 59.
- sec: 0 to 60.
- ms: 0 to 999.
- weekday: 0 to 6, Sunday is 0.

```

typedef struct {
    const proven_u8str_view_t *month_names;
    const proven_u8str_view_t *month_short_names;
    const proven_u8str_view_t *weekday_names;
    const proven_u8str_view_t *weekday_short_names;
} proven_time_locale_t;

```

proven\_time\_locale\_en is the default English locale.

## Functions

| API | Intent | Return |
|-----|--------|--------|
|-----|--------|--------|

|                                                  |                                                                                                                 |                    |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------|
| proven_time_u8_fmt(alloc, str, dt, locale, fmt)  | Append formatted datetime to U8 string.                                                                         | proven_err_t.      |
| proven_time_u16_fmt(alloc, str, dt, locale, fmt) | Append formatted datetime to U16 string unless U16 is disabled.                                                 | proven_err_t.      |
| proven_time_now()                                | Current timestamp in nanoseconds.                                                                               | proven_time_t.     |
| proven_time_breakdown(time_ns)                   | Convert epoch nanoseconds to broken-down UTC time.                                                              | proven_datetime_t. |
| proven_time_monotonic_now()                      | The monotonic clock: nanoseconds from a per-process origin. Subtract two readings for a duration; never a date. | proven_time_t.     |
| proven_time_now_datetime()                       | Current local broken-down time.                                                                                 | proven_datetime_t. |
| proven_time_sleep(ms)                            | Sleep for milliseconds.                                                                                         | void.              |

Datetime format keys:

- {year}, {month}, {day}, {hour}, {min}, {sec}, {ms}, {wday\_num}
- {Month}, {mon} using locale
- {Weekday}, {wday} using locale

Example:

```
proven_result_u8str_t r = proven_u8str_create(alloc, 32);
if (proven_is_ok(r.err)) {
    proven_u8str_t s = r.value;

    proven_datetime_t now = proven_time_now_datetime();
    proven_err_t e = proven_time_u8_fmt(
        alloc,
        &s,
        now,
        &proven_time_locale_en,
        "{year}-{month:0>2}-{day:0>2} {hour:0>2}:{min:0>2}:{sec:0>2}"
    );
    if (proven_is_ok(e)) {
        proven_println("{f}", PROVEN_ARG(proven_u8str_as_view(&s)));
    }

    proven_u8str_destroy(alloc, &s);
}
```

## 5. Examples and misuse cases

### Single writes can be partial

Wrong:

```
proven_result_size_t w = proven_fs_write(file, data);
/* wrong: one write may be partial, and w.value is never looked at */
```

Correct:

```
proven_result_file_t f = proven_fs_open(alloc, PROVEN_LIT("out.txt"),
                                         PROVEN_FS_WRITE | PROVEN_FS_CREATE | PROVEN_FS_TRUNC);
if (proven_is_ok(f.err)) {
    proven_mem_view_t data = proven_mem_view_from_u8(PROVEN_LIT("payload"));
    proven_err_t e = proven_fs_write_all(f.value, data); /* loops until done */
    proven_err_t ce = proven_fs_close(f.value);          /* and the close can fail too */
    if (proven_is_ok(e)) e = ce;
}
```

```
(void)e;
}
```

## Directory listings need special destruction

Wrong:

```
proven_result_array_t r = proven_fs_list(alloc, PROVEN_LIT("."));
PROVEN_ARRAY_DESTROY(&r.value); /* wrong: entry names leak */
```

Correct:

```
proven_result_array_t r = proven_fs_list(alloc, PROVEN_LIT("."));
if (proven_is_ok(r.err)) {
    proven_fs_list_destroy(alloc, &r.value);
}
```

## Do not use a mapping after destroy

Wrong:

```
proven_mmap_destroy(&map);
use_bytes(map.ptr, map.size); /* wrong: mapping has been released */
```

## Use buffered sysio scanning for streams

For repeated reading from stdin or pipes, prefer:

```
proven_sysio_scanner_t scanner = {0};
proven_err_t e = proven_sysio_scanner_init(&scanner, proven_sysio_stdin(), alloc, 4096);
if (proven_is_ok(e)) {
    int value = 0;
    e = proven_sysio_scanner_scan(&scanner, "{}", PROVEN_SCAN_ARG(&value));
    proven_sysio_scanner_deinit(&scanner);
}
```

## Environment values are owned strings

Wrong:

```
proven_result_u8str_t env = proven_env_get(alloc, PROVEN_LIT("PATH"));
use_path(proven_u8str_as_view(&env.value)); /* wrong if env.err was not checked, and it leaks */
```

Correct:

```
proven_result_u8str_t env = proven_env_get(alloc, PROVEN_LIT("HOME"));
if (proven_is_ok(env.err)) {
    proven_u8str_view_t home = proven_u8str_as_view(&env.value);
    proven_println("HOME={}", PROVEN_ARG(home));
    proven_u8str_destroy(alloc, &env.value);
}
```

## Worked example: reading and writing whole files

Compiled and run by the test suite. This is the whole-file API most callers actually want, including the atomic rewrite that preserves the target's permissions.

```
/*
 * The whole-file API: one call in, one call out. It exists because the
 * open/read-loop/close dance is where most file-handling bugs live - a forgotten
 * close, a partial read treated as EOF, a truncated file left behind by a failed
 * write. If you are reading or writing a file in its entirety, this is the API.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* A relative path in the current directory: the example must not depend on a
     * writable /tmp, and it removes what it creates before returning. */
```

```

proven_u8str_view_t path = PROVEN_LIT("proven_example_wholefile.tmp");
proven_u8str_view_t text = PROVEN_LIT("first line\nsecond line\n");

/* --- write it in one call ----- */
/* Not atomic: a concurrent reader can see this file half-written. Fine here,
 * because nobody else is looking at it yet. */
proven_err_t err = proven_fs_write_file(alloc, path, proven_mem_view_from_u8(text));
EXAMPLE_REQUIRE(proven_is_ok(err), "writing the whole file should succeed");
if (!proven_is_ok(err)) return 1;

/* --- read it back as raw bytes ----- */
/* proven_fs_read_all reads to EOF rather than to a pre-measured size, so it
 * also works on a pipe or a /proc entry, whose size cannot be known up front. */
proven_result_mem_mut_t raw = proven_fs_read_all(alloc, path);
EXAMPLE_REQUIRE(proven_is_ok(raw.err), "reading the whole file should succeed");
if (proven_is_ok(raw.err)) {
    EXAMPLE_REQUIRE(raw.value.size == text.size, "read_all should return every byte written");
    /* The block is plain allocator memory - hand it back to the allocator that
     * produced it. There is no proven_fs_read_all_destroy. */
    alloc.free_fn(alloc.ctx, raw.value.ptr);
}

/* --- read it back as a string ----- */
/* This is the one most callers want: the result is NUL-terminated, so it can
 * be handed to a view, to as_cstr, or to the scanner with no second copy. The
 * terminator slot is reserved up front, so it costs no extra allocation. */
proven_result_u8str_t s = proven_fs_read_all_u8str(alloc, path);
EXAMPLE_REQUIRE(proven_is_ok(s.err), "reading the whole file as a string should succeed");
if (!proven_is_ok(s.err)) {
    (void)proven_fs_remove(alloc, path);
    return 1;
}
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&s.value), text),
    "the file's contents should come back unchanged");
printf("read back %zu bytes: %s", (size_t)proven_u8str_as_view(&s.value).size,
    proven_u8str_as_cstr(&s.value));
proven_u8str_destroy(alloc, &s.value);

/* --- stat, and the perms round-trip ----- */
proven_fs_stat_t st = {0};
err = proven_fs_stat(alloc, path, &st);
EXAMPLE_REQUIRE(proven_is_ok(err), "stat on a file we just wrote should succeed");
EXAMPLE_REQUIRE(st.type == PROVEN_FS_TYPE_FILE, "a regular file should stat as a FILE");
EXAMPLE_REQUIRE(st.size == text.size, "stat should report the size we wrote");

/* `perms` carries the nine permission bits and nothing else, so it can be fed
 * straight back to chmod. That is the whole point of the field: read a file's
 * mode, and later restore it. (It used to carry the raw POSIX st_mode, whose
 * file-type bits chmod rejects - so this obvious round-trip failed.) */
err = proven_fs_chmod(alloc, path, st.perms);
EXAMPLE_REQUIRE(proven_is_ok(err), "a stat's perms must be accepted back by chmod");

/* Now make the file owner-only, so the next check has something to prove. */
proven_fs_perms_t private_perms = PROVEN_FS_PERM_OWNER_R | PROVEN_FS_PERM_OWNER_W;
err = proven_fs_chmod(alloc, path, private_perms);
EXAMPLE_REQUIRE(proven_is_ok(err), "restricting the file to its owner should succeed");

/* What the platform keeps of that: all nine bits on POSIX, only the owner-write
 * bit (the READONLY attribute) on Windows. The rewrite must keep exactly this. */
proven_fs_stat_t st1 = {0};

```

```

err = proven_fs_stat(alloc, path, &st1);
EXAMPLE_REQUIRE(proven_is_ok(err), "stat after chmod should succeed");

/* --- rewrite it atomically ----- */
/* A sibling temp file plus a rename: a concurrent reader sees either the whole
 * old file or the whole new one, never a half-written mix. Atomic for readers,
 * not durable across power loss. When it must be, proven_fs_write_file_durable asks. */
proven_u8str_view_t text2 = PROVEN_LIT("replacement\n");
err = proven_fs_write_file_atomic(alloc, path, proven_mem_view_from_u8(text2));
EXAMPLE_REQUIRE(proven_is_ok(err), "the atomic rewrite should succeed");

proven_fs_stat_t st2 = {0};
err = proven_fs_stat(alloc, path, &st2);
EXAMPLE_REQUIRE(proven_is_ok(err), "stat after the atomic rewrite should succeed");
EXAMPLE_REQUIRE(st2.size == text2.size, "the file should now hold the replacement text");
/* The rename writes a *new* inode over the old name, so the permissions would
 * be lost unless they were copied across. They are: rewriting a 0600 file does
 * not republish it as 0644. */
EXAMPLE_REQUIRE(st2.perms == st1.perms,
    "the atomic rewrite must preserve the target's permissions");

/* --- clean up ----- */
err = proven_fs_remove(alloc, path);
EXAMPLE_REQUIRE(proven_is_ok(err), "removing the temp file should succeed");

return EXAMPLE_OK();
}

```

### Worked example: open, read, write, close

Compiled and run by the test suite. Note the read loop: a read at end-of-file returns `PROVEN_ERR_EOF`, not a zero-byte success, so a loop that only checks for zero bytes never terminates the way its author expected.

```

/*
 * The open/read/write/close path, for when the whole-file API (ex_05_fs_wholefile)
 * is not enough: you are streaming, or you want to own the buffer.
 *
 * The one thing to get right here: a single read or write moves *up to* the
 * requested number of bytes, not exactly that many. Treating one short read as
 * end-of-file is the classic way to silently lose the tail of a file.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    proven_u8str_view_t path = PROVEN_LIT("proven_example_stream.tmp");
    proven_u8str_view_t text = PROVEN_LIT("streamed bytes, read back in chunks\n");

    /* --- write ----- */
    /* CREATE makes the file if it is absent; TRUNC empties it if it is not. The
     * allocator is only used to convert the path for the platform call. */
    proven_result_file_t out = proven_fs_open(alloc, path,
        PROVEN_FS_WRITE | PROVEN_FS_CREATE | PROVEN_FS_TRUNC);
    EXAMPLE_REQUIRE(proven_is_ok(out.err), "opening the file for writing should succeed");
    if (!proven_is_ok(out.err)) return 1;

    /* write_all loops for us. proven_fs_write does one attempt and may write less,
     * which is almost never what a caller means. */
    proven_err_t err = proven_fs_write_all(out.value, proven_mem_view_from_u8(text));
}

```

```

/* The close is part of the write, and on a network or quota-enforced filesystem it is
 * the ONLY place the failure appears: the bytes were buffered, write() said yes, and
 * close() is where the disk finally says no. Close on the failure path too - the
 * handle is ours either way - but do not throw the answer away. */
proven_err_t cerr = proven_fs_close(out.value);
if (proven_is_ok(err)) err = cerr;

EXAMPLE_REQUIRE(proven_is_ok(err), "writing the whole buffer should succeed");
if (!proven_is_ok(err)) {
    (void)proven_fs_remove(alloc, path);
    return 1;
}

/* --- read ----- */
proven_result_file_t in = proven_fs_open(alloc, path, PROVEN_FS_READ);
EXAMPLE_REQUIRE(proven_is_ok(in.err), "opening the file for reading should succeed");
if (!proven_is_ok(in.err)) {
    (void)proven_fs_remove(alloc, path);
    return 1;
}

/* size is a hint for sizing the buffer, not a promise about how many bytes any
 * one read will hand over - and it is 0 for anything that is not a regular
 * file (a pipe, a device, a /proc entry). The loop below does not rely on it. */
proven_result_size_t sz = proven_fs_size(in.value);
EXAMPLE_REQUIRE(proven_is_ok(sz.err), "querying the size of an open file should succeed");
EXAMPLE_REQUIRE(sz.value == text.size, "the file should be as long as what we wrote");

proven_byte_t buf[128];
proven_size_t total = 0;

/* The partial-read loop. Each pass asks for whatever is left of the buffer and
 * advances by however much actually arrived: a short read is normal, not the
 * end of the file. The end of the file is a distinct status - PROVEN_ERR_EOF
 * with zero bytes - so the loop terminates on that, and on nothing else. The
 * loop also stops if the source outgrows the buffer; noticing that is the
 * caller's business (here it cannot happen, but a growing file could). */
for (;;) {
    if (total == sizeof buf) break; /* buffer full: caller decides what to do */

    proven_mem_mut_t dest = { .ptr = buf + total, .size = sizeof buf - total };
    proven_result_size_t r = proven_fs_read(in.value, dest);
    if (r.err == PROVEN_ERR_EOF) break;
    if (!proven_is_ok(r.err)) {
        (void)proven_fs_close(in.value);
        (void)proven_fs_remove(alloc, path);
        EXAMPLE_REQUIRE(false, "reading from the open file should not fail");
        return 1;
    }
    total += r.value;
}

(void)proven_fs_close(in.value);

EXAMPLE_REQUIRE(total == text.size, "the loop should have read every byte in the file");
proven_u8str_view_t got = { .ptr = buf, .size = total };
EXAMPLE_REQUIRE(proven_u8str_view_eq(got, text), "the bytes should come back unchanged");

printf("read %zu bytes in chunks: %.*s", (size_t)total, (int)total, (const char *)buf);

```

```

/* --- clean up ----- */
err = proven_fs_remove(alloc, path);
EXAMPLE_REQUIRE(proven_is_ok(err), "removing the temp file should succeed");

return EXAMPLE_OK();
}

```

## Reading a directory one entry at a time

proven\_fs\_list reads the **whole** directory before you see any of it, and it allocates a string for every name. Measured on 50,000 entries: **189 ms, +4.2 MB resident, 50,008 allocations**, with nothing visible until the last entry was read. That is fine for a config directory and useless for a mail spool.

proven\_fs\_dir\_open / \_next / \_close walks the same directory one entry at a time and **allocates nothing per entry**.

| API                               | Intent                                                                                                                                                    | Return               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| proven_fs_dir_open(scratch, path) | Open a streaming iterator. scratch is for the path conversion only.                                                                                       | proven_result_dir_t. |
| proven_fs_dir_next(&dir, &entry)  | The next entry. PROVEN_ERR_EOF when there are no more; PROVEN_ERR_INVALID_ENCODING for one entry whose name is not valid text, and the next call goes on. | proven_err_t.        |
| proven_fs_dir_close(&dir)         | Release the iterator.                                                                                                                                     | void.                |

```

typedef struct {
    proven_u8str_view_t name; /* BORROWED: points into the iterator's own storage,
                               valid only until the next proven_fs_dir_next */
    proven_fs_type_t type; /* FILE / DIR / OTHER - and it follows symlinks */
} proven_fs_dir_entry_t;

```

Use it like this:

```

proven_result_dir_t d = proven_fs_dir_open(alloc, PROVEN_LIT("."));
if (proven_is_ok(d.err)) {
    proven_fs_dir_t dir = d.value;
    proven_fs_dir_entry_t entry;
    for (;;) {
        proven_err_t e = proven_fs_dir_next(&dir, &entry);
        if (e == PROVEN_ERR_EOF) break;
        if (e == PROVEN_ERR_INVALID_ENCODING) continue; /* a name that is not text: skip it */
        if (!proven_is_ok(e)) break; /* a real error: report it */
        proven_println("{} ", PROVEN_ARG(entry.name));
    }
    proven_fs_dir_close(&dir);
}

```

**The name is borrowed, and it dies at the next call.** This is what makes the whole thing cost no allocations - and a dangling pointer the moment you keep it.

Wrong:

```

proven_u8str_view_t names[100];
int n = 0;
while (proven_is_ok(proven_fs_dir_next(&dir, &entry)))
    names[n++] = entry.name; /* wrong: every entry aliases the same storage, and the
                             next _next overwrites it. All 100 end up equal - to
                             whatever the last entry happened to be. */

```

Correct: copy the bytes (`proven_u8str_create_from_view`) for the ones you need to keep - or use `proven_fs_list`, which does exactly that for every entry and charges you for it.

**PROVEN\_ERR\_EOF is the end; anything else is a failure.** A loop that stops on "not OK" treats a permission error as a complete listing.

```
while (proven_is_ok(proven_fs_dir_next(&dir, &entry))) { ... }
/* wrong: an I/O error ends the loop exactly like the end of the directory does,
   and you cannot tell whether you saw everything. */
```

## Walking a tree

`proven_fs_dir_*` walks ONE directory. `proven_fs_walk` walks a tree - and it is worth saying exactly what it refuses to do, because those refusals are the feature:

|                         |                                                                                                                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |                                                                                                                                                                                                                 |
| It cannot <b>loop</b>   | It never descends <i>through</i> a symlink.                                                                                                                                                                     |
| It cannot <b>escape</b> | Same rule: a link out of the tree is reported, not followed.                                                                                                                                                    |
| It cannot <b>lie</b>    | A directory it cannot read comes back as an <i>error</i> naming that directory, and the walk goes on. A tree walker that silently skips an unreadable subtree is how a backup misses files and reports success. |
| It cannot <b>bloat</b>  | One open handle per <i>level</i> of the current path, plus one reused path buffer. Memory is a function of depth, not of how many files there are.                                                              |

A symlinked directory is still **reported** - it exists, type is DIR, `is_symlink` is true - it is simply not entered. Hiding it would be its own kind of lie. If you *want* to follow it, you have the path: open a second walk on it, and you own the cycle question.

## The structure you receive

```
typedef struct {
    proven_u8str_view_t path; /* the whole path, from the root you passed in.
                               BORROWED: it points into the walk's one reused
                               buffer and is valid only until the next call. */
    proven_u8str_view_t name; /* the last component of `path`. Same lifetime. */
    proven_fs_type_t    type; /* FILE / DIR / OTHER - and it FOLLOWS symlinks,
                               exactly as proven_fs_stat does. */
    proven_size_t       size; /* bytes, for a regular file; 0 otherwise. */
    proven_size_t       depth; /* 0 for an entry directly inside the root. */
    bool                is_symlink; /* reached through a symlink. `type` describes
                                     the TARGET; the walk does not enter it. */
} proven_fs_walk_entry_t;
```

`entry.path` and `entry.name` are borrowed from the walk's one reused buffer and are valid **until the next call**. Copy them if you need them to outlive the step; that is the price of a walk of a million entries costing one allocation instead of a million.

Wrong - the same trap the directory iterator has, for the same reason:

```
proven_u8str_view_t found[100];
int n = 0;
while (proven_is_ok(proven_fs_walk_next(&walk, &entry)))
    found[n++] = entry.path; /* wrong: every entry aliases ONE buffer. When the loop
                               ends, all 100 point at whatever the last path was. */
```

Two limits, both of which say so rather than going quiet:

- `max_depth` - how far to descend. A directory *at* the limit is still reported (it is an entry); it is not entered.
- `PROVEN_FS_WALK_DEPTH_LIMIT` (256) - how deep the walk's own stack goes, ever. A directory past it comes back as `PROVEN_ERR_OUT_OF_BOUNDS`, naming the directory.

## Compiled and run by the test suite:

```
/*
 * Walking a tree.
 *
 * The three things a recursive walker gets wrong, and what this one does instead:
 *
 * It loops.      A symlink pointing at an ancestor is a cycle. This walk never descends
 *                THROUGH a symlink - the symlinked directory is still reported, it is
 *                simply not entered - so a cycle is impossible, and so is walking out of
 *                the tree you asked about and into the rest of the filesystem.
 *
 * It lies.       A directory it cannot read gets skipped, and the walk reports success.
 *                That is how a backup misses a subtree. Here the error comes back from
 *                proven_fs_walk_next, with the entry naming the directory, and the walk
 *                carries on from the next sibling. You decide what to do about it.
 *
 * It bloats.     Reading a whole directory into memory before yielding anything makes a
 *                walk of a big tree cost a big allocation. This one holds one open handle
 *                per LEVEL of the current path and one reused path buffer - so its memory
 *                is a function of depth, not of how many files there are.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* A small tree to walk: a file, a directory, a file inside it. */
    EXAMPLE_REQUIRE(proven_is_ok(proven_fs_mkdir(alloc, PROVEN_LIT("ex_walk"))), "mkdir");
    EXAMPLE_REQUIRE(proven_is_ok(proven_fs_mkdir(alloc, PROVEN_LIT("ex_walk/inner"))), "mkdir inner");
    EXAMPLE_REQUIRE(proven_is_ok(proven_fs_write_file(alloc, PROVEN_LIT("ex_walk/top.txt"),
        proven_mem_view_from_u8(PROVEN_LIT("top"))), "write top.txt");
    EXAMPLE_REQUIRE(proven_is_ok(proven_fs_write_file(alloc, PROVEN_LIT("ex_walk/inner/deep.txt"),
        proven_mem_view_from_u8(PROVEN_LIT("deep"))), "write deep.txt");

    proven_result_walk_t walk = proven_fs_walk_open(alloc, PROVEN_LIT("ex_walk"),
        PROVEN_FS_WALK_UNLIMITED);
    EXAMPLE_REQUIRE(proven_is_ok(walk.err), "the walk should open");

    proven_size_t files = 0;
    proven_size_t dirs = 0;
    proven_size_t unreadable = 0;
    proven_size_t total_bytes = 0;

    for (;;) {
        proven_fs_walk_entry_t entry = {0};
        proven_err_t err = proven_fs_walk_next(&walk.value, &entry);

        if (err == PROVEN_ERR_EOF) break;

        if (!proven_is_ok(err)) {
            /* A directory that could not be read, or one deeper than the walk's stack. It is
             * REPORTED, not skipped - `entry.path` says which one - and the walk goes on. A
             * tool that copies a tree should fail here; one that reports on a tree should
             * count it and say so. What it must not do is pretend it did not happen. */
            unreadable++;
            continue;
        }

        /* `entry.path` and `entry.name` are borrowed: they point into the walk's one reused
         * buffer and are valid until the next call. Copy them if you need them longer. */
    }
}
```

```

    if (entry.type == PROVEN_FS_TYPE_DIR) {
        dirs++;
    } else if (entry.type == PROVEN_FS_TYPE_FILE) {
        files++;
        total_bytes += entry.size;
    }
}

proven_fs_walk_close(&walk.value);

EXAMPLE_REQUIRE(files == 2, "two files: top.txt and inner/deep.txt");
EXAMPLE_REQUIRE(dirs == 1, "one directory: inner");
EXAMPLE_REQUIRE(unreadable == 0, "and nothing in this tree is unreadable");
EXAMPLE_REQUIRE(total_bytes == 7, "three bytes plus four");

/* Depth-limited: max_depth 0 reports what is directly inside the root and descends
 * nowhere. The directory at the limit is still an entry, so it is still reported. */
walk = proven_fs_walk_open(alloc, PROVEN_LIT("ex_walk"), 0);
EXAMPLE_REQUIRE(proven_is_ok(walk.err), "the shallow walk should open");

proven_size_t shallow = 0;
for (;;) {
    proven_fs_walk_entry_t entry = {0};
    proven_err_t err = proven_fs_walk_next(&walk.value, &entry);
    if (err == PROVEN_ERR_EOF) break;
    if (proven_is_ok(err)) shallow++;
}
proven_fs_walk_close(&walk.value);

EXAMPLE_REQUIRE(shallow == 2, "top.txt and inner - but nothing inside inner");

(void)proven_fs_remove(alloc, PROVEN_LIT("ex_walk/inner/deep.txt"));
(void)proven_fs_remove(alloc, PROVEN_LIT("ex_walk/top.txt"));
(void)proven_fs_remove(alloc, PROVEN_LIT("ex_walk/inner"));
(void)proven_fs_remove(alloc, PROVEN_LIT("ex_walk"));

return EXAMPLE_OK();
}

```

## Streams: writers and readers

The formatter's only sink used to be a `proven_u8str_t`. A file was a `proven_file_t`. The in-memory scanner read a view; the file scanner read something else again. **Four types, four function families, no common interface** - so you could not write one `serialize(sink, value)` that worked over both memory and a file, you could not format into a file at all, and there was no way to read a file line by line.

A **writer** is a byte sink. A **reader** is a byte source. Both are small vtables passed by value, exactly like `proven_allocator_t`, and for the same reason: the caller decides where the bytes go, and nothing is hidden.

| API                                                                | Intent                                                  |
|--------------------------------------------------------------------|---------------------------------------------------------|
| <code>proven_writer_from_file(&amp;file)</code>                    | Unbuffered sink over an open file.                      |
| <code>proven_writer_from_u8str(&amp;state, &amp;str, alloc)</code> | Appends to an owned string.                             |
| <code>proven_writer_from_buffer(&amp;state)</code>                 | Fixed caller memory. Allocates nothing, ever.           |
| <code>proven_writer_buffered(&amp;state, inner, buf)</code>        | Accumulates small writes; <b>you</b> supply the buffer. |

|                                                                                |                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_fprint(w, fmt, ...) / proven_fprintln</code>                      | Format straight into a writer. No allocation.                                                                                                                                                                   |
| <code>proven_reader_from_file(&amp;file) / _from_view(&amp;state, view)</code> | Byte sources.                                                                                                                                                                                                   |
| <code>proven_reader_buffered(&amp;state, inner, buf)</code>                    | Buffered source; required for line reading.                                                                                                                                                                     |
| <code>proven_reader_read_line(&amp;state)</code>                               | One line, without the newline.                                                                                                                                                                                  |
| <code>proven_writer_is_valid(w) / proven_reader_is_valid(r)</code>             | Did the constructor succeed? A zeroed handle is invalid, and every constructor returns one on bad arguments - so this is the check, not a NULL test.                                                            |
| <code>proven_fwrite_fmt(w, scratch, fmt, ...)</code>                           | <code>proven_fprint</code> with a scratch buffer <b>you</b> size. <code>proven_fprint</code> uses a 512-byte stack buffer and returns <code>OUT_OF_BOUNDS</code> for a longer line; this is how you format one. |
| <code>proven_fmt_to_writer_impl(w, scratch, fmt, args, n)</code>               | The function the two macros above expand to. Call it directly only if you are building your own variadic wrapper; the macros exist so you do not have to count arguments.                                       |

Four rules worth stating plainly, because each of them is a way this could have been designed badly:

- **Buffering uses memory you supply.** `proven_writer_buffered` takes a `proven_mem_mut_t`, the way `proven_arena_create` does. There is no hidden global buffer, which means there is also no destructor to flush it for you - **you must flush before the buffer goes out of scope**. In exchange, your logging path never allocates, and a program logging its way out of an out-of-memory condition can still log.
- **A full sink refuses; it does not truncate.** A fixed buffer that fills up returns `PROVEN_ERR_OUT_OF_BOUNDS` and records overflowed. A sink that silently drops the end of your data is worse than one that says it cannot take it.
- **A line too long for the reader's buffer is an error**, not a truncated line. A truncated line handed back as if it were whole is a corruption the caller has no way to detect. The buffer is yours; size it for the input you expect.
- **A partial write is a fact, not a failure to be papered over.** A `write_fn` returns a `proven_result_size_t`: how many bytes the sink took, *and* what went wrong. A pipe, a socket, or a filling disk really does accept 4096 of your 6000 bytes and then fail, and a trait that says "consume it all or fail" simply makes such a sink impossible to write correctly. `proven_writer_write` still means all-or-nothing (it loops); `proven_writer_write_partial` is there when you need to see how far you got. The buffered writer keeps only the tail the sink did **not** take - the first version kept the whole buffer and re-sent it, so a failing sink received the accepted prefix twice.
- **A writer that has failed stays failed.** Once a writer has lost bytes - a buffered writer whose sink died, a fixed buffer that overflowed - the stream it was producing has a hole in it that the receiver cannot see, so every later write and flush returns the error. A shorter chunk that *would* fit is refused too: writing it would put it after the hole, and the result would look like complete output. There is no `clear()`: if you have a recovery story it involves a new writer over a new sink, not pretending this one is fine. (Before this, `flush` answered `PROVEN_OK` after a failed write - the buffer was empty, so there was nothing left to fail on - and "write, write, write, check the flush", which is how almost everyone uses a buffered writer, reported success on a full disk.)

A reader's rule is the mirror image: **a read that fails is an error, never an end of file.**

proven\_reader\_read returns PROVEN\_ERR\_IO, not a clean zero-byte EOF, when the source breaks - because a file cut short by a disk error and a file that simply ended are the same thing to a caller who cannot tell them apart, and only one of them is safe to act on.

What it costs, measured over 10,000 lines to stdout:

|                                         | write() syscalls | malloc()       |
|-----------------------------------------|------------------|----------------|
| proven_println                          | 10,000           | 0 (was 10,000) |
| buffered writer, 8 KiB of caller memory | 24               | 0              |

The malloc() column is for lines that fit the 512-byte stack buffer, which is the ordinary case. A line longer than that falls back to the heap for that one call rather than being refused.

proven\_println is deliberately still one syscall per line: buffering it would need hidden global state. A caller who wants the 24 builds a buffered writer and says so.

### Worked example: one serializer, three destinations, and reading it back

Compiled and run by the test suite. Note that render\_row does not know where its bytes are going - that is the entire point.

```
/*
 * Writers and readers: one interface for "where bytes go" and one for "where bytes
 * come from".
 *
 * The point is that the code below - render_row - does not know and does not care
 * whether it is writing to a string, to a fixed buffer, or to a file. That was
 * impossible before: the formatter's only sink was a proven_u8str_t.
 */

/* One serializer. It takes a sink, not a destination. */
static proven_err_t render_row(proven_writer_t w, int id, const char *name) {
    proven_fmt_result_t r = proven_fprintln(w, "{:>4} | {}", PROVEN_ARG(id), PROVEN_ARG(name));
    return r.err;
}

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- the same code, into a growing string ----- */
    proven_result_u8str_t s = proven_u8str_create(alloc, 16);
    EXAMPLE_REQUIRE(proven_is_ok(s.err), "string create");

    proven_writer_u8str_t s_state;
    proven_writer_t to_string = proven_writer_from_u8str(&s_state, &s.value, alloc);
    EXAMPLE_REQUIRE(proven_is_ok(render_row(to_string, 7, "ada")), "render into a string");

    printf("into a string:\n%s", proven_u8str_as_cstr(&s.value));
    proven_u8str_destroy(alloc, &s.value);

    /* --- the same code, into memory you own: zero allocations ----- */
    proven_byte_t fixed[64];
    proven_writer_buf_t b_state = { .buf = { .ptr = fixed, .size = sizeof fixed } };
    proven_writer_t to_buffer = proven_writer_from_buffer(&b_state);
    EXAMPLE_REQUIRE(proven_is_ok(render_row(to_buffer, 8, "grace")), "render into a buffer");
    EXAMPLE_REQUIRE(b_state.len > 0, "the buffer received the row");

    /* A full buffer REFUSES; it does not truncate. A sink that silently drops the
     * end of your data is worse than one that says it cannot take it. */
}
```

```

proven_byte_t tiny[4];
proven_writer_buf_t t_state = { .buf = { .ptr = tiny, .size = sizeof tiny } };
proven_writer_t to_tiny = proven_writer_from_buffer(&t_state);
EXAMPLE_REQUIRE(proven_writer_write_str(to_tiny, PROVEN_LIT("far too long")) == PROVEN_ERR_OUT_OF_BOUNDS,
    "a full buffer refuses rather than truncating");
EXAMPLE_REQUIRE(t_state.overflowed, "and it records that it did");

/* --- the same code, into a file, buffered ----- */
/*
 * The buffer is memory YOU supply, exactly like an arena's. This library has no
 * hidden global state, so it cannot flush for you at exit - which is why you must
 * flush before the buffer goes out of scope. In exchange, your logging path never
 * allocates: ten thousand lines here cost 0 mallocs and a couple of dozen write
 * syscalls, where ten thousand proven_printf calls cost 10,000 syscalls.
 */
proven_u8str_view_t path = PROVEN_LIT("example_stream_rows.txt");
proven_result_file_t f = proven_fs_open(alloc, path,
    (proven_fs_mode_t)(PROVEN_FS_WRITE | PROVEN_FS_CREATE | PROVEN_FS_TRUNC));
EXAMPLE_REQUIRE(proven_is_ok(f.err), "open the output file");
proven_file_t file = f.value;

proven_byte_t out_buf[4096];
proven_writer_buffered_t w_state;
proven_writer_t to_file = proven_writer_buffered(&w_state,
    proven_writer_from_file(&file),
    (proven_mem_mut_t){ .ptr = out_buf, .size = sizeof out_buf });

for (int i = 0; i < 3; ++i) {
    EXAMPLE_REQUIRE(proven_is_ok(render_row(to_file, i, "row")), "render into the file");
}
EXAMPLE_REQUIRE(proven_is_ok(proven_writer_flush(to_file)),
    "flush: nothing is written until you say so");
/* And the close, which is the last thing that can tell you the write did not land. */
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_close(file)), "closing the written file");

/* --- reading it back, a line at a time ----- */
/* Reading a file line by line was simply not possible before: the only route was
 * loading the entire file into memory and splitting it by hand. */
proven_result_file_t rf = proven_fs_open(alloc, path, PROVEN_FS_READ);
EXAMPLE_REQUIRE(proven_is_ok(rf.err), "reopen for reading");
proven_file_t rfile = rf.value;

proven_byte_t in_buf[128];
proven_reader_buffered_t r_state;
(void)proven_reader_buffered(&r_state, proven_reader_from_file(&rfile),
    (proven_mem_mut_t){ .ptr = in_buf, .size = sizeof in_buf });

int lines = 0;
for (;;) {
    proven_result_u8str_view_t line = proven_reader_read_line(&r_state);
    if (line.err == PROVEN_ERR_EOF) break;
    EXAMPLE_REQUIRE(proven_is_ok(line.err), "read a line");
    /* The view points INTO the reader's buffer, and is valid only until the next
     * call. Copy it if it has to outlive that. */
    printf("line %d: %.*s\n", lines, (int)line.val.size, (const char *)line.val.ptr);
    ++lines;
}
EXAMPLE_REQUIRE(lines == 3, "three rows in, three lines out");

(void)proven_fs_close(rfile);

```

```

    (void)proven_fs_remove(alloc, path);

    return EXAMPLE_OK();
}

```

## The structures you hold

The two **handles** are passed by value and are cheap:

```

typedef struct {
    void *ctx;
    proven_result_size_t (*write_fn)(void *ctx, proven_mem_view_t chunk);
    proven_err_t (*flush_fn)(void *ctx); /* may be NULL: this sink holds nothing back */
} proven_writer_t;

typedef struct {
    void *ctx;
    proven_result_size_t (*read_fn)(void *ctx, proven_mem_mut_t dest);
} proven_reader_t;

```

write\_fn reports **how much went out even when it then fails**, and that is not a nicety: a write to a pipe or a full disk really does put some bytes out and then fail. A buffered writer built on the tidier "all or nothing" lie kept its whole buffer on failure and re-sent it on the next flush - a 6000-byte payload arrived as 10,096 bytes with the first 4096 **duplicated**. Losing data is bad; silently doubling it is worse, because the receiver cannot tell.

Everything else is **caller-owned state** -- proven\_writer\_buf\_t, proven\_writer\_u8str\_t, proven\_writer\_buffered\_t, proven\_reader\_view\_t, proven\_reader\_buffered\_t. They allocate nothing, they have no destroy, and **they must not be copied or moved** while a handle points into them.

## Cautions, and what goes wrong

**A buffered writer that is never flushed is output that never happened.** There is no hidden state, so there is no destructor to flush it for you - and nothing here registers an atexit handler, because a library that owns your process is a library you cannot reason about.

Wrong:

```

proven_writer_buffered_t st;
proven_writer_t w = proven_writer_buffered(&st, inner, buf);
(void)proven_fprintln(w, "the important line");
return; /* wrong: the buffer dies with the frame, and so does the line */

```

Correct - flush before the buffer or the inner sink goes away:

```

proven_byte_t buf[256];
proven_sysio_out_t out;
proven_writer_t w = proven_sysio_stdout_buffered(&out,
    (proven_mem_mut_t){ .ptr = buf, .size = sizeof buf });
(void)proven_fprintln(w, "the important line");
(void)proven_writer_flush(w); /* now it has happened */

```

**The line a reader hands you points into its buffer.** It is valid only until the next call. That is what makes reading a million lines cost one buffer instead of a million allocations - and it is a dangling pointer the moment you keep it.

Wrong:

```

proven_u8str_view_t lines[100];
for (int i = 0; i < 100; ++i) {
    proven_result_u8str_view_t ln = proven_reader_read_line(&st);
    lines[i] = ln.val; /* wrong: every entry aliases the SAME buffer, and the
                        next read_line overwrites what the last one returned */
}

```

Correct: copy the bytes you need to keep (into a `proven_u8str_t`, an arena, wherever), before you call again.

**A flush is not a durability barrier.** `proven_writer_flush` pushes a buffered writer's bytes to the thing behind it; getting a *file's* bytes onto the disk is `proven_fs_sync`. They are different operations, and one word could not honestly mean both - which is why the old `proven_sysio_flush`, which claimed to be both and was neither, is gone.

**A line longer than the buffer is refused, not truncated.** `PROVEN_ERR_OUT_OF_BOUNDS` - and the reader then stays wedged on that line: there is no resync, because a line you cannot hold is not a line you can skip past without deciding what to do with the bytes. Size the buffer for the input you expect. (A line that *exactly fills* the buffer is fine: the newline does not have to fit too, and neither does a final line with no newline at all.)

## The standard streams

`stream.h` has writers, readers, buffered writers and a line reader. `sysio.h` has `stdin`, `stdout` and `stderr`. Until they were introduced to each other, two things were simply not possible - and one call was a lie.

**Reading stdin a line at a time had no route.** The most common thing a program does with `stdin`, and the choices were the token scanner or reading the whole of a stream that may never end. The bridge fixes that: a standard handle is parked in caller-owned storage, so the line reader has something stable to point at.

```
proven_byte_t buf[4096];
proven_sysio_lines_t lines;
if (proven_is_ok(proven_sysio_stdin_lines(&lines, (proven_mem_mut_t){ .ptr = buf, .size = sizeof buf }))) {
    for (;;) {
        proven_result_u8str_view_t line = proven_sysio_read_line(&lines);
        if (line.err == PROVEN_ERR_EOF) break;
        if (!proven_is_ok(line.err)) break; /* OUT_OF_BOUNDS: a line longer than `buf` */
        /* `line.val` points INTO `buf` and is valid only until the next call. */
        proven_println("{", PROVEN_ARG(line.val));
    }
}
```

It inherits the line reader's properties, which is the point of not writing a second one: the view costs no allocation, `"\r\n"` is handled, a final line with no trailing newline is still returned, and a line longer than your buffer is `PROVEN_ERR_OUT_OF_BOUNDS` - never a silently truncated line.

**The formatter could not be aimed at a standard stream.** `proven_fprintln` takes a writer; `stdout` was not one. Now it is, and it can be a *buffered* one - so a thousand small lines cost one syscall instead of a thousand.

| <code>proven_sysio_stdout_writer(&amp;st)</code>             | An unbuffered writer over <code>stdout</code> . Every write is a write syscall.                                      |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <code>proven_sysio_stderr_writer(&amp;st)</code>             | The same for <code>stderr</code> - which is what you want for an error: it is out before the next line of code runs. |
| <code>proven_sysio_stdin_reader(&amp;st)</code>              | A reader over <code>stdin</code> .                                                                                   |
| <code>proven_sysio_stdout_buffered(&amp;out, buf)</code>     | <code>stdout</code> behind a buffered writer over a buffer you own.                                                  |
| <code>proven_sysio_file_buffered(&amp;out, file, buf)</code> | The same over any open file.                                                                                         |

**And flush means something now.** `proven_sysio_flush` used to claim to flush a buffer that did not exist: a no-op on POSIX, a *disk sync* on Windows. It is **deleted**. Pushing a buffered writer's bytes to

the OS is `proven_writer_flush`; pushing the OS's bytes to the disk is `proven_fs_sync`. They are different operations and now say so.

**You must flush a buffered writer.** Nothing reaches the terminal until the buffer fills or you flush it, and nothing in this library registers an `atexit` handler to do it behind your back - a library that owns your process is a library you cannot reason about. Buffered output that is never flushed is output that never happened. The direct calls (`proven_print`, `proven_println`, `proven_eprint`) remain unbuffered for exactly this reason: what they write is on its way out before they return.

### The structures you hold

```
typedef struct {
    proven_file_t      file;
    bool               console; /* a Windows console: see the next section */
    proven_sysio_carry_t carry; /* text in flight at the console edge; not yours to touch */
} proven_sysio_std_t;
/* Storage for a standard handle, so a writer or reader has something stable to
   point at. proven_writer_from_file takes a proven_file_t * and the file must
   outlive the writer - so it cannot be a temporary. This is that storage. */

typedef struct {
    proven_sysio_std_t      std;
    proven_writer_buffered_t buffered;
} proven_sysio_out_t; /* a buffered writer over a standard stream or a file */

typedef struct {
    proven_sysio_std_t      std;
    proven_reader_buffered_t buffered;
} proven_sysio_lines_t; /* a line reader over a standard stream or a file */
```

All three are [caller-owned state](#).

### Cautions, and what goes wrong

**These state structs contain a pointer to themselves.** The writer you get back addresses `&std->std.file` *inside* the struct you passed. Copy the struct, or return it by value, and the writer still points at the original - which may be a dead frame.

Wrong - and an audit reproduced exactly this as a heap-use-after-free:

```
proven_sysio_out_t out;
proven_writer_t w = proven_sysio_stdout_buffered(&out, buf);

proven_sysio_out_t copy = out; /* wrong: `w` still points into `out` */
/* ... `out` goes out of scope ... */
(void)proven_writer_write_str(w, PROVEN_LIT("boom")); /* writes through dead storage */
```

The one exception is `proven_sysio_lines_t`: `proven_sysio_read_line` re-binds it on every call, so a line reader **may** be moved. That is a deliberate courtesy, because it takes its state by pointer - the shape that says "relocatable" - and the library should not lay a trap in the shape of a promise.

**A zero-initialised `proven_file_t` is not an invalid handle - on POSIX it is fd 0, which is `stdin`.** The library cannot tell a handle you forgot to fill in from one that legitimately refers to fd 0.

```
proven_sysio_out_t out;
proven_file_t f = {0}; /* wrong: this is stdin */
proven_writer_t w = proven_sysio_file_buffered(&out, f, buf); /* writes to fd 0 */
```

**Buffered stdout and unbuffered stderr do not interleave in the order you wrote them.**

Anything you buffer sits in your buffer while stderr goes straight out. Flush before you print an error that is supposed to appear after your output.

## UTF-16 text in and out, and the Windows console

Two problems share this section because they share a cause: text is not bytes everywhere.

**UTF-16 text had no way in or out.** A `proven_u16str_t` - the names a Windows wide API hands back, text from a UTF-16 file - could not be printed, written to a file, or read a line at a time. Now `u16` text goes through every channel `u8` text does. Output goes through the formatter, so `PROVEN_ARG` on a `proven_u16str_view_t` works in `proven_println`, `proven_eprintln`, `proven_fprintln` into a file or any other writer, and `proven_u8str_append_fmt`, and writes UTF-8. When the consumer wants UTF-16 *bytes* - a Windows tool that reads UTF-16LE files, say - `proven_writer_write_u16` writes them in the byte order you name, and a byte order mark only when you ask with `proven_writer_write_bom`.

Input is `proven_u16_reader_t`: a reader over any byte source that decodes UTF-8, UTF-16LE or UTF-16BE into your buffer of code units, a line or a chunk at a time. `PROVEN_TEXT_AUTO` lets a leading byte order mark decide and consumes it; with no mark the text is UTF-8. It carries a character split between two reads of the source, and it keeps the byte line reader's rules: `"\r\n"` loses its `'\r'`, a final line needs no newline, a line longer than the buffer is `PROVEN_ERR_OUT_OF_BOUNDS`. It is strict, like [utf.h](#): malformed input stops it with `PROVEN_ERR_INVALID_ENCODING` after the valid text before it has been delivered, and so does a source that ends inside a character.

**A Windows console is not a file.** It decodes the bytes `WriteFile` gives it in its code page - 949 on Korean Windows - so UTF-8 written to it was mojibake unless someone had run `chcp 65001`, and `ReadFile` on it returned code-page bytes, not UTF-8. Changing the console's code page from inside a library would change it for every other program sharing that console, after this one exits. So the sysio layer does the other thing: when a standard handle is a console (`GetConsoleMode` accepts it), its writers and readers talk to it in UTF-16 through `WriteConsoleW` and `ReadConsoleW`, converting at the edge. That covers `proven_print` and `proven_eprint`, the stdout and stderr writers, the buffered writers, the `u8` and `u16` line readers and the token scanner. A UTF-8 character split across two writes - a buffered writer flushes at its buffer size, not at a character boundary - is carried in the state struct and completed by the next write. `Ctrl+Z` at the start of a console line is end of input, as it is for every Windows console program. Files, pipes and redirected streams are not consoles and still receive the exact bytes, and on POSIX nothing is a console.

Reading stdin as UTF-16, a line at a time:

```
#ifndef PROVEN_NO_U16STR
proven_u16 wbuf[512];
proven_sysio_u16_lines_t in;
if (proven_is_ok(proven_sysio_stdin_u16_lines(&in, wbuf, 512))) { /* stdin is read as UTF-8 */
    for (;;) {
        proven_result_u16str_view_t line = proven_sysio_read_u16_line(&in);
        if (line.err == PROVEN_ERR_EOF) break;
        if (!proven_is_ok(line.err)) break; /* OUT_OF_BOUNDS, or INVALID_ENCODING */
        /* `line.val` points INTO `wbuf` until the next call - e.g. hand it to a wide API. */
        (void)proven_eprintln("{} code units", PROVEN_ARG(line.val.size));
    }
}
#endif
```

|                                                      |                                                     |
|------------------------------------------------------|-----------------------------------------------------|
| <code>proven_arg_u16(view) / PROVEN_ARG(view)</code> | u16 text as a formatter argument, written as UTF-8. |
|------------------------------------------------------|-----------------------------------------------------|

|                                                                        |                                                                                                        |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>proven_writer_write_u16(w, text, enc)</code>                     | u16 text to any writer as UTF-8, UTF-16LE or UTF-16BE. Validated first: malformed text writes nothing. |
| <code>proven_writer_write_bom(w, enc)</code>                           | The byte order mark for enc, only when you ask.                                                        |
| <code>proven_u16_reader_init(&amp;st, reader, enc, buf, cap)</code>    | A u16 reader over any reader, through cap code units of yours (at least 2).                            |
| <code>proven_u16_reader_read_line(&amp;st)</code>                      | The next line as a u16 view into your buffer.                                                          |
| <code>proven_u16_reader_read(&amp;st, dest, cap)</code>                | Up to cap code units, never half a surrogate pair.                                                     |
| <code>proven_sysio_u16_lines_open(&amp;st, file, enc, buf, cap)</code> | The same reader over a file or a standard stream.                                                      |
| <code>proven_sysio_stdin_u16_lines(&amp;st, buf, cap)</code>           | Over stdin, read as UTF-8 (a console is read as UTF-16).                                               |
| <code>proven_sysio_read_u16_line(&amp;st)</code>                       | The next line; the struct may be moved between calls.                                                  |

### The structures you hold

```
typedef struct {
    proven_reader_t inner; proven_text_encoding_t enc; /* AUTO resolved from the BOM */
    proven_u16 *buf; proven_size_t cap, len, cursor; /* your buffer, in code units */
    proven_byte_t raw[1024]; proven_size_t raw_pos, raw_len; /* bytes read; [raw_pos, raw_len) not yet
decoded */
    proven_u16 peek[2]; proven_size_t peek_len; /* one character of lookahead */
    bool bom_checked, eof; proven_err_t err; /* err is sticky */
} proven_u16_reader_t;

typedef struct {
    proven_sysio_std_t std;
    proven_u16_reader_t reader;
} proven_sysio_u16_lines_t; /* a u16 line reader over a standard stream or a file */
```

Both are [caller-owned state](#). Nothing is allocated: the raw bytes are staged in the struct, the decoded text in your buffer.

### Cautions, and what goes wrong

`proven_writer_from_file` **on a console handle does not convert**. It is a file writer and writes the bytes as they are - mojibake on a Windows console in a non-UTF-8 code page. Make console writers and readers with the `proven_sysio_*` calls, which detect the console.

```
proven_file_t out = proven_sysio_stdout();
proven_writer_t w = proven_writer_from_file(&out); /* wrong for a console: bytes, not text */
proven_sysio_std_t st;
proven_writer_t ok = proven_sysio_stdout_writer(&st); /* right: converts on a console */
```

**Strict means strict on the console too.** Malformed UTF-8 sent to a console is refused with `PROVEN_ERR_INVALID_ENCODING` after the valid text before it has been shown. Nothing is replaced with a question mark. A flush is not the end of a text - a buffered writer drains at its buffer size, and you may flush between any two bytes - so the bytes of a character still open at a flush are kept and the next write completes it, or is refused if it cannot. `proven_print` and `proven_eprint` write a whole formatted text, so for them a text that ends inside a character is refused outright.

**An explicit encoding does not strip a byte order mark.** Opened as `PROVEN_TEXT_UTF16LE`, a file that starts with `FF FE` delivers the character `U+FEFF` as the first unit of its first line. Open it as `PROVEN_TEXT_AUTO` when you do not control who wrote it.

**A surrogate pair is one character.** `proven_writer_write_u16` refuses a text that ends with the high half of a pair even if the next call would have supplied the low half - keep a pair in one call. `proven_u16_reader_read` needs room for two units for the same reason.

**Width counts UTF-8 bytes.** `{:>10}` around a `u16` argument pads to ten UTF-8 bytes, exactly as it does for a `u8` view - not ten characters, and not ten terminal columns; a Hangul syllable is three bytes and two columns.

### Worked example: u16 text to the standard streams, to a UTF-16LE file, and back

```
#include <string.h>

/*
 * UTF-16 text in and out: to the standard streams, to a file in the encoding
 * its reader expects, and back in a line at a time.
 *
 * Output goes through the formatter, so every sink that takes u8 text takes u16
 * text too: PROVEN_ARG picks the u16 renderer for a proven_u16str_view_t and
 * writes UTF-8. When a consumer wants UTF-16 bytes - a Windows tool that reads
 * UTF-16LE files, say - a writer writes them in the byte order you name.
 *
 * Input is a reader that decodes any of the three encodings into your u16
 * buffer. It carries a character split between two reads of the source, and it
 * keeps the byte line reader's rules: "\r\n" loses its '\r', the last line needs
 * no newline, and a line too long for the buffer is an error, not a cut line.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* Two Hangul syllables ("hello"), a space, and an emoji - which is a surrogate pair. */
    static const proven_u16 hello[] = { 0xC548, 0xB155, ' ', 0xD83D, 0xDE42 };
    proven_u16str_view_t text = { hello, 5 };

    /* --- to the standard streams ----- */

    /* UTF-8 on stdout and stderr. On a Windows console the sysio layer hands
     * it to the console as UTF-16, so it shows correctly whatever the code page. */
    proven_err_t err = proven_println("stdout: {}", PROVEN_ARG(text));
    EXAMPLE_REQUIRE(proven_is_ok(err), "u16 text prints like u8 text");
    err = proven_eprintln("stderr: {}", proven_arg_u16(text));
    EXAMPLE_REQUIRE(proven_is_ok(err), "and to stderr");

    /* A buffered writer takes it too; flush it, or it never happened. */
    proven_sysio_out_t out;
    proven_byte_t buf[256];
    proven_writer_t w = proven_sysio_stdout_buffered(&out, (proven_mem_mut_t){ buf, sizeof buf });
    EXAMPLE_REQUIRE(proven_is_ok(proven_fprintln(w, "buffered: {}", PROVEN_ARG(text)).err), "fprintln takes
u16");
    EXAMPLE_REQUIRE(proven_is_ok(proven_writer_flush(w)), "and the flush sends it");

    /* Strict: an unpaired surrogate fails the whole line instead of printing junk. */
    static const proven_u16 torn_units[] = { 'o', 'k', 0xD83D };
    proven_u16str_view_t torn = { torn_units, 3 };
    err = proven_println("{} ", PROVEN_ARG(torn));
    EXAMPLE_REQUIRE(err == PROVEN_ERR_INVALID_ENCODING, "half a surrogate pair is refused");
}
```

```

/* --- to a file, as UTF-16LE with a BOM ----- */

proven_u8str_view_t path = PROVEN_LIT("proven_example_u16.txt");
proven_result_file_t f = proven_fs_open(alloc, path, PROVEN_FS_WRITE | PROVEN_FS_CREATE |
PROVEN_FS_TRUNC);
EXAMPLE_REQUIRE(proven_is_ok(f.err), "creating the file must succeed");
if (!proven_is_ok(f.err)) return EXAMPLE_OK();

proven_sysio_out_t fout;
proven_writer_t fw = proven_sysio_file_buffered(&fout, f.value, (proven_mem_mut_t){ buf, sizeof buf });
/* The BOM is written only because it is asked for. */
err = proven_writer_write_bom(fw, PROVEN_TEXT_UTF16LE);
for (int i = 0; i < 2 && proven_is_ok(err); ++i) {
    err = proven_writer_write_u16(fw, text, PROVEN_TEXT_UTF16LE);
    if (proven_is_ok(err)) err = proven_writer_write_u16(fw, (proven_u16str_view_t){ u"\r\n", 2 },
PROVEN_TEXT_UTF16LE);
}
if (proven_is_ok(err)) err = proven_writer_flush(fw);
EXAMPLE_REQUIRE(proven_is_ok(err), "two lines of UTF-16LE are written");
(void)proven_fs_close(f.value);

/* --- back in, a line at a time ----- */

f = proven_fs_open(alloc, path, PROVEN_FS_READ);
EXAMPLE_REQUIRE(proven_is_ok(f.err), "opening the file again must succeed");
if (proven_is_ok(f.err)) {
    /* AUTO: the BOM says UTF-16LE, and is consumed. The buffer is in code
    * units and must hold the longest line. */
    proven_u16 line_buf[64];
    proven_sysio_u16_lines_t lines;
    err = proven_sysio_u16_lines_open(&lines, f.value, PROVEN_TEXT_AUTO, line_buf, 64);
    EXAMPLE_REQUIRE(proven_is_ok(err), "the line reader opens");
    int n = 0;
    for (;;) {
        proven_result_u16str_view_t line = proven_sysio_read_u16_line(&lines);
        if (line.err == PROVEN_ERR_EOF) break;
        EXAMPLE_REQUIRE(proven_is_ok(line.err), "each line reads");
        if (!proven_is_ok(line.err)) break;
        /* The view points into line_buf until the next call. */
        EXAMPLE_REQUIRE(line.val.size == 5 && memcmp(line.val.ptr, hello, sizeof hello) == 0,
            "each line is the text, without its CR LF");
        ++n;
    }
    EXAMPLE_REQUIRE(n == 2 && lines.reader.enc == PROVEN_TEXT_UTF16LE, "two lines, read as UTF-16LE");
    (void)proven_fs_close(f.value);
}
(void)proven_fs_remove(alloc, path);

/* --- any reader, any encoding ----- */

/* The same reader over bytes already in memory, here UTF-8. read() hands out
* code units whatever lines they belong to, and never half a pair. */
proven_reader_view_t src;
proven_u16 rbuf[16];
proven_u16_reader_t rd;
err = proven_u16_reader_init(&rd, proven_reader_from_view(&src, PROVEN_LIT("A\xF0\x9F\x99\x82\nB")),
    PROVEN_TEXT_UTF8, rbuf, 16);
EXAMPLE_REQUIRE(proven_is_ok(err), "a u16 reader over UTF-8 bytes");
proven_result_u16str_view_t first = proven_u16_reader_read_line(&rd);

```

```

EXAMPLE_REQUIRE(proven_is_ok(first.err) && first.val.size == 3, "'A' and the emoji's two units");
proven_u16 rest[4];
proven_result_size_t got = proven_u16_reader_read(&rd, rest, 4);
EXAMPLE_REQUIRE(proven_is_ok(got.err) && got.value == 1 && rest[0] == 'B', "the last line, without a
newline");
EXAMPLE_REQUIRE(proven_u16_reader_read(&rd, rest, 4).err == PROVEN_ERR_EOF, "then the end");

return EXAMPLE_OK();
}

```

The console path cannot be shown by a program that also runs on Linux. It was checked on Windows 11 by `b039-console-check.c`, which makes its own console in code page 949, writes through each sysio path and reads the screen buffer back, and places key events in the console's input buffer to read them through the line readers and the scanner.

## Randomness, by use case

There is no single "random". There are two jobs that look identical and are not:

| Your job                                                       | Use                                                                                                                                              | Why                                                                                                                |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| A key, a token, a nonce - anything an attacker must not guess. | <code>proven_random_bytes</code> , or a <code>proven_chacha_rng_t</code> seeded from it.                                                         | Only a cryptographic source is unguessable.                                                                        |
| The same, on a target with no OS.                              | <code>proven_chacha_rng_t</code> , seeded from the board's own entropy.                                                                          | ChaCha20 is pure arithmetic; it needs no OS. It is only as unguessable as its seed.                                |
| A simulation, a test, a game, a sample.                        | <code>proven_xoshiro256ss_t</code> .                                                                                                             | Fast, and <b>reproducible</b> : the same seed replays the same run, which is what makes a failing test debuggable. |
| A number in a range, a shuffle, a float in [0,1).              | <code>proven_rng_below</code> , <code>proven_rng_range</code> , <code>proven_rng_f64</code> , <code>proven_rng_shuffle</code> - over any source. | % n is biased, and everyone writes it anyway. These are not.                                                       |

The two requirements are in direct opposition. Reproducible means predictable, and predictable is exactly what a token must not be: a few outputs of `proven_xoshiro256ss_t` reveal its entire state and therefore every number it will ever produce. That is a feature for a simulation you need to replay and a catastrophe for a session token - so the two carry names that cannot be confused, and the choice is visible at the call site rather than buried in how something was seeded.

**The trait is infallible.** `proven_rng_t` is a source of random bytes, and drawing from a valid one cannot fail. That is not a simplification; it is where the failure went. Asking an operating system for entropy *can* fail, so that failure is confined to exactly one place -- seeding - which you check once, at startup. Every draw downstream is total.

|                                                          |                                                                                                                                      |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_random_bytes(buf, len)</code>               | The OS CSPRNG. Returns <code>false</code> on failure; do not use <code>buf</code> then. <code>len == 0</code> is a successful no-op. |
| <code>proven_random_u64()</code>                         | One strong word from the OS, or <code>0</code> on failure.                                                                           |
| <code>proven_random_u64_checked(&amp;out)</code>         | One strong word, and <code>false</code> when there is none. The form for a token or a key.                                           |
| <code>proven_chacha_rng_seed_from_entropy(&amp;g)</code> | Seed the cryptographic generator from the entropy source. <b>This is the call that can fail.</b>                                     |

|                                                     |                                                                                                                            |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <code>proven_random_set_source(fn, ctx)</code>      | Install the entropy source. The OS is already installed on a hosted target; a bare-metal target installs its board's TRNG. |
| <code>proven_chacha_rng_seed(&amp;g, seed32)</code> | Seed it from 32 bytes you supply - a hardware entropy source on a board. Never the clock.                                  |
| <code>proven_xoshiro256ss_seed(&amp;g, seed)</code> | Seed the reproducible generator. Even seed 0 is fine: it is expanded through SplitMix64.                                   |

### Where entropy comes from

Everything above is pure arithmetic - the generators, the helpers, and `proven_random_bytes` itself.

What differs by platform is the **entropy source** behind it, because that is the one thing a program cannot compute for itself.

- **Hosted:** the OS CSPRNG is installed for you - `getrandom` on Linux, `getentropy` on the BSDs and macOS, `BCryptGenRandom` on Windows, and `/dev/urandom` where none of those exist. You call nothing.
- **Bare metal:** there is no source until you install one. A board *has* real entropy - an on-chip TRNG, a ring oscillator, an ADC's noise floor - and the library cannot know where.

```
/* On a board: hand the library its hardware entropy, once, at startup.
 * (A listing, not a fragment: it defines a function, and `hardware_rng_read` is
 * whatever your SoC calls its entropy register.) */
static bool board_trng(void *ctx, void *buf, proven_size_t len) {
    (void)ctx;
    /* read the SoC's entropy register into buf; return false if it is not ready */
    return hardware_rng_read(buf, len);
}

proven_random_set_source(board_trng, NULL);

/* From here everything above works unchanged - including the one call that turns a few
 * hundred bytes of hardware entropy into an endless cryptographic stream. */
proven_chacha_rng_t g;
if (!proven_chacha_rng_seed_from_entropy(&g)) {
    /* the TRNG was not ready. The generator is INERT - it yields zeros and an invalid
     * trait - so ignoring this does not get you plausible-looking bytes. */
}
```

With no source installed, `proven_random_bytes` returns **false**. It does not fall back to a clock-seeded PRNG, because that looks like success and is a security hole nothing reports - a refusal is a fact a caller can act on.

There is deliberately **no built-in RDRAND / RNDR backend**. On a hosted target the OS already mixes the CPU's instruction into its own pool, so calling it directly buys nothing and costs you that mixing; and a raw hardware instruction used as the *sole* source is exactly the arrangement people have argued about for a decade. If you want it, it is four lines behind this hook - and then the choice is visibly yours.

### The structures you hold

All three are [caller-owned state](#): they allocate nothing, there is nothing to destroy, and **copying one clones its sequence**.

```
typedef struct { const proven_rng_vtable_t *vt; void *ctx; } proven_rng_t;
/* The trait: two pointers, held by value. `ctx` points at one of the generators
   below, which must outlive it. Drawing from a VALID one cannot fail - that is
   the whole design: the failure lives in seeding, not in drawing. */

typedef struct { proven_u64 s[4]; } proven_xoshiro256ss_t;
/* 256 bits of state. Reproducible, and NOT secret-grade. */
```

```

typedef struct {
    proven_u32    state[16]; /* the ChaCha state: constants, key, counter, nonce */
    proven_byte_t block[64]; /* the keystream block currently being handed out */
    proven_size_t used;      /* how much of it is spent */
    proven_u32    seeded;    /* set only by seeding. A zero-initialised struct is
                             the shape of "never seeded", and must stay inert. */
} proven_chacha_rng_t;

```

## Reference

| API                                                     | Intent                                                                                              | Return                                                                                                              |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| proven_random_bytes(buf, len)                           | Fill from the entropy source (the OS by default). <b>The one call that can fail.</b>                | bool. On false, buf is unspecified and must not be used. len == 0 succeeds.                                         |
| proven_random_u64()                                     | One strong word from the same source.                                                               | proven_u64, or 0 on failure - which is also a valid draw, so use proven_random_bytes when you must tell them apart. |
| proven_random_u64_checked(&out)                         | One strong word, for a secret.                                                                      | bool. On false, out is 0 and must not be used: a token of 0 is the predictable secret this exists to prevent.       |
| proven_random_set_source(fn, ctx)                       | Install the entropy source. Not needed on a hosted target; this is how a board hands over its TRNG. | void.                                                                                                               |
| proven_xoshiro256ss_seed(&g, seed)                      | Seed the reproducible generator. Any seed is fine - even 0; it is expanded through SplitMix64.      | void.                                                                                                               |
| proven_xoshiro256ss_next(&g)                            | The next word. The hot path: call it directly, not through the trait.                               | proven_u64.                                                                                                         |
| proven_xoshiro256ss_rng(&g)                             | View it as a proven_rng_t, for the helpers.                                                         | proven_rng_t.                                                                                                       |
| proven_chacha_rng_seed(&g, seed32)                      | Seed the cryptographic generator from 32 bytes of <b>real entropy</b> you supply.                   | void.                                                                                                               |
| proven_chacha_rng_seed_from_entropy(&g)                 | Seed it from the installed source. <b>Check this.</b>                                               | bool. On false the generator is left INERT - it yields zeros and an invalid trait.                                  |
| proven_chacha_rng_next/_fill                            | Draw. Cannot fail once seeded.                                                                      | proven_u64 / void.                                                                                                  |
| proven_chacha_rng(&g)                                   | View it as a proven_rng_t.                                                                          | proven_rng_t - <b>invalid</b> if the generator was never successfully seeded.                                       |
| proven_rng_u64(rng) /<br>proven_rng_fill(rng, buf, len) | Draw through the trait, from whichever generator.                                                   | proven_u64 / void. 0 / no-op for an invalid source.                                                                 |
| proven_rng_below(rng, bound)                            | Uniform in [0, bound), <b>unbiased.</b>                                                             | proven_u64; 0 when bound == 0.                                                                                      |

|                                                 |                                                                                          |                            |
|-------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|
| proven_rng_range(rng, lo, hi)                   | Uniform in [lo, hi], inclusive.<br>The full INT64_MIN..INT64_MAX span does not overflow. | proven_i64; lo if hi < lo. |
| proven_rng_f64(rng)                             | Uniform in [0, 1). 53 bits;<br>never returns 1.0.                                        | double.                    |
| proven_rng_shuffle(rng, base, count, elem_size) | An unbiased Fisher-Yates permutation, in place.                                          | void.                      |

```

/* A session token must not be guessable, so it may not be the 0 that proven_random_u64
 * returns when the entropy source fails. The checked form says so instead. */
proven_u64 token;
if (!proven_random_u64_checked(&token)) {
    proven_eprintln("no entropy source: refusing to issue a session token");
} else {
    proven_println("token {:#x}", PROVEN_ARG(token));
}

```

### Cautions, and what goes wrong

**Never generate a secret with proven\_xoshiro256ss\_t.** It is fast *because* it is predictable: a handful of its outputs reveal its entire 256-bit state, and from the state every number it will ever produce. The two generators carry names that cannot be confused for exactly this reason.

Wrong - a session token an attacker can compute after watching a few:

```

proven_xoshiro256ss_t g;
proven_xoshiro256ss_seed(&g, 12345);
proven_u64 session_token = proven_xoshiro256ss_next(&g); /* wrong: predictable */

```

### Never seed the cryptographic generator from the clock, a counter, or a serial number.

ChaCha20 is exactly as unguessable as its seed. A clock-derived seed produces a stream that looks perfectly random and is not - which is worse than an obvious failure, because nothing reports it.

```

proven_byte_t seed[32] = { 0 };
memcpy(seed, &now_ns, sizeof now_ns); /* wrong: ~20 bits of real entropy, and guessable */
proven_chacha_rng_seed(&g, seed);

```

**Check the seeding.** It is the only thing here that can fail, which is precisely why ignoring it is tempting. If you do, the generator is inert and hands you zeros - a visibly dead value, by design, rather than a plausible one.

Wrong:

```

proven_chacha_rng_t g;
proven_chacha_rng_seed_from_entropy(&g); /* wrong: the bool was the point */
proven_chacha_rng_fill(&g, key, 32); /* key is now 32 zero bytes */

```

Correct:

```

proven_chacha_rng_t g;
if (!proven_chacha_rng_seed_from_entropy(&g)) {
    /* No entropy. There is nothing safe to do here except refuse to continue. */
} else {
    proven_byte_t key[32];
    proven_chacha_rng_fill(&g, key, sizeof key); /* cannot fail: it is seeded */
}

```

**% n is biased, and everyone writes it anyway.** Unless  $n$  divides  $2^{64}$  the low values come up more often - invisible in a spot check, real in a shuffle or a sample.

```

proven_u64 die = proven_rng_u64(rng) % 6 + 1; /* wrong: 1 and 2 are slightly likelier */

```

Correct: `proven_rng_below(rng, 6) + 1.`

**Do not copy a seeded generator** unless you mean to clone its stream. Two "independent" generators copied from one produce identical output - which is a feature for replaying a simulation and a catastrophe for issuing two tokens.

Compiled and run by the test suite:

```
/*
 * Randomness, by use case. There is no single "random": there are two jobs that look
 * identical and are not, and picking the wrong one is the whole danger.
 *
 * A key, a token, a nonce - anything an attacker must not guess - needs a CRYPTOGRAPHIC
 * source. A simulation, a test, a game needs a REPRODUCIBLE one, because a failing run you
 * cannot replay is a failing run you cannot debug. The two requirements are in direct
 * opposition: reproducible means predictable, and predictable is exactly what a token must
 * not be. So the library gives them different names, and the choice is visible here at the
 * call site rather than buried in how something was seeded.
 */

int main(void) {
    /* ---- Job 1: a secret. The OS CSPRNG - and the one place randomness can fail. ---- */
    proven_byte_t key[32];
    EXAMPLE_REQUIRE(proven_random_bytes(key, sizeof key),
        "the OS must give us strong bytes on a hosted platform");

    /* ---- Job 2: lots of cryptographic bytes, or any at all on a board with no OS.
     * ChaCha20 is pure arithmetic: seed it once from real entropy and it needs nothing from
     * the operating system afterwards - no syscall per draw, and it works on bare metal.
     * Seeding is the ONLY step that can fail, so it is the only one you have to check. ---- */
    proven_chacha_rng_t crypto;
    EXAMPLE_REQUIRE(proven_chacha_rng_seed_from_entropy(&crypto), "seed the CSPRNG from the OS, once");

    proven_byte_t token[16];
    proven_chacha_rng_fill(&crypto, token, sizeof token); /* cannot fail: it is seeded */

    /* ---- Job 3: a REPRODUCIBLE run. xoshiro256** is fast and replays exactly from its seed,
     * which is what makes a failing simulation debuggable. It is NOT secret-grade: a few of
     * its outputs reveal its whole state. Never hand it a token to generate. ---- */
    proven_xoshiro256ss_t sim;
    proven_xoshiro256ss_seed(&sim, 12345);

    proven_xoshiro256ss_t replay;
    proven_xoshiro256ss_seed(&replay, 12345);
    EXAMPLE_REQUIRE(proven_xoshiro256ss_next(&sim) == proven_xoshiro256ss_next(&replay),
        "the same seed replays the same run - that is the whole point");

    /* ---- The helpers work over ANY source, through the proven_rng_t trait. ---- */
    proven_rng_t rng = proven_xoshiro256ss_rng(&sim);

    /* A number in a range. `rng_u64() % 6` is what everyone writes, and it is BIASED unless
     * the bound divides 2^64 - the low values come up more often. This one is not. */
    for (int i = 0; i < 100; ++i) {
        proven_u64 die = proven_rng_below(rng, 6) + 1;
        EXAMPLE_REQUIRE(die >= 1 && die <= 6, "a die roll is 1..6, uniformly");
    }

    proven_i64 temperature = proven_rng_range(rng, -40, 85);
    EXAMPLE_REQUIRE(temperature >= -40 && temperature <= 85, "an inclusive range, both ends");

    double p = proven_rng_f64(rng);
    EXAMPLE_REQUIRE(p >= 0.0 && p < 1.0, "a double in [0, 1) - never 1.0");
}
```

```

/* An unbiased shuffle: Fisher-Yates over the unbiased index above. The `% n` version of
 * this loop measurably favours some orderings. */
int deck[10];
for (int i = 0; i < 10; ++i) deck[i] = i;
proven_rng_shuffle(rng, deck, 10, sizeof deck[0]);

int sum = 0;
for (int i = 0; i < 10; ++i) sum += deck[i];
EXAMPLE_REQUIRE(sum == 45, "a shuffle is a permutation: every card is still there, once");

/* The cryptographic generator satisfies the same trait, so the same helpers work over it
 * when the choice must be unguessable rather than merely uniform. */
proven_rng_t secure = proven_chacha_rng(&crypto);
proven_u64 unguessable_index = proven_rng_below(secure, 1000);
EXAMPLE_REQUIRE(unguessable_index < 1000, "the helpers do not care which source they draw from");

(void)token;
(void)key;
return EXAMPLE_OK();
}

```

### Worked example: replacing a file so a power cut cannot corrupt it

Writing over a file in place has a failure mode that testing never shows and production eventually does: the machine loses power halfway through, and the file that remains is neither the old one nor the new one. The fix is a four-step recipe, and every step of it is load-bearing:

1. Write the new contents to a **temporary file** beside the real one.
2. `proven_fs_sync()` - the new file's bytes are now on the storage device, not merely in the operating system's cache.
3. `proven_fs_rename()` - the name flips to the new file in one indivisible step. A reader sees the whole old file or the whole new one, never a mixture.
4. `proven_fs_sync_dir()` - the rename itself is now on the device.

Skip step 2 and the rename can publish a file whose contents never arrived. Skip step 4 and the contents are safe under a name that is not.

The same example covers the record-level calls that go with it:

| Call                                                         | What it is for                                                                                                                              |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_fs_pread</code> / <code>proven_fs_pwrite</code> | Read or write at an absolute offset <b>without moving the file position</b> - which is what makes one handle safe to share between threads. |
| <code>proven_fs_seek</code> / <code>proven_fs_tell</code>    | Move the position, and ask where it is. Seeking from the end with a negative offset finds the last record without knowing the length.       |
| <code>proven_fs_truncate</code>                              | Set the length directly. One call, and the filesystem adjusts a number; the alternative is copying the part you keep.                       |
| <code>proven_fs_lock</code>                                  | An <b>advisory</b> lock: it excludes other processes that also ask for one, and does not affect a program that never asks.                  |
| <code>proven_fs_copy</code>                                  | Duplicate the bytes into a second, independent file.                                                                                        |
| <code>proven_fs_link</code>                                  | A second <b>name</b> for the same file (a hard link). No original: the data lives until the last name goes. Same filesystem only.           |

|                                       |                                                                                                                                                                                            |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proven_fs_symlink                     | A small file holding a path (a symbolic link). May cross filesystems, and may point at nothing.                                                                                            |
| proven_fs_is_absolute                 | Does this path start from the root? The rule differs per platform, which is why it is a call.                                                                                              |
| proven_fs_mkdir / proven_fs_mkdir_all | Create one directory, or a directory with every missing level above it. The first says PROVEN_ERR_EXISTS for a name that is taken; the second accepts a level that is already a directory. |
| proven_fs_rmdir                       | Remove an <b>empty</b> directory. A non-empty one is refused with PROVEN_ERR_INVALID_STATE, so a recursive delete stays an explicit decision.                                              |

```

/*
 * Updating a file so that a power cut cannot leave it half-written.
 *
 * The recipe is old and every part of it is load-bearing:
 *
 * 1. write the new contents to a TEMPORARY file beside the real one,
 * 2. proven_fs_sync - the new file's bytes are now on the device,
 * 3. proven_fs_rename - the name flips to the new file in one step; a reader
 *    sees either the whole old file or the whole new one,
 * 4. proven_fs_sync_dir - the rename itself is now on the device.
 *
 * Skip step 2 and the rename can publish a file whose contents never arrived.
 * Skip step 4 and the contents are safe under a name that may not be. Neither
 * failure shows up in testing; both show up in production, once.
 *
 * The same program also shows the record-level calls - seek/tell, pread/pwrite,
 * truncate - and the advisory lock that stops two copies of the program doing
 * all this at the same time.
 */

typedef struct {
    proven_u32 id;
    proven_u32 score;
} record_t;

static proven_err_t write_records(proven_file_t f, const record_t *recs, proven_size_t n) {
    proven_mem_view_t view = { .ptr = (const proven_byte_t *)recs, .size = n * sizeof recs[0] };
    return proven_fs_write_all(f, view);
}

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    proven_u8str_view_t live = PROVEN_LIT("proven_example_durable.dat");
    proven_u8str_view_t temp = PROVEN_LIT("proven_example_durable.dat.new");
    proven_u8str_view_t here = PROVEN_LIT(".");

    /* is_absolute answers a question worth asking before you join paths or
     * resolve one against a base directory: does this path already start from
     * the root? The rule differs per platform - a leading '/' here, a drive
     * letter or a UNC prefix on Windows - which is exactly why it is a call and
     * not a comparison against '/'. */
    EXAMPLE_REQUIRE(!proven_fs_is_absolute(live), "the working paths in this example are relative");
    EXAMPLE_REQUIRE(proven_fs_is_absolute(PROVEN_LIT("/etc/hosts")), "a leading slash is absolute on POSIX");

    static const record_t initial[] = {

```

```

    { .id = 1, .score = 10 }, { .id = 2, .score = 20 },
    { .id = 3, .score = 30 }, { .id = 4, .score = 40 },
};

/* --- an ordinary first write ----- */

proven_result_file_t f = proven_fs_open(alloc, live, PROVEN_FS_WRITE | PROVEN_FS_CREATE |
PROVEN_FS_TRUNC);
EXAMPLE_REQUIRE(proven_is_ok(f.err), "creating the data file must succeed");
if (!proven_is_ok(f.err)) return 1;

proven_err_t err = write_records(f.value, initial, 4);
EXAMPLE_REQUIRE(proven_is_ok(err), "writing the initial records must succeed");
err = proven_fs_close(f.value);
EXAMPLE_REQUIRE(proven_is_ok(err), "closing after a write must succeed");

/* --- an advisory lock, so two copies do not interleave ----- */

proven_result_file_t rw = proven_fs_open(alloc, live, PROVEN_FS_READ | PROVEN_FS_WRITE);
EXAMPLE_REQUIRE(proven_is_ok(rw.err), "reopening for update must succeed");
if (!proven_is_ok(rw.err)) return 1;

/* An EXCLUSIVE lock keeps every other process that also asks for one out.
 * "Advisory" means exactly that: it stops cooperating programs, and a
 * program that never asks for the lock is not affected. `wait = false`
 * returns immediately rather than blocking - the right choice when you have
 * something else to do, and the only safe choice when the other holder
 * might be waiting on you. */
err = proven_fs_lock(rw.value, PROVEN_FS_LOCK_EXCLUSIVE, false);
EXAMPLE_REQUIRE(proven_is_ok(err), "taking the exclusive lock must succeed when nobody holds it");

/* --- reading and writing one record, by offset ----- */

/* pread reads at an absolute offset and does NOT move the file position.
 * That is what makes it safe to use from two threads sharing one handle:
 * there is no shared cursor for them to race on. */
record_t third = {0};
proven_mem_mut_t into = { .ptr = (proven_byte_t *)&third, .size = sizeof third };
proven_result_size_t got = proven_fs_pread(rw.value, into, 2 * sizeof(record_t));
EXAMPLE_REQUIRE(proven_is_ok(got.err) && got.value == sizeof third, "reading record 2 must succeed");
EXAMPLE_REQUIRE(third.id == 3 && third.score == 30, "and yield the record that was written there");

/* tell reports the position; after a pread it has not moved. */
proven_result_u64_t pos = proven_fs_tell(rw.value);
EXAMPLE_REQUIRE(proven_is_ok(pos.err) && pos.val == 0, "pread must not move the file position");

/* pwrite updates that record in place, again without touching the cursor. */
third.score = 99;
proven_mem_view_t out_view = { .ptr = (const proven_byte_t *)&third, .size = sizeof third };
proven_result_size_t put = proven_fs_pwrite(rw.value, out_view, 2 * sizeof(record_t));
EXAMPLE_REQUIRE(proven_is_ok(put.err) && put.value == sizeof third, "writing record 2 back must succeed");

/* seek is the cursor-moving alternative, and it returns the position it
 * arrived at. Seeking from the END with a negative offset is how you find
 * the last record without knowing the file length first. */
proven_result_u64_t last = proven_fs_seek(rw.value, -(proven_i64)sizeof(record_t), PROVEN_FS_SEEK_END);
EXAMPLE_REQUIRE(proven_is_ok(last.err), "seeking to the last record must succeed");
EXAMPLE_REQUIRE(last.val == 3 * sizeof(record_t), "which is three records in");

pos = proven_fs_tell(rw.value);

```

```

EXAMPLE_REQUIRE(proven_is_ok(pos.err) && pos.val == last.val, "tell agrees with the seek result");

/* truncate sets the length directly. Dropping the last record is one call
 * and O(1); the old way - read everything, write back the part you keep -
 * was an O(n) copy for an operation the filesystem does by adjusting a
 * number. */
err = proven_fs_truncate(rw.value, 3 * sizeof(record_t));
EXAMPLE_REQUIRE(proven_is_ok(err), "truncating to three records must succeed");
proven_result_size_t size = proven_fs_size(rw.value);
EXAMPLE_REQUIRE(proven_is_ok(size.err) && size.value == 3 * sizeof(record_t),
    "the file is now exactly three records long");

/* Release the lock explicitly. Closing the handle would also drop it, but
 * saying so keeps the critical section visible in the code. */
err = proven_fs_lock(rw.value, PROVEN_FS_LOCK_UNLOCK, false);
EXAMPLE_REQUIRE(proven_is_ok(err), "releasing the lock must succeed");
err = proven_fs_close(rw.value);
EXAMPLE_REQUIRE(proven_is_ok(err), "closing the update handle must succeed");

/* --- the durable replace ----- */

static const record_t replacement[] = {
    { .id = 1, .score = 11 }, { .id = 2, .score = 22 },
};

/* 1. Write the new contents beside the old file. CREATE_NEW refuses if the
 * temporary name already exists, which is how a leftover from a crashed
 * run is noticed instead of silently reused. */
proven_result_file_t tmp = proven_fs_open(alloc, tmp,
    PROVEN_FS_WRITE | PROVEN_FS_CREATE | PROVEN_FS_TRUNC);
EXAMPLE_REQUIRE(proven_is_ok(tmp.err), "creating the temporary file must succeed");
if (!proven_is_ok(tmp.err)) return 1;

err = write_records(tmp.value, replacement, 2);
EXAMPLE_REQUIRE(proven_is_ok(err), "writing the new contents must succeed");

/* 2. Push those bytes all the way to the storage device. This is expensive
 * and meant to be: you are buying the guarantee that the data exists
 * after a power cut, and the price is a real trip to the device. */
err = proven_fs_sync(tmp.value);
EXAMPLE_REQUIRE(proven_is_ok(err), "syncing the new file's data must succeed");

err = proven_fs_close(tmp.value);
EXAMPLE_REQUIRE(proven_is_ok(err), "closing the temporary file must succeed");

/* 3. Flip the name. A rename within one directory is atomic: any reader
 * sees the old file or the new one, never a partial write. */
err = proven_fs_rename(alloc, tmp, live);
EXAMPLE_REQUIRE(proven_is_ok(err), "renaming the temporary file over the live one must succeed");

/* 4. Make the rename itself durable. Until the directory reaches the device,
 * the new contents are safe under a name that might not be. */
err = proven_fs_sync_dir(alloc, here);
/* Windows has no directory handle to flush, and says so: UNSUPPORTED, not a
 * false OK. There the rename's durability is the file system's to decide. */
EXAMPLE_REQUIRE(proven_is_ok(err) || err == PROVEN_ERR_UNSUPPORTED,
    "syncing the directory must succeed, or say it cannot");

proven_result_file_t check = proven_fs_open(alloc, live, PROVEN_FS_READ);
EXAMPLE_REQUIRE(proven_is_ok(check.err), "the live path must now open");

```

```

size = proven_fs_size(check.value);
EXAMPLE_REQUIRE(proven_is_ok(size.err) && size.value == 2 * sizeof(record_t),
    "and hold exactly the replacement records");
err = proven_fs_close(check.value);
EXAMPLE_REQUIRE(proven_is_ok(err), "closing the verification handle must succeed");

/* --- copies, and the two kinds of link ----- */

/* copy duplicates the bytes: two independent files from here on. The
 * allocator is for the temporary buffer the copy moves data through. */
proven_u8str_view_t backup = PROVEN_LIT("proven_example_durable.bak");
err = proven_fs_copy(alloc, live, backup);
EXAMPLE_REQUIRE(proven_is_ok(err), "copying the file must succeed");

/* A HARD link is a second name for the same file. There is no original: the
 * data lives until the last name is removed. Both names must be on the same
 * filesystem, because a name and its data cannot span two. */
proven_u8str_view_t hard = PROVEN_LIT("proven_example_durable.hard");
err = proven_fs_link(alloc, live, hard);
EXAMPLE_REQUIRE(proven_is_ok(err), "creating a hard link must succeed");

proven_fs_stat_t st_live = {0}, st_hard = {0};
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_stat(alloc, live, &st_live)), "stat of the live name");
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_stat(alloc, hard, &st_hard)), "stat of the hard link");
EXAMPLE_REQUIRE(st_live.ino == st_hard.ino && st_live.dev == st_hard.dev,
    "both names refer to the same file, which is what a hard link means");

/* A SYMBOLIC link is a small file holding a path. It may point at something
 * on another filesystem, and it may point at nothing at all - following it
 * then fails, which a hard link can never do. */
proven_u8str_view_t soft = PROVEN_LIT("proven_example_durable.link");
err = proven_fs_symlink(alloc, live, soft);
EXAMPLE_REQUIRE(proven_is_ok(err), "creating a symbolic link must succeed");

proven_fs_stat_t st_soft = {0};
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_stat(alloc, soft, &st_soft)),
    "stat follows the symbolic link to its target");
EXAMPLE_REQUIRE(st_soft.size == st_live.size, "so it reports the target's size");

/* --- directories, and cleaning up ----- */

proven_u8str_view_t dir = PROVEN_LIT("proven_example_durable_dir");
err = proven_fs_mkdir(alloc, dir);
EXAMPLE_REQUIRE(proven_is_ok(err), "creating a directory must succeed");

/* Asking for a directory that is already there is refused, and the refusal says so:
 * PROVEN_ERR_EXISTS, where a missing parent is PROVEN_ERR_NOT_FOUND. A caller can tell
 * "nothing to do" from "something is wrong" without a second look at the disk. */
err = proven_fs_mkdir(alloc, dir);
EXAMPLE_REQUIRE(err == PROVEN_ERR_EXISTS, "a second mkdir of the same name must say it exists");

/* proven_fs_mkdir_all makes every missing level of a path, and a level that is already
 * a directory is not an error - so it can simply be called before writing into a tree.
 * A FILE in the way is still PROVEN_ERR_EXISTS. */
proven_u8str_view_t deep = PROVEN_LIT("proven_example_durable_dir/a/b");
err = proven_fs_mkdir_all(alloc, deep);
EXAMPLE_REQUIRE(proven_is_ok(err), "creating two missing levels at once must succeed");
err = proven_fs_mkdir_all(alloc, deep);
EXAMPLE_REQUIRE(proven_is_ok(err), "and asking again must succeed too");
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_rmdir(alloc, deep)), "removing the inner level must succeed");

```

```

EXAMPLE_REQUIRE(proven_is_ok(proven_fs_rmdir(alloc, PROVEN_LIT("proven_example_durable_dir/a"))),
    "removing the level above it must succeed");

/* rmdir removes an EMPTY directory only. That refusal is a feature: a
 * recursive delete is a decision the caller should have to make explicitly,
 * not something a stray path argument can trigger. The refusal has its own code,
 * PROVEN_ERR_INVALID_STATE: nothing failed, the directory has to be emptied first. */
proven_u8str_view_t inside = PROVEN_LIT("proven_example_durable_dir/file.txt");
proven_result_file_t child = proven_fs_open(alloc, inside, PROVEN_FS_WRITE | PROVEN_FS_CREATE);
EXAMPLE_REQUIRE(proven_is_ok(child.err), "creating a file inside it must succeed");
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_close(child.value)), "closing it must succeed");

err = proven_fs_rmdir(alloc, dir);
EXAMPLE_REQUIRE(err == PROVEN_ERR_INVALID_STATE, "removing a non-empty directory must be refused");

EXAMPLE_REQUIRE(proven_is_ok(proven_fs_remove(alloc, inside)), "removing the file must succeed");
err = proven_fs_rmdir(alloc, dir);
EXAMPLE_REQUIRE(proven_is_ok(err), "and then the empty directory can be removed");

printf("durable replace complete: %zu byte(s) live\n", (size_t)size.value);

(void)proven_fs_remove(alloc, soft);
(void)proven_fs_remove(alloc, hard);
(void)proven_fs_remove(alloc, backup);
(void)proven_fs_remove(alloc, live);
return EXAMPLE_OK();
}

```

Wrong - the in-place rewrite the recipe exists to replace:

```

proven_result_file_t f = proven_fs_open(alloc, live, PROVEN_FS_WRITE | PROVEN_FS_TRUNC);
proven_err_t e = proven_fs_write_all(f.value, new_contents); /* wrong */

```

Between the truncate and the last byte of the write, the file on disk is incomplete. A crash there does not lose the update; it loses the data that was already there.

Wrong - renaming without syncing first:

```

proven_err_t e = proven_fs_close(tmp.value); /* no proven_fs_sync */
e = proven_fs_rename(alloc, temp, live); /* wrong */

```

Closing a file does not put its bytes on the device. The rename can be durable while the contents it points at are not.

Wrong - assuming the lock stops everyone:

```

proven_err_t e = proven_fs_lock(f.value, PROVEN_FS_LOCK_EXCLUSIVE, true);
/* another program writes the file without ever calling proven_fs_lock */

```

An advisory lock is a convention between programs that all use it. It is not permission control.

### Worked example: readers, writers, and the standard streams

A **writer** is "somewhere bytes go" and a **reader** is "somewhere bytes come from". Each is two pointers: a small table of functions, and the state those functions work on. The whole value of the arrangement is that code written against a writer does not know whether the bytes end up in a file, in a string, on the terminal, or in a test buffer - and code written against a reader can be tested against a string in memory instead of a file on disk.

The example puts the pieces together:

- `proven_reader_from_view()` makes a reader over bytes you already have, which is how a parser gets tested without touching the filesystem. `proven_reader_read()` fills up to the buffer size and reports what it got; a short read is normal, and end of input is `PROVEN_ERR_EOF`, not a zero-byte success.

- `proven_sysio_stdout_writer()` and `proven_sysio_stderr_writer()` are the unbuffered standard streams, and `proven_sysio_stdin_reader()` the standard input. `proven_writer_write()` means "all of it, or an error"; `proven_writer_write_partial()` moves what it can and reports the count, for callers doing their own retry or back-pressure.
- `proven_reader_is_valid()` and `proven_writer_is_valid()` catch a handle that was never built, at the boundary rather than at the first read or write.
- `proven_sysio_file_buffered()` wraps an open file in a buffered writer over a buffer **you** supply, so the memory cost of buffering is a number you chose. It must be flushed: nothing here flushes on your behalf at exit.
- `proven_sysio_lines_open()` reads that file back one line at a time, through the same kind of caller-supplied buffer. Size it for the longest line you expect - a longer one is `PROVEN_ERR_OUT_OF_BOUNDS`, never a silently cut line.
- `proven_scan_fmt_from_file()` (which calls `proven_sysio_scan_chunk_impl()`) pulls typed values straight out of a file handle when the input has a known shape rather than being free text.

```

/*
 * A writer is "somewhere bytes go" and a reader is "somewhere bytes come from",
 * each of them two pointers: a small table of functions, and the state those
 * functions work on. That is all. The value of the arrangement is that code
 * written against a writer does not know or care whether the bytes end up in a
 * file, in a string, on the terminal, or in a test buffer - and code written
 * against a reader can be tested against a string in memory instead of a file
 * on disk.
 *
 * This program shows both, and the standard-stream and file plumbing that hangs
 * off them:
 *
 * - a reader over a view (an in-memory string), which is how you test a
 *   parser without touching the filesystem,
 * - the unbuffered stdout and stderr writers, and the difference between
 *   "write all of this" and "write what you can",
 * - a buffered writer over a file, which turns many small writes into few
 *   large ones,
 * - a line reader over that same file,
 * - reading formatted values straight out of a file handle.
 *
 * A note that costs people an afternoon: the state structs below must stay
 * where they are declared. A writer holds a pointer INTO its state struct, so
 * copying the struct leaves the copy inert and the original addressed.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    /* --- 1. a reader over bytes you already have ----- */

    /* The parser below does not know this is a string. It would work the same
     * over a file, a pipe or a socket - which is exactly why the test can use
     * the cheap one. */
    proven_reader_view_t src_state;
    proven_reader_t src = proven_reader_from_view(&src_state, PROVEN_LIT("id=41\nid=42\n"));

    /* is_valid asks whether the reader was actually built - a zero-initialised
     * handle is not a reader, and this is the check that says so at the
     * boundary instead of at the first read. */
    EXAMPLE_REQUIRE(proven_reader_is_valid(src), "a reader made from a view must be usable");
    proven_reader_t never_made = {0};
    EXAMPLE_REQUIRE(!proven_reader_is_valid(never_made), "a zero-initialised reader handle is not");

```

```

/* read fills up to dest.size bytes and reports how many it actually got.
 * A short read is normal - it is not end of file, and treating it as one is
 * the classic way to lose the tail of an input. End of file has its own
 * code, PROVEN_ERR_EOF. */
proven_byte_t chunk[5];
proven_mem_mut_t into = { .ptr = chunk, .size = sizeof chunk };
proven_result_size_t got = proven_reader_read(src, into);
EXAMPLE_REQUIRE(proven_is_ok(got.err), "the first read must succeed");
EXAMPLE_REQUIRE(got.value == 5, "and fill the buffer from the view");

proven_size_t total = got.value;
for (;;) {
    got = proven_reader_read(src, into);
    if (got.err == PROVEN_ERR_EOF) {
        break;
    }
    EXAMPLE_REQUIRE(proven_is_ok(got.err), "reads before end of file must succeed");
    total += got.value;
}
EXAMPLE_REQUIRE(total == 12, "the loop must consume the whole view, however it was chunked");

/* --- 2. the standard streams ----- */

/* The state struct holds the handle the writer points at, so it has to
 * outlive the writer. Declaring the two next to each other is the habit
 * that keeps that true. */
proven_sysio_std_t out_state;
proven_writer_t out = proven_sysio_stdout_writer(&out_state);
EXAMPLE_REQUIRE(proven_writer_is_valid(out), "the stdout writer must be usable");

/* write means "all of it, or an error". It loops internally, because a
 * single system-level write may move fewer bytes than asked. */
proven_err_t err = proven_writer_write(out, proven_mem_view_from_u8(PROVEN_LIT("stream example:
start\n")));
EXAMPLE_REQUIRE(proven_is_ok(err), "writing a whole line to stdout must succeed");

/* write_partial is the honest low-level twin: it moves what it can and
 * reports the count. Use it when you are managing your own retry or
 * back-pressure; use write when you just want the bytes out. */
proven_result_size_t part = proven_writer_write_partial(out, proven_mem_view_from_u8(PROVEN_LIT("partial
write\n")));
EXAMPLE_REQUIRE(proven_is_ok(part.err), "a partial write to a terminal or pipe must succeed");
EXAMPLE_REQUIRE(part.value > 0, "and report how many bytes it moved");

/* stderr is unbuffered on purpose: a diagnostic is out before the next line
 * of code runs, which is what you need when the next line is the one that
 * crashes. */
proven_sysio_std_t err_state;
proven_writer_t diag = proven_sysio_stderr_writer(&err_state);
EXAMPLE_REQUIRE(proven_writer_is_valid(diag), "the stderr writer must be usable");
err = proven_writer_write(diag, proven_mem_view_from_u8(PROVEN_LIT("stream example: diagnostics go
here\n")));
EXAMPLE_REQUIRE(proven_is_ok(err), "writing to stderr must succeed");

/* A reader over stdin is built the same way. This example does not read
 * from it - a test run has no one typing - but building it shows the shape,
 * and it is the handle a filter program would loop over. */
proven_sysio_std_t in_state;
proven_reader_t stdin_reader = proven_sysio_stdin_reader(&in_state);

```

```

EXAMPLE_REQUIRE(proven_reader_is_valid(stdin_reader), "the stdin reader must be usable");

/* --- 3. buffered output to a file ----- */

proven_u8str_view_t path = PROVEN_LIT("proven_example_streams.txt");
proven_result_file_t f = proven_fs_open(alloc, path, PROVEN_FS_WRITE | PROVEN_FS_CREATE |
PROVEN_FS_TRUNC);
EXAMPLE_REQUIRE(proven_is_ok(f.err), "creating the output file must succeed");
if (!proven_is_ok(f.err)) return 1;

/* The buffer is yours: the library does not allocate one behind your back,
 * so the memory cost of buffering is a number you chose and can see. Sixty
 * lines through a 256-byte buffer is a handful of writes instead of sixty. */
proven_byte_t outbuf[256];
proven_sysio_out_t file_out;
proven_writer_t w = proven_sysio_file_buffered(&file_out, f.value,
                                              (proven_mem_mut_t){ .ptr = outbuf, .size = sizeof
outbuf });
EXAMPLE_REQUIRE(proven_writer_is_valid(w), "the buffered file writer must be usable");

for (int i = 0; i < 20; ++i) {
    proven_fmt_result_t line_out = proven_fprintln(w, "reading {} = {}",
                                                  proven_arg_i32(i), proven_arg_i32(i * i));
    EXAMPLE_REQUIRE(proven_is_ok(line_out.err), "writing a formatted line must succeed");
}

/* Buffered output that is never flushed is output that never happened.
 * Nothing here flushes at exit on your behalf. */
err = proven_writer_flush(w);
EXAMPLE_REQUIRE(proven_is_ok(err), "the flush is what actually writes the file");
err = proven_fs_close(f.value);
EXAMPLE_REQUIRE(proven_is_ok(err), "closing the file must succeed");

/* --- 4. reading it back a line at a time ----- */

proven_result_file_t rf = proven_fs_open(alloc, path, PROVEN_FS_READ);
EXAMPLE_REQUIRE(proven_is_ok(rf.err), "reopening for reading must succeed");
if (!proven_is_ok(rf.err)) return 1;

/* Size the buffer for the longest LINE you expect. A longer line is
 * reported as OUT_OF_BOUNDS - never silently cut in half, which is the
 * failure that turns one bad record into two plausible ones. */
proven_byte_t linebuf[128];
proven_sysio_lines_t lines;
err = proven_sysio_lines_open(&lines, rf.value, (proven_mem_mut_t){ .ptr = linebuf, .size = sizeof
linebuf });
EXAMPLE_REQUIRE(proven_is_ok(err), "opening a line reader over the file must succeed");

proven_size_t count = 0;
for (;;) {
    proven_result_u8str_view_t line = proven_sysio_read_line(&lines);
    if (line.err == PROVEN_ERR_EOF) {
        break;
    }
    EXAMPLE_REQUIRE(proven_is_ok(line.err), "reading a line must succeed until end of file");
    /* The view points into linebuf and is good only until the next call.
     * That is what makes a million lines cost one buffer instead of a
     * million allocations - and why you copy a line you want to keep. */
    if (count == 0) {
        EXAMPLE_REQUIRE(proven_u8str_view_eq(line.val, PROVEN_LIT("reading 0 = 0")),

```

```

        "the first line reads back exactly as it was written");
    }
    ++count;
}
EXAMPLE_REQUIRE(count == 20, "every line written must be read back");
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_close(rf.value)), "closing the read handle must succeed");

/* --- 5. reading values, not lines, from a file ----- */

/* When the input is a known shape rather than free text, scanning it
 * directly saves writing the split-and-convert loop by hand. This reads one
 * chunk from the file handle and pulls the two numbers out of it. */
proven_result_file_t sf = proven_fs_open(alloc, path, PROVEN_FS_READ);
EXAMPLE_REQUIRE(proven_is_ok(sf.err), "opening the file for scanning must succeed");

proven_i32 index = -1, square = -1;
err = proven_scan_fmt_from_file(sf.value, "reading {} = {}",
                                proven_scan_arg_i32(&index), proven_scan_arg_i32(&square));
EXAMPLE_REQUIRE(proven_is_ok(err), "scanning the first record must succeed");
EXAMPLE_REQUIRE(index == 0 && square == 0, "and produce the values that were written");
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_close(sf.value)), "closing the scan handle must succeed");

printf("streams: %zu byte(s) read from the view, %zu line(s) round-tripped\n",
       (size_t)total, (size_t)count);

(void)proven_fs_remove(alloc, path);
return EXAMPLE_OK();
}

```

Wrong - copying a state struct after making a writer from it:

```

proven_sysio_out_t state;
proven_writer_t w = proven_sysio_file_buffered(&state, file, buf);
proven_sysio_out_t moved = state;    /* wrong: w still points into `state` */

```

The writer holds a pointer **into** the struct. Leave the state where you declared it, for as long as the writer lives.

Wrong - forgetting the flush:

```

proven_fmt_result_t r = proven_fprintln(w, "done");
return 0;                /* wrong: the buffer never reached the file */

```

### Worked example: mapping a file into memory, and making the change durable

A memory-mapped file is a file the processor reads as memory: the operating system arranges for its pages to appear at an address, and a read happens without a read call. It suits random access into a large file, especially one several processes share; it does not suit streaming, where a buffered reader is simpler and does not tie the file's size to your address space.

The distinction that decides whether your writes survive:

| Mapping             | Writes go                                             | proven_mmap_sync()                                                     |
|---------------------|-------------------------------------------------------|------------------------------------------------------------------------|
| PROVEN_MMAP_SHARED  | to the file                                           | pushes them to the storage device                                      |
| PROVEN_MMAP_PRIVATE | to a private copy (copy-on-write) - this process only | returns PROVEN_ERR_UNSUPPORTED, because there is nothing to write back |

That refusal is deliberate. A caller who believed the mapping was shared finds out at the sync, rather than when the data turns out to be missing.

The example ends with the calendar formatter, because the record it writes carries a date:

`proven_time_u8_fmt()` for the UTF-8 form, and `proven_time_u16_fmt()` for the UTF-16 form - the latter being right in exactly one situation, handing text to a system call that takes wide strings.

```
#include <string.h>

/*
 * A memory-mapped file is a file the processor reads as memory: the operating
 * system arranges for the pages to appear at an address, and reads happen
 * without a read call. It is the right tool for one shape of problem - random
 * access into a large file that several processes look at - and the wrong tool
 * for streaming, where a buffered reader is simpler and does not tie a file's
 * size to your address space.
 *
 * The part that is easy to get wrong is durability, and it is the reason this
 * example exists:
 *
 * PROVEN_MMAP_SHARED - writes go to the file. proven_mmap_sync pushes them
 *                      to the storage device.
 * PROVEN_MMAP_PRIVATE - writes are copy-on-write: they exist in this process
 *                      and nowhere else. There is nothing to write back, and
 *                      asking to sync one says so instead of quietly doing
 *                      nothing.
 *
 * This example also shows the calendar formatter alongside it, because the
 * record it writes carries a timestamp - and a timestamp written for a Windows
 * API is the one place the UTF-16 formatter is the right call.
 */

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();
    proven_u8str_view_t path = PROVEN_LIT("proven_example_mmap.dat");

    /* A mapping cannot extend a file, so the file has to be the size you intend
     * to map before you map it. */
    static const char initial[] = "record 0: pending  \n";
    proven_result_file_t create = proven_fs_open(alloc, path,
                                                PROVEN_FS_WRITE | PROVEN_FS_CREATE | PROVEN_FS_TRUNC);
    EXAMPLE_REQUIRE(proven_is_ok(create.err), "creating the backing file must succeed");
    if (!proven_is_ok(create.err)) return 1;

    proven_mem_view_t seed = { .ptr = (const proven_byte_t *)initial, .size = sizeof initial - 1 };
    EXAMPLE_REQUIRE(proven_is_ok(proven_fs_write_all(create.value, seed)), "writing the record must succeed");
    EXAMPLE_REQUIRE(proven_is_ok(proven_fs_close(create.value)), "closing it must succeed");

    /* --- a shared mapping: writes reach the file ----- */

    proven_result_file_t f = proven_fs_open(alloc, path, PROVEN_FS_READ | PROVEN_FS_WRITE);
    EXAMPLE_REQUIRE(proven_is_ok(f.err), "opening the file for mapping must succeed");
    if (!proven_is_ok(f.err)) return 1;

    /* size 0 means "to the end of the file". */
    proven_result_mmap_t m = proven_mmap_create(f.value, 0, 0,
                                                PROVEN_MMAP_READ | PROVEN_MMAP_WRITE,
                                                PROVEN_MMAP_SHARED);
    EXAMPLE_REQUIRE(proven_is_ok(m.err), "mapping the file must succeed");
    if (!proven_is_ok(m.err)) {
        (void)proven_fs_close(f.value);
        return 1;
    }
}
```

```

proven_mmap_t map = m.value;

/* Reading is just reading memory - no call, no copy. as_view hands back the
 * whole mapping as a byte view, so the ordinary view helpers apply. */
proven_u8str_view_t contents = proven_mmap_as_view(map);
EXAMPLE_REQUIRE(contents.size == sizeof initial - 1, "the mapping covers the whole file");
EXAMPLE_REQUIRE(proven_u8str_view_starts_with(contents, PROVEN_LIT("record 0:")),
    "and shows the bytes that were written");

/* Writing is writing memory. The status field is a fixed width on purpose:
 * a mapping cannot make the file longer, so an in-place edit has to fit the
 * space that is already there. */
proven_size_t at = proven_u8str_view_find(contents, 0, PROVEN_LIT("pending"));
EXAMPLE_REQUIRE(at != PROVEN_SIZE_MAX, "the status field must be found");
memcpy((proven_byte_t *)map.ptr + at, "done ", 7);

/* sync is the durability step: without it the change is in the page cache,
 * where it survives this program exiting but not the machine losing power. */
EXAMPLE_REQUIRE(proven_is_ok(proven_mmap_sync(&map)), "syncing a shared mapping must succeed");
EXAMPLE_REQUIRE(proven_is_ok(proven_mmap_destroy(&map)), "unmapping must succeed");
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_close(f.value)), "closing the mapped file must succeed");

/* The edit is in the file, not merely in this process's memory. */
proven_result_u8str_t back = proven_fs_read_all_u8str(alloc, path);
EXAMPLE_REQUIRE(proven_is_ok(back.err), "reading the file back must succeed");
EXAMPLE_REQUIRE(proven_u8str_view_find(proven_u8str_as_view(&back.value), 0, PROVEN_LIT("done")) !=
PROVEN_SIZE_MAX,
    "the mapped write reached the file");
proven_u8str_destroy(alloc, &back.value);

/* --- a private mapping: writes go nowhere ----- */

proven_result_file_t pf = proven_fs_open(alloc, path, PROVEN_FS_READ);
EXAMPLE_REQUIRE(proven_is_ok(pf.err), "reopening for a private mapping must succeed");

proven_result_mmap_t pm = proven_mmap_create(pf.value, 0, 0, PROVEN_MMAP_READ, PROVEN_MMAP_PRIVATE);
EXAMPLE_REQUIRE(proven_is_ok(pm.err), "a private read mapping must succeed");
proven_mmap_t priv = pm.value;

/* Asking to sync a private mapping is refused rather than accepted and
 * ignored. The refusal is the useful behaviour: a caller who believed the
 * mapping was shared finds out here, not when the data is missing. */
EXAMPLE_REQUIRE(proven_mmap_sync(&priv) == PROVEN_ERR_UNSUPPORTED,
    "a private mapping has nothing to write back, and says so");

EXAMPLE_REQUIRE(proven_is_ok(proven_mmap_destroy(&priv)), "unmapping the private mapping must succeed");
EXAMPLE_REQUIRE(proven_is_ok(proven_fs_close(pf.value)), "closing it must succeed");

/* --- the timestamp, in both string types ----- */

proven_datetime_t now = proven_time_now_datetime();

proven_result_u8str_t stamp = proven_u8str_create(alloc, 64);
EXAMPLE_REQUIRE(proven_is_ok(stamp.err), "creating the timestamp string must succeed");
EXAMPLE_REQUIRE(proven_is_ok(proven_time_u8_fmt(alloc, &stamp.value, now, &proven_time_locale_en,
    "{year}-{month:0>2}-{day:0>2}")),
    "formatting the date as UTF-8 must succeed");
EXAMPLE_REQUIRE(proven_u8str_as_view(&stamp.value).size == 10, "a YYYY-MM-DD date is ten characters");

/* The UTF-16 form of the same call, for the one place it is the right one:

```

```

    * handing the text straight to a system call that takes wide strings. */
    proven_result_u16str_t wide = proven_u16str_create(alloc, 64);
    EXAMPLE_REQUIRE(proven_is_ok(wide.err), "creating the wide timestamp string must succeed");
    EXAMPLE_REQUIRE(proven_is_ok(proven_time_u16_fmt(alloc, &wide.value, now, &proven_time_locale_en,
        "{year}-{month:0>2}-{day:0>2}")),
        "formatting the same date as UTF-16 must succeed");
    EXAMPLE_REQUIRE(proven_u16str_len(&wide.value) == 10, "ten code units, one per character here");
    EXAMPLE_REQUIRE(proven_u16str_as_ptr(&wide.value)[4] == (proven_u16)'-',
        "and the same layout as the UTF-8 form");

    printf("mapped record updated; stamped %s\n", proven_u8str_as_cstr(&stamp.value));

    proven_u16str_destroy(alloc, &wide.value);
    proven_u8str_destroy(alloc, &stamp.value);
    (void)proven_fs_remove(alloc, path);
    return EXAMPLE_OK();
}

```

Wrong - expecting a mapping to extend the file:

```

proven_result_mmap_t m = proven_mmap_create(f.value, 0, 1 << 20, ...); /* file is 22 bytes */
memcpy((char *)m.value.ptr + 4096, data, n);                          /* wrong */

```

Set the file's length first - proven\_fs\_truncate() does it in one call - and map what exists.

### Worked example: where randomness comes from

The example above picks the right generator. This one is the layer underneath: the source of the bytes, and how to write code that does not care which source it got.

- **proven\_rng\_t is the interface.** A function that takes one works with the operating system's generator, with ChaCha20, with xoshiro, and with a fake you wrote for a test, unchanged. proven\_rng\_is\_valid() says whether a source was actually built; drawing from one that was not returns 0 rather than inventing a number, so check once at the boundary instead of trusting every draw. proven\_rng\_u64() draws one 64-bit word and proven\_rng\_fill() fills a whole buffer in one call.
- **A fixed seed makes a test reproducible.** proven\_chacha\_rng\_seed() takes the seed bytes directly, so proven\_chacha\_rng\_next() walks a known sequence -- which is what turns "fails once a week" into a failure you can replay. In production the seed must come from real entropy, because anyone who learns it knows every byte that follows.
- **proven\_random\_u64()** draws a single strong word straight from the entropy source. Right for a one-off - a hash key at start-up, an identifier - and wrong in a loop, where each call costs a trip to the operating system. It returns 0 when the source fails, so for a token or a key use proven\_random\_u64\_checked(), which says so.
- **proven\_random\_set\_source()** installs the entropy source itself. A hosted program already has the operating system's and should leave it alone; a bare-metal program has none, and this is where its hardware source goes.

```
#include <string.h>
```

```

/*
 * The other example shows which generator to pick. This one is about the layer
 * underneath: where the randomness comes FROM, and how to write code that does
 * not care.
 *
 * proven_rng_t is a source of random bytes as a pair of pointers - a small
 * table of functions and the generator state they work on. Code that takes a
 * proven_rng_t works with the OS generator, with ChaCha20, with xoshiro, and
 * with a fake you wrote for a test, without a line of change.
 */

```

```

* proven_random_set_source is the layer below THAT: where the raw entropy a
* generator is seeded from comes from. A hosted program already has one - the
* operating system's - and should leave it alone. A bare-metal program has
* none, and this is the hook where its hardware source is installed.
*
* The fixed-seed part matters more than it looks: a cryptographic generator
* seeded from a KNOWN seed produces a known sequence, which is what makes a
* test that involves randomness reproducible instead of "fails once a week".
*/

/* A source of "entropy" that is not random at all: it counts. Nothing like this
* belongs in a real program - see the counter-example in the chapter - but it
* is exactly the right shape for showing how the hook works, and for a test
* that must produce the same bytes every run. */
static bool counting_entropy(void *ctx, void *buf, proven_size_t len) {
    proven_u8 *next = (proven_u8 *)ctx;
    proven_u8 *out = (proven_u8 *)buf;
    for (proven_size_t i = 0; i < len; ++i) {
        out[i] = (*next)++;
    }
    return true;
}

/* A function written against the trait. It never learns which generator it got. */
static proven_u64 roll_total(proven_rng_t rng, int rolls) {
    proven_u64 sum = 0;
    for (int i = 0; i < rolls; ++i) {
        sum += proven_rng_below(rng, 6) + 1;
    }
    return sum;
}

int main(void) {
    /* --- 1. a source you can check before you use it ----- */

    proven_rng_t nothing = {0};
    EXAMPLE_REQUIRE(!proven_rng_is_valid(nothing), "a zero-initialised source is not a generator");

    /* Drawing from an invalid source does not crash and does not invent a
    * number: it returns 0. That is a defined, boring answer - but a stream of
    * zeros is not randomness, so check the source once when you receive it
    * rather than trusting every draw. */
    EXAMPLE_REQUIRE(proven_rng_u64(nothing) == 0, "an invalid source yields 0, not a fabricated value");

    /* --- 2. a cryptographic generator from a KNOWN seed ----- */

    /* proven_chacha_rng_seed takes the seed bytes directly, so the sequence is
    * reproducible. That is what you want in a test and never in production:
    * anyone who learns the seed knows every byte the generator will produce. */
    proven_byte_t seed[PROVEN_CHACHA_SEED_SIZE];
    memset(seed, 0xA5, sizeof seed);

    proven_chacha_rng_t a, b;
    proven_chacha_rng_seed(&a, seed);
    proven_chacha_rng_seed(&b, seed);

    /* next returns one 64-bit word at a time. Two generators given the same
    * seed walk the same sequence - which is the property the test relies on. */
    proven_u64 first = proven_chacha_rng_next(&a);
    EXAMPLE_REQUIRE(first == proven_chacha_rng_next(&b), "the same seed replays the same sequence");

```

```

EXAMPLE_REQUIRE(proven_chacha_rng_next(&a) == proven_chacha_rng_next(&b), "and keeps replaying it");

/* --- 3. using it through the trait ----- */

proven_rng_t rng = proven_chacha_rng(&a);
EXAMPLE_REQUIRE(proven_rng_is_valid(rng), "a seeded generator makes a valid source");

proven_u64 word = proven_rng_u64(rng);
(void)word; /* any 64-bit value is a legal answer; there is nothing to assert about it */

/* fill is the bulk form: one call for a whole buffer, rather than a loop
 * over 64-bit words that has to deal with the remainder itself. */
proven_byte_t nonce[12] = {0};
proven_rng_fill(rng, nonce, sizeof nonce);

bool all_zero = true;
for (proven_size_t i = 0; i < sizeof nonce; ++i) {
    if (nonce[i] != 0) all_zero = false;
}
EXAMPLE_REQUIRE(!all_zero, "filling from a seeded generator must produce something");

/* The same function, driven by two different generators. This is the only
 * reason the trait exists. */
proven_chacha_rng_t c;
proven_chacha_rng_seed(&c, seed);
proven_u64 crypto_total = roll_total(proven_chacha_rng(&c), 50);

proven_xoshiro256ss_t fast;
proven_xoshiro256ss_seed(&fast, 7);
proven_u64 fast_total = roll_total(proven_xoshiro256ss_rng(&fast), 50);

EXAMPLE_REQUIRE(crypto_total >= 50 && crypto_total <= 300, "50 dice must total between 50 and 300");
EXAMPLE_REQUIRE(fast_total >= 50 && fast_total <= 300, "whichever generator produced them");

/* --- 4. one strong word, without holding a generator ----- */

/* proven_random_u64 draws straight from the entropy source. Convenient for
 * a one-off - a table's hash key at startup, a request id - and the wrong
 * tool for a loop, because each call costs a trip to the operating system.
 * For bulk output, seed a generator once and draw from that. */
proven_u64 one_off = proven_random_u64();
proven_u64 another = proven_random_u64();
EXAMPLE_REQUIRE(one_off != another || one_off != 0,
    "two draws from the OS source are essentially never the same value");

/* --- 5. installing an entropy source ----- */

/* On a hosted target the operating system's source is already installed and
 * you should leave it there. This hook exists for the bare-metal case,
 * where the library cannot know that the board's entropy lives in a
 * particular hardware register. Here it is installed with a deliberately
 * fake source, purely to show the mechanism and to prove the switch took
 * effect - a real one must be genuine hardware entropy. */
proven_u8 counter = 0;
proven_random_set_source(counting_entropy, &counter);

proven_byte_t drawn[4] = {0};
EXAMPLE_REQUIRE(proven_random_bytes(drawn, sizeof drawn), "the installed source must answer");
EXAMPLE_REQUIRE(drawn[0] == 0 && drawn[1] == 1 && drawn[2] == 2 && drawn[3] == 3,
    "and it is the source we installed that answered");

```

```

/* Put the platform default back. Leaving a test source installed is how a
 * program ends up generating predictable keys in production. */
proven_random_set_source(NULL, NULL);

proven_byte_t real[8] = {0};
EXAMPLE_REQUIRE(proven_random_bytes(real, sizeof real), "the OS source is back and working");

printf("random: first word %llu, dice totals %llu and %llu\n",
       (unsigned long long)first, (unsigned long long)crypto_total, (unsigned long long)fast_total);

return EXAMPLE_OK();
}

```

Wrong - installing something that merely looks random:

```

static bool clock_entropy(void *ctx, void *buf, proven_size_t len) {
    proven_time_t t = proven_time_now(); /* wrong: predictable */
    memcpy(buf, &t, len < sizeof t ? len : sizeof t);
    return true;
}
proven_random_set_source(clock_entropy, NULL);

```

A clock, a serial number, an uninitialised buffer or a pseudo-random generator all produce something that passes a glance and is guessable. If a board has no real entropy, install nothing: a refusal is a fact the caller can act on, and silent predictability is not.

Wrong - leaving a test source installed:

```

proven_random_set_source(counting_entropy, &counter);
/* ... the rest of the program, now generating predictable keys ... */

```

Restore the platform default with `proven_random_set_source(NULL, NULL)` as soon as the test is over.

## Chapter 9: Networking

**Part V - Talking to the operating system. Prerequisites: Part II (1, 2, 3) and Chapter 5 section 4 (the two clocks) and its section on streams. After this chapter** you can open a TCP or UDP socket, put a time limit on everything that waits, tell a refused connection from a lost one, serve many sockets from one thread, and write a protocol that does not care what carries it.

This chapter covers `net.h`. Like Chapter 5 it needs an operating system and is not part of a [freestanding](#) build; a hosted build can leave it out with `PROVEN_NO_NET`.

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#### 1. Three rules

BSD sockets are forty years old and every one of their sharp edges has a name. `net.h` is a thin layer whose whole job is to take three of them away.

**Every call that can wait takes a deadline.** `recv` on a plain socket waits as long as the peer keeps it waiting. A program built that way works on a desk and stops in production the first time a client opens a connection and says nothing - and one such client is enough to stop a server that handles one connection at a time. Here there is no way to write that program: a read, a write, an accept and a connect each take a deadline argument, and you must say something.

**A failure says which failure.** "The call failed" is not enough to decide what to do next. Whether the other side said no, went away, or finished properly are three different situations with three different responses:

| What happened                                                 | Code                                | Sensible response                                            |
|---------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------|
| The peer finished sending                                     | <code>PROVEN_ERR_EOF</code>         | Stop reading; this is the normal end                         |
| The deadline passed                                           | <code>PROVEN_ERR_TIMEOUT</code>     | Retry, give up, or close - nothing is broken                 |
| Nothing listens at that address                               | <code>PROVEN_ERR_REFUSED</code>     | Try another address, or report; a retry gets the same answer |
| The peer vanished, or the connection was aborted              | <code>PROVEN_ERR_RESET</code>       | Close; what was in flight may be lost                        |
| No route to the host or network                               | <code>PROVEN_ERR_UNREACHABLE</code> | Report, or retry later                                       |
| The address to bind is taken, or the system is out of sockets | <code>PROVEN_ERR_BUSY</code>        | Back off, or choose another port                             |
| The address may not be bound                                  | <code>PROVEN_ERR_PERMISSION</code>  | Report; a retry will not help                                |
| A host name has no address; a Unix-domain path is missing     | <code>PROVEN_ERR_NOT_FOUND</code>   | Report                                                       |

|                                                            |                          |                                |
|------------------------------------------------------------|--------------------------|--------------------------------|
| A datagram did not fit the buffer, or is too large to send | PROVEN_ERR_OUT_OF_BOUNDS | Use a larger buffer; send less |
| The socket is not open                                     | PROVEN_ERR_INVALID_STATE | Fix the sequence of calls      |

PROVEN\_ERR\_IO is what is left when none of these fits.

**Nothing is hidden.** No call in this header allocates. A socket is a small value you own and must close. A reader or writer over a socket uses state you declare. The one piece of process-wide state - starting Winsock on Windows - is done once, on first use, and is not yours to manage.

## 2. Addresses

A `proven_net_addr_t` says where a socket is or where it should go. It is a plain value: copy it, compare it, store it in a table.

```
typedef struct {
    proven_net_family_t family; /* IPV4, IPV6 or UNIX; NONE when unset */
    proven_u16 port;           /* host byte order: the number you would write down */
    proven_u32 scope_id;       /* IPv6 zone for a link-local address; 0 otherwise */
    proven_byte_t ip[16];      /* network order: 4 bytes for IPV4, 16 for IPV6 */
    proven_u8 path_len;        /* Unix-domain only */
    char path[104];            /* Unix-domain only */
} proven_net_addr_t;
```

## Reference

| API                                                               | Intent                                                                                                       | Return                                                                                                                        |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_net_addr_ipv4(a, b, c, d, port)</code>               | An IPv4 address from its four numbers.                                                                       | <code>proven_net_addr_t</code> .                                                                                              |
| <code>proven_net_addr_ipv6(ip[16], port, scope_id)</code>         | An IPv6 address from sixteen bytes in network order.                                                         | <code>proven_net_addr_t</code> .                                                                                              |
| <code>proven_net_addr_loopback(family, port)</code>               | This machine only: 127.0.0.1 or ::1.                                                                         | <code>proven_net_addr_t</code> ; family NONE if family has no loopback.                                                       |
| <code>proven_net_addr_any(family, port)</code>                    | Every interface: 0.0.0.0 or ::.                                                                              | <code>proven_net_addr_t</code> .                                                                                              |
| <code>proven_net_addr_unix(path, &amp;out)</code>                 | A Unix-domain address from a filesystem path.                                                                | <code>proven_err_t</code> : INVALID_ARG for an empty path or one with a NUL, OUT_OF_BOUNDS past PROVEN_NET_UNIX_PATH_MAX.     |
| <code>proven_net_addr_parse(text, port, &amp;out)</code>          | Parse a literal: 192.0.2.1, 2001:db8::1, [2001:db8::1], fe80::1%3. Never touches the network.                | <code>proven_err_t</code> : INVALID_FORMAT when text is not an address; out is then untouched.                                |
| <code>proven_net_addr_format(&amp;addr, out, &amp;written)</code> | Write 192.0.2.1:80, [2001:db8::1]:80, or the path. No NUL. PROVEN_NET_ADDR_TEXT_MAX bytes are always enough. | <code>proven_err_t</code> : OUT_OF_BOUNDS when out is too small.                                                              |
| <code>proven_net_addr_eq(&amp;a, &amp;b)</code>                   | Same family, address, port and zone - or the same path.                                                      | bool.                                                                                                                         |
| <code>proven_net_resolve(host, port, out, cap, &amp;count)</code> | Look a name up through the system resolver; a literal is answered without a query.                           | <code>proven_err_t</code> : NOT_FOUND, TIMEOUT (the name servers did not answer), INVALID_ARG for an empty or over-long name. |

Two decisions in the parser are deliberate. An IPv4 literal is exactly four decimal numbers: 127.1 and 2130706433 are refused although older parsers accept both, and so is 010.0.0.1, which those parsers read as octal *eight*. And IPv6 has many spellings going in but one coming out

- lowercase, no leading zeros, the longest run of zero groups as :: (RFC 5952) - so two equal addresses print as equal text and the text can be a map key or a log field you search for.

### Cautions, and what goes wrong

`proven_net_resolve` **blocks and has no deadline**. It is the system resolver, which offers none, and on a broken network it can take tens of seconds. It is the one call in this header that waits without a limit, and the header says so rather than pretend.

Wrong - resolving inside the loop that serves every other connection:

```
for (;;) {
    wait_for_events();
    if (client_wants_upstream)
        proven_net_resolve(name, 443, addrs, 4, &n);    /* wrong: every client waits on DNS */
}
```

Correct - resolve where waiting is acceptable: at startup, or on a job ([Chapter 6](#)), and hand the address to the loop.

**Binding to "any" publishes the service.** `proven_net_addr_any` listens on every interface the machine has, including the ones that face a network. A tool meant for your own machine binds `proven_net_addr_loopback`.

Compiled and run by the test suite:

```
/*
 * Addresses are plain values: build one, parse one from text, print it, compare two.
 * Nothing in this program opens a socket or asks the network anything.
 */

static bool prints_as(const proven_net_addr_t *addr, proven_u8str_view_t want) {
    proven_byte_t text[PROVEN_NET_ADDR_TEXT_MAX];
    proven_size_t n = 0;
    if (proven_net_addr_format(addr, (proven_mem_mut_t){ text, sizeof text }, &n) != PROVEN_OK) return false;
    return proven_u8str_view_eq((proven_u8str_view_t){ text, n }, want);
}

int main(void) {
    /* Built from parts. The port is the number you would write down - host byte order. */
    proven_net_addr_t web = proven_net_addr_ipv4(192, 0, 2, 10, 80);
    EXAMPLE_REQUIRE(prints_as(&web, PROVEN_LIT("192.0.2.10:80")), "an IPv4 address prints as address:port");

    /* The two fixed addresses a program needs most: this machine only, and every interface. */
    proven_net_addr_t local = proven_net_addr_loopback(PROVEN_NET_FAMILY_IPV4, 8080);
    proven_net_addr_t everywhere = proven_net_addr_any(PROVEN_NET_FAMILY_IPV6, 8080);
    EXAMPLE_REQUIRE(prints_as(&local, PROVEN_LIT("127.0.0.1:8080")), "loopback is 127.0.0.1");
    EXAMPLE_REQUIRE(prints_as(&everywhere, PROVEN_LIT(":::8080")), "any is ::, and IPv6 is bracketed before a port");

    /* Parsed from text. Many spellings of one IPv6 address parse to the same bytes, and there
     * is exactly one spelling on the way out, so the text can be compared and used as a key. */
    proven_net_addr_t a, b;
    EXAMPLE_REQUIRE(proven_net_addr_parse(PROVEN_LIT("2001:DB8:0:0:0:0:0:1"), 443, &a) == PROVEN_OK, "the long form parses");
    EXAMPLE_REQUIRE(proven_net_addr_parse(PROVEN_LIT("[2001:db8::1]"), 443, &b) == PROVEN_OK, "the short, bracketed form parses");
    EXAMPLE_REQUIRE(proven_net_addr_eq(&a, &b), "and they are the same address");
}
```

```

EXAMPLE_REQUIRE(prints_as(&a, PROVEN_LIT("[2001:db8::1]:443")), "printed in the one canonical form");

/* What is not an address is refused - including what other parsers quietly accept. */
proven_net_addr_t bad;
EXAMPLE_REQUIRE(proven_net_addr_parse(PROVEN_LIT("010.0.0.1"), 80, &bad) == PROVEN_ERR_INVALID_FORMAT,
    "a leading zero is refused, not read as octal");
EXAMPLE_REQUIRE(proven_net_addr_parse(PROVEN_LIT("example.com"), 80, &bad) == PROVEN_ERR_INVALID_FORMAT,
    "a name is not a literal: that is proven_net_resolve's job");

/* Sixteen raw bytes, in network order, when the address comes from a packet or a file. */
proven_byte_t raw[16] = { 0xfe, 0x80, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1 };
proven_net_addr_t link_local = proven_net_addr_ipv6(raw, 22, 3);
EXAMPLE_REQUIRE(prints_as(&link_local, PROVEN_LIT("[fe80::1%3]:22")), "a link-local address carries its
zone");

/* A Unix-domain address is a filesystem path. */
proven_net_addr_t sock;
EXAMPLE_REQUIRE(proven_net_addr_unix(PROVEN_LIT("/run/app.sock"), &sock) == PROVEN_OK, "a path makes an
address");
EXAMPLE_REQUIRE(prints_as(&sock, PROVEN_LIT("/run/app.sock")), "and prints as the path");

/* Resolving a literal is answered without a query, so this line works with no network.
 * A real name goes to the system resolver, blocks, and has no deadline: call it where
 * that is acceptable. */
proven_net_addr_t found[4];
proven_size_t count = 0;
EXAMPLE_REQUIRE(proven_net_resolve(PROVEN_LIT("192.0.2.10"), 80, found, 4, &count) == PROVEN_OK && count
== 1,
    "a literal resolves to itself");
EXAMPLE_REQUIRE(proven_net_addr_eq(&found[0], &web), "the same address as the one built by hand");

return EXAMPLE_OK();
}

```

### 3. Deadlines

A deadline is a moment, not a duration: a reading of the monotonic clock ([Chapter 5 section 4](#)) at which a call must give up.

|                            | Meaning                                                             |
|----------------------------|---------------------------------------------------------------------|
| proven_net_deadline_in(ms) | The deadline ms milliseconds from now.                              |
| PROVEN_NET_NO_DEADLINE     | Wait as long as it takes. A silent peer then waits with you.        |
| PROVEN_NET_DONT_WAIT       | Do what can be done right now; anything else is PROVEN_ERR_TIMEOUT. |

It is absolute so that one deadline can bound a whole exchange. "This request gets five seconds" is one value passed to connect, to each write and to each read; with a per-call timeout the same five seconds would have to be recomputed after every step, and usually is not.

It is on the monotonic clock so that setting the system time does not move it.

And a deadline that passes breaks nothing. PROVEN\_ERR\_TIMEOUT means "not yet"; the socket is exactly as usable as it was. Whether to try again, give up, or close is yours to decide.

Wrong - a timeout treated as a dead connection:

```

proven_result_size_t r = proven_net_read(&conn, buf, proven_net_deadline_in(100));
if (r.err != PROVEN_OK) proven_net_close(&conn); /* wrong: TIMEOUT is not a failure of the connection */

```

Correct - decide what silence means for this protocol:

```

if (r.err == PROVEN_ERR_TIMEOUT) { send_keepalive(&conn); continue; }
if (r.err == PROVEN_ERR_EOF)      { finish(&conn); break; }
if (r.err != PROVEN_OK)           { proven_net_close(&conn); break; }

```

## 4. TCP

A TCP connection is a stream of bytes in each direction. It has no message boundaries: what one side sends in a single write the other may receive in three reads, or two writes in one read.

### Reference

| API                                                                                                              | Intent                                                                                                              | Return                                                                                                       |
|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>proven_net_listen(at, backlog, &amp;listener, &amp;bound)</code>                                           | Listen at <code>at</code> . Port 0 lets the OS choose; <code>bound</code> (optional) is the address actually bound. | <code>proven_err_t</code> : BUSY (taken), PERMISSION, INVALID_ARG (not this machine's address), UNSUPPORTED. |
| <code>proven_net_accept(&amp;listener, until, &amp;conn, &amp;peer)</code>                                       | Take the next connection. <code>peer</code> is optional.                                                            | <code>proven_err_t</code> : TIMEOUT when none arrived.                                                       |
| <code>proven_net_listener_addr(&amp;listener, &amp;out)</code>                                                   | Where a listener is bound.                                                                                          | <code>proven_err_t</code> .                                                                                  |
| <code>proven_net_listener_close(&amp;listener)</code>                                                            | Stop listening. Accepted connections are unaffected.                                                                | <code>proven_err_t</code> .                                                                                  |
| <code>proven_net_connect(to, until, &amp;conn)</code>                                                            | Connect. On failure there is nothing to close.                                                                      | <code>proven_err_t</code> : REFUSED, TIMEOUT, UNREACHABLE, NOT_FOUND (Unix-domain path).                     |
| <code>proven_net_read(&amp;conn, dest, until)</code>                                                             | Read what has arrived, up to <code>dest.size</code> .                                                               | <code>proven_result_size_t</code> : EOF at the end - never zero bytes of success; TIMEOUT; RESET.            |
| <code>proven_net_write(&amp;conn, src, until)</code>                                                             | Write what fits now. <code>.value</code> is how much went, also on failure.                                         | <code>proven_result_size_t</code> .                                                                          |
| <code>proven_net_write_all(&amp;conn, src, until)</code>                                                         | Write all of <code>src</code> , or fail saying how far it got.                                                      | <code>proven_result_size_t</code> : <code>.value == src.size</code> exactly when <code>.err</code> is OK.    |
| <code>proven_net_shutdown_write(&amp;conn)</code>                                                                | "Nothing more to send." The peer reads EOF; this side can still read.                                               | <code>proven_err_t</code> .                                                                                  |
| <code>proven_net_close(&amp;conn)</code>                                                                         | Close. Safe on a connection that is not open.                                                                       | <code>proven_err_t</code> .                                                                                  |
| <code>proven_net_conn_local_addr(&amp;conn, &amp;out),<br/>proven_net_conn_peer_addr(&amp;conn, &amp;out)</code> | The two ends of a connection.                                                                                       | <code>proven_err_t</code> .                                                                                  |
| <code>proven_net_conn_set_nodelay(&amp;conn, on)</code>                                                          | Send small writes at once instead of gathering them.                                                                | <code>proven_err_t</code> : UNSUPPORTED on a Unix-domain connection.                                         |
| <code>proven_net_listener_is_open(&amp;l),<br/>proven_net_conn_is_open(&amp;c)</code>                            | Whether the value holds an open socket.                                                                             | <code>bool</code> .                                                                                          |

`proven_net_listener_t` and `proven_net_conn_t` are small values you own. Close each one you opened, and do not copy one that is open: two copies are two owners of one socket.

The same calls serve **Unix-domain** stream sockets - pass an address made by `proven_net_addr_unix`. Listening creates the path; closing the listener does not remove it, so a server removes a leftover path before it listens.

### Cautions, and what goes wrong

**One read is not one message.** A stream delivers bytes, in order, in whatever pieces the network made of them.

Wrong - assuming the request arrives in one read:

```
proven_result_size_t r = proven_net_read(&conn, buf, until);
handle_request(buf, r.value);    /* wrong: this may be half a request, or one and a half */
```

Correct - read until the protocol says the message is complete: a length you were told, a delimiter you were promised, or `PROVEN_ERR_EOF`. The worked example below reads to the end.

**One write may not send everything.** `proven_net_write` reports how much went. Ignoring that count silently truncates the message whenever the buffer happens to be full - which is to say, under load, in production. `proven_net_write_all` is the call that keeps going; and when it fails, its count is what tells you whether resending would duplicate data.

**A reset is not an end.** `PROVEN_ERR_EOF` means the peer said it was finished and everything it sent has been read. `PROVEN_ERR_RESET` means it went away without saying so, and some of what was in flight - in either direction - may be gone. A file transfer that ends in `EOF` is complete; one that ends in `RESET` is not, however many bytes arrived.

**Closing is not the same as finishing.** To be sure the peer received everything, say you are done with `proven_net_shutdown_write`, then read until `EOF`, then close. Closing a connection that still has unread data in it is how the *other* side gets `PROVEN_ERR_RESET`.

Compiled and run by the test suite:

```
/*
 * A TCP exchange, start to finish, with a deadline on everything that can wait.
 *
 * Both ends live in this one program, which works because the kernel completes a connection as
 * soon as the listener's backlog takes it - before accept is called. A real client and server
 * are these same calls in two processes.
 */

int main(void) {
    /* Port 0: the OS picks a free one, and `at` says which. Loopback: this machine only. */
    proven_net_listener_t listener;
    proven_net_addr_t at;
    proven_err_t err = proven_net_listen(proven_net_addr_loopback(PROVEN_NET_FAMILY_IPV4, 0), 16, &listener,
    &at);
    if (err == PROVEN_ERR_PERMISSION || err == PROVEN_ERR_UNSUPPORTED) {
        printf("no sockets in this environment; nothing to show\n");
        return EXAMPLE_OK();
    }
    EXAMPLE_REQUIRE(err == PROVEN_OK && proven_net_listener_is_open(&listener), "the listener opens");
    EXAMPLE_REQUIRE(at.port != 0, "and reports the port it was given");

    /* One deadline for the whole exchange: five seconds from now, however many calls it takes. */
    proven_net_deadline_t until = proven_net_deadline_in(5000);

    proven_net_conn_t client, server;
    proven_net_addr_t peer;
    EXAMPLE_REQUIRE(proven_net_connect(at, until, &client) == PROVEN_OK, "the client connects");
    EXAMPLE_REQUIRE(proven_net_accept(&listener, until, &server, &peer) == PROVEN_OK, "the server accepts");
    EXAMPLE_REQUIRE(proven_net_conn_is_open(&client) && proven_net_conn_is_open(&server), "two open ends");
```

```

/* Each end knows both addresses. The accepted peer is the client's own local address. */
proven_net_addr_t client_local, client_peer, listening;
EXAMPLE_REQUIRE(proven_net_conn_local_addr(&client, &client_local) == PROVEN_OK &&
    proven_net_conn_peer_addr(&client, &client_peer) == PROVEN_OK, "the client's two
addresses");
EXAMPLE_REQUIRE(proven_net_addr_eq(&client_local, &peer), "the server saw the client's address");
EXAMPLE_REQUIRE(proven_net_listener_addr(&listener, &listening) == PROVEN_OK &&
    proven_net_addr_eq(&client_peer, &listening), "the client's peer is where the server
listens");

/* A request written in two small pieces: without this they may wait to be sent together. */
EXAMPLE_REQUIRE(proven_net_conn_set_nodelay(&client, true) == PROVEN_OK, "small writes go at once");

/* write_all sends everything or says how far it got. */
proven_mem_view_t request = proven_mem_view_from_u8(PROVEN_LIT("what time is it?"));
proven_result_size_t sent = proven_net_write_all(&client, request, until);
EXAMPLE_REQUIRE(sent.err == PROVEN_OK && sent.value == request.size, "the whole request went");

/* "I have nothing more to send." The server will read the request, then end of input. */
EXAMPLE_REQUIRE(proven_net_shutdown_write(&client) == PROVEN_OK, "the client half-closes");

/* The server reads until the end. A read returns what has arrived - a stream has no message
* boundaries - so a loop is the only correct way to read "all of it". */
proven_byte_t buf[64];
proven_size_t got = 0;
for (;;) {
    proven_result_size_t r = proven_net_read(&server, (proven_mem_mut_t){ buf + got, sizeof buf - got },
until);
    if (r.err == PROVEN_ERR_EOF) break; /* the client finished */
    EXAMPLE_REQUIRE(r.err == PROVEN_OK, "a read that is not the end succeeds");
    if (r.err != PROVEN_OK) break;
    got += r.value;
}
EXAMPLE_REQUIRE(got == request.size, "the server received the whole request");

/* A single write may send only part; the count says how much. Here it all fits. */
proven_mem_view_t reply = proven_mem_view_from_u8(PROVEN_LIT("time to close"));
proven_result_size_t w = proven_net_write(&server, reply, until);
EXAMPLE_REQUIRE(w.err == PROVEN_OK && w.value == reply.size, "a short reply goes in one write");

proven_result_size_t r = proven_net_read(&client, (proven_mem_mut_t){ buf, sizeof buf }, until);
EXAMPLE_REQUIRE(r.err == PROVEN_OK && r.value == reply.size, "the client reads the reply after its half-
close");

/* Nothing more is coming, and the client will not wait for ever to find that out. A timeout
* is not damage: the connection is exactly as usable as before. */
r = proven_net_read(&client, (proven_mem_mut_t){ buf, sizeof buf }, proven_net_deadline_in(50));
EXAMPLE_REQUIRE(r.err == PROVEN_ERR_TIMEOUT, "silence is a timeout, not a hang");

/* Every socket is closed by its owner. */
EXAMPLE_REQUIRE(proven_net_close(&server) == PROVEN_OK, "the server closes its end");
r = proven_net_read(&client, (proven_mem_mut_t){ buf, sizeof buf }, until);
EXAMPLE_REQUIRE(r.err == PROVEN_ERR_EOF, "which the client reads as the end");
EXAMPLE_REQUIRE(proven_net_close(&client) == PROVEN_OK, "the client closes");
EXAMPLE_REQUIRE(proven_net_listener_close(&listener) == PROVEN_OK, "the listener closes");

/* With the listener gone, the same address refuses - a different answer from a timeout. */
proven_net_conn_t nobody;
EXAMPLE_REQUIRE(proven_net_connect(at, proven_net_deadline_in(15000), &nobody) == PROVEN_ERR_REFUSED,

```

```

        "nothing listening is PROVEN_ERR_REFUSED");

    return EXAMPLE_OK();
}

```

## 5. UDP

UDP sends messages - datagrams - one at a time, each on its own. A datagram arrives whole or not at all; datagrams may arrive out of order, twice, or never, and nothing tells the sender which.

### Reference

| API                                                                    | Intent                                                                                  | Return                                                                                                            |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <code>proven_net_udp_open(at, &amp;udp, &amp;bound)</code>             | Open a socket bound to at. A client binds <code>proven_net_addr_any(family, 0)</code> . | <code>proven_err_t</code> : BUSY, PERMISSION, UNSUPPORTED (a Unix-domain address).                                |
| <code>proven_net_udp_send_to(&amp;udp, to, data, until)</code>         | Send one datagram: all of data or nothing.                                              | <code>proven_err_t</code> : OUT_OF_BOUNDS when data is larger than a datagram can be.                             |
| <code>proven_net_udp_rcv_from(&amp;udp, dest, &amp;from, until)</code> | Receive one datagram; from is optional.                                                 | <code>proven_result_size_t</code> : OUT_OF_BOUNDS with <code>.value == dest.size</code> when it was cut; TIMEOUT. |
| <code>proven_net_udp_addr(&amp;udp, &amp;out)</code>                   | Where the socket is bound.                                                              | <code>proven_err_t</code> .                                                                                       |
| <code>proven_net_udp_close(&amp;udp)</code>                            | Close. Safe on a socket that is not open.                                               | <code>proven_err_t</code> .                                                                                       |
| <code>proven_net_udp_is_open(&amp;udp)</code>                          | Whether the value holds an open socket.                                                 | <code>bool</code> .                                                                                               |

### Cautions, and what goes wrong

**"Sent" means it left, not that it arrived.** A successful `proven_net_udp_send_to` has handed the datagram to the network. A protocol that needs an answer waits for one with a deadline and sends again when it does not come.

**A zero-length datagram is a message.** On a stream, zero bytes would mean the end, which is why `proven_net_read` never returns it. A datagram can really be empty, so `proven_net_udp_rcv_from` returning success with `.value 0` is not an end and not an error.

**A buffer that is too small loses the rest.** The part of a datagram that did not fit is not kept for the next receive - it is gone.

Wrong - a buffer sized by guess, and the error ignored:

```

proven_byte_t buf[64];
proven_result_size_t r = proven_net_udp_rcv_from(&udp, buf_mem, &from, until);
parse(buf, r.value);      /* wrong: on OUT_OF_BOUNDS this is the first 64 bytes of something longer */

```

Correct - size the buffer for the largest message the protocol allows, and treat `PROVEN_ERR_OUT_OF_BOUNDS` as a malformed packet.

Compiled and run by the test suite:

```

/*
 * UDP: messages, not a stream. Each send is one datagram and each receive is one datagram,
 * with the sender's address attached - and no promise that it arrives.
 */

int main(void) {
    proven_net_udp_t server, client;

```

```

    proven_net_addr_t server_at, client_at;
    proven_err_t err = proven_net_udp_open(proven_net_addr_loopback(PROVEN_NET_FAMILY_IPV4, 0), &server,
&server_at);
    if (err == PROVEN_ERR_PERMISSION || err == PROVEN_ERR_UNSUPPORTED) {
        printf("no sockets in this environment; nothing to show\n");
        return EXAMPLE_OK();
    }
    EXAMPLE_REQUIRE(err == PROVEN_OK && proven_net_udp_is_open(&server), "the server's socket opens");
    EXAMPLE_REQUIRE(proven_net_udp_open(proven_net_addr_loopback(PROVEN_NET_FAMILY_IPV4, 0), &client, NULL) ==
PROVEN_OK,
        "and the client's");
    EXAMPLE_REQUIRE(proven_net_udp_addr(&client, &client_at) == PROVEN_OK && client_at.port != 0,
        "a socket can be asked where it is bound");

    proven_net_deadline_t until = proven_net_deadline_in(2000);

    /* Two sends are two datagrams. Success means each left this machine. */
    EXAMPLE_REQUIRE(proven_net_udp_send_to(&client, server_at, proven_mem_view_from_u8(PROVEN_LIT("first")),
until) == PROVEN_OK &&
        proven_net_udp_send_to(&client, server_at, proven_mem_view_from_u8(PROVEN_LIT("second
message")), until) == PROVEN_OK,
        "two datagrams leave");

    /* Each receive is one whole datagram and says who sent it. */
    proven_byte_t buf[32];
    proven_net_addr_t from;
    proven_result_size_t r = proven_net_udp_rcv_from(&server, (proven_mem_mut_t){ buf, sizeof buf }, &from,
until);
    EXAMPLE_REQUIRE(r.err == PROVEN_OK && r.value == 5, "the first datagram, alone: five bytes");
    EXAMPLE_REQUIRE(proven_net_addr_eq(&from, &client_at), "from the client's address");

    /* A buffer that is too small cuts the datagram and says so. The rest of it is gone - the
    * next receive is the next datagram - so size the buffer for the largest message. */
    proven_byte_t small[6];
    r = proven_net_udp_rcv_from(&server, (proven_mem_mut_t){ small, sizeof small }, &from, until);
    EXAMPLE_REQUIRE(r.err == PROVEN_ERR_OUT_OF_BOUNDS && r.value == sizeof small,
        "fourteen bytes into six: PROVEN_ERR_OUT_OF_BOUNDS, six kept");

    /* The reply goes to the address the datagram came from. */
    EXAMPLE_REQUIRE(proven_net_udp_send_to(&server, from, proven_mem_view_from_u8(PROVEN_LIT("ack")), until)
== PROVEN_OK,
        "the server answers the sender");
    r = proven_net_udp_rcv_from(&client, (proven_mem_mut_t){ buf, sizeof buf }, NULL, until);
    EXAMPLE_REQUIRE(r.err == PROVEN_OK && r.value == 3, "and the client receives it");

    /* Nothing else is coming. A lost datagram looks exactly like this, which is why every
    * receive has a deadline and every UDP protocol has a retry. */
    r = proven_net_udp_rcv_from(&client, (proven_mem_mut_t){ buf, sizeof buf }, NULL,
proven_net_deadline_in(50));
    EXAMPLE_REQUIRE(r.err == PROVEN_ERR_TIMEOUT, "silence is a timeout");

    EXAMPLE_REQUIRE(proven_net_udp_close(&client) == PROVEN_OK && proven_net_udp_close(&server) == PROVEN_OK,
        "both sockets are closed by their owner");
    return EXAMPLE_OK();
}

```

## 6. Many sockets, one thread

A deadline bounds a wait on one socket. A server waits on all of its sockets at once and serves whichever needs it: that is `proven_net_poll`.

## Reference

| API                                                                                                                    | Intent                                                                                                    | Return                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>proven_net_listener_handle(&amp;l),<br/>proven_net_conn_handle(&amp;c),<br/>proven_net_udp_handle(&amp;u)</code> | A socket as the poll sees it,<br>whatever its kind.                                                       | <code>proven_net_handle_t</code> ; not valid for<br>a socket that is not open.                                                                                                                     |
| <code>proven_net_poll(items, count,<br/>until, &amp;ready)</code>                                                      | Wait until at least one item is<br>ready. Up to<br><code>PROVEN_NET_POLL_INLINE_MAX</code> (64)<br>items. | <code>proven_err_t</code> : <code>TIMEOUT</code> when none<br>was ( <code>ready</code> is 0); <code>OUT_OF_BOUNDS</code> for<br>too many items; <code>INVALID_ARG</code> for<br>an invalid handle. |
| <code>proven_net_poll_with(scratch,<br/>items, count, until, &amp;ready)</code>                                        | The same for any number of<br>items, using memory you<br>supply.                                          | <code>proven_err_t</code> , and <code>OUT_OF_BOUNDS</code><br>when <code>scratch</code> is too small.                                                                                              |
| <code>proven_net_poll_scratch_size(count)</code>                                                                       | How much scratch count items<br>need.                                                                     | <code>proven_size_t</code> ; <code>SIZE_MAX</code> when it<br>cannot be represented.                                                                                                               |

```
typedef struct {  
    proven_net_handle_t handle;  
    proven_u8 want;    /* PROVEN_NET_READABLE and/or PROVEN_NET_WRITABLE */  
    proven_u8 got;     /* filled in: zero, or what is ready - which may include PROVEN_NET_FAILED */  
} proven_net_poll_item_t;
```

**Ready means "will not wait", not "will succeed".** A readable connection may hold data, or the end, or an error; a readable listener has a connection to accept. So the call that follows a ready report takes `PROVEN_NET_DONT_WAIT`, and its result is what you act on. An item reported `PROVEN_NET_FAILED` is in error or hung up; the next read or write on it says which.

**Ask for writability only when you have something to write.** An idle connection is almost always writable, so a loop that always asks is woken at once, every time, and spins a processor doing nothing.

Wrong:

```
items[i].want = PROVEN_NET_READABLE | PROVEN_NET_WRITABLE;    /* wrong when nothing is queued: never sleeps */
```

Correct - `PROVEN_NET_WRITABLE` only for a connection whose last write returned less than you asked or timed out, and only until that data has gone.

`proven_net_poll_with` exists because the operating system wants an array of its own and this library does not allocate one behind your back: you give it working memory, sized by `proven_net_poll_scratch_size`, and reuse it for the next call. Readiness here is `poll` on POSIX and `WSAPoll` on Windows - the right tool for hundreds of sockets, not for hundreds of thousands.

Compiled and run by the test suite:

```
/*  
 * One thread, many sockets: wait until any of them needs attention, then serve the ones  
 * that do. This is the loop inside every single-threaded server.  
 *  
 * "Ready" means the next call of that kind will not wait - so each socket reported ready is  
 * handled with PROVEN_NET_DONT_WAIT, and the loop never blocks anywhere but in the poll.  
 */  
  
int main(void) {  
    proven_net_listener_t listener;  
    proven_net_addr_t at;  
    proven_err_t err = proven_net_listen(proven_net_addr_loopback(PROVEN_NET_FAMILY_IPV4, 0), 16, &listener,  
&at);  
    if (err == PROVEN_ERR_PERMISSION || err == PROVEN_ERR_UNSUPPORTED) {
```

```

    printf("no sockets in this environment; nothing to show\n");
    return EXAMPLE_OK();
}
EXAMPLE_REQUIRE(err == PROVEN_OK, "the listener opens");

/* Two clients connect and each sends one message. */
proven_net_conn_t clients[2];
for (int i = 0; i < 2; ++i) {
    EXAMPLE_REQUIRE(proven_net_connect(at, proven_net_deadline_in(5000), &clients[i]) == PROVEN_OK, "a
client connects");
}
EXAMPLE_REQUIRE(proven_net_write_all(&clients[0], proven_mem_view_from_u8(PROVEN_LIT("from the first")),
proven_net_deadline_in(5000)).err == PROVEN_OK &&
    proven_net_write_all(&clients[1], proven_mem_view_from_u8(PROVEN_LIT("from the second")),
proven_net_deadline_in(5000)).err == PROVEN_OK,
    "each sends a message");

/* The server: a listener and up to two connections, in one list. */
proven_net_conn_t served[2] = {0};
proven_size_t served_count = 0;
proven_size_t bytes_seen = 0;

proven_net_deadline_t until = proven_net_deadline_in(5000);
while (bytes_seen < 14 + 15) {
    proven_net_poll_item_t items[3];
    proven_size_t count = 0;
    items[count++] = (proven_net_poll_item_t){ .handle = proven_net_listener_handle(&listener), .want =
PROVEN_NET_READABLE };
    for (proven_size_t i = 0; i < served_count; ++i) {
        items[count++] = (proven_net_poll_item_t){ .handle = proven_net_conn_handle(&served[i]), .want =
PROVEN_NET_READABLE };
    }

    proven_size_t ready = 0;
    err = proven_net_poll(items, count, until, &ready);
    EXAMPLE_REQUIRE(err == PROVEN_OK && ready > 0, "something becomes ready before the deadline");
    if (err != PROVEN_OK) break;

    /* A readable listener has a connection waiting. */
    if (items[0].got & PROVEN_NET_READABLE) {
        EXAMPLE_REQUIRE(served_count < 2 &&
            proven_net_accept(&listener, PROVEN_NET_DONT_WAIT, &served[served_count], NULL) ==
PROVEN_OK,
            "the accept that poll predicted does not wait");
        served_count++;
    }
    /* A readable connection has data, or its end. `items[1 + i]` was built from `served[i]`. */
    for (proven_size_t i = 1; i < count; ++i) {
        if (!items[i].got) continue;
        proven_byte_t buf[64];
        proven_result_size_t r = proven_net_read(&served[i - 1], (proven_mem_mut_t){ buf, sizeof buf },
PROVEN_NET_DONT_WAIT);
        EXAMPLE_REQUIRE(r.err == PROVEN_OK, "the read that poll predicted does not wait");
        bytes_seen += r.value;
    }
}
EXAMPLE_REQUIRE(served_count == 2 && bytes_seen == 29, "both clients were accepted and both messages read,
in one thread");

/* proven_net_poll takes up to PROVEN_NET_POLL_INLINE_MAX sockets. A server with more passes

```

```

    * its own working memory: the OS wants an array, and this library does not allocate one. */
    proven_net_poll_item_t one = { .handle = proven_net_conn_handle(&served[0]), .want =
PROVEN_NET_WRITABLE };
    proven_byte_t scratch[256];
    proven_size_t ready = 0;
    EXAMPLE_REQUIRE(proven_net_poll_scratch_size(1) <= sizeof scratch, "the scratch for one item is small");
    EXAMPLE_REQUIRE(proven_net_poll_with((proven_mem_mut_t){ scratch, sizeof scratch }, &one, 1,
PROVEN_NET_DONT_WAIT, &ready) == PROVEN_OK &&
        (one.got & PROVEN_NET_WRITABLE), "an idle connection is writable at once");

    /* A UDP socket waits in the same list as everything else. */
    proven_net_udp_t udp;
    EXAMPLE_REQUIRE(proven_net_udp_open(proven_net_addr_loopback(PROVEN_NET_FAMILY_IPV4, 0), &udp, NULL) ==
PROVEN_OK, "a UDP socket");
    proven_net_poll_item_t quiet = { .handle = proven_net_udp_handle(&udp), .want = PROVEN_NET_READABLE };
    EXAMPLE_REQUIRE(proven_net_poll(&quiet, 1, proven_net_deadline_in(30), &ready) == PROVEN_ERR_TIMEOUT &&
ready == 0,
        "nothing arrives on it: PROVEN_ERR_TIMEOUT, the idle tick of an event loop");
    (void)proven_net_udp_close(&udp);

    for (int i = 0; i < 2; ++i) {
        (void)proven_net_close(&clients[i]);
        (void)proven_net_close(&served[i]);
    }
    (void)proven_net_listener_close(&listener);
    return EXAMPLE_OK();
}

```

## 7. Transports

A protocol - a line-based command set, an HTTP exchange - should be written once and not know what carries its bytes. Today that is a TCP connection. Under TLS it is an encrypted session over one. In a test it is two memory buffers and no network at all. `proven_transport_t` is that seam.

```

typedef struct {
    void *ctx;
    proven_result_size_t (*read_fn)(void *ctx, proven_mem_mut_t dest, proven_net_deadline_t until);
    proven_result_size_t (*write_fn)(void *ctx, proven_mem_view_t src, proven_net_deadline_t until);
    proven_err_t (*shutdown_fn)(void *ctx); /* may be NULL */
    proven_err_t (*close_fn)(void *ctx); /* may be NULL */
} proven_transport_t;

```

It is a small vtable passed by value, the same shape as `proven_allocator_t` ([Chapter 2](#)) and `proven_writer_t`, and for the same reason: the caller decides, and nothing is hidden. To supply your own, fill in the struct. The contract is that of the TCP calls: a read never succeeds with zero bytes, and a write reports the bytes that went even when it then fails.

### Reference

| API                                                                                       | Intent                                                                    | Return                                                                                 |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <code>proven_net_conn_transport(&amp;conn)</code>                                         | A connection as a transport. Closing the transport closes the connection. | <code>proven_transport_t</code> .                                                      |
| <code>proven_transport_is_valid(t)</code>                                                 | Whether it has a read and a write function.                               | <code>bool</code> .                                                                    |
| <code>proven_transport_read(t, dest, until), proven_transport_write(t, src, until)</code> | As <code>proven_net_read</code> and <code>proven_net_write</code> .       | <code>proven_result_size_t</code> ; <code>INVALID_ARG</code> for an invalid transport. |

|                                                            |                                                                                                                                      |                                                              |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| proven_transport_write_all(t, src, until)                  | Write everything, or fail with the count. A transport that takes nothing and reports no error is PROVEN_ERR_IO, not an endless loop. | proven_result_size_t.                                        |
| proven_transport_shutdown(t),<br>proven_transport_close(t) | End the sending side; close. PROVEN_OK when the transport has nothing to do.                                                         | proven_err_t.                                                |
| proven_transport_reader(&state, t, timeout_ms)             | A proven_reader_t over a transport, for the line reader and anything else that reads a stream.                                       | proven_reader_t; not valid if state is NULL or t is invalid. |
| proven_transport_writer(&state, t, timeout_ms)             | A proven_writer_t over a transport, so proven_fprint can send.                                                                       | proven_writer_t.                                             |

The reader and writer adapters connect a socket to everything in [Chapter 5's stream section](#). A reader has nowhere to pass a deadline, so each read or write through an adapter gets its own: `timeout_ms` from the moment it starts, or no deadline when that is 0. The `proven_transport_stream_t` state is [caller-owned](#): it must outlive the reader or writer and must not move.

The writer sends each piece as it is given. Wrap it in `proven_writer_buffered` when a message is formatted in many small pieces, so that it leaves in few packets - and flush before you wait for the reply.

Compiled and run by the test suite:

```
/*
 * A protocol written once, against proven_transport_t, and run over a TCP connection.
 *
 * The function below does not know what carries its bytes. Today it is a socket; under TLS it
 * will be an encrypted session over a socket; in a test it can be two memory buffers. That is
 * what the interface is for.
 */

/* Send one request line and read one reply line, each with a five-second limit per step. */
static proven_err_t ask(proven_transport_t t, const char *question, proven_u8str_view_t expected) {
    if (!proven_transport_is_valid(t)) return PROVEN_ERR_INVALID_ARG;

    /* The formatter writes straight to the transport. */
    proven_transport_stream_t out_state;
    proven_writer_t out = proven_transport_writer(&out_state, t, 5000);
    proven_err_t err = proven_fprintln(out, "ASK {}", PROVEN_ARG(question)).err;
    if (err != PROVEN_OK) return err;

    /* The line reader reads from it, into a buffer this function owns. */
    proven_transport_stream_t in_state;
    proven_byte_t line_buf[128];
    proven_reader_buffered_t lines;
    (void)proven_reader_buffered(&lines, proven_transport_reader(&in_state, t, 5000),
        (proven_mem_mut_t){ line_buf, sizeof line_buf });
    proven_result_u8str_view_t reply = proven_reader_read_line(&lines);
    if (reply.err != PROVEN_OK) return reply.err;
    return proven_u8str_view_eq(reply.val, expected) ? PROVEN_OK : PROVEN_ERR_INVALID_FORMAT;
}
```

```

int main(void) {
    proven_net_listener_t listener;
    proven_net_addr_t at;
    proven_err_t err = proven_net_listen(proven_net_addr_loopback(PROVEN_NET_FAMILY_IPV4, 0), 4, &listener,
&at);
    if (err == PROVEN_ERR_PERMISSION || err == PROVEN_ERR_UNSUPPORTED) {
        printf("no sockets in this environment; nothing to show\n");
        return EXAMPLE_OK();
    }
    EXAMPLE_REQUIRE(err == PROVEN_OK, "the listener opens");

    proven_net_conn_t client_conn, server_conn;
    EXAMPLE_REQUIRE(proven_net_connect(at, proven_net_deadline_in(5000), &client_conn) == PROVEN_OK &&
        proven_net_accept(&listener, proven_net_deadline_in(5000), &server_conn, NULL) ==
PROVEN_OK,
        "a connected pair");

    /* A connection as a transport. From here on, neither side names a socket. */
    proven_transport_t client = proven_net_conn_transport(&client_conn);
    proven_transport_t server = proven_net_conn_transport(&server_conn);
    proven_net_deadline_t until = proven_net_deadline_in(5000);

    /* The server's half, done by hand: the reply is written before the client asks, which is
    * fine in one thread because the bytes simply wait in the connection. */
    proven_result_size_t w = proven_transport_write_all(server, proven_mem_view_from_u8(PROVEN_LIT("ANSWER
42\n")), until);
    EXAMPLE_REQUIRE(w.err == PROVEN_OK && w.value == 10, "the server's reply is on its way");

    EXAMPLE_REQUIRE(ask(client, "the question", PROVEN_LIT("ANSWER 42")) == PROVEN_OK,
        "the protocol function sends its line and reads the reply");

    /* And the server reads what the client wrote, with the raw calls. */
    proven_byte_t buf[64];
    proven_result_size_t r = proven_transport_read(server, (proven_mem_mut_t){ buf, sizeof buf }, until);
    EXAMPLE_REQUIRE(r.err == PROVEN_OK &&
        proven_u8str_view_eq((proven_u8str_view_t){ buf, r.value }, PROVEN_LIT("ASK the
question\n")),
        "the request line arrived as written");

    /* A single write reports how much went; here, all of it. */
    w = proven_transport_write(server, proven_mem_view_from_u8(PROVEN_LIT("BYE\n")), until);
    EXAMPLE_REQUIRE(w.err == PROVEN_OK && w.value == 4, "one more line");

    /* Ending and closing go through the same interface. */
    EXAMPLE_REQUIRE(proven_transport_shutdown(server) == PROVEN_OK, "the server ends its sending side");
    r = proven_transport_read(client, (proven_mem_mut_t){ buf, sizeof buf }, until);
    EXAMPLE_REQUIRE(r.err == PROVEN_OK && r.value == 4, "the client reads the last line");
    r = proven_transport_read(client, (proven_mem_mut_t){ buf, sizeof buf }, until);
    EXAMPLE_REQUIRE(r.err == PROVEN_ERR_EOF, "and then the end");

    EXAMPLE_REQUIRE(proven_transport_close(client) == PROVEN_OK && proven_transport_close(server) ==
PROVEN_OK,
        "closing a transport closes what carries it");
    EXAMPLE_REQUIRE(!proven_net_conn_is_open(&client_conn), "the connection underneath is closed");
    (void)proven_net_listener_close(&listener);
    return EXAMPLE_OK();
}

```

## 8. What the platforms do differently

The calls behave the same on Linux, the BSDs, macOS and Windows. These are the differences the library absorbs, written down because they are the ones that bite when sockets are used directly:

| The trap                                                                                            | What <code>net.h</code> does                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A write to a closed connection raises <code>SIGPIPE</code> and kills the process (POSIX).           | Every send suppresses the signal; the write returns <code>PROVEN_ERR_RESET</code> . No signal handler is installed for you.                                                      |
| Winsock must be started before any call, name resolution included.                                  | Started once, on first use.                                                                                                                                                      |
| On Windows <code>SO_REUSEADDR</code> lets another process bind your port and take your connections. | A listener uses <code>SO_REUSEADDR</code> on POSIX and <code>SO_EXCLUSIVEADDRUSE</code> on Windows: a second listener on one address is <code>PROVEN_ERR_BUSY</code> everywhere. |
| An IPv6 listener also accepts IPv4 on some systems and not on others.                               | IPv6 sockets are IPv6-only everywhere. A dual-stack server opens two listeners.                                                                                                  |
| A non-blocking Windows send accepts a buffer of any size and queues it all.                         | Sends are offered in bounded pieces, so a full buffer pushes back - and a deadline can fire - as it does elsewhere.                                                              |
| Windows reports a peer's orderly close to <code>WSAPoll</code> as a hang-up, not as readable.       | Reported as readable, as on POSIX: the read that follows returns <code>PROVEN_ERR_EOF</code> .                                                                                   |
| A UDP send to a closed port makes a <i>later</i> receive on the same Windows socket fail.           | That report is turned off; a receive answers for the datagram it received.                                                                                                       |
| A child process inherits open sockets.                                                              | Every socket is created not inheritable (close-on-exec).                                                                                                                         |
| Winsock answers a Unix-domain connect to a missing path as "refused".                               | <code>PROVEN_ERR_NOT_FOUND</code> , as on POSIX.                                                                                                                                 |

Two things differ and stay different. Windows waits about two seconds before it reports a refused loopback connection, where POSIX answers at once - allow for it in the deadline. And Unix-domain sockets need Windows 10 version 1803 or later; where the family is missing, `proven_net_listen` and `proven_net_connect` return `PROVEN_ERR_UNSUPPORTED`.

## 9. What is not here

- **TLS.** The transport interface is where it will attach; this version has none, and a connection made here is not encrypted.
- **A deadline on name resolution.** See section 2.
- **Scale beyond `poll`.** No `epoll`, `kqueue` or completion ports.
- **Unix-domain datagrams, raw sockets, multicast, socket options beyond `TCP_NODELAY`.**
- **HTTP.** The message codec is [Chapter 10](#) the client and server that drive it over these sockets are being built.

## Chapter 10: URLs and HTTP Messages

**Part V - Talking to the operating system. Prerequisites: Part II (1, 2, 3). Chapter 9 is where the bytes come from, but nothing here needs it. After this chapter** you can take a URL apart, turn a request path into one that is safe to open, parse an HTTP/1.1 request or response as it arrives, tell how long its body is, decode that body, and write a message that nobody else's input can corrupt.

This chapter covers `url.h` and `http.h`. Both are pure text handling: no socket, no file, no allocation. Unlike Chapter 9 they are available in a [freestanding](#) build, and `PROVEN_NO_NET` does not remove them.

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### 1. A codec, not a server

`http.h` does not read from anything and does not write to anything. You hand it bytes you already have and it tells you what they say; you hand it memory and it writes a message into it. What carries those bytes - a blocking socket, an event loop, a TLS session, a test - is not its business, and that is the point: the part of HTTP that is hard to get right is the same in all of them, and here it is written once.

Two properties follow, and the rest of the chapter leans on both.

**A prefix is never an error.** Bytes arrive in whatever pieces the network makes. If what has arrived so far is the beginning of a valid message, the parser answers `PROVEN_ERR_NEED_MORE` - never a failure, never a premature success. So the loop around it is always the same: parse; if it needs more, read more; otherwise act.

**Ambiguity is refused, not resolved.** An HTTP message is routinely read by two programs - a proxy in front and a server behind. Wherever the two read the same bytes differently, one can be shown a request the other never sees. That is *request smuggling*, and it lives entirely in the places where a lenient parser quietly picks an interpretation. This one does not pick:

| What is refused                                                             | Why it matters                                                               |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| A line ending that is a bare LF, or a bare CR                               | One parser sees one line where another sees two                              |
| Whitespace between a header name and its colon                              | <code>Content-Length : 5</code> is a header to one parser and not to another |
| A header line starting with a space or tab (obsolete folding)               | A continuation to one parser, a new field to another                         |
| More than one <code>Content-Length</code> , or one that is not plain digits | Two lengths means two opinions about where the body ends                     |
| <code>Transfer-Encoding</code> together with <code>Content-Length</code>    | The classic: each parser believes a different one                            |

|                                                                    |                                                                   |
|--------------------------------------------------------------------|-------------------------------------------------------------------|
| Any transfer coding but a single chunked                           | A parser that skips an unknown coding frames the body differently |
| A chunk size with whitespace, a sign, a 0x, or more than 16 digits | Chunk-size parsing is the other classic                           |

Each row is something deployed software accepts and published attacks use. A refusal is an ordinary error value - answer 400 and close.

## 2. URLs

`proven_url_parse` splits an absolute URL into views of the text you gave it:

```
https://user:pw@example.com:8443/docs/a%20b?q=1#top
\___/  \____/ \_____/ \___/ \_____/ \_/ \_/
scheme userinfo host port path query fragment

typedef struct {
    proven_u8str_view_t scheme, userinfo, host, path, query, fragment;
    proven_u16 port;
    bool has_userinfo, has_port, has_query, has_fragment;
    bool host_is_ipv6; /* written in brackets; `host` is without them */
} proven_url_t;
```

**Parsing does not decode.** Every component comes back as written, percent signs included.

Decoding is a separate step taken on the one piece you are about to use - because a URL decoded as a whole cannot be taken apart again. A `%2F` that was data inside a path segment becomes a `/` that separates segments; a `%26` inside a query value becomes an `&` that ends it.

### Reference

| API                                                                                                  | Intent                                                                                                                              | Return                                                                                                         |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>proven_url_parse(text, &amp;url)</code>                                                        | Split <code>scheme://authority/path?query#fragment</code> . ASCII only, no spaces, every <code>%</code> followed by two hex digits. | <code>proven_err_t</code> : <code>INVALID_FORMAT</code> for anything else; <code>url</code> is then untouched. |
| <code>proven_url_default_port(scheme)</code>                                                         | 80 for <code>http/ws</code> , 443 for <code>https/wss</code> , 21 for <code>ftp</code> .                                            | <code>proven_u16</code> ; 0 for an unknown scheme.                                                             |
| <code>proven_url_effective_port(&amp;url)</code>                                                     | The port to connect to: the one written, else the scheme's.                                                                         | <code>proven_u16</code> .                                                                                      |
| <code>proven_url_scheme_is(&amp;url, scheme)</code>                                                  | Compare the scheme without case.                                                                                                    | <code>bool</code> .                                                                                            |
| <code>proven_url_split_target(target, &amp;path, &amp;query, &amp;has_query)</code>                  | Split an HTTP request target: <code>/a?b</code> , <code>http://h/a?b</code> , or <code>*</code> .                                   | <code>proven_err_t</code> : <code>INVALID_FORMAT</code> otherwise.                                             |
| <code>proven_url_query_iter(query),<br/>proven_url_query_next(&amp;it, &amp;name, &amp;value)</code> | Walk <code>name=value&amp;...</code> , yielding encoded views. Empty pairs are skipped.                                             | iterator; <code>bool</code> (false at the end).                                                                |
| <code>proven_url_percent_decode(in, out, &amp;written)</code>                                        | <code>%XX</code> to bytes. <code>out</code> may be <code>in</code> . <code>+</code> stays <code>+</code> .                          | <code>proven_err_t</code> : <code>INVALID_FORMAT</code> for a bad escape; <code>OUT_OF_BOUNDS</code> .         |
| <code>proven_url_form_decode(in, out, &amp;written)</code>                                           | The same, and <code>+</code> is a space: for query names and values.                                                                | <code>proven_err_t</code> .                                                                                    |
| <code>proven_url_encode_component(in, out, &amp;written)</code>                                      | Encode for ONE component: all but letters, digits and <code>-._~</code> become <code>%XX</code> .                                   | <code>proven_err_t</code> : <code>OUT_OF_BOUNDS</code> ; <code>OVERFLOW</code> .                               |

|                                           |                                  |                                             |
|-------------------------------------------|----------------------------------|---------------------------------------------|
| proven_url_encode_path(in, out, &written) | The same, keeping /.             | proven_err_t.                               |
| proven_url_form_encode(in, out, &written) | The same, writing a space as +.  | proven_err_t.                               |
| proven_url_encoded_size_max(n)            | Bytes always enough to encode n. | proven_size_t; SIZE_MAX if unrepresentable. |

This is RFC 3986, not the URL algorithm browsers implement. A browser repairs what it is given - backslashes become slashes, spaces are encoded, a missing scheme is guessed. Nothing here is repaired: text that would need it is `PROVEN_ERR_INVALID_FORMAT`. A program that accepts URLs typed by people should say so to them rather than guess.

### Cautions, and what goes wrong

#### Decode a component, never the whole URL.

Wrong:

```
proven_url_percent_decode(whole_url, buf, &n);    /* wrong: "a%2Fb" and "a/b" are now the same path */
proven_url_parse(decoded, &url);
```

Correct - parse first, then decode the one component you need, with the decoder that matches it: `proven_url_form_decode` for a query name or value, `proven_url_path_resolve` for a path.

**Decoded text is bytes.** `proven_url_percent_decode` returns whatever was encoded: a NUL, a newline, bytes that are not UTF-8. It is the sender's data. Validate it for what you are about to do with it.

**The userinfo is not the host.** In `http://example.com@evil.test/` the host is `evil.test`. The parser gets this right - the host begins after the *last* `@` - but a program that shows a URL to a person, or checks it against an allow-list, must compare `url.host`, never a prefix of the text.

Compiled and run by the test suite:

```
/*
 * URLs: take one apart, decode only the piece you are about to use, and never let a path
 * from the network reach the filesystem without resolving it first.
 */

static bool view_is(proven_u8str_view_t v, const char *text) {
    return proven_u8str_view_eq(v, proven_u8str_view_from_cstr(text));
}

int main(void) {
    /* Parsing gives views of what was written. Nothing is decoded and nothing is copied. */
    proven_url_t url;
    proven_err_t err = proven_url_parse(PROVEN_LIT("https://example.com:8443/docs/a%20b?q=caf%C3%A9+au+lait&
page=2#top"), &url);
    EXAMPLE_REQUIRE(err == PROVEN_OK, "an absolute URL parses");
    EXAMPLE_REQUIRE(view_is(url.host, "example.com") && url.has_port && url.port == 8443, "host and port");
    EXAMPLE_REQUIRE(view_is(url.path, "/docs/a%20b"), "the path is still encoded");
    EXAMPLE_REQUIRE(url.has_query && url.has_fragment && view_is(url.fragment, "top"), "query and fragment are
there");

    /* Scheme names compare without case; a URL with no port gets the scheme's own. */
    proven_url_t plain;
    EXAMPLE_REQUIRE(proven_url_parse(PROVEN_LIT("HTTP://example.com/"), &plain) == PROVEN_OK, "another URL");
    EXAMPLE_REQUIRE(proven_url_scheme_is(&plain, PROVEN_LIT("http")), "HTTP is http");
    EXAMPLE_REQUIRE(proven_url_effective_port(&plain) == 80 && proven_url_effective_port(&url) == 8443,
        "the port to connect to: the default, or the one written");
    EXAMPLE_REQUIRE(proven_url_default_port(PROVEN_LIT("wss")) == 443, "wss is 443");
```

```

/* What is not an absolute URL is refused, not repaired. */
proven_url_t bad;
EXAMPLE_REQUIRE(proven_url_parse(PROVEN_LIT("example.com/path"), &bad) == PROVEN_ERR_INVALID_FORMAT, "no
scheme: refused");
EXAMPLE_REQUIRE(proven_url_parse(PROVEN_LIT("http://example.com/a b"), &bad) == PROVEN_ERR_INVALID_FORMAT,
"a raw space: refused");

/* A query is walked pair by pair; each name and value is decoded on its own, as form data,
 * where '+' means a space. Decoding the whole query first would turn an encoded '&' inside
 * a value into a separator. */
proven_url_query_iter_t it = proven_url_query_iter(url.query);
proven_u8str_view_t name, value;
proven_byte_t text[64];
proven_size_t n = 0;
EXAMPLE_REQUIRE(proven_url_query_next(&it, &name, &value) && view_is(name, "q"), "the first pair is q");
EXAMPLE_REQUIRE(proven_url_form_decode(value, (proven_mem_mut_t){ text, sizeof text }, &n) == PROVEN_OK &&
view_is((proven_u8str_view_t){ text, n }, "caf\xc3\xa9 au lait"), "its value, decoded");
EXAMPLE_REQUIRE(proven_url_query_next(&it, &name, &value) && view_is(name, "page") && view_is(value, "2"),
"the second pair");
EXAMPLE_REQUIRE(!proven_url_query_next(&it, &name, &value), "and no third");

/* Plain percent-decoding leaves '+' alone, and yields BYTES - any bytes. */
EXAMPLE_REQUIRE(proven_url_percent_decode(PROVEN_LIT("a%2Fb+c"), (proven_mem_mut_t){ text, sizeof text },
&n) == PROVEN_OK &&
view_is((proven_u8str_view_t){ text, n }, "a/b+c"), "%2F becomes a slash; '+' stays");

/* Going the other way: encode a value so that it cannot end the component it goes into. */
proven_mem_view_t file_name = proven_mem_view_from_u8(PROVEN_LIT("report 100%/final?.txt"));
proven_byte_t enc[96];
EXAMPLE_REQUIRE(proven_url_encoded_size_max(file_name.size) <= sizeof enc, "three bytes per byte is always
enough");
EXAMPLE_REQUIRE(proven_url_encode_component(file_name, (proven_mem_mut_t){ enc, sizeof enc }, &n) ==
PROVEN_OK &&
view_is((proven_u8str_view_t){ enc, n }, "report%20100%25%2Ffinal%3F.txt"), "one path
segment: even the slash is encoded");
EXAMPLE_REQUIRE(proven_url_encode_path(proven_mem_view_from_u8(PROVEN_LIT("/my docs/a b.txt")),
(proven_mem_mut_t){ enc, sizeof enc }, &n) == PROVEN_OK &&
view_is((proven_u8str_view_t){ enc, n }, "/my%20docs/a%20b.txt"), "a whole path: the
slashes stay");
EXAMPLE_REQUIRE(proven_url_form_encode(proven_mem_view_from_u8(PROVEN_LIT("a b&c")), (proven_mem_mut_t)
{ enc, sizeof enc }, &n) == PROVEN_OK &&
view_is((proven_u8str_view_t){ enc, n }, "a+b%26c"), "form data: a space is '+'");

/* A server's side of it. The request target is split first... */
proven_u8str_view_t path, query;
bool has_query = false;
EXAMPLE_REQUIRE(proven_url_split_target(PROVEN_LIT("/static/css/./img/logo%20v2.png?v=3"), &path, &query,
&has_query) == PROVEN_OK &&
has_query && view_is(query, "v=3"), "target into path and query");

/* ...and the path is resolved before anything else looks at it: decoded once, dot segments
 * removed, and refused if it would leave the root. */
proven_byte_t clean[128];
EXAMPLE_REQUIRE(proven_url_path_resolve(path, (proven_mem_mut_t){ clean, sizeof clean }, &n) == PROVEN_OK
&&
view_is((proven_u8str_view_t){ clean, n }, "/static/img/logo v2.png"), "a clean path under
the root");

/* The requests a file server exists to refuse. */
EXAMPLE_REQUIRE(proven_url_path_resolve(PROVEN_LIT("../etc/passwd"), (proven_mem_mut_t){ clean, sizeof

```

```

clean }, &n) == PROVEN_ERR_PERMISSION,
    "climbing above the root");
EXAMPLE_REQUIRE(proven_url_path_resolve(PROVEN_LIT("/%2e%2e/etc/passwd"), (proven_mem_mut_t){ clean,
sizeof clean }, &n) == PROVEN_ERR_PERMISSION,
    "the same climb with the dots encoded");
EXAMPLE_REQUIRE(proven_url_path_resolve(PROVEN_LIT("../%2f..%2fetc/passwd"), (proven_mem_mut_t){ clean,
sizeof clean }, &n) == PROVEN_ERR_INVALID_FORMAT,
    "an encoded slash");
EXAMPLE_REQUIRE(proven_url_path_resolve(PROVEN_LIT("/file%00.png"), (proven_mem_mut_t){ clean, sizeof
clean }, &n) == PROVEN_ERR_INVALID_FORMAT,
    "an encoded NUL");
EXAMPLE_REQUIRE(proven_url_path_resolve(PROVEN_LIT("/%c0%ae%c0%ae/"), (proven_mem_mut_t){ clean, sizeof
clean }, &n) == PROVEN_ERR_INVALID_FORMAT,
    "an overlong encoding of a dot");

return EXAMPLE_OK();
}

```

### 3. A path that is safe to use

A request path names a file, and the sender chooses it. GET ../../etc/passwd is the oldest attack on a file server and it has many spellings. proven\_url\_path\_resolve is the one function to put between a request path and anything that opens files:

| API                                              | Intent                                                                                                                                                                                         | Return                                                                                                                                     |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| proven_url_path_resolve(raw_path, out, &written) | Decode once, refuse what has no place in a file name, resolve . and ... The result starts with /, has no dot or empty segments, and keeps a trailing /. out needs at most raw_path.size bytes. | proven_err_t: PERMISSION when a .. would climb above the root; INVALID_FORMAT for the refusals below; OUT_OF_BOUNDS when out is too small. |

What it does, in order - and the order is the security property:

1. **Decodes once.** %2e%2e becomes .. and is then treated as ... And %252e%252e becomes the text %2e%2e, which is a strange file name and nothing more - decoding again would turn it into a traversal, which is the *double decoding* bug.
2. **Refuses** an encoded slash (%2F - it would split one segment into two after the fact), a backslash written or encoded (a separator on Windows), any control character including NUL (file%00.png truncates the name in C), and anything that is not valid UTF-8 once decoded - which is what catches %c0%ae, an *overlong* encoding of a dot that some decoders accept.
3. **Resolves** . and .. on the decoded text. A .. with nothing left to remove is PROVEN\_ERR\_PERMISSION: the path is well formed and asks for what it may not have.

Wrong - checking for .. before decoding:

```

if (contains(raw_path, "..")) return 400;          /* wrong: "%2e%2e" sails through */
proven_url_percent_decode(raw_path, buf, &n);
open_under_root(buf, n);

```

Wrong - decoding twice "to be safe":

```

proven_url_percent_decode(raw, a, &n);
proven_url_percent_decode(a_view, b, &m);          /* wrong: "%252e%252e" has just become ".." */

```

Correct:

```

proven_err_t err = proven_url_path_resolve(raw_path, clean_mem, &n);
if (err == PROVEN_ERR_PERMISSION) return 403;      /* or 400 */

```

```

if (err != PROVEN_OK)                return 400;
open_under_root(clean, n);

```

**What it does not do.** It stops traversal by path *syntax*. It does not know your filesystem: a symbolic link inside the root can still point outside it, and Windows gives some file names a meaning of their own (CON, a trailing dot, name:stream). Check those where the file is opened. The example above shows the resolver; the file-opening half belongs to [Chapter 5](#).

## 4. Parsing a head

A *head* is everything before the body: the request line or status line, the header fields, and the empty line that ends them.

```

typedef struct { proven_u8str_view_t name, value; } proven_http_header_t;

typedef struct {
    proven_http_method_t method;        /* GET, POST, ...; OTHER for a token the enum has no name for */
    proven_u8str_view_t method_text;
    proven_u8str_view_t target;          /* "/a?b", "http://h/a", "*", or "host:port" */
    proven_u8 version_minor;            /* 1 or 0 */
    proven_http_header_t *headers;      /* the array you supplied */
    proven_size_t header_count;
} proven_http_request_t;

typedef struct {
    proven_u16 status;
    proven_u8str_view_t reason;         /* may be empty; means nothing to a program */
    proven_u8 version_minor;
    proven_http_header_t *headers;
    proven_size_t header_count;
} proven_http_response_t;

```

Every view points into the buffer you parsed. Nothing is copied, so the buffer must outlive the parsed head and must not move - and the array for the fields is yours, too.

## Reference

| API                                                                        | Intent                                                                                                                                   | Return                                                                                                                                                                |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proven_http_parse_request(data, headers, cap, max_head, &req, &head_size)  | Parse a request head from the front of data. head_size is where the body starts. max_head 0 means PROVEN_HTTP_DEFAULT_MAX_HEAD (16 KiB). | proven_err_t: NEED_MORE (read more); OUT_OF_BOUNDS (head over the limit, or more fields than cap: 431); INVALID_FORMAT (400); UNSUPPORTED (not HTTP/1.0 or 1.1: 505). |
| proven_http_parse_response(data, headers, cap, max_head, &res, &head_size) | The same for a response.                                                                                                                 | as above.                                                                                                                                                             |
| proven_http_header_find(headers, count, name, &value)                      | The first field of that name, without case. value may be NULL.                                                                           | bool.                                                                                                                                                                 |
| proven_http_header_count(headers, count, name)                             | How many fields have that name.                                                                                                          | proven_size_t.                                                                                                                                                        |
| proven_http_header_has_token(headers, count, name, token)                  | Whether any such field lists token in a comma-separated value: Connection: keep-alive, Upgrade lists upgrade.                            | bool.                                                                                                                                                                 |

|                                                                                |                                                                                                                                 |                                     |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| proven_http_method_from_text(text),<br>proven_http_method_text(method)         | Between a method token and the enum. Case-sensitive: get is not GET.                                                            | enum; view (empty for OTHER).       |
| proven_http_reason_phrase(status)                                              | The standard phrase: Not Found.                                                                                                 | view; Unknown for a code with none. |
| proven_http_request_keep_alive(&req),<br>proven_http_response_keep_alive(&res) | Whether the connection may carry another message: HTTP/1.1 unless Connection: close; HTTP/1.0 only with Connection: keep-alive. | bool.                               |

**The limit is checked while the head is still incomplete.** If `max_head` bytes have arrived and hold no end of head, the answer is `PROVEN_ERR_OUT_OF_BOUNDS` at once - not `NEED_MORE`. A client that sends a header line for ever costs you `max_head` bytes and no more.

Empty lines before a request line are skipped (clients have long sent a stray CRLF after a body) and counted in `head_size`.

### Cautions, and what goes wrong

**NEED\_MORE is not a failure.**

Wrong:

```
if (proven_http_parse_request(data, h, 32, 0, &req, &n) != PROVEN_OK) return send_400(); /* wrong: a slow client gets a 400 */
```

Correct - three outcomes, three responses: `NEED_MORE` reads more (with a deadline, from [Chapter 9](#)); `OUT_OF_BOUNDS`, `INVALID_FORMAT` and `UNSUPPORTED` answer 431, 400 and 505 and close; `PROVEN_OK` goes on.

**A field can repeat.** `proven_http_header_find` returns the first. For a field where repetition changes the meaning, ask `proven_http_header_count` - which is exactly how this library refuses a second Content-Length.

**Views die with the buffer.** If you read the next request into the same buffer, every view from the previous head now points at something else. Copy what you need to keep.

## 5. How long is the body

After the head, the single most important question is where the message ends, because the next message starts there. The head answers it:

```
typedef struct {
    proven_http_body_kind_t kind; /* NONE, LENGTH, CHUNKED, or UNTIL_CLOSE (responses only) */
    proven_u64 length;           /* for LENGTH */
} proven_http_framing_t;
```

| API                                                          | Intent                                                                                                    | Return                                                                                                                                                         |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| proven_http_request_framing(&req, &framing)                  | Chunked, a length, or no body.                                                                            | proven_err_t: <code>INVALID_FORMAT</code> (400) for the ambiguous cases of section 1; <code>UNSUPPORTED</code> (501) for a transfer coding other than chunked. |
| proven_http_response_framing(&res, request_method, &framing) | The same, knowing what was asked: a response to HEAD, a 1xx, 204 or 304, and a successful CONNECT have no | as above.                                                                                                                                                      |

|  |                                                                |  |
|--|----------------------------------------------------------------|--|
|  | body. With neither header the body runs until the peer closes. |  |
|--|----------------------------------------------------------------|--|

A response needs the request's method because the same headers mean different things: a response to HEAD carries the Content-Length the GET would have had, and no body at all.

## 6. Decoding a body

```
proven_http_body_t body; /* caller-owned; nothing to destroy */
proven_http_body_init(&body, framing, max_body_bytes);
proven_http_body_feed(&body, in, &consumed, &payload, &done);
```

| API                                                          | Intent                                                                      | Return                                                                                                          |
|--------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| proven_http_body_init(&body, framing, max_body)              | Begin. max_body is how much you will accept - there is no default.          | proven_err_t: OUT_OF_BOUNDS (413) when a Content-Length already exceeds it.                                     |
| proven_http_body_feed(&body, in, &consumed, &payload, &done) | Consume a prefix of in; payload is the body bytes in it, as a view INTO in. | proven_err_t: INVALID_FORMAT (bad chunk framing); OUT_OF_BOUNDS (over the limit); INVALID_STATE after an error. |
| proven_http_body_end(&body)                                  | The peer closed: was the body complete?                                     | PROVEN_OK, or PROVEN_ERR_NEED_MORE when the message was cut short.                                              |
| proven_http_body_received(&body)                             | Body bytes delivered so far.                                                | proven_u64.                                                                                                     |

feed copies nothing. For a chunked body it steps over the chunk framing and hands back the data between it, one contiguous piece per call - so the loop is: feed what you have; use payload; advance by consumed; when everything is consumed and done is still false, read more. When done turns true, bytes of in past consumed belong to the **next** message.

**State your limit.** max\_body is the answer to "how much may a stranger make me hold". A Content-Length over it is refused before a byte of body is read; a chunk whose size would cross it is refused when the size is read, not after its data has arrived.

**A close is not always an end.** A body that runs until close ends, properly, when the peer closes. Any other body that is interrupted by a close is *truncated* - proven\_http\_body\_end returns PROVEN\_ERR\_NEED\_MORE - and must not be treated as whole. Half a file that looks like a whole file is the worst possible outcome of a download.

**After an error the connection is finished.** Once framing is wrong, where the next message begins is unknown. Answer if you can, and close.

Compiled and run by the test suite - a request arriving in four awkward pieces:

```
/*
 * A server's view of one request: bytes arrive in pieces, the head is parsed when it is all
 * there, the framing says how long the body is, and the body is decoded as it comes.
 *
 * No socket appears here. The codec takes bytes you have; where they came from is your affair,
 * which is why this same code runs under a blocking read, an event loop, or a test like this.
 */

/* The request, as the network might deliver it: split in awkward places. */
static const char *const pieces[] = {
    "POST /upload/notes.txt?overwrite=1 HTTP",
    "P/1.1\r\nHost: example.com\r\nTransfer-Encoding: chunk",
```

```

"ked\r\nConnection: keep-alive, TE\r\nX-Tag: a\r\nX-Tag: b\r\n\r\n5\r\nhel",
"lo\r\n7;note=x\r\n, world\r\n0\r\n\r\nGET /next HTTP/1.1\r\n\r\n",
};

int main(void) {
    proven_byte_t buf[512]; /* the read buffer: the head must fit in it */
    proven_size_t have = 0;
    proven_size_t next_piece = 0;

    proven_http_header_t fields[16];
    proven_http_request_t req;
    proven_size_t head_size = 0;

    /* Read until the head is complete. NEED_MORE is not an error: it means exactly that. */
    for (;;) {
        proven_err_t err = proven_http_parse_request((proven_mem_view_t){ buf, have }, fields, 16, sizeof buf,
        &req, &head_size);
        if (err == PROVEN_OK) break;
        EXAMPLE_REQUIRE(err == PROVEN_ERR_NEED_MORE, "an incomplete head asks for more; anything else is a
        400, 431 or 505");
        if (err != PROVEN_ERR_NEED_MORE) return EXAMPLE_OK();
        /* "read": append the next piece. */
        proven_size_t len = proven_cstr_len(pieces[next_piece]);
        for (proven_size_t i = 0; i < len; ++i) buf[have + i] = (proven_byte_t)pieces[next_piece][i];
        have += len;
        next_piece++;
    }

    /* The head, as views into `buf`. */
    EXAMPLE_REQUIRE(req.method == PROVEN_HTTP_POST && req.version_minor == 1, "POST, HTTP/1.1");
    EXAMPLE_REQUIRE(proven_http_method_from_text(req.method_text) == PROVEN_HTTP_POST &&
        proven_u8str_view_eq(proven_http_method_text(req.method), PROVEN_LIT("POST")), "the
        method, both ways");

    proven_u8str_view_t host;
    EXAMPLE_REQUIRE(proven_http_header_find(fields, req.header_count, PROVEN_LIT("host"), &host) &&
        proven_u8str_view_eq(host, PROVEN_LIT("example.com")), "header names compare without
        case");
    EXAMPLE_REQUIRE(proven_http_header_count(fields, req.header_count, PROVEN_LIT("X-Tag")) == 2, "a field may
        repeat");
    EXAMPLE_REQUIRE(proven_http_header_has_token(fields, req.header_count, PROVEN_LIT("Connection"),
        PROVEN_LIT("te")),
        "a token inside a comma-separated list");
    EXAMPLE_REQUIRE(proven_http_request_keep_alive(&req), "the connection stays open for another request");

    /* Where the request points. Split, then resolve - before the path is used for anything. */
    proven_u8str_view_t path, query;
    bool has_query = false;
    proven_byte_t clean[128];
    proven_size_t clean_len = 0;
    EXAMPLE_REQUIRE(proven_url_split_target(req.target, &path, &query, &has_query) == PROVEN_OK &&
        proven_url_path_resolve(path, (proven_mem_mut_t){ clean, sizeof clean }, &clean_len) ==
        PROVEN_OK &&
        proven_u8str_view_eq((proven_u8str_view_t){ clean, clean_len }, PROVEN_LIT("/upload/
        notes.txt")),
        "the target's path, resolved");

    /* How long is the body? The head decides, and an ambiguous head is an error, not a guess. */
    proven_http_framing_t framing;
    EXAMPLE_REQUIRE(proven_http_request_framing(&req, &framing) == PROVEN_OK && framing.kind ==

```

```

PROVEN_HTTP_BODY_CHUNKED,
    "this body is chunked");

/* Decode it. The limit is the caller's statement of how much a peer may send. */
proven_http_body_t body;
EXAMPLE_REQUIRE(proven_http_body_init(&body, framing, 1024) == PROVEN_OK, "at most 1 KiB of body");

proven_byte_t content[64];
proven_size_t content_len = 0;
proven_size_t pos = head_size;          /* the body starts where the head ended */
bool done = false;
while (!done) {
    if (pos == have) {
        /* Everything read so far is used up: "read" more. A real server would stop here
        * with proven_http_body_end if the peer had closed instead. */
        EXAMPLE_REQUIRE(next_piece < sizeof pieces / sizeof pieces[0], "more input exists");
        proven_size_t len = proven_cstr_len(pieces[next_piece]);
        for (proven_size_t i = 0; i < len; ++i) buf[have + i] = (proven_byte_t)pieces[next_piece][i];
        have += len;
        next_piece++;
    }
    proven_size_t used = 0;
    proven_mem_view_t payload;
    proven_err_t err = proven_http_body_feed(&body, (proven_mem_view_t){ buf + pos, have - pos }, &used,
    &payload, &done);
    EXAMPLE_REQUIRE(err == PROVEN_OK, "the chunk framing is well formed");
    if (err != PROVEN_OK) break;
    /* `payload` is a view into `buf`: body bytes with the chunk framing stepped over. */
    for (proven_size_t i = 0; i < payload.size; ++i) content[content_len + i] = payload.ptr[i];
    content_len += payload.size;
    pos += used;
}
EXAMPLE_REQUIRE(proven_u8str_view_eq((proven_u8str_view_t){ content, content_len }, PROVEN_LIT("hello,
world")),
    "two chunks, joined");
EXAMPLE_REQUIRE(proven_http_body_received(&body) == 12 && proven_http_body_end(&body) == PROVEN_OK,
"twelve bytes, complete");

/* What is left in the buffer is the next request, untouched. */
proven_http_request_t next;
EXAMPLE_REQUIRE(proven_http_parse_request((proven_mem_view_t){ buf + pos, have - pos }, fields, 16, 0,
&next, &head_size) == PROVEN_OK &&
    proven_u8str_view_eq(next.target, PROVEN_LIT("/next")), "the decoder stopped exactly where
the next request begins");

return EXAMPLE_OK();
}

```

## 7. Writing a message

The writers append to memory you supply. Each takes the buffer and a length, writes at that length, and advances it; on failure the length does not move and nothing before it is touched.

| API                                                                        | Appends                                                                |
|----------------------------------------------------------------------------|------------------------------------------------------------------------|
| <code>proven_http_write_request_line(out, &amp;len, method, target)</code> | GET /path HTTP/1.1 CRLF                                                |
| <code>proven_http_write_status_line(out, &amp;len, status, reason)</code>  | HTTP/1.1 404 Not Found CRLF; an empty reason sends the standard phrase |

|                                                       |                                                           |
|-------------------------------------------------------|-----------------------------------------------------------|
| proven_http_write_header(out, &len, name, value)      | Name: value CRLF                                          |
| proven_http_write_header_u64(out, &len, name, number) | Content-Length: 1234 CRLF                                 |
| proven_http_write_head_end(out, &len)                 | the empty line that ends the head                         |
| proven_http_write_chunk_begin(out, &len, size)        | the line that starts a chunk of size bytes (not 0)        |
| proven_http_write_chunk_end(out, &len)                | the CRLF after a chunk's data                             |
| proven_http_write_last_chunk(out, &len)               | the zero chunk and empty trailers that end a chunked body |

All return `proven_err_t: PROVEN_ERR_OUT_OF_BOUNDS` when it does not fit, and `PROVEN_ERR_INVALID_ARG` when what you asked it to write would not be the thing it is named for.

**That second error is the important one.** A header value almost always comes from somewhere else - a file name, a redirect target, something the client sent. If it contains a CR and an LF, writing it verbatim ends the field and starts a new one that the *sender* chose:

Location: /home

Set-Cookie: session=attacker <- this line came from inside the "value"

That is *response splitting*, the writing twin of request smuggling. `proven_http_write_header` refuses any value containing CR, LF, NUL or another control character, and any name that is not a token; `proven_http_write_request_line` refuses a target with a space or a line break. The check is not optional and has no "raw" variant.

Wrong - one check at the end:

```
proven_http_write_status_line(out, &len, 200, PROVEN_LIT(""));
proven_http_write_header(out, &len, name, user_value);      /* refused: len did not move */
err = proven_http_write_head_end(out, &len);                /* OK - and the header is silently missing */
```

Correct - check each call, or chain them so the first failure stops the rest, as the example does. The writers are `[[nodiscard]]`, so the compiler objects to the first form.

A body of known length is `Content-Length` and then the bytes, which you write yourself. A body of unknown length is `Transfer-Encoding: chunked` and then, for each piece, `chunk_begin`, the data, `chunk_end` - and `last_chunk` at the end.

Compiled and run by the test suite:

```
/*
 * Writing a response and reading one back.
 *
 * Each writer appends at `len` and moves it. A value that could break the message - a line
 * break in a header - is refused instead of written, so a string taken from a request cannot
 * be used to add a header of the sender's choosing.
 */

int main(void) {
    proven_byte_t msg[512];
    proven_mem_mut_t out = { msg, sizeof msg };
    proven_size_t len = 0;

    /* The date header: the wall clock, in the one form HTTP writes. */
    proven_byte_t date[PROVEN_HTTP_DATE_SIZE];
    proven_time_t now = proven_time_now();
    EXAMPLE_REQUIRE(proven_http_date_format(now, date) == PROVEN_OK, "now, as an HTTP date");

    /* A response whose length is not known in advance goes out in chunks. */
    proven_err_t err = proven_http_write_status_line(out, &len, 200, PROVEN_LIT("")); /* "" : the standard
phrase */
```

```

    if (err == PROVEN_OK) err = proven_http_write_header(out, &len, PROVEN_LIT("Date"), (proven_u8str_view_t)
{ date, sizeof date });
    if (err == PROVEN_OK) err = proven_http_write_header(out, &len, PROVEN_LIT("Content-Type"),
PROVEN_LIT("text/plain; charset=utf-8"));
    if (err == PROVEN_OK) err = proven_http_write_header(out, &len, PROVEN_LIT("Transfer-Encoding"),
PROVEN_LIT("chunked"));
    if (err == PROVEN_OK) err = proven_http_write_header_u64(out, &len, PROVEN_LIT("X-Request-Id"), 4711);
    if (err == PROVEN_OK) err = proven_http_write_head_end(out, &len);
    EXAMPLE_REQUIRE(err == PROVEN_OK, "the head is written");
    proven_size_t head_len = len;

    static const char *const parts[] = { "written in ", "two chunks" };
    for (proven_size_t i = 0; i < 2 && err == PROVEN_OK; ++i) {
        proven_size_t n = proven_cstr_len(parts[i]);
        err = proven_http_write_chunk_begin(out, &len, n);
        if (err != PROVEN_OK || n > sizeof msg - len) { err = PROVEN_ERR_OUT_OF_BOUNDS; break; }
        for (proven_size_t k = 0; k < n; ++k) msg[len + k] = (proven_byte_t)parts[i][k]; /* the chunk's data
*/
        len += n;
        err = proven_http_write_chunk_end(out, &len);
    }
    if (err == PROVEN_OK) err = proven_http_write_last_chunk(out, &len);
    EXAMPLE_REQUIRE(err == PROVEN_OK, "the body is written");

    /* A header value with a line break in it is not written at all. */
    proven_size_t before = len;
    EXAMPLE_REQUIRE(proven_http_write_header(out, &len, PROVEN_LIT("Location"), PROVEN_LIT("/home\r\nSet-
Cookie: session=stolen"))) == PROVEN_ERR_INVALID_ARG &&
        len == before, "response splitting is refused, and nothing was appended");

    /* --- the client's side: the same bytes, read back ----- */

    proven_http_header_t fields[16];
    proven_http_response_t res;
    proven_size_t head_size = 0;
    EXAMPLE_REQUIRE(proven_http_parse_response((proven_mem_view_t){ msg, len }, fields, 16, 0, &res,
&head_size) == PROVEN_OK &&
        head_size == head_len, "the head parses, and ends where it was written to end");
    EXAMPLE_REQUIRE(res.status == 200 && proven_u8str_view_eq(res.reason, proven_http_reason_phrase(200)),
"200 OK");
    EXAMPLE_REQUIRE(proven_http_response_keep_alive(&res), "an HTTP/1.1 response leaves the connection open");

    /* The body's framing depends on the request it answers: a response to HEAD has none. */
    proven_http_framing_t framing;
    EXAMPLE_REQUIRE(proven_http_response_framing(&res, PROVEN_HTTP_HEAD, &framing) == PROVEN_OK &&
framing.kind == PROVEN_HTTP_BODY_NONE,
        "had this answered HEAD, no body would follow");
    EXAMPLE_REQUIRE(proven_http_response_framing(&res, PROVEN_HTTP_GET, &framing) == PROVEN_OK && framing.kind
== PROVEN_HTTP_BODY_CHUNKED,
        "it answers GET, and the body is chunked");

    proven_http_body_t body;
    proven_byte_t text[64];
    proven_size_t text_len = 0;
    proven_size_t pos = head_size;
    bool done = false;
    EXAMPLE_REQUIRE(proven_http_body_init(&body, framing, sizeof text) == PROVEN_OK, "accept at most what the
buffer holds");
    while (!done && pos < len) {
        proven_size_t used = 0;

```

```

    proven_mem_view_t payload;
    if (proven_http_body_feed(&body, (proven_mem_view_t){ msg + pos, len - pos }, &used, &payload,
&done) != PROVEN_OK) break;
    for (proven_size_t i = 0; i < payload.size; ++i) text[text_len + i] = payload.ptr[i];
    text_len += payload.size;
    pos += used;
}
EXAMPLE_REQUIRE(done && pos == len && proven_u8str_view_eq((proven_u8str_view_t){ text, text_len },
PROVEN_LIT("written in two chunks")),
    "the body, decoded");

/* The date comes back as the time it was made from, to the second. */
proven_u8str_view_t date_text;
proven_time_t parsed = 0;
EXAMPLE_REQUIRE(proven_http_header_find(fields, res.header_count, PROVEN_LIT("Date"), &date_text) &&
    proven_http_date_parse(date_text, now, &parsed) == PROVEN_OK && parsed == now - now %
1000000000,
    "the Date header parses back to the same second");

/* A request is written the same way. */
len = 0;
err = proven_http_write_request_line(out, &len, PROVEN_LIT("GET"), PROVEN_LIT("/search?q=a%20b"));
if (err == PROVEN_OK) err = proven_http_write_header(out, &len, PROVEN_LIT("Host"),
PROVEN_LIT("example.com"));
if (err == PROVEN_OK) err = proven_http_write_head_end(out, &len);
EXAMPLE_REQUIRE(err == PROVEN_OK && proven_u8str_view_eq((proven_u8str_view_t){ msg, len },
    PROVEN_LIT("GET /search?q=a%20b HTTP/1.1\r\nHost: example.com\r\n\r\n")), "a request head,
byte for byte");

return EXAMPLE_OK();
}

```

## 8. Dates

HTTP writes one date format and must read three.

| API                                            | Intent                                                                                              | Return                                  |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------|
| proven_http_date_format(wall_ns, out[29])      | Sun, 06 Nov 1994 08:49:37 GMT.<br>Always 29 bytes<br>(PROVEN_HTTP_DATE_SIZE), always GMT, no NUL.   | proven_err_t: INVALID_ARG before 1970.  |
| proven_http_date_parse(text, now_ns, &wall_ns) | Read that form, the obsolete Sunday, 06-Nov-94 08:49:37 GMT, and asctime's Sun Nov 6 08:49:37 1994. | proven_err_t: INVALID_FORMAT; OVERFLOW. |

Pass the **wall clock** (proven\_time\_now), never the monotonic one, which has no date ([Chapter 5 section 4](#)).

The parser checks that the date exists - 30 February is refused, and so is a weekday that does not match its date. now\_ns is used for one rule only: a two-digit year, which the obsolete form has, is read as the most recent year with those digits that is not more than 50 years in the future.

PROVEN\_ERR\_OVERFLOW is a valid date that proven\_time\_t cannot hold: its nanoseconds end in April 2262. Servers send such dates on purpose - Expires in the year 9999 means "never" - so treat that answer as "far in the future", not as a malformed header.

## 9. What is not here

- **A client or a server.** The drivers that read, write, time out and keep connections alive are built on this codec and Chapter 9, and will have their own chapter.
- **HTTP/2 and HTTP/3.** A head that says HTTP/2.0 is `PROVEN_ERR_UNSUPPORTED`.
- **Content codings.** `gzip` and the rest: the library has no `DEFLATE` yet.
- **Trailers.** After a chunked body they are checked and skipped, not returned.
- **Cookies, authentication, multipart bodies, ranges.** Headers are text to this layer.
- **Relative URLs** and resolving one against a base.

## Chapter 6: Execution, Aliases, PAL, Freestanding, and Cross Builds

**Part VI - Going further. Prerequisites: Parts II-V.** After this chapter you can run work on more than one thread without the memory model surprising you, write a state machine that reads like a loop, and build for a target that has no operating system.

This chapter covers `coro.h`, `job.h`, `alias_xcv.h`, thread-safety and pointer provenance, PAL contracts, freestanding mode, and platform build notes. This is the hardest material in the manual, and it is last on purpose - nothing in Parts I-V depends on it.

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### 1. Stackless coroutines

The coroutine macros implement stackless state machines using `__LINE__` labels inside a switch. A coroutine function should return `proven_i32`: 0 means yielded or waiting, 1 means done.

#### Structure

```
typedef struct {
    proven_i32 state;
} proven_coro_t;
```

#### Macros

| Macro                                   | Intent                                                   |
|-----------------------------------------|----------------------------------------------------------|
| <code>PROVEN_CORO_INIT(c)</code>        | Set state to zero before first use.                      |
| <code>PROVEN_CORO_BEGIN(c)</code>       | Begin coroutine switch block.                            |
| <code>PROVEN_CORO_YIELD(c)</code>       | Save resume point and return 0.                          |
| <code>PROVEN_CORO_AWAIT(c, cond)</code> | Return 0 until <code>cond</code> is true, then continue. |
| <code>PROVEN_CORO_END(c)</code>         | Mark done and return 1.                                  |
| <code>PROVEN_CORO_IS_DONE(c)</code>     | Check whether state is -1.                               |

#### How it expands (and the one rule that matters)

These macros are the classic Protothreads/Duff's-device trick. `BEGIN` opens a switch (`state`), each `YIELD/AWAIT` records `__LINE__` as the resume label and returns, and `END` closes the switch. Conceptually:

```
proven_i32 f(Ctx *c) {
    switch (c->coro.state) {          /* PROVEN_CORO_BEGIN */
    case 0:
        /* ... code before the first yield ... */
        c->coro.state = 42; return 0; case 42:; /* PROVEN_CORO_YIELD on line 42 */
        /* ... code after the yield ... */
    }
    c->coro.state = -1; return 1; /* PROVEN_CORO_END */
}
```

The function **returns** at each yield and is **re-entered from the top**, jumping straight to the resume case. The consequence - the single rule to remember:

**Local (stack) variables do NOT survive a yield.** Anything whose value must persist across a YIELD/AWAIT must live in the coroutine's own struct (or be static), because the locals are re-created on every call.

Other constraints that follow from the expansion: two coroutine macros must not share a source line (they'd collide on `__LINE__`), and a YIELD/AWAIT must not sit inside a switch of your own (it would land in the wrong case). The coroutine has no stack, so it cannot yield from a helper function it calls - only from its own body.

### Counter-example - a local across a yield

```
static proven_i32 bad(Ctx *c) {
    PROVEN_CORO_BEGIN(&c->coro);
    int i = 0;                      /* WRONG: a local */
    while (i < 3) {
        i += 1;
        PROVEN_CORO_YIELD(&c->coro); /* on re-entry `i` is re-initialized to 0 */
    }
    PROVEN_CORO_END(&c->coro);      /* loop never ends: infinite yields */
}
/* RIGHT: put `i` in Ctx (like `value` in the Counter example below). */
```

Example (a sketch: a coroutine is a whole function, so it cannot be shown as a statement fragment - the compiled version of this shape is the worked example in [section 8](#), `manual/examples/ex_06_coro.c`):

```
typedef struct Counter {
    proven_coro_t coro;
    int value;
} Counter;

static proven_i32 counter_next(Counter *c) {
    PROVEN_CORO_BEGIN(&c->coro);
    c->value = 0;
    while (c->value < 3) {
        c->value += 1;
        PROVEN_CORO_YIELD(&c->coro);
    }
    PROVEN_CORO_END(&c->coro);
}

Counter c = {0};
PROVEN_CORO_INIT(&c.coro);
while (!counter_next(&c)) {
    use_value(c.value);
}
```

## 2. Job system

The job system owns worker threads and a bounded MPMC-style queue. Producers submit `routine(arg)` work items. Workers execute queued jobs until the system is closed and drained.

### Structures

```
typedef struct {
    void (*routine)(void *arg);
    void *arg;
} proven_job_t;

typedef struct proven_job_sys proven_job_sys_t;

proven_job_sys_t is opaque.
```

## Functions

| API                                                                                  | Intent                                                                                                                        | Return                                                                                                 |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>proven_job_system_init(alloc, num_workers, max_queue_capacity, out_sys)</code> | Allocate and start a job system. Queue capacity must be a power of two.                                                       | <code>proven_err_t</code> .                                                                            |
| <code>proven_job_system_close(sys)</code>                                            | Stop accepting new jobs.                                                                                                      | <code>void</code> .                                                                                    |
| <code>proven_job_system_destroy(sys)</code>                                          | Close if needed, drain queue, join workers, free resources.                                                                   | <code>void</code> .                                                                                    |
| <code>proven_job_submit(sys, routine, arg)</code>                                    | Submit one job. Thread-safe with other submitters.                                                                            | true if queued, false if full or closed.                                                               |
| <code>proven_job_execute_one(sys)</code>                                             | Let the calling thread execute one available job.                                                                             | true if a job ran.                                                                                     |
| <code>proven_job_submit_ex(sys, routine, arg)</code>                                 | Submit, saying why a refusal happened.                                                                                        | PROVEN_OK; PROVEN_ERR_AGAIN when full (retry or help);<br>PROVEN_ERR_INVALID_STATE when closed (stop). |
| <code>proven_job_group_init(&amp;group)</code>                                       | Start counting a batch of jobs.                                                                                               | <code>void</code> .                                                                                    |
| <code>proven_job_group_submit(sys, &amp;group, routine, arg)</code>                  | Submit a job the group counts. A refused job is not counted.                                                                  | as <code>submit_ex</code> .                                                                            |
| <code>proven_job_group_pending(&amp;group)</code>                                    | Jobs of the group not finished yet.                                                                                           | <code>proven_size_t</code> .                                                                           |
| <code>proven_job_group_wait(sys, &amp;group)</code>                                  | Return when every counted job has run; the caller runs queued jobs while it waits. What the jobs wrote is visible afterwards. | <code>void</code> .                                                                                    |

```

/* Render 16 tiles on the pool and wait for all of them. render_tile is the
 * program's own job routine; the wait runs queued tiles itself while it waits. */
void render_tile(void *tile);
static int tiles[16];
proven_job_sys_t *pool = NULL;
if (proven_is_ok(proven_job_system_init(alloc, 4, 64, &pool))) {
    proven_job_group_t frame;
    proven_job_group_init(&frame);
    for (int i = 0; i < 16; ++i) {
        proven_err_t e = proven_job_group_submit(pool, &frame, render_tile, &tiles[i]);
        if (e == PROVEN_ERR_AGAIN) { (void)proven_job_execute_one(pool); --i; continue; } /* full: help,
retry */
        if (e != PROVEN_OK) break; /* closed */
    }
    proven_job_group_wait(pool, &frame);
    proven_println("{ } tiles still pending", PROVEN_ARG(proven_job_group_pending(&frame)));
    proven_job_system_close(pool);
    if (proven_job_submit_ex(pool, render_tile, &tiles[0]) == PROVEN_ERR_INVALID_STATE) {
        proven_println("closed: no more work is taken"); /* not "full - try again" */
    }
    proven_job_system_destroy(pool);
}

```

Important constraints:

- `proven_job_system_close()` may race with submitters. A submission admitted before `close` either commits or returns `false` before `close` returns; later submissions are rejected.
- `proven_job_system_destroy()` must not race with producer threads still calling `proven_job_submit()`.
- To use `close` as the stop signal, call `close`, join every producer, and only then `destroy`.
- Queue sequence counters assume they do not wrap beyond signed pointer-difference range during one job-system lifetime.

### Concurrency model

The queue is a fixed-size ring buffer of `max_queue_capacity` slots (must be a power of two, so a slot index is `seq & (capacity - 1)`). Producers and consumers do not lock the whole queue; they coordinate with **atomic sequence counters**:

- A producer (`proven_job_submit`) atomically claims the next enqueue position. If claiming would overrun the slowest un-consumed slot, the queue is full and `submit` returns `false` rather than waiting for capacity or overwriting pending work. Posting the worker wake may briefly enter platform synchronization.
- A consumer (a worker thread, or your own thread via `proven_job_execute_one`) atomically claims the next dequeue position, reads the `routine/arg`, marks the slot reusable, and runs the routine outside the claim.

An idle worker blocks on a platform counting semaphore instead of repeatedly polling and yielding. A successful `submit` publishes its slot before posting one permit. A permit remains available when no worker is waiting, so a wake cannot be lost between an empty-queue observation and the wait. `close` first stops admission and waits for already-admitted submissions to finish, then posts one final permit per worker. If an external `proven_job_execute_one` consumes a job before a worker consumes its permit, the retained permit is merely stale: the worker performs an empty recheck and parks again, or exits after `close`.

Because `submit` and `execute` are MPMC-safe, *many* threads may submit and *many* may consume at once. What the library does **not** do for you: it does not synchronize the *data your jobs touch*. Two jobs that write the same variable still need their own locking/atomics - the queue only orders the handoff, not the work. For the thread-safety of the allocators those jobs use, the pointer-provenance hazards of sharing allocations, and the lock-free toolbox (CAS, ABA, tagged pointers, hazard pointers, epoch-based reclamation), see Chapter 2 section 7 "Allocator thread-safety & provenance".

Lifecycle state machine (drive it in this order):

```
init ---> running --(close)--> closed --(destroy: drain + join)--> freed
```

- `init` reserves `num_workers` OS threads and the slot buffer up front.
- `close` flips a flag so every later `submit` returns `false`; already-queued and in-flight submissions settle before `close` returns, and every accepted job still runs.
- `destroy` closes if needed, then **blocks the calling thread until the queue is empty and every worker has joined**, then frees everything.

**Memory visibility.** A worker joining inside `destroy` is a synchronization point: every memory effect of every job is visible to the thread that called `destroy` *after* `destroy` returns. So the safe pattern for collecting results is "submit all -> close -> destroy -> read results". Reading a job's output from another thread *before* that join needs your own synchronization.

### Counter-examples

```
/* WRONG: capacity must be a power of two. */
proven_job_system_init(alloc, 4, 100, &sys); /* 100 is not 2^n */

/* WRONG: reading a result before the join makes it visible. */
int total = 0;
```

```
(void)proven_job_submit(sys, accumulate, &total);
proven_println("total = {}", PROVEN_ARG(total)); /* data race: job may still be running */
/* RIGHT: */
proven_job_system_close(sys);
proven_job_system_destroy(sys); /* joins workers -> writes now visible */
proven_println("total = {}", PROVEN_ARG(total));
```

```
/* WRONG: ignoring the submit result drops work silently when the ring is full. */
proven_job_submit(sys, work, arg); /* [[nodiscard]]: a false return means NOT queued */
```

Example (a sketch, because the job routine has to be a function of its own; the compiled version, with the atomics a shared counter really needs, is the worked example in [section 8](#), manual/examples/ex\_06\_job.c):

```
static void increment(void *arg) {
    int *p = arg;
    *p += 1;
}

proven_job_sys_t *sys = NULL;
proven_err_t e = proven_job_system_init(alloc, 2, 64, &sys);
if (!proven_is_ok(e)) return e;

int value = 0;
if (!proven_job_submit(sys, increment, &value)) {
    proven_job_system_close(sys);
    proven_job_system_destroy(sys);
    return PROVEN_ERR_BUSY;
}

proven_job_system_close(sys);
proven_job_system_destroy(sys);
```

The lifecycle itself, with no routine in sight, is ordinary statement code:

```
proven_job_sys_t *sys = NULL;
proven_err_t e = proven_job_system_init(alloc, 2, 64, &sys); /* capacity: power of two */
if (proven_is_ok(e)) {
    /* ... submit work here, from this thread or from producers you own ... */
    proven_job_system_close(sys); /* every later submit now returns false */
    proven_job_system_destroy(sys); /* drains the queue, joins the workers */
}
```

### 3. Threads, allocators, and pointer provenance

**This section moved here from Chapter 2.** It was the hardest material in the manual sitting in one of the first chapters a reader opens, and nothing in Chapters 1-5 depends on it. If you are sharing proven objects between threads, or building a lock-free structure on top of one, this is the section you need; if you are not, you can skip it entirely and nothing later will suffer.

The `proven_allocator_t` trait is just a `ctx` plus three function pointers; it contains **no synchronization of its own**. Whether concurrent use is safe depends entirely on the concrete allocator behind the trait, and on how you share the *objects* the allocator produces. This section states the guarantees, then explains the deeper pointer-provenance hazards and the lock-free concepts you would need if you build concurrent structures on top.

#### 3.1 What is and isn't thread-safe

| Allocator | Concurrent use of one instance | Why |
|-----------|--------------------------------|-----|
|-----------|--------------------------------|-----|

|                         |                                               |                                                                                                                                                       |
|-------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| proven_heap_allocator() | <b>Safe</b> for concurrent alloc/realloc/free | The trait is stateless (ctx == NULL); it forwards to the platform aligned_alloc/posix_memalign/_aligned_malloc+free, which are thread-safe since C11. |
| proven_arena_t          | <b>Not safe</b>                               | proven_arena_alloc_* does a non-atomic read-modify-write of arena->offset.                                                                            |
| proven_pool_t           | <b>Not safe</b>                               | proven_pool_* pop/push the free-list (bin, bin_len) non-atomically.                                                                                   |

Two rules follow:

1. **"Allocator safe" is not "object safe."** Even with the heap allocator, the containers built on it (proven\_u8str\_t, proven\_array\_t, proven\_map\_t, ...) add no internal locks. A proven\_u8str\_t that one thread appends to while another reads it is a data race regardless of how thread-safe the allocator is. Shared mutable objects always need *your* synchronization.
2. **Arena and pool must not be shared.** Concurrent proven\_arena\_alloc can tear offset and hand the *same bytes* to two threads or drop an allocation entirely; concurrent pool pop/push can hand out the same slot twice or underflow bin\_len. A data race is undefined behavior on its own - see section 7.3 for why the consequences are worse than "a wrong number."

### 3.2 Pointer provenance in one paragraph

In C's abstract machine every pointer carries **provenance**: the identity of the storage instance it was derived from. Two rules matter here: (a) using a pointer outside the lifetime or bounds of its provenance object is undefined; and (b) the optimizer is allowed to assume that **pointers with different provenance do not alias**, and to reorder or elide memory accesses on that basis. Allocation, reallocation, and freeing all create and destroy provenance - which is exactly why they interact badly with unsynchronized sharing.

### 3.3 Where allocation + provenance bite under threads

- **realloc always relocates here.** The platform realloc is *allocate-new + copy + free-old* (an aligned block cannot be resized in place portably), so a successful grow **always** returns a new object with new provenance and ends the old one. Any retained copy of the old pointer - a cached element pointer, a borrowed proven\_mem\_view\_t/proven\_u8str\_view\_t - is now dangling. In a single thread you avoid this by not holding a view across a grow; under threads another thread can grow a shared container at *any* instant, leaving your view pointing into freed storage (rule (a): UB, plus a data race).
- **Address reuse / ABA.** free then alloc (especially the pool's free-list recycling) can return the *same address* for a *different* object with *fresh* provenance. A thread still holding the old pointer assumes the old provenance; by rule (b) the compiler may treat the two as non-aliasing even though the bytes coincide, and without a happens-before edge the new object's writes need not be visible. Same address, different provenance - still UB.
- **uintptr\_t round-trips + torn reads.** The arena converts backing.ptr to an integer for alignment math. It is careful to derive the *result* pointer by offsetting the original backing.ptr (preserving its provenance) rather than fabricating a pointer from the integer - the provenance-correct technique. But if offset is read torn under a race, the computed pointer can land *outside* the backing object, i.e. an access with no valid provenance for that location. The race produces not just a wrong offset but a pointer with no right to point there.
- **Data race x provenance reasoning = miscompilation, not just wrong values.** Because the compiler applies single-threaded, provenance-based non-aliasing reasoning within each thread,

a racy allocator can let the optimizer "prove" non-aliasing that does not hold at runtime - yielding torn pointers, double allocations the compiler believes cannot overlap, or dropped stores. The failure mode is structural, not a flaky value.

### 3.4 The lock-free toolbox (concepts, if you build concurrency on top)

proven does **not** implement any lock-free allocator or safe-memory-reclamation scheme. If you build concurrent data structures over these allocators, you supply the following yourself (<stdatomic.h> is available - the job system in Chapter 6 uses it for its queue indices). These are the standard pieces and how they relate to the provenance hazards above.

- **CAS (compare-and-swap).** The atomic primitive behind almost all lock-free code:  
`atomic_compare_exchange_strong(&p, &expected, desired)` atomically sets `p = desired` only if `p == expected`, otherwise reports the current value. It lets one thread publish a change only if no other thread changed the word first.

text /\* Treiber-stack style push (sketch): `node_t` and `n` are illustrative, so this is a listing rather than a compiled block. \*/  
`_Atomic(node_t *) head; node_t *old = atomic_load(&head); do { n->next = old; } while (!atomic_compare_exchange_weak(&head, &old, n));`

- **The ABA problem.** CAS compares *values*, not *history*. A lock-free pop reads `head == A`, plans to install `A->next`, then CASes. If, in between, other threads pop `A`, pop its successor, free them, and push `A` back (its address reused), the CAS still sees `head == A` and *succeeds* - installing a pointer to freed memory. The value matched (`A->B->A`) but the world changed. The pool's free-list is a textbook ABA candidate if naively made lock-free.
- **Tagged pointers / version counters.** Pack a monotonically increasing tag next to the pointer and CAS them together (a double-width CAS, or by stealing the low alignment bits / high bits). Every successful update bumps the tag, so an `A->B->A` sequence comes back with a *different* tag and the CAS fails - ABA detected. Costs/limits: bit-stealing needs guaranteed alignment and shrinks the usable address range; a full-width tag needs hardware double-word CAS (e.g. `cmpxchg16b`); the tag can in principle wrap. **\*\*Crucially, a tag fixes ABA *detection* on the shared word - it does not restore provenance.\*\*** The reused address is still a new object; the tag only stops you from *acting* on the stale view.
- **Hazard pointers (safe memory reclamation).** Each thread owns a few single-writer/multi-reader "hazard" slots. Before dereferencing a shared pointer it publishes that pointer into a slot and re-validates; a thread that wants to free an object first scans every hazard slot and **defers** the free (a retire list) if anyone is protecting it. This bounds memory and prevents use-after-free and reclamation-ABA, at the cost of a store + fence per protected access and a fixed number of simultaneously protected pointers per thread.
- **Epoch-based reclamation (EBR).** A global epoch counter; a thread "pins" the current epoch while touching shared structures, and retired memory is tagged with the epoch it was retired in. Memory retired in epoch *e* is freed only once every thread has been observed past *e* (a grace period of a couple of epochs). Cheaper per operation than hazard pointers (just a pinned flag), but a thread that stalls while pinned blocks all reclamation - unbounded memory growth. Variants: quiescent-state (QSBR) and interval-based reclamation.
- **How these tie back to provenance.** Reclamation schemes (hazard pointers, EBR) exist precisely to keep a freed object's *storage alive* - its provenance valid -- until no thread can still reference it. CAS + a tag keep the shared *word* consistent but say nothing about the lifetime of what it points to. That is why a correct lock-free stack typically needs **both**: a tag (for ABA on the head word) *and* an SMR scheme (for safe reclamation of popped nodes). A proven pool or arena gives you neither, so concurrency must be layered above them, not assumed.

### 3.5 Safe patterns with proven

- **Per-thread arena/pool.** Give each thread its own `proven_arena_t` / `proven_pool_t`. No sharing means no race and no cross-thread provenance - the simplest correct design, and usually the fastest.
- **Heap for cross-thread alloc/free, but synchronize the objects.** It is fine for thread A to allocate and thread B to free via `proven_heap_allocator()`; it is *not* fine for both to mutate the same `proven_array_t`/`proven_u8str_t` without a lock.
- **Hand off ownership with a happens-before edge.** Transfer a pointer to another thread through a mutex, atomic with release/acquire ordering, or a queue (like the job system) so that only one thread observes it at a time. This keeps each allocation's provenance confined to a single thread and makes the producer's writes visible to the consumer.
- **Do not pass borrowed views across threads** unless the owner is guaranteed not to grow/move/free for the whole duration of the borrow - and remember a grow here always relocates.
- **If you must share an arena/pool, wrap it** in your own mutex (or build a real lock-free allocator with the tools in section 7.4); the built-in ones assume a single owner at a time.

## 4. Alias layer

### Why a second set of names exists

`proven_u8str_view_slice` is 26 characters. In a file that calls twenty such functions, the prefix is a third of the line, and the part that varies - the part you actually read - is squeezed into what is left.

Prefixes are not decoration in C. The language has one global namespace for functions, so a library that exports `slice` or `create` will eventually collide with something else the program links. The prefix is what makes a C library safe to combine with others, and it is why every serious C library has one.

`alias_xcv.h` is the compromise: a shorter `xcv_` spelling for every public symbol, in a header you opt into. Include it and `xcv_u8str_view_slice` works; do not, and the short names do not exist to collide with anything.

**The canonical names remain the source of truth** - for the ABI, for this manual, and for the tests. The alias layer is a spelling, not an API: nothing is exported under the short name that is not exported under the long one, and a completeness gate (`tests/test_docs_alias_completeness.c`) fails the build if a public function is ever added without its alias, because an alias layer with holes is worse than none - a caller who adopts it discovers the gaps one compile error at a time.

Wrong - expecting the aliases without including the header:

```
#include "proven.h"
xcv_u8str_view_t v; /* wrong: proven.h does not pull in the alias layer */
```

Wrong - mixing the two spellings for the same idea in one file:

```
proven_result_u8str_t s = xcv_u8str_create(alloc, 32); /* wrong: pick one and stay with it */
```

Both compile. Neither is a good idea: a reader now has to know both vocabularies to follow one function.

Include it after the canonical headers:

```
#include "proven.h"
#include "proven/alias_xcv.h"
```

```
xcv_allocator_t heap = xcv_heap_allocator();
(void)heap;
xcv_println("{} ", XCV_ARG(123));
```

Alias design rules:

- Aliases are preprocessor conveniences, not a separate ABI.

- The canonical API remains the source of truth.
- Treat `alias_xcv.h` as a template if a project wants its own shorter local prefix.
- Keep alias tests synchronized when public API names are added or removed.

Alias macro families:

- Formatting: `XCV_ARG`, `XCV_ARG_CSTR`, `XCV_ARG_CSTR_N`, `XCV_FMT_IS_OK`.
- Containers: `XCV_ARRAY_*`, `XCV_RING_*`, `XCV_MAP_*`, `XCV_LIST_*`.
- Errors and arithmetic: `XCV_ERR_*`, `XCV_IS_OK`, `XCV_CKD_*`.
- Coroutines: `XCV_CORO_*`.
- Filesystem and sysio: `xcv_fs_*`, `xcv_print*`, `xcv_scan*` aliases.
- Types and allocators: `xcv_*_t` type aliases and allocator aliases.

For an exhaustive source-grounded alias table, see [Chapter 7: Alias Index](#).

## 5. PAL contract

### Why the syscalls are quarantined

Every library that touches an operating system has to decide where the OS-specific code goes. The usual answer is "wherever it is needed", with `#ifdef _WIN32` sprinkled through the source. That works, and it has two costs that compound: nobody can tell what the library actually requires from the OS, and porting means auditing every file.

This library puts all of it in one directory. `platform/` **is the only place that makes a syscall**.

Everything in `src/proven/` is portable C that calls through the PAL, which means:

- **The requirement list is a file listing.** What this library needs from an operating system is exactly the set of `proven_sys_*` functions - nothing hidden, nothing discovered late.
- **Porting is bounded.** A new target reimplements `platform/`. Nothing in `src/proven/` changes, and the tests that exercise the portable half keep passing.
- **Freestanding is the same mechanism, not a special case.** Build without the hosted PAL files and the portable core is what remains. That is why [the freestanding guide](#) is a list of which files to leave out rather than a separate port.

The PAL is **internal**. `proven_sys_*` functions are not the public API and their signatures may change; the public wrappers in `fs.h`, `sysio.h`, `time.h` and `random.h` are what you call. The symbols gate knows this - it exempts the PAL from the "must be documented" requirement precisely because it is not a surface you are meant to program against.

Wrong - calling the PAL directly from application code:

```
proven_sys_fs_open(path, flags); /* wrong: internal. Use proven_fs_open. */
```

You lose the error translation, the argument validation, and any guarantee that the call keeps working next release.

PAL areas:

- `proven_sys_mem`: heap allocation backend.
- `proven_sys_fs`: files, directories, paths, links, permissions, locks.
- `proven_sys_time`: the wall clock, the monotonic clock, sleep, time breakdown.
- `proven_sys_net`: sockets, readiness, name resolution.
- `proven_sys_env`: environment variables.
- `proven_sys_thread`: threads and synchronization for the job system.
- `proven_sys_io`: standard stream I/O.
- `proven_sys_math`: math helpers where needed.

Public application code should prefer high-level APIs:

- `proven_fs_*` for filesystem operations.

- `proven_sysio_*` and `print/scan` macros for console I/O.
- `proven_time_*` for time.
- `proven_job_*` for threaded jobs.

PAL internal exception: thread lifecycle code may need internal platform allocation to keep opaque OS metadata alive across entry points. That exception stays inside PAL code and should not leak into core container APIs.

## 6. Freestanding subset

`PROVEN_FREESTANDING` builds a reduced subset for OS-free or libc-minimal targets. The current freestanding build also defines:

```
-DPROVEN_FREESTANDING -DPROVEN_FMT_NO_FLOAT -DPROVEN_NO_U16STR -ffreestanding
```

Available modules in the current freestanding configuration:

- `types`
- `error`
- `memory`
- `align`
- `allocator`
- `arena`
- `pool`
- `buffer`
- `array`
- `list`
- `ring`
- `map`
- `algorithm`
- `u8str`
- `coro`
- `scan`
- `fmt` without floating-point formatting
- `panic`

Excluded or stubbed modules:

- `heap`: compiled, but `proven_heap_allocator()` returns an invalid zero allocator.
- `u16str`: excluded by `PROVEN_NO_U16STR` in the current freestanding build.
- `time`: partial. `src/proven/time.c` is compiled, so `proven_time_breakdown()` and the datetime formatters are available, but the clock backend (`platform/proven_sys_time.c`) is not - so `proven_time_now()`, `proven_time_now_datetime()`, and `proven_time_sleep()` have no implementation to link against.
- `fs`, `sysio`, `mmap`, `job`, `net`: hosted/PAL services excluded from the current freestanding subset. A hosted build can also leave `net` out with `-DPROVEN_NO_NET`.

See `manual-freestanding.md` for the exact source list and command examples.

## 7. Cross compilation

### Why the build compiles for targets it cannot run

A library that claims to be portable is making a claim that decays silently. Code that assumed 64-bit pointers, or that `char` is signed, or that unaligned loads are fine, compiles perfectly on the machine it was written on and breaks on an ARM board six months later - and the commit that broke it looked harmless.

`./nob` cross compiles the library for every target in a matrix without running anything. The configured MinGW lanes additionally link a smoke executable. That sounds weak and is not: **most portability failures are compile-time failures**. A type that is not the width you assumed, a missing intrinsic, an alignment requirement the target actually enforces, a header that does not exist there - all of them fail the compiler, on this machine, in seconds, in the commit that introduced them.

What a compile-only check cannot catch is behaviour: endianness bugs, alignment faults that only happen at run time, and anything timing-dependent. Those need real hardware or an emulator, and the matrix does not pretend otherwise. It is the cheap half of portability testing, run on every build, rather than the expensive half run never.

Cross builds are also where the freestanding profile is exercised for real targets - Cortex-M, RISC-V and wasm32 among them - so the "no operating system" claim is checked by a compiler rather than by a paragraph in a guide.

Wrong - treating a green cross matrix as proof the library runs on that target:

```
/* wrong: ./nob cross compiles and links objects. It does not execute anything.
   Endianness, alignment faults and real timing still need the hardware. */
```

The build driver command is:

```
./nob cross -build-root build-out/proven_c_lib
```

Current target categories:

- Native GCC hosted.
- Native Clang hosted when available.
- Linux AArch64 hosted compile-only.
- Linux ARM hard-float hosted compile-only.
- Linux i686 hosted compile-only through `i686-linux-gnu-gcc` OR `gcc -m32`.
- Windows x86\_64 and i686 WinAPI compile-and-link smoke checks through MinGW.
- ARM Cortex-M freestanding compile-only.
- RISC-V ELF freestanding compile-only.

Rules:

- Missing optional compilers are skipped.
- Real compile errors fail the command.
- Cross builds are compile-only except for the configured MinGW compile-and-link smoke; none is runtime verification.
- Runtime behavior still needs a target runner, emulator, device, or OS environment.

## 8. Examples and misuse cases

### Coroutine macros must not share one source line

Wrong:

```
PROVEN_CORO_YIELD(&c->coro); PROVEN_CORO_YIELD(&c->coro);
/* wrong: both use the same __LINE__ value */
```

Correct:

```
PROVEN_CORO_YIELD(&c->coro);
PROVEN_CORO_YIELD(&c->coro);
```

### Coroutine local variables are ordinary locals

A stackless coroutine returns to the caller. Any local variable whose value must survive across yields should live in the coroutine state object, not as an automatic local inside the coroutine function.

Wrong:

```
static proven_i32 next(Task *t) {
    int temporary = 0;
    PROVEN_CORO_BEGIN(&t->coro);
    temporary = 42;
    PROVEN_CORO_YIELD(&t->coro);
    use_int(temporary); /* wrong assumption: ordinary local lifetime/state is not persistent */
    PROVEN_CORO_END(&t->coro);
}
```

Correct:

```
typedef struct Task {
    proven_coro_t coro;
    int temporary;
} Task;
```

### Job destroy must not race with submitters

Wrong:

```
/* thread A */
proven_job_system_destroy(sys);

/* thread B at the same time */
proven_job_submit(sys, work, arg); /* wrong: external synchronization missing */
```

Correct:

```
proven_job_system_close(sys); /* may race with submit and makes it return false */
join_producer_threads();      /* no producer can access sys after this point */
proven_job_system_destroy(sys);
```

### Job argument lifetime

Wrong:

```
void submit_bad(proven_job_sys_t *sys) {
    int value = 10;
    proven_job_submit(sys, work, &value);
} /* wrong: value may be dead before work runs */
```

Correct:

```
JobData *data = allocate_job_data();
proven_job_submit(sys, work_and_free_data, data);
```

### Alias layer should not hide canonical docs

Wrong:

```
/* Document only xcv_map_t and forget proven_map_t. */
```

Correct: document canonical proven\_ API first, then mention aliases as optional local spelling.

### Worked example: the bounded job system

Compiled and run by the test suite. Note the ordering the contract requires: submit, then close, then destroy. proven\_job\_system\_destroy must not race with proven\_job\_submit.

```
#include <stdatomic.h>
```

```
/*
 * The job system: worker threads plus a bounded atomic MPMC queue. Idle workers
 * park on platform synchronization. It orders the *handoff* of work - it does
 * not synchronize the data the work touches. That is why the counter below is
 * an atomic and not a plain int: two jobs incrementing the same variable is a
 * data race unless the caller says otherwise.
 */
```

```

*
* The lifecycle is a straight line, and it is not optional:
*
*   init -> submit... -> close -> destroy
*
* destroy must not race with submit. Nothing in the library enforces that; the
* caller has to stop its producers first. Here there is only one producer - this
* thread - so "stop the producers" means "finish the submit loop before closing".
*/

#define JOB_COUNT 64

static void increment(void *arg) {
    atomic_int *counter = arg;
    /* relaxed is enough: we only need the total to be right, not to order anything
     * against it. The join inside destroy is what publishes the result to us. */
    atomic_fetch_add_explicit(counter, 1, memory_order_relaxed);
}

int main(void) {
    proven_allocator_t alloc = proven_heap_allocator();

    proven_job_sys_t *sys = NULL;
    /* Queue capacity must be a power of two - the ring maps a sequence number to a
     * slot with a mask. Sized above JOB_COUNT so a submit cannot find the ring
     * full even if every worker is still starting up. */
    proven_err_t err = proven_job_system_init(alloc, 4, 128, &sys);
    EXAMPLE_REQUIRE(proven_is_ok(err), "starting a job system with 4 workers should succeed");
    if (!proven_is_ok(err)) return 1;

    /* Lives until after destroy: a job's arg must outlive the job, and jobs run
     * until destroy has drained the queue. */
    atomic_int counter = 0;

    proven_size_t submitted = 0;
    for (proven_size_t i = 0; i < JOB_COUNT; ++i) {
        /* submit returns false when the ring is full or the system is closed. It
         * never waits for queue capacity and never drops work silently; its wake
         * path may briefly enter platform synchronization. Ignoring the answer is
         * how you lose jobs, which is why it is [[nodiscard]]. */
        if (!proven_job_submit(sys, increment, &counter)) {
            /* A real caller would back off and retry, or run the job inline with
             * proven_job_execute_one. Here a full ring means the sizing above is
             * wrong, so say so rather than paper over it. */
            EXAMPLE_REQUIRE(false, "the queue was sized to hold every job");
            break;
        }
        ++submitted;
    }

    /* This thread is the only producer, and it is done submitting - so it is safe
     * to close. close makes every later submit fail; jobs already queued still run. */
    proven_job_system_close(sys);

    /* destroy blocks until the queue is empty and every worker has been joined.
     * That join is the synchronization point: after destroy returns, every memory
     * effect of every job is visible to this thread. Reading `counter` before this
     * line would be reading a value the workers are still writing. */
    proven_job_system_destroy(sys);
}

```

```

int ran = atomic_load(&counter);
EXAMPLE_REQUIRE(submitted == JOB_COUNT, "every job should have been accepted");
EXAMPLE_REQUIRE(ran == (int)submitted, "every submitted job should have run exactly once");

printf("submitted %zu jobs, %d ran\n", (size_t)submitted, ran);

return EXAMPLE_OK();
}

```

### Worked example: a stackless coroutine

Compiled and run by the test suite. The rule that catches everyone: locals do not survive a yield, so every piece of state the coroutine needs across a suspension has to live in its struct.

```

/*
 * A stackless coroutine is a switch statement in disguise: BEGIN opens a
 * switch on the saved state, each YIELD records __LINE__ as a resume label and
 * *returns*, and the next call re-enters the function from the top and jumps
 * straight back to that label.
 *
 * Everything that follows comes from that one fact:
 *
 * - Locals do NOT survive a yield. The function returned; its stack frame is
 *   gone. Anything that must persist lives in the coroutine's own struct - which
 *   is what `value` and `remaining` are doing below.
 * - Two coroutine macros must not share a source line (they would collide on
 *   __LINE__).
 * - It cannot yield from a helper it calls: there is no stack to suspend.
 *
 * The payoff is that a suspended coroutine costs exactly its struct - four bytes
 * of state plus whatever you put next to it - and no thread, no stack, no context
 * switch.
 */

typedef struct {
    proven_coro_t coro;
    /* The generator's state. These would be `int i` locals in a normal loop; here
     * they have to be fields, or they would be reset to their initial values on
     * every resume and the loop would never end. */
    int value;
    int remaining;
} squares_t;

/* A coroutine returns proven_i32: 0 = suspended (call me again), 1 = done. */
static proven_i32 squares_next(squares_t *g) {
    PROVEN_CORO_BEGIN(&g->coro);

    g->remaining = 5;
    g->value = 1;

    while (g->remaining > 0) {
        g->value = g->value * g->value;
        PROVEN_CORO_YIELD(&g->coro);    /* the caller reads g->value here */
        g->value = g->value + 1;          /* resumes exactly on this line */
        g->remaining -= 1;
    }

    PROVEN_CORO_END(&g->coro);
}

int main(void) {

```

```

proven_allocator_t alloc = proven_heap_allocator();

/* The coroutine owns no memory, so there is nothing to destroy - but the values
 * it produces have to go somewhere, and that string does have an owner. */
proven_result_u8str_t out = proven_u8str_create(alloc, 32);
EXAMPLE_REQUIRE(proven_is_ok(out.err), "creating the output string should succeed");
if (!proven_is_ok(out.err)) return 1;

squares_t gen = {0};
PROVEN_CORO_INIT(&gen.coro); /* unconditional, exactly once, before the first call */

int produced = 0;
int last = 0;

/* Drive it to completion. squares_next returns 1 on the call that runs off the
 * end of the body - that call produces no value, so the loop body only runs
 * while it returned 0. */
while (!squares_next(&gen)) {
    proven_fmt_result_t r = proven_u8str_append_fmt_grow(alloc, &out.value, "{} ",
                                                         PROVEN_ARG(gen.value));
    EXAMPLE_REQUIRE(PROVEN_FMT_IS_OK(r), "appending a generated value should succeed");
    last = gen.value;
    ++produced;
}

/* Done is sticky: the state is -1 and stays there. Calling it again would just
 * return 1 without re-running the body. */
EXAMPLE_REQUIRE(PROVEN_CORO_IS_DONE(&gen.coro), "the generator should have finished");
EXAMPLE_REQUIRE(squares_next(&gen) == 1, "a finished coroutine stays finished");

/* 1, then (1+1)^2 = 4, then (4+1)^2 = 25, then 676, then 458329. */
EXAMPLE_REQUIRE(produced == 5, "the generator yields once per iteration");
EXAMPLE_REQUIRE(last == 458329, "the state carried across every yield");
EXAMPLE_REQUIRE(proven_u8str_view_eq(proven_u8str_as_view(&out.value),
                                     PROVEN_LIT("1 4 25 676 458329 ")),
                "the generated sequence should be exactly this");

printf("squares: %s\n", proven_u8str_as_cstr(&out.value));

proven_u8str_destroy(alloc, &out.value);
return EXAMPLE_OK();
}

```

## Proven Freestanding Mode (v0.11.0)

**Part VI - Going further. Prerequisites:** Parts II-V, and [Chapter 6](#). After this guide you can build the library for a target with no operating system and no libc, and you will know exactly which modules survive that and which do not.

This guide describes the current `PROVEN_FREESTANDING` configuration as implemented by `nob.c` and the public headers.

**A note on the shape of this guide.** Unlike the chapters, roughly half of it is procedure - flag lists, file listings, and two compile commands. Those sections are deliberately short: a compile command explained at length is a compile command nobody reads. The sections that carry a *decision* -- why this mode exists (section 0), what "excluded" really means (section 3), what your panic handler has to do (section 5), what you lose by dropping float (section 8), why hosted calls fail at link time (section 9), why the lifetime rules matter more here (section 10), and how the claims in this guide are verified (section 11) - are written out in full.

### 0. Why this mode exists, and why the whole library is shaped by it

Freestanding mode is for firmware, kernels, bootloaders, hypervisors, and anywhere else that has no operating system underneath it - no `malloc`, no `open`, no `printf`, often no `libc` at all.

The C standard has a name for this. A **hosted** implementation gives you the whole standard library and starts your program at `main`. A **freestanding** implementation guarantees only a handful of headers - `<stddef.h>`, `<stdint.h>`, `<limits.h>` and a few others - and is what a compiler targeting bare metal gives you. `-ffreestanding` tells the compiler to assume exactly that.

Most C libraries cannot be used here at all, and the reason is rarely a big one. It is a `malloc` buried three calls down, or a `printf` in an error path, or a global that is initialised by something that never runs. One hidden dependency on the host is enough to make a library unusable on a microcontroller.

**This is the constraint that produced most of this library's design decisions**, and it is worth seeing that connection, because it explains choices that otherwise look like taste:

| Design decision                                             | Read the earlier chapters as   | And the reason is                                                                                                                              |
|-------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| The allocator is a parameter ( <a href="#">Ch 2</a> )       | "so you can swap the strategy" | There is no <code>malloc</code> here. An arena over a static array is the only allocator, and code that took one as a parameter already works. |
| Errors are returned values ( <a href="#">Ch 1</a> )         | "so you cannot ignore them"    | There is no <code>errno</code> and nothing to unwind to. A return value is the only mechanism that exists.                                     |
| Panics go through a hook ( <a href="#">Ch 1 section 6</a> ) | "so you can override it"       | There is no <code>abort()</code> and no <code>stderr</code> . The default traps because that is all it can portably do.                        |
| Syscalls live only in <code>platform/</code>                | "separation of concerns"       | It is the only directory that has to be replaced for a new target. Everything in <code>src/proven/</code> is portable by construction.         |

|                                   |                              |                                                                     |
|-----------------------------------|------------------------------|---------------------------------------------------------------------|
| Views instead of C strings (Ch 3) | "so lengths cannot get lost" | strlen is libc. A view brings its length with it and needs nothing. |
|-----------------------------------|------------------------------|---------------------------------------------------------------------|

In other words: freestanding support is not a feature bolted onto the side. The rest of the manual has been describing a library built to survive here, and this guide is where you find out what survives and what does not.

### What you give up

Everything that needs the operating system, and nothing else:

- **No heap.** `proven_heap_allocator` does not exist. You supply memory - a static array is usual - and put an arena or a pool over it.
- **No filesystem, no standard streams, no clock, no OS randomness.** `fs.h`, `sysio.h`, `mmap.h`, `time.h` and the OS entropy source are all hosted services. `proven_chacha_rng_t` still works if you seed it yourself, which is what `proven_random_set_source` is for.
- **No float formatting by default.** `PROVEN_FMT_NO_FLOAT` drops it, because the float path is the largest piece of code in the formatter and most firmware never prints a double. section 8 says how to turn it back on.
- **No UTF-16 strings.** `PROVEN_NO_U16STR` is on; there is no Windows API here to talk to.

What you keep is the whole portable core: errors, views, checked arithmetic, alignment, arenas, pools, buffers, strings, arrays, maps, lists, rings, sorting, searching, hashing, encoding, coroutines, and integer formatting and scanning.

### What still bites you

The lifetime rules do not relax because the target got smaller - if anything they matter more, because there is no operating system to notice a mistake and no allocator that will refuse to reuse memory you still hold. section 10 is short and is the section people skip.

## 1. Build profile

The build driver uses these freestanding flags:

```
-std=c23
-ffreestanding
-DPROVEN_FREESTANDING
-DPROVEN_FMT_NO_FLOAT
-DPROVEN_NO_U16STR
```

The driver still probes `-std=c23` first and falls back to `-std=c2x` when the compiler uses the transitional spelling. The freestanding source contract stays C23-first even though the build wrapper accepts either flag spelling.

The hosted `./nob freestanding` command also links its local freestanding test executables statically on the build host. Cross freestanding targets in `./nob cross` use compile-only object checks.

Run the project freestanding check:

```
cc nob.c -o nob
./nob freestanding -build-root build-out/proven_c_lib
```

Run the cross-build matrix:

```
./nob cross -build-root build-out/proven_c_lib
```

## 2. Available source files

In the current freestanding build, `nob.c` compiles the portable source files except hosted-only modules.

#### Included core modules:

```
src/proven/memory.c
src/proven/arena.c
src/proven/pool.c
src/proven/buffer.c
src/proven/heap.c
src/proven/u8str.c
src/proven/array.c
src/proven/ring.c
src/proven/map.c
src/proven/algorithm.c
src/proven/hash.c
src/proven/hash_legacy.c
src/proven/utf.c
src/proven/url.c
src/proven/http.c
src/proven/encode.c
src/proven/random.c
src/proven/float_decimal.c
src/proven/float_parse.c
src/proven/float_format.c
src/proven/time.c
src/proven/fmt.c
src/proven/scan.c
src/proven/panic.c
platform/proven_sys_math.c
```

#### Excluded hosted modules:

```
src/proven/u16str.c
src/proven/fs.c
src/proven/stream.c
src/proven/sysio.c
src/proven/mmap.c
src/proven/job.c
platform/proven_sys_fs.c
platform/proven_sys_thread.c
platform/proven_sys_io.c
platform/proven_sys_env.c
platform/proven_sys_time.c
platform/proven_sys_random.c
platform/proven_sys_mem.c
```

Do not add excluded hosted modules to a bare-metal build unless you also provide a real platform backend for the target.

### 3. Module availability

#### How to read this table, and what "excluded" means

A module is excluded here for exactly one reason: it needs something only an operating system can provide. It is not that the code is untested on small targets or that somebody has not got round to it - `fs.h` needs a filesystem, `sysio.h` needs standard streams, `mmap.h` needs virtual memory, `job.h` needs threads. There is nothing to port.

The interesting rows are the two that are neither available nor excluded:

- **heap.h is a stub.** `proven_heap_allocator()` still exists and still compiles, and it returns an allocator whose function pointers are null - one that `proven_alloc_is_valid` reports as invalid. It does not silently allocate from somewhere else and it does not fail to link. That choice matters: code written against the allocator trait keeps compiling for a bare-metal target, and the place it breaks

is the one line that asked for a heap, at run time, with a check you can write. section 4 shows what to pass instead.

- **time.h is limited.** The datetime formatting is pure arithmetic over a number you supply, so it compiles. What is missing is the number: reading a clock is a syscall, so `proven_time_now` has no backend here. Format timestamps you got from your own hardware timer.

Everything marked Available is the portable core, and "available" means the same tests that run on a hosted build run against it - the freestanding profile is built and checked by `./nob freestanding` on every release, not asserted in this table.

| Module                                                                                     | Status in current freestanding profile | Notes                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>types.h</code> , <code>error.h</code> , <code>align.h</code> , <code>memory.h</code> | Available                              | Requires fixed-width integer and <code>uintptr_t</code> support.                                                                                                                                                                                                               |
| <code>allocator.h</code>                                                                   | Available                              | Trait only. Caller supplies backing allocators.                                                                                                                                                                                                                                |
| <code>alloc_check.h</code>                                                                 | Available                              | Pure bookkeeping over a caller-supplied record; reports through <code>proven_panic</code> .                                                                                                                                                                                    |
| <code>arena.h</code>                                                                       | Available                              | Primary allocator for static memory regions.                                                                                                                                                                                                                                   |
| <code>pool.h</code>                                                                        | Available                              | Uses a caller-provided base allocator, often an arena.                                                                                                                                                                                                                         |
| <code>buffer.h</code>                                                                      | Available                              | Fixed-capacity byte buffer.                                                                                                                                                                                                                                                    |
| <code>u8str.h</code>                                                                       | Available                              | U8 strings and views.                                                                                                                                                                                                                                                          |
| <code>u16str.h</code>                                                                      | Excluded                               | Current profile defines <code>PROVEN_NO_U16STR</code> .                                                                                                                                                                                                                        |
| <code>array.h</code> , <code>list.h</code> , <code>ring.h</code> , <code>map.h</code>      | Available                              | No hidden OS dependency.                                                                                                                                                                                                                                                       |
| <code>algorithm.h</code>                                                                   | Available                              | Sort/search helpers for arrays.                                                                                                                                                                                                                                                |
| <code>hash.h</code>                                                                        | Available                              | FNV-1a, SipHash-2-4, CRC-32, SHA-256 - byte-exact, no OS dependency. CRC-32 uses eight 256-entry tables (8 KiB of read-only data) by default; build the library with <code>-DPROVEN_CRC32_SMALL=1</code> to keep one 1 KiB table, about 5x slower on long inputs, same output. |
| <code>hash_legacy.h</code>                                                                 | Available                              | SHA-1 and MD5, for formats that fix them (the WebSocket accept key, old checksum files). Same shape as SHA-256; no table larger than MD5's 64 constants.                                                                                                                       |
| <code>url.h</code> , <code>http.h</code>                                                   | Available                              | URL parsing and percent-coding, and the HTTP/1.1 message codec. Pure text handling; no socket, no                                                                                                                                                                              |

|                             |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                                                  | <p>allocation. The transport is yours to supply - on a target with no sockets (wasm32, bare metal) the codec still parses and writes messages.</p> <p><code>proven_http_date_format</code> takes a time you provide; the freestanding clock returns 0.</p>                                                                                                                                                                                                                                                                                                         |
| <code>encode.h</code>       | Available                                        | Hex and Base64 - pure computation, no OS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>utf.h</code>          | Available                                        | <p>Strict UTF-8 &lt;-&gt; UTF-16 transcoding - pure computation, no OS.</p> <p><code>proven_utf8_append_to_u16str</code> is excluded with <code>PROVEN_NO_U16STR</code>; the rest works on raw <code>proven_u16</code> arrays.</p>                                                                                                                                                                                                                                                                                                                                 |
| <code>fmt.h</code>          | Available without float                          | Current profile defines <code>PROVEN_FMT_NO_FLOAT</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>scan.h</code>         | Available                                        | Scanner for memory views.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>float_parse.h</code>  | Available                                        | <p><code>proven_strtod</code>, <code>proven_parse_double_ascii</code> and <code>proven_parse_f64_ascii</code> all compile here: the decimal-to-binary64 engine is integer-only and needs no libc. The one difference is that the freestanding build does not set <code>errno</code> on overflow or underflow - the returned <code>proven_err_t</code> carries that instead, which is the value to check on a target with no <code>errno</code> at all. This is separate from <code>fmt.h</code>'s float <b>formatting</b>, which the profile does compile out.</p> |
| <code>float_format.h</code> | Available without <code>fmt.h</code> integration | <p>The binary64 formatter is integer-only and compiles, but the current profile defines <code>PROVEN_FMT_NO_FLOAT</code>, so <code>{f}</code> will not render a float. Call the <code>float_format.h</code> entry points directly if you need digits on a target where you have decided the code size is worth it.</p>                                                                                                                                                                                                                                             |

|                                               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| time.h                                        | Limited   | Core datetime formatting can compile; real PAL time is excluded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| heap.h                                        | Stub      | proven_heap_allocator() returns an invalid allocator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| fs.h, stream.h, net.h, mmap.h, sysio.h, job.h | Excluded  | Require hosted PAL services.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| random.h                                      | Available | The generators and helpers are pure arithmetic.<br>proven_random_bytes works here too, but only once you install an entropy source with proven_random_set_source - a board's TRNG, ring oscillator, or ADC noise floor. With none installed it returns <b>false</b> rather than falling back to a clock-seeded PRNG, which would look like success and be a security hole nothing reports.<br>proven_chacha_rng_seed_from_entropy then turns the board's entropy into an endless cryptographic stream. proven_random_u64 returns 0 with no source, silently; for a token or key use proven_random_u64_checked, which returns false. |
| coro.h                                        | Available | Macro-only stackless coroutine support.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| panic.h                                       | Available | Override for target-specific trap/reset behavior.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

#### 4. Minimal static arena setup

There is no heap, so the memory comes from a static block and an arena is laid over it. The arena does not own that block: `alloc` bumps an offset, `free` is a no-op, and `reset` rewinds the whole thing at once.

This is the pattern the whole library was shaped to allow, and it is worth seeing why it works. Every function that can allocate takes a `proven_allocator_t` as a parameter, so code written for a hosted build against `proven_heap_allocator()` runs here unchanged once you hand it an arena instead. Nothing in `ustr.h`, `array.h` or `map.h` knows or cares that the memory came from a static array in `.bss` rather than from `malloc`.

Three decisions to make when you size that block, none of which the library can make for you:

- **How big.** The arena cannot grow, so its size is a hard limit you are choosing at compile time. Too small and allocations start returning `PROVEN_ERR_NOMEM`; too large and you have spent RAM the rest of the firmware needed. This is the trade you are being asked to make explicitly rather than discovering it as heap fragmentation months later.

- **When to reset.** An arena's whole advantage is freeing everything at once. The natural points are a loop iteration, a received packet, a command - anywhere a batch of work has a clear end. Everything allocated during that batch dies at the reset, so **nothing may outlive it**.
- **Alignment of the backing array.** `alignas(max_align_t)` on the declaration, as below. Without it the array may start at an address that cannot hold a `double`, and the arena has nothing to fix that with.

Wrong - a view built in the arena that outlives the reset:

```
proven_u8str_view_t label = build_label(arena_alloc); /* lives in the arena */
proven_arena_reset(&arena);
send(label); /* wrong: those bytes are free space now, and the next allocation takes them */
```

On a hosted build this often survives long enough to look fine in testing. Here the next allocation gets the same bytes immediately, so it does not.

```
#include "proven/types.h"
#include "proven/memory.h"
#include "proven/arena.h"
#include "proven/array.h"
#include "proven/panic.h"

/* static storage: no OS, no heap, known at link time */
static alignas(PROVEN_MAX_ALIGN) proven_byte_t storage[4096];

proven_arena_t arena = proven_arena_create((proven_mem_mut_t){
    .ptr = storage,
    .size = sizeof storage,
});
proven_allocator_t arena_alloc = proven_arena_as_allocator(&arena);

proven_result_array_t r = PROVEN_ARRAY_INIT(arena_alloc, proven_i32, 16);
if (proven_is_ok(r.err)) {
    proven_array_t values = r.value;
    proven_err_t e = PROVEN_ARRAY_PUSH(&values, proven_i32, 42);
    (void)e;
    PROVEN_ARRAY_DESTROY(&values); /* arena free is a no-op */
}

proven_arena_destroy(&arena); /* also a no-op: the caller owns `storage` */
```

Common mistake:

```
proven_allocator_t heap = proven_heap_allocator();
heap.alloc_fn(heap.ctx, 64, 8); /* wrong: heap is invalid in PROVEN_FREESTANDING */
```

Correct:

```
proven_allocator_t heap = proven_heap_allocator();
if (!proven_alloc_is_valid(heap)) {
    /* use an arena, pool, or target-provided allocator instead */
    proven_panic("no heap on this target");
}
```

## 5. Panic handler override

### Why installing one is not optional here

On a hosted system a panic that traps is survivable: the process dies, the operating system cleans up, and something restarts it. On bare metal there is nothing underneath. A trap halts the core, and whatever the device was doing - holding a motor at speed, keeping a radio link, driving a heater - it is still doing when the CPU stops.

So the default handler is a placeholder for the one you must write, and what yours does is a product decision rather than a programming one. The usual shapes:

- **Put the hardware in a safe state, then halt.** Motors off, outputs to a known level, then spin or wait for a debugger. Correct for anything with a physical actuator.
- **Record and reset.** Write a reason code to a register or a reserved RAM area that survives reset, then trigger the watchdog. The next boot reports why the last one ended.
- **Halt loudly.** Blink an LED in a recognisable pattern. On a board with no console this is the entire diagnostic channel, and it is worth more than it sounds.

The rule from [Chapter 1 section 6](#) applies with more force here: **the handler must not return**. If it does, `proven_arena_alloc_or_panic` proceeds with a block that was never allocated, and the failure moves from "the device stopped" to "the device is writing through a null pointer".

Wrong - a handler that logs and returns:

```
static void my_panic(const char *msg) {
    uart_write(msg); /* wrong: this returns, and the caller uses a block it never got */
}
```

`proven_arena_alloc_or_panic()` and related panic paths call `proven_panic()`, which dispatches to the handler installed with `proven_set_panic_handler()`. The default handler traps.

The handler is a whole function, so this is a listing rather than a fragment:

```
#include "proven/panic.h"

static void my_panic(const char *msg) {
    (void)msg;
    /* optional: write msg to UART or crash log */
    for (;;) {
        /* or reset the MCU */
    }
}

/* install once during startup; pass NULL to restore the default */
proven_set_panic_handler(my_panic);
```

A production panic handler should not return. If it returns, the allocation result that triggered the panic is not guaranteed to be valid.

## 6. Example Cortex-M compile command

```
arm-none-eabi-gcc \
-std=c23 \
-mcpu=cortex-m4 -mthumb \
-ffreestanding -nostdlib \
-DPROVEN_FREESTANDING \
-DPROVEN_FMT_NO_FLOAT \
-DPROVEN_NO_U16STR \
-Iinclude -Iplatform \
-c src/proven/memory.c -o memory.o
```

A real firmware build should compile the included freestanding modules listed above, compile your target startup code, and link with your linker script.

## 7. Example RISC-V ELF compile command

```
riscv64-elf-gcc \
-std=c23 \
-ffreestanding -nostdlib \
-DPROVEN_FREESTANDING \
-DPROVEN_FMT_NO_FLOAT \
```

```
-DPROVEN_NO_U16STR \
-Iinclude -Iplatform \
-c src/proven/arena.c -o arena.o
```

If your toolchain uses the `riscv64-unknown-elf-gcc` name, use that compiler instead. The project cross matrix checks both names when available.

## 8. Formatting in freestanding mode

### What is left, and why that is usually enough

The formatter is portable computation - it builds bytes in a destination you supply - so almost all of it survives here. What is gone is one placeholder type: `double`.

That sounds like a big loss and rarely is. Correct float formatting means emitting the shortest decimal that reads back as the same value, on every input including subnormals, and doing that requires big-integer arithmetic and lookup tables. It is the largest single piece of code in the formatter. Most firmware formats integers, string views, characters and pointers - a sensor reading is a scaled integer, a status line is text - so the profile drops the float path by default and the binary is smaller for it.

What remains: `{}` for integers of every width, views, characters, booleans and pointers; the whole spec grammar (width, fill, alignment, base, sign); `PROVEN_ARG_OF` for your own types; and the scanner in full, including float *parsing*, which does not carry the same weight.

If you do need floats on a small target, the switch is `PROVEN_FMT_NO_FLOAT` and section 8a covers the capacity knob that goes with it. Measure the size change before deciding - on a part with 32 KB of flash it is not a rounding error.

Note the other consequence: **there is no** `proven_printf` **here**. Formatting appends into a `proven_u8str_t` or a buffer, and getting those bytes to a UART or an RTT channel is your platform code, because a freestanding build has no standard output to assume.

Float formatting is disabled by `PROVEN_FMT_NO_FLOAT`. Integer and string-view formatting remain available.

Correct (`alloc` here is an arena allocator, as in section 4 - there is no heap):

```
proven_result_u8str_t r = proven_u8str_create(alloc, 32);
if (proven_is_ok(r.err)) {
    proven_fmt_result_t f = proven_u8str_append_fmt_grow(alloc, &r.value,
                                                         "value={}", PROVEN_ARG(123));
    (void)f;
    proven_u8str_destroy(alloc, &r.value);
}
```

Wrong:

```
proven_u8str_append_fmt_grow(alloc, &s, "{}", PROVEN_ARG(3.14));
/* wrong in the current freestanding profile: float args are excluded, so
   PROVEN_ARG has no _Generic association for double and this fails to compile */
```

### 8a. Tuning the float big-integer capacity

The exact decimal-to-binary64 fallback and the big-integer division helper use a fixed-size big integer whose capacity is set by `PROVEN_FLOAT_BIGINT_LIMBS` (default 160, in `proven/float_config.h`). This is the dominant factor in their stack usage: each big integer is  $8 * \text{PROVEN\_FLOAT\_BIGINT\_LIMBS}$  bytes, and a division uses several plus 32-bit scratch.

Targets with tight stacks can lower it, for example:

```
-DPROVEN_FLOAT_BIGINT_LIMBS=48
```

The kept-significand cap (`PROVEN_FLOAT_MAX_SIGNIFICAND_DIGITS`) is derived from the capacity, so a smaller value never overflows: the parser still rounds correctly for inputs up to the derived number of significant digits and stays within one ULP for longer inputs. The Clinger and Eisel-Lemire fast paths do not use the big integer, so typical short inputs are unaffected. Keep the default (160) when exact rounding of pathological long decimals matters, since a binary64 rounding boundary can need up to 767 significant digits.

## 9. Avoid hosted APIs

### Why this is a link error rather than a run-time surprise

The hosted modules are not compiled into a freestanding build at all. Calling one is therefore a **link error** - an undefined symbol, at build time, naming the function you should not have used.

That is the design working. The alternative, which many embedded libraries choose, is to provide stubs that return an error at run time; then a call to `proven_fs_open` on a microcontroller compiles, links, ships, and fails in the field on a path nobody exercised. Here it cannot get out of the build.

The practical consequence for your own code: if a module is shared between a hosted tool and firmware, the hosted-only calls need to be behind `#ifndef PROVEN_FREESTANDING`, and the link error tells you exactly which ones you missed.

Do not call these in the current freestanding profile:

```
proven_fs_open(...);
proven_println(...);
proven_env_get(...);
proven_mmap_create(...);
proven_job_system_init(...);
```

They require hosted PAL files that are intentionally excluded.

### What your environment must provide

Exactly what any freestanding C program compiled by GCC or Clang needs, and nothing else: `memcpy`, `memmove`, `memset` **and** `memcmp`, which the compiler may call for copies and zeroing whatever the source says, and the compiler's own support library (`libgcc`: on Cortex-M the `__aeabi_*` division and soft-float helpers). Your `newlib`, `picolibc` or vendor HAL already has the four; if you have none, they are a few lines each. No `strlen`, no `malloc`, no `stdio`, no startup files.

No C library **headers** either. The library's freestanding sources include only the headers a freestanding compiler supplies itself (`<stddef.h>`, `<stdint.h>`, `<limits.h>` and their kind); where one calls `memcpy` it declares it. That was not true before v0.9.0: three files included `<string.h>` or `<math.h>`, which the embedded toolchains happen to ship and a bare Clang does not. The `wasm32` target below is what found it, and what now keeps it found.

That is checked, not hoped for: `./nob cross links` every freestanding object of the library, for Cortex-M4 and RISC-V, with a program that supplies only those four functions, using `-nostdlib -nostartfiles -static -lgcc` (`tests/test_portability_freestanding_nocrt_link.c`). A static link fails on any unresolved symbol, so a clean link is the evidence. It is not run - there is no board here - so behaviour on the target is still yours to test.

`wasm32` gets the same link with Clang and `wasm-ld`, which also refuses an undefined symbol. Its support library is compiler-rt's builtins rather than `libgcc`, and the one helper the library needs from it is `__multi3`, the 128-bit multiply. A build host need not have those builtins installed for `wasm32`, so the link program supplies that one function itself; your own `wasm32` build links the builtins. `wasm32` has no sockets and no filesystem here: it is the same freestanding subset as the other targets, not a hosted port.

## 10. Lifetime rules still apply

### The section people skip, and why it costs more here

Freestanding does not relax any of the ownership rules in [Chapter 0 section 5](#). If anything it sharpens them, for three reasons that all point the same way:

- **There is no allocator between you and the memory.** A dangling pointer into a heap block often survives for a while because the allocator has not handed that block out again. An arena hands the same bytes out on the very next allocation after a reset, so a stale view is overwritten immediately and deterministically.
- **There is nothing to catch you.** No MMU trap on a small target, no operating system to kill the process, no sanitizer in the field. A use-after-reset does not crash - it reads someone else's data and carries on, and the symptom appears somewhere unrelated.
- **The consequences are physical.** The wrong byte here can be a motor command or a radio packet.

The rules are the same ones you already know: a view is valid only while its owner is, an owned object is destroyed exactly once with the allocator that made it, and caller-owned state structs must not be copied. What changes is the margin for error, which is zero.

Freestanding does not relax container rules.

Wrong:

```
int *p = PROVEN_ARRAY_GET_MUT(&arr, int, 0);
PROVEN_ARRAY_PUSH(&arr, int, 9);
*p = 1; /* wrong: push may have moved the array */
```

Wrong:

```
proven_u8str_view_t key = make_stack_view();
PROVEN_MAP_SET_U8_BORROWED(&map, key, int, 1);
return; /* wrong if map survives after key bytes go out of scope */
```

## 11. Verification

### Why this guide is checked rather than believed

Everything above is a claim about what compiles and links without an operating system, and claims like that rot quietly. One `#include <stdio.h>` added to a portable source file in an unrelated commit and the freestanding profile is broken - on a host build nothing would notice, because `stdio.h` is right there.

So the profile is built on every release rather than described. `./nob freestanding` compiles the portable core with `-ffreestanding` and the profile's defines, links the checks below **statically** on the build host, and runs them. `./nob cross` then compiles the same profile for real embedded targets - Cortex-M, RISC-V and wasm32 among them - as a compile-only matrix.

The two checks are chosen for what they would catch:

- The **heap stub** check proves `proven_heap_allocator()` still exists and returns something `proven_alloc_is_valid` rejects. If somebody made it fail to link instead, trait-based code would stop compiling for bare metal - the failure section 3 explains this design avoids.
- The **compile check** builds a representative program against the profile, which catches a hosted header sneaking into a portable file - when the toolchain lacks that header. On the host and on the embedded toolchains `<string.h>` is always there, so it is the wasm32 target, which has no C library headers at all, that makes this check complete.

Neither runs on the actual hardware, and this guide does not claim otherwise: alignment faults, endianness and timing need a board. What is verified is the part that can be, on every build, rather than the whole thing on no build.

The project freestanding command builds and runs these local checks on the build host:

```
tests/test_portability_freestanding_heap_stub
tests/test_portability_compile_freestanding
tests/test_portability_compile_nofloat
tests/test_portability_compile_noul6str
tests/test_portability_freestanding
```

The cross command performs compile-only checks for available embedded compilers:

```
freestanding-arm-cortex-m4      arm-none-eabi-gcc -mcpu=cortex-m4 -mthumb
freestanding-riscv64-elf        riscv64-elf-gcc
freestanding-riscv64-unknown-elf riscv64-unknown-elf-gcc
freestanding-wasm32             clang --target=wasm32
```

Every target is reported as PASS, FAIL or SKIP. A missing toolchain is a SKIP - a gap in the evidence, not a pass; the freestanding targets may be skipped, the native host compilers and both Windows targets may not. Real compile or link failures fail the command. Runtime behavior still needs validation on the target or emulator.

## Appendix A: `alias_xcv.h` Alias Index (Chapter 7)

**This is a lookup table, not reading material.** It is the one file in the manual with no prose to work through: 500-odd rows mapping every `xcv_` short name to its canonical `proven_` name. Skip it on a first read and come back when you meet an `xcv_` spelling you do not recognise, or when you are deciding whether to adopt the short names in your own code.

The file keeps its `manual-07` number because that number is a stable identifier used by links and by the build; its position in the reading order is "appendix", which is what this title says.

This appendix is generated from `include/proven/alias_xcv.h`. The alias layer is optional. The canonical `proven_` and `PROVEN_` names remain the source of truth for ABI, documentation, and tests.

### Usage rule

Include aliases after the normal headers:

```
#include "proven.h"
#include "proven/alias_xcv.h"
```

Do not mix alias documentation with canonical API documentation. Use this index only as a spelling map.

### Alias table

727 aliases: 557 lowercase `xcv_` names and 170 uppercase `xcv_` names. The table is generated from `include/proven/alias_xcv.h`; `tests/test_docs_alias_completeness` fails the build if a public function has no alias. That gate compares the headers with each other, not with this appendix - the table had fallen 61 rows behind the header before it was regenerated, so regenerate it from the header rather than editing rows by hand.

There is deliberately no line-number column. It was wrong after every alias that got inserted above it, which is worse than having no column at all.

| Alias macro                     | Expands to                         |
|---------------------------------|------------------------------------|
| <code>XCV_ARG</code>            | <code>PROVEN_ARG</code>            |
| <code>XCV_ARG_CSTR</code>       | <code>proven_arg_cstr</code>       |
| <code>XCV_ARG_DATETIME</code>   | <code>proven_arg_datetime</code>   |
| <code>XCV_ARG_F64</code>        | <code>proven_arg_f64</code>        |
| <code>XCV_ARG_FN</code>         | <code>PROVEN_ARG_FN(f)</code>      |
| <code>XCV_ARG_I32</code>        | <code>proven_arg_i32</code>        |
| <code>XCV_ARG_I64</code>        | <code>proven_arg_i64</code>        |
| <code>XCV_ARG_NONE</code>       | <code>proven_arg_none</code>       |
| <code>XCV_ARG_PTR</code>        | <code>proven_arg_ptr</code>        |
| <code>XCV_ARG_STR_VIEW</code>   | <code>proven_arg_str_view</code>   |
| <code>XCV_ARG_U16</code>        | <code>proven_arg_u16</code>        |
| <code>XCV_ARG_U32</code>        | <code>proven_arg_u32</code>        |
| <code>XCV_ARG_U64</code>        | <code>proven_arg_u64</code>        |
| <code>XCV_ARRAY_DESTROY</code>  | <code>PROVEN_ARRAY_DESTROY</code>  |
| <code>XCV_ARRAY_GET</code>      | <code>PROVEN_ARRAY_GET</code>      |
| <code>XCV_ARRAY_GET_MUT</code>  | <code>PROVEN_ARRAY_GET_MUT</code>  |
| <code>XCV_ARRAY_INIT</code>     | <code>PROVEN_ARRAY_INIT</code>     |
| <code>XCV_ARRAY_IS_VALID</code> | <code>proven_array_is_valid</code> |
| <code>XCV_ARRAY_POP</code>      | <code>PROVEN_ARRAY_POP</code>      |

|                          |                             |
|--------------------------|-----------------------------|
| XCV_ARRAY_PUSH           | PROVEN_ARRAY_PUSH           |
| XCV_CKD_ADD              | PROVEN_CKD_ADD              |
| XCV_CKD_MUL              | PROVEN_CKD_MUL              |
| XCV_CKD_SUB              | PROVEN_CKD_SUB              |
| XCV_CONTAINER_OF         | PROVEN_CONTAINER_OF         |
| XCV_CORO_AWAIT           | PROVEN_CORO_AWAIT           |
| XCV_CORO_BEGIN           | PROVEN_CORO_BEGIN           |
| XCV_CORO_END             | PROVEN_CORO_END             |
| XCV_CORO_INIT            | PROVEN_CORO_INIT            |
| XCV_CORO_IS_DONE         | PROVEN_CORO_IS_DONE         |
| XCV_CORO_YIELD           | PROVEN_CORO_YIELD           |
| XCV_DEFAULT_ALIGNMENT    | PROVEN_DEFAULT_ALIGNMENT    |
| XCV_ERR_AGAIN            | PROVEN_ERR_AGAIN            |
| XCV_ERR_BUSY             | PROVEN_ERR_BUSY             |
| XCV_ERR_EOF              | PROVEN_ERR_EOF              |
| XCV_ERR_EXISTS           | PROVEN_ERR_EXISTS           |
| XCV_ERR_INVALID_ARG      | PROVEN_ERR_INVALID_ARG      |
| XCV_ERR_INVALID_ENCODING | PROVEN_ERR_INVALID_ENCODING |
| XCV_ERR_INVALID_FORMAT   | PROVEN_ERR_INVALID_FORMAT   |
| XCV_ERR_INVALID_STATE    | PROVEN_ERR_INVALID_STATE    |
| XCV_ERR_IO               | PROVEN_ERR_IO               |
| XCV_ERR_LAST             | PROVEN_ERR_LAST             |
| XCV_ERR_NEED_MORE        | PROVEN_ERR_NEED_MORE        |
| XCV_ERR_NOMEM            | PROVEN_ERR_NOMEM            |
| XCV_ERR_NOT_FOUND        | PROVEN_ERR_NOT_FOUND        |
| XCV_ERR_OUT_OF_BOUNDS    | PROVEN_ERR_OUT_OF_BOUNDS    |
| XCV_ERR_OVERFLOW         | PROVEN_ERR_OVERFLOW         |
| XCV_ERR_PERMISSION       | PROVEN_ERR_PERMISSION       |
| XCV_ERR_REFUSED          | PROVEN_ERR_REFUSED          |
| XCV_ERR_RESERVED_END     | PROVEN_ERR_RESERVED_END     |
| XCV_ERR_RESET            | PROVEN_ERR_RESET            |
| XCV_ERR_TIMEOUT          | PROVEN_ERR_TIMEOUT          |
| XCV_ERR_UNREACHABLE      | PROVEN_ERR_UNREACHABLE      |
| XCV_ERR_UNSUPPORTED      | PROVEN_ERR_UNSUPPORTED      |
| XCV_ERR_UNTRUSTED        | PROVEN_ERR_UNTRUSTED        |
| XCV_FS_APPEND            | PROVEN_FS_APPEND            |
| XCV_FS_CREATE            | PROVEN_FS_CREATE            |
| XCV_FS_LOCK_EXCLUSIVE    | PROVEN_FS_LOCK_EXCLUSIVE    |
| XCV_FS_LOCK_SHARED       | PROVEN_FS_LOCK_SHARED       |
| XCV_FS_LOCK_UNLOCK       | PROVEN_FS_LOCK_UNLOCK       |
| XCV_FS_PATH_SEP          | PROVEN_FS_PATH_SEP          |
| XCV_FS_PERM_DEFAULT      | PROVEN_FS_PERM_DEFAULT      |
| XCV_FS_PERM_GROUP_R      | PROVEN_FS_PERM_GROUP_R      |
| XCV_FS_PERM_GROUP_W      | PROVEN_FS_PERM_GROUP_W      |
| XCV_FS_PERM_GROUP_X      | PROVEN_FS_PERM_GROUP_X      |

|                             |                                |
|-----------------------------|--------------------------------|
| XCV_FS_PERM_OTHER_R         | PROVEN_FS_PERM_OTHER_R         |
| XCV_FS_PERM_OTHER_W         | PROVEN_FS_PERM_OTHER_W         |
| XCV_FS_PERM_OTHER_X         | PROVEN_FS_PERM_OTHER_X         |
| XCV_FS_PERM_OWNER_R         | PROVEN_FS_PERM_OWNER_R         |
| XCV_FS_PERM_OWNER_W         | PROVEN_FS_PERM_OWNER_W         |
| XCV_FS_PERM_OWNER_X         | PROVEN_FS_PERM_OWNER_X         |
| XCV_FS_READ                 | PROVEN_FS_READ                 |
| XCV_FS_TRUNC                | PROVEN_FS_TRUNC                |
| XCV_FS_TYPE_DIR             | PROVEN_FS_TYPE_DIR             |
| XCV_FS_TYPE_FILE            | PROVEN_FS_TYPE_FILE            |
| XCV_FS_TYPE_OTHER           | PROVEN_FS_TYPE_OTHER           |
| XCV_FS_WRITE                | PROVEN_FS_WRITE                |
| XCV_HTTP_BODY_CHUNKED       | PROVEN_HTTP_BODY_CHUNKED       |
| XCV_HTTP_BODY_LENGTH        | PROVEN_HTTP_BODY_LENGTH        |
| XCV_HTTP_BODY_NONE          | PROVEN_HTTP_BODY_NONE          |
| XCV_HTTP_BODY_UNTIL_CLOSE   | PROVEN_HTTP_BODY_UNTIL_CLOSE   |
| XCV_HTTP_CONNECT            | PROVEN_HTTP_CONNECT            |
| XCV_HTTP_DATE_SIZE          | PROVEN_HTTP_DATE_SIZE          |
| XCV_HTTP_DEFAULT_MAX_HEAD   | PROVEN_HTTP_DEFAULT_MAX_HEAD   |
| XCV_HTTP_DELETE             | PROVEN_HTTP_DELETE             |
| XCV_HTTP_GET                | PROVEN_HTTP_GET                |
| XCV_HTTP_HEAD               | PROVEN_HTTP_HEAD               |
| XCV_HTTP_MAX_CHUNK_LINE     | PROVEN_HTTP_MAX_CHUNK_LINE     |
| XCV_HTTP_MAX_TRAILER_BYTES  | PROVEN_HTTP_MAX_TRAILER_BYTES  |
| XCV_HTTP_METHOD_OTHER       | PROVEN_HTTP_METHOD_OTHER       |
| XCV_HTTP_OPTIONS            | PROVEN_HTTP_OPTIONS            |
| XCV_HTTP_PATCH              | PROVEN_HTTP_PATCH              |
| XCV_HTTP_POST               | PROVEN_HTTP_POST               |
| XCV_HTTP_PUT                | PROVEN_HTTP_PUT                |
| XCV_HTTP_TRACE              | PROVEN_HTTP_TRACE              |
| XCV_INDEX_NOT_FOUND         | PROVEN_INDEX_NOT_FOUND         |
| XCV_IS_OK                   | PROVEN_IS_OK                   |
| XCV_KEY_TYPE_INT            | PROVEN_KEY_TYPE_INT            |
| XCV_KEY_TYPE_U8_BORROWED    | PROVEN_KEY_TYPE_U8_BORROWED    |
| XCV_KEY_TYPE_U8_OWNED       | PROVEN_KEY_TYPE_U8_OWNED       |
| XCV_LIST_ENTRY              | PROVEN_LIST_ENTRY              |
| XCV_LIST_FOR_EACH           | PROVEN_LIST_FOR_EACH           |
| XCV_LIST_FOR_EACH_SAFE      | PROVEN_LIST_FOR_EACH_SAFE      |
| XCV_LIT                     | PROVEN_LIT                     |
| XCV_MAP_DESTROY             | PROVEN_MAP_DESTROY             |
| XCV_MAP_GET_INT             | PROVEN_MAP_GET_INT             |
| XCV_MAP_GET_MUT_INT         | PROVEN_MAP_GET_MUT_INT         |
| XCV_MAP_GET_MUT_U8_BORROWED | PROVEN_MAP_GET_MUT_U8_BORROWED |
| XCV_MAP_GET_U8_BORROWED     | PROVEN_MAP_GET_U8_BORROWED     |
| XCV_MAP_GET_U8_OWNED        | PROVEN_MAP_GET_U8_OWNED        |

|                                      |                                         |
|--------------------------------------|-----------------------------------------|
| XCV_MAP_INIT_INT                     | PROVEN_MAP_INIT_INT                     |
| XCV_MAP_INIT_U8_BORROWED             | PROVEN_MAP_INIT_U8_BORROWED             |
| XCV_MAP_INIT_U8_OWNED                | PROVEN_MAP_INIT_U8_OWNED                |
| XCV_MAP_IS_VALID                     | proven_map_is_valid                     |
| XCV_MAP_REMOVE_INT                   | PROVEN_MAP_REMOVE_INT                   |
| XCV_MAP_REMOVE_U8_BORROWED           | PROVEN_MAP_REMOVE_U8_BORROWED           |
| XCV_MAP_REMOVE_U8_OWNED              | PROVEN_MAP_REMOVE_U8_OWNED              |
| XCV_MAP_SET_INT                      | PROVEN_MAP_SET_INT                      |
| XCV_MAP_SET_U8_BORROWED              | PROVEN_MAP_SET_U8_BORROWED              |
| XCV_MAP_SET_U8_OWNED                 | PROVEN_MAP_SET_U8_OWNED                 |
| XCV_MAP_SET_WITH_SCRATCH_INT         | PROVEN_MAP_SET_WITH_SCRATCH_INT         |
| XCV_MAP_SET_WITH_SCRATCH_U8_BORROWED | PROVEN_MAP_SET_WITH_SCRATCH_U8_BORROWED |
| XCV_MAX_ALIGN                        | PROVEN_MAX_ALIGN                        |
| XCV_MMAP_EXEC                        | PROVEN_MMAP_EXEC                        |
| XCV_MMAP_PRIVATE                     | PROVEN_MMAP_PRIVATE                     |
| XCV_MMAP_READ                        | PROVEN_MMAP_READ                        |
| XCV_MMAP_SHARED                      | PROVEN_MMAP_SHARED                      |
| XCV_MMAP_WRITE                       | PROVEN_MMAP_WRITE                       |
| XCV_NET_ADDR_TEXT_MAX                | PROVEN_NET_ADDR_TEXT_MAX                |
| XCV_NET_DONT_WAIT                    | PROVEN_NET_DONT_WAIT                    |
| XCV_NET_FAILED                       | PROVEN_NET_FAILED                       |
| XCV_NET_FAMILY_IPV4                  | PROVEN_NET_FAMILY_IPV4                  |
| XCV_NET_FAMILY_IPV6                  | PROVEN_NET_FAMILY_IPV6                  |
| XCV_NET_FAMILY_NONE                  | PROVEN_NET_FAMILY_NONE                  |
| XCV_NET_FAMILY_UNIX                  | PROVEN_NET_FAMILY_UNIX                  |
| XCV_NET_NO_DEADLINE                  | PROVEN_NET_NO_DEADLINE                  |
| XCV_NET_POLL_INLINE_MAX              | PROVEN_NET_POLL_INLINE_MAX              |
| XCV_NET_READABLE                     | PROVEN_NET_READABLE                     |
| XCV_NET_UNIX_PATH_MAX                | PROVEN_NET_UNIX_PATH_MAX                |
| XCV_NET_WRITABLE                     | PROVEN_NET_WRITABLE                     |
| XCV_OK                               | PROVEN_OK                               |
| XCV_RING_DESTROY                     | PROVEN_RING_DESTROY                     |
| XCV_RING_INIT                        | PROVEN_RING_INIT                        |
| XCV_RING_IS_VALID                    | proven_ring_is_valid                    |
| XCV_RING_POP                         | PROVEN_RING_POP                         |
| XCV_RING_PUSH                        | PROVEN_RING_PUSH                        |
| XCV_SCAN_ARG                         | PROVEN_SCAN_ARG(x)                      |
| XCV_SCAN_ARG_LONG                    | PROVEN_SCAN_ARG_LONG(ptr)               |
| XCV_SCAN_ARG_TYPE_F64                | PROVEN_SCAN_ARG_TYPE_F64                |
| XCV_SCAN_ARG_TYPE_I32                | PROVEN_SCAN_ARG_TYPE_I32                |
| XCV_SCAN_ARG_TYPE_I64                | PROVEN_SCAN_ARG_TYPE_I64                |
| XCV_SCAN_ARG_TYPE_INT                | PROVEN_SCAN_ARG_TYPE_INT                |
| XCV_SCAN_ARG_TYPE_LLONG              | PROVEN_SCAN_ARG_TYPE_LLONG              |
| XCV_SCAN_ARG_TYPE_LONG               | PROVEN_SCAN_ARG_TYPE_LONG               |
| XCV_SCAN_ARG_TYPE_NONE               | PROVEN_SCAN_ARG_TYPE_NONE               |

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| XCV_SCAN_ARG_TYPE_SHORT          | PROVEN_SCAN_ARG_TYPE_SHORT                 |
| XCV_SCAN_ARG_TYPE_STR_VIEW       | PROVEN_SCAN_ARG_TYPE_STR_VIEW              |
| XCV_SCAN_ARG_TYPE_U32            | PROVEN_SCAN_ARG_TYPE_U32                   |
| XCV_SCAN_ARG_TYPE_U64            | PROVEN_SCAN_ARG_TYPE_U64                   |
| XCV_SCAN_ARG_TYPE_UINT           | PROVEN_SCAN_ARG_TYPE_UINT                  |
| XCV_SCAN_ARG_TYPE_ULLONG         | PROVEN_SCAN_ARG_TYPE_ULLONG                |
| XCV_SCAN_ARG_TYPE_ULONG          | PROVEN_SCAN_ARG_TYPE_ULONG                 |
| XCV_SCAN_ARG_TYPE_USHORT         | PROVEN_SCAN_ARG_TYPE_USHORT                |
| XCV_SCAN_ARG_ULONG               | PROVEN_SCAN_ARG_ULONG(ptr)                 |
| XCV_U16_LIT                      | PROVEN_U16_LIT                             |
| XCV_VERSION_ENCODE               | PROVEN_VERSION_ENCODE(major, minor, patch) |
| XCV_VERSION_MAJOR                | PROVEN_VERSION_MAJOR                       |
| XCV_VERSION_MINOR                | PROVEN_VERSION_MINOR                       |
| XCV_VERSION_NUM                  | PROVEN_VERSION_NUM                         |
| XCV_VERSION_PATCH                | PROVEN_VERSION_PATCH                       |
| XCV_VERSION_STRING               | PROVEN_VERSION_STRING                      |
| xcv_alloc_check_entry_t          | proven_alloc_check_entry_t                 |
| xcv_alloc_check_live             | proven_alloc_check_live                    |
| xcv_alloc_check_owns             | proven_alloc_check_owns                    |
| xcv_alloc_check_t                | proven_alloc_check_t                       |
| xcv_alloc_check_wrap             | proven_alloc_check_wrap                    |
| xcv_alloc_checked                | proven_alloc_checked                       |
| xcv_alloc_fn_t                   | proven_alloc_fn_t                          |
| xcv_alloc_is_valid               | proven_alloc_is_valid                      |
| xcv_allocator_t                  | proven_allocator_t                         |
| xcv_arena_alloc                  | proven_arena_alloc                         |
| xcv_arena_alloc_aligned          | proven_arena_alloc_aligned                 |
| xcv_arena_alloc_aligned_or_panic | proven_arena_alloc_aligned_or_panic        |
| xcv_arena_alloc_or_panic         | proven_arena_alloc_or_panic                |
| xcv_arena_alloc_trait            | proven_arena_alloc_trait                   |
| xcv_arena_as_allocator           | proven_arena_as_allocator                  |
| xcv_arena_create                 | proven_arena_create                        |
| xcv_arena_destroy                | proven_arena_destroy                       |
| xcv_arena_free_trait             | proven_arena_free_trait                    |
| xcv_arena_realloc_aligned        | proven_arena_realloc_aligned               |
| xcv_arena_realloc_trait          | proven_arena_realloc_trait                 |
| xcv_arena_reset                  | proven_arena_reset                         |
| xcv_arena_t                      | proven_arena_t                             |
| xcv_arg_bool                     | proven_arg_bool                            |
| xcv_arg_char                     | proven_arg_char                            |
| xcv_arg_cstr                     | proven_arg_cstr                            |
| xcv_arg_cstr_n                   | proven_arg_cstr_n                          |
| xcv_arg_custom                   | proven_arg_custom                          |
| xcv_arg_datetime                 | proven_arg_datetime                        |
| xcv_arg_f64                      | proven_arg_f64                             |

|                         |                            |
|-------------------------|----------------------------|
| xcv_arg_fn              | proven_arg_fn              |
| xcv_arg_i32             | proven_arg_i32             |
| xcv_arg_i64             | proven_arg_i64             |
| xcv_arg_identity        | proven_arg_identity        |
| xcv_arg_none            | proven_arg_none            |
| xcv_arg_ptr             | proven_arg_ptr             |
| xcv_arg_str_view        | proven_arg_str_view        |
| xcv_arg_t               | proven_arg_t               |
| xcv_arg_type_t          | proven_arg_type_t          |
| xcv_arg_u16             | proven_arg_u16             |
| xcv_arg_u32             | proven_arg_u32             |
| xcv_arg_u64             | proven_arg_u64             |
| xcv_arg_ucstr           | proven_arg_ucstr           |
| xcv_array_binary_search | proven_array_binary_search |
| xcv_array_clear         | proven_array_clear         |
| xcv_array_create        | proven_array_create        |
| xcv_array_destroy       | proven_array_destroy       |
| xcv_array_extend        | proven_array_extend        |
| xcv_array_get           | proven_array_get           |
| xcv_array_get_mut       | proven_array_get_mut       |
| xcv_array_insert        | proven_array_insert        |
| xcv_array_is_valid      | proven_array_is_valid      |
| xcv_array_linear_search | proven_array_linear_search |
| xcv_array_pop           | proven_array_pop           |
| xcv_array_push          | proven_array_push          |
| xcv_array_remove_at     | proven_array_remove_at     |
| xcv_array_reserve       | proven_array_reserve       |
| xcv_array_sort          | proven_array_sort          |
| xcv_array_swap_remove   | proven_array_swap_remove   |
| xcv_array_t             | proven_array_t             |
| xcv_array_truncate      | proven_array_truncate      |
| xcv_base64_decode       | proven_base64_decode       |
| xcv_base64_decoded_size | proven_base64_decoded_size |
| xcv_base64_encode       | proven_base64_encode       |
| xcv_base64_encoded_size | proven_base64_encoded_size |
| xcv_base64url_encode    | proven_base64url_encode    |
| xcv_buf_append          | proven_buf_append          |
| xcv_buf_create          | proven_buf_create          |
| xcv_buf_destroy         | proven_buf_destroy         |
| xcv_buf_t               | proven_buf_t               |
| xcv_byte_t              | proven_byte_t              |
| xcv_c_lib               | proven_c_lib               |
| xcv_chacha_rng          | proven_chacha_rng          |
| xcv_chacha_rng_fill     | proven_chacha_rng_fill     |
| xcv_chacha_rng_next     | proven_chacha_rng_next     |

|                                        |                                           |
|----------------------------------------|-------------------------------------------|
| xcv_chacha_rng_seed                    | proven_chacha_rng_seed                    |
| xcv_chacha_rng_seed_from_entropy       | proven_chacha_rng_seed_from_entropy       |
| xcv_compare_fn_t                       | proven_compare_fn_t                       |
| xcv_coro_t                             | proven_coro_t                             |
| xcv_crc32                              | proven_crc32                              |
| xcv_crc32_update                       | proven_crc32_update                       |
| xcv_cstr_len                           | proven_cstr_len                           |
| xcv_datetime_t                         | proven_datetime_t                         |
| xcv_entropy_fn                         | proven_entropy_fn                         |
| xcv_env_get                            | proven_env_get                            |
| xcv_eprint                             | proven_eprint                             |
| xcv_eprintln                           | proven_eprintln                           |
| xcv_err_t                              | proven_err_t                              |
| xcv_file_t                             | proven_file_t                             |
| xcv_float_format_f32_policy            | proven_float_format_f32_policy            |
| xcv_float_format_f64_policy            | proven_float_format_f64_policy            |
| xcv_float_format_options_fixed_default | proven_float_format_options_fixed_default |
| xcv_float_format_options_scientific    | proven_float_format_options_scientific    |
| xcv_float_format_options_shortest      | proven_float_format_options_shortest      |
| xcv_fmt_put                            | proven_fmt_put                            |
| xcv_fmt_to_writer_impl                 | proven_fmt_to_writer_impl                 |
| xcv_fprint                             | proven_fprint                             |
| xcv_free_fn_t                          | proven_free_fn_t                          |
| xcv_fs_chmod                           | proven_fs_chmod                           |
| xcv_fs_close                           | proven_fs_close                           |
| xcv_fs_copy                            | proven_fs_copy                            |
| xcv_fs_dir_close                       | proven_fs_dir_close                       |
| xcv_fs_dir_next                        | proven_fs_dir_next                        |
| xcv_fs_dir_open                        | proven_fs_dir_open                        |
| xcv_fs_entry_t                         | proven_fs_entry_t                         |
| xcv_fs_handle_t                        | proven_fs_handle_t                        |
| xcv_fs_is_absolute                     | proven_fs_is_absolute                     |
| xcv_fs_is_staging_name                 | proven_fs_is_staging_name                 |
| xcv_fs_link                            | proven_fs_link                            |
| xcv_fs_list                            | proven_fs_list                            |
| xcv_fs_list_destroy                    | proven_fs_list_destroy                    |
| xcv_fs_lock                            | proven_fs_lock                            |
| xcv_fs_lock_type_t                     | proven_fs_lock_type_t                     |
| xcv_fs_mkdir                           | proven_fs_mkdir                           |
| xcv_fs_mkdir_all                       | proven_fs_mkdir_all                       |
| xcv_fs_mode_t                          | proven_fs_mode_t                          |
| xcv_fs_open                            | proven_fs_open                            |
| xcv_fs_perms_t                         | proven_fs_perms_t                         |
| xcv_fs_pread                           | proven_fs_pread                           |
| xcv_fs_pwrite                          | proven_fs_pwrite                          |

|                           |                              |
|---------------------------|------------------------------|
| xcv_fs_read               | proven_fs_read               |
| xcv_fs_read_all           | proven_fs_read_all           |
| xcv_fs_read_all_bounded   | proven_fs_read_all_bounded   |
| xcv_fs_read_all_u8str     | proven_fs_read_all_u8str     |
| xcv_fs_remove             | proven_fs_remove             |
| xcv_fs_rename             | proven_fs_rename             |
| xcv_fs_rmdir              | proven_fs_rmdir              |
| xcv_fs_seek               | proven_fs_seek               |
| xcv_fs_size               | proven_fs_size               |
| xcv_fs_stat               | proven_fs_stat               |
| xcv_fs_stat_t             | proven_fs_stat_t             |
| xcv_fs_symlink            | proven_fs_symlink            |
| xcv_fs_sync               | proven_fs_sync               |
| xcv_fs_sync_dir           | proven_fs_sync_dir           |
| xcv_fs_tell               | proven_fs_tell               |
| xcv_fs_truncate           | proven_fs_truncate           |
| xcv_fs_type_t             | proven_fs_type_t             |
| xcv_fs_walk_close         | proven_fs_walk_close         |
| xcv_fs_walk_next          | proven_fs_walk_next          |
| xcv_fs_walk_open          | proven_fs_walk_open          |
| xcv_fs_write              | proven_fs_write              |
| xcv_fs_write_all          | proven_fs_write_all          |
| xcv_fs_write_file         | proven_fs_write_file         |
| xcv_fs_write_file_atomic  | proven_fs_write_file_atomic  |
| xcv_fs_write_file_durable | proven_fs_write_file_durable |
| xcv_hash_bytes            | proven_hash_bytes            |
| xcv_hash_keyed            | proven_hash_keyed            |
| xcv_heap_allocator        | proven_heap_allocator        |
| xcv_hex_decode            | proven_hex_decode            |
| xcv_hex_decoded_size      | proven_hex_decoded_size      |
| xcv_hex_encode            | proven_hex_encode            |
| xcv_hex_encoded_size      | proven_hex_encoded_size      |
| xcv_http_body_end         | proven_http_body_end         |
| xcv_http_body_feed        | proven_http_body_feed        |
| xcv_http_body_init        | proven_http_body_init        |
| xcv_http_body_kind_t      | proven_http_body_kind_t      |
| xcv_http_body_received    | proven_http_body_received    |
| xcv_http_body_t           | proven_http_body_t           |
| xcv_http_date_format      | proven_http_date_format      |
| xcv_http_date_parse       | proven_http_date_parse       |
| xcv_http_framing_t        | proven_http_framing_t        |
| xcv_http_header_count     | proven_http_header_count     |
| xcv_http_header_find      | proven_http_header_find      |
| xcv_http_header_has_token | proven_http_header_has_token |
| xcv_http_header_t         | proven_http_header_t         |

|                              |                                 |
|------------------------------|---------------------------------|
| xcv_http_method_from_text    | proven_http_method_from_text    |
| xcv_http_method_t            | proven_http_method_t            |
| xcv_http_method_text         | proven_http_method_text         |
| xcv_http_parse_request       | proven_http_parse_request       |
| xcv_http_parse_response      | proven_http_parse_response      |
| xcv_http_reason_phrase       | proven_http_reason_phrase       |
| xcv_http_request_framing     | proven_http_request_framing     |
| xcv_http_request_keep_alive  | proven_http_request_keep_alive  |
| xcv_http_request_t           | proven_http_request_t           |
| xcv_http_response_framing    | proven_http_response_framing    |
| xcv_http_response_keep_alive | proven_http_response_keep_alive |
| xcv_http_response_t          | proven_http_response_t          |
| xcv_http_write_chunk_begin   | proven_http_write_chunk_begin   |
| xcv_http_write_chunk_end     | proven_http_write_chunk_end     |
| xcv_http_write_head_end      | proven_http_write_head_end      |
| xcv_http_write_header        | proven_http_write_header        |
| xcv_http_write_header_u64    | proven_http_write_header_u64    |
| xcv_http_write_last_chunk    | proven_http_write_last_chunk    |
| xcv_http_write_request_line  | proven_http_write_request_line  |
| xcv_http_write_status_line   | proven_http_write_status_line   |
| xcv_i16                      | proven_i16                      |
| xcv_i32                      | proven_i32                      |
| xcv_i64                      | proven_i64                      |
| xcv_i8                       | proven_i8                       |
| xcv_intptr_t                 | proven_intptr_t                 |
| xcv_is_ok                    | proven_is_ok                    |
| xcv_is_pow2                  | proven_is_pow2                  |
| xcv_job_execute_one          | proven_job_execute_one          |
| xcv_job_group_init           | proven_job_group_init           |
| xcv_job_group_pending        | proven_job_group_pending        |
| xcv_job_group_submit         | proven_job_group_submit         |
| xcv_job_group_t              | proven_job_group_t              |
| xcv_job_group_wait           | proven_job_group_wait           |
| xcv_job_submit               | proven_job_submit               |
| xcv_job_submit_ex            | proven_job_submit_ex            |
| xcv_job_sys                  | proven_job_sys                  |
| xcv_job_sys_t                | proven_job_sys_t                |
| xcv_job_system_close         | proven_job_system_close         |
| xcv_job_system_destroy       | proven_job_system_destroy       |
| xcv_job_system_init          | proven_job_system_init          |
| xcv_job_t                    | proven_job_t                    |
| xcv_key_type_t               | proven_key_type_t               |
| xcv_list_init                | proven_list_init                |
| xcv_list_insert_after        | proven_list_insert_after        |
| xcv_list_is_empty            | proven_list_is_empty            |

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|------------------------------|---------------------------------|
| xcv_list_node_t              | proven_list_node_t              |
| xcv_list_push_back           | proven_list_push_back           |
| xcv_list_remove              | proven_list_remove              |
| xcv_list_t                   | proven_list_t                   |
| xcv_map_create               | proven_map_create               |
| xcv_map_create_trusted       | proven_map_create_trusted       |
| xcv_map_create_with_capacity | proven_map_create_with_capacity |
| xcv_map_destroy              | proven_map_destroy              |
| xcv_map_get                  | proven_map_get                  |
| xcv_map_get_mut              | proven_map_get_mut              |
| xcv_map_hash                 | proven_map_hash                 |
| xcv_map_is_valid             | proven_map_is_valid             |
| xcv_map_iter_init            | proven_map_iter_init            |
| xcv_map_iter_next            | proven_map_iter_next            |
| xcv_map_iter_t               | proven_map_iter_t               |
| xcv_map_key_t                | proven_map_key_t                |
| xcv_map_len                  | proven_map_len                  |
| xcv_map_remove               | proven_map_remove               |
| xcv_map_reserve              | proven_map_reserve              |
| xcv_map_set                  | proven_map_set                  |
| xcv_map_set_u8_owned         | proven_map_set_u8_owned         |
| xcv_map_set_with_scratch     | proven_map_set_with_scratch     |
| xcv_map_t                    | proven_map_t                    |
| xcv_md5                      | proven_md5                      |
| xcv_md5_final                | proven_md5_final                |
| xcv_md5_init                 | proven_md5_init                 |
| xcv_md5_to_hex               | proven_md5_to_hex               |
| xcv_md5_update               | proven_md5_update               |
| xcv_mem_align_up             | proven_mem_align_up             |
| xcv_mem_copy                 | proven_mem_copy                 |
| xcv_mem_move                 | proven_mem_move                 |
| xcv_mem_mut_from_owned       | proven_mem_mut_from_owned       |
| xcv_mem_mut_slice_checked    | proven_mem_mut_slice_checked    |
| xcv_mem_mut_slice_unchecked  | proven_mem_mut_slice_unchecked  |
| xcv_mem_mut_t                | proven_mem_mut_t                |
| xcv_mem_t                    | proven_mem_t                    |
| xcv_mem_view_from_owned      | proven_mem_view_from_owned      |
| xcv_mem_view_from_u8         | proven_mem_view_from_u8         |
| xcv_mem_view_slice_checked   | proven_mem_view_slice_checked   |
| xcv_mem_view_slice_unchecked | proven_mem_view_slice_unchecked |
| xcv_mem_view_t               | proven_mem_view_t               |
| xcv_memcmp                   | proven_memcmp                   |
| xcv_mmap_as_view             | proven_mmap_as_view             |
| xcv_mmap_create              | proven_mmap_create              |
| xcv_mmap_destroy             | proven_mmap_destroy             |

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|---------------------------|------------------------------|
| xcv_mmap_flags_t          | proven_mmap_flags_t          |
| xcv_mmap_prot_t           | proven_mmap_prot_t           |
| xcv_mmap_sync             | proven_mmap_sync             |
| xcv_mmap_t                | proven_mmap_t                |
| xcv_net_accept            | proven_net_accept            |
| xcv_net_addr_any          | proven_net_addr_any          |
| xcv_net_addr_eq           | proven_net_addr_eq           |
| xcv_net_addr_format       | proven_net_addr_format       |
| xcv_net_addr_ipv4         | proven_net_addr_ipv4         |
| xcv_net_addr_ipv6         | proven_net_addr_ipv6         |
| xcv_net_addr_loopback     | proven_net_addr_loopback     |
| xcv_net_addr_parse        | proven_net_addr_parse        |
| xcv_net_addr_t            | proven_net_addr_t            |
| xcv_net_addr_unix         | proven_net_addr_unix         |
| xcv_net_close             | proven_net_close             |
| xcv_net_conn_handle       | proven_net_conn_handle       |
| xcv_net_conn_is_open      | proven_net_conn_is_open      |
| xcv_net_conn_local_addr   | proven_net_conn_local_addr   |
| xcv_net_conn_peer_addr    | proven_net_conn_peer_addr    |
| xcv_net_conn_set_nodelay  | proven_net_conn_set_nodelay  |
| xcv_net_conn_t            | proven_net_conn_t            |
| xcv_net_conn_transport    | proven_net_conn_transport    |
| xcv_net_connect           | proven_net_connect           |
| xcv_net_deadline_in       | proven_net_deadline_in       |
| xcv_net_deadline_t        | proven_net_deadline_t        |
| xcv_net_event_t           | proven_net_event_t           |
| xcv_net_family_t          | proven_net_family_t          |
| xcv_net_handle_t          | proven_net_handle_t          |
| xcv_net_listen            | proven_net_listen            |
| xcv_net_listener_addr     | proven_net_listener_addr     |
| xcv_net_listener_close    | proven_net_listener_close    |
| xcv_net_listener_handle   | proven_net_listener_handle   |
| xcv_net_listener_is_open  | proven_net_listener_is_open  |
| xcv_net_listener_t        | proven_net_listener_t        |
| xcv_net_poll              | proven_net_poll              |
| xcv_net_poll_item_t       | proven_net_poll_item_t       |
| xcv_net_poll_scratch_size | proven_net_poll_scratch_size |
| xcv_net_poll_with         | proven_net_poll_with         |
| xcv_net_read              | proven_net_read              |
| xcv_net_resolve           | proven_net_resolve           |
| xcv_net_shutdown_write    | proven_net_shutdown_write    |
| xcv_net_udp_addr          | proven_net_udp_addr          |
| xcv_net_udp_close         | proven_net_udp_close         |
| xcv_net_udp_handle        | proven_net_udp_handle        |
| xcv_net_udp_is_open       | proven_net_udp_is_open       |

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|--------------------------|-----------------------------|
| xcv_net_udp_open         | proven_net_udp_open         |
| xcv_net_udp_recv_from    | proven_net_udp_recv_from    |
| xcv_net_udp_send_to      | proven_net_udp_send_to      |
| xcv_net_udp_t            | proven_net_udp_t            |
| xcv_net_write            | proven_net_write            |
| xcv_net_write_all        | proven_net_write_all        |
| xcv_panic                | proven_panic                |
| xcv_parse_double_ascii   | proven_parse_double_ascii   |
| xcv_parse_f64_ascii      | proven_parse_f64_ascii      |
| xcv_pool_as_allocator    | proven_pool_as_allocator    |
| xcv_pool_destroy         | proven_pool_destroy         |
| xcv_pool_init            | proven_pool_init            |
| xcv_pool_t               | proven_pool_t               |
| xcv_print                | proven_print                |
| xcv_println              | proven_println              |
| xcv_ptrdiff_t            | proven_ptrdiff_t            |
| xcv_random_bytes         | proven_random_bytes         |
| xcv_random_set_source    | proven_random_set_source    |
| xcv_random_u64           | proven_random_u64           |
| xcv_random_u64_checked   | proven_random_u64_checked   |
| xcv_range_contains_ptr   | proven_range_contains_ptr   |
| xcv_reader_buffered      | proven_reader_buffered      |
| xcv_reader_from_file     | proven_reader_from_file     |
| xcv_reader_from_view     | proven_reader_from_view     |
| xcv_reader_is_valid      | proven_reader_is_valid      |
| xcv_reader_read          | proven_reader_read          |
| xcv_reader_read_line     | proven_reader_read_line     |
| xcv_realloc_fn_t         | proven_realloc_fn_t         |
| xcv_result_array_t       | proven_result_array_t       |
| xcv_result_buf_t         | proven_result_buf_t         |
| xcv_result_cstr_t        | proven_result_cstr_t        |
| xcv_result_f64_t         | proven_result_f64_t         |
| xcv_result_file_t        | proven_result_file_t        |
| xcv_result_i64_t         | proven_result_i64_t         |
| xcv_result_map_t         | proven_result_map_t         |
| xcv_result_mem_mut_t     | proven_result_mem_mut_t     |
| xcv_result_mem_view_t    | proven_result_mem_view_t    |
| xcv_result_mmap_t        | proven_result_mmap_t        |
| xcv_result_ring_t        | proven_result_ring_t        |
| xcv_result_size_t        | proven_result_size_t        |
| xcv_result_u16str_t      | proven_result_u16str_t      |
| xcv_result_u16str_view_t | proven_result_u16str_view_t |
| xcv_result_u64_t         | proven_result_u64_t         |
| xcv_result_u8str_t       | proven_result_u8str_t       |
| xcv_result_u8str_view_t  | proven_result_u8str_view_t  |

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|----------------------------|-------------------------------|
| xcv_ring_create            | proven_ring_create            |
| xcv_ring_destroy           | proven_ring_destroy           |
| xcv_ring_is_valid          | proven_ring_is_valid          |
| xcv_ring_pop               | proven_ring_pop               |
| xcv_ring_push              | proven_ring_push              |
| xcv_ring_t                 | proven_ring_t                 |
| xcv_rng_below              | proven_rng_below              |
| xcv_rng_f64                | proven_rng_f64                |
| xcv_rng_fill               | proven_rng_fill               |
| xcv_rng_is_valid           | proven_rng_is_valid           |
| xcv_rng_range              | proven_rng_range              |
| xcv_rng_shuffle            | proven_rng_shuffle            |
| xcv_rng_u64                | proven_rng_u64                |
| xcv_scan_arg_f64           | proven_scan_arg_f64           |
| xcv_scan_arg_i32           | proven_scan_arg_i32           |
| xcv_scan_arg_i64           | proven_scan_arg_i64           |
| xcv_scan_arg_identity      | proven_scan_arg_identity      |
| xcv_scan_arg_int           | proven_scan_arg_int           |
| xcv_scan_arg_llong         | proven_scan_arg_llong         |
| xcv_scan_arg_long          | proven_scan_arg_long          |
| xcv_scan_arg_none          | proven_scan_arg_none          |
| xcv_scan_arg_short         | proven_scan_arg_short         |
| xcv_scan_arg_str_view      | proven_scan_arg_str_view      |
| xcv_scan_arg_t             | proven_scan_arg_t             |
| xcv_scan_arg_type_t        | proven_scan_arg_type_t        |
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| xcv_scan_arg_u64           | proven_scan_arg_u64           |
| xcv_scan_arg_uint          | proven_scan_arg_uint          |
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| xcv_scan_arg_ulong         | proven_scan_arg_ulong         |
| xcv_scan_arg_ushort        | proven_scan_arg_ushort        |
| xcv_scan_f64               | proven_scan_f64               |
| xcv_scan_fmt               | proven_scan_fmt               |
| xcv_scan_fmt_cursor        | proven_scan_fmt_cursor        |
| xcv_scan_fmt_from_file     | proven_scan_fmt_from_file     |
| xcv_scan_fmt_from_stdin    | proven_scan_fmt_from_stdin    |
| xcv_scan_fmt_internal      | proven_scan_fmt_internal      |
| xcv_scan_fmt_internal_view | proven_scan_fmt_internal_view |
| xcv_scan_i64               | proven_scan_i64               |
| xcv_scan_i64_hex           | proven_scan_i64_hex           |
| xcv_scan_init              | proven_scan_init              |
| xcv_scan_skip_until        | proven_scan_skip_until        |
| xcv_scan_skip_until_number | proven_scan_skip_until_number |
| xcv_scan_skip_whitespace   | proven_scan_skip_whitespace   |
| xcv_scan_str               | proven_scan_str               |

|                             |                                |
|-----------------------------|--------------------------------|
| xcv_scan_t                  | proven_scan_t                  |
| xcv_scan_u64                | proven_scan_u64                |
| xcv_scan_u64_hex            | proven_scan_u64_hex            |
| xcv_set_panic_handler       | proven_set_panic_handler       |
| xcv_sha1                    | proven_sha1                    |
| xcv_sha1_final              | proven_sha1_final              |
| xcv_sha1_init               | proven_sha1_init               |
| xcv_sha1_to_hex             | proven_sha1_to_hex             |
| xcv_sha1_update             | proven_sha1_update             |
| xcv_sha256                  | proven_sha256                  |
| xcv_sha256_final            | proven_sha256_final            |
| xcv_sha256_init             | proven_sha256_init             |
| xcv_sha256_to_hex           | proven_sha256_to_hex           |
| xcv_sha256_update           | proven_sha256_update           |
| xcv_size_t                  | proven_size_t                  |
| xcv_strtod                  | proven_strtod                  |
| xcv_sysio_file_buffered     | proven_sysio_file_buffered     |
| xcv_sysio_lines_open        | proven_sysio_lines_open        |
| xcv_sysio_lines_t           | proven_sysio_lines_t           |
| xcv_sysio_out_t             | proven_sysio_out_t             |
| xcv_sysio_print_impl        | proven_sysio_print_impl        |
| xcv_sysio_read_line         | proven_sysio_read_line         |
| xcv_sysio_read_u16_line     | proven_sysio_read_u16_line     |
| xcv_sysio_scan_chunk_impl   | proven_sysio_scan_chunk_impl   |
| xcv_sysio_scanner_deinit    | proven_sysio_scanner_deinit    |
| xcv_sysio_scanner_init      | proven_sysio_scanner_init      |
| xcv_sysio_scanner_scan_impl | proven_sysio_scanner_scan_impl |
| xcv_sysio_std_t             | proven_sysio_std_t             |
| xcv_sysio_stderr            | proven_sysio_stderr            |
| xcv_sysio_stderr_writer     | proven_sysio_stderr_writer     |
| xcv_sysio_stdin             | proven_sysio_stdin             |
| xcv_sysio_stdin_lines       | proven_sysio_stdin_lines       |
| xcv_sysio_stdin_reader      | proven_sysio_stdin_reader      |
| xcv_sysio_stdin_u16_lines   | proven_sysio_stdin_u16_lines   |
| xcv_sysio_stdout            | proven_sysio_stdout            |
| xcv_sysio_stdout_buffered   | proven_sysio_stdout_buffered   |
| xcv_sysio_stdout_writer     | proven_sysio_stdout_writer     |
| xcv_sysio_u16_lines_open    | proven_sysio_u16_lines_open    |
| xcv_sysio_u16_lines_t       | proven_sysio_u16_lines_t       |
| xcv_text_encoding_t         | proven_text_encoding_t         |
| xcv_time_breakdown          | proven_time_breakdown          |
| xcv_time_locale_en          | proven_time_locale_en          |
| xcv_time_locale_t           | proven_time_locale_t           |
| xcv_time_monotonic_now      | proven_time_monotonic_now      |
| xcv_time_now                | proven_time_now                |

|                                   |                                      |
|-----------------------------------|--------------------------------------|
| xcv_time_now_datetime             | proven_time_now_datetime             |
| xcv_time_sleep                    | proven_time_sleep                    |
| xcv_time_t                        | proven_time_t                        |
| xcv_time_u16_fmt                  | proven_time_u16_fmt                  |
| xcv_time_u8_fmt                   | proven_time_u8_fmt                   |
| xcv_transport_close               | proven_transport_close               |
| xcv_transport_is_valid            | proven_transport_is_valid            |
| xcv_transport_read                | proven_transport_read                |
| xcv_transport_reader              | proven_transport_reader              |
| xcv_transport_shutdown            | proven_transport_shutdown            |
| xcv_transport_stream_t            | proven_transport_stream_t            |
| xcv_transport_t                   | proven_transport_t                   |
| xcv_transport_write               | proven_transport_write               |
| xcv_transport_write_all           | proven_transport_write_all           |
| xcv_transport_writer              | proven_transport_writer              |
| xcv_u16                           | proven_u16                           |
| xcv_u16_reader_init               | proven_u16_reader_init               |
| xcv_u16_reader_read               | proven_u16_reader_read               |
| xcv_u16_reader_read_line          | proven_u16_reader_read_line          |
| xcv_u16_reader_t                  | proven_u16_reader_t                  |
| xcv_u16str_append                 | proven_u16str_append                 |
| xcv_u16str_append_grow            | proven_u16str_append_grow            |
| xcv_u16str_append_partial         | proven_u16str_append_partial         |
| xcv_u16str_as_ptr                 | proven_u16str_as_ptr                 |
| xcv_u16str_create                 | proven_u16str_create                 |
| xcv_u16str_create_from_view       | proven_u16str_create_from_view       |
| xcv_u16str_destroy                | proven_u16str_destroy                |
| xcv_u16str_len                    | proven_u16str_len                    |
| xcv_u16str_t                      | proven_u16str_t                      |
| xcv_u16str_view_t                 | proven_u16str_view_t                 |
| xcv_u32                           | proven_u32                           |
| xcv_u64                           | proven_u64                           |
| xcv_u8                            | proven_u8                            |
| xcv_u8str_append                  | proven_u8str_append                  |
| xcv_u8str_append_byte             | proven_u8str_append_byte             |
| xcv_u8str_append_fmt              | proven_u8str_append_fmt              |
| xcv_u8str_append_fmt_grow         | proven_u8str_append_fmt_grow         |
| xcv_u8str_append_fmt_trunc        | proven_u8str_append_fmt_trunc        |
| xcv_u8str_append_fmt_with_scratch | proven_u8str_append_fmt_with_scratch |
| xcv_u8str_append_grow             | proven_u8str_append_grow             |
| xcv_u8str_append_partial          | proven_u8str_append_partial          |
| xcv_u8str_as_cstr                 | proven_u8str_as_cstr                 |
| xcv_u8str_as_view                 | proven_u8str_as_view                 |
| xcv_u8str_borrow                  | proven_u8str_borrow                  |
| xcv_u8str_create                  | proven_u8str_create                  |

|                               |                                  |
|-------------------------------|----------------------------------|
| xcv_u8str_create_from_view    | proven_u8str_create_from_view    |
| xcv_u8str_destroy             | proven_u8str_destroy             |
| xcv_u8str_fmt_internal        | proven_u8str_fmt_internal        |
| xcv_u8str_insert              | proven_u8str_insert              |
| xcv_u8str_insert_grow         | proven_u8str_insert_grow         |
| xcv_u8str_is_valid            | proven_u8str_is_valid            |
| xcv_u8str_mut_t               | proven_u8str_mut_t               |
| xcv_u8str_remove              | proven_u8str_remove              |
| xcv_u8str_replace_at          | proven_u8str_replace_at          |
| xcv_u8str_replace_at_grow     | proven_u8str_replace_at_grow     |
| xcv_u8str_replace_first       | proven_u8str_replace_first       |
| xcv_u8str_reserve             | proven_u8str_reserve             |
| xcv_u8str_reset               | proven_u8str_reset               |
| xcv_u8str_t                   | proven_u8str_t                   |
| xcv_u8str_view_cmp            | proven_u8str_view_cmp            |
| xcv_u8str_view_cmp_ptr        | proven_u8str_view_cmp_ptr        |
| xcv_u8str_view_contains       | proven_u8str_view_contains       |
| xcv_u8str_view_ends_with      | proven_u8str_view_ends_with      |
| xcv_u8str_view_eq             | proven_u8str_view_eq             |
| xcv_u8str_view_find           | proven_u8str_view_find           |
| xcv_u8str_view_find_last      | proven_u8str_view_find_last      |
| xcv_u8str_view_from_cstr      | proven_u8str_view_from_cstr      |
| xcv_u8str_view_is_well_formed | proven_u8str_view_is_well_formed |
| xcv_u8str_view_remove_prefix  | proven_u8str_view_remove_prefix  |
| xcv_u8str_view_remove_suffix  | proven_u8str_view_remove_suffix  |
| xcv_u8str_view_slice          | proven_u8str_view_slice          |
| xcv_u8str_view_split          | proven_u8str_view_split          |
| xcv_u8str_view_split_next     | proven_u8str_view_split_next     |
| xcv_u8str_view_split_t        | proven_u8str_view_split_t        |
| xcv_u8str_view_starts_with    | proven_u8str_view_starts_with    |
| xcv_u8str_view_t              | proven_u8str_view_t              |
| xcv_u8str_view_to_cstr        | proven_u8str_view_to_cstr        |
| xcv_u8str_view_trim           | proven_u8str_view_trim           |
| xcv_u8str_view_trim_end       | proven_u8str_view_trim_end       |
| xcv_u8str_view_trim_start     | proven_u8str_view_trim_start     |
| xcv_uintptr_align_up          | proven_uintptr_align_up          |
| xcv_uintptr_t                 | proven_uintptr_t                 |
| xcv_url_default_port          | proven_url_default_port          |
| xcv_url_effective_port        | proven_url_effective_port        |
| xcv_url_encode_component      | proven_url_encode_component      |
| xcv_url_encode_path           | proven_url_encode_path           |
| xcv_url_encoded_size_max      | proven_url_encoded_size_max      |
| xcv_url_form_decode           | proven_url_form_decode           |
| xcv_url_form_encode           | proven_url_form_encode           |
| xcv_url_parse                 | proven_url_parse                 |

|                           |                              |
|---------------------------|------------------------------|
| xcv_url_path_resolve      | proven_url_path_resolve      |
| xcv_url_percent_decode    | proven_url_percent_decode    |
| xcv_url_query_iter        | proven_url_query_iter        |
| xcv_url_query_iter_t      | proven_url_query_iter_t      |
| xcv_url_query_next        | proven_url_query_next        |
| xcv_url_scheme_is         | proven_url_scheme_is         |
| xcv_url_split_target      | proven_url_split_target      |
| xcv_url_t                 | proven_url_t                 |
| xcv_utf16_append_to_u8str | proven_utf16_append_to_u8str |
| xcv_utf16_to_utf8         | proven_utf16_to_utf8         |
| xcv_utf16_to_utf8_partial | proven_utf16_to_utf8_partial |
| xcv_utf16_to_utf8_size    | proven_utf16_to_utf8_size    |
| xcv_utf8_append_to_u16str | proven_utf8_append_to_u16str |
| xcv_utf8_char_t           | proven_utf8_char_t           |
| xcv_utf8_decode_next      | proven_utf8_decode_next      |
| xcv_utf8_to_utf16         | proven_utf8_to_utf16         |
| xcv_utf8_to_utf16_partial | proven_utf8_to_utf16_partial |
| xcv_utf8_to_utf16_size    | proven_utf8_to_utf16_size    |
| xcv_utf_step_t            | proven_utf_step_t            |
| xcv_writer_buffered       | proven_writer_buffered       |
| xcv_writer_flush          | proven_writer_flush          |
| xcv_writer_from_buffer    | proven_writer_from_buffer    |
| xcv_writer_from_file      | proven_writer_from_file      |
| xcv_writer_from_u8str     | proven_writer_from_u8str     |
| xcv_writer_is_valid       | proven_writer_is_valid       |
| xcv_writer_write          | proven_writer_write          |
| xcv_writer_write_bom      | proven_writer_write_bom      |
| xcv_writer_write_partial  | proven_writer_write_partial  |
| xcv_writer_write_str      | proven_writer_write_str      |
| xcv_writer_write_u16      | proven_writer_write_u16      |
| xcv_xoshiro256ss_next     | proven_xoshiro256ss_next     |
| xcv_xoshiro256ss_rng      | proven_xoshiro256ss_rng      |
| xcv_xoshiro256ss_seed     | proven_xoshiro256ss_seed     |

